A TASTING TOUR OF PHYSICS

Assume a Spherical Cow

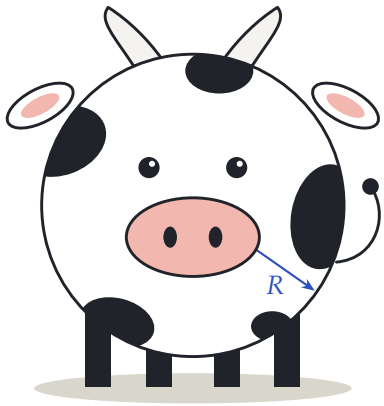


Figure 0. A cow, to leading order.

Draft edition

A dairy farmer, worried about falling milk production, asks the local university for help. A team of physicists takes the problem away. Weeks later they return, triumphant: "We have a solution. It works only for spherical cows in a vacuum."

— folklore

This is a draft. Corrections, comments, and better cows are welcome.

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Preface

Why the cow

The joke on the previous page is told against physicists, but physicists tell it about themselves, and they tell it with a certain pride. The spherical cow is not a failure of nerve. It is the whole method.

A real cow is a hopeless object. It has a nonuniform density, a complicated shape, four stomachs, and opinions. If you want to know how fast it falls off a cliff, how long it takes to cool after it dies, or how much it weighs on the Moon, almost none of that matters. A sphere of water with the right mass gets you most of the answer. The skill — and it is a skill, the central one this book tries to teach — is knowing *which* details to throw away, throwing them away without guilt, and then knowing how to put them back when the answer stops being good enough.

You will see this move in every chapter. Near the bottom of any potential well, everything is a spring (Chapter 3). A gas of 10^{23} molecules is described by three numbers (Chapter 9). A star is a ball of gas in balance between gravity and pressure (Chapter 27). The universe, on large enough scales, is the same everywhere (Chapter 28). Each of these is a spherical cow, and each one works astonishingly well.

What this book is

This book is a tasting platter of physics. It was inspired by Evan Chen's *An Infinitely Large Napkin*, a light but mostly self-contained tour of higher mathematics, written out of the frustration of trying to explain category theory to friends over a forty-minute lunch. The same frustration exists in physics. Someone asks what a quantum field is, and you start explaining, and you realise they need to know what a harmonic oscillator is, and then that they need to know what a Lagrangian is, and by then lunch is over.

So each chapter here is a single plate: one area of physics, served so you can taste what it is actually like. In each chapter you should get

- the main characters and their precise definitions,
- one *prototype* — a single example so central that every other idea in the chapter can be tested against it,
- the famous results, stated precisely, with a sense of *why* they are true,
- the orders of magnitude, so you know what the numbers actually are,
- a set of problems that climb from routine to genuinely hard, and
- directions to the real literature, so you know exactly where to go next.

There are thirty-three plates. Some are foundations that everything else rests on (mechanics, electromagnetism, quantum mechanics, statistical mechanics). Some are entire research fields

compressed to a chapter (condensed matter, cosmology, particle physics). Some are crossovers where physics meets medicine, the Earth, life, or computers. Taken together they cover, at a first-taste level, roughly everything you would find in the physics section of a university library or a preprint server.

What this book is not

It is not a replacement for a textbook. A chapter here cannot do what a semester of Griffiths or a summer with Taylor can. The amount of depth is not comparable, and it is not meant to be. Physics is learned by doing problems, many of them, over a long time, and the honest advice is still the old advice: pick up the standard books and work through them with a pencil.

What this book tries to do is make that possible. Its goal is that after any chapter you can open the serious book on that subject and *recognise the landscape*: you know what the main objects are, you have met the prototype, you know which results are coming and roughly why they hold. The rest is filling in. Every chapter ends with a short guide to the literature for exactly this reason.

Three mottos

There are three ideas that run through everything here. They are worth stating once, plainly.

MOTTO 1 **Physical intuition supersedes derivations.** A derivation tells you that something is true. An explanation tells you why it had to be. Where a derivation teaches nothing, it is skipped; where it contains an idea worth seeing, it stays.

MOTTO 2 **Estimate before you calculate.** Before solving anything, guess the answer to within a factor of ten using units, limiting cases, and orders of magnitude. If the exact answer disagrees wildly with the guess, one of them is wrong, and finding out which one is how you learn.

MOTTO 3 **Learn the prototype, then test everything against it.** For each new definition or theorem, ask what it says about the prototype. If the prototype is a good one, the definition will seem natural and the theorem will seem obvious.

Who this is for

The book was written with a curious, mathematically capable high-school student or early undergraduate in mind: someone comfortable with calculus who wants to know what the rest of physics is actually about. If you are further along, you may still find it useful as a way to see subjects you never studied, or to remember why the subjects you did study fit together. If you are earlier along, Chapter 1 tries to supply the mathematics you will need, though it moves

quickly.

Acknowledgements

This book owes its structure and spirit to Evan Chen's *Napkin*, whose shape it borrows without apology, and its sense of what a solid foundation looks like to Kevin Zhou's writing on learning physics and his short list of canonical texts. Its voice owes a great deal to the *Feynman Lectures on Physics*, to Purcell's *Electricity and Magnetism*, and to David Morin's problem books, which are proof that problems can be a joy. Every textbook recommended at the end of a chapter is recommended because it taught somebody, somewhere, how this subject really works.

Advice for the reader

Prerequisites

The mathematical prerequisite is calculus: you should be able to differentiate and integrate without fear, and know what a differential equation is even if you cannot yet solve many of them. Multivariable calculus, linear algebra, and complex numbers are used freely in later chapters; Chapter 1 introduces them quickly, and you should treat it as a reference to come back to rather than a wall to climb first.

The physical prerequisite is harder to state, and matters more. Call it *physical maturity*. It means being willing to

- reason in **limiting cases** (what happens as the mass goes to infinity? as the temperature goes to zero?),
- check every equation against its **units** and its **symmetries** before you trust it,
- **estimate** answers to within a factor of ten before computing them,
- move back and forth between a **physical picture** and the equations that encode it, and
- accept that every model is a **useful lie**, and that knowing how it lies is part of knowing it.

Nobody starts with all of this. The book tries to teach it as it goes.

Deciding what to read

There is no need to read this book in order. The chapters run as a loose climb — the foundations first, the frontier last — but almost every chapter can be started directly. Pick a chapter that sounds interesting, look at what it *builds on* (listed at the top of every chapter and in the map below), and follow those links back only as far as you need to.

If you want a suggestion, here are a few tasting menus:

- **The quantum menu:** 1 → 2 → 3 → 4 → 14, then any of 18, 19, or 20.
- **The cosmic menu:** 2 → 10 → 26 → 27 → 28 → 29.
- **The everyday menu:** 2 → 7 → 11 → 15 → 25.
- **The deep-end menu:** 3 → 4 → 10 → 14 → 30 → 32 → 33.
- **The life menu:** 7 → 9 → 21 → 23 → 24.

The boxes

The book uses a few recurring boxes. They mean the same thing everywhere.

PROTOTYPE The prototypical example of the chapter. Every definition and theorem that follows should be tested against it.

KEY IDEA The one idea from a section that you should remember a year from now.

Definition 0.1. A precise definition of a new character.

Law 0.2. A law of nature or a major result, stated precisely. (Physics has fewer theorems than mathematics and more laws.)

Example 0.3. A worked example. I am, like the author of the *Napkin*, pathologically fond of examples, and some pages are mostly boxes. I hope you will not mind.

Exercise 0.4. Inline exercises are short and should be done on the spot. If you cannot do one, the explanation before it probably failed; read back.

Problems, and how hard they are

Every chapter ends with problems, and the problems are the most important part of the chapter. They climb in difficulty, and the difficulty is marked in the most physicist way available: by *order of perturbation theory*.

In physics, when a problem is too hard to solve exactly, we look for a small parameter ϵ and expand: the answer is the solvable part, plus a correction of order ϵ, plus a smaller correction of order ϵ^2, and so on. Most of physics lives in the first two terms. But some effects — quantum tunnelling is the classic example — behave like $e^{-1/\epsilon}$. Such a function is smaller than *every* power of ϵ, so it is invisible at every order of the expansion. You cannot reach it by working harder at the obvious approach; you need a new idea.

So:

$O(\epsilon)$ **First order.** Apply the ideas of the chapter directly. If one of these defeats you, re-read the relevant section.

$O(\epsilon^2)$ **Second order.** Combine several ideas, keep careful track of a correction, or carry a calculation through several steps. These are where most of the learning happens.

$e^{-1/\epsilon}$ **Nonperturbative.** No amount of grinding on the obvious approach will work; you need a genuinely new idea, a clever change of viewpoint, or a trick you have not seen. Some of these are famous. Do not be discouraged if you need hints, or days.

Two further marks indicate importance: a ★ marks a result that later chapters rely on, and a † marks an aside worth knowing but not needed elsewhere.

Every nonperturbative problem has a hint in Appendix C, and every problem has a solution in Appendix D. Use them the way you would use a friend who knows the answer: only after you have honestly tried, and one step at a time.

Pencil and paper

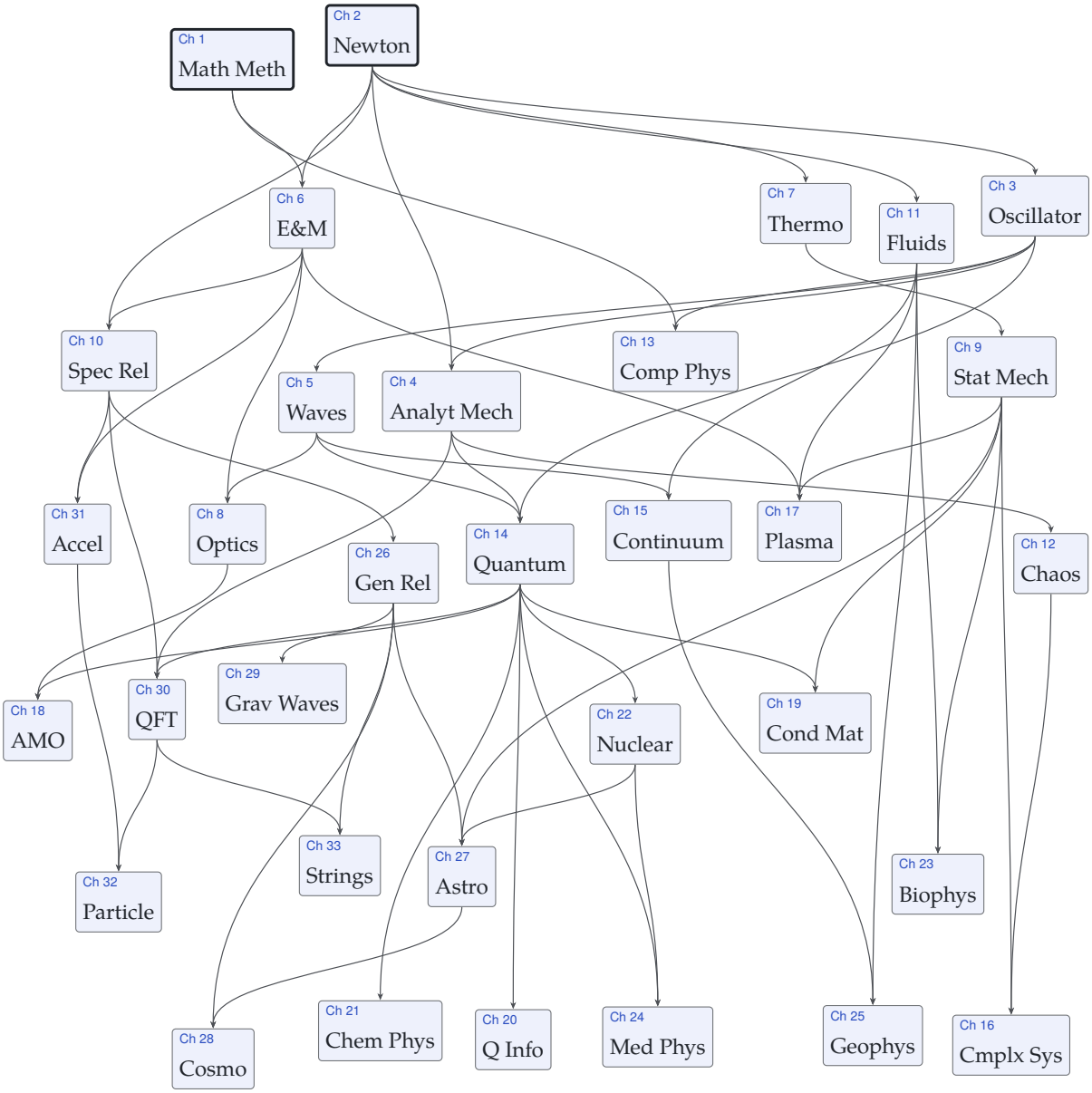
Read this book with a pencil and paper. You cannot keep it all in your head; nobody can. Check the algebra that is skipped, sketch the graphs that are described, and try every inline exercise before reading on.

Conventions

- Units are SI unless stated otherwise. From Chapter 30 onward we sometimes set $\hbar = c = 1$, and say so loudly when we do.
- Vectors are bold, $\mathbf{F}$; their magnitudes are plain, F. A hat marks a unit vector, $\hat{\mathbf{r}}$.
- Dots are time derivatives: $\dot{x} = \mathrm{d}x/\,\mathrm{d}t$.
- The symbol $\approx$ means "approximately equal to"; $\sim$ means "of the same order of magnitude as," i.e. equal up to a factor of a few; $\propto$ means "proportional to."
- Appendix A collects physical constants and useful orders of magnitude.

The map

Here is how the chapters lean on each other. An arrow runs from a chapter to each chapter that builds on it; the two boxes with heavy borders need nothing at all. I suggest that you simply **pick a chapter you find interesting, and walk the arrows back only as far as you need to.** (Every box in the chart is a link to its chapter.)



CHAPTER 1

Mathematical methods

BUILDS ON nothing beyond calculus — this chapter is a toolkit to return to

The working toolkit the rest of the book quietly runs on. Read it quickly once; come back whenever a later chapter reaches for a tool you have not used before.

PROTOTYPE The exponential $e^{\lambda t}$, the function that is its own derivative up to a constant. It solves every linear differential equation with constant coefficients, it turns rotation into multiplication when λ is imaginary, and it is the building block of Fourier analysis, damping, growth, decay, and quantum mechanics.

Mathematics is the language of physics, but physicists speak it with an accent. We care less about whether a series converges than about how many terms we need; less about existence theorems than about estimates. This chapter collects the tools in the order a physicist reaches for them — and the first tools are not equations at all.

§1.1 Dimensional analysis

Every physical quantity carries *dimensions*: mass $[M]$, length $[L]$, time $[T]$, and a few others (charge, temperature). An equation that adds metres to seconds is not wrong in a subtle way; it is nonsense. This simple constraint is astonishingly powerful.

Example 1.1 (The pendulum, without physics)**.** The period T of a pendulum could depend on its length L, the bob's mass m, the gravitational acceleration g, and the amplitude θ_0. Try $T \propto L^a m^b g^c$. Dimensions: $[T] = [L]^a [M]^b [LT^{-2}]^c$. Matching powers: $b = 0$ (nothing else carries mass), $-2c = 1$, $a + c = 0$. So

$$T = f(\theta_0)\sqrt{L/g}$$

for some function f of the dimensionless amplitude. We have found, without any dynamics, that the period does not depend on mass and scales as $\sqrt{L}$. The full theory (Chapter 3) only supplies $f(0) = 2\pi$.

The general statement is the **Buckingham π theorem**: if a physical law involves n quantities built from k independent dimensions, it can be rewritten as a relation among $n-k$ dimensionless combinations. In the pendulum, $n = 5$ quantities (T, L, m, g, θ_0) and $k = 3$ dimensions give two dimensionless groups, $T\sqrt{g/L}$ and θ_0, so one must be a function of the other.

Example 1.2 (G. I. Taylor and the atomic bomb)**.** In 1950 the physicist G. I. Taylor estimated the energy of the first atomic explosion from photographs of the fireball published in a magazine. The radius R of a strong blast wave can depend only on the energy E, the time t, and the air density ρ. The only combination with dimensions of length is

$$R \sim \left(\frac{E\,t^2}{\rho}\right)^{1/5}.$$

Reading $R \approx 140\,\mathrm{m}$ at $t \approx 25\,\mathrm{ms}$ off the photographs and taking $\rho \approx 1.2\,\mathrm{kg/m^3}$ gives $E \sim 10^{14}\,\mathrm{J}$, about 20 kilotons of TNT — close to the classified value.

KEY IDEA Before solving any problem, find the dimensionless combinations. Often the answer is "a number of order one, times the only combination with the right units," and the hard calculation only supplies the number.

§1.2 Orders of magnitude

Physicists estimate constantly, and they estimate in powers of ten. The goal of an estimate is not two significant figures; it is the right power of ten, and the confidence to know you have it.

Example 1.3 (How many atoms are in you?)**.** You weigh about 70 kg and are mostly water. A mole of water is 18 g and contains 6×10^{23} molecules, each with three atoms. So you contain about

$$\frac{70\,000\,\mathrm{g}}{18\,\mathrm{g}} \times 6 \times 10^{23} \times 3 \approx 7 \times 10^{27} \text{ atoms.}$$

The technique, often called a *Fermi problem* after Enrico Fermi who loved them, is to break an unknown quantity into factors you can each estimate within a factor of three. Errors in independent factors tend to partly cancel, so the product is usually better than any single guess. Appendix A lists numbers worth knowing by heart; a working physicist carries a few dozen of them.

A useful habit is to think logarithmically. Over a range of a factor 10^6, "halfway" is 10^3, not 5×10^5. When you do not know a quantity but know it lies between a and b, guess the geometric mean $\sqrt{ab}$.

§1.3 Approximation and the small parameter

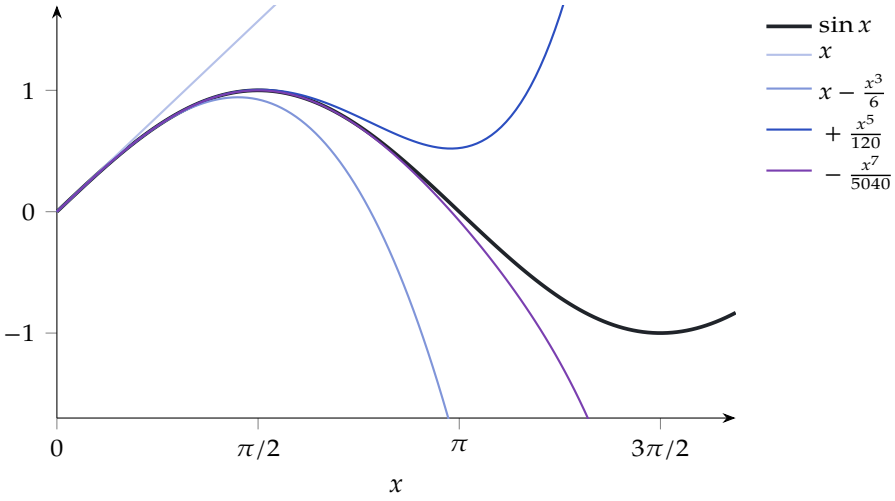
The most important formula in physics may be the Taylor series:

$$f(x_0+\epsilon) = f(x_0) + f'(x_0)\,\epsilon + \tfrac{1}{2}f''(x_0)\,\epsilon^2 + \tfrac{1}{6}f'''(x_0)\,\epsilon^3 + \cdots$$

When ϵ is small, each term is smaller than the one before, and we stop after one or two. A few expansions are used so often they should become reflexes:

$$(1+\epsilon)^n \approx 1 + n\epsilon + \tfrac{n(n-1)}{2}\epsilon^2, \qquad e^{\epsilon} \approx 1 + \epsilon + \tfrac{1}{2}\epsilon^2,$$
$$\sin\epsilon \approx \epsilon - \tfrac{1}{6}\epsilon^3, \qquad \cos\epsilon \approx 1 - \tfrac{1}{2}\epsilon^2,$$
$$\ln(1+\epsilon) \approx \epsilon - \tfrac{1}{2}\epsilon^2, \qquad \frac{1}{1-\epsilon} \approx 1 + \epsilon + \epsilon^2.$$

The binomial one, $(1+\epsilon)^n \approx 1 + n\epsilon$, works for *any* n, including negative and fractional powers. It turns $\sqrt{1.02}$ into 1.01, $1/1.03$ into 0.97, and $1/\sqrt{1 - v^2/c^2}$ into $1 + v^2/2c^2$ (Chapter 10).



sin x and its first four Taylor polynomials. Each extra term keeps the approximation good over a wider range of x.

Example 1.4 (How far is the horizon?)**.** Standing with your eyes at height h above a sphere of radius R, the distance d to the horizon satisfies $(R+h)^2 = R^2 + d^2$, so $d^2 = 2Rh + h^2$. Since $h \ll R$, drop h^2: $d \approx \sqrt{2Rh}$. For $h = 1.7$ m and $R = 6.4 \times 10^6$ m, $d \approx 4.7$ km.

The general strategy — find a small parameter, solve the problem with it set to zero, then compute corrections order by order — is called *perturbation theory*, and it is the reason this book grades its problems by order of ϵ. Its limitation is also worth knowing now: a function like $e^{-1/\epsilon}$ has every Taylor coefficient zero at $\epsilon = 0$. No finite number of corrections will ever see it.

§1.4 Vectors and index notation

A vector is a quantity with magnitude and direction, like a displacement, a velocity, or a force. In components, $\mathbf{a} = (a_1, a_2, a_3)$. The two products are

$$\mathbf{a}\cdot\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = ab\cos\theta, \qquad |\mathbf{a}\times\mathbf{b}| = ab\sin\theta,$$

with $\mathbf{a}\times\mathbf{b}$ perpendicular to both, by the right-hand rule. The dot product measures how much two vectors point the same way (work is $\mathbf{F}\cdot d\mathbf{r}$); the cross product measures how much they twist around each other (torque is $\mathbf{r}\times\mathbf{F}$).

Physicists rarely write out components. Instead we use *index notation* with the *Einstein summation convention*: a repeated index is summed. Then

$$\mathbf{a}\cdot\mathbf{b} = a_i b_i, \qquad (\mathbf{a}\times\mathbf{b})_i = \varepsilon_{ijk}\, a_j b_k,$$

where the *Kronecker delta* δ_{ij} is 1 if $i = j$ and 0 otherwise, and the *Levi-Civita symbol* ε_{ijk} is $+1$ for (ijk) an even permutation of (123), -1 for odd, and 0 if any index repeats. One identity does most of the work:

$$\varepsilon_{ijk}\varepsilon_{ilm} = \delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}.$$

Exercise 1.5. Use it to prove $\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{c}(\mathbf{a}\cdot\mathbf{b})$, the "BAC minus CAB" rule.

Index notation will come back in relativity (Chapter 10), where indices go up and down, and in general relativity (Chapter 26), where they carry the whole theory.

§1.5 Calculus in many dimensions

A *field* assigns a number (a scalar field, like temperature $T(\mathbf{r})$) or a vector (a vector field, like the wind $\mathbf{v}(\mathbf{r})$) to every point in space. The derivative operator $\nabla = (\partial_x, \partial_y, \partial_z)$ acts on fields in three ways:

- The **gradient** ∇T points in the direction T increases fastest, with magnitude equal to that rate. Heat flows down ∇T; a ball rolls down ∇U, since $\mathbf{F} = -\nabla U$.
- The **divergence** $\nabla\cdot\mathbf{v}$ measures how much the field spreads out from a point: the net outflow per unit volume. It is positive at a source (a tap), negative at a sink (a drain).
- The **curl** $\nabla\times\mathbf{v}$ measures how much the field swirls around a point. Drop a tiny paddle wheel in the flow; the curl is twice its angular velocity.

Two combinations deserve names. The **Laplacian** $\nabla^2 T = \nabla\cdot\nabla T$ compares the value at a point with the average of its neighbours: $\nabla^2 T > 0$ means the point is colder than its surroundings. That one fact is why it appears in the heat equation, the wave equation, and the Schrödinger equation. And two identities hold for every smooth field: $\nabla\times(\nabla T) = 0$ and $\nabla\cdot(\nabla\times\mathbf{v}) = 0$.

The integral theorems turn local information into global information:

Law 1.6 (The fundamental theorems of vector calculus)**.** For a volume V with closed boundary surface ∂V, and a surface S with boundary curve ∂S,

$$\int_V (\nabla\cdot\mathbf{v})\, \mathrm{d}V = \oint_{\partial V} \mathbf{v}\cdot\mathrm{d}\mathbf{A} \qquad \text{(divergence theorem)},$$

$$\int_S (\nabla\times\mathbf{v})\cdot\mathrm{d}\mathbf{A} = \oint_{\partial S} \mathbf{v}\cdot\mathrm{d}\mathbf{l} \qquad \text{(Stokes' theorem)}.$$

Both say the same thing: adding up local sources (or local swirls) inside a region gives the total flow out through (or around) its edge, because the contributions of interior neighbours cancel in pairs. They are the reason Maxwell's equations (Chapter 6) can be written in either a local, differential form or a global, integral form.

§1.6 Complex numbers

A complex number $z = x + iy$ with $i^2 = -1$ is a point in the plane. The single most useful fact about complex numbers is **Euler's formula**,

$$e^{i\theta} = \cos\theta + i\sin\theta,$$

which you can check by comparing Taylor series. It says that $e^{i\theta}$ is the point at angle θ on the unit circle, so any complex number can be written $z = re^{i\theta}$, and multiplying complex numbers multiplies their lengths and *adds their angles*. Multiplication by i is rotation by 90°.

Physicists use complex numbers for anything that oscillates. A quantity $A\cos(\omega t + \varphi)$ is the real part of $\tilde{A}e^{i\omega t}$, where the complex amplitude $\tilde{A} = Ae^{i\varphi}$ (a *phasor*) stores both size and phase. Because $\frac{\mathrm{d}}{\mathrm{d}t}e^{i\omega t} = i\omega e^{i\omega t}$, differential equations become algebra. In quantum mechanics (Chapter 14) the complex numbers stop being a bookkeeping trick and become essential: the wavefunction is genuinely complex.

§1.7 Ordinary differential equations

Most laws of physics are differential equations: they say how something changes from moment to moment. Three facts cover a remarkable fraction of physics.

First-order linear equations $\dot{x} = -x/\tau$ give exponential decay, $x = x_0 e^{-t/\tau}$. The time constant τ is the time to fall by a factor e; the half-life is $\tau \ln 2$. Radioactive decay, a capacitor discharging, a cup of coffee cooling, and the attenuation of light in water all obey this law.

Linear equations with constant coefficients, such as $a\ddot{x} + b\dot{x} + cx = 0$, are solved by guessing $x = e^{\lambda t}$, which turns the equation into a polynomial $a\lambda^2 + b\lambda + c = 0$. Real roots give growth or decay, imaginary roots give oscillation, complex roots give damped oscillation. Chapter 3 works through every case.

Superposition: if x_1 and x_2 solve a linear homogeneous equation, so does any combination $c_1x_1 + c_2x_2$. Linearity is what lets us break complicated problems into simple pieces — normal modes, Fourier components, Green's functions — and add the answers. Most of the hardest problems in physics are hard precisely because they are nonlinear (Chapter 12).

Remark 1.7. An nth-order equation needs n pieces of initial information: Newton's second law is second order, so you need both the initial position and the initial velocity. In quantum mechanics, the Schrödinger equation is first order in time, so the wavefunction now determines the wavefunction forever.

§1.8 Linear algebra

A *vector space* is any collection of objects you can add and scale: arrows in space, but also functions, quantum states, and the displacements of all the atoms in a crystal. A *linear map* (a matrix, once you choose a basis) takes vectors to vectors and respects addition and scaling.

The most important question to ask of a linear map M is: which vectors does it merely stretch? An *eigenvector* $\mathbf{v}$ satisfies

$$M\mathbf{v} = \lambda\mathbf{v},$$

with *eigenvalue* λ. The eigenvalues solve $\det(M - \lambda I) = 0$. In the basis of eigenvectors (when one exists), M is diagonal and the problem falls apart into independent one-dimensional problems. This single idea is behind normal modes of vibration (Chapter 5), principal axes of a spinning body (Chapter 2), stability of fixed points (Chapter 12), and the entirety of quantum measurement (Chapter 14).

Theorem 1.8 (Spectral theorem)**.** A real symmetric matrix ($M = M^T$), or more generally a complex Hermitian one ($M = M^\dagger$, the conjugate transpose), has real eigenvalues, and its eigenvectors can be chosen to form an orthonormal basis.

Physics loves this theorem because the matrices that appear in nature are usually symmetric or Hermitian: inertia tensors, coupling matrices of springs, and every quantum observable. Real eigenvalues are measurable frequencies and energies; orthogonal eigenvectors are independent modes.

Example 1.9 (Two coupled pendulums)**.** Two identical pendulums joined by a weak spring obey $\ddot{\mathbf{x}} = -K\mathbf{x}$ with $K = \begin{pmatrix} \omega_0^2 + \kappa & -\kappa \\ -\kappa & \omega_0^2 + \kappa \end{pmatrix}$. The eigenvectors are $(1, 1)$, swinging together with $\omega^2 = \omega_0^2$, and $(1, -1)$, swinging opposite with $\omega^2 = \omega_0^2 + 2\kappa$. Start one pendulum alone and it is a superposition of both modes; their slightly different frequencies beat, and the energy sloshes slowly back and forth between the two pendulums.

§1.9 Fourier analysis

KEY IDEA Any reasonable function can be built out of sines and cosines. Breaking a signal into frequencies is often the single most clarifying thing you can do to it.

A function with period $2L$ has a **Fourier series**

$$f(x) = \sum_{n=-\infty}^{\infty} c_n \, e^{in\pi x/L}, \qquad c_n = \frac{1}{2L} \int_{-L}^{L} f(x) \, e^{-in\pi x/L} \, \mathrm{d}x.$$

The coefficient formula works because the exponentials are orthogonal: integrating one against another over a period gives zero unless they are the same. This is "Fourier's trick," and it is exactly the same move as finding the components of a vector by taking dot products with orthonormal basis vectors. Functions are vectors; sines are a basis.

For non-periodic functions the sum becomes an integral, the **Fourier transform**:

$$\tilde{f}(k) = \int_{-\infty}^{\infty} f(x) \, e^{-ikx} \, \mathrm{d}x, \qquad f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(k) \, e^{ikx} \, \mathrm{d}k.$$

One fact about it reaches across all of physics: *a function cannot be narrow in both x and k.* A pulse of width Δx is made of wavenumbers spread over $\Delta k \gtrsim 1/\Delta x$. A short musical note has an uncertain pitch; a short laser pulse has a broad spectrum; and in quantum mechanics, where momentum is $p = \hbar k$, this becomes the uncertainty principle $\Delta x \, \Delta p \gtrsim \hbar$.

§1.10 Probability

Physics deals with randomness in two ways: from ignorance (we cannot track 10^{23} molecules, Chapter 9) and, in quantum mechanics, apparently from nature itself. Either way the tools are the same.

A random variable X with probability density $p(x)$ has *mean* $\langle X\rangle = \int x\,p(x)\,\mathrm{d}x$ and *variance* $\sigma^2 = \langle X^2\rangle - \langle X\rangle^2$; the standard deviation σ measures its spread. The most important distribution is the **Gaussian**,

$$p(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\,e^{-(x-\mu)^2/2\sigma^2},$$

and the reason is the **central limit theorem**: the sum of many independent random contributions, whatever their individual distributions, is approximately Gaussian. Measurement errors, the speeds of gas molecules along one axis, and the position of a diffusing particle are all Gaussian for this reason.

A consequence worth memorising: the sum of N independent contributions, each of typical size σ, has a typical size $\sqrt{N}\,\sigma$, not $N\sigma$. A random walk of N steps gets $\sqrt{N}$ steps from home. Relative fluctuations therefore shrink as $1/\sqrt{N}$, which is why a gas of 10^{24} molecules has a pressure steady to one part in 10^{12}.

§1.11 Symmetry, briefly

A *symmetry* is a transformation that leaves something unchanged: rotating a sphere, translating an infinite plane, shifting the origin of time. The set of symmetries of an object always forms a *group*: you can compose two symmetries, undo any symmetry, and do nothing.

Symmetry is the most powerful idea in modern physics, for two reasons that this book will develop. First, by **Noether's theorem** (Chapter 4), every continuous symmetry of the laws of physics comes with a conserved quantity: time-translation symmetry gives energy conservation, space-translation gives momentum, rotation gives angular momentum. Second, demanding that a theory have certain symmetries can nearly determine its form; in particle physics (Chapter 32), the forces of nature are almost forced on us by symmetry. The mathematics of how a group can act on a vector space — *representation theory* — is why electrons have spin $\tfrac{1}{2}$ and why the periodic table looks the way it does.

§1.12 The delta function and Green's functions

Physicists often need a point: a point charge, a point mass, a sudden kick. The **Dirac delta** $\delta(x)$ is zero everywhere except at $x = 0$, where it is infinitely tall in such a way that $\int \delta(x)\,\mathrm{d}x = 1$. Its defining property is

$$\int f(x)\,\delta(x-a)\,\mathrm{d}x = f(a).$$

It is not a function but a limit of ever-narrower spikes, and it is perfectly safe as long as it ends up inside an integral.

The delta function gives a beautiful strategy for linear problems. Find the response $G(t)$ of a system to a single sharp kick at $t = 0$; this is the **Green's function**. Any force is a superposition

of kicks, $F(t) = \int F(t')\,\delta(t-t')\,\mathrm{d}t'$, so by linearity the response to any force is

$$x(t) = \int G(t-t')\,F(t')\,\mathrm{d}t'.$$

Solve the problem once, for a kick, and you have solved it for every force. The electric potential of any charge distribution, the sound from any source, and the propagators of quantum field theory (Chapter 30) are all Green's functions.

§1.13 Problems

1.1 $O(\epsilon)$ ★ Use dimensional analysis to find how the speed of deep-water waves depends on their wavelength λ and on g. Why can't the density of water appear? Do long or short waves travel faster?

1.2 $O(\epsilon)$ Estimate the number of heartbeats in a human lifetime, the number of seconds in a year to two significant figures ($\pi \times 10^7$ is a famous approximation), and the number of grains of sand on all the world's beaches. Write each to the nearest power of ten.

1.3 $O(\epsilon)$ Expand $\dfrac{1}{\sqrt{1-x}}$ to second order in x, and use it to evaluate $1/\sqrt{0.98}$ without a calculator. How large is the error?

1.4 $O(\epsilon)$ Find the eigenvalues and eigenvectors of $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and of $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Interpret the second matrix geometrically, and explain why it has no real eigenvectors.

1.5 $O(\epsilon^2)$ ★ Compute the Gaussian integral $I = \int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x$ by evaluating I^2 as a double integral in polar coordinates. Then deduce $\int_{-\infty}^{\infty} x^2 e^{-ax^2}\,\mathrm{d}x$ by differentiating with respect to a. (Both appear constantly in statistical mechanics and quantum mechanics.)

1.6 $O(\epsilon^2)$ Compute the Fourier series of a square wave that equals $+1$ on $(0,\pi)$ and -1 on $(-\pi,0)$. Show that only odd harmonics appear, with amplitudes falling like $1/n$. Explain in words why a sharp corner needs high frequencies.

1.7 $O(\epsilon^2)$ Using index notation, prove that $\nabla\cdot(\nabla\times\mathbf{v}) = 0$ and $\nabla\times(\nabla\times\mathbf{v}) = \nabla(\nabla\cdot\mathbf{v}) - \nabla^2\mathbf{v}$. The second identity is how you get a wave equation out of Maxwell's equations.

1.8 $O(\epsilon^2)$ The surface of the ocean bulges near a submarine mountain because the mountain's extra mass pulls water toward it. Use dimensional analysis and a rough estimate to decide whether this bulge is closer to a millimetre, a metre, or a kilometre for a mountain 2 km tall and 20 km wide.

1.9 $O(\epsilon^2)$ ★ (*Random walks.*) A drunkard takes N steps of length ℓ in random directions on a line. Show that $\langle x\rangle = 0$ but $\langle x^2\rangle = N\ell^2$. A molecule of perfume in still air takes steps of about 0.1 μm roughly 10^{10} times per second. How long does it take to diffuse across a 5 m room? (Why, then, can you smell perfume across a room within a minute?)

1.10 $e^{-1/\epsilon}$ ★ (*Stirling's approximation.*) Write $N! = \int_0^\infty x^N e^{-x}\,dx$. The integrand is sharply peaked; expand the logarithm of the integrand to second order about its maximum and do the resulting Gaussian integral to show

$$N! \approx \sqrt{2\pi N}\, N^N e^{-N}.$$

This trick — approximate a sharply peaked integrand by a Gaussian — is called Laplace's method, and it underlies most of statistical mechanics.

1.11 $e^{-1/\epsilon}$ (*A nonperturbative function.*) Show that every derivative of $f(\epsilon) = e^{-1/\epsilon}$ (for $\epsilon > 0$, with $f(0) = 0$) vanishes at $\epsilon = 0$, so its Taylor series about zero is identically zero even though f is not. Explain why this means no perturbative expansion in ϵ can ever detect such an effect. (Quantum tunnelling, Chapter 14, has exactly this form, with $\epsilon \propto \hbar$.)

1.12 $e^{-1/\epsilon}$ (*Buckingham at the breakfast table.*) A drop of water falls from a tap. Its size depends on the surface tension σ (force per length), the density ρ, g, and the tap radius r. Form the dimensionless groups and show that for small taps the drop mass is $m \propto \sigma r/g$, independent of ρ. Then argue from a force balance why this is true, and estimate the size of a drop from a kitchen tap.

Further reading

- **Mary L. Boas, *Mathematical Methods in the Physical Sciences*** — The standard first book on mathematics for physicists. Clear, complete, full of practice problems.
- **H. M. Schey, *Div, Grad, Curl, and All That*** — A short, friendly, physically motivated introduction to vector calculus.
- **Sanjoy Mahajan, *Street-Fighting Mathematics* and *The Art of Insight in Science and Engineering*** — The definitive books on estimation, dimensional analysis, and approximation. Both are freely available online.
- **Gilbert Strang, *Introduction to Linear Algebra*** — Linear algebra taught with the geometric intuition a physicist needs.
- **Riley, Hobson, and Bence, *Mathematical Methods for Physics and Engineering*** — A comprehensive reference for when you need the next level.

CHAPTER 2

Newtonian mechanics

BUILDS ON nothing — this is the trunk of the tree

Why things move: forces, orbits, spinning tops, and the clockwork universe that started all of physics.

> **PROTOTYPE** The Moon as a falling apple. The same force that pulls an apple to the ground holds the Moon in its orbit; the Moon is falling toward the Earth all the time, and missing. Newton checked this with one division, which we will redo in §2.7.1.

Classical mechanics is the physics of objects that are larger than atoms, slower than light, and far from black holes. That covers nearly everything you have ever touched, and it is still the first thing every physicist learns, because its ideas — force, momentum, energy, angular momentum, the equation of motion — reappear, transformed, in every later theory.

§2.1 Describing motion

The position of a particle is a vector $\mathbf{r}(t)$. Its *velocity* is $\mathbf{v} = \dot{\mathbf{r}}$ and its *acceleration* is $\mathbf{a} = \dot{\mathbf{v}} = \ddot{\mathbf{r}}$. For constant acceleration, integrating twice gives the formulas everyone meets first:

$$\mathbf{v} = \mathbf{v}_0 + \mathbf{a}t, \qquad \mathbf{r} = \mathbf{r}_0 + \mathbf{v}_0 t + \tfrac{1}{2}\mathbf{a}t^2, \qquad v^2 = v_0^2 + 2\,\mathbf{a}\cdot(\mathbf{r}-\mathbf{r}_0).$$

Example 2.1 (Projectiles)**.** Throw a ball at speed v_0 and angle θ above flat ground, ignoring air. Horizontally it coasts, $x = v_0\cos\theta\, t$; vertically it decelerates, $y = v_0\sin\theta\, t - \tfrac{1}{2}gt^2$. It lands when $y = 0$, at $t = 2v_0\sin\theta/g$, having travelled

$$R = \frac{v_0^2 \sin 2\theta}{g}.$$

The range is greatest at 45°, and complementary angles (30° and 60°) give the same range. The two motions are independent: a bullet fired horizontally and one dropped from the same height hit the ground at the same moment.

§2.2 Newton's three laws

Law 2.2 (Newton's laws of motion)**.** (1) In an *inertial frame*, a body with no net force on it moves in a straight line at constant speed.

(2) The net force on a body equals the rate of change of its momentum: $\mathbf{F} = \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t}$, with $\mathbf{p} = m\mathbf{v}$. For constant mass, $\mathbf{F} = m\mathbf{a}$.

(3) Forces come in pairs: if A pushes on B with force $\mathbf{F}$, then B pushes on A with $-\mathbf{F}$.

The first law is not a special case of the second. It defines the stage on which the second law is true: an inertial frame is one in which free particles move uniformly. The Earth's surface is approximately inertial; a car going around a corner is not. The third law is what makes momentum conservation work (§2.4).

The second law is the engine. It is a second-order differential equation: give it the forces and the initial position and velocity, and it predicts the entire future. This determinism — the clockwork universe — survived for two centuries until quantum mechanics (Chapter 14) softened it.

KEY IDEA To solve a mechanics problem: draw every body separately, draw every force acting *on* it (a free-body diagram), write $\mathbf{F} = m\mathbf{a}$ for each body along convenient axes, and add constraints that link the bodies together. Then count: you need as many equations as unknowns.

§2.3 A field guide to forces

Fundamentally there are only a few forces (gravity, electromagnetism, and the nuclear forces; Chapter 32). But in everyday mechanics we use effective descriptions — spherical cows — of the electromagnetic forces between enormous numbers of atoms.

- **Weight**: near the Earth's surface, $\mathbf{F} = m\mathbf{g}$ with $g \approx 9.8\,\mathrm{m/s^2}$.
- **Normal force**: whatever perpendicular push a surface must supply to stop you passing through it. It is not mg in general; it is whatever the constraint demands.
- **Tension**: the pull along a taut string. For a massless string over a frictionless pulley, the tension is the same everywhere.
- **Friction**: a static friction force up to $\mu_s N$ that opposes the onset of sliding, and a kinetic friction $\mu_k N$ (usually $\mu_k < \mu_s$) that opposes sliding. Remarkably, friction does not depend on the area of contact. The reason is that surfaces touch only at microscopic asperities whose true contact area is proportional to the load, not the apparent area.
- **Springs**: $F = -kx$, Hooke's law, the subject of Chapter 3.
- **Drag**: a body moving through a fluid feels a force opposing its motion. At low speeds in viscous fluids it is linear, $F = 6\pi\eta r v$ for a sphere (Stokes' law). At everyday speeds in air

it is quadratic,

$$F_{\text{drag}} \approx \tfrac{1}{2} C_D \, \rho \, A \, v^2,$$

with $C_D \sim 1$. The crossover is governed by the Reynolds number (Chapter 11).

Example 2.3 (Terminal velocity)**.** A falling object speeds up until drag balances weight: $mg = \tfrac{1}{2} C_D \rho A v_t^2$. A skydiver ($m = 75\,\text{kg}$, $A \approx 0.7\,\text{m}^2$, $C_D \approx 1$, $\rho = 1.2\,\text{kg/m}^3$) reaches $v_t \approx 40\,\text{m/s}$, about 140 km/h. Because $v_t \propto \sqrt{m/A}$ and for a body of size L this ratio grows like $\sqrt{L}$, small animals fall slowly: a mouse survives a fall down a mine shaft, a horse does not.

§2.4 Momentum

For a system of particles, the internal forces cancel in pairs by Newton's third law. Adding the second law over all particles gives

$$\mathbf{F}_{\text{ext}} = \frac{\mathrm{d}\mathbf{P}}{\mathrm{d}t}, \qquad \mathbf{P} = \sum_i m_i \mathbf{v}_i = M \, \dot{\mathbf{R}},$$

where $\mathbf{R} = \sum m_i \mathbf{r}_i / M$ is the *centre of mass*. The centre of mass moves as if all the mass were concentrated there and all the external forces acted there. A firework that explodes in mid-air sends its pieces flying, but their centre of mass keeps following the original parabola.

Law 2.4 (Conservation of momentum)**.** If the net external force on a system vanishes, its total momentum is constant.

Collisions are the classic application. In every collision momentum is conserved. In an *elastic* collision kinetic energy is also conserved; in an *inelastic* one some becomes heat or deformation; in a *perfectly inelastic* one the bodies stick. Collisions look simplest in the centre-of-mass frame, where the total momentum is zero: an elastic collision just rotates the two velocities without changing their sizes.

Example 2.5 (The rocket equation)**.** A rocket expels exhaust backward at speed u relative to itself. In time $\mathrm{d}t$ it loses mass $\mathrm{d}m$ (negative) and gains speed $\mathrm{d}v$. Conservation of momentum gives $m \, \mathrm{d}v = -u \, \mathrm{d}m$, so

$$\Delta v = u \ln \frac{m_{\text{initial}}}{m_{\text{final}}}.$$

This is *Tsiolkovsky's rocket equation*, and the logarithm is the cruelty of spaceflight. Reaching low Earth orbit takes $\Delta v \approx 9.4\,\text{km/s}$ (including losses). With chemical exhaust at $u \approx 4.4\,\text{km/s}$, the mass ratio is $e^{2.1} \approx 8$: about 88% of a rocket on the launch pad must be fuel.

§2.5 Energy

The *work* done by a force along a path is $W = \int \mathbf{F} \cdot \mathrm{d}\mathbf{r}$. Integrating Newton's second law along the path gives the **work–energy theorem**:

$$W_{\text{net}} = \Delta K, \qquad K = \tfrac{1}{2}mv^2.$$

Some forces — gravity, springs, electrostatics — have the special property that the work they do depends only on the endpoints, not the path. These *conservative* forces can be written as $\mathbf{F} = -\nabla U$ for a *potential energy* $U(\mathbf{r})$, and then

$$E = K + U = \text{constant}.$$

(A force is conservative exactly when $\nabla \times \mathbf{F} = 0$; friction is the standard non-conservative example.) Near the Earth, $U = mgh$; for a spring, $U = \tfrac{1}{2}kx^2$; for gravity in general, $U = -GMm/r$.

KEY IDEA In one dimension, the potential energy curve $U(x)$ tells you everything qualitative about the motion. A particle with energy E is confined to where $U(x) \le E$; it turns around where $U = E$; it speeds up in valleys and slows on hills. Stable equilibria are minima of U, and near any of them the motion is a harmonic oscillation (Chapter 3).

Energy conservation is often the fastest route to an answer, because it skips the details of the path. How fast is a roller coaster at the bottom? $v = \sqrt{2gh}$, whatever the shape of the track. Deeper down, energy conservation is a consequence of the laws of physics being the same at all times (Chapter 4).

§2.6 Rotation

Rotation has its own versions of everything. A body spinning about a fixed axis at *angular velocity* ω has

$$K = \tfrac{1}{2}I\omega^2, \qquad L = I\omega, \qquad \tau = I\dot{\omega},$$

where the *moment of inertia* $I = \sum m_i r_i^2$ measures how far the mass sits from the axis. It plays the role of mass for rotation. For a thin hoop $I = MR^2$; a uniform disc $\tfrac{1}{2}MR^2$; a solid sphere $\tfrac{2}{5}MR^2$; a rod about its centre $\tfrac{1}{12}ML^2$. The *parallel axis theorem*, $I = I_{\text{cm}} + Md^2$, shifts the axis a distance d from the centre of mass.

The *torque* $\tau = \mathbf{r} \times \mathbf{F}$ is the rotational version of force, and *angular momentum* $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ is the rotational version of momentum, with $\tau = \dot{\mathbf{L}}$.

Law 2.6 (Conservation of angular momentum)**.** If the net external torque on a system vanishes, its total angular momentum is constant.

A figure skater pulling in her arms decreases I, so ω must increase. A collapsing star does the same thing on a grander scale, which is why neutron stars can spin hundreds of times per second (Chapter 27).

Example 2.7 (Rolling). A ball rolling without slipping has $v = \omega R$ and kinetic energy $\frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$. Released from rest at height h,

$$v = \sqrt{\frac{2gh}{1 + I/MR^2}}.$$

A solid sphere ($I/MR^2 = \frac{2}{5}$) beats a disc ($\frac{1}{2}$), which beats a hoop (1) — independently of their masses and sizes. Some of the energy has to go into spinning, and the hoop wastes the most.

In three dimensions, $\mathbf{L}$ and ω need not be parallel: they are related by a 3×3 symmetric matrix, the *inertia tensor*, $\mathbf{L} = I\omega$. By the spectral theorem (Chapter 1) it has three perpendicular *principal axes*, and spinning about any of them is steady. A spinning gyroscope with a torque applied does not fall over; its angular momentum vector *precesses*, swinging sideways at a rate $\Omega = \tau/L$.

§2.7 Gravity and orbits

Law 2.8 (Newton's law of gravitation). Two point masses attract each other with a force

$$F = \frac{Gm_1m_2}{r^2}, \qquad G = 6.674 \times 10^{-11}\ \mathrm{N\,m^2/kg^2}.$$

Newton also proved the **shell theorem**: a spherically symmetric body attracts outside masses as if all its mass were at its centre, and a spherical shell exerts no net force on anything inside it. (This took him years; with Gauss's law of Chapter 6 it takes a line.) So at the Earth's surface $g = GM_\oplus/R_\oplus^2$, and weighing the Earth amounts to measuring G, which Cavendish did in 1798.

2.7.1 The Moon is a falling apple

The Moon is $r = 3.84 \times 10^8$ m away and goes around in $T = 27.3$ days. Its centripetal acceleration is

$$a = \frac{4\pi^2 r}{T^2} \approx 2.7 \times 10^{-3}\ \mathrm{m/s^2}.$$

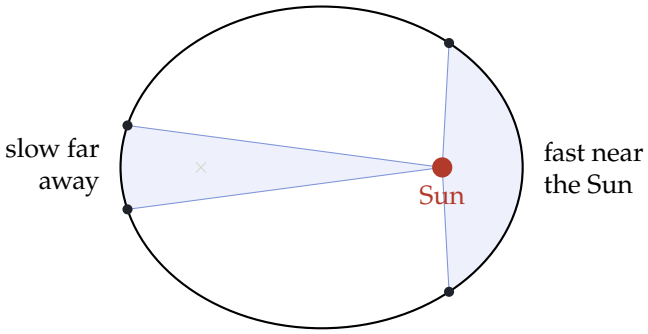
The ratio $g/a \approx 3600 = 60^2$ — and the Moon is 60 Earth radii away. The force falls off exactly as the inverse square of distance. One law describes the apple and the Moon: the heavens and the Earth obey the same physics. It is hard to overstate how radical that was.

2.7.2 Kepler's laws

For a circular orbit, gravity supplies the centripetal force: $GMm/r^2 = mv^2/r$, so $v = \sqrt{GM/r}$ and the period obeys

$$T^2 = \frac{4\pi^2}{GM}\, r^3.$$

This is Kepler's third law, true also for ellipses with r replaced by the semi-major axis. Kepler's first two laws — orbits are ellipses with the Sun at a focus, and the line to the planet sweeps equal areas in equal times — also follow from Newton. The equal-area law is just conservation of angular momentum, and holds for any central force.



Kepler's second law. The planet takes the same time to cover each shaded stretch of its orbit, so the two shaded areas are equal: it moves fast near the Sun and slowly far from it.

The cleanest way to see the shapes of orbits is the **effective potential**. Because angular momentum L is conserved, the radial motion behaves like a one-dimensional problem with

$$U_{\text{eff}}(r) = -\frac{GMm}{r} + \frac{L^2}{2mr^2}.$$

The second term is the "centrifugal barrier." Negative-energy orbits are trapped between two turning points — ellipses. Zero energy gives a parabola; positive energy, a hyperbola that escapes. The *escape velocity* from radius r is $v_{\text{esc}} = \sqrt{2GM/r}$, which is 11.2 km/s from the Earth's surface.

§2.8 Non-inertial frames

Newton's laws hold in inertial frames. If you insist on working in an accelerating frame, you can rescue them by adding *fictitious forces* — forces with no physical source, which exist only because your frame is accelerating. In a frame rotating at angular velocity Ω,

$$m\mathbf{a}_{\text{rot}} = \mathbf{F} - 2m\,\Omega \times \mathbf{v}_{\text{rot}} - m\,\Omega \times (\Omega \times \mathbf{r}).$$

The last term is the *centrifugal* force, pushing outward from the axis. The middle term is the *Coriolis* force, which deflects moving objects sideways — to the right in the northern hemisphere. Over the scale of a hurricane it is decisive (Chapter 25); in a bathtub it is utterly negligible, whatever you have heard. The *Foucault pendulum*, whose plane of swing slowly rotates over the day, is a direct demonstration that the Earth turns.

That fictitious forces are proportional to mass, just like gravity, is a clue Einstein took seriously. It leads, through the equivalence principle, to general relativity (Chapter 26).

§2.9 Where Newton breaks

Newtonian mechanics is a spherical cow too. It fails, in a controlled way, in three directions:

- at speeds approaching c, where special relativity takes over (Chapter 10), with corrections of order v^2/c^2;
- at small scales, where quantum mechanics takes over (Chapter 14), when the action involved is comparable to $\hbar$;
- in strong gravity, where general relativity takes over (Chapter 26), with corrections of order GM/rc^2.

In each case the new theory reduces to Newton's in the appropriate limit. That is a pattern worth noticing: good theories are not overthrown, they are *contained*.

§2.10 Problems

2.1 $O(\epsilon)$ ★ Show that the range of a projectile on flat ground is greatest at 45°. If the projectile is launched from a cliff of height h, is the optimal angle more or less than 45°? Explain without calculating.

2.2 $O(\epsilon)$ A block rests on a ramp with coefficient of static friction μ_s. Show that it begins to slide when the ramp angle exceeds $\theta_c = \arctan \mu_s$, regardless of the block's mass. How could you use this to measure μ_s?

2.3 $O(\epsilon)$ Two masses $m_1 > m_2$ hang on either side of a massless pulley (an Atwood machine). Find their acceleration and the tension in the string. Check the limits $m_2 \to 0$ and $m_1 = m_2$.

2.4 $O(\epsilon)$ A raindrop of radius 1 mm falls from a cloud. Estimate its terminal velocity using quadratic drag. Why don't raindrops hurt?

2.5 $O(\epsilon^2)$ ★ Derive Kepler's third law for circular orbits, and use it to find the radius of a geostationary orbit (one with a period of one sidereal day, 23.93 hours). Express the answer in Earth radii.

2.6 $O(\epsilon^2)$ ★ A hoop, a disc, and a solid sphere race from rest down the same incline, rolling without slipping. Show that the order of finish is independent of their masses and radii. Now add a frictionless block that slides. Who wins?

2.7 $O(\epsilon^2)$ Using the rocket equation, compare a single-stage rocket with a two-stage rocket in which each stage carries half the fuel and the empty first stage is discarded. Why are real launch vehicles multi-stage?

2.8 $O(\epsilon^2)$ A ball is dropped from a 100 m tower at the equator. Because the top of the tower moves eastward slightly faster than its base, the ball lands slightly east of the point directly below. Estimate the deflection, either in the inertial frame or using the Coriolis force. (Answer: about 2 cm.)

2.9 $O(\epsilon^2)$ Compute the escape velocity from the Moon. The typical thermal speed of a gas molecule is $\sqrt{3k_BT/m}$. Use this to explain why the Moon has no atmosphere while

the Earth keeps its nitrogen but has lost almost all its hydrogen. (Rule of thumb: a planet keeps a gas for billions of years only if the thermal speed is below about a sixth of the escape velocity.)

2.10 $e^{-1/\epsilon}$ ★ (*The leaning tower of books.*) You stack identical books of length L at the edge of a table, each overhanging the one below. Show that with n books the maximum total overhang is

$$\frac{L}{2}\left(1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{n}\right),$$

and conclude that the overhang can be made arbitrarily large. How many books do you need for the top book to lie entirely beyond the table's edge?

Hint: Build the stack from the top down. At each stage the centre of mass of everything above must lie exactly over the edge of the book below.

2.11 $e^{-1/\epsilon}$ (*The bead on a spinning hoop.*) A bead slides without friction on a vertical circular hoop of radius R that spins about its vertical diameter at angular velocity Ω. Find the equilibrium positions of the bead. Show that the bottom is stable for $\Omega^2 < g/R$, but that above this critical speed the bottom becomes unstable and two new stable equilibria appear symmetrically on either side. This sudden change in the number of equilibria is a *bifurcation* (Chapter 12), and a toy model of spontaneous symmetry breaking (Chapter 32).

2.12 $e^{-1/\epsilon}$ (*The tennis racket theorem.*) A rigid body has three principal moments of inertia $I_1 < I_2 < I_3$. Euler's equations for torque-free rotation are $I_1\dot{\omega}_1 = (I_2 - I_3)\omega_2\omega_3$ and cyclic permutations. Show that rotation about the axis of largest or smallest moment is stable to small perturbations, but rotation about the intermediate axis is unstable: a small wobble grows exponentially. Try it with a phone or a tennis racket tossed in the air; astronauts on the International Space Station have filmed a spinning wing nut flipping over and over (the Dzhanibekov effect).

Hint: Linearize about $\omega = (0, \Omega, 0)$ and look for solutions $\propto e^{\lambda t}$.

Further reading

- **Kleppner and Kolenkow, *An Introduction to Mechanics*** — The classic first rigorous mechanics course, with problems that build real skill.
- **David Morin, *Introduction to Classical Mechanics*** — Hundreds of beautiful problems with full solutions; the best book for learning to solve problems.
- **John R. Taylor, *Classical Mechanics*** — The smoothest route to intermediate mechanics, from Newton to Lagrange and beyond.
- ***The Feynman Lectures on Physics*, Volume I** — Free online. Read the chapters on mechanics for the physical insight no textbook quite matches.
- **Any calculus-based introductory text** — Halliday–Resnick, Young–Freedman, or similar. They are all about equally good; what matters is doing the problems.

CHAPTER 3

The harmonic oscillator

BUILDS ON Newtonian mechanics (Chapter 2)

The most important toy model in physics — and a case study in why throwing almost everything away is the right move.

PROTOTYPE A mass m on a spring of stiffness k: $m\ddot{x} = -kx$. Keep it in mind for every equation in this chapter. You will meet it again in nearly every chapter of this book.

§3.1 One equation

A system is a *simple harmonic oscillator* when the force pulling it back to equilibrium grows in exact proportion to how far it has strayed: $F = -kx$. Fed into Newton's second law, that single line is the whole chapter in compressed form:

$$m\ddot{x} = -kx \qquad \Longleftrightarrow \qquad \ddot{x} = -\omega^2 x, \qquad \omega = \sqrt{\frac{k}{m}}.$$

You do not need to grind through the differential equation to guess its solution. What function is its own second derivative, up to a negative constant? A sinusoid. So

$$x(t) = A\cos(\omega t + \varphi),$$

where the amplitude A and phase φ are fixed by how the motion started. The *period* is $T = 2\pi/\omega$ and the frequency is $f = 1/T = \omega/2\pi$.

The striking fact, hiding in plain sight, is that ω does not depend on A: a large swing and a tiny one take *exactly* the same time. This property is called *isochronism*. Hold onto it — it is special, it is the reason clocks work, and it is fragile.

Example 3.1 (Orders of magnitude). A car's suspension: $m \sim 300\,\text{kg}$ per wheel, $k \sim 3 \times 10^4\,\text{N/m}$, so $f \sim 1.6\,\text{Hz}$ — the gentle bounce you feel over a speed bump. A quartz crystal in a watch is engineered to ring at exactly $32768\,\text{Hz} = 2^{15}\,\text{Hz}$, so that fifteen halvings give one tick per second. A carbon–oxygen bond in a CO molecule vibrates at about $6.4 \times 10^{13}\,\text{Hz}$. Same equation, eighteen orders of magnitude.

KEY IDEA The defining feature is not the spring, the sine, or the energy. It is *linearity*: restoring influence proportional to displacement. Every system that shares this property

is, mathematically, the same system.

§3.2 Why it is everywhere

Here is the reason one model earns a whole chapter. Take *any* system sitting in a smooth potential energy $U(x)$, resting at a stable equilibrium x_0. Zoom in close, and ask what the potential looks like. Taylor expansion answers:

$$U(x) = U(x_0) + U'(x_0)\,(x-x_0) + \tfrac{1}{2}U''(x_0)\,(x-x_0)^2 + \tfrac{1}{6}U'''(x_0)\,(x-x_0)^3 + \cdots$$

The constant term only sets the zero of energy. The linear term vanishes, because $U'(x_0) = 0$ is what "equilibrium" means. So the first term that does any physics is the quadratic one. Keep it, throw the rest away:

$$U(x) \approx \tfrac{1}{2}\,k_{\text{eff}}\,(x-x_0)^2, \qquad k_{\text{eff}} = U''(x_0).$$

That is a spring — not *like* a spring; a spring, with a stiffness you compute by differentiating twice. Stability requires $U''(x_0) > 0$ (a minimum, not a maximum), and then

$$\omega = \sqrt{\frac{U''(x_0)}{m}}.$$

This is the single most reused approximation in physics, and it is exactly the spirit of this book. A real potential is a mess. Near the bottom, we assume a spherical cow.

> **PROTOTYPE, GENERALIZED** A bead in a bowl, the length of a chemical bond, an atom in a crystal, a buoy bobbing on water, the current in a circuit, a planet nudged off its orbit, the electromagnetic field in a cavity — all the same oscillator, all with $\omega = \sqrt{k_{\text{eff}}/m_{\text{eff}}}$. Learn it once and you have met it a hundred times.

Example 3.2 (The pendulum)**.** A pendulum of length L has potential $U(\theta) = -mgL\cos\theta$, with a minimum at $\theta = 0$. Then $U''(0) = mgL$, and the "mass" for the angle coordinate is the moment of inertia mL^2, since the kinetic energy is $\tfrac{1}{2}mL^2\dot{\theta}^2$. So

$$\omega = \sqrt{\frac{U''(0)}{mL^2}} = \sqrt{\frac{g}{L}}, \qquad T = 2\pi\sqrt{\frac{L}{g}}.$$

Notice what vanished: the mass, and the amplitude. A one-metre pendulum has a period of almost exactly two seconds, which is not a coincidence: the metre was once proposed to be defined this way. A pendulum clock keeps time *because* its frequency ignores how hard it is pushed — until the swing is no longer small, and the terms we discarded come back to bite (Problem 3.5).

Exercise 3.3. Two atoms interact through the Lennard–Jones potential $U(r) = U_0[(a/r)^{12} - 2(a/r)^6]$. Show that the equilibrium separation is $r = a$, and that small vibrations have angular frequency $\omega = \sqrt{72\,U_0/(\mu a^2)}$, where μ is the reduced mass. This is, to a first approximation, the "note" a molecule rings at.

§3.3 Energy, and a picture worth more than the algebra

The total energy is kinetic plus potential,

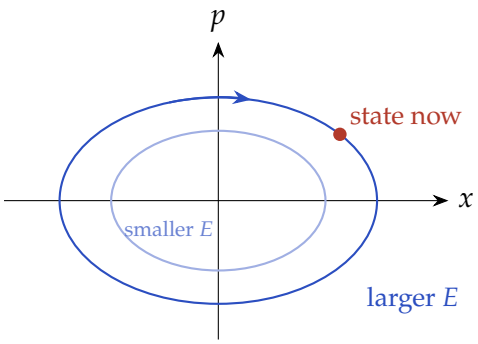
$$E = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}kx^2 = \tfrac{1}{2}kA^2,$$

and it is constant: energy sloshes back and forth between motion and stretch, twice per period. On average the two are equal, $\langle K\rangle = \langle U\rangle = E/2$. (This equal split is the seed of the *equipartition theorem* of Chapter 9.)

Now plot the state of the oscillator not as a function of time, but as a point (x, p) with $p = m\dot{x}$. This plane is called *phase space*, and drawing it is the single most useful habit you can carry into the rest of physics. Fixed energy means

$$\frac{x^2}{2E/k} + \frac{p^2}{2mE} = 1,$$

an ellipse. The state goes around the ellipse, clockwise, at a constant rate.



Rescale the axes so the ellipse becomes a circle, and you see what simple harmonic motion really is: *uniform circular motion seen edge-on*. The projection of a point moving steadily around a circle is a sinusoid. Every fact about the oscillator — the amplitude-independent period, the sinusoidal shape, the quarter-cycle lag between x and p — is obvious the moment you see the circle, and mysterious if you only see the sine.

This picture also explains why physicists write oscillations as complex exponentials. A point going around a circle is $e^{i\omega t}$ in the complex plane, and its real part is $\cos\omega t$. Writing $x(t) = \mathrm{Re}\,[\tilde{A}e^{i\omega t}]$, with a complex amplitude $\tilde{A} = Ae^{i\varphi}$ carrying both size and phase, turns differentiation into multiplication by $i\omega$. Calculus becomes algebra. (See Chapter 1.)

Exercise 3.4. Using the circle picture and no integrals, explain why the oscillator spends more of its time near the turning points than near the centre. The quantum oscillator of Chapter 14 agrees, at least for highly excited states.

§3.4 Damping

Real oscillators lose energy. The simplest model of friction is a drag force proportional to velocity, $-b\dot{x}$. Writing $b = 2m\gamma$,

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0.$$

Try $x = e^{\lambda t}$. Then $\lambda^2 + 2\gamma\lambda + \omega_0^2 = 0$, so $\lambda = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$. Three regimes appear, depending on which is bigger, γ or ω_0:

- **Underdamped** ($\gamma < \omega_0$): $x(t) = Ae^{-\gamma t}\cos(\omega_d t + \varphi)$ with $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$. It still rings, inside a decaying envelope. A plucked guitar string.
- **Critically damped** ($\gamma = \omega_0$): $x(t) = (A + Bt)e^{-\omega_0 t}$. The fastest possible return to rest without overshoot. A well-designed car suspension, or a door closer.
- **Overdamped** ($\gamma > \omega_0$): a sum of two decaying exponentials; it oozes back to equilibrium without oscillating. A spring immersed in honey.

The energy of a weakly damped oscillator decays as $e^{-2\gamma t}$. A useful single number for "how good an oscillator is" is the *quality factor*

$$Q = \frac{\omega_0}{2\gamma} \approx 2\pi \times \frac{\text{energy stored}}{\text{energy lost per cycle}}.$$

Roughly, Q is the number of radians the oscillator rings before it has lost most of its energy. A pendulum clock has $Q \sim 10^4$, a quartz crystal $\sim 10^5$, the atomic transition in a modern optical clock $\sim 10^{17}$.

§3.5 Driving, and resonance

Now push. Add a sinusoidal driving force $F_0 \cos \omega t$:

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = \frac{F_0}{m}\cos \omega t.$$

Whatever the initial condition, the damped free motion dies away and the system settles into a *steady state* oscillating at the *drive's* frequency ω, not its own. The complex-exponential trick makes this a two-line calculation: try $x = \mathrm{Re}[\tilde{A}e^{i\omega t}]$ and find

$$\tilde{A} = \frac{F_0/m}{\omega_0^2 - \omega^2 + 2i\gamma\omega}, \qquad A = |\tilde{A}| = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}.$$

Read the formula in its limits — this is always the first thing to do with a formula.

- **Slow drive** ($\omega \ll \omega_0$): $A \approx F_0/k$. The mass just follows the force, as if the spring were being stretched statically. In phase.
- **Fast drive** ($\omega \gg \omega_0$): $A \approx F_0/(m\omega^2)$. The spring is irrelevant; the mass is too heavy to respond much. Exactly out of phase: the mass moves opposite the force.
- **Resonance** ($\omega \approx \omega_0$): $A \approx F_0/(2m\gamma\omega_0)$, which can be enormous when the damping is small. The response lags the force by exactly a quarter cycle, which is precisely the phase at which the force does the most work.

The resonance peak has width $\Delta\omega \approx 2\gamma = \omega_0/Q$. High-$Q$ systems respond enormously, but only in a narrow band. That is why a radio tuned to one station ignores the others, and why a singer can shatter a wine glass ($Q \sim 10^3$) but not a paper cup.

KEY IDEA Linearity bought us resonance for free. Because the equation is linear, a complicated push can be split into pure tones — this is Fourier's whole program — each tone answered separately, and the answers added back up. The oscillator is where you first see *why* decomposing into frequencies is the natural language of physics.

Example 3.5 (Why the sky is blue, in one line)**.** An electron bound in an air molecule is, to a first approximation, a damped oscillator with ω_0 in the ultraviolet. Visible light drives it far below resonance, where $A \approx F_0/k$ is independent of frequency. But the power it re-radiates goes like $\ddot{x}^2 \propto \omega^4 A^2$ (Chapter 6). Blue light is scattered $(700/450)^4 \approx 6$ times more strongly than red. Look away from the Sun and you see the scattered blue.

§3.6 Beyond the spherical cow: nonlinearity

Two ways an oscillator can stop being simple are worth meeting now.

3.6.1 Anharmonicity

If you keep the cubic and quartic terms in the Taylor expansion, the frequency starts to depend on the amplitude: isochronism breaks. For a pendulum the period lengthens with amplitude, $T \approx T_0(1 + \theta_0^2/16)$. Large anharmonicity, as we will see in Chapter 12, is one road to chaos.

3.6.2 Parametric resonance

Here is a different way to pump energy in: instead of pushing the oscillator, wiggle one of its *parameters*. A child on a swing stands up at the bottom of each swing and squats at the top, changing the effective length of the pendulum twice per period. The equation becomes

$$\ddot{x} + \omega_0^2 \left[1 + h\cos(2\omega_0 t)\right] x = 0,$$

the *Mathieu equation*, and for modulation at twice the natural frequency the amplitude grows *exponentially*, roughly as $e^{h\omega_0 t/4}$. Unlike ordinary resonance, the growth needs some initial wiggle to amplify: a swing exactly at rest cannot be pumped this way. Parametric amplifiers are used in quantum optics and in the readout of superconducting qubits (Chapter 20).

§3.7 Where this goes

The reason to spend real time here is the payoff downstream. The same model, with almost no new ideas, is waiting in chapter after chapter.

- **Waves** (Chapter 5) are coupled oscillators, infinitely many of them, and their normal modes are oscillators in disguise.
- **Quantum mechanics** (Chapter 14): quantize the oscillator and its energy comes in evenly spaced rungs, $E_n = (n + \frac{1}{2})\hbar\omega$, including a zero-point energy that can never be removed.

- **Statistical mechanics** and **condensed matter** (Chapters 9 and 19): a solid is a lattice of atomic oscillators, and counting their quanta explains why heat capacities vanish at low temperatures.
- **Electromagnetism** (Chapter 6): an inductor and capacitor form an oscillator; so does each mode of the field in a box.
- **Quantum field theory** (Chapter 30): a free field is nothing but infinitely many oscillators, one per wavelength, and their quanta *are* particles. A photon is one rung of excitation of one mode of the electromagnetic field.

Much of the rest of physics is this chapter, repeated.

§3.8 Problems

3.1 $O(\epsilon)$ ★ A mass m hangs from a vertical spring and stretches it by d at rest. Show that it oscillates about its new equilibrium with $\omega = \sqrt{g/d}$, and explain in one sentence why gravity dropped out of the frequency.

3.2 $O(\epsilon)$ † An LC circuit has charge obeying $L\ddot{Q} + Q/C = 0$. Read off its frequency, and give the dictionary between this circuit and the mass on a spring: what plays the role of mass? Of stiffness? Of velocity? Of kinetic energy?

3.3 $O(\epsilon)$ A 0.5 kg mass on a spring with $k = 200\,\mathrm{N/m}$ is pulled 4 cm from equilibrium and released. Find the period, the total energy, and the maximum speed.

3.4 $O(\epsilon^2)$ A uniform solid ball of radius r rolls without slipping inside a large spherical bowl of radius $R \gg r$. Show that small oscillations about the bottom have $\omega = \sqrt{5g/7R}$. Where did the factor $5/7$ come from, and why does a sliding (frictionless) ball give $\sqrt{g/R}$ instead?

3.5 $O(\epsilon^2)$ ★ Keep the quartic term you discarded in the pendulum potential. Show that the period grows with amplitude as $T \approx T_0\left(1 + \theta_0^2/16\right)$ for small θ_0. How much does a grandfather clock with a 5° swing lose per day relative to an ideal one?

Hint: Write the period as an integral using energy conservation, $T = 4\int_0^{\theta_0} \mathrm{d}\theta/\dot{\theta}$, and expand.

3.6 $O(\epsilon^2)$ Imagine a straight tunnel drilled through the Earth (assumed uniform) between any two points, not necessarily through the centre. Show that an object dropped into the frictionless tunnel undergoes simple harmonic motion, and that the one-way trip takes about 42 minutes regardless of the tunnel's length. Compare with the period of a satellite skimming the surface.

3.7 $O(\epsilon^2)$ ★ For the driven damped oscillator, show that the steady-state amplitude peaks not exactly at ω_0 but at $\omega_{\max} = \sqrt{\omega_0^2 - 2\gamma^2}$, and that the peak amplitude divided by the static response F_0/k is approximately Q for large Q. Show also that the average power absorbed peaks exactly at ω_0.

3.8 $O(\epsilon^2)$ † A car's suspension is critically damped. The car is loaded with four passengers, increasing its mass by 25%. Is the suspension now underdamped or overdamped? Estimate the new Q.

3.9 $e^{-1/\epsilon}$ ★ The quantum oscillator's ground state has energy $\frac{1}{2}\hbar\omega$, not zero. Using only the uncertainty principle $\Delta x\,\Delta p \gtrsim \hbar/2$, show that the energy of a quantum oscillator cannot be less than about $\frac{1}{2}\hbar\omega$, so this zero-point energy cannot be removed. You have just done real physics with one inequality.

3.10 $e^{-1/\epsilon}$ (*Huygens' tautochrone.*) The pendulum is not isochronous because its bob moves on a circle. Find the curve on which a bead sliding without friction under gravity has a period *exactly* independent of amplitude.

Hint: Isochronism means the restoring force is exactly proportional to the distance s measured *along* the curve. If the slope angle is ϕ, you need $\sin\phi \propto s$. The answer is a cycloid, and Huygens built clocks around it in 1673.

3.11 $e^{-1/\epsilon}$ (*Pumping a swing.*) Model a child pumping a swing as a pendulum whose length changes abruptly: from $L+\delta$ to $L-\delta$ at the bottom of each swing (standing up) and back at each turning point (squatting). Using conservation of angular momentum through the sudden shortening at the bottom, show that the energy is multiplied by a fixed factor $\approx 1 + 6\delta/L$ every half period, so the amplitude grows exponentially. Why can't the child start the swing from rest this way?

Further reading

- **John R. Taylor, *Classical Mechanics*** — Chapter 5 is the definitive undergraduate treatment of oscillations, damping, resonance, and Fourier series, with a superb problem set.
- **A. P. French, *Vibrations and Waves*** — A whole short book on this chapter and the next, written with great physical care.
- **David Morin, *Introduction to Classical Mechanics*** — Chapter 4. Harder problems, beautifully solved.
- **Griffiths and Schroeter, *Introduction to Quantum Mechanics*** — Section 2.3 does the quantum oscillator twice, by brute force and with ladder operators. Do both.

CHAPTER 4

Analytical mechanics

BUILDS ON Newtonian mechanics (Chapter 2), the harmonic oscillator (Chapter 3)

The same physics as Newton, reborn as a single principle — nature is lazy — and the doorway through which every modern theory enters.

PROTOTYPE A ball thrown from point A at time t_1 to point B at time t_2. Of all the paths it could conceivably take between those two events, it takes the one that makes a single number, the *action*, stationary.

Newton's mechanics is about forces and vectors. It works, but it gets awkward fast: a bead on a curved wire, a double pendulum, or a spinning top all require you to find constraint forces you do not actually care about. In the eighteenth and nineteenth centuries Lagrange and Hamilton reformulated mechanics in terms of energies and a variational principle. The new formulation is easier to use, makes symmetries and conservation laws obvious, and — this is the real reason it matters — is the language in which quantum mechanics, field theory, and general relativity are written.

§4.1 The principle of stationary action

Describe a system by any convenient set of *generalized coordinates* $q_1, \dots, q_n$: angles, distances, whatever fits the problem. Define the **Lagrangian**

$$L(q, \dot{q}, t) = T - V,$$

kinetic minus potential energy, and the **action** of a path $q(t)$ as

$$S[q] = \int_{t_1}^{t_2} L(q(t), \dot{q}(t), t)\ \mathrm{d}t.$$

The action is a *functional*: feed it a whole path, and it returns a number.

Law 4.1 (Hamilton's principle)**.** The path a system actually follows between fixed endpoints $q(t_1)$ and $q(t_2)$ is one for which the action is stationary: changing the path by a small amount $\delta q(t)$, vanishing at the endpoints, changes S only at second order.

To turn this into an equation, vary the path, $q \to q + \delta q$, and expand:

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q}\,\delta q + \frac{\partial L}{\partial \dot{q}}\,\delta \dot{q} \right) \mathrm{d}t = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}} \right) \delta q\ \mathrm{d}t,$$

where we integrated the second term by parts and used $\delta q(t_1) = \delta q(t_2) = 0$. For this to vanish for *every* δq, the bracket must vanish at every moment:

Law 4.2 (The Euler–Lagrange equations)**.**

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} = \frac{\partial L}{\partial q_i}, \qquad i = 1, \dots, n.$$

Example 4.3 (Recovering Newton)**.** For a particle in a potential, $L = \frac{1}{2}m\dot{x}^2 - V(x)$. Then $\frac{\partial L}{\partial \dot{x}} = m\dot{x}$ and $\frac{\partial L}{\partial x} = -V'(x)$, so the Euler–Lagrange equation is $m\ddot{x} = -V'(x)$. Newton's second law, recovered from a minimization principle.

Example 4.4 (The pendulum, with no forces in sight)**.** With angle θ as the coordinate, $T = \frac{1}{2}m\ell^2\dot{\theta}^2$ and $V = -mg\ell\cos\theta$. The Euler–Lagrange equation gives $m\ell^2\ddot{\theta} = -mg\ell\sin\theta$ immediately. The tension in the string never appears, because the coordinate θ automatically respects the constraint. This is the practical superpower of Lagrangian mechanics: *choose coordinates that respect the constraints, and the constraint forces disappear*.

Example 4.5 (Polar coordinates, and a force that comes from nowhere)**.** A particle in a central potential $V(r)$, in the plane: $L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2) - V(r)$. The r equation is

$$m\ddot{r} = mr\dot{\phi}^2 - V'(r),$$

and the term $mr\dot{\phi}^2$ is the centrifugal force, produced automatically by the kinetic energy in curvilinear coordinates. The ϕ equation says $\frac{\mathrm{d}}{\mathrm{d}t}(mr^2\dot{\phi}) = 0$: angular momentum is conserved, because L does not depend on ϕ.

§4.2 Momenta, symmetry, and Noether's theorem

The quantity $p_i = \frac{\partial L}{\partial \dot{q}_i}$ is the *generalized momentum* conjugate to q_i. For an ordinary coordinate it is ordinary momentum; for an angle it is angular momentum. If L happens not to depend on some q_i (a *cyclic* coordinate), the Euler–Lagrange equation says $\dot{p}_i = 0$: that momentum is conserved.

This is a special case of the most beautiful theorem in classical physics.

Theorem 4.6 (Noether's theorem)**.** For every continuous symmetry of the action there is a conserved quantity.

- Symmetry under translation in time $\Longrightarrow$ conservation of energy.
- Symmetry under translation in space $\Longrightarrow$ conservation of momentum.
- Symmetry under rotation $\Longrightarrow$ conservation of angular momentum.

Read that list again. Energy is conserved *because* the laws of physics are the same today as they were yesterday. Momentum is conserved *because* the laws are the same here as over there. These conservation laws are not separate facts about nature; they are shadows of its symmetries. When a symmetry is broken — a ball rolling on a surface whose shape changes with time, say — the corresponding quantity is no longer conserved.

The time-translation case deserves an explicit formula. If L has no explicit time dependence, then

$$H = \sum_i p_i \dot{q}_i - L$$

is conserved. For ordinary systems, $H = T + V$, the total energy.

KEY IDEA Symmetries give conservation laws. In modern physics this runs in the other direction too: we *postulate* symmetries and deduce the laws. Electric charge is conserved because of a symmetry of the quantum phase of the electron (Chapter 30), and the forces of the Standard Model are dictated by symmetry (Chapter 32).

§4.3 Variational principles everywhere

Hamilton's principle is one example of a pattern that runs through all of physics: *the actual configuration extremizes something*.

- **Fermat's principle** (Chapter 8): light travels between two points along the path of stationary travel time. Snell's law of refraction falls out immediately.
- **The brachistochrone**: the curve of fastest descent between two points under gravity is not a straight line but a cycloid. Johann Bernoulli posed the problem in 1696; Newton reportedly solved it overnight.
- **Geodesics**: the shortest path between two points on a curved surface. In general relativity (Chapter 26), freely falling objects follow geodesics of spacetime — which is to say, they extremize their proper time.
- **Soap films** minimize their area, and a hanging chain minimizes its potential energy (the curve is a catenary, $y = a\cosh(x/a)$).

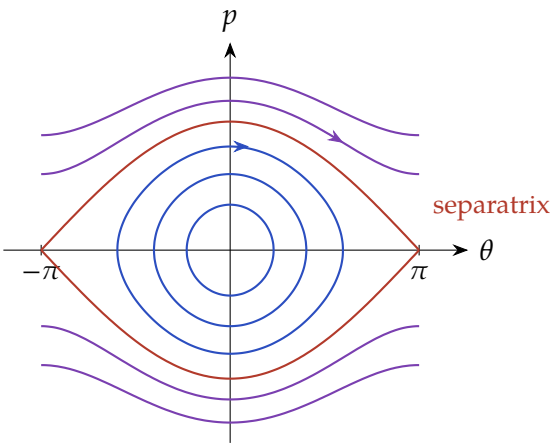
§4.4 Hamiltonian mechanics

Lagrange's equations are n second-order equations. Hamilton rewrote them as $2n$ first-order equations by treating the momenta p_i as independent variables alongside the q_i. Trading $\dot{q}$ for p by a *Legendre transform* gives the **Hamiltonian** $H(q,p) = \sum p_i \dot{q}_i - L$, and the equations of motion become beautifully symmetric:

Law 4.7 (Hamilton's equations)**.**

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

For the harmonic oscillator, $H = p^2/2m + \frac{1}{2}kx^2$, and Hamilton's equations give $\dot{x} = p/m$, $\dot{p} = -kx$: the familiar ellipse in phase space (Chapter 3). In general, the state of a system is a single point in the $2n$-dimensional *phase space*, and Hamilton's equations define a flow: every point moves along a definite trajectory, and trajectories never cross.



The phase space of a pendulum. Each curve is a path of constant energy. Low energies give closed loops (blue: swinging back and forth); high energies give wavy lines (purple: spinning over the top); the separatrix divides the two.

Theorem 4.8 (Liouville's theorem)**.** The Hamiltonian flow preserves volume in phase space. A blob of initial conditions may stretch and fold, but its volume never changes.

Liouville's theorem has big consequences. A Hamiltonian system cannot have an attractor — a region that sucks in all nearby trajectories — because that would shrink phase-space volume; attractors need friction (Chapter 12). And it is the reason statistical mechanics can assign equal probability to equal volumes of phase space (Chapter 9): the dynamics never makes some regions "more likely" than others.

§4.5 Poisson brackets: the bridge to quantum mechanics

For any two functions $f(q,p)$ and $g(q,p)$ on phase space, define the **Poisson bracket**

$$\{f,g\} = \sum_i \left(\frac{\partial f}{\partial q_i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q_i} \right).$$

Then the time evolution of *any* quantity is

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \{f, H\},$$

so f is conserved exactly when its bracket with H vanishes. The basic brackets are $\{q_i, p_j\} = \delta_{ij}$, and for angular momentum $\{L_x, L_y\} = L_z$ and cyclic permutations.

Now the astonishing part. In 1925 Dirac noticed that quantum mechanics has exactly the same structure, with the Poisson bracket replaced by a commutator of operators:

$$\{f,g\} \quad\longrightarrow\quad \frac{1}{i\hbar}[\hat{f},\hat{g}], \qquad \text{so} \qquad \{x,p\} = 1 \;\longrightarrow\; [\hat{x},\hat{p}] = i\hbar.$$

Every classical structure in this section survives quantization almost unchanged. That is why we learn Hamiltonian mechanics: it is the classical shadow of quantum mechanics (Chapter 14).

§4.6 Small oscillations and normal modes

Near a stable equilibrium, any system with n degrees of freedom has, to second order,

$$L = \tfrac{1}{2}\,\dot{\mathbf{q}}^{T} M\,\dot{\mathbf{q}} - \tfrac{1}{2}\,\mathbf{q}^{T} K\,\mathbf{q},$$

with M and K symmetric matrices. The equations of motion $M\ddot{\mathbf{q}} = -K\mathbf{q}$ are solved by $\mathbf{q} = \mathbf{a}\,e^{i\omega t}$ if

$$\det(K - \omega^2 M) = 0.$$

The n solutions ω_k are the *normal mode frequencies*, and in the coordinates of the eigenvectors the system falls apart into n independent harmonic oscillators. Every vibrating system — a molecule, a bridge, a crystal, a drum — is a collection of independent oscillators in the right coordinates. This is the spherical cow of Chapter 3, multiplied.

§4.7 Adiabatic invariants

Suppose a parameter of a system changes slowly compared to its natural period: a pendulum whose string is slowly shortened, say. The energy is not conserved — work is being done — but something else nearly is. For any periodic motion, the **action variable**

$$J = \oint p\,\mathrm{d}q,$$

the area enclosed by the orbit in phase space, is an *adiabatic invariant*: it changes by an amount that vanishes faster than any power of the slowness. For an oscillator, the ellipse has area $2\pi E/\omega$, so E/ω stays constant: shorten a pendulum slowly and its energy rises in proportion to its frequency.

This had a remarkable historical role. In early quantum theory, the quantities that were quantized ($E = n\hbar\omega$ for oscillators, $J = nh$ in general) had to be adiabatic invariants, or else slowly changing the parameters could carry a system between allowed states into forbidden ones.

§4.8 Why the action? A glimpse of the path integral

Why should nature extremize anything? The deepest answer comes from quantum mechanics. In Feynman's formulation, a quantum particle does not take one path: it takes *all* of them, and each path contributes a complex number $e^{iS/\hbar}$ to the probability amplitude. For macroscopic objects, $S \gg \hbar$, and the phase $S/\hbar$ swings wildly between neighbouring paths, so their contributions cancel — except near paths where S is stationary, where neighbouring paths have nearly the same phase and add up. The classical path is the one that survives the cancellation.

KEY IDEA Classical mechanics is the $\hbar \to 0$ limit of quantum mechanics, and the principle of stationary action is what that limit looks like. Nature is not lazy; it is democratic, and the classical path wins by constructive interference.

§4.9 Problems

4.1 $O(\epsilon)$ Write down the Lagrangian of an Atwood machine (two masses on a pulley) using a single coordinate, derive its acceleration, and confirm it agrees with Newton's laws. Notice that you never needed the tension.

4.2 $O(\epsilon)$ ★ For a particle moving in a central potential in three dimensions, use spherical coordinates to show that $L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2) - V(r)$, and identify which momentum is immediately conserved. Why are all three components of angular momentum conserved, even though only one is manifest?

4.3 $O(\epsilon)$ Write the Hamiltonian of the harmonic oscillator, derive Hamilton's equations, and verify that the phase-space trajectories are ellipses traversed clockwise. What is the area of the ellipse in terms of E and ω?

4.4 $O(\epsilon^2)$ Redo the bead on a rotating hoop (Problem 2.11) using a Lagrangian in the rotating frame. The Lagrangian has no explicit time dependence, so H is conserved; show that H is *not* the bead's energy, and explain why the energy is not conserved even though H is.

4.5 $O(\epsilon^2)$ ★ (*The double pendulum.*) Two equal masses m hang in series from equal massless rods of length ℓ. Write the Lagrangian, linearize for small angles, and show that the normal mode frequencies are $\omega^2 = (2 \pm \sqrt{2})\,g/\ell$. Describe each mode. (For large angles, the double pendulum is chaotic; see Chapter 12.)

4.6 $O(\epsilon^2)$ ★ Compute the Poisson brackets $\{L_x, L_y\}$, $\{L_z, L^2\}$, and $\{x, L_z\}$, where $\mathbf{L} = \mathbf{r} \times \mathbf{p}$. These become the angular momentum commutation relations of quantum mechanics, and they are why a quantum particle cannot have a definite L_x and L_y at the same time.

4.7 $O(\epsilon^2)$ A spherical pendulum (a mass on a rod free to swing in any direction) has two conserved quantities. Identify them, reduce the motion to an effective one-dimensional problem in the polar angle θ, and find the angular frequency of a steady conical orbit at angle θ_0.

4.8 $O(\epsilon^2)$ (*Fermat implies Snell.*) Light travels at speed c/n_1 in one medium and c/n_2 in another, separated by a flat interface. Minimize the travel time between a point in each medium to derive Snell's law, $n_1 \sin\theta_1 = n_2 \sin\theta_2$.

4.9 $e^{-1/\epsilon}$ ★ (*The brachistochrone.*) Find the curve $y(x)$ down which a bead, starting from rest, slides frictionlessly from one point to a lower one in the least time. Show that the time is $\int \sqrt{(1 + y'^2)/(2gy)}\,dx$ (with y measured downward), use the fact that the integrand

does not depend on x to obtain a first integral, and show that the solution is a cycloid. Compare with Problem 3.10.

4.10 $e^{-1/\epsilon}$ (*The Kapitza pendulum.*) The pivot of a pendulum is shaken vertically as $a\cos\Omega t$, with $a \ll \ell$ and Ω much larger than the pendulum's natural frequency. Separating the motion into a slow part and a fast jitter and averaging over the fast jitter, show that the slow motion feels an effective potential

$$U_{\text{eff}}(\theta) = -mg\ell\cos\theta + \frac{ma^2\Omega^2}{4}\sin^2\theta,$$

and conclude that the *inverted* position $\theta = \pi$ becomes stable when $a^2\Omega^2 > 2g\ell$. A broomstick balanced on a vibrating saw stays upright.

Hint: Write $\theta = \bar{\theta} + \xi$ with ξ small and fast; solve for ξ to leading order, then average the force over one fast period.

4.11 $e^{-1/\epsilon}$ (*An adiabatic piston.*) A ball bounces elastically back and forth between two walls a distance ℓ apart. One wall is moved slowly, at a speed much smaller than the ball's. Show that $v\ell$ is constant, so the ball's energy scales as $1/\ell^2$. Interpreting a gas as many such balls, show that slowly compressing a one-dimensional gas gives $P\ell^3 =$ constant, and compare with the adiabatic law $PV^\gamma =$ constant of Chapter 7. What is γ for a one-dimensional gas?

Further reading

- **John R. Taylor, *Classical Mechanics*** — Chapters 6, 7, 11, and 13 give the gentlest complete introduction to Lagrangians, Hamiltonians, and normal modes.
- ***The Feynman Lectures on Physics*, Volume II, Chapter 19** — "The Principle of Least Action," a famous lecture that explains why the action matters better than any textbook.
- **L. D. Landau and E. M. Lifshitz, *Mechanics*** — Short, dense, and magnificent. Derives all of mechanics from symmetry and the action. Read it after a first course.
- **Helliwell and Sahakian, *Modern Classical Mechanics*** — A graduate-level treatment with modern examples and close attention to the bridge to quantum mechanics.

CHAPTER 5

Waves

BUILDS ON the harmonic oscillator (Chapter 3)

Everything that wiggles and carries the wiggle somewhere else: plucked strings, sound, ocean swell, light, and — in disguise — the electron.

> **PROTOTYPE** A taut string, plucked. A wave is a pattern of displacement that travels along it while each bit of string merely moves up and down. Information and energy move; matter stays put.

A single oscillator stays where it is. Couple many of them together — each one tugging on its neighbours — and the motion can travel. That is all a wave is. The idea is simple, but its reach is enormous: sound, light, earthquakes, water waves, and the quantum wavefunction all obey versions of the equation we derive below.

§5.1 From oscillators to waves

Picture a line of beads of mass m, spaced a distance a apart, joined by springs of stiffness K. Let y_j be the displacement of bead j. Each bead is pulled by its two neighbours:

$$m\ddot{y}_j = K(y_{j+1} - y_j) - K(y_j - y_{j-1}) = K\,(y_{j+1} - 2y_j + y_{j-1}).$$

That combination $y_{j+1} - 2y_j + y_{j-1}$ is a discrete second derivative. Now let the spacing shrink, $a \to 0$, keeping the mass per length $\mu = m/a$ and the tension-like quantity Ka fixed. The displacement becomes a smooth field $y(x,t)$ and $y_{j+1} - 2y_j + y_{j-1} \approx a^2\,\partial^2 y/\partial x^2$. We get

$$\frac{\partial^2 y}{\partial t^2} = v^2\,\frac{\partial^2 y}{\partial x^2}.$$

This is the **wave equation**. For a string under tension T with mass per length μ, a direct force balance on a small segment gives the same equation with

$$v = \sqrt{\frac{T}{\mu}}.$$

The pattern "speed $= \sqrt{\text{stiffness}/\text{inertia}}$" recurs everywhere: sound in a gas, $v = \sqrt{\gamma P/\rho}$; waves on deep water, $\sqrt{g/k}$; light in vacuum, $1/\sqrt{\mu_0\epsilon_0}$.

§5.2 Solving the wave equation

Theorem 5.1 (d'Alembert's solution)**.** Every solution of the one-dimensional wave equation has the form

$$y(x,t) = f(x - vt) + g(x + vt)$$

for some functions f and g: a shape moving right at speed v plus a shape moving left.

Exercise 5.2. Verify by direct differentiation that $f(x-vt)$ solves the wave equation for any twice-differentiable f. Notice that its shape never changes as it moves. That is special: it is because v does not depend on wavelength (see §5.5).

The most important shape is the sinusoid:

$$y = A\cos(kx - \omega t), \qquad k = \frac{2\pi}{\lambda}, \quad \omega = 2\pi f, \quad \omega = vk.$$

The *wavenumber* k counts radians per metre just as ω counts radians per second. In complex form we write $y = \operatorname{Re}[Ae^{i(kx-\omega t)}]$, and because the wave equation is linear, Fourier analysis (Chapter 1) builds every other wave out of these.

§5.3 Superposition: interference, beats, and standing waves

Because the wave equation is linear, waves pass through each other and simply add. Where crests meet crests the waves reinforce (*constructive interference*); where crests meet troughs they cancel (*destructive interference*). This is the defining behaviour of waves, and it will be the decisive evidence that light (Chapter 8) and electrons (Chapter 14) are waves.

Beats. Add two waves of nearby frequencies f_1 and f_2 at one point:

$$\cos\omega_1 t + \cos\omega_2 t = 2\cos\Big(\frac{\omega_1 - \omega_2}{2}t\Big)\cos\Big(\frac{\omega_1 + \omega_2}{2}t\Big).$$

You hear the average pitch, swelling and fading $|f_1 - f_2|$ times per second. Piano tuners use beats to match strings to better than a tenth of a hertz.

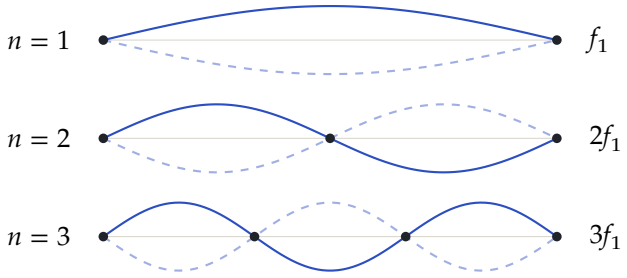
Standing waves. Two equal waves going in opposite directions add to

$$A\cos(kx - \omega t) + A\cos(kx + \omega t) = 2A\cos(kx)\cos(\omega t),$$

a pattern that does not travel at all: every point oscillates in place, with fixed *nodes* where $\cos kx = 0$. A string clamped at both ends ($y = 0$ at $x = 0$ and $x = L$) can only hold standing waves that fit:

$$\lambda_n = \frac{2L}{n}, \qquad f_n = \frac{nv}{2L}, \qquad n = 1, 2, 3, \ldots$$

These are the *normal modes* of the string, the continuum version of Chapter 3's coupled oscillators. A plucked string vibrates in many modes at once; the fundamental f_1 sets the pitch, and the mix of harmonics $2f_1, 3f_1, \ldots$ sets the timbre — why a violin and a flute playing the same note sound different.



The first three standing waves on a string fixed at both ends, drawn at two moments half a period apart. The dots are nodes, points that never move.

KEY IDEA Confinement gives quantization. A wave trapped in a region can only take certain discrete frequencies, because it has to fit. The same fact, applied to the electron's wave in an atom, is why atomic energy levels are discrete (Chapter 14).

§5.4 Energy, impedance, and reflection

A wave carries energy. For a sinusoidal wave on a string the power transmitted is

$$P = \tfrac{1}{2} Z \omega^2 A^2, \qquad Z = \sqrt{T\mu} = \mu v,$$

where Z is the *impedance* of the string: its resistance to being shaken. Notice that $P \propto A^2$; this is general. The energy of a wave goes as the square of its amplitude.

When a wave reaches a junction between two strings of different impedance, part reflects and part transmits. Requiring the string to be continuous and the transverse forces to balance at the junction gives

$$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \qquad t = \frac{2Z_1}{Z_1 + Z_2}$$

for the reflected and transmitted amplitudes. A wave hitting a much heavier string ($Z_2 \gg Z_1$) reflects almost entirely and flips upside down; hitting a light string, it reflects without flipping. When $Z_1 = Z_2$ nothing reflects: the impedances are *matched*.

Impedance matching is everywhere once you look. Ultrasound scanners use gel because air and skin have wildly different acoustic impedances (Chapter 24). The anti-reflection coating on glasses is a quarter-wavelength layer chosen to cancel reflections. Your middle ear is an impedance matcher between air and the fluid of the cochlea. And transmission lines carrying signals must be terminated in a matched resistor to avoid echoes.

§5.5 Dispersion

For a string, $\omega = vk$ exactly, and every wavelength travels at the same speed. Most media are not so simple: the relation $\omega(k)$, called the *dispersion relation*, is not a straight line. Then two velocities matter:

$$v_{\text{phase}} = \frac{\omega}{k}, \qquad v_{\text{group}} = \frac{\mathrm{d}\omega}{\mathrm{d}k}.$$

The phase velocity is the speed of an individual crest. The group velocity is the speed of the *envelope* of a wave packet — and so the speed at which energy and information travel.

Example 5.3 (Deep-water waves)**.** For gravity waves on deep water, $\omega = \sqrt{gk}$. Then $v_{\text{phase}} = \sqrt{g/k}$ and $v_{\text{group}} = \frac{1}{2}\sqrt{g/k}$: the packet moves at half the speed of its crests. Watch a group of waves spread from a stone dropped in a pond: individual ripples appear at the back of the group, run forward through it, and vanish at the front. Because long waves travel faster, a storm thousands of kilometres away is announced by long, slow swell arriving days ahead of the shorter waves — which is how surf forecasters work.

Example 5.4 (The beaded string revisited)**.** Return to the discrete chain of beads. Trying $y_j = Ae^{i(kja-\omega t)}$ gives

$$\omega = 2\sqrt{K/m}\ \left|\sin(ka/2)\right|.$$

For long wavelengths ($ka \ll 1$) this is $\omega \approx vk$, the continuum wave equation. But for short wavelengths the dispersion relation bends over and there is a *maximum frequency* $2\sqrt{K/m}$: a lattice cannot vibrate faster than its atoms can respond. Wavelengths shorter than $2a$ are meaningless — they describe the same bead motions as longer ones. This is our first glimpse of the *Brillouin zone*, and of phonons in crystals (Chapter 19).

A wave packet in a dispersive medium spreads, because its different wavelength components travel at different speeds. A packet of width Δx contains wavenumbers spread over $\Delta k \sim 1/\Delta x$ (Chapter 1); a narrow packet therefore spreads fast. This is why a quantum particle's wavefunction spreads out over time (Chapter 14), and why fibre-optic signals must be periodically regenerated.

§5.6 Waves in two and three dimensions

In higher dimensions the wave equation becomes

$$\frac{\partial^2 \psi}{\partial t^2} = v^2 \nabla^2 \psi.$$

Plane waves $e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$ travel in the direction of the wave vector $\mathbf{k}$. *Spherical waves* spreading from a point have amplitude falling as $1/r$, so their intensity falls as $1/r^2$: the energy is spread over a sphere of area $4\pi r^2$. That is the reason for every inverse-square law of radiation.

Huygens' principle says that every point on a wavefront acts as a source of new wavelets, and the next wavefront is their envelope. It explains why waves bend around obstacles — *diffraction*. A wave passing through an opening of width d spreads out by an angle

$$\theta \sim \frac{\lambda}{d}.$$

Sound ($\lambda \sim 1\,\text{m}$) diffracts around doorways, so you can hear around corners; light ($\lambda \sim 0.5\,\mu\text{m}$) does not, so you cannot see around them. The same formula sets the resolution of every telescope and microscope (Chapter 8).

§5.7 The Doppler effect

A source of sound at frequency f moving toward you at speed v_s squeezes its wavefronts together; one moving away stretches them. If you, the observer, move at speed v_o toward the source, you run into crests more often. Altogether, for sound of speed v,

$$f' = f\,\frac{v + v_o}{v - v_s}$$

(with velocities counted positive toward each other). The rising-then-falling pitch of a passing siren is the textbook example; radar speed guns, Doppler ultrasound of blood flow (Chapter 24), and the redshift of receding galaxies (Chapter 28) are the same physics. For light the formula must be modified by relativity (Chapter 10), because there is no medium to move relative to.

When the source moves faster than the waves, $v_s > v$, the wavefronts pile up into a cone — a *shock wave* — of half-angle $\sin\theta = v/v_s$. This is the sonic boom of a supersonic jet, the bow wave of a boat, and the Cherenkov light of a particle faster than light in water (Chapter 31).

§5.8 Problems

5.1 $O(\epsilon)$ ★ A guitar string of length 65 cm has fundamental frequency 110 Hz (the note A_2). Find the wave speed. If the string has mass per length 6 g/m, find its tension. By what factor must the tension change to raise the pitch one octave?

5.2 $O(\epsilon)$ Two tuning forks sound at 440 Hz and 443 Hz. How many beats per second do you hear, and what pitch? If a small piece of wax is stuck to the second fork, the beats slow down. Was the wax's effect to raise or lower its frequency?

5.3 $O(\epsilon)$ A string is fixed at one end and free (attached to a massless ring on a frictionless pole) at the other. Find its allowed wavelengths and show that only odd harmonics appear. This is why a clarinet, a pipe closed at one end, sounds "hollow."

5.4 $O(\epsilon)$ An ambulance siren sounds at 700 Hz while driving at 30 m/s. What frequencies do you hear as it approaches and recedes? (Take $v = 343$ m/s.) How many semitones does the pitch drop? (A semitone is a factor $2^{1/12}$.)

5.5 $O(\epsilon^2)$ ★ A wave on a string of impedance Z_1 meets a string of impedance Z_2. Derive the reflection and transmission amplitudes from continuity of displacement and of the transverse force, and show that the reflected and transmitted powers add up to the incident power.

5.6 $O(\epsilon^2)$ Show that the group velocity of deep-water waves is half their phase velocity. A storm 2000 km away generates swell with periods from 8 to 20 seconds. How long after the first long-period swell arrives do the 8-second waves arrive?

5.7 $O(\epsilon^2)$ ★ A wave packet is a Gaussian of width σ in x: $y(x,0) = e^{-x^2/2\sigma^2}\cos k_0 x$. Compute its Fourier transform and show it is a Gaussian in k of width $1/\sigma$. Conclude that $\Delta x\,\Delta k \geq \tfrac{1}{2}$ for this packet, and argue that no packet can do better. (With $p = \hbar k$, this is the uncertainty principle.)

5.8 $O(\epsilon^2)$ ★ For the beaded string, derive the dispersion relation $\omega = 2\sqrt{K/m}\,|\sin(ka/2)|$. Find the group velocity as a function of k, and show that it vanishes at $k = \pi/a$. What does a standing wave at that wavenumber look like?

5.9 $O(\epsilon^2)$ A jet flies at Mach 2 at an altitude of 10 km. Find the half-angle of its shock cone, and how long after it passes directly overhead you hear the sonic boom.

5.10 $e^{-1/\epsilon}$ (*Green's law.*) A tsunami travels across the open ocean (depth 4 km) with an amplitude of half a metre. Shallow-water waves travel at $v = \sqrt{gh}$. Assuming the energy flux is conserved as the wave moves into shallower water without reflecting, and that the wave's length scales with its speed, show that its amplitude grows as $h^{-1/4}$. Estimate the amplitude in water 10 m deep, and the speed in the open ocean.

Hint: The energy flux is (energy per area) × (group velocity), and energy per area scales as amplitude squared.

5.11 $e^{-1/\epsilon}$ (*Why waves break.*) In shallow water the wave speed depends on the local depth, $v \approx \sqrt{g(h+\eta)}$, where η is the surface height. Explain why crests move faster than troughs, and show that a wave approaching a beach must eventually steepen until its front becomes vertical. This is the simplest example of a nonlinear wave forming a shock.

5.12 $e^{-1/\epsilon}$ (*Kelvin's wake.*) Behind a duck, a boat, or a ship moving at constant speed on deep water, the wake is confined to a wedge of half-angle $\arcsin(1/3) \approx 19.5^\circ$ — *independent of speed*. Derive this.

Hint: The boat continuously emits waves of all wavelengths. A wave that keeps up with the boat must have phase velocity $u\cos\theta$, where θ is the angle of its crest normal to the boat's path, but its energy travels at the group velocity, which is half the phase velocity. Find the maximum angle at which energy emitted at earlier times can be found.

Further reading

- **A. P. French, *Vibrations and Waves*** — The standard short introduction, very physical.
- **Frank S. Crawford, *Waves* (Berkeley Physics Course, Vol. 3)** — Full of home experiments and physical insight.
- **Howard Georgi, *The Physics of Waves*** — A clean, systematic treatment built on normal modes and symmetry. Freely available online.
- ***The Feynman Lectures on Physics*, Volume I, Chapters 47–51** — Sound, the wave equation, beats, modes, and harmonics.

CHAPTER 6

Electromagnetism

BUILDS ON mathematical methods (Chapter 1), Newtonian mechanics (Chapter 2)

The first field theory, the first great unification, and still the template for every theory that came after.

> **PROTOTYPE** A single point charge. At rest, it surrounds itself with an electric field that falls off as $1/r^2$. Moving steadily, it adds a magnetic field. Shaken back and forth, it sends ripples of field outward at the speed of light — and that is what light *is*.

Electricity and magnetism look like separate things — static cling and compass needles — and for most of history they were treated separately. Over the nineteenth century Faraday and then Maxwell showed that they are two faces of a single *electromagnetic field*, and that ripples in this field travel at a speed that matched the measured speed of light. Electromagnetism explains almost everything in daily life that is not gravity: chemistry, friction, the solidity of matter, light, and every electrical device.

§6.1 Charge and Coulomb's law

Electric charge comes in two signs, is conserved, and is quantized in units of the electron charge $e = 1.602 \times 10^{-19}\,\mathrm{C}$. Like charges repel, opposite charges attract, with a force

$$F = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}, \qquad \frac{1}{4\pi\epsilon_0} \approx 8.99 \times 10^9\,\mathrm{N\,m^2/C^2}.$$

This is the same inverse-square law as gravity, but vastly stronger: two electrons repel each other electrically about 4×10^{42} times more strongly than they attract gravitationally. Gravity dominates the universe only because matter is so nearly neutral that the electric forces cancel.

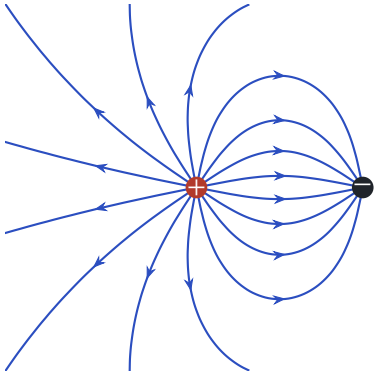
§6.2 The electric field

Instead of saying that charges act on each other at a distance, we say each charge creates an *electric field* $\mathbf{E}(\mathbf{r})$ filling space, and a charge q placed at $\mathbf{r}$ feels a force $q\mathbf{E}(\mathbf{r})$. For a point charge,

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0 r^2}\,\hat{\mathbf{r}},$$

and fields of many charges simply add. At first the field looks like bookkeeping. By the end of this chapter it will carry energy, momentum, and waves of its own, and it will be impossible to think of it as anything but real.

A pair of opposite charges $\pm q$ separated by $\mathbf{d}$ forms an *electric dipole* with moment $\mathbf{p} = q\mathbf{d}$. Far away its field falls off as $1/r^3$ rather than $1/r^2$, because the two charges nearly cancel. Most molecules are neutral, so their electric influence on each other is a dipole effect or weaker — which is why intermolecular forces are short-ranged (Chapter 21).



The field lines of an electric dipole. Each line leaves the positive charge and, unless it escapes off the page, ends on the negative one. The field is strong where the lines crowd together.

§6.3 Gauss's law

Define the *flux* of $\mathbf{E}$ through a surface as $\int \mathbf{E} \cdot \mathrm{d}\mathbf{A}$: roughly, the number of field lines piercing it. For a point charge at the centre of a sphere, the flux is $E \cdot 4\pi r^2 = q/\epsilon_0$, independent of r — the $1/r^2$ of the field exactly cancels the r^2 of the area. That is not a coincidence of the sphere; it is true for any closed surface.

Law 6.1 (Gauss's law)**.** For any closed surface S enclosing total charge Q_{enc},

$$\oint_S \mathbf{E} \cdot \mathrm{d}\mathbf{A} = \frac{Q_{\text{enc}}}{\epsilon_0} \qquad \Longleftrightarrow \qquad \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}.$$

The two forms are related by the divergence theorem (Chapter 1). Charges are the sources of the electric field.

With enough symmetry, Gauss's law hands you the field in one line. A uniformly charged sphere looks, from outside, exactly like a point charge (this is the electric shell theorem). An infinite line of charge density λ has $E = \lambda/(2\pi\epsilon_0 r)$. An infinite plane of charge density σ has $E = \sigma/(2\epsilon_0)$, independent of distance.

Conductors. In a conductor, charges move freely until they feel no force. So in equilibrium $\mathbf{E} = 0$ inside a conductor, any excess charge sits on its surface, and the field just outside is perpendicular to the surface. A hollow conductor shields its interior from outside fields: this is a *Faraday cage*, which is why you are safe in a car struck by lightning and why your phone loses signal in a lift.

§6.4 Potential and energy

Because the electrostatic field is conservative ($\nabla \times \mathbf{E} = 0$), we can write $\mathbf{E} = -\nabla V$ for an *electric potential* V, measured in volts. A charge q moving through a potential difference ΔV gains energy $q\Delta V$. One *electron-volt*, $1\,\text{eV} = 1.6 \times 10^{-19}\,\text{J}$, is the energy an electron gains falling through one volt; it is the natural unit of atomic physics. Chemical bonds are a few eV; visible photons are about 2–3 eV.

Combining $\mathbf{E} = -\nabla V$ with Gauss's law gives *Poisson's equation* $\nabla^2 V = -\rho/\epsilon_0$, and in empty space *Laplace's equation* $\nabla^2 V = 0$. Solutions of Laplace's equation have no local maxima or minima: the value at any point is the average of its surroundings.

A *capacitor* — two conductors holding charges $\pm Q$ — stores energy. For parallel plates of area A and separation d, $C = Q/V = \epsilon_0 A/d$ and the stored energy is $\frac{1}{2}CV^2$. Where is this energy? The beautiful answer is: in the field. The energy density of an electric field is

$$u_E = \tfrac{1}{2}\epsilon_0 E^2,$$

and integrating it over the gap reproduces $\frac{1}{2}CV^2$ exactly.

§6.5 Currents and circuits

A current I is charge flow per time. In most materials, current is proportional to voltage, $V = IR$ (*Ohm's law*), with resistance $R = \rho_{\text{r}} L/A$ set by the resistivity of the material. A resistor dissipates power $P = IV = I^2R$ as heat. Microscopically, electrons are accelerated by the field and constantly scattered by the lattice, reaching a drift speed of only millimetres per second — even though the signal travels at nearly the speed of light (Chapter 19).

Circuits obey Kirchhoff's rules: current is conserved at every junction, and the voltage changes around any loop sum to zero (energy conservation). A capacitor discharging through a resistor obeys $R\dot{Q} = -Q/C$, so the charge decays exponentially with time constant $\tau = RC$ — the first-order equation of Chapter 1 again. A camera flash charges its capacitor over a second or two and dumps it in a millisecond.

§6.6 Magnetism

Moving charges feel a second force. The full force on a charge is the **Lorentz force**

$$\mathbf{F} = q\,(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

The magnetic part is always perpendicular to the velocity, so *magnetic forces do no work*; they bend paths but never change speeds. A charge in a uniform field $\mathbf{B}$ therefore circles at the *cyclotron frequency*

$$\omega_c = \frac{qB}{m},$$

independent of its speed. Faster particles circle in bigger orbits in the same time. This is the principle of the cyclotron (Chapter 31), of mass spectrometers, and of the auroras, where solar-wind particles spiral down the Earth's field lines (Chapter 17).

Magnetic fields are made by currents. A long straight wire carrying current I has field lines circling it with

$$B = \frac{\mu_0 I}{2\pi r}, \qquad \mu_0 = 4\pi \times 10^{-7}\,\mathrm{T\,m/A}.$$

A solenoid with n turns per metre has a uniform field $B = \mu_0 n I$ inside it. Both follow from Ampère's law, $\oint \mathbf{B}\cdot \mathrm{d}\mathbf{l} = \mu_0 I_{\text{enc}}$, the magnetic analogue of Gauss's law. And unlike electric fields, magnetic field lines never begin or end:

$$\nabla \cdot \mathbf{B} = 0.$$

There are no magnetic charges ("monopoles"), as far as anyone has found. The simplest source of magnetism is a current loop, a *magnetic dipole*; a bar magnet is a vast number of aligned atomic dipoles, ultimately due to electron spin (Chapters 14 and 19).

§6.7 Induction

In 1831 Faraday discovered that a *changing* magnetic field creates an electric field.

Law 6.2 (Faraday's law)**.** The electromotive force around a loop equals minus the rate of change of magnetic flux through it:

$$\oint \mathbf{E}\cdot \mathrm{d}\mathbf{l} = -\frac{\mathrm{d}\Phi_B}{\mathrm{d}t} \qquad \Longleftrightarrow \qquad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

The minus sign is *Lenz's law*: induced currents flow so as to oppose the change that made them. Drop a magnet down a copper pipe and it falls in slow motion, braked by the currents it induces. Faraday's law runs the modern world: every generator spins a coil in a magnetic field, every transformer couples two coils through a changing flux, and every electric guitar pickup, induction hob, and wireless charger is Faraday's law in a box.

A coil opposes changes in its own current, with *inductance* L defined by $\mathcal{E} = -L\,\mathrm{d}I/\mathrm{d}t$. It stores energy $\tfrac{1}{2}LI^2$, which again lives in the field, with density

$$u_B = \frac{B^2}{2\mu_0}.$$

An inductor and a capacitor together make an oscillator, $L\ddot{Q} = -Q/C$, ringing at $\omega = 1/\sqrt{LC}$ (Chapter 3). Every radio tuner is one.

§6.8 Maxwell's equations

By the 1860s, the laws of Gauss, Faraday, and Ampère were known, but Maxwell noticed that Ampère's law was inconsistent. Charge a capacitor: current flows in the wire but not across the gap. A loop around the wire can be spanned by a surface that cuts the wire (enclosing current I) or by one that passes through the gap (enclosing nothing). Ampère's law gives two different answers.

Maxwell's fix was to add a term: a changing electric field acts as a source of magnetic field, just as a changing magnetic field is a source of electric field. With this *displacement current* the equations become complete.

Law 6.3 (Maxwell's equations)**.**

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \qquad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

$$\nabla \cdot \mathbf{B} = 0, \qquad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

Together with the Lorentz force law, these are all of classical electromagnetism.

Four lines. Everything in this chapter, and all of optics, follows from them.

§6.9 Light

In empty space ($\rho = 0$, $\mathbf{J} = 0$), take the curl of Faraday's law and use the curl of Ampère–Maxwell:

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t}(\nabla \times \mathbf{B}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

With $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E}$, this is the wave equation (Chapter 5),

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}, \qquad v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 2.998 \times 10^8 \,\mathrm{m/s}.$$

Maxwell computed this speed from the constants of electricity and magnetism, measured on lab benches with batteries and coils, and found it equal to the measured speed of light. "We can scarcely avoid the inference," he wrote, "that light consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena." Light is an electromagnetic wave.

The waves are transverse: $\mathbf{E}$, $\mathbf{B}$, and the direction of travel are mutually perpendicular, with $E = cB$. They carry energy at a rate per area given by the **Poynting vector**

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B},$$

and they carry momentum: light falling on an absorbing surface pushes it with pressure I/c, where I is the intensity. Sunlight at the Earth has $I \approx 1360\,\mathrm{W/m^2}$, a pressure of about $5\,\mu\mathrm{Pa}$ — tiny, but enough to sail a spacecraft. The electromagnetic spectrum runs from radio waves (metres to kilometres) through microwaves, infrared, the narrow visible band (400–700 nm), ultraviolet, X-rays, and gamma rays. It is all the same physics at different wavelengths.

§6.10 Radiation

A charge at rest has a static field; a charge moving uniformly has a field that moves with it. Only an *accelerating* charge radiates. Think of the field lines as stiff wires attached to the charge: jerk

the charge, and a kink travels outward along the lines at speed c. That kink is the radiation, and its field falls off as $1/r$ rather than $1/r^2$, so it carries energy all the way to infinity. The total power radiated by a charge q with acceleration a (at speeds well below c) is given by the **Larmor formula**

$$P = \frac{q^2 a^2}{6\pi\epsilon_0 c^3}.$$

An oscillating charge (a dipole antenna) has $a \propto \omega^2$, so it radiates power $\propto \omega^4$. That steep frequency dependence is why the sky is blue (Chapter 3), why radio antennas must be roughly as long as the wavelength they emit, and why the classical atom is doomed (Problem 6.10).

§6.11 Potentials and gauge freedom

Because $\nabla \cdot \mathbf{B} = 0$, we can write $\mathbf{B} = \nabla \times \mathbf{A}$ for a *vector potential* $\mathbf{A}$, and then Faraday's law lets us write $\mathbf{E} = -\nabla\phi - \partial\mathbf{A}/\partial t$. But the potentials are not unique: the substitution

$$\mathbf{A} \to \mathbf{A} + \nabla\chi, \qquad \phi \to \phi - \frac{\partial\chi}{\partial t}$$

leaves $\mathbf{E}$ and $\mathbf{B}$ unchanged for *any* function $\chi(\mathbf{r}, t)$. This redundancy is called *gauge freedom*. In classical physics it looks like a curiosity. In quantum mechanics it becomes the organizing principle of all of fundamental physics: the electromagnetic field exists *because* the quantum phase of charged particles can be changed independently at every point in space (Chapter 30), and the strong and weak forces come from the same idea with bigger symmetry groups (Chapter 32).

§6.12 Fields in matter

In a material, the field polarizes atoms and molecules, creating tiny dipoles that partly cancel the applied field. The material is described by a *relative permittivity* ϵ_r (water: about 80, which is why it dissolves salts so well) and a relative permeability μ_r. Light then travels at c/n with refractive index $n = \sqrt{\epsilon_r \mu_r}$; the frequency dependence of n produces rainbows (Chapter 8). Magnetic materials range from feebly diamagnetic (everything, including frogs, which can be levitated in a 16 T magnet) to strongly ferromagnetic (iron, where quantum effects align electron spins over whole domains; Chapter 19).

§6.13 Electromagnetism and relativity

Here is a puzzle. A charge moves parallel to a current-carrying wire and feels a magnetic force. Jump into the charge's rest frame: now it is not moving, so there is no magnetic force — but it must still be pushed, because whether it hits something cannot depend on your point of view. The resolution is that in the charge's frame the wire is electrically charged, because the moving positive ions and the moving electrons are length-contracted by different amounts (Chapter 10). *Magnetism is the relativistic correction to electrostatics.* It is detectable at the tiny speeds of electrons in wires only because the wire contains so many of them that their enormous electric forces otherwise cancel perfectly.

KEY IDEA Electric and magnetic fields are not separate things. They are components of a single object — the electromagnetic field tensor of Chapter 10 — that mix into each other when you change your frame of reference. Maxwell's equations were relativistic decades before relativity; Einstein's 1905 paper was titled "On the Electrodynamics of Moving Bodies."

§6.14 Problems

6.1 $O(\epsilon)$ ★ Compute the ratio of the electric repulsion to the gravitational attraction between two protons. Does it depend on their separation? Use the result to explain why gravity, the weakest force, dominates astronomy.

6.2 $O(\epsilon)$ ★ A sphere of radius R carries charge Q spread uniformly through its volume. Use Gauss's law to find $\mathbf{E}$ inside and outside. Sketch $E(r)$. Compare with the gravitational field inside a uniform Earth, which Problem 3.6 used.

6.3 $O(\epsilon)$ A parallel-plate capacitor has plates of area $1\ \mathrm{m}^2$ separated by 1 mm of air, charged to 1000 V. Find its capacitance, charge, and stored energy, both from $\tfrac{1}{2}CV^2$ and by integrating $\tfrac{1}{2}\epsilon_0 E^2$ over the gap.

6.4 $O(\epsilon)$ A defibrillator stores 200 J in a 100 μF capacitor. What is the voltage? If it discharges through the chest (resistance about $50\ \Omega$), what is the time constant?

6.5 $O(\epsilon)$ Find the cyclotron frequency and the radius of the orbit of an electron with kinetic energy 1 keV in a 1 T field. Repeat for a proton of the same energy.

6.6 $O(\epsilon^2)$ ★ Starting from Maxwell's equations in vacuum, derive the wave equation for $\mathbf{B}$, and show for a plane wave that $\mathbf{E} \perp \mathbf{B} \perp \mathbf{k}$ and $E = cB$.

6.7 $O(\epsilon^2)$ A magnet falls through a vertical copper pipe and quickly reaches a slow terminal velocity. Explain qualitatively why, using Faraday's and Lenz's laws, and argue that the braking force is proportional to the velocity. How should the terminal velocity depend on the pipe's conductivity and wall thickness?

6.8 $O(\epsilon^2)$ A solar sail is a $100\ \mathrm{m} \times 100\ \mathrm{m}$ reflecting sheet with total mass 100 kg, at the Earth's distance from the Sun. Find the radiation force on it, and compare with the Sun's gravitational pull. Why does the ratio not depend on the distance from the Sun?

6.9 $O(\epsilon^2)$ A point charge q sits a distance d above an infinite grounded conducting plane. Show that the field above the plane is the same as if the plane were replaced by an *image charge* $-q$ at distance d below it, and find the force on q. Why does this trick work?

6.10 $O(\epsilon^2)$ ★ (*The doom of the classical atom.*) In a classical picture of hydrogen, an electron circles the proton at radius $a_0 = 0.53 \times 10^{-10}\ \mathrm{m}$. Use the Larmor formula to find the power it radiates, and estimate how long it takes to spiral into the nucleus. (Answer:

about 10^{-11} s.) Something is badly wrong with classical physics; Chapter 14 explains what.

6.11 $e^{-1/e}$ (*Earnshaw's theorem.*) Prove that no arrangement of static charges can hold a test charge in stable equilibrium in empty space. That is, the potential energy $qV(\mathbf{r})$ cannot have a local minimum where there is no charge.

Hint: At a minimum, $\nabla^2 V > 0$ in every direction. What does Laplace's equation say? (Ion traps evade this theorem by using oscillating fields — compare the Kapitza pendulum, Problem 4.10.)

6.12 $e^{-1/e}$ (*Magnetism from relativity.*) A line of positive charges (density λ) moves right at speed v, and a line of negative charges (density $-\lambda$) moves left at speed v; the wire is neutral and carries current $I = 2\lambda v$. A test charge q moves right at speed v at distance r. In the lab frame, compute the magnetic force on it. Then transform to the test charge's rest frame, use length contraction to find the net charge density of the wire there, and compute the purely electric force. Show the two agree (after transforming the force), to leading order in v/c.

Further reading

- **Purcell and Morin, *Electricity and Magnetism*** — The Berkeley course text. Builds magnetism from relativity, as in §6.13, and is full of insight.
- **David J. Griffiths, *Introduction to Electrodynamics*** — The best undergraduate electromagnetism book by a wide margin; it opens with an excellent crash course in vector calculus.
- ***The Feynman Lectures on Physics*, Volume II** — Feynman at his best: fields, potentials, and the reality of the field.
- **Andrew Zangwill, *Modern Electrodynamics*** — A modern graduate text with careful explanations and many applications.

CHAPTER 7

Thermodynamics

BUILDS ON Newtonian mechanics (Chapter 2)

Engines, entropy, and why time only ever runs one way — a theory so general that it survived the discovery of atoms, quantum mechanics, and relativity without changing a word.

PROTOTYPE The Carnot engine: an idealized engine that takes heat from a hot reservoir, turns part of it into work, and dumps the rest into a cold reservoir, all as gently and reversibly as possible. Its efficiency, $1 - T_c/T_h$, is a hard ceiling on every engine that has ever been or ever will be built.

Thermodynamics is strange among physical theories. It says nothing about what matter is made of. It was built in the early nineteenth century by engineers trying to make better steam engines, at a time when most of them believed heat was a fluid. And yet its laws are among the most reliable in all of science. Einstein said it was the only physical theory of universal content which he was convinced would never be overthrown. Its power comes from the fact that it does not care about details: it is the ultimate spherical cow.

§7.1 Temperature

Two bodies in contact eventually stop exchanging heat: they reach *thermal equilibrium*. The **zeroth law** says that if A is in equilibrium with B, and B with C, then A is in equilibrium with C. That transitivity is what makes temperature meaningful: it is the property that is equal for bodies in equilibrium, and a thermometer is a body B that we bring into contact with others.

The natural temperature scale is the *absolute* or Kelvin scale, whose zero is the lowest possible temperature: $T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273.15$. Room temperature is about 300 K. A useful number to carry: at room temperature $k_\mathrm{B}T \approx \frac{1}{40}\,\mathrm{eV}$, where $k_\mathrm{B} = 1.38 \times 10^{-23}\,\mathrm{J/K}$ is Boltzmann's constant. Chemical bonds (a few eV) are therefore stable at room temperature, while hydrogen bonds ($\sim 0.2\,\mathrm{eV}$) are only just stable — which is why life, built from both, works at the temperatures it does.

§7.2 The ideal gas

A dilute gas obeys the **ideal gas law**,

$$PV = Nk_\mathrm{B}T = nRT,$$

where N is the number of molecules, n the number of moles, and $R = N_A k_\mathrm{B} = 8.31\,\mathrm{J/(mol\,K)}$.

Kinetic theory explains it. Model the gas as point molecules bouncing elastically off the walls. A molecule hitting a wall of area A transfers momentum $2mv_x$; working out the rate of collisions gives the pressure $P = \frac{1}{3}(N/V)\, m\langle v^2\rangle$. Comparing with the ideal gas law,

$$\langle \tfrac{1}{2}mv^2 \rangle = \tfrac{3}{2}\, k_{\mathrm{B}} T.$$

Temperature is a measure of the average kinetic energy of the molecules. Nitrogen molecules in the air around you move at about 500 m/s, roughly the speed of sound — which is not a coincidence, since sound is carried by these collisions.

§7.3 The first law

Energy is conserved, but in thermodynamics it can enter a system in two ways: as *work*, through organized motion like a piston, or as *heat*, through disorganized molecular jostling across a boundary.

Law 7.1 (The first law of thermodynamics)**.**

$$\Delta U = Q - W,$$

where U is the internal energy of the system, Q the heat added to it, and W the work done by it. For a gas, $W = \int P \, \mathrm{d}V$.

U is a *state function*: it depends only on the current state of the system. Heat and work are not; they depend on how you got there. It makes sense to ask how much energy a gas has, but not how much "heat" it contains.

For an ideal monatomic gas, $U = \frac{3}{2}Nk_{\mathrm{B}}T$. The *heat capacity* at constant volume is $C_V = \mathrm{d}U/\,\mathrm{d}T = \frac{3}{2}Nk_{\mathrm{B}}$. At constant pressure the gas also does work as it expands, so it needs more heat per degree: $C_P = C_V + Nk_{\mathrm{B}}$. The ratio $\gamma = C_P/C_V$ is $\frac{5}{3}$ for monatomic gases and $\frac{7}{5}$ for diatomic ones like air.

7.3.1 Processes

On a P–V diagram, the work done by a gas is the area under its path. The standard processes:

- **Isothermal** (constant T, in contact with a reservoir): PV is constant, and the work done in expanding from V_1 to V_2 is $Nk_{\mathrm{B}}T\ln(V_2/V_1)$, all supplied as heat.
- **Adiabatic** (no heat exchange, $Q = 0$): the work comes out of the internal energy, so the gas cools as it expands. Combining $\mathrm{d}U = -P\,\mathrm{d}V$ with the ideal gas law gives

$$PV^{\gamma} = \text{constant}, \qquad TV^{\gamma-1} = \text{constant}.$$

- **Isobaric** (constant P) and **isochoric** (constant V).

Adiabatic processes are everywhere. Air rising in the atmosphere expands and cools at about 10°C per kilometre, which is why mountaintops are cold (Chapter 25). A bicycle pump gets hot. A diesel engine needs no spark plug: compressing air by a factor of twenty heats it enough to ignite the fuel.

§7.4 The second law

The first law says energy is conserved. It does not say which way things go. Heat flows from hot coffee into the room, never from the room into the coffee, though both directions conserve energy. Something else is at work.

Law 7.2 (The second law of thermodynamics)**.** Two equivalent statements:

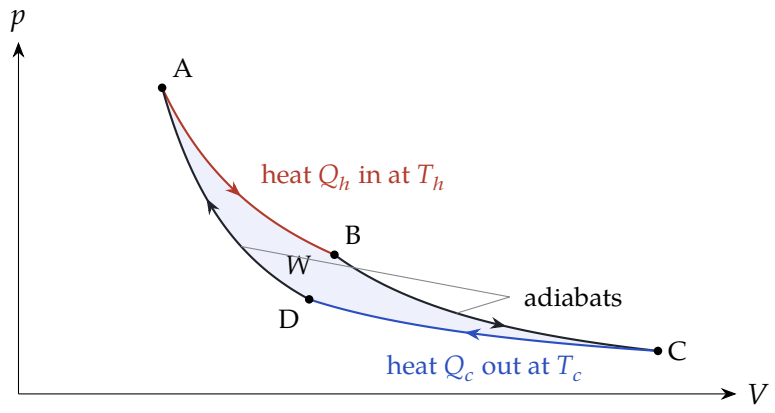
- (**Clausius**) No process can have as its sole result the transfer of heat from a colder body to a hotter one.
- (**Kelvin–Planck**) No process can have as its sole result the complete conversion of heat into work.

7.4.1 Heat engines and the Carnot limit

A *heat engine* takes heat Q_h from a hot reservoir at T_h, does work W, and dumps heat Q_c into a cold reservoir at T_c. By the first law $W = Q_h - Q_c$, and its efficiency is $\eta = W/Q_h$. Kelvin–Planck says Q_c can never be zero: every engine must waste some heat.

How much? Carnot imagined the gentlest possible engine: a gas taken around a cycle of two isotherms and two adiabats, always infinitesimally close to equilibrium, so that it could be run backwards. Working through the cycle gives $Q_c/Q_h = T_c/T_h$, so

$$\eta_{\text{Carnot}} = 1 - \frac{T_c}{T_h}.$$



The Carnot cycle for an ideal gas: two isotherms joined by two adiabats. The shaded area is the work W done in one trip around.

Theorem 7.3 (Carnot's theorem)**.** No engine operating between two reservoirs can be more efficient than a reversible one, and all reversible engines operating between the same two reservoirs have the same efficiency.

The proof is a beautiful piece of reasoning: if some engine beat Carnot's, use its work output to drive a Carnot engine *backwards* as a refrigerator. The combination would move heat from cold to hot with no other effect, violating Clausius's statement.

A power station with steam at 800 K and cooling water at 300 K can never exceed 63% efficiency; real ones reach about 40%. The waste is not bad engineering. It is a law of nature.

Run backwards, an engine becomes a refrigerator or a heat pump. A heat pump warming a house to 293 K from outdoor air at 273 K could in principle deliver $T_h/(T_h - T_c) \approx 15$ joules of heat for every joule of electricity, which is why heat pumps are so much better than electric heaters.

§7.5 Entropy

Carnot's result $Q_h/T_h = Q_c/T_c$ suggests that the quantity Q/T is somehow conserved in reversible processes. Clausius made this into a new state function.

Definition 7.4 (Entropy)**.** The change in *entropy* of a system along a reversible path is

$$\mathrm{d}S = \frac{\delta Q_{\text{rev}}}{T}.$$

Because S is a state function, ΔS between two states is the same whatever path is actually taken; to compute it, invent any reversible path between them.

Law 7.5 (The second law, in terms of entropy)**.** The entropy of an isolated system never decreases: $\Delta S \geq 0$, with equality only for reversible processes.

Heat Q flowing from a hot body to a cold one changes the total entropy by $Q/T_c - Q/T_h > 0$: allowed. The reverse would decrease it: forbidden. Every irreversible process — friction, mixing, free expansion, heat flowing down a temperature gradient — creates entropy, and that entropy can never be destroyed.

KEY IDEA The second law is the only fundamental law of physics that distinguishes past from future. The equations of mechanics and electromagnetism run equally well backwards. Eggs unscramble in principle; they just never do. The arrow of time is the direction in which entropy increases.

What *is* entropy? Thermodynamics cannot say. Statistical mechanics (Chapter 9) can: entropy is the logarithm of the number of microscopic arrangements consistent with what you see, $S = k_{\mathrm{B}} \ln \Omega$, and the second law holds because there are overwhelmingly more disordered arrangements than ordered ones. Why the universe started in such a low-entropy state that it has been running downhill ever since is a question for cosmology (Chapter 28), and not a fully answered one.

§7.6 Free energy

At constant temperature, a system can exchange heat with its surroundings, so its own entropy can go down as long as the surroundings' goes up by more. The bookkeeping is done by *ther-*

modynamic potentials:

$$F = U - TS \qquad \text{(Helmholtz free energy: fixed } T \text{ and } V\text{)},$$
$$G = U - TS + PV \qquad \text{(Gibbs free energy: fixed } T \text{ and } P\text{)},$$
$$H = U + PV \qquad \text{(enthalpy: heat absorbed at fixed } P\text{)}.$$

KEY IDEA At fixed temperature and pressure, a system evolves to minimize its Gibbs free energy $G = H - TS$. Low energy is favoured, but so is high entropy, and temperature sets the exchange rate between them.

At low temperature energy wins, and things order: water freezes, magnets magnetize, proteins fold. At high temperature entropy wins, and things disorder: ice melts, magnets lose their magnetism, proteins unfold. Every phase transition is a contest between H and TS. Chemistry runs on ΔG: a reaction proceeds spontaneously when it lowers G (Chapter 21). The free energy is also the maximum work you can extract from a system at fixed temperature — the "free" part of its energy.

Because these potentials are state functions, their mixed second derivatives are equal, giving the *Maxwell relations*, for example $(\partial S/\partial V)_T = (\partial P/\partial T)_V$. They relate quantities that are hard to measure (entropy changes) to quantities that are easy (pressures and temperatures), and they are how thermodynamics extracts surprising predictions from almost nothing.

§7.7 Phases and phase transitions

A *phase diagram* shows which phase of a substance is stable at each temperature and pressure. Along the lines between phases, two phases coexist; at the *triple point* all three do; beyond the *critical point* the distinction between liquid and gas disappears entirely.

Crossing a phase boundary absorbs or releases *latent heat* L without changing the temperature: a pot of water boils at 100°C however high the flame, the extra energy going into breaking molecules free. The slope of a phase boundary is fixed by the **Clausius–Clapeyron equation**,

$$\frac{\mathrm{d}P}{\mathrm{d}T} = \frac{L}{T\,\Delta V},$$

where ΔV is the change in volume. Since vapour takes up far more volume than liquid, the boiling point is very sensitive to pressure: on Everest, water boils at about 70°C, and pasta never really cooks. Water is anomalous in that ice is less dense than liquid water, so $\Delta V < 0$ for melting and the melting line tilts backwards: squeezing ice can melt it. (This effect is far too small to explain ice skating, despite the textbook myth; the slipperiness of ice comes mostly from a thin premelted surface layer and frictional heating.)

§7.8 The third law

Law 7.6 (The third law of thermodynamics)**.** As $T \to 0$, the entropy of a system approaches a constant (zero, for a perfect crystal), and absolute zero cannot be reached in a finite number of steps.

Each cooling step removes a fraction of the remaining entropy, so you approach zero asymptotically, like Zeno's runner. The record low temperatures, achieved with laser cooling and evaporative cooling of atoms (Chapter 18), are below a nanokelvin.

§7.9 Radiation has a temperature too

A hot object glows. A cavity in equilibrium at temperature T fills with electromagnetic radiation whose energy density depends only on T — not on the material of the walls, as Kirchhoff showed by a clever second-law argument. Using only thermodynamics and the fact that radiation pressure is one third of its energy density, Boltzmann proved that the radiated power per unit area is

$$j = \sigma T^4, \qquad \sigma = 5.67 \times 10^{-8}\ \mathrm{W/(m^2 K^4)}$$

(Problem 7.11). But thermodynamics cannot tell you the *spectrum* of the radiation — how much energy is at each frequency. Every classical attempt gave nonsense, predicting infinite energy at high frequencies. Fixing that required Planck's quantum hypothesis in 1900, and it was the first crack in classical physics (Chapters 9 and 14).

§7.10 Problems

7.1 $O(\epsilon)$ One mole of an ideal gas at 300 K expands isothermally to twice its volume. How much work does it do, and how much heat does it absorb? What is its change in entropy?

7.2 $O(\epsilon)$ ★ A coal power station runs between 800 K and 300 K. Find its maximum possible efficiency. If it produces 1 GW of electricity at 40% efficiency, how much heat does it dump into its cooling river every second? Estimate the temperature rise of a river flowing at $100\ \mathrm{m^3/s}$.

7.3 $O(\epsilon)$ An ideal heat pump warms a house to 20°C when it is 0°C outside. Compute its maximum coefficient of performance (heat delivered per unit work). Why does a heat pump become less effective in very cold weather?

7.4 $O(\epsilon^2)$ A diesel engine compresses air adiabatically from 1 atm and 300 K by a factor of 20 in volume. Find the final temperature and pressure, taking $\gamma = 1.4$. Why does this ignite the fuel without a spark?

7.5 $O(\epsilon^2)$ One kilogram of water at 80°C is mixed with one kilogram at 20°C in an insulated container. Find the final temperature and the total change in entropy, and verify that it is positive.

Hint: Imagine each kilogram changing temperature slowly and reversibly; then for each one, $\Delta S = mc\ln(T_f/T_i)$.

7.6 $O(\epsilon^2)$ Use the Clausius–Clapeyron equation, treating the vapour as an ideal gas and neglecting the liquid's volume, to show that the vapour pressure obeys $P \propto e^{-L/RT}$ (with L per mole). Use $L = 40.7\,\mathrm{kJ/mol}$ to estimate the boiling point of water at the summit of Everest, where the pressure is about 0.33 atm.

7.7 $O(\epsilon^2)$ ★ Two identical bodies with constant heat capacity C start at temperatures T_1 and T_2. A reversible engine runs between them until they reach a common temperature. Show that the final temperature is $\sqrt{T_1 T_2}$ and that the work extracted is $C(\sqrt{T_1} - \sqrt{T_2})^2$. Compare with simply putting them in contact.

7.8 $O(\epsilon^2)$ An ideal gas expands freely into a vacuum, doubling its volume, inside an insulated box. No heat flows and no work is done, so $\Delta U = 0$ and T is unchanged. Find ΔS by inventing a reversible path between the same states. Why is the process irreversible?

7.9 $O(\epsilon^2)$ (*Rubber bands.*) Stretch a rubber band quickly and touch it to your lip: it warms. Hang a weight from a rubber band and heat the band with a hair dryer: it contracts, lifting the weight. Explain both observations by modelling rubber as a tangle of polymer chains whose tension comes mainly from entropy, not energy. (See also Chapter 23.)

7.10 $e^{-1/\epsilon}$ ★ (*Landauer's principle.*) Maxwell imagined a demon that watches molecules at a trapdoor between two gas chambers, opening it only for fast molecules going one way, apparently making one side hot and the other cold with no work. Show that if the demon must eventually erase its memory, and erasing one bit of information into an environment at temperature T increases the environment's entropy by at least $k_{\mathrm{B}}\ln 2$, the second law is saved. How much energy must a computer dissipate, at minimum, to erase a gigabyte at room temperature? Compare with what real computers dissipate.

7.11 $e^{-1/\epsilon}$ ★ (*The T^4 law from pure thermodynamics.*) For radiation in a cavity, the energy density u depends only on T, and the pressure is $P = u/3$. Starting from $\mathrm{d}U = T\,\mathrm{d}S - P\,\mathrm{d}V$ and the Maxwell relation $(\partial S/\partial V)_T = (\partial P/\partial T)_V$, derive

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial P}{\partial T}\right)_V - P,$$

and apply it to $U = uV$ to show that $u \propto T^4$. Boltzmann did this in 1884, sixteen years before anyone understood why.

Further reading

- **Daniel V. Schroeder, *An Introduction to Thermal Physics*** — An extremely clean, focused, and clear introduction to thermodynamics and statistical mechanics together. The standard recommendation, deservedly.
- **Enrico Fermi, *Thermodynamics*** — A short, classic Dover book of lectures. Thermodynamics as a logical structure, with no microscopic crutches.

- **Blundell and Blundell, *Concepts in Thermal Physics*** — Broad and modern, with many applications to astrophysics, information, and biology.
- ***The Feynman Lectures on Physics*, Volume I, Chapters 39–46** — Kinetic theory and the laws of thermodynamics, including the famous ratchet-and-pawl.

CHAPTER 8

Optics

BUILDS ON waves (Chapter 5), electromagnetism (Chapter 6)

How light bends, focuses, interferes with itself, and paints the sky — and the experiment that proved, twice, that light is stranger than anyone thought.

> **PROTOTYPE** The double slit. Shine light through two narrow parallel slits onto a screen and you do not get two bright lines: you get a whole row of bright and dark fringes. The light from the two slits interferes. Light is a wave.

Light is an electromagnetic wave (Chapter 6) with a wavelength between about 400 and 700 nm. That one sentence contains all of optics, but it contains it in two very different regimes. When the objects light meets are much larger than its wavelength — lenses, mirrors, raindrops — light travels in straight lines called *rays*, and we can do *geometric optics*. When they are comparable to the wavelength — slits, gratings, thin films, the edges of shadows — the wave nature takes over, and we need *wave optics*. Knowing which regime you are in is half the battle.

§8.1 Rays: reflection and refraction

In a medium, light travels slower than in vacuum, at speed c/n, where n is the *refractive index*: about 1.0003 for air, 1.33 for water, 1.5 for glass, 2.42 for diamond. All of geometric optics follows from one principle.

Law 8.1 (Fermat's principle)**.** Light travels between two points along a path for which the travel time is stationary — usually a minimum.

For reflection from a mirror this gives the familiar rule that the angle of incidence equals the angle of reflection. For refraction at a boundary between media it gives

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \qquad \text{(Snell's law)},$$

with angles measured from the normal. Light bends *toward* the normal entering a slower medium. A lifeguard running along a beach and then swimming to a drowning swimmer, trying to minimize time, faces exactly the same problem and should follow the same law.

Example 8.2 (The bent straw and the shallow pool)**.** Looking straight down into water of depth d, the bottom appears at depth $d/n \approx 0.75d$. Rays from the bottom bend away from the normal as they leave the water, and your brain traces them back along straight lines.

Pools are deeper than they look.

Going from a slower to a faster medium, light bends away from the normal, and beyond the *critical angle* $\sin\theta_c = n_2/n_1$ there is no refracted ray at all: the light is entirely reflected. **Total internal reflection** traps light in optical fibres, which carry most of the world's internet traffic by bouncing light along glass threads thinner than a hair. It also makes diamonds sparkle: with $n = 2.42$, the critical angle is only $24°$, so light that enters a well-cut diamond bounces around inside many times before escaping.

Mirages come from a refractive index that varies smoothly. Air just above a hot road is less dense, so its index is slightly lower. Rays from the sky heading toward the road curve upward, and you see an image of the sky on the road — a "puddle" that recedes as you approach.

§8.2 Lenses and instruments

A lens is a piece of glass shaped so that refraction brings parallel rays to a focus at a distance f, the *focal length*. For a thin lens, an object at distance s forms an image at distance s' with

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}, \qquad m = -\frac{s'}{s},$$

where m is the magnification (negative meaning inverted). Opticians measure lens power in *diopters*, $1/f$ in inverse metres; powers of thin lenses in contact add.

Your **eye** is a lens (mostly the cornea, with the crystalline lens for fine adjustment) that focuses images onto the retina about 2 cm behind. To focus on near objects, muscles squeeze the lens to shorten its focal length. A short-sighted eye focuses distant objects in front of the retina and needs a diverging lens; a long-sighted one, a converging lens.

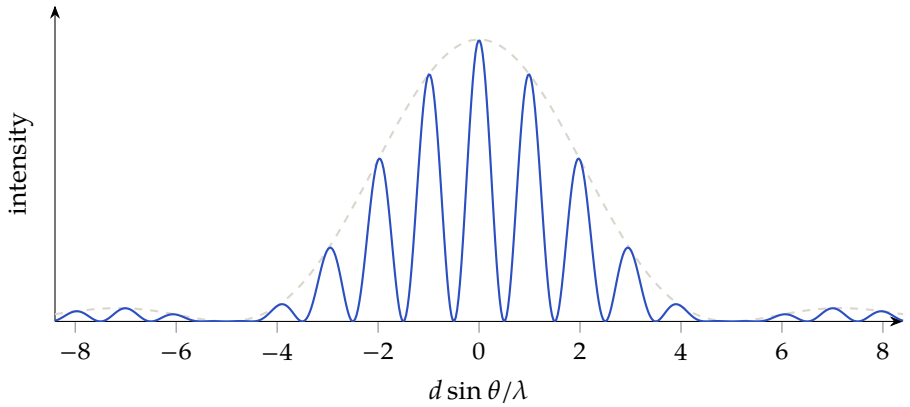
Instruments combine lenses. A **magnifying glass** lets you hold an object closer than your eye could otherwise focus. A **microscope** uses a short-focus objective to form a large real image, then examines it with a magnifier. A **telescope** uses a long-focus objective (lens or mirror) to form an image of a distant object and a short-focus eyepiece to view it, with angular magnification $f_{\text{objective}}/f_{\text{eyepiece}}$. But the most important job of a big telescope is not magnification. It is to collect light, and to beat diffraction (§8.4).

§8.3 Interference

Now the waves. In 1801 Thomas Young passed light through two narrow slits separated by a distance d. At an angle θ, the paths from the two slits to a distant screen differ in length by $d\sin\theta$. Where this is a whole number of wavelengths, crests meet crests and the light is bright; where it is a half-integer, crests meet troughs and the light cancels:

$$d\sin\theta = m\lambda \quad \text{(bright)}, \qquad d\sin\theta = (m + \tfrac{1}{2})\lambda \quad \text{(dark)}.$$

On a screen a distance L away the fringes are spaced by $\lambda L/d$. With red laser light, slits 0.1 mm apart, and a screen 2 m away, the fringes are about a centimetre apart: you can do this experiment on a kitchen table.



The two-slit pattern for slits a fifth as wide as their separation. Bright fringes fall where $d \sin\theta$ is a whole number of wavelengths, under the envelope (dashed) of each slit's own diffraction. Fringe number 5 is missing, because it lands on a zero of the envelope.

KEY IDEA Interference requires waves; particles cannot cancel. Young's experiment settled, for a century, that light is a wave. Then the photoelectric effect (§8.8) unsettled it, and the resolution — that light is something new, a quantum field — is the subject of Chapters 14 and 30.

Thin films. The colours of soap bubbles and oil slicks come from interference between light reflected from the top and bottom surfaces of a film. The two reflections differ in path by twice the film thickness, and there is an extra half-wavelength shift on reflection whenever light reflects off a medium of higher index. Different thicknesses reinforce different colours, so the film paints a contour map of its own thickness. A soap bubble about to pop turns black: the film is so thin that the two reflections, differing only by the half-wavelength flip, cancel.

Interferometers exploit interference to measure tiny distances. A Michelson interferometer splits a beam, sends the halves down two perpendicular arms, and recombines them; a change in one arm's length by a quarter wavelength turns a bright output dark. Michelson and Morley used one in 1887 to look for the Earth's motion through the "ether" and found nothing (Chapter 10). LIGO, with 4 km arms, now detects changes in arm length a thousand times smaller than a proton (Chapter 29).

§8.4 Diffraction

Light passing through a single slit of width a also spreads out. By Huygens' principle every point in the slit is a source, and pairing up points half a slit apart shows that the light cancels completely at angles where

$$a \sin\theta = m\lambda, \qquad m = \pm 1, \pm 2, \ldots$$

The central bright band has angular half-width λ/a: the narrower the slit, the wider the spread. Squeeze a beam and it fans out.

For a circular aperture of diameter D — the pupil of your eye, the mirror of a telescope — the first dark ring is at $\theta = 1.22\lambda/D$. Two point sources (two stars, two headlights) closer

together than this blur into one. This **Rayleigh criterion** sets a fundamental limit on every optical instrument:

$$\theta_{\min} \approx 1.22\,\frac{\lambda}{D}.$$

Your eye ($D \approx 3\,\text{mm}$) resolves about one arcminute; the Hubble Space Telescope ($D = 2.4\,\text{m}$) about 0.05 arcseconds. A light microscope cannot resolve detail much smaller than half a wavelength, about 200 nm — which is why we need electron microscopes and X-rays to see atoms (Chapter 19).

A **diffraction grating** — thousands of slits — sharpens Young's fringes into very narrow lines at $d\sin\theta = m\lambda$. Because the angle depends on λ, a grating spreads light into a spectrum, and the resolving power improves with the number of slits N: it can distinguish wavelengths differing by $\lambda/(mN)$. Spectroscopy with gratings is how we know what stars are made of (Chapter 27) and how atomic energy levels were first mapped (Chapter 18).

Remark 8.3. There is a deep and useful fact hiding here: the far-field diffraction pattern of an aperture is the Fourier transform of the aperture's shape (Chapter 1). A narrow slit gives a wide pattern because a narrow function has a wide Fourier transform. A lens, in fact, *performs* a Fourier transform: the pattern in its focal plane is the transform of the light falling on it.

§8.5 Polarization

Light is a transverse wave, so its electric field can point in any direction perpendicular to its motion. Light whose field oscillates along a fixed line is *linearly polarized*; light from the Sun or a bulb is *unpolarized*, a random jumble of directions. A polarizing filter transmits only one component: unpolarized light loses half its intensity, and polarized light at angle θ to the filter's axis is transmitted with intensity

$$I = I_0\cos^2\theta \qquad \text{(Malus's law)}.$$

Light reflected from water or glass is partly polarized parallel to the surface, and at *Brewster's angle*, $\tan\theta_B = n_2/n_1$ (53° for water), it is completely polarized. Polarized sunglasses block exactly this horizontally polarized glare. Scattered skylight is also polarized, most strongly at 90° from the Sun, and bees navigate by it. Liquid-crystal displays work by electrically rotating the polarization of light between crossed polarizers.

§8.6 Light in matter: colour

The refractive index depends on wavelength, usually increasing toward the blue. This *dispersion* is why a prism spreads white light into a spectrum and why simple lenses have coloured fringes.

The rainbow is dispersion in raindrops. Sunlight enters a spherical drop, reflects once off the back, and leaves. The exit angle depends on where the ray entered, but it has a stationary value near 42° from the antisolar point, so light piles up at that angle (Problem 8.10). Because n varies with colour, red lands at 42° and violet at 40°: the bow is a circle centred on the shadow of your head, red on the outside. Two internal reflections give the fainter secondary bow at 51°, with colours reversed.

Scattering colours the sky. Molecules far smaller than the wavelength scatter light with intensity $\propto 1/\lambda^4$ (*Rayleigh scattering*; see Chapter 3), so the sky is blue, and the setting Sun, seen through a long path of air that has scattered its blue away, is red. Water droplets in clouds are larger than the wavelength and scatter all colours almost equally (*Mie scattering*), so clouds are white.

Light passing through an absorbing medium decays exponentially, $I = I_0 e^{-\alpha x}$ (the *Beer–Lambert law*). Water absorbs red light slightly, which is why deep water and glaciers look blue, and why it is blue-green and dim a few tens of metres down.

§8.7 Coherence and lasers

Interference requires *coherence*: the waves being combined must keep a fixed phase relationship. Sunlight is coherent only over a few micrometres, which is why you see thin-film colours in soap bubbles but not in window panes. A **laser** produces light that is coherent over metres or kilometres, nearly monochromatic, and tightly collimated. It works by *stimulated emission*: a photon passing an excited atom can trigger it to emit a second photon that is an exact copy — same frequency, direction, phase, and polarization. Pump a medium so that more atoms are excited than not (a *population inversion*), put it between two mirrors, and a cascade of identical photons builds up. Einstein predicted stimulated emission in 1917; the first laser was built in 1960. Chapter 18 explains how it works.

§8.8 Photons

By 1900 the wave theory of light was complete. Then it broke. When light shines on a metal it can knock out electrons — the *photoelectric effect* — and the electrons' energies depend not on the light's intensity but on its frequency. Below a threshold frequency, no electrons come out however bright the light; above it, electrons come out instantly however dim. In 1905 Einstein explained this by proposing that light comes in lumps, *photons*, each with energy

$$E = hf = \hbar\omega, \qquad h = 6.626 \times 10^{-34}\,\mathrm{J\,s}.$$

A visible photon carries about 2–3 eV.

So is light a wave or a particle? Do the double-slit experiment with light so dim that only one photon is in the apparatus at a time. Each photon arrives at a single point on the screen, like a particle. But as thousands of dots accumulate, they build up the interference pattern, like a wave. Each photon somehow went through both slits. This is not a paradox to be resolved by better experiments; it is the central fact of quantum mechanics, and Chapter 14 is about learning to live with it.

§8.9 Problems

8.1 $O(\epsilon)$ A swimming pool is 2.0 m deep. How deep does it appear when viewed from directly above? A fish looks up at the surface. Within what cone can it see the outside world, and what does it see outside that cone?

8.2 $O(\epsilon)$ Find the critical angle for total internal reflection from diamond to air, and from glass ($n = 1.5$) to air. An optical fibre has a glass core ($n = 1.48$) clad in glass of $n = 1.46$. What is the maximum angle to the fibre axis at which light is trapped?

8.3 $O(\epsilon)$ An object is 30 cm in front of a converging lens of focal length 10 cm. Locate the image, find its magnification, and say whether it is real or virtual, upright or inverted. Repeat with the object at 5 cm.

8.4 $O(\epsilon)$ ★ A red laser (633 nm) shines on two slits 0.10 mm apart. Find the fringe spacing on a screen 2.0 m away. What happens to the pattern if the whole apparatus is submerged in water?

8.5 $O(\epsilon^2)$ ★ Estimate the smallest detail you could resolve on the Moon by eye, through a backyard telescope ($D = 20$ cm), and with Hubble. Could a spy satellite with a 2.4 m mirror at an altitude of 250 km read a car's license plate? (Ignore the atmosphere, then explain why you shouldn't.)

8.6 $O(\epsilon^2)$ A soap film ($n = 1.33$) in air reflects strongly at 550 nm when viewed at normal incidence. Find the smallest possible thickness of the film. Why does the film look black just before it breaks?

8.7 $O(\epsilon^2)$ Find Brewster's angle for light reflecting from water. You are on a boat; at what height above the horizon should the Sun be for polarizing sunglasses to cut the glare from the water most effectively?

8.8 $O(\epsilon^2)$ The yellow sodium doublet consists of two lines at 589.0 and 589.6 nm. How many slits must a diffraction grating have to resolve them in first order?

8.9 $O(\epsilon^2)$ Prove the law of reflection from Fermat's principle. Then use Fermat's principle to explain qualitatively why a hot road shows a mirage of the sky.

8.10 $e^{-1/\epsilon}$ ★ (*Why the rainbow is at* 42°.) A ray enters a spherical raindrop at angle of incidence i, refracts to angle r ($\sin i = n \sin r$), reflects once off the back, and exits. Show that its total deviation is $D = 180^\circ + 2i - 4r$. Find the angle of incidence at which D is stationary, and show that for $n = 4/3$ the rainbow appears 42° from the antisolar point. Why does light concentrate at the stationary angle?

Hint: Near a stationary point, a wide range of incidence angles all give nearly the same exit angle.

8.11 $e^{-1/\epsilon}$ (*Arago's spot.*) In 1818 Poisson tried to ridicule Fresnel's wave theory of light: if it were right, he argued, the shadow of an opaque disc should have a bright spot at its very centre. Arago did the experiment and found the spot. Explain why the wave theory predicts it.

Hint: Every point on the circular edge of the disc is the same distance from the centre of the shadow.

8.12 $e^{-1/\epsilon}$ (*You can't out-shine the Sun.*) With lenses and mirrors you can focus sunlight onto a small spot. Show, using the second law of thermodynamics, that no passive optical

system can heat an object above the temperature of the Sun's surface. Then explain the optical reason: a lens cannot make the Sun look *brighter* (more power per solid angle per area), only bigger, and at best can surround the target with the Sun's surface on all sides.

Further reading

- **Eugene Hecht,** ***Optics*** — The standard undergraduate textbook, encyclopaedic and well illustrated.
- ***The Feynman Lectures on Physics*****, Volume I, Chapters 26–36** — Optics from Fermat's principle to polarization and colour vision.
- **Richard Feynman,** ***QED: The Strange Theory of Light and Matter*** — Four popular lectures explaining reflection, refraction, and interference as photons adding up arrows. Essential, and readable in an evening.
- **Lipson, Lipson, and Lipson,** ***Optical Physics*** — Modern optics built on Fourier methods, through lasers and holography.

CHAPTER 9

Statistical mechanics

BUILDS ON thermodynamics (Chapter 7)

How countless blind little parts conspire into exact laws — and why the most powerful tool for understanding matter is knowing that you don't know.

PROTOTYPE A box of gas. It holds 10^{23} molecules, each with its own position and velocity, colliding billions of times a second. Yet its behaviour is captured perfectly by three numbers: pressure, volume, temperature. Statistical mechanics explains how so much ignorance produces so much certainty.

Thermodynamics (Chapter 7) describes heat, work, and entropy without ever asking what they are. Statistical mechanics answers the question. It starts from the microscopic laws — Newton's or Schrödinger's — adds one assumption about probability, and derives thermodynamics. Along the way it explains why heat capacities change with temperature, why stars glow the colours they do, why metals conduct, why magnets lose their magnetism when heated, and why there is an arrow of time.

§9.1 Counting

A *microstate* is a complete specification of the system: every molecule's position and velocity. A *macrostate* is what we can actually observe: total energy, volume, pressure. Enormous numbers of microstates correspond to each macrostate. Call that number the *multiplicity*, Ω.

Example 9.1 (Coins). Flip N coins. There is one microstate with all heads, but $\binom{N}{N/2} \approx 2^N\sqrt{2/\pi N}$ with exactly half heads. For $N = 100$, that is about 10^{29}. The distribution of the number of heads is peaked at $N/2$ with width $\sqrt{N}/2$; the *relative* width is $1/\sqrt{N}$. For $N = 10^{23}$, the fraction of heads is 0.5 to within one part in 10^{11}, every time you look. You would wait far longer than the age of the universe to see a deviation of one part in a million.

This is the whole secret. For large systems, the overwhelming majority of microstates look the same macroscopically. The macroscopic world is predictable not despite the chaos underneath, but because of it.

Law 9.2 (The fundamental assumption). An isolated system in equilibrium is equally likely to be in any of its accessible microstates.

(Liouville's theorem, Chapter 4, makes this natural: the dynamics never favours one region of phase space over another of equal volume.)

§9.2 Entropy and temperature

Boltzmann's great insight was that the entropy of thermodynamics is just the logarithm of the multiplicity:

$$S = k_{\mathrm{B}} \ln \Omega.$$

This is carved on his tombstone in Vienna. The logarithm makes entropy additive: two independent systems have $\Omega = \Omega_1 \Omega_2$, so $S = S_1 + S_2$. And it explains the second law. A system wanders randomly among its microstates; it ends up in the macrostate with the most of them simply because that macrostate is overwhelmingly more probable. Entropy increases because there are more ways to be disordered than ordered.

Now put two systems in thermal contact, sharing a fixed total energy $U = U_1 + U_2$. The most probable split of energy maximizes $\Omega_1(U_1)\Omega_2(U_2)$, i.e. maximizes $S_1 + S_2$:

$$\frac{\partial S_1}{\partial U_1} = \frac{\partial S_2}{\partial U_2}.$$

The quantity that is equal for two systems in equilibrium is, by the zeroth law, temperature. So

Definition 9.3 (Temperature)**.**

$$\frac{1}{T} = \frac{\partial S}{\partial U}.$$

A hot system is one whose entropy grows only a little when you add energy; a cold one gains a lot. Heat flows from hot to cold because that increases the total entropy. Temperature, which seemed so concrete, is really a statement about how multiplicity changes with energy.

§9.3 The Boltzmann factor

Most of the time we study a small system in contact with a large *reservoir* at temperature T. What is the probability that the small system is in a particular state s with energy E_s? The reservoir has energy $U_0 - E_s$, and the probability is proportional to the number of reservoir microstates, $\Omega_R(U_0 - E_s) = e^{S_R(U_0 - E_s)/k_{\mathrm{B}}}$. Expanding S_R to first order in the small energy E_s, and using $\partial S_R / \partial U = 1/T$:

Law 9.4 (The Boltzmann distribution)**.** A system in equilibrium with a reservoir at temperature T is found in a state of energy E_s with probability

$$P(s) = \frac{e^{-E_s/k_{\mathrm{B}}T}}{Z}, \qquad Z = \sum_s e^{-E_s/k_{\mathrm{B}}T}.$$

This is the single most useful formula in statistical mechanics. States with energy much more than $k_{\mathrm{B}}T$ above the ground state are exponentially suppressed; states within $k_{\mathrm{B}}T$ are easily

reached. Thermal energy $k_{\mathrm{B}}T$ is the currency, and the Boltzmann factor $e^{-E/k_{\mathrm{B}}T}$ is the exchange rate.

Example 9.5 (The thinning atmosphere)**.** A molecule at height h has extra energy mgh, so the density of the atmosphere falls as $e^{-mgh/k_{\mathrm{B}}T}$. The *scale height* $k_{\mathrm{B}}T/mg$ is about 8 km for air. Every 8 km up, the air thins by a factor of e; at the top of Everest you breathe about a third as much oxygen.

The same factor controls chemical reaction rates (Arrhenius: rate $\propto e^{-E_a/k_{\mathrm{B}}T}$, Chapter 21), the thermal emission of electrons from hot metals, the ionization of gases in stars (Chapter 17), and the fraction of molecules fast enough to escape a planet.

§9.4 The partition function

The normalizing sum $Z = \sum_s e^{-\beta E_s}$, with $\beta = 1/k_{\mathrm{B}}T$, is called the *partition function*. It looks like a mere normalization, but it is a generating function for all of thermodynamics:

$$\langle E\rangle = -\frac{\partial \ln Z}{\partial \beta}, \qquad F = -k_{\mathrm{B}}T \ln Z, \qquad S = -\frac{\partial F}{\partial T}, \qquad P = -\frac{\partial F}{\partial V}.$$

Compute Z, and every thermodynamic property follows by differentiation.

Example 9.6 (The two-state system)**.** A system with two states, energies 0 and ε (a spin in a magnetic field, say), has $Z = 1 + e^{-\beta\varepsilon}$ and $\langle E\rangle = \varepsilon/(e^{\beta\varepsilon}+1)$. At low temperature it sits in the ground state; at high temperature it is equally likely to be in either, so $\langle E\rangle \to \varepsilon/2$. The heat capacity $\mathrm{d}\langle E\rangle/\,\mathrm{d}T$ has a peak near $k_{\mathrm{B}}T \approx \varepsilon/2$ and vanishes at both ends. A system with finitely many states can only absorb so much energy.

§9.5 Equipartition, and its failure

For a classical system whose energy contains a quadratic term — $\frac{1}{2}mv_x^2$, $\frac{1}{2}kx^2$, $\frac{1}{2}I\omega^2$ — a short Gaussian integral shows:

Theorem 9.7 (Equipartition)**.** In classical equilibrium at temperature T, every quadratic degree of freedom has average energy $\frac{1}{2}k_{\mathrm{B}}T$.

A monatomic gas has three (motion in x, y, z), so $U = \frac{3}{2}Nk_{\mathrm{B}}T$, as kinetic theory found. A diatomic molecule adds two rotations, giving $\frac{5}{2}Nk_{\mathrm{B}}T$, and should add two more for vibration (kinetic and potential), giving $\frac{7}{2}Nk_{\mathrm{B}}T$. A solid has three kinetic and three potential terms per atom, giving the Dulong–Petit heat capacity $3Nk_{\mathrm{B}}$.

But measured heat capacities of air are $\frac{5}{2}$, not $\frac{7}{2}$; hydrogen's drops to $\frac{3}{2}$ at low temperatures; and the heat capacities of solids vanish as $T \to 0$. Degrees of freedom *freeze out*. Classical physics has no explanation. The answer is that energy comes in quanta: an oscillator of frequency ω can only absorb energy in lumps of $\hbar\omega$ (Chapter 14), and when $k_{\mathrm{B}}T \ll \hbar\omega$ the Boltzmann factor

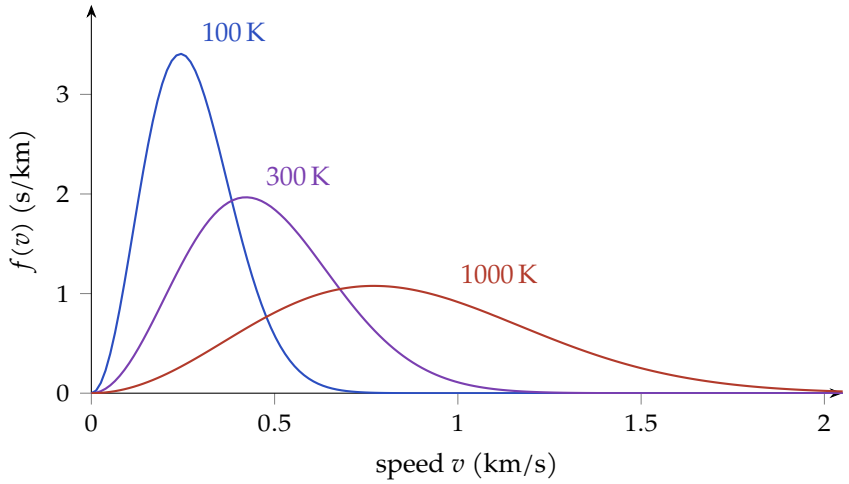
$e^{-\hbar\omega/k_B T}$ makes even one quantum unlikely. Molecular vibrations need $k_B T \sim 0.3\,\text{eV}$, far above room temperature, so they are frozen. Equipartition is the high-temperature limit of quantum statistics.

§9.6 The speeds of molecules

The Boltzmann factor $e^{-mv^2/2k_B T}$, together with the number of velocity vectors with speed between v and $v + \mathrm{d}v$ ($\propto 4\pi v^2\,\mathrm{d}v$), gives the **Maxwell–Boltzmann distribution** of molecular speeds:

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2\, e^{-mv^2/2k_B T}.$$

It rises from zero, peaks at $v_p = \sqrt{2k_B T/m}$, and has a long exponential tail. That tail matters: it is the few fastest molecules that escape planetary atmospheres, and the few fastest protons in the Sun's core that undergo fusion (Chapter 22).



Maxwell–Boltzmann speed distributions for nitrogen at three temperatures. Hotter gas has a lower, broader peak at higher speed; the area under each curve is 1.

§9.7 Quantum statistics

In quantum mechanics, identical particles are *truly* identical: swapping two electrons does not give a new microstate. And particles come in two kinds (Chapter 14). **Fermions** (electrons, protons, neutrons) obey the Pauli exclusion principle: at most one per quantum state. **Bosons** (photons, helium-4 atoms) can pile into the same state, and in fact prefer to. Working out the average number of particles in a single-particle state of energy ε gives

$$\bar{n}_{\text{FD}}(\varepsilon) = \frac{1}{e^{(\varepsilon-\mu)/k_B T} + 1}, \qquad \bar{n}_{\text{BE}}(\varepsilon) = \frac{1}{e^{(\varepsilon-\mu)/k_B T} - 1},$$

where μ is the *chemical potential*, set by the number of particles. When the states are sparsely occupied ($\bar{n} \ll 1$), both reduce to the classical Boltzmann factor. When they are crowded, the difference of a single ± 1 produces entirely different physics: metals, white dwarfs, lasers, superfluids.

9.7.1 Blackbody radiation

Photons are bosons with $\mu = 0$ (they can be created and destroyed freely). Counting the electromagnetic modes in a box (each one a harmonic oscillator) and occupying them with $\bar{n}_{\mathrm{BE}}$ gives **Planck's law** for the energy density per unit frequency:

$$u(\nu) = \frac{8\pi h\nu^3}{c^3}\,\frac{1}{e^{h\nu/k_{\mathrm{B}}T} - 1}.$$

Classical equipartition would give each mode an energy $k_{\mathrm{B}}T$; since there are infinitely many high-frequency modes, the total energy diverges. This is the *ultraviolet catastrophe*. Planck's quantization cuts the high-frequency modes off, because exciting them costs a quantum $h\nu \gg k_{\mathrm{B}}T$. The spectrum peaks at a wavelength given by **Wien's law**,

$$\lambda_{\max} T \approx 2.9\,\mathrm{mm\,K},$$

and integrating it gives the Stefan–Boltzmann law $j = \sigma T^4$ with σ now computed from h, c, and k_{B}. The Sun (5800 K) peaks in the visible, at about 500 nm — and our eyes evolved to see there. You (310 K) glow at about 10 μm, in the infrared that thermal cameras see. The universe itself is filled with blackbody radiation at 2.725 K, the cosmic microwave background, the most perfect blackbody ever measured (Chapter 28).

9.7.2 Degenerate fermions

Put N electrons in a box at zero temperature. By the exclusion principle they cannot all sit in the ground state; they fill the lowest states one by one up to the *Fermi energy*

$$E_F = \frac{\hbar^2}{2m}\left(3\pi^2 n\right)^{2/3},$$

where n is the number density. For the conduction electrons in copper, $E_F \approx 7\,\mathrm{eV}$, equivalent to a temperature of 80 000 K. So at room temperature the electron gas in a metal is "cold": only the electrons within $k_{\mathrm{B}}T$ of the Fermi energy can be excited at all. This is why electrons contribute so little to the heat capacity of metals, which baffled classical physicists. The same *degeneracy pressure*, the resistance of fermions to being squeezed, holds up white dwarf stars against gravity, and neutron degeneracy holds up neutron stars (Chapter 27).

9.7.3 Bose–Einstein condensation

Bosons do the opposite. Cool a gas of bosons below a critical temperature and a macroscopic fraction of them suddenly piles into the single lowest-energy state: a *Bose–Einstein condensate*, a quantum wave large enough to see. The condition is that the thermal de Broglie wavelength, $\lambda_T = h/\sqrt{2\pi m k_{\mathrm{B}}T}$, becomes comparable to the spacing between atoms: the atoms' quantum waves overlap. Einstein predicted it in 1925; it was achieved in dilute atomic gases in 1995, at temperatures around 100 nK (Chapter 18). Liquid helium-4 becomes a superfluid below 2.17 K, flowing without viscosity, for closely related reasons.

§9.8 Phase transitions and the Ising model

Why does a magnet suddenly lose its magnetism above a certain temperature? The simplest model of collective behaviour, the **Ising model**, gives the answer. Place spins $s_i = \pm 1$ on a lattice, with energy

$$E = -J \sum_{\langle ij \rangle} s_i s_j,$$

summed over neighbouring pairs. With $J > 0$, aligned neighbours lower the energy. At low temperature energy wins, the spins align, and the system is magnetized. At high temperature entropy wins, and the spins point randomly. In between there is a sharp *critical temperature* T_c.

The **mean-field approximation** is a beautiful spherical cow. Replace each spin's neighbours by their average magnetization m. Each spin then sees an effective field zJm (with z neighbours) and, as a two-state system, has average value $\tanh(zJm/k_{\mathrm{B}}T)$. Self-consistency demands

$$m = \tanh\!\left(\frac{zJm}{k_{\mathrm{B}}T}\right).$$

For $k_{\mathrm{B}}T > zJ$ the only solution is $m = 0$; below, two magnetized solutions appear. So $k_{\mathrm{B}}T_c \approx zJ$, and just below the transition $m \propto (T_c - T)^{1/2}$.

Mean-field theory gets the qualitative story right and the details wrong. The true exponent in three dimensions is about 0.33, not 0.5. Strangely, the same exponent appears at the liquid–gas critical point of water, in binary alloys, and in many other systems that seem to have nothing in common. This *universality* is explained by the *renormalization group*: near a critical point, fluctuations exist at every length scale, and the microscopic details wash out as you zoom out. It is one of the deepest ideas in modern physics, connecting statistical mechanics to quantum field theory (Chapter 30).

§9.9 Fluctuations and Brownian motion

Under a microscope, pollen grains in water jiggle ceaselessly. In 1905 Einstein explained this *Brownian motion* as the grain being kicked by water molecules, and predicted that it diffuses with

$$\langle x^2 \rangle = 2Dt, \qquad D = \frac{k_{\mathrm{B}}T}{6\pi\eta r},$$

where η is the viscosity and r the grain's radius. This *Einstein relation* links a fluctuation (random kicks) to dissipation (viscous drag), a deep and general connection. In 1908 Jean Perrin measured $\langle x^2 \rangle$, extracted Boltzmann's constant and hence Avogadro's number, and so convinced the last sceptics that atoms are real. Random walks of this kind govern how molecules move inside cells (Chapter 23) and how prices move in markets (Chapter 16).

§9.10 Problems

9.1 $O(\epsilon)$ Flip 100 fair coins. Using the Gaussian approximation to the binomial distribution, estimate the probability of exactly 50 heads, and of 60 or more heads.

9.2 $O(\epsilon)$ ★ Derive the barometric formula $n(h) = n_0 e^{-mgh/k_{\mathrm{B}}T}$ for an isothermal atmosphere, and compute the scale height for nitrogen at 250 K. Why is the atmosphere's helium fraction larger at high altitudes?

9.3 $O(\epsilon)$ A two-state system has energy splitting $\varepsilon = 0.1\,\mathrm{eV}$. Find the ratio of populations of the upper and lower states at 300 K and at 3000 K. At what temperature is the upper state 10% occupied?

9.4 $O(\epsilon)$ Explain why the molar heat capacity of air at room temperature is about $\tfrac{5}{2}R$, and predict how it changes at very high temperatures and at very low temperatures.

9.5 $O(\epsilon^2)$ From the Maxwell–Boltzmann distribution, find the most probable speed, the mean speed, and the root-mean-square speed of nitrogen molecules at 300 K, and show that they are in the ratio $\sqrt{2} : \sqrt{8/\pi} : \sqrt{3}$.

9.6 $O(\epsilon^2)$ ★ Use Wien's law to find the peak wavelength of the radiation from the Sun's surface, a human body, a stove element glowing red, and the cosmic microwave background. Explain why stars hotter than the Sun look blue and cooler stars look red, but no star looks green.

9.7 $O(\epsilon^2)$ The Sun has radius 7.0×10^{8} m and surface temperature 5800 K. Find its luminosity from the Stefan–Boltzmann law, and the intensity of sunlight at the Earth, 1.5×10^{11} m away. Then estimate the Earth's temperature as a black sphere in equilibrium (Chapter 25 refines this).

9.8 $O(\epsilon^2)$ ★ Copper has one conduction electron per atom and a density of $8.9\,\mathrm{g/cm^3}$. Compute its Fermi energy and Fermi temperature. Explain why the classical expectation that electrons should add $\tfrac{3}{2}k_{\mathrm{B}}$ per electron to the heat capacity of a metal fails by a factor of about $k_{\mathrm{B}}T/E_F$.

9.9 $O(\epsilon^2)$ Solve the mean-field Ising equation graphically, and show that just below T_c the magnetization is $m \approx \sqrt{3(T_c - T)/T}$. What does mean-field theory predict for the magnetization exactly at T_c in a small applied field?

9.10 $O(\epsilon^2)$ A sphere of radius $0.5\,\mu\mathrm{m}$ is suspended in water ($\eta = 10^{-3}\,\mathrm{Pa\,s}$) at 300 K. Find its diffusion constant and its typical displacement after one second and after one hour. How long does it take to diffuse 1 mm?

9.11 $e^{-1/\epsilon}$ ★ (*Making a condensate.*) A Bose–Einstein condensate forms when $n\lambda_T^3 \approx 2.6$, where $\lambda_T = h/\sqrt{2\pi m k_{\mathrm{B}} T}$. Estimate the condensation temperature for rubidium-87 atoms at a density of $10^{20}\,\mathrm{m^{-3}}$. Compare with liquid helium-4, of density $145\,\mathrm{kg/m^3}$, and with the observed superfluid transition at 2.17 K. Why does a gas of rubidium have to be so much colder?

9.12 $e^{-1/\epsilon}$ (*Hotter than infinity.*) For N two-state spins with energies 0 and ε, compute $S(U)$ using $\Omega = \binom{N}{n}$ with n excited spins, and from it $1/T = \partial S/\partial U$. Show that when more than half the spins are excited, T is negative. Explain why a system at negative tempera-

ture is *hotter* than one at any positive temperature, in the sense that heat flows from it into any positive-temperature system. (Lasers need exactly such a population inversion.)

9.13 $e^{-1/\epsilon}$ (*The Gibbs paradox.*) Treating N gas molecules as distinguishable, the entropy of an ideal gas comes out as $S = Nk_B \ln(V T^{3/2}) + \text{const}$. Show that this is not extensive: doubling the system (two identical boxes side by side) does not double S, and removing a partition between two boxes of the *same* gas would appear to increase entropy. Show that dividing Ω by $N!$ fixes both problems. Classical physics has no reason to divide by $N!$; quantum mechanics, where identical particles are genuinely indistinguishable, does.

Further reading

- **Daniel V. Schroeder, *An Introduction to Thermal Physics*** — The best introduction, building entropy and temperature from counting Einstein solids exactly as this chapter does.
- **Kittel and Kroemer, *Thermal Physics*** — A classic, quantum-first approach.
- **James P. Sethna, *Entropy, Order Parameters, and Complexity*** — Modern statistical mechanics with outstanding exercises, from information theory to the renormalization group. Free online.
- **Mehran Kardar, *Statistical Physics of Particles* and *Statistical Physics of Fields*** — The graduate standard, the second volume covering phase transitions and the renormalization group.

CHAPTER 10

Special relativity

BUILDS ON Newtonian mechanics (Chapter 2), electromagnetism (Chapter 6)

Space and time, caught trading places at high speed — and the discovery that they were never really separate.

> **PROTOTYPE** The light clock: two mirrors facing each other, with a pulse of light bouncing between them. Each round trip is one tick. Watch a light clock fly past you and its light zigzags along a longer path; since light always moves at speed c, each tick takes longer. Moving clocks run slow.

Special relativity is what happens when you take one experimental fact completely seriously: the speed of light is the same for everyone. The consequences — time dilation, length contraction, the relativity of simultaneity, $E = mc^2$ — are counterintuitive only because our intuitions were built at speeds a millionth of the speed of light. Underneath, the theory is not complicated. It is geometry.

§10.1 The problem

Newtonian mechanics obeys the principle of relativity: the laws of physics are the same in every inertial frame, and you cannot tell by any mechanical experiment whether your train is moving smoothly or standing still. Velocities simply add: throw a ball forward at 10 m/s from a train moving at 30 m/s, and it moves at 40 m/s relative to the ground.

But Maxwell's equations (Chapter 6) predict that light travels at $c = 1/\sqrt{\mu_0 \epsilon_0}$, with no mention of any frame. Relative to what? Nineteenth-century physicists assumed relative to a medium, the "luminiferous ether," and that the Earth must be moving through it. In 1887 Michelson and Morley used an interferometer (Chapter 8) sensitive enough to detect the Earth's orbital motion through the ether. They found nothing. Light moved at the same speed in every direction, in every season.

§10.2 Einstein's postulates

In 1905 Einstein took the null result at face value.

Law 10.1 (The postulates of special relativity). (1) The laws of physics are the same in all inertial frames.

(2) The speed of light in vacuum is the same, c, in all inertial frames, regardless of the motion of the source or observer.

Each postulate seems reasonable alone. Together they seem impossible: if I chase a light beam at half the speed of light, how can it still recede from me at c? Something about our ideas of space and time must give. It turns out to be the idea that time is the same for everyone.

§10.3 Simultaneity is relative

A train moves past a platform. Lightning strikes both ends of the train, and the flashes reach an observer standing at the middle of the platform at the same moment; she concludes the strikes were simultaneous. A passenger at the middle of the train is moving toward the front flash and away from the rear one, so he sees the front flash first. In his frame, light travels at c in both directions and he is standing still at the middle, so he must conclude the front strike happened *first*.

Both are right. *Whether two separated events are simultaneous depends on the observer's frame.* Everything else in special relativity follows from this.

§10.4 Time dilation

Take a light clock with mirrors a distance L apart. At rest, a tick takes $\Delta\tau = 2L/c$. Now let the clock move sideways at speed v. In our frame, the light travels diagonally, covering a distance $c\Delta t/2$ on each half-trip while the clock moves $v\Delta t/2$. Pythagoras gives $(c\Delta t/2)^2 = L^2 + (v\Delta t/2)^2$, so

$$\Delta t = \gamma\,\Delta\tau, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \geq 1.$$

Moving clocks run slow by the factor γ, the *Lorentz factor*. By the first postulate this applies to every clock, including atoms, muons, and hearts — otherwise you could compare a light clock with a mechanical clock and detect absolute motion. The time τ measured by a clock in its own rest frame is called its *proper time*.

At everyday speeds $\gamma \approx 1 + v^2/2c^2$ (Chapter 1), and the effect is tiny: on a jet at 250 m/s, about a nanosecond per hour. But it is measurable. Atomic clocks flown around the world in 1971 confirmed it, and GPS satellites must correct for it every day (Problem 10.4).

Example 10.2 (Muons reach the ground)**.** Cosmic rays striking the upper atmosphere make muons, unstable particles that live 2.2 μs at rest. Even at nearly the speed of light, a muon should travel only about 660 m before decaying — yet they rain down at sea level, 15 km below. At $v = 0.998c$, $\gamma \approx 16$: in our frame their clocks run sixteen times slow, and they survive long enough.

§10.5 Length contraction

From the muon's own point of view, its clock is fine. So how does it reach the ground? In its frame the atmosphere rushes past at $0.998c$, and is *contracted* along the direction of motion:

$$L = \frac{L_0}{\gamma}.$$

The 15 km of atmosphere is less than a kilometre thick to the muon. Time dilation in one frame is length contraction in the other; the two observers agree on what happens (the muon reaches the ground) while disagreeing about why. Only lengths along the motion contract; perpendicular lengths are unchanged.

§10.6 The Lorentz transformation

The full rule for converting the coordinates (t, x) of an event in one frame to (t', x') in a frame moving at speed v along x is the **Lorentz transformation**:

$$t' = \gamma\left(t - \frac{vx}{c^2}\right), \qquad x' = \gamma\,(x - vt), \qquad y' = y, \quad z' = z.$$

At low speeds it reduces to the Galilean $t' = t$, $x' = x - vt$. The new ingredient is the term vx/c^2 in the time: clocks that are synchronized in one frame are out of sync in another, by an amount that grows with their separation. That is the relativity of simultaneity, in a formula.

Velocities no longer simply add. An object moving at u in one frame moves at

$$u' = \frac{u + v}{1 + uv/c^2}$$

in a frame moving at $-v$. Add $0.8c$ to $0.8c$ and you get $0.976c$. Add anything to c and you get c. Nothing with mass can be accelerated to the speed of light.

§10.7 Spacetime

In 1908 Minkowski, Einstein's former teacher, recast special relativity as geometry. "Henceforth space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality."

An *event* is a point in four-dimensional spacetime. Different observers disagree about time intervals and about distances, but they all agree on the **spacetime interval**

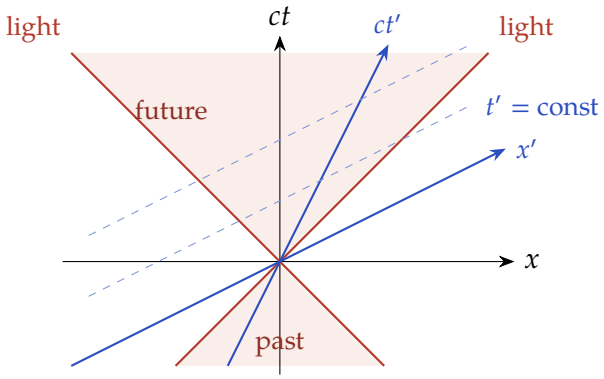
$$\Delta s^2 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2.$$

It plays the role that distance plays in ordinary geometry: just as rotating your axes changes Δx and Δy but not $\Delta x^2 + \Delta y^2$, a Lorentz transformation changes Δt and Δx but not Δs^2. A Lorentz boost is a rotation in spacetime — a hyperbolic one, because of that minus sign.

The sign of the interval classifies pairs of events:

- **Timelike** ($\Delta s^2 > 0$): one event can influence the other. All observers agree on their order, and $\Delta s/c$ is the proper time measured by a clock that travels between them.
- **Lightlike** ($\Delta s^2 = 0$): they can be connected only by a light signal.
- **Spacelike** ($\Delta s^2 < 0$): no signal can connect them, and different observers disagree about which happened first. Causality is safe precisely because such events cannot influence each other.

On a *spacetime diagram*, with time up and space across, light travels on 45° lines, and each event has a *light cone*: the future cone of everything it can affect, and the past cone of everything that could have affected it.



A spacetime diagram. Light moves along the 45° lines, which bound the future and past of the event at the origin. An observer moving at $v = c/2$ has tilted axes ct' and x'; the dashed lines are events that observer calls simultaneous.

KEY IDEA In ordinary geometry, the straight line is the *shortest* path between two points. In spacetime, because of the minus sign, the straight worldline is the *longest*: an unaccelerated clock records more proper time between two events than any other clock that travels between them. This one fact dissolves the twin paradox (Problem 10.9).

§10.8 Four-vectors, energy, and momentum

Just as the position of an event is a *four-vector* $x^\mu = (ct, x, y, z)$, other quantities combine into four-vectors whose "length" (in the sense of the interval) is invariant. The most important is the **four-momentum**

$$p^\mu = \left(\frac{E}{c}, \mathbf{p}\right), \qquad E = \gamma mc^2, \quad \mathbf{p} = \gamma m\mathbf{v}.$$

Its invariant length gives the most important equation in relativity:

$$E^2 = (pc)^2 + (mc^2)^2.$$

The energy of a particle at rest is not zero but

$$E = mc^2.$$

Mass is a form of energy, and a very concentrated one: one kilogram is 9×10^{16} J, roughly the annual output of a few large power stations. The kinetic energy is $(\gamma - 1)mc^2$, which at low speed expands to $\frac{1}{2}mv^2 + \frac{3}{8}mv^4/c^2 + \cdots$: Newton's kinetic energy plus relativistic corrections. As $v \to c$, $\gamma \to \infty$, and it would take infinite energy to reach c.

$E = mc^2$ also means that the mass of a bound system is *less* than the sum of its parts, by its binding energy divided by c^2. For a chemical bond the difference is a part in 10^{10}, unmeasurable. For a nucleus it is nearly a percent, and releasing it powers the Sun and nuclear reactors (Chapter 22). For a proton, almost all of whose mass comes from the energy of the quarks and gluons inside it, the "parts" weigh about 1% of the whole (Chapter 32).

Massless particles have $m = 0$, so $E = pc$, and they must move at exactly c. Light carries momentum E/c, which is the radiation pressure of Chapter 6. When a photon scatters off an electron it loses energy and its wavelength increases by $\Delta\lambda = (h/m_e c)(1 - \cos\theta)$, the *Compton effect*, which convinced physicists in 1923 that photons carry momentum like particles.

§10.9 Collisions and the creation of matter

In any collision the total four-momentum is conserved. But mass is not! Kinetic energy can become rest energy: collide particles hard enough and new particles appear. That is what particle accelerators do (Chapter 31). The useful energy for making new particles is the energy in the centre-of-mass frame, which is fixed by the invariant

$$s = \Big(\sum E\Big)^2 - \Big(\sum \mathbf{p}\,c\Big)^2.$$

A beam hitting a stationary target wastes most of its energy on moving the debris forward; two beams colliding head-on waste none. That is why modern accelerators are colliders.

§10.10 The relativistic Doppler effect

A source receding at speed v emits light that we receive at a lower frequency,

$$f_{\text{obs}} = f_{\text{src}}\sqrt{\frac{1 - v/c}{1 + v/c}}.$$

Unlike sound (Chapter 5), there is no medium, so only the relative velocity matters; and time dilation adds a *transverse* Doppler shift even for sideways motion. (The redshifts of distant galaxies are mostly not Doppler shifts at all but the stretching of space itself; Chapter 28.)

§10.11 Electromagnetism, finally at home

Maxwell's equations were already relativistic. In four-vector language, the charge density and current combine into a four-vector $J^\mu = (c\rho, \mathbf{J})$, the potentials into $A^\mu = (\phi/c, \mathbf{A})$, and the electric and magnetic fields into a single antisymmetric *field tensor* $F^{\mu\nu}$, whose six independent components are the three components of $\mathbf{E}$ and the three of $\mathbf{B}$. All four of Maxwell's equations collapse into two:

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu, \qquad \partial_{[\lambda} F_{\mu\nu]} = 0.$$

Under a Lorentz boost, the components of $F^{\mu\nu}$ mix, which is the precise sense in which a pure electric field in one frame is partly magnetic in another (Chapter 6). Charge conservation, $\partial_\mu J^\mu = 0$, is now a single equation.

§10.12 Where this goes

Special relativity has two great descendants. Combine it with gravity and you need curved spacetime: general relativity (Chapter 26). Combine it with quantum mechanics and particle number stops being conserved — $E = mc^2$ lets particles be created and destroyed — so you need quantum field theory (Chapter 30), in which antimatter appears as a requirement rather than a guess.

§10.13 Problems

10.1 $O(\epsilon)$ ★ Muons are produced 15 km up, moving at $0.998c$, and have a rest-frame lifetime of 2.2 μs. What fraction would reach the ground without time dilation? With it? Describe the same calculation from the muon's frame.

10.2 $O(\epsilon)$ Find the rest energy of one gram of matter in joules, and in tons of TNT ($1\,\text{ton} = 4.2\times10^9\,\text{J}$). The Hiroshima bomb released about 15 kilotons. How much mass was converted to energy?

10.3 $O(\epsilon)$ A spaceship moving at $0.8c$ fires a probe forward at $0.8c$ relative to itself. How fast does the probe move relative to the Earth? What would Galileo have said?

10.4 $O(\epsilon^2)$ ★ GPS satellites orbit at a speed of about 3.9 km/s. By how many microseconds per day do their clocks fall behind clocks on the ground, due to special relativity alone? If uncorrected, roughly how quickly would position errors accumulate? (General relativity gives a larger effect in the *opposite* direction; see Chapter 26.)

10.5 $O(\epsilon^2)$ A charged pion at rest ($m_\pi c^2 = 139.6\,\text{MeV}$) decays into a muon ($m_\mu c^2 = 105.7\,\text{MeV}$) and a massless neutrino. Use conservation of energy and momentum to find the muon's kinetic energy.

10.6 $O(\epsilon^2)$ ★ Antiprotons can be made by firing protons at protons at rest: $p+p \to p+p+p+\bar{p}$. Find the threshold kinetic energy of the beam proton. (Answer: $6m_pc^2 \approx 5.6\,\text{GeV}$.) Why is it so much more than the $2m_pc^2$ of new rest mass? This is why the Bevatron at Berkeley, which first made antiprotons in 1955, was designed for 6 GeV.

10.7 $O(\epsilon^2)$ Show directly from the Lorentz transformation that $c^2t^2 - x^2$ is the same in both frames. Then show that the transformation can be written as $ct' = ct\cosh\phi - x\sinh\phi$, $x' = x\cosh\phi - ct\sinh\phi$, with *rapidity* $\tanh\phi = v/c$, and that rapidities simply add under successive boosts.

10.8 $O(\epsilon^2)$ Derive the Compton formula $\Delta\lambda = (h/m_ec)(1-\cos\theta)$ for a photon scattering off an electron at rest, using conservation of four-momentum.

Hint: Write $p'_e = p_\gamma + p_e - p'_\gamma$ and square both sides; the invariant $p \cdot p$ of each particle is m^2c^2.

10.9 $e^{-1/\epsilon}$ ★ (*The twin paradox.*) One twin stays on Earth; the other travels at $0.8c$ to a star 4 light-years away and immediately returns. Show that the travelling twin ages 6 years while the Earth twin ages 10. The travelling twin can argue that it was the Earth that moved away and came back, so the *Earth* twin should be younger. Draw a spacetime diagram, including the travelling twin's lines of simultaneity just before and just after the turnaround, and explain precisely where the symmetry breaks.

10.10 $e^{-1/\epsilon}$ (*The barn and the pole.*) A 20 m pole is carried at $0.866c$ ($\gamma = 2$) toward a 10 m barn with doors at both ends. In the barn's frame the pole contracts to 10 m and fits inside, so both doors can be shut simultaneously for an instant. In the pole's frame the *barn* contracts to 5 m, so the pole cannot possibly fit. Resolve the paradox. What happens if the far door is solid and stays shut?

10.11 $e^{-1/\epsilon}$ (*To the centre of the galaxy in a lifetime.*) A rocket accelerates at a constant $g = 9.8\,\mathrm{m/s^2}$ in its own instantaneous rest frame. One can show that after proper time τ it has travelled a distance $x = (c^2/g)\,[\cosh(g\tau/c) - 1]$ in the original frame. How far does it get in 10 years of the crew's time? In 20? How many years of crew time does it take to reach the centre of the galaxy, 26 000 light-years away (accelerating halfway, then decelerating)? What is the catch?

Further reading

- **Taylor and Wheeler, *Spacetime Physics*** — The classic introduction that teaches relativity as geometry from the first page.
- **David Morin, *Special Relativity for the Enthusiastic Beginner*** — Short, clear, and packed with problems and paradoxes, fully solved.
- **A. P. French, *Special Relativity*** — The MIT introductory text, careful about experiments.
- **Bernard Schutz, *A First Course in General Relativity*** — Chapters 1–2 give a tensor-based treatment of special relativity that leads directly into general relativity.

CHAPTER 11

Fluid dynamics

BUILDS ON Newtonian mechanics (Chapter 2)

The unreasonably hard physics of everything that flows — rivers, blood, weather, wings — governed by an equation nobody can fully solve.

PROTOTYPE A sphere moving through a fluid. Move it slowly through honey and the flow is smooth, symmetric, and reversible. Move it fast through air and a wake forms, vortices peel off in alternating rows, and finally the wake dissolves into turbulence. One equation describes all of it, and one dimensionless number — the Reynolds number — tells you which regime you are in.

A fluid is a substance that cannot resist a shearing force at rest: push sideways on a layer of water and it flows, however gently you push. That includes liquids, gases, and on long enough timescales, glaciers and the Earth's mantle. Fluid dynamics is Newton's second law applied to a continuous medium, and its central equation, the Navier–Stokes equation, has been known since the 1840s. It is still not understood. Whether its solutions always stay smooth is one of the seven Millennium Prize Problems, with a million dollars attached.

§11.1 The continuum

A glass of water contains 10^{25} molecules. We do not track them. Instead we treat the fluid as a *continuum* with a density $\rho(\mathbf{r}, t)$, a velocity field $\mathbf{v}(\mathbf{r}, t)$, and a pressure $P(\mathbf{r}, t)$, defined by averaging over regions large compared to the molecular spacing but small compared to anything we care about. This is a spherical cow that works superbly for water in pipes and fails for gas in a near-vacuum, where molecules travel macroscopic distances between collisions.

Pressure in a fluid at rest acts equally in all directions — it is a scalar — and always perpendicular to any surface.

§11.2 Fluids at rest

A thin horizontal slab of fluid must support the weight of the fluid above it. Balancing forces gives $\mathrm{d}P/\,\mathrm{d}z = -\rho g$, so in a liquid of constant density,

$$P = P_0 + \rho g h$$

at depth h. Every 10 m of water adds about one atmosphere. Pressure depends only on depth, not on the shape of the container: a narrow tube of water 10 m tall exerts the same pressure on

its base as a lake 10 m deep. *Pascal's principle* says pressure applied anywhere in an enclosed fluid is transmitted everywhere, which is how hydraulic brakes and jacks multiply force.

Law 11.1 (Archimedes' principle)**.** A body immersed in a fluid feels an upward buoyant force equal to the weight of the fluid it displaces.

The proof is beautiful: replace the body by the fluid it displaced. That fluid was in equilibrium, so the surrounding pressure must have pushed up on it with exactly its weight — and the surrounding pressure does not know what has replaced it. A steel ship floats because its hull displaces a large volume of water; an iceberg floats with about 89% submerged because ice is 89% as dense as seawater.

§11.3 Describing flow

A flowing fluid has a velocity field $\mathbf{v}(\mathbf{r}, t)$. The curves everywhere tangent to it are *streamlines*. There are two ways to describe change: at a fixed point (the *Eulerian* view, $\partial/\partial t$) or following a parcel of fluid as it moves (the *Lagrangian* view). They are related by the **material derivative**

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla.$$

Even in a steady flow, a parcel accelerates as it is carried into faster or slower regions; the term $\mathbf{v} \cdot \nabla \mathbf{v}$ captures this, and because it is quadratic in $\mathbf{v}$, it is the source of nearly everything difficult about fluids.

Mass is conserved, which gives the **continuity equation**

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0.$$

Liquids, and gases moving much slower than sound, are nearly *incompressible*, and then simply $\nabla \cdot \mathbf{v} = 0$. In a pipe of varying cross-section A, this means Av is constant: squeeze the end of a garden hose and the water speeds up.

§11.4 Ideal flow and Bernoulli

Ignore viscosity for a moment. Newton's second law for a fluid parcel, pushed by pressure differences and pulled by gravity, is **Euler's equation**,

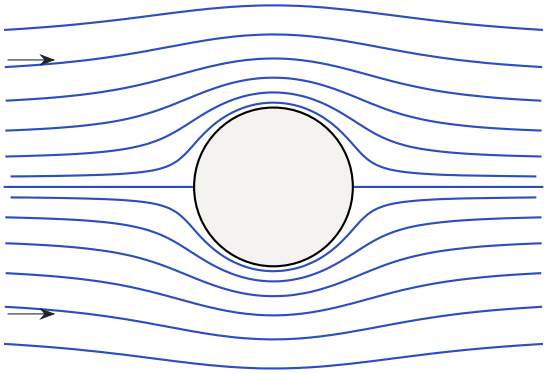
$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla P + \rho \mathbf{g}.$$

For steady, incompressible flow, integrating along a streamline gives

Law 11.2 (Bernoulli's equation)**.** Along a streamline of steady, incompressible, inviscid flow,

$$P + \tfrac{1}{2}\rho v^2 + \rho g h = \text{constant}.$$

It is energy conservation per unit volume: where the fluid speeds up, its pressure drops. Water escaping from a hole at depth h in a tank leaves at $v = \sqrt{2gh}$ (*Torricelli's law*), exactly the speed of something dropped from height h. A Venturi meter measures flow speed from the pressure drop in a constriction. A shower curtain billows inward because the moving air and water inside lower the pressure.



Ideal (frictionless) flow past a cylinder. Streamlines crowd together over the top and bottom, where the fluid is fastest and, by Bernoulli, the pressure is lowest. The pattern is the same in front and behind, so the pressure pushes equally on both sides: ideal flow predicts no drag at all. This is d'Alembert's paradox, and viscosity resolves it.

Remark 11.3. Bernoulli's equation is often used to "explain" how wings make lift: air over the curved top travels farther, so must go faster, so has lower pressure. The second step is false — there is no reason the air above and below must arrive at the trailing edge together, and in fact the upper air arrives first. The honest explanation is that a wing deflects air downward, and by Newton's third law the air pushes the wing up; the pressure differences are real but are the *mechanism* of that deflection, not an independent cause. See §11.9.

§11.5 Vorticity

The *vorticity* $\omega = \nabla \times \mathbf{v}$ measures local spinning of the fluid (Chapter 1). A flow with zero vorticity is *irrotational*, and for an ideal fluid, **Kelvin's circulation theorem** says vorticity can be neither created nor destroyed: it is carried along with the fluid, and vortex lines move as if frozen in. A smoke ring is a vortex ring that holds together for metres because of this. A tornado or a draining bathtub concentrates vorticity, spinning faster as it narrows, like a skater pulling in her arms (Chapter 2). In real fluids, vorticity is born at solid boundaries, where viscosity acts.

§11.6 Viscosity and the Navier–Stokes equation

Real fluids resist shearing. If adjacent layers slide past each other with a velocity gradient $\mathrm{d}v/\,\mathrm{d}y$, the shear stress between them is

$$\tau = \eta \,\frac{\mathrm{d}v}{\mathrm{d}y},$$

where η is the *viscosity*: about 10^{-3} Pa s for water, 2×10^{-5} for air, 10 for honey, and 10^{21} for the Earth's mantle. At a solid surface, a viscous fluid does not slip: the fluid touching the wall moves with it. Adding viscous forces to Euler's equation gives

Law 11.4 (The Navier–Stokes equation)**.** For an incompressible fluid,

$$\rho\left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v}\cdot\nabla\mathbf{v}\right) = -\nabla P + \eta\nabla^2\mathbf{v} + \rho\mathbf{g}, \qquad \nabla\cdot\mathbf{v} = 0.$$

It is Newton's second law, nothing more: mass times acceleration on the left, pressure, viscous, and gravitational forces on the right. The trouble is the nonlinear term $\mathbf{v}\cdot\nabla\mathbf{v}$.

§11.7 The Reynolds number

Compare the two forces that fight over a flow of speed v around an object of size L. Inertia (the nonlinear term) scales as $\rho v^2/L$; viscous force scales as $\eta v/L^2$. Their ratio is the **Reynolds number**:

$$\mathrm{Re} = \frac{\rho v L}{\eta} = \frac{vL}{\nu},$$

where $\nu = \eta/\rho$ is the *kinematic viscosity*.

KEY IDEA Two flows with the same Reynolds number and the same geometry are the same flow, just rescaled. A model plane in a wind tunnel, a real plane in the sky, and a model in a water tank all behave alike if their Reynolds numbers match. This *dynamic similarity* is dimensional analysis (Chapter 1) at its most profitable, and it is why engineers can design aircraft at all.

The range of Reynolds numbers in nature is staggering:

- a swimming bacterium: $\mathrm{Re} \sim 10^{-5}$;
- a sperm cell: 10^{-2};
- a falling raindrop: 10^{3};
- a human swimmer: 10^{6};
- an airliner wing: 10^{8};
- the atmosphere: 10^{10} and beyond.

Low Reynolds number ($\mathrm{Re} \ll 1$): inertia is irrelevant, and the Navier–Stokes equation becomes linear. Drag on a sphere is *Stokes' law*, $F = 6\pi\eta r v$. Flows are smooth and, remarkably, *reversible*: stir dye into corn syrup carefully, then unstir it, and the dye blob reassembles. A bacterium that stops swimming coasts for about a tenth of an atomic diameter. Life at low Reynolds number is so alien that swimming by any reciprocal motion, like a scallop opening and closing, gets you nowhere (Problem 11.10). Bacteria swim by rotating corkscrew flagella (Chapter 23).

High Reynolds number: inertia dominates, viscosity matters only in thin layers near surfaces, and flows become unstable — they shed vortices, form wakes, and eventually go turbulent.

§11.8 Flow in pipes

Steady viscous flow through a pipe of radius R and length ℓ has a parabolic velocity profile, fastest in the middle and zero at the walls. The volume flow rate under a pressure difference ΔP is **Poiseuille's law**:

$$Q = \frac{\pi R^4 \, \Delta P}{8 \eta \ell}.$$

That fourth power is dramatic. Halve the radius of a pipe and, at fixed pressure, the flow drops sixteenfold. This is why narrowing of the arteries is so dangerous, and why your body regulates blood flow by small changes in the radius of arterioles (Chapter 23). Above $\mathrm{Re} \approx 2000$, pipe flow becomes turbulent and the drag rises sharply.

§11.9 Drag, wakes, and lift

At high Reynolds number, viscosity is confined to a thin *boundary layer* on an object's surface, of thickness $\sim L/\sqrt{\mathrm{Re}}$. Outside it, the flow is nearly ideal. The boundary layer can *separate* from the surface, leaving a broad, low-pressure wake behind the object, and the pressure difference between front and back is what causes most drag: $F \approx \frac{1}{2} C_D \rho A v^2$ with C_D of order one (Chapter 2).

A golf ball's dimples are a clever trick. They trip the boundary layer into turbulence, which clings to the ball longer before separating, shrinking the wake and roughly halving the drag. A smooth golf ball would travel about half as far.

Lift on a wing comes from *circulation*, a net swirl $\Gamma = \oint \mathbf{v} \cdot d\mathbf{l}$ around the wing. The sharp trailing edge forces the flow to leave smoothly from it (the *Kutta condition*), which fixes the circulation, and the lift per unit span is

$$L' = \rho v \Gamma \qquad \text{(Kutta–Joukowski theorem)}.$$

Equivalently, the wing turns a large mass of air downward. A spinning ball has circulation too, and so feels a sideways force — the *Magnus effect* that curves a football free kick or a tennis topspin shot.

§11.10 Turbulence

At high enough Reynolds number, flow becomes *turbulent*: chaotic, three-dimensional, full of eddies of every size, and far better at mixing than smooth flow. Turbulence is everywhere — in rivers, chimney smoke, the atmosphere, the Sun, the coffee you stir — and it is the great unsolved problem of classical physics.

The best partial understanding comes from a picture of Richardson: big eddies break into smaller ones, which break into smaller ones still, passing energy down a *cascade* until, at small enough scales, viscosity turns it into heat. In 1941 Kolmogorov argued that in the middle of the cascade, the statistics can depend only on the rate of energy transfer per unit mass, ε. Dimensional analysis then fixes the energy spectrum,

$$E(k) \propto \varepsilon^{2/3} k^{-5/3},$$

and the smallest eddies have size $\eta_K = (\nu^3/\varepsilon)^{1/4}$. Both predictions are remarkably close to measurements, though not exactly right, and the corrections are a subject of active research.

§11.11 Surface tension

Molecules at a liquid's surface are missing neighbours on one side, so creating surface costs energy: the *surface tension* γ, about $0.072\,\mathrm{N/m}$ for water. A curved surface therefore has a pressure jump across it, the *Laplace pressure*, $\Delta P = 2\gamma/R$ for a sphere of radius R. Small drops are spherical, water striders walk on ponds, water climbs narrow glass tubes (*capillarity*), and a falling stream of water breaks into droplets. Whether gravity or surface tension wins is set by the *capillary length* $\sqrt{\gamma/\rho g} \approx 2.7\,\mathrm{mm}$ for water: drops smaller than this are round, puddles larger than this are flat.

§11.12 Problems

11.1 $O(\epsilon)$ ★ Find the pressure at the bottom of the Mariana Trench, 11 km deep. Express it in atmospheres. How much does the density of water increase there, given its bulk modulus of 2.2 GPa?

11.2 $O(\epsilon)$ Show that an iceberg floats with $\rho_{\text{ice}}/\rho_{\text{sea}} \approx 89\%$ of its volume submerged. When it melts, does the sea level rise? (For a floating cube of ice in a glass of fresh water? For an iceberg in seawater?)

11.3 $O(\epsilon)$ Estimate the Reynolds numbers of a bacterium ($2\,\mu\mathrm{m}$, $30\,\mu\mathrm{m/s}$), a person swimming, a car on a motorway, and blood in the human aorta (diameter 2.5 cm, speed $0.3\,\mathrm{m/s}$, $\nu \approx 3\times10^{-6}\,\mathrm{m^2/s}$). Which flows are turbulent?

11.4 $O(\epsilon^2)$ ★ A cylindrical tank of cross-section A, filled to height h_0, drains through a small hole of area a at the bottom. Using Torricelli's law, show that the time to empty is $(A/a)\sqrt{2h_0/g}$. Why does the water level fall faster at the beginning?

11.5 $O(\epsilon^2)$ Fog droplets have radius about $10\,\mu\mathrm{m}$. Find their terminal velocity in air using Stokes' law, and check that the Reynolds number is small. Why do clouds appear to float?

11.6 $O(\epsilon^2)$ A coronary artery narrows by 20% in radius due to plaque. At fixed pressure difference, by what factor does the blood flow fall? By how much must the heart raise the pressure difference to restore the flow?

11.7 $O(\epsilon^2)$ ★ (*Newton's bucket.*) A bucket of water rotates steadily at angular velocity Ω. Show that the water's surface is a paraboloid, $z = \Omega^2 r^2/2g$. (Newton used this experiment to argue for absolute space: how does the water "know" it is rotating? Einstein later took the question seriously; see Chapter 26.) Rotating liquid mercury makes perfect parabolic telescope mirrors.

11.8 $O(\epsilon^2)$ Water rises to height $h = 2\gamma/(\rho g r)$ in a clean glass tube of radius r. Derive this, and evaluate it for $r = 0.1\,\mathrm{mm}$. Xylem vessels in trees are about $20\,\mu\mathrm{m}$ in radius. Can capillarity alone lift water to the top of a 100 m redwood? (It cannot; trees pull water up by evaporation creating tension, a negative pressure, in the water column.)

11.9 $O(\epsilon^2)$ Stirring a cup of tea dissipates about $10^{-2}\,\mathrm{W}$ in 0.3 kg of water. Estimate the Kolmogorov scale, the size of the smallest eddies. Repeat for the atmosphere, which dissipates about $10^{-3}\,\mathrm{W/kg}$.

11.10 $e^{-1/\epsilon}$ ★ (*The scallop theorem.*) At $\mathrm{Re} \ll 1$ the Navier–Stokes equation becomes the Stokes equation $\eta \nabla^2 \mathbf{v} = \nabla P$, which contains no time derivative. Argue that the flow at each instant is determined entirely by the body's instantaneous shape and velocity of deformation, not by its history. Conclude that a swimmer whose stroke is reciprocal — the same sequence of shapes forwards and then backwards, at any speeds — returns exactly to its starting point. How does a bacterium with a rotating helical flagellum evade the theorem?

11.11 $e^{-1/\epsilon}$ (*The Magnus effect.*) A football is kicked at 25 m/s with a spin of 10 revolutions per second about a vertical axis. Argue that the sideways force should scale as $F \sim \rho\,(\omega r)\,v\,\pi r^2$, up to a factor of order one, with ball radius 11 cm and mass 0.43 kg. Estimate how far the ball bends sideways over a 25 m free kick, taking the prefactor to be about $\tfrac{1}{2}$. Is the effect large enough to matter?

11.12 $e^{-1/\epsilon}$ (*Why a stream breaks into drops.*) A long cylinder of liquid of radius R is perturbed so its radius varies as $R + \delta \cos kz$. Keeping the volume fixed, show that the surface area *decreases* if the wavelength is longer than the cylinder's circumference, $2\pi/k > 2\pi R$. Since surface tension acts to reduce area, such perturbations grow, and the stream breaks into drops. This is the *Rayleigh–Plateau instability*; the fastest-growing wavelength is about $9R$.

Further reading

- **T. E. Faber, *Fluid Dynamics for Physicists*** — A careful, physical introduction written for physicists rather than engineers.
- **D. J. Acheson, *Elementary Fluid Dynamics*** — A clear mathematical introduction with excellent treatment of vorticity and stability.
- **Benny Lautrup, *Physics of Continuous Matter*** — Fluids and solids together, with exotic and everyday applications.
- **E. M. Purcell, "Life at Low Reynolds Number"** — A famous 1976 lecture, short and delightful, freely available online.
- ***The Feynman Lectures on Physics*, Volume II, Chapters 40–41** — "The Flow of Dry Water" and "The Flow of Wet Water."

CHAPTER 12

Nonlinear dynamics and chaos

BUILDS ON analytical mechanics (Chapter 4)

When dead-simple rules turn impossible to predict — and why the impossibility has a beautiful, universal structure of its own.

PROTOTYPE The logistic map, $x_{n+1} = r\,x_n(1 - x_n)$: a one-line model of a population that grows when small and starves when large. As the growth rate r is turned up, the population settles down, then oscillates between two values, then four, then eight, and then — with no randomness anywhere in the rule — becomes chaotic. Its physical cousin is the double pendulum, whose tumbling no one can predict for more than a few swings.

Nearly every equation solved in an introductory course is linear. That is not because nature is linear; it is because linear equations are the ones we can solve. Superposition lets us break them into simple pieces (Chapter 1). Nonlinear equations offer no such mercy, and for centuries they were treated as hopeless special cases. In the last sixty years, aided by computers and geometric thinking, physicists and mathematicians discovered that nonlinear systems have a rich, beautiful, and in some ways universal behaviour — including *chaos*, deterministic behaviour so sensitive that it is unpredictable in practice.

§12.1 Flows on a line

Start with the simplest dynamical system: one variable obeying $\dot{x} = f(x)$. There is no need to solve it. Plot $f(x)$; where $f > 0$, x increases, where $f < 0$ it decreases. The *fixed points*, where $f(x^*) = 0$, are equilibria. A fixed point is *stable* if nearby points flow toward it, which happens when $f'(x^*) < 0$; linearizing, a small deviation δ obeys $\dot{\delta} = f'(x^*)\,\delta$ and decays exponentially.

Example 12.1 (Logistic growth)**.** A population with limited food obeys $\dot{N} = rN(1 - N/K)$. There are fixed points at $N = 0$ (unstable: any population grows) and $N = K$, the carrying capacity (stable). Every positive population approaches K along an S-shaped curve. In one dimension, that is all that can happen: trajectories move monotonically to fixed points or to infinity. No oscillation, no chaos.

Bifurcations. As a parameter changes, fixed points can appear, vanish, or exchange stability. These qualitative changes are called *bifurcations*, and there are only a few basic kinds.

- **Saddle-node**: $\dot{x} = r + x^2$. For $r < 0$ there are two fixed points, one stable and one unstable; at $r = 0$ they collide and annihilate; for $r > 0$ there are none, and x runs away. This is how

a system can lose its equilibrium suddenly — how a slowly pushed object suddenly starts to slide, or a lake suddenly turns murky.

- **Pitchfork**: $\dot{x} = rx - x^3$. For $r < 0$ the only fixed point is $x = 0$, stable. For $r > 0$ it becomes unstable and two new stable fixed points appear at $\pm\sqrt{r}$. The system must choose one, breaking the symmetry $x \to -x$ of the equation. The bead on a spinning hoop (Problem 2.11), the buckling of a column, a ferromagnet below its Curie temperature (Chapter 9), and the Higgs field (Chapter 32) all do this.

Saddle-node bifurcations come with *hysteresis*: turn the parameter back, and the system does not retrace its path. Many sudden transitions in climate, ecosystems, and engineering are of this type (Chapter 25).

§12.2 Flows in the plane

With two variables, $\dot{x} = f(x, y), \dot{y} = g(x, y)$, much more can happen. Near a fixed point, linearize: $(\delta\dot{x}, \delta\dot{y}) = J\,(\delta x, \delta y)$ with J the *Jacobian* matrix of partial derivatives. Its eigenvalues (Chapter 1) classify the fixed point:

- both real and negative: a stable *node*;
- real with opposite signs: a *saddle*, attracting along one direction and repelling along another;
- complex with negative real part: a stable *spiral* — the damped oscillator of Chapter 3;
- purely imaginary: a *centre*, surrounded by closed orbits — the frictionless oscillator.

Example 12.2 (The nonlinear pendulum)**.** The pendulum $\ddot{\theta} = -(g/\ell)\sin\theta$, written as a flow in $(\theta, \dot{\theta})$, has centres at $\theta = 0, \pm 2\pi, \ldots$ (hanging down) and saddles at $\theta = \pm\pi$ (balanced upright). Small swings circle the centres; large enough energy carries the pendulum over the top, rotating forever. The two kinds of motion are separated by the *separatrix*, the trajectory that creeps toward the upright position and takes infinitely long to arrive. The whole phase portrait can be drawn from the energy alone, without solving anything.

A genuinely new behaviour appears in two dimensions: the **limit cycle**, an isolated closed orbit that nearby trajectories spiral toward. A limit cycle is a self-sustained oscillation whose amplitude is set by the system itself, not by initial conditions. The van der Pol oscillator, $\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$, has negative damping for small x (pumping energy in) and positive damping for large x (draining it), and settles onto one. Heartbeats, the squeal of a violin string, the flashing of a firefly, and the ticking of a pendulum clock's escapement are limit cycles. A stable fixed point can give birth to a limit cycle as a parameter changes; this is a *Hopf bifurcation*, and it is how a steady flow past a cylinder begins to shed vortices rhythmically (Chapter 11).

But in two dimensions, chaos is impossible.

Theorem 12.3 (Poincaré–Bendixson)**.** A bounded trajectory of a smooth autonomous flow in the plane approaches either a fixed point, or a closed orbit, or a cycle of fixed points

connected by trajectories.

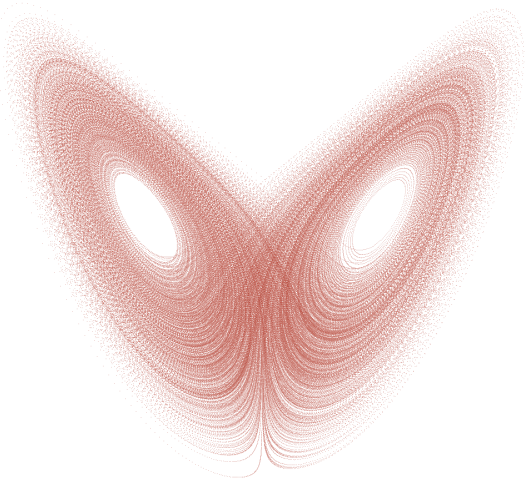
Trajectories cannot cross, and in a plane a curve that cannot cross itself has little room to misbehave. Chaos needs a third dimension.

§12.3 Chaos

In 1963 the meteorologist Edward Lorenz simplified a model of convection in the atmosphere down to three equations:

$$\dot{x} = \sigma(y - x), \qquad \dot{y} = x(\rho - z) - y, \qquad \dot{z} = xy - \beta z,$$

with $\sigma = 10$, $\beta = 8/3$, $\rho = 28$. Running it on an early computer, he restarted a calculation from numbers printed to three decimal places instead of six. Within a few simulated months the two runs had nothing to do with each other.



The Lorenz attractor, projected onto the (x, z) plane.

This is the defining feature of chaos: **sensitive dependence on initial conditions**. Two trajectories starting a distance δ_0 apart separate exponentially,

$$\delta(t) \approx \delta_0 \, e^{\lambda t},$$

where $\lambda > 0$ is the *Lyapunov exponent*. Any uncertainty in the initial state, however small, grows until it is as large as the system itself, after a *predictability horizon*

$$t_{\text{pred}} \approx \frac{1}{\lambda} \ln \frac{L}{\delta_0}.$$

The logarithm is cruel: making measurements a thousand times more precise buys only $\ln 1000 \approx 7$ more Lyapunov times. For the atmosphere, $1/\lambda$ is a day or two, so detailed weather forecasts

are limited to about two weeks no matter how good our computers become. Lorenz called it the *butterfly effect*.

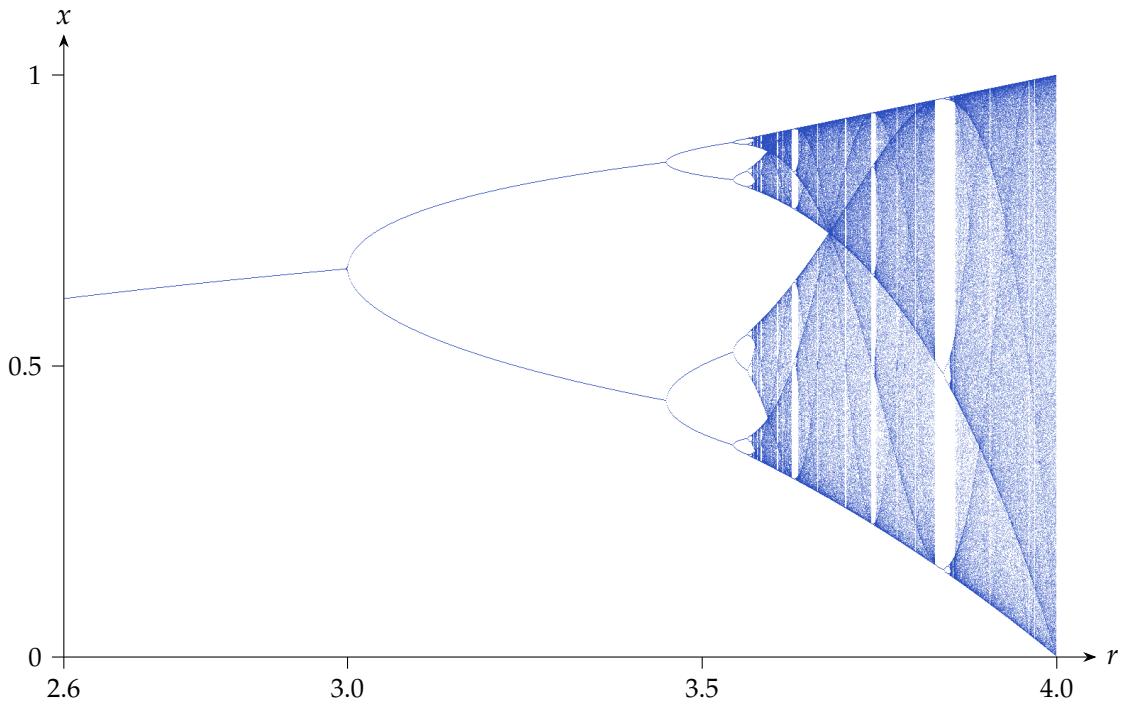
Yet the motion is not random. The Lorenz trajectory never settles down and never repeats, but it stays forever on a butterfly-shaped set called a **strange attractor**. Volumes in its phase space shrink (the system is dissipative), so trajectories are squeezed onto a set of zero volume; but nearby trajectories also separate, so the set must be endlessly stretched and folded, like dough being kneaded. The result is a *fractal*, with dimension about 2.06: more than a surface, less than a solid.

KEY IDEA Determinism does not imply predictability. A system can be governed by perfectly definite laws, with no randomness anywhere, and still be unpredictable in practice, because it amplifies every uncertainty exponentially. Chaos is how deterministic physics manufactures apparent randomness.

§12.4 Maps, and the road to chaos

Many questions about flows reduce to iterated *maps* $x_{n+1} = F(x_n)$, for example by recording a trajectory once per orbit. The logistic map is the simplest that shows everything.

- For $1 < r < 3$, x_n converges to the fixed point $x^* = 1 - 1/r$.
- At $r = 3$ the fixed point loses stability (the slope of the map there passes through -1), and a stable cycle of period 2 appears.
- At $r \approx 3.449$ the 2-cycle gives way to a 4-cycle, then an 8-cycle at 3.544, a 16-cycle at 3.564, and so on, the intervals shrinking geometrically.
- At $r_\infty \approx 3.5699$ the period becomes infinite: chaos. Beyond it, chaotic bands are interrupted by *windows* of periodic behaviour, the largest a period-3 window near $r = 3.83$.



The long-run values of the logistic map as the parameter r increases: period doubling, then chaos.

In 1975 Mitchell Feigenbaum, playing with a pocket calculator, noticed that the ratios of successive intervals between period doublings converge to a constant,

$$\delta = \lim_{n\to\infty} \frac{r_n - r_{n-1}}{r_{n+1} - r_n} = 4.6692\ldots$$

and then discovered that the *same* constant appears for every map with a single smooth hump, not just the logistic map. It has since been measured in convecting fluids, dripping taps, electronic circuits, and lasers. This **universality** is the same phenomenon as in critical phenomena (Chapter 9), explained by the same renormalization ideas: near the onset of chaos, the details of the system wash out.

§12.5 Hamiltonian chaos

Systems without friction cannot have attractors, because Liouville's theorem forbids phase-space volumes from shrinking (Chapter 4). They can still be chaotic. Poincaré discovered this in 1890 while studying the *three-body problem* — the motion of three masses under gravity — for a prize offered by the King of Sweden. He found that the problem could not be solved in closed form, because some trajectories tangle with each other in unimaginably complicated ways. His prize essay contained a mistake, and fixing it gave birth to chaos theory.

For nearly integrable systems, the **KAM theorem** (Kolmogorov, Arnold, Moser) says most regular orbits survive small perturbations, but chaotic regions grow between them as the perturbation increases. The solar system is a case in point: it is chaotic, with a Lyapunov time of about five million years for the inner planets. We cannot predict where the Earth will be in its orbit a hundred million years from now, though the orbit itself is very likely stable for billions of

years. Saturn's moon Hyperion tumbles chaotically, and the gaps in the asteroid belt are where chaos has cleared the orbits out.

§12.6 Fractals

A *fractal* is a shape with structure at every scale. Its *dimension* can be found by counting how many boxes of size ϵ it takes to cover it: if $N(\epsilon) \propto \epsilon^{-D}$, the dimension is D. A line has $D = 1$ and a filled square $D = 2$, but the *Cantor set* — remove the middle third of a segment, then the middle thirds of what remains, forever — has $D = \ln 2 / \ln 3 \approx 0.63$. Coastlines, clouds, river networks, lungs, and the boundaries of chaotic regions are approximately fractal, and Mandelbrot's question "How long is the coast of Britain?" has the answer that it depends on your ruler.

§12.7 Solitons and patterns

Nonlinearity does not only destroy order; it can create it. In 1834 the engineer John Scott Russell followed, on horseback, a single hump of water that a stopped barge had sent rolling along a canal; it travelled for miles without changing shape. Ordinarily, dispersion would spread such a pulse out (Chapter 5), and nonlinearity would steepen it into a breaking wave. In a *soliton* the two effects exactly balance. Solitons appear in optical fibres, where they carry pulses without distortion, in plasmas, and in quantum field theory.

Nonlinear systems driven away from equilibrium also form spontaneous *patterns*: the hexagonal convection cells in a pan of heated oil, the stripes of a zebra (which Turing proposed in 1952 arise from chemicals that react and diffuse at different rates), the spirals of certain chemical reactions, and the regular ripples of sand dunes.

§12.8 Synchronization

In 1665 Christiaan Huygens, ill in bed, noticed that two pendulum clocks hanging from the same beam always ended up swinging in exact opposition, however he started them. Weakly coupled oscillators with nearby frequencies tend to *lock* together. Thousands of fireflies in Southeast Asian trees flash in unison; pacemaker cells in the heart fire together; audiences clapping fall into rhythm; and the Moon's rotation is locked to its orbit. In the *Kuramoto model*, many oscillators with a spread of natural frequencies couple to their average, and above a critical coupling strength a synchronized group spontaneously forms — a phase transition in time.

§12.9 Problems

12.1 $O(\epsilon)$ ★ Find the fixed points of $\dot{x} = x - x^3$ and determine their stability. Sketch $x(t)$ for several initial conditions. Which of these behaviours could a linear equation produce?

12.2 $O(\epsilon)$ For $\dot{x} = r + x^2$, find the fixed points as a function of r, determine their stability, and draw the bifurcation diagram. For r slightly positive, show that x spends a long time near $x = 0$ (a "bottleneck"), and estimate how long, using $\int \mathrm{d}x/(r + x^2)$.

12.3 $O(\epsilon)$ Classify the fixed point at the origin of $\dot{x} = y$, $\dot{y} = -x - 0.2y$. What physical system is this?

12.4 $O(\epsilon^2)$ ★ For the logistic map, find the fixed points and show that $x^* = 1 - 1/r$ is stable for $1 < r < 3$. Then find the period-2 orbit for $r > 3$ by solving $F(F(x)) = x$ (factor out the fixed points), and show that it is stable for $3 < r < 1 + \sqrt{6} \approx 3.449$.

12.5 $O(\epsilon^2)$ The doubling map $x_{n+1} = 2x_n \bmod 1$ shifts the binary expansion of x one place left at each step. Show that its Lyapunov exponent is $\ln 2$, and interpret this: each iteration, one more binary digit of the initial condition becomes visible in the motion. Why does this make long-term prediction hopeless?

12.6 $O(\epsilon^2)$ Weather forecasts have a Lyapunov time of roughly 1.5 days. Initial measurements are accurate to about one part in 10^5 of the relevant scale. Estimate the predictability horizon. By how much would it improve if measurements became a thousand times better?

12.7 $O(\epsilon^2)$ Draw the phase portrait of the frictionless pendulum using energy conservation. Find the energy of the separatrix, and show that the period of a pendulum whose amplitude approaches π diverges logarithmically.

12.8 $O(\epsilon^2)$ For the Lorenz equations, show that the origin is a stable fixed point for $\rho < 1$. For $\rho > 1$, show that two more fixed points appear at $x = y = \pm\sqrt{\beta(\rho - 1)}$, $z = \rho - 1$. (These are the centres of the two "wings.")

12.9 $O(\epsilon^2)$ Find the fractal dimension of the Cantor set, and of the Koch snowflake curve (in which each segment is replaced by four segments one third as long). Show that the Koch snowflake has infinite perimeter but finite area.

12.10 $e^{-1/\epsilon}$ ★ Explain, using the Poincaré–Bendixson theorem, why a pendulum with friction cannot be chaotic, but a pendulum with friction that is also driven by a periodic torque can be. (What is the dimension of the driven pendulum's phase space?) How many dimensions does a discrete map need to be chaotic?

12.11 $e^{-1/\epsilon}$ (*Phase locking.*) Two oscillators with natural frequencies ω_1 and ω_2 are coupled: $\dot{\theta}_1 = \omega_1 + K\sin(\theta_2 - \theta_1)$, $\dot{\theta}_2 = \omega_2 + K\sin(\theta_1 - \theta_2)$. Write an equation for the phase difference $\phi = \theta_1 - \theta_2$ and show that the oscillators lock (with ϕ constant) if and only if $|\omega_1 - \omega_2| \le 2K$. What happens just outside this range?

12.12 $e^{-1/\epsilon}$ (*Feigenbaum's constant by hand.*) The first period-doubling points of the logistic map are $r_1 = 3$, $r_2 = 1 + \sqrt{6} \approx 3.4495$, $r_3 \approx 3.5441$, $r_4 \approx 3.5644$. Compute the ratios of successive intervals and compare with $\delta = 4.669$. Assuming the intervals shrink geometrically by exactly δ, predict the accumulation point r_∞ and compare with 3.5699. Then explain why the same δ appears in the dripping of a tap.

Further reading

- **Steven H. Strogatz,** ***Nonlinear Dynamics and Chaos*** — The best introduction by far: geometric, physical, applied, and fun. This chapter follows its outline.
- **James Gleick,** ***Chaos: Making a New Science*** — The popular history of how chaos was discovered.
- **Edward Ott,** ***Chaos in Dynamical Systems*** — A more advanced and complete treatment.
- **Cvitanović et al.,** ***Chaos: Classical and Quantum*** (**ChaosBook.org**) — A free online book that goes very deep, including periodic orbit theory and quantum chaos.

CHAPTER 13

Computational physics

BUILDS ON mathematical methods (Chapter 1), the harmonic oscillator (Chapter 3)

When the equations won't solve, make the computer play them out — and learn the surprising ways a computer can lie to you.

PROTOTYPE Euler's method applied to the harmonic oscillator. Step forward in time with $x_{n+1} = x_n + hv_n$ and $v_{n+1} = v_n - h\omega^2 x_n$. The obvious method, applied to the simplest system in physics, spirals *outward*: the energy grows every step, and a simulated planet escapes its star. Understanding why — and how to fix it — is the beginning of computational wisdom.

Most of the equations of physics cannot be solved with pen and paper. The three-body problem has no closed-form solution (Chapter 12); the Navier–Stokes equation is barely understood (Chapter 11); the Schrödinger equation for anything bigger than hydrogen is hopeless analytically (Chapter 21). Computation has become a third way of doing physics, alongside theory and experiment. It is used to simulate galaxies, design aircraft, predict the weather, fold proteins, and compute the mass of the proton from first principles. But a computer is a very fast idiot, and doing it well requires knowing exactly how numerical methods fail.

§13.1 Numbers in a computer

Computers store real numbers in *floating point*: a sign, about 16 significant decimal digits, and an exponent. The smallest number ϵ with $1 + \epsilon \neq 1$ is the *machine epsilon*, about 2.2×10^{-16} for standard double precision. Every arithmetic operation can introduce a relative error of this size.

Usually this is harmless. But subtracting two nearly equal numbers destroys significant digits: if a and b agree to 12 digits, $a - b$ has only 4 correct ones. This *catastrophic cancellation* is the most common way numerical calculations silently go wrong. Even $0.1 + 0.2$ is not exactly 0.3 in a computer, because 0.1 has no finite binary expansion.

Example 13.1 (A cancellation, and its cure)**.** Evaluate $(1 - \cos x)/x^2$ at $x = 10^{-8}$. The true answer is very close to $\frac{1}{2}$. But $\cos(10^{-8}) = 1 - 5\times10^{-17}$, which rounds to exactly 1 in double precision, so the computer returns 0. Rewriting with the identity $1 - \cos x = 2\sin^2(x/2)$ avoids the subtraction and gives 0.5 to full precision. Algebra that is trivial on paper can matter enormously in a computer.

§13.2 Derivatives, integrals, and roots

The derivative $f'(x)$ can be approximated by the *central difference*

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h},$$

with a *truncation error* of order h^2 (Taylor-expand to check). Smaller h reduces truncation error but increases round-off error, since $f(x+h)$ and $f(x-h)$ nearly cancel. The best h balances the two, around $h \sim \epsilon^{1/3} \approx 10^{-5}$, giving about ten correct digits rather than sixteen.

Integrals are approximated by sums. The trapezoid rule has error $O(h^2)$; Simpson's rule, fitting parabolas, has error $O(h^4)$; Gaussian quadrature chooses the sample points cleverly and is exact for polynomials of high degree.

To solve $f(x) = 0$, *Newton's method* iterates

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)},$$

following the tangent line to its root. Near a simple root it converges *quadratically*: the number of correct digits roughly doubles each step. Far from a root it can wander off, so it is often combined with the slow but safe *bisection* method.

§13.3 Ordinary differential equations

Newton's laws, circuits, chemical kinetics, and population dynamics are all ordinary differential equations $\dot{\mathbf{y}} = \mathbf{f}(\mathbf{y}, t)$. The simplest method, **Euler's method**, steps forward along the tangent:

$$\mathbf{y}_{n+1} = \mathbf{y}_n + h\,\mathbf{f}(\mathbf{y}_n, t_n).$$

Its error per step is $O(h^2)$, so its accumulated error over a fixed time is $O(h)$: a *first-order* method. Worse, it can be qualitatively wrong. For the oscillator, each Euler step multiplies the energy by exactly $1 + \omega^2 h^2$ (Problem 13.3). Over many periods the orbit spirals out. For a planet, the simulated orbit would steadily expand.

The workhorse for general problems is the **fourth-order Runge–Kutta method** (RK4), which samples $\mathbf{f}$ four times per step and combines the samples so that errors cancel up to fourth order. Its accumulated error is $O(h^4)$: halving the step improves the answer sixteenfold.

But for long simulations of conservative systems — planets, molecules, particle beams — even RK4 slowly drifts in energy. The fix is a **symplectic integrator**, one that respects the geometry of Hamiltonian mechanics and preserves phase-space volume exactly (Liouville's theorem, Chapter 4). The simplest is *velocity Verlet* (also called leapfrog):

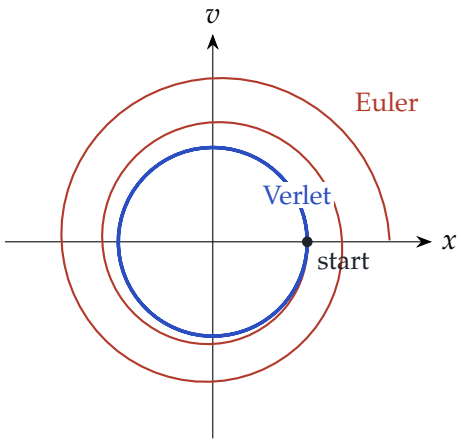
```
import numpy as np

def verlet(x, v, accel, h, steps):
    a = accel(x)
    for _ in range(steps):
        x = x + h*v + 0.5*h*h*a        # drift, using current acceleration
        a_new = accel(x)
```

```
        v = v + 0.5*h*(a + a_new)      # kick, using the average acceleration
        a = a_new
    return x, v

# The Earth around the Sun, in AU and years (so GM = 4 pi^2)
accel = lambda r: -4*np.pi**2 * r / np.linalg.norm(r)**3
x, v = verlet(np.array([1.0, 0.0]), np.array([0.0, 2*np.pi]), accel, 1e-3, 100000)
```

Verlet is only second order, but its energy error does not grow: it oscillates within a fixed band forever, because the method exactly conserves a slightly modified "shadow" Hamiltonian (Problem 13.11). For long-time behaviour, respecting the structure of the physics beats raw accuracy.



Two orbits of the harmonic oscillator in phase space, with the same step size $h = 0.1/\omega$. Euler's method spirals outward, gaining energy every step; velocity Verlet stays on its loop indefinitely.

KEY IDEA Build the physics into the algorithm. A method that respects a conservation law or a symmetry of the true dynamics will remain qualitatively right long after a more "accurate" method has drifted into nonsense.

§13.4 Matrices and eigenvalues

Many problems become linear algebra once space is discretized. Replace a continuous function by its values at N grid points, and derivatives become matrices. The Schrödinger equation for a particle in a box (Chapter 14), with $\hbar = m = 1$, becomes a matrix eigenvalue problem, which a library solves in one line:

```
import numpy as np
N = 1000                        # interior grid points
h = 1.0 / (N + 1)               # box of length 1
# -(1/2) psi'' as a tridiagonal matrix
H = (np.diag(np.full(N, 1.0)) - 0.5*np.diag(np.ones(N-1), 1)
     - 0.5*np.diag(np.ones(N-1), -1)) / h**2
E = np.linalg.eigvalsh(H)[:4]
print(E / (np.pi**2 / 2))       # close to 1, 4, 9, 16
```

Change the diagonal to include a potential $V(x_i)$ and you can find the energy levels of any one-dimensional potential. Dense linear algebra costs $O(N^3)$, so exploiting structure — sparsity, symmetry, bandedness — is the difference between a calculation that takes a second and one that takes a year.

§13.5 Partial differential equations

Fields evolving in space and time — heat, waves, fluids, quantum wavefunctions — obey partial differential equations. Discretize space with spacing Δx and time with step Δt. For the heat equation $\partial_t u = D\,\partial_x^2 u$, the simplest explicit scheme updates each point from its neighbours. It is *stable* only if

$$\frac{D\,\Delta t}{\Delta x^2} \le \frac{1}{2};$$

otherwise round-off errors grow exponentially and the solution explodes into a checkerboard of enormous numbers (Problem 13.8). For the wave equation, stability requires the *Courant–Friedrichs–Lewy* (CFL) condition $c\,\Delta t/\Delta x \le 1$: information cannot travel more than one grid cell per time step. Implicit methods, which solve a linear system at each step, are stable for any time step and are used when the explicit limit is too restrictive.

Another powerful approach expands the field in Fourier modes, where derivatives become multiplication. This is practical because of the **fast Fourier transform**, which computes a discrete Fourier transform of N points in $O(N \log N)$ operations instead of $O(N^2)$. It is one of the most important algorithms ever devised: it runs inside every phone, MRI scanner (Chapter 24), and spectral simulation.

§13.6 Monte Carlo methods

Some problems are best solved by gambling. To estimate π, throw random points into a unit square and count the fraction landing inside the quarter circle: it approaches $\pi/4$. The statistical error falls as $1/\sqrt{N}$ (Chapter 1), which is slow — a million points give only three digits. But the error is $1/\sqrt{N}$ *in any number of dimensions*. A grid-based integral in d dimensions needs exponentially many points as d grows, so for the very high-dimensional integrals of statistical mechanics and quantum field theory, Monte Carlo is the only option.

The **Metropolis algorithm** (1953) samples configurations with Boltzmann probabilities $e^{-E/k_{\mathrm{B}}T}$ (Chapter 9) without ever computing the partition function. Propose a small random change; if it lowers the energy, accept it; if it raises the energy by ΔE, accept it with probability $e^{-\Delta E/k_{\mathrm{B}}T}$. Repeated many times, this visits states with exactly the right probabilities. Here is the Ising model:

```
import numpy as np
rng = np.random.default_rng()
L, T = 32, 2.3                      # lattice size, temperature (J = k_B = 1)
s = rng.choice([-1, 1], size=(L, L))
for sweep in range(2000):
    for _ in range(L*L):
        i, j = rng.integers(L, size=2)
        nb = s[(i+1) % L, j] + s[i-1, j] + s[i, (j+1) % L] + s[i, j-1]
```

```
        dE = 2 * s[i, j] * nb      # energy cost of flipping spin (i, j)
        if dE <= 0 or rng.random() < np.exp(-dE / T):
            s[i, j] = -s[i, j]
print(abs(s.mean()))               # magnetization per spin
```

Run it at several temperatures, and the magnetization collapses near $T_c = 2.269$, the exact value found by Onsager in 1944 — noticeably below the mean-field estimate of 4. Monte Carlo methods compute the properties of materials, the behaviour of neutrons in reactors (where the method was born, at Los Alamos), the pricing of financial options, and, on the world's largest computers, the masses of the proton and neutron from the theory of quarks and gluons (*lattice QCD*, Chapter 32).

§13.7 Many bodies

Simulating N particles that all interact requires computing N^2 pairwise forces per step, which becomes impossible for large N. *Tree codes* group distant particles together and treat a cluster as a single mass, reducing the cost to $O(N \log N)$ — a spherical cow inside a computer. Simulations of the universe now follow 10^{12} particles from shortly after the Big Bang to today, and reproduce the cosmic web of galaxies and voids (Chapter 28).

Molecular dynamics applies the same idea to atoms, with forces from potentials like Lennard-Jones (Chapter 3) integrated by Verlet's method. Its curse is the time step: bonds vibrate every 10^{-14} s, so the time step must be a femtosecond, and simulating one millisecond of a protein's life takes 10^{12} steps.

§13.8 Data, fitting, and learning

Experiments produce data with noise, and computers turn data into knowledge. The basic tool is the *least-squares fit*: choose the parameters of a model to minimize

$$\chi^2 = \sum_i \frac{(y_i - f(x_i))^2}{\sigma_i^2},$$

the squared deviations weighted by the measurement errors. A good fit has χ^2 roughly equal to the number of data points; much larger means the model is wrong, much smaller means the errors were overestimated. The uncertainties of the fitted parameters come from how fast χ^2 rises as you move them away from the best fit.

Machine learning extends this to models with millions of parameters. Neural networks are flexible function approximators trained by minimizing a loss function, like χ^2 on steroids. In physics they classify particle collisions at the LHC (Chapter 32), emulate expensive simulations, find gravitational-wave signals (Chapter 29), and — in AlphaFold — predict protein structures (Chapter 23). They are powerful, but they interpolate better than they extrapolate, and they do not by themselves explain anything.

§13.9 How not to fool yourself

Every simulation should be checked. Test the code on a problem with a known answer. Halve the step size and confirm the answer converges at the expected rate. Monitor conserved quantities — energy, momentum, probability — as diagnostics. And always ask whether the result makes physical sense. The spherical cow is not only a way of thinking; it is the best way to test code.

§13.10 Problems

13.1 $O(\epsilon)$ In a programming language of your choice, check whether `0.1 + 0.2 == 0.3`. Then find the machine epsilon by starting with $\epsilon = 1$ and halving it until $1 + \epsilon$ equals 1. Explain both results.

13.2 $O(\epsilon)$ Evaluate $(1-\cos x)/x^2$ for $x = 10^{-1}, 10^{-3}, \dots, 10^{-9}$ directly and using $2\sin^2(x/2)/x^2$. At what x does the direct formula fail, and why there?

13.3 $O(\epsilon^2)$ ★ Show that one step of Euler's method applied to $\dot{x} = v$, $\dot{v} = -\omega^2 x$ multiplies $E = \tfrac{1}{2}v^2 + \tfrac{1}{2}\omega^2 x^2$ by exactly $1 + \omega^2 h^2$. If you take 100 steps per period, by what factor does the energy grow each period? How many steps per period would you need to keep the energy within 1% over 100 periods?

13.4 $O(\epsilon^2)$ ★ Show that the error of the central-difference derivative is about $h^2 |f'''|/6 + \epsilon |f|/h$, and that this is minimized for $h \sim \epsilon^{1/3}$. Verify numerically for $f = \sin$ at $x = 1$ by plotting the error against h on log-log axes. Explain the V shape.

13.5 $O(\epsilon^2)$ Apply Newton's method to $f(x) = x^2 - 2$ starting from $x_0 = 1$. Show that the iteration is $x_{n+1} = \tfrac{1}{2}(x_n + 2/x_n)$ (the Babylonian method) and that the number of correct digits roughly doubles each step.

13.6 $O(\epsilon^2)$ ★ Simulate the Earth's orbit for 100 years with Euler's method, RK4, and velocity Verlet, using the same step size. Plot the energy against time for each. Describe the long-term behaviour of each energy error.

13.7 $O(\epsilon^2)$ Use the finite-difference code in this chapter to compute the four lowest energy levels of a particle in a box with $N = 10, 100, 1000$ grid points. Show that the error falls as h^2. Then add a harmonic potential $V = \tfrac{1}{2}x^2$ (centred in a wider box) and check that the levels approach $n + \tfrac{1}{2}$.

13.8 $O(\epsilon^2)$ For the explicit scheme $u_j^{n+1} = u_j^n + r(u_{j+1}^n - 2u_j^n + u_{j-1}^n)$ with $r = D\Delta t/\Delta x^2$, substitute a Fourier mode $u_j^n = g^n e^{ikj\Delta x}$ and find the growth factor g. Show that $|g| \le 1$ for all k only if $r \le \tfrac{1}{2}$. Which mode goes unstable first?

13.9 $O(\epsilon^2)$ Estimate π by Monte Carlo with $N = 10^2, 10^4, 10^6$ points and confirm that the error falls as $1/\sqrt{N}$. Then explain why integrating a smooth function over a 10-dimensional

cube with 10 grid points per dimension would need 10^{10} evaluations, while Monte Carlo gets 1% accuracy with 10^4.

13.10 $O(\epsilon^2)$ ★ Run the Metropolis Ising code at temperatures from 1.5 to 3.5. Plot the magnetization and estimate T_c. Compare with Onsager's exact result, $2/\ln(1+\sqrt{2}) = 2.269$, and with mean-field theory. Why is the transition rounded in a finite lattice?

13.11 $e^{-1/\epsilon}$ ★ (*Why Verlet doesn't drift.*) For the oscillator $\ddot{x} = -\omega^2 x$ with unit mass, write one step of velocity Verlet as a linear map $(x_n, v_n) \mapsto (x_{n+1}, v_{n+1})$. Show that its matrix has determinant exactly 1 (it preserves phase-space area), and that it exactly conserves the "shadow energy"

$$\tilde{H} = \tfrac{1}{2}v^2 + \tfrac{1}{2}\omega^2 x^2\Big(1 - \frac{\omega^2 h^2}{4}\Big).$$

Conclude that the true energy error stays bounded for all time if $\omega h < 2$. What happens if $\omega h > 2$?

Hint: Check that $M^T S M = S$ for $S = \mathrm{diag}(\omega^2(1 - \omega^2 h^2/4),\, 1)$.

13.12 $e^{-1/\epsilon}$ (*Trusting chaos.*) Integrate the Lorenz equations twice from the same initial condition, once with step $h = 10^{-3}$ and once with $h = 10^{-3}(1 + 10^{-12})$. Show that the two trajectories diverge completely within a few dozen time units. In what sense, then, is *any* simulation of a chaotic system "correct"? Look up the *shadowing lemma*, and explain why statistical properties of the attractor can still be computed reliably.

Further reading

- **Mark Newman, *Computational Physics*** — An excellent, friendly introduction in Python, covering everything in this chapter. Much of it is free online.
- **Press, Teukolsky, Vetterling, and Flannery, *Numerical Recipes*** — The classic practical reference: what algorithms exist, when they fail, and why.
- **Werner Krauth, *Statistical Mechanics: Algorithms and Computations*** — Monte Carlo and molecular dynamics, beautifully explained.
- **Hairer, Lubich, and Wanner, *Geometric Numerical Integration*** — The advanced reference on symplectic and structure-preserving methods.

CHAPTER 14

Quantum mechanics

BUILDS ON the harmonic oscillator (Chapter 3), analytical mechanics (Chapter 4), waves (Chapter 5)

The rulebook for the very small, where certainty runs out, particles behave like waves, and the most successful theory in the history of science refuses to tell us what it means.

PROTOTYPE A particle in a box. An electron confined between two impenetrable walls cannot have just any energy: its wave has to fit, so its energies come in a discrete ladder $E_n \propto n^2$, starting not at zero but at a minimum "zero-point" energy. Its little sister is the *qubit*: a system with only two states, which can nevertheless be in any superposition of them.

Quantum mechanics is the theory of atoms, molecules, solids, nuclei, light, and everything made of them — which is to say everything. It is not an optional refinement of classical physics for small things; it is the actual physics, of which classical mechanics is an approximation valid when actions are much larger than Planck's constant. Its predictions have been confirmed to as many as twelve significant figures. Its interpretation is still argued about by serious people. Both facts are worth keeping in mind.

§14.1 The crisis

Around 1900, classical physics was failing in several places at once.

- **Blackbody radiation.** Classical statistical mechanics predicted infinite energy in the high-frequency modes of a hot cavity. Planck fixed it in 1900 by assuming that oscillators exchange energy only in lumps $E = h\nu$ (Chapter 9).
- **The photoelectric effect.** Light knocks electrons out of metals with energies set by its frequency, not its intensity. Einstein explained it in 1905 with photons of energy $h\nu$ (Chapter 8).
- **Atomic spectra.** Hot hydrogen emits light only at specific wavelengths, following a simple numerical pattern no one could explain.
- **Atomic stability.** An electron orbiting a nucleus should radiate its energy away and spiral in within 10^{-11} seconds (Problem 6.10). Atoms should not exist.

In 1924 de Broglie proposed the symmetric idea: if light waves behave like particles, perhaps particles behave like waves, with wavelength

$$\lambda = \frac{h}{p}.$$

For an electron with a few electron-volts of energy, λ is about a nanometre, comparable to the spacing of atoms in a crystal. In 1927 Davisson and Germer scattered electrons off nickel and saw diffraction peaks. Electrons fired one at a time through a double slit build up an interference pattern dot by dot, just as photons do. Matter is wavelike.

§14.2 The wavefunction

A quantum particle is described not by a position but by a *wavefunction* $\psi(x,t)$, a complex-valued function of space.

Law 14.1 (The Born rule)**.** The probability of finding the particle between x and $x+\mathrm{d}x$ is $|\psi(x,t)|^2\,\mathrm{d}x$. In particular, $\int |\psi|^2\,\mathrm{d}x = 1$.

Two features matter enormously. First, ψ is complex: it has a magnitude and a phase. Second, quantum mechanics is linear: if ψ_1 and ψ_2 are possible states, so is any *superposition* $c_1\psi_1 + c_2\psi_2$. In the double slit, $\psi = \psi_1 + \psi_2$ is the sum of the waves from each slit, and the probability is

$$|\psi_1+\psi_2|^2 = |\psi_1|^2 + |\psi_2|^2 + 2\,\mathrm{Re}(\psi_1^*\psi_2).$$

The last term is interference. *Amplitudes add, not probabilities.* This one rule is the source of almost everything strange about quantum mechanics. If you put a detector at the slits to find out which way the particle went, the interference term disappears and you get two blobs, as you would for bullets.

§14.3 The Schrödinger equation

The wavefunction evolves according to the **Schrödinger equation**:

$$i\hbar\,\frac{\partial\psi}{\partial t} = \hat{H}\psi, \qquad \hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}),$$

where $\hbar = h/2\pi = 1.055\times10^{-34}\,\mathrm{J\,s}$. The operator $\hat{H}$, the *Hamiltonian*, is the quantum version of the energy (Chapter 4), with momentum replaced by the operator $\hat{p} = -i\hbar\,\partial/\partial x$. For a free particle, a plane wave $e^{i(kx-\omega t)}$ solves it with $p=\hbar k$ and $E=\hbar\omega=\hbar^2k^2/2m$: de Broglie's relation and Einstein's, together.

Because the equation is first order in time, the wavefunction now determines the wavefunction forever. Quantum evolution is perfectly deterministic. The randomness enters only at measurement.

The most useful solutions are *stationary states*, $\psi = \phi(x)e^{-iEt/\hbar}$, whose probability density does not change in time. Substituting gives the **time-independent Schrödinger equation**

$$-\frac{\hbar^2}{2m}\phi'' + V\phi = E\phi.$$

It is an eigenvalue equation (Chapter 1): we look for the energies E for which a well-behaved solution exists. Any other state is a superposition of stationary states, each rotating at its own frequency $E/\hbar$.

§14.4 The particle in a box

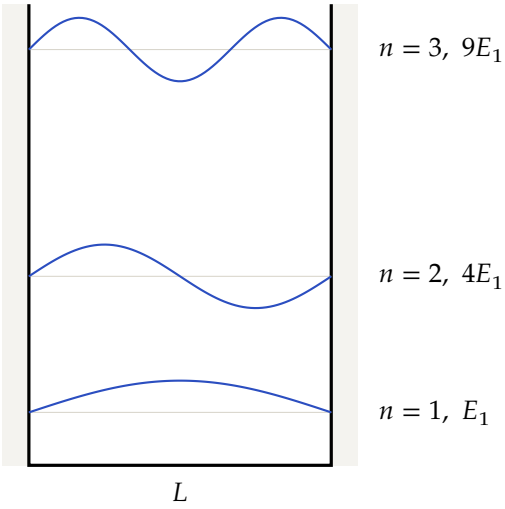
Put a particle in a one-dimensional box of length L with infinitely high walls, so $\phi = 0$ at $x = 0$ and $x = L$. Inside, $\phi'' = -k^2\phi$ with $k^2 = 2mE/\hbar^2$, so $\phi = A\sin kx$, and fitting the wave in requires $kL = n\pi$:

$$\phi_n(x) = \sqrt{\frac{2}{L}}\sin\frac{n\pi x}{L}, \qquad E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \qquad n = 1, 2, 3, \ldots$$

These are exactly the standing waves on a string (Chapter 5). Three lessons generalize:

- **Confinement gives quantization.** The energy is discrete because the wave has to fit.
- **There is a zero-point energy.** The lowest energy is not zero. A confined particle cannot sit still.
- **Smaller boxes mean bigger energies**, as $1/L^2$. An electron in an atom-sized box ($0.1\,\text{nm}$) has energies of tens of eV; a proton in a nucleus-sized box ($10^{-15}\,\text{m}$) has energies of tens of MeV. That factor of a million is the ratio between chemical and nuclear energy.

Quantum dots — semiconductor crystals a few nanometres across — are boxes for electrons, and their colour is set by their size: the same material glows red or blue depending on how big the dot is.



The three lowest states of a particle in a box, each drawn at the height of its energy. Each is a standing wave that fits a whole number of half-wavelengths between the walls, and the energies grow as n^2.

§14.5 The uncertainty principle

A wave packet localized within Δx must contain wavenumbers spread over $\Delta k \gtrsim 1/\Delta x$ (Chapter 1). With $p = \hbar k$:

Law 14.2 (The Heisenberg uncertainty principle)**.**

$$\Delta x\,\Delta p \geq \frac{\hbar}{2}.$$

This is not a statement about clumsy measurements. It says that a particle *does not have* a precise position and momentum at the same time, because a wave cannot be both localized and of definite wavelength.

The uncertainty principle is also the physicist's best estimation tool. It explains why atoms do not collapse. Confine an electron within radius r of a proton and its momentum must be at least $\sim \hbar/r$, giving kinetic energy $\sim \hbar^2/2mr^2$. The potential energy is $-e^2/4\pi\epsilon_0 r$. Squeezing lowers the potential energy but raises the kinetic energy, and the total

$$E(r) \approx \frac{\hbar^2}{2mr^2} - \frac{e^2}{4\pi\epsilon_0 r}$$

is minimized at the *Bohr radius* $a_0 = 4\pi\epsilon_0\hbar^2/me^2 = 0.053\,\text{nm}$, with $E = -13.6\,\text{eV}$. These are exactly the true size and binding energy of hydrogen. Matter is stable, and has the size it does, because of the uncertainty principle.

§14.6 Tunnelling

A classical ball without enough energy to climb a hill is reflected. A quantum particle can pass through. Inside a barrier where $V > E$, the Schrödinger equation gives exponentially decaying solutions $e^{-\kappa x}$ with $\kappa = \sqrt{2m(V-E)}/\hbar$ instead of oscillations, so the wavefunction leaks through a barrier of width a with transmission probability

$$T \approx e^{-2\kappa a}.$$

Tunnelling is everywhere. It lets alpha particles escape nuclei, with half-lives from microseconds to billions of years set by tiny differences in barrier thickness (Chapter 22). It lets protons in the Sun's core fuse despite having a thousand times too little energy to climb their electric repulsion (Problem 14.13). It is how a scanning tunnelling microscope images individual atoms, how flash memory stores bits, and how some enzymes speed up reactions.

KEY IDEA Tunnelling probability goes like $e^{-2\kappa a}$, and $\kappa \propto 1/\hbar$. As a function of $\hbar$, it is $e^{-c/\hbar}$: smaller than any power of $\hbar$. It vanishes at every order of an expansion around classical physics. That is why this book's hardest problems are marked $e^{-1/\epsilon}$: like tunnelling, they cannot be reached by perturbing the obvious.

§14.7 The quantum harmonic oscillator

For $V = \frac{1}{2}m\omega^2x^2$ (Chapter 3), there is an elegant algebraic solution. Define the *ladder operators*

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right), \qquad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right).$$

Using $[\hat{x},\hat{p}] = i\hbar$ (§14.8), one finds $[\hat{a},\hat{a}^\dagger] = 1$ and

$$\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \tfrac{1}{2}\right).$$

Then $\hat{a}^\dagger$ raises the energy by exactly $\hbar\omega$ and $\hat{a}$ lowers it. The ladder must stop at a ground state with $\hat{a}|0\rangle = 0$, which is a Gaussian, and the energies are

$$E_n = \left(n + \tfrac{1}{2}\right)\hbar\omega.$$

Evenly spaced rungs, with a zero-point energy $\frac{1}{2}\hbar\omega$. This is the most important exactly solvable problem in physics. Molecular vibrations are quantized this way (Chapter 18); so are lattice vibrations, whose quanta are *phonons* (Chapter 19); and in quantum field theory, $\hat{a}^\dagger$ becomes the operator that creates a particle (Chapter 30).

§14.8 The formalism

By the late 1920s, Dirac and von Neumann had distilled quantum mechanics into a short list of rules. They are worth stating cleanly.

Law 14.3 (The postulates of quantum mechanics)**.** (1) The state of a system is a unit vector $|\psi\rangle$ in a complex vector space with an inner product (a *Hilbert space*).

(2) Every observable is a Hermitian operator $\hat{A}$. The possible results of measuring it are its eigenvalues a.

(3) If the system is in state $|\psi\rangle$, the probability of getting result a is $|\langle a|\psi\rangle|^2$, where $|a\rangle$ is the corresponding eigenvector. Immediately after the measurement, the state is $|a\rangle$.

(4) Between measurements, the state evolves by the Schrödinger equation, $i\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\psi\rangle = \hat{H}|\psi\rangle$.

The wavefunction $\psi(x)$ is just the component of $|\psi\rangle$ along the position eigenstate $|x\rangle$. Hermitian operators have real eigenvalues and orthogonal eigenvectors (Chapter 1), which is why measurement results are real numbers and why distinct outcomes are mutually exclusive. The average of many measurements is the *expectation value* $\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle$.

The deepest new feature is that operators need not commute. Position and momentum satisfy

$$[\hat{x},\hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar,$$

exactly the classical Poisson bracket $\{x,p\} = 1$ multiplied by $i\hbar$ (Chapter 4). Two observables can have definite values simultaneously only if they commute, and in general

$$\Delta A\,\Delta B \geq \tfrac{1}{2}\left|\langle[\hat{A},\hat{B}]\rangle\right|,$$

which gives the Heisenberg principle as a special case.

How does classical mechanics emerge? **Ehrenfest's theorem** says that expectation values obey $\frac{\mathrm{d}}{\mathrm{d}t}\langle\hat{x}\rangle = \langle\hat{p}\rangle/m$ and $\frac{\mathrm{d}}{\mathrm{d}t}\langle\hat{p}\rangle = -\langle V'(\hat{x})\rangle$: Newton's laws on average. When the wave packet is narrow compared to the scale on which V varies, the averages follow classical trajectories.

§14.9 Spin and the two-level system

In 1922 Stern and Gerlach sent a beam of silver atoms through a nonuniform magnetic field. Classically, randomly oriented magnetic moments should spread the beam into a smear. Instead it split cleanly into *two*. Electrons carry an intrinsic angular momentum, *spin*, whose component along any axis can only be $+\hbar/2$ or $-\hbar/2$.

Spin-$\frac{1}{2}$ is the simplest quantum system, and the cleanest place to see quantum mechanics work. Its states live in a two-dimensional space with basis $|\uparrow\rangle$, $|\downarrow\rangle$ (spin up or down along z). The general state is

$$|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

The spin components are $\hat{S}_i = \frac{\hbar}{2}\sigma_i$, with the *Pauli matrices*

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The state "spin up along x" is $(|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$. Measure its z-component and you get up or down with probability $\frac{1}{2}$ each. Send the up-along-z survivors through an x-measurement, then a z-measurement again, and half of them come out *down*: the x-measurement erased the z-information. No classical arrow behaves like this.

In a magnetic field, a spin precesses around the field at the *Larmor frequency*, proportional to the field. For protons in a 3 T magnet it is 128 MHz, which is the radio frequency that magnetic resonance imaging listens to (Chapter 24). A two-level system of any kind — spin, an atom with two relevant levels, a superconducting circuit — is a *qubit*, the unit of quantum information (Chapter 20).

§14.10 Angular momentum and the hydrogen atom

Orbital angular momentum $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ has components that do not commute: $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$. So we can know $\hat{L}^2$ and one component, say $\hat{L}_z$, but not all three. The algebra alone forces the eigenvalues:

$$L^2 = \ell(\ell+1)\hbar^2, \quad \ell = 0, 1, 2, \ldots; \qquad L_z = m\hbar, \quad m = -\ell, \ldots, \ell.$$

(Spin is the same algebra allowing half-integer ℓ, which orbital motion cannot.)

For the **hydrogen atom**, $V = -e^2/4\pi\epsilon_0 r$. Because the potential is spherically symmetric, the angular part of the wavefunction is fixed by angular momentum, and the radial equation can be solved exactly. The energies are

$$E_n = -\frac{13.6\,\text{eV}}{n^2}, \qquad n = 1, 2, 3, \ldots,$$

with $\ell = 0, \ldots, n-1$ and $m = -\ell, \ldots, \ell$. Each level has n^2 states (times two for spin). States with $\ell = 0, 1, 2, 3$ are called s, p, d, f *orbitals*, and their shapes are the familiar spheres, dumbbells, and cloverleaves of chemistry. When an electron drops from level n to n', it emits a photon with energy $E_n - E_{n'}$. The visible *Balmer* lines (down to $n' = 2$) are the red, blue-green, and violet lines that had puzzled spectroscopists for decades. Refinements — fine structure, the Lamb shift, hyperfine splitting — are the subject of Chapter 18.

§14.11 Identical particles

All electrons are identical, not just similar. Swapping two of them cannot change any measurable quantity, so the wavefunction can at most change sign. Nature uses both options:

- **Bosons** (integer spin: photons, gluons, helium-4 atoms) have symmetric wavefunctions, $\psi(x_1, x_2) = +\psi(x_2, x_1)$.
- **Fermions** (half-integer spin: electrons, protons, neutrons, quarks) have antisymmetric ones, $\psi(x_1, x_2) = -\psi(x_2, x_1)$.

The connection between spin and symmetry is the *spin-statistics theorem*, which follows from relativistic quantum field theory (Chapter 30). For fermions, if both particles are in the same state, the antisymmetric wavefunction vanishes:

Law 14.4 (The Pauli exclusion principle)**.** No two identical fermions can occupy the same quantum state.

This one principle is responsible for the periodic table. Electrons fill an atom's orbitals from the bottom up, two per orbital (one of each spin), and the chemistry of an element is set by its outermost, partly filled shell. It makes matter take up space: you cannot squeeze two atoms into each other because their electrons would have to share states. It holds up white dwarfs and neutron stars (Chapter 27). And bosons' opposite preference, to bunch into the same state, gives lasers and Bose–Einstein condensates (Chapter 9).

§14.12 Approximation methods

Almost nothing beyond the box, the oscillator, and hydrogen can be solved exactly. Physicists live on approximations — the spherical cows of quantum mechanics.

Perturbation theory. If $\hat{H} = \hat{H}_0 + \lambda \hat{H}'$, with $\hat{H}_0$ solvable and λ small, the energies shift by

$$E_n \approx E_n^{(0)} + \lambda \langle n|\hat{H}'|n\rangle + \lambda^2 \sum_{m \neq n} \frac{\left|\langle m|\hat{H}'|n\rangle\right|^2}{E_n^{(0)} - E_m^{(0)}} + \cdots$$

The first-order shift is just the average of the perturbation in the unperturbed state. The second-order shift always lowers the ground state: perturbations push levels apart.

The variational principle. For *any* normalized trial state, $\langle \psi|\hat{H}|\psi\rangle \geq E_0$. Guess a state with some adjustable parameters, minimize its energy, and you get an upper bound on the ground state energy — often an excellent one. This is how quantum chemistry computes molecules (Chapter 21).

Fermi's golden rule. A perturbation can cause transitions between states. The rate from state i to a continuum of final states f is

$$\Gamma = \frac{2\pi}{\hbar} \left|\langle f|\hat{H}'|i\rangle\right|^2 \rho(E_f),$$

where ρ is the density of final states. It governs how atoms absorb and emit light, how nuclei decay, and how particles scatter.

§14.13 Entanglement

Two spin-$\frac{1}{2}$ particles can be in the state

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle).$$

Neither particle has a definite spin, yet their spins are perfectly anticorrelated along *every* axis. The state cannot be written as (state of particle 1) times (state of particle 2): the particles are *entangled*.

In 1935 Einstein, Podolsky, and Rosen argued that this proved quantum mechanics incomplete: measuring particle 1 seems to instantly fix particle 2, however far away, so particle 2 must have had a definite spin all along, carried by some "hidden variable." In 1964 John Bell proved that *any* theory with such local hidden variables obeys an inequality that quantum mechanics violates (Chapter 20). Experiments, culminating in loophole-free tests in 2015 and the 2022 Nobel Prize, side with quantum mechanics. Nature is not locally realistic. Yet entanglement cannot be used to send signals faster than light: the correlations only become visible when the two sides compare notes by ordinary means.

§14.14 What does it mean?

Quantum mechanics has two rules for how states change: smooth, deterministic Schrödinger evolution, and abrupt, random collapse at measurement. But measuring devices are made of atoms, which obey the Schrödinger equation. When, exactly, does collapse happen? This is the *measurement problem*, and it has several proposed resolutions:

- **Copenhagen**: the wavefunction is a tool for predicting measurement outcomes, not a description of reality; asking what happens between measurements is not meaningful.
- **Many worlds**: there is no collapse. The wavefunction of the universe evolves smoothly, and every outcome happens in a different branch.
- **Pilot waves** (de Broglie–Bohm): particles have definite positions at all times, guided by the wavefunction, at the cost of explicit nonlocality.
- **Objective collapse**: the Schrödinger equation is slightly wrong, and wavefunctions spontaneously collapse at a rate that grows with size.

What all agree on is *decoherence*: when a quantum system interacts with a large environment, the interference terms between macroscopically different states become unobservable extremely fast. This explains why we never see a superposed cat, though not, by itself, why we see one outcome rather than another. Most physicists, most of the time, follow the advice attributed to David Mermin: shut up and calculate. The calculations work.

§14.15 Problems

14.1 $O(\epsilon)$ ★ Compute the de Broglie wavelength of an electron with kinetic energy 100 eV and of a 0.15 kg baseball at 40 m/s. Why can electron beams be diffracted by crystals, while baseballs are never diffracted by doorways?

14.2 $O(\epsilon)$ Normalize $\psi(x) = A e^{-|x|/a}$, and find the probability that the particle is within a of the origin. Compute $\langle x \rangle$ and $\langle x^2 \rangle$.

14.3 $O(\epsilon)$ Find the energies of the three lowest levels of an electron in a one-dimensional box of width 1 nm, and the wavelength of a photon emitted in the $2 \to 1$ transition. Is it visible?

14.4 $O(\epsilon)$ Sodium has a work function of 2.3 eV. What is the maximum kinetic energy of electrons ejected by 400 nm light? By 600 nm light?

14.5 $O(\epsilon^2)$ ★ Using the uncertainty principle estimate $E(r) \approx \hbar^2/2mr^2 - e^2/4\pi\epsilon_0 r$, find the size and energy of the hydrogen ground state. Repeat for a "muonic hydrogen" atom in which the electron is replaced by a muon, 207 times heavier. How much smaller is it?

14.6 $O(\epsilon^2)$ An electron with energy 1 eV meets a rectangular barrier of height 5 eV and width 0.5 nm. Estimate the tunnelling probability. By what factor does it change if the width is doubled? If the electron is replaced by a proton?

14.7 $O(\epsilon^2)$ ★ Using $[\hat{a}, \hat{a}^\dagger] = 1$, show that $[\hat{H}, \hat{a}^\dagger] = \hbar\omega\, \hat{a}^\dagger$. Deduce that if $|n\rangle$ has energy E, then $\hat{a}^\dagger|n\rangle$ has energy $E + \hbar\omega$. Then solve $\hat{a}|0\rangle = 0$ as a differential equation in position space to find the ground-state wavefunction.

14.8 $O(\epsilon^2)$ Find the eigenvalues and eigenvectors of σ_x, σ_y, and σ_z. A spin is prepared up along x. Find the probabilities of each outcome when measuring along z, and along an axis at angle θ to x in the x–z plane. Describe the outcome statistics of the sequence $z \to x \to z$.

14.9 $O(\epsilon^2)$ ★ Verify $[\hat{x}, \hat{p}] = i\hbar$ by acting on an arbitrary test function $f(x)$. Then use $[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$ to show $[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z$.

14.10 $O(\epsilon^2)$ An anharmonic oscillator has $\hat{H} = \hat{H}_{\mathrm{SHO}} + \lambda \hat{x}^4$. Using $\langle x^4 \rangle = 3(\hbar/2m\omega)^2$ in the ground state, find the first-order shift of the ground state energy. When is first-order perturbation theory trustworthy here?

14.11 $O(\epsilon^2)$ Use a Gaussian trial function $\psi \propto e^{-r^2/2b^2}$ for the hydrogen atom to show that the variational energy is minimized at $E = -\frac{8}{3\pi} \times 13.6\,\mathrm{eV} \approx -11.5\,\mathrm{eV}$. Why is this above the true ground state energy, and why is the Gaussian a worse guess than $e^{-r/a}$?

14.12 $e^{-1/\epsilon}$ ★ A particle moves in an attractive delta-function potential $V(x) = -\alpha\,\delta(x)$. Show that there is exactly one bound state, with $\psi \propto e^{-\kappa|x|}$ and energy $E = -m\alpha^2/2\hbar^2$.

Hint: Away from $x = 0$ the particle is free. Integrate the Schrödinger equation across $x = 0$ to find the jump in ψ'.

14.13 $e^{-1/\epsilon}$ ★ (*Why the Sun shines.*) Two protons must approach to about 1 fm to fuse. Show that the Coulomb barrier there is about 1 MeV, while the thermal energy in the Sun's core ($T \approx 1.5\times10^7$ K) is about 1 keV. Classically, fusion is impossible. Quantum mechanically, the tunnelling probability at energy E is about $e^{-\sqrt{E_G/E}}$, where the *Gamow energy* is $E_G = 2\mu c^2(\pi\alpha)^2 \approx 490\,\text{keV}$ for two protons ($\mu = m_p/2$, $\alpha \approx 1/137$). Evaluate this at $E = 1\,\text{keV}$ and $E = 5\,\text{keV}$. Combining it with the Boltzmann tail, explain why fusion happens mainly at energies of several $k_\text{B}T$, and why the Sun burns so slowly.

14.14 $e^{-1/\epsilon}$ (*The bomb tester.*) A Mach–Zehnder interferometer splits a single photon into two paths and recombines them so that, by interference, it always exits through detector A, never B. A bomb that explodes if a photon touches it may or may not be placed in one path. Show that if a bomb is present, there is a 25% chance that detector B clicks — certifying that a live bomb is in the path, even though the photon never touched it. (Elitzur and Vaidman, 1993.) By using the quantum Zeno effect, the success rate can be pushed arbitrarily close to 100%.

Further reading

- **Griffiths and Schroeter, *Introduction to Quantum Mechanics* (3rd ed.)** — The most accessible introduction, built on elementary exercises plus a rich variety of harder problems.
- **John S. Townsend, *A Modern Approach to Quantum Mechanics*** — Starts with spin and Stern–Gerlach rather than wavefunctions, which many find clearer.
- ***The Feynman Lectures on Physics*, Volume III** — Quantum mechanics from amplitudes and two-state systems, with unmatched physical insight.
- **R. Shankar, *Principles of Quantum Mechanics*** — A thorough, more mathematical treatment, with an excellent review of linear algebra.
- **Robert Littlejohn's lecture notes (Berkeley Physics 221)** — The clearest graduate-level treatment, freely available online.

CHAPTER 15

Continuum mechanics and acoustics

BUILDS ON waves (Chapter 5), fluid dynamics (Chapter 11)

Elastic solids, why bridges stand and glass shatters, why giants are impossible, and the physics of every sound you have ever heard.

PROTOTYPE A vibrating drumhead: a stretched elastic membrane, fixed at its rim. Strike it and it rings in a set of normal modes, like a string — but unlike a string, its overtones are not whole-number multiples of the fundamental, which is why most drums do not have a clear pitch.

Fluid dynamics (Chapter 11) treats matter that flows. This chapter is about matter that resists: solids that stretch, bend, buckle, and break, and the waves that travel through them. It is an old subject — Hooke, Euler, and Cauchy laid most of it down — and an unglamorous one, but every building, bone, bridge, and musical instrument depends on it. And it is full of beautiful spherical cows.

§15.1 Stress and strain

Pull on a rod and it stretches. The natural measures are intensive. The *stress* $\sigma = F/A$ is force per area, measured in pascals; the *strain* $\varepsilon = \Delta L/L$ is fractional change in length, dimensionless. For small strains, most solids obey Hooke's law:

$$\sigma = E\,\varepsilon,$$

with *Young's modulus* E. Steel has $E \approx 200\,\text{GPa}$, bone 20 GPa, wood 10 GPa, rubber about 0.01 GPa.

A stretched rod also gets thinner. The ratio of sideways contraction to lengthwise stretch is *Poisson's ratio* ν, about 0.3 for most metals and 0.5 for rubber (which keeps its volume as it deforms). Two more moduli describe other deformations: the *shear modulus* G, for sliding layers past each other, and the *bulk modulus* K, for uniform compression. For an isotropic material only two of E, ν, G, K are independent; for example $G = E/2(1+\nu)$.

In three dimensions, stress is not a vector but a *tensor*. At any point, the force per area on a small surface depends on which way the surface faces, and can have components both normal to it (pressure or tension) and along it (shear). It takes a 3×3 symmetric matrix σ_{ij} to describe it. Its eigenvectors are the *principal stress directions*, along which there is no shear (Chapter 1). Strain is a tensor too, and Hooke's law becomes a linear relation between the two.

§15.2 Why moduli are what they are

Here is a spherical cow worth having. A solid is atoms joined by bonds of energy $U_0 \sim 1\,\mathrm{eV}$ and length $a \sim 0.2\,\mathrm{nm}$. Stretching a bond by δ costs energy $\sim U_0(\delta/a)^2$ (it is a spring near its minimum; Chapter 3). Per unit volume, the elastic energy is $\tfrac{1}{2}E\varepsilon^2$ with $\varepsilon = \delta/a$, and there is one bond per volume a^3. Matching,

$$E \sim \frac{U_0}{a^3} \approx \frac{1.6\times10^{-19}\,\mathrm{J}}{(2\times10^{-10}\,\mathrm{m})^3} \approx 20\,\mathrm{GPa}.$$

That is the right order of magnitude for ordinary solids, from nothing but the size and strength of a chemical bond. Rubber, with E ten thousand times smaller, must be stiff for a completely different reason — and it is: rubber's elasticity is entropic, from the tangled polymer chains resisting being straightened (Chapter 23).

§15.3 Bending, buckling, and scaling

A beam bends with a stiffness proportional to EI, where $I = \int y^2\,\mathrm{d}A$ is the *second moment of area* of its cross-section. A rectangular beam of width b and depth h has $I = bh^3/12$: doubling the depth makes it eight times stiffer. This is why floor joists are placed on edge, why I-beams put most of their material far from the centre, and why bones and bamboo are hollow tubes. Material near the middle of a beam is barely stressed and mostly adds weight.

A thin column under compression does not crush; it *buckles* sideways, suddenly, at the **Euler load**

$$F_c = \frac{\pi^2 EI}{L^2}.$$

Long columns are disproportionately weak. Buckling is a pitchfork bifurcation (Chapter 12): below F_c the straight column is stable; above it, it must choose to bend one way or the other.

Galileo noticed the most important scaling law in structural engineering. Scale an animal up by a factor s in every direction. Its weight grows as s^3, but the cross-section of its bones, which sets their strength, grows only as s^2. The stress in its bones grows as s. A giant ten times human size would break its legs standing up. This *square–cube law* is why elephants have proportionally thick legs, why the largest animals live in water, why insects can lift many times their weight, and why King Kong is a physical impossibility.

§15.4 Strength and fracture

How strong *should* a material be? To pull a solid apart you must stretch its bonds by roughly their own length, a strain of order one, so the theoretical strength is a sizeable fraction of E, roughly $E/10$. Real materials are typically a hundred to a thousand times weaker. Why?

In 1920, A. A. Griffith found the answer: cracks. A crack of length a concentrates stress at its tip. It grows if the elastic energy released by extending it exceeds the surface energy γ needed to make new surface. Balancing the two gives a fracture stress

$$\sigma_f \approx \sqrt{\frac{2E\gamma}{\pi a}}.$$

Larger flaws mean weaker materials. Freshly drawn glass fibres, free of surface scratches, approach the theoretical strength; a scratched window pane is far weaker, which is why glaziers score glass before snapping it. Metals avoid brittle fracture because they can deform plastically: planes of atoms slip past each other by moving *dislocations*, line defects in the crystal (Chapter 19), blunting crack tips and absorbing energy. That is the difference between a material that is *strong* (resists stress) and one that is *tough* (resists cracks).

§15.5 Elastic waves

A solid carries two kinds of waves. In **longitudinal** or *P waves* (primary), the material compresses and expands along the direction of travel; in **transverse** or *S waves* (secondary), it shears sideways. Their speeds are

$$v_P = \sqrt{\frac{K + \frac{4}{3}G}{\rho}}, \qquad v_S = \sqrt{\frac{G}{\rho}},$$

always following the pattern $\sqrt{\text{stiffness}/\text{inertia}}$ (Chapter 5). P waves are faster, which is why they arrive first after an earthquake; the delay before the S waves tells a seismometer how far away the quake was. And since fluids cannot resist shear ($G = 0$), *S waves cannot travel through liquids*. The shadow that the Earth's core casts for S waves is how we know the outer core is molten iron (Chapter 25). Along a surface there are also *Rayleigh waves*, rolling like ocean swell, which do much of an earthquake's damage.

§15.6 Sound

In a fluid only the bulk modulus resists compression, so sound travels at $v = \sqrt{K/\rho}$. For a gas, Newton assumed the compressions were isothermal, $K = P$, and got 280 m/s for air — fifteen percent too slow. Laplace realized that sound oscillates too fast for heat to flow, so the compressions are adiabatic (Chapter 7), $K = \gamma P$:

$$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma k_{\mathrm{B}} T}{m}} \approx 343\,\mathrm{m/s}$$

at room temperature. The speed depends on temperature but not on pressure, and it is comparable to the speed of the molecules themselves. Sound travels at about 1500 m/s in water and 5000 m/s in steel. In helium it is nearly three times faster than in air, which raises the resonances of your vocal tract and gives the famous squeaky voice.

The ear responds to an astonishing range of intensities, from about $10^{-12}\,\mathrm{W/m^2}$ (the threshold of hearing, a pressure wave that moves the eardrum less than an atom's width) to about $1\,\mathrm{W/m^2}$ (the threshold of pain). So intensity is measured logarithmically, in *decibels*:

$$\beta = 10 \log_{10} \frac{I}{10^{-12}\,\mathrm{W/m^2}}\ \mathrm{dB}.$$

Every 10 dB is a factor of ten in intensity. A whisper is about 30 dB, conversation 60 dB, a rock concert 110 dB. Two identical sources together are only 3 dB louder than one. The ear perceives a 10 dB increase as roughly "twice as loud" — one of many examples of our senses being approximately logarithmic.

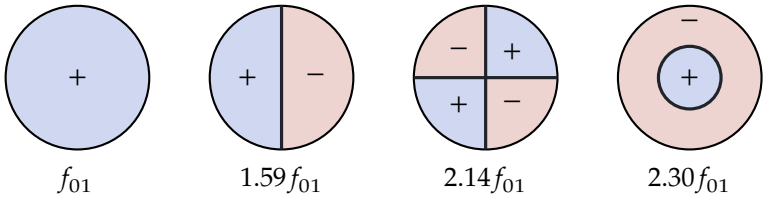
§15.7 Resonators and music

Musical instruments are resonators. A string or an open pipe has harmonics at whole-number multiples of its fundamental (Chapter 5), which is why strings and flutes sound sweetly pitched. A pipe closed at one end has only odd harmonics. Blowing across the mouth of a bottle excites a different kind of resonator, the *Helmholtz resonator*: the plug of air in the neck acts as a mass, bouncing on the springy air in the body. Its frequency is

$$f = \frac{v}{2\pi}\sqrt{\frac{A}{V\ell}},$$

for neck area A, neck length ℓ, and body volume V. The body of a guitar, the hollow of a violin, and the bass reflex port of a loudspeaker are Helmholtz resonators.

A **drumhead** is a two-dimensional string. A circular membrane of radius R, tension per length T, and mass per area σ has wave speed $c = \sqrt{T/\sigma}$, and its modes are described by Bessel functions, the circular cousins of sines. The fundamental is $f_{01} = 2.405\,c/2\pi R$, and the next modes are at 1.59, 2.14, 2.30, 2.65, ... times the fundamental — not whole numbers. The overtones clash, and the sound has no clear pitch. The patterns of nodes can be seen by sprinkling sand on a vibrating plate, which collects along the nodal lines in *Chladni figures*.



The four lowest modes of a circular drumhead, seen from above. Shaded regions move in opposite directions (+ and −), and the black nodal lines stay still. The frequencies are not whole-number multiples of the fundamental.

In 1966 Mark Kac asked: can one hear the shape of a drum? That is, do the frequencies determine the shape? You can hear its area and its perimeter (Problem 15.11). But in 1992 Gordon, Webb, and Wolpert constructed two differently shaped drums with exactly the same frequencies. You cannot, in general, hear the shape of a drum.

§15.8 Between solid and liquid

Many materials are neither. Silly Putty bounces like a solid but flows into a puddle overnight: it is *viscoelastic*, with a relaxation time that separates elastic from viscous behaviour. Glaciers and the Earth's mantle are solids on human timescales and fluids on geological ones, which is why continents can drift (Chapter 25). Cornstarch in water ("oobleck") thickens when struck and flows when stroked; ketchup does the opposite. These *non-Newtonian fluids*, whose viscosity depends on how fast they are sheared, include blood, paint, toothpaste, and most of the soft matter of Chapter 19.

§15.9 Problems

15.1 $O(\epsilon)$ ★ A steel wire 2 m long and 1 mm in diameter supports a 100 kg load. What is the stress? Would mild steel, with a yield strength of about 250 MPa, survive? Piano wire, with a yield strength near 2 GPa, would; how much does it stretch?

15.2 $O(\epsilon)$ Compute the speed of sound in air at 20°C both isothermally and adiabatically. Then compute it in helium and explain why a helium-filled voice sounds higher, even though the vocal cords vibrate at the same rate.

15.3 $O(\epsilon)$ An orchestra of ten identical violins plays a note at 70 dB. How loud is a single violin? How many violins would you need to reach 90 dB?

15.4 $O(\epsilon)$ Find the first three resonant frequencies of a 1 m organ pipe open at both ends, and of one closed at one end.

15.5 $O(\epsilon^2)$ ★ Blowing across a wine bottle with a 0.75 L body and a neck 8 cm long and 3 cm^2 in area makes a hum. Estimate its pitch. How does the pitch change as you drink the wine?

15.6 $O(\epsilon^2)$ ★ Using the bond-stretching argument in this chapter, estimate Young's modulus for a solid with bonds of energy 3 eV and length 0.15 nm. Compare with diamond, $E \approx 1000\,\mathrm{GPa}$.

15.7 $O(\epsilon^2)$ A tree trunk is a vertical column loaded by its own weight. Greenhill showed in 1881 that a uniform column buckles under its own weight if it is taller than about $0.8\,(E/\rho g)^{1/3} D^{2/3}$, where D is its diameter. Estimate the maximum height of a tree with a 1 m-diameter trunk ($E \approx 10\,\mathrm{GPa}$, $\rho \approx 700\,\mathrm{kg/m^3}$). Real trees obey this $h \propto D^{2/3}$ scaling remarkably well, with a safety factor of about four.

15.8 $O(\epsilon^2)$ Scale a human up tenfold in every dimension. By what factor do the weight, the bone cross-section, and the stress in the bones change? By what factor would the bones need to be thickened to keep the stress the same, and what would the giant look like?

15.9 $O(\epsilon^2)$ After an earthquake a seismometer records the S waves arriving 8 s after the P waves. Taking $v_P = 6\,\mathrm{km/s}$ and $v_S = 3.5\,\mathrm{km/s}$, how far away was the quake? How many stations do you need to locate it?

15.10 $e^{-1/\epsilon}$ ★ (*Griffith's criterion.*) A thin plate under tensile stress σ contains a crack of length $2a$. The crack relieves stress in a region of area roughly πa^2 per unit thickness, releasing elastic energy about $\pi a^2 \sigma^2 / E$, while costing surface energy $4a\gamma$ (two faces, each of length $2a$). Find the critical crack length beyond which the crack grows spontaneously, and hence the fracture stress. Estimate the strength of glass ($E = 70\,\mathrm{GPa}$, $\gamma = 1\,\mathrm{J/m^2}$) with 1 μm surface flaws.

15.11 $e^{-1/\epsilon}$ (*Hearing the area of a drum.*) For a membrane of any shape with area A and wave speed c, the number of modes with frequency below f grows, for large f, as $N(f) \approx \pi A f^2 / c^2$ (Weyl's law). Derive this by counting the modes of a square membrane, whose modes have $f = \frac{c}{2L}\sqrt{m^2 + n^2}$, as lattice points in a quarter circle. Conclude that a listener with perfect pitch could, in principle, hear the area of any drum.

15.12 $e^{-1/\epsilon}$ The overtones of a circular drumhead are inharmonic, yet orchestral timpani have a clear pitch. The reason is that the air enclosed in the kettle and the surrounding air load the membrane, lowering the frequencies of different modes by different amounts. Look up the frequencies of the $(1,1)$, $(2,1)$, $(3,1)$, and $(4,1)$ modes of a real timpani, and show that they are close to the ratios $2:3:4:5$ — a nearly harmonic series, which the ear fuses into a single definite pitch. Why are the symmetric $(0,n)$ modes unimportant for the pitch?

Further reading

- **Benny Lautrup,** ***Physics of Continuous Matter*** — Elasticity and fluids together, with the physicist's eye for estimates and exotic examples.
- **J. E. Gordon,** ***Structures: Or Why Things Don't Fall Down*** **and** ***The New Science of Strong Materials*** — Two wonderful popular books by an engineer, full of insight about strength, cracks, and design.
- **Landau and Lifshitz,** ***Theory of Elasticity*** — Short and rigorous; for when you want the tensors done properly.
- **Fletcher and Rossing,** ***The Physics of Musical Instruments*** — The definitive treatment of strings, pipes, plates, and drums.

CHAPTER 16

Complex systems and networks

BUILDS ON statistical mechanics (Chapter 9), nonlinear dynamics (Chapter 12)

How structure and surprise emerge from many interacting parts — in epidemics, earthquakes, flocks, markets, and the internet — and how far the physicist's toolkit reaches beyond physics.

PROTOTYPE A power law, $P(x) \propto x^{-\alpha}$. The number of earthquakes releasing energy E, the sizes of cities, the frequencies of words, the numbers of links to web pages: all follow distributions with no typical scale, where enormous events are rare but nowhere near as rare as a bell curve would suggest. A power law is the fingerprint of a system where the parts talk to each other.

In 1972 the condensed matter physicist Philip Anderson wrote a short essay titled "More Is Different." His point was that knowing the laws governing the parts does not tell you the behaviour of the whole: at each level of complexity, new properties *emerge* that need new concepts. Water molecules do not know about waves; neurons do not know about thoughts. A *complex system* is one in which many parts interact to produce collective behaviour that no single part controls or contains. Statistical mechanics (Chapter 9) was the first science of such systems. Its ideas — phase transitions, critical points, universality, fluctuations — turn out to apply far beyond gases and magnets.

§16.1 Power laws and scale invariance

A Gaussian distribution has a characteristic scale: almost all values lie within a few standard deviations of the mean. Human heights are Gaussian; you will never meet someone ten metres tall. A *power-law* distribution $P(x) \propto x^{-\alpha}$ has no such scale. It looks the same at every magnification: the ratio $P(2x)/P(x) = 2^{-\alpha}$ does not depend on x. Plotted on logarithmic axes, it is a straight line of slope $-\alpha$.

Power laws appear everywhere once you look:

- **Earthquakes** obey the Gutenberg–Richter law: each step of one in magnitude (a factor of about 30 in energy) makes earthquakes about ten times rarer.
- **Words** obey Zipf's law: the rth most common word in a language appears with frequency roughly $\propto 1/r$. "The" is about twice as common as "of," three times as common as "and."
- **City sizes, personal wealth** (Pareto's law), the sizes of solar flares, forest fires, and moon craters, and the numbers of citations to scientific papers.

Heavy tails have consequences. If $\alpha \leq 2$, the mean of the distribution is infinite; if $\alpha \leq 3$, the variance is. Averages computed from data then never settle down, because they are dominated by the single largest event so far. Financial returns have fat tails, which is why "once-in-a-century" market crashes happen every decade or so.

Why do power laws appear? There are several mechanisms, and each is a lesson. Systems tuned to a *critical point* have fluctuations at every scale (Chapter 9). *Rich-get-richer* processes, where growth is proportional to current size, produce them naturally (§16.3). And *multiplicative* random processes produce log-normal distributions that can masquerade as power laws over a few decades. A word of caution: many claimed power laws in the literature dissolve under careful statistics. A straight line on a log-log plot over one decade is not evidence of anything.

§16.2 Random walks and diffusion, again

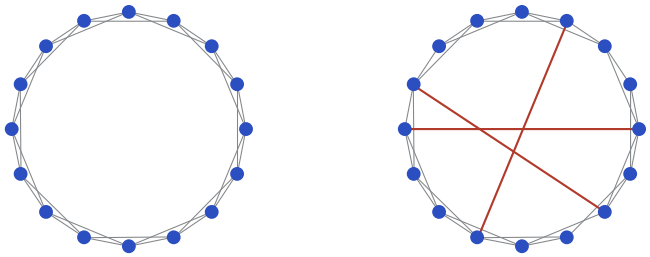
A random walker taking steps of fixed size spreads as $\sqrt{t}$ (Chapter 1). This *normal diffusion* describes perfume in a room, proteins in a cell (Chapter 23), and heat in a solid. But if the step lengths themselves follow a heavy-tailed distribution, occasional enormous jumps dominate and the walker spreads faster than $\sqrt{t}$: a *Lévy flight*. Albatrosses, sharks, and human travellers show Lévy-like patterns, perhaps because such searches are efficient when food is sparse and patchy.

§16.3 Networks

Many complex systems are best described as *networks*: nodes joined by links. Friends in a social network, routers on the internet, proteins that bind each other, airports joined by flights, neurons joined by synapses. The simplest statistic is the *degree distribution* $P(k)$, the fraction of nodes with k links.

Random graphs. In the Erdős–Rényi model, each pair of N nodes is linked independently with probability p. The mean degree is $c = p(N-1)$, and the degree distribution is Poisson. As c increases past 1, something dramatic happens: a *giant component* suddenly appears, a connected cluster containing a finite fraction of all nodes (Problem 16.5). Below $c = 1$, the network is a dust of small islands; above, most nodes can reach most others. This is a phase transition, with all the trappings of Chapter 9.

Small worlds. In 1967 Stanley Milgram asked people in Nebraska to forward a letter to a stranger in Boston via acquaintances; the letters that arrived took about six steps. In 1998 Watts and Strogatz explained how: a network that is mostly local and cliquish (your friends know each other) needs only a few random long-range links to make the typical distance between any two nodes grow only logarithmically with N.



regular ring three links rewired

Left: sixteen people, each linked to the four nearest. Getting from one side of the ring to the other takes four steps. Right: rewire just three links at random, and the shortcuts bring most pairs much closer. This is the Watts–Strogatz small world.

Scale-free networks. Real networks are not Poisson. The web, citation networks, and airline networks have a few enormous *hubs* and a power-law degree distribution, $P(k) \propto k^{-\gamma}$ with γ typically between 2 and 3. In 1999 Barabási and Albert proposed a mechanism: networks grow, and new nodes prefer to link to nodes that are already well connected. This *preferential attachment* generates a power law with $\gamma = 3$ (Problem 16.9). Scale-free networks are robust to random failures, since most nodes are unimportant, but fragile to targeted attacks on their hubs.

A curious consequence of heavy-tailed degree distributions is the *friendship paradox*: on average, your friends have more friends than you do. You are more likely to be friends with someone popular, simply because popular people have more friends to be.

§16.4 Percolation

Pour water on coffee grounds. Will it find a path through? Model the grounds as a lattice in which each site is open with probability p. For small p the open sites form isolated clusters; above a sharp *percolation threshold* p_c, an infinite cluster spans the system. On a square lattice $p_c \approx 0.593$ for sites and exactly $\frac{1}{2}$ for bonds. Percolation is perhaps the simplest phase transition of all — it is purely geometric — and near p_c it shows the same power laws, fractal clusters, and universal exponents as magnets and fluids at their critical points. It models forest fires, the flow of oil through porous rock, the conductivity of composite materials, and the spread of disease.

§16.5 Epidemics

The **SIR model** divides a population into susceptible (s), infected (i), and recovered (r) fractions:

$$\dot{s} = -\beta s i, \qquad \dot{i} = \beta s i - \gamma i, \qquad \dot{r} = \gamma i.$$

Each infected person meets and infects others at rate β and recovers at rate γ. The key number is the *basic reproduction number* $R_0 = \beta/\gamma$, the average number of people one infected person infects in a fully susceptible population. Early on, $\dot{i} = (\beta - \gamma)i$, so the epidemic grows exponentially if $R_0 > 1$ and fizzles if $R_0 < 1$. A threshold, again.

The epidemic grows only while each case produces more than one new case, $R_0 s > 1$. Once the susceptible fraction falls below $1/R_0$, the epidemic declines. If that fraction is reached by

vaccination, the disease cannot spread at all:

$$\text{herd immunity threshold} = 1 - \frac{1}{R_0}.$$

For influenza, with $R_0 \approx 1.5$, that is a third of the population; for measles, with $R_0 \approx 15$, it is 93%. On real contact networks with heavy-tailed degree distributions, highly connected "super-spreaders" dominate transmission, and vaccinating hubs is far more effective than vaccinating at random.

§16.6 Self-organized criticality

Critical points usually require tuning: a magnet is critical only at exactly T_c. Yet power laws are everywhere in nature, where no one is tuning anything. In 1987 Bak, Tang, and Wiesenfeld proposed that some systems drive *themselves* to a critical state. Their model is a *sandpile*. Drop grains one at a time onto a grid; when a site holds four grains, it topples, sending one grain to each neighbour, which may topple in turn. The pile builds up to a state where a single added grain can trigger an avalanche of any size, and the avalanche sizes follow a power law — with no tuning at all. Whether real earthquakes, forest fires, extinctions, and neural activity are self-organized critical is debated, but the idea changed how physicists think about where power laws come from.

§16.7 Collective motion

A flock of starlings turns as one, though no bird is in charge. In the *Vicsek model*, self-propelled particles move at constant speed and, at each step, align their direction with the average of their neighbours plus some noise. At low noise, the whole flock spontaneously chooses a direction and moves together; at high noise it is a disordered swarm. The transition between them is a phase transition of a new kind, one that is possible only because the particles are active, constantly consuming energy. This is the field of *active matter*, which also describes bacterial colonies, crowds, and the cytoskeleton of cells.

Traffic is collective motion too. On a crowded road, a single driver tapping the brakes can set off a *phantom jam*, a stop-and-go wave that travels backward along the highway at about 20 km/h long after its cause has driven away. It has been reproduced by asking drivers to circle a ring road at constant speed: they cannot.

§16.8 Econophysics

Physicists have applied statistical mechanics to markets and economies, sometimes fruitfully. Stock prices follow roughly a random walk (Bachelier proposed this in 1900, five years before Einstein's Brownian motion), but returns have fat power-law tails, and volatility clusters in time. A toy model of an economy — agents who meet in random pairs and exchange random amounts of money, conserving the total — relaxes to an *exponential* distribution of wealth, exactly the Boltzmann distribution with money playing the role of energy (Problem 16.10). Adding savings or rich-get-richer effects produces a Pareto power-law tail, as observed for the wealthiest few

percent in real economies. Such models illuminate mechanisms; they should not be mistaken for economies.

§16.9 Information

Many complex systems process information, and Shannon gave it a measure in 1948. For a source emitting symbols with probabilities p_i, the *information entropy* is

$$H = -\sum_i p_i \log_2 p_i \quad \text{bits per symbol},$$

the minimum average number of bits needed to encode its output. A fair coin has $H = 1$ bit; a coin that lands heads 90% of the time has only 0.47; English text, because of its redundancy, has about one bit per character rather than the $\log_2 27 \approx 4.75$ of random letters. Shannon's formula is Boltzmann's $S = -k_\mathrm{B} \sum p_i \ln p_i$ in different units — a coincidence that turned out to be deep. Information is physical: erasing it costs energy (Problem 7.10 in Chapter 7).

§16.10 Problems

16.1 $O(\epsilon)$ ★ Compute the herd immunity threshold for $R_0 = 1.5$, 3, and 15. Explain why a vaccine that is 90% effective cannot by itself stop measles, whatever fraction of people receive it.

16.2 $O(\epsilon)$ English has about 50 000 words in common use. If word frequencies follow Zipf's law exactly, $f_r \propto 1/r$, what fraction of a typical text is made of the 100 most common words?

Hint: $\sum_{r=1}^{n} 1/r \approx \ln n + 0.577$.

16.3 $O(\epsilon)$ For a power-law probability density $P(x) = Cx^{-\alpha}$ on $x \geq 1$, show that the mean is finite only for $\alpha > 2$ and the variance only for $\alpha > 3$. What does it mean, in practice, for a variance to be infinite?

16.4 $O(\epsilon^2)$ ★ In the SIR model, divide the equations for $\dot{s}$ and $\dot{r}$ to show that $s = e^{-R_0 r}$ throughout the epidemic. Deduce the *final size equation* $r_\infty = 1 - e^{-R_0 r_\infty}$, and solve it numerically for $R_0 = 2$. What fraction of the population is never infected?

16.5 $O(\epsilon^2)$ ★ In a large Erdős–Rényi graph with mean degree c, let u be the probability that a node is *not* in the giant component. A node is outside it only if none of its neighbours are in it. Argue that $u = e^{-c(1-u)}$, so the giant component fraction $S = 1 - u$ satisfies $S = 1 - e^{-cS}$. Show that a nonzero solution exists only for $c > 1$. Compare with the SIR final size equation.

16.6 $O(\epsilon^2)$ Prove the friendship paradox: if you pick a random *friendship* and look at one end, the expected degree is $\langle k^2 \rangle / \langle k \rangle \geq \langle k \rangle$. When is the inequality an equality?

16.7 $O(\epsilon^2)$ The energy released by an earthquake grows as $10^{1.5M}$ with magnitude M, and the number of quakes falls as 10^{-M}. Show that the total energy released by all quakes of magnitude near M grows with M. What does this imply about which earthquakes release most of the Earth's seismic energy?

16.8 $O(\epsilon^2)$ Compute the Shannon entropy of a fair six-sided die, a fair coin, and a coin with $p = 0.9$. A compression program reduces a large English text file to 25% of its size, from 8 bits per character. Estimate the information entropy of English.

16.9 $e^{-1/\epsilon}$ ★ (*Preferential attachment.*) A network grows by adding one node at a time, each bringing m new links that attach to existing nodes with probability proportional to their degree. In a continuum approximation, the degree of node i added at time t_i obeys $\mathrm{d}k_i/\,\mathrm{d}t = k_i/2t$. Solve this, and deduce that the fraction of nodes with degree greater than k is $\propto k^{-2}$, so $P(k) \propto k^{-3}$. Why are the oldest nodes the biggest hubs?

16.10 $e^{-1/\epsilon}$ (*Money as energy.*) N agents each start with \$100. At each step two random agents meet, pool their money, and split it at a uniformly random fraction. Total money is conserved. Argue, by analogy with the derivation of the Boltzmann distribution, that the equilibrium distribution of wealth is exponential, $P(m) \propto e^{-m/T}$, and identify the "temperature" T. Simulate it to check. Why don't real economies look like this at the top?

16.11 $e^{-1/\epsilon}$ (*The sandpile.*) Implement the Bak–Tang–Wiesenfeld sandpile on a 50×50 grid with grains falling off the edges. After the pile reaches its steady state, record the size of the avalanche triggered by each added grain. Plot the distribution of sizes on log-log axes and estimate its exponent. Then explain qualitatively why the pile must evolve to a state that is critical, rather than one that is too flat (no large avalanches) or too steep (huge avalanches only).

Further reading

- **M. E. J. Newman,** ***Networks*** — The standard textbook on network science, from a physicist.
- **Albert-László Barabási,** ***Network Science*** — An accessible, beautifully illustrated textbook, free online.
- **Melanie Mitchell,** ***Complexity: A Guided Tour*** — A thoughtful popular overview of complex systems science.
- **Per Bak,** ***How Nature Works*** — The case for self-organized criticality, by its originator.
- **P. W. Anderson, "More Is Different" (*Science*, 1972)** — Four pages that every physicist should read.

CHAPTER 17

Plasma physics

BUILDS ON electromagnetism (Chapter 6), statistical mechanics (Chapter 9), fluid dynamics (Chapter 11)

The fourth state of matter — the stuff of stars, lightning, auroras, and fusion reactors — and most of the visible universe.

PROTOTYPE A magnetized plasma: a gas of free electrons and ions threaded by a magnetic field. Each particle spirals tightly around a field line, free to slide along it but unable to cross. The field lines behave like wires on which the plasma is strung, and the plasma in turn drags the field lines along with it.

Heat a solid and it melts; heat the liquid and it boils; heat the gas enough and its atoms lose their electrons. The result is a *plasma*: a gas of charged particles, overall neutral, in which electromagnetic forces dominate. On Earth plasmas are exotic — lightning, neon signs, flames at their hottest. In the universe they are the norm: stars, the solar wind, the interstellar medium, and accretion disks are all plasma, and over 99% of the visible matter in the universe is in this state. Plasma physics is electromagnetism and statistical mechanics working together, and its central practical goal, controlled fusion, is one of the great engineering challenges of the century.

§17.1 When does a gas become a plasma?

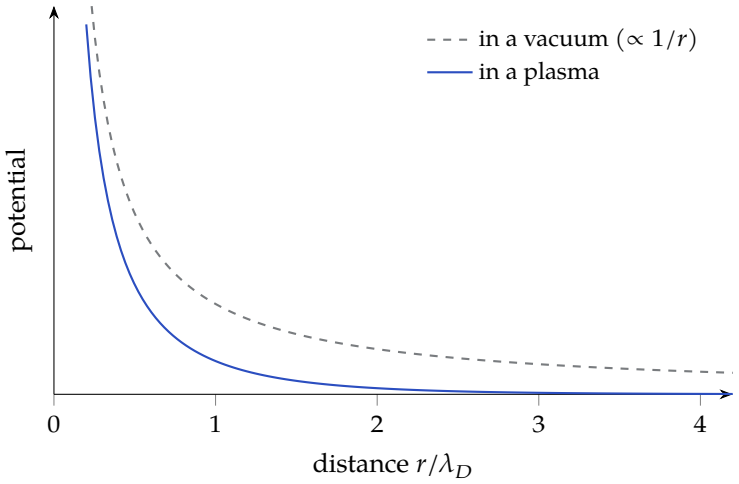
Ionizing hydrogen takes 13.6 eV, which corresponds to a temperature of about 160 000 K. Yet hydrogen in stellar atmospheres is half ionized at only about 10 000 K. Why so low? Because of entropy. A free electron has a vast amount of phase space available to it, while a bound one is confined to a single state. The balance is described by the *Saha equation*, which, like every phase transition, weighs energy against entropy (Chapter 7): in a dilute gas the free-electron entropy wins at temperatures well below $E_{\text{ion}}/k_{\text{B}}$. The same reasoning tells cosmologists when the early universe became transparent (Chapter 28).

§17.2 Debye shielding

The defining property of a plasma is that it screens electric fields. Place a positive charge in a plasma: electrons crowd around it and ions are pushed away, and from a distance the charge is hidden. Solving Poisson's equation with Boltzmann-distributed electrons gives a screened potential

$$\phi(r) = \frac{q}{4\pi\epsilon_0 r}\, e^{-r/\lambda_D}, \qquad \lambda_D = \sqrt{\frac{\epsilon_0 k_{\text{B}} T}{n e^2}},$$

where λ_D is the *Debye length*. Beyond a few Debye lengths, any charge imbalance is wiped out: the plasma is *quasi-neutral*. Hotter plasmas screen less effectively (fast electrons are hard to corral); denser ones screen better.



The potential around a charge in a vacuum (dashed) and in a plasma (solid). In the plasma, a cloud of opposite charge gathers round and cancels the field beyond a few Debye lengths.

A gas of charged particles behaves as a plasma, rather than a collection of individual charges, when it is much larger than λ_D and when there are many particles within a Debye sphere, $n\lambda_D^3 \gg 1$. Then collective behaviour dominates over individual collisions.

§17.3 The plasma frequency

Displace all the electrons in a slab of plasma slightly relative to the ions. The resulting charge separation creates an electric field that pulls them back; they overshoot, and oscillate. A short calculation using Gauss's law gives the *plasma frequency*

$$\omega_p = \sqrt{\frac{ne^2}{\epsilon_0 m_e}}.$$

It depends only on the electron density: numerically, $f_p \approx 9\sqrt{n}\,\text{Hz}$ with n in m^{-3}.

The plasma frequency is a cutoff for electromagnetic waves. Light in a plasma obeys

$$\omega^2 = \omega_p^2 + c^2k^2.$$

Waves below ω_p cannot propagate and are reflected. The ionosphere, with f_p of a few megahertz, reflects AM and shortwave radio back to the ground, which is how radio signals reach over the horizon at night; FM and TV signals, at higher frequencies, pass straight through to space. Metals are plasmas of conduction electrons with $\hbar\omega_p \sim 10\,\text{eV}$, in the ultraviolet: they reflect visible light, which is why they are shiny, but are transparent to X-rays.

KEY IDEA The dispersion relation $\omega^2 = \omega_p^2 + c^2k^2$ has exactly the form of $E^2 = m^2c^4 + p^2c^2$ for a massive particle (Chapter 10). Inside a plasma, the photon behaves as if it had a mass $\hbar\omega_p/c^2$. Philip Anderson noticed in 1963 that this mechanism, generalized, could give mass to the carriers of a force. It became the Higgs mechanism (Chapter 32).

§17.4 Particles in magnetic fields

A charged particle in a uniform magnetic field gyrates around a field line at the cyclotron frequency $\omega_c = qB/m$ (Chapter 6) with *Larmor radius* $r_L = mv_\perp/qB$, while moving freely along the field. Its path is a helix. Electrons spiral tightly and fast; ions loosely and slowly.

Add other forces and the guiding centre of the helix *drifts*. With an electric field perpendicular to $\mathbf{B}$, the particle speeds up on one half of its orbit and slows on the other, so its orbit is larger on one side, and it walks sideways at

$$\mathbf{v}_E = \frac{\mathbf{E}\times\mathbf{B}}{B^2}.$$

Remarkably, this drift is the same for every particle, whatever its charge or mass, so electrons and ions drift together and no current flows. Gradients and curvature of $\mathbf{B}$ produce other drifts that *do* depend on charge, separating electrons from ions — which, as we will see, is the root of the difficulty of confining a plasma.

17.4.1 Magnetic mirrors

A gyrating particle has a magnetic moment $\mu = mv_\perp^2/2B$, which is an adiabatic invariant (Chapter 4): it is conserved as the particle moves into regions of slowly changing field. As the particle spirals into a stronger field, $v_\perp$ must increase to keep μ fixed, and since a magnetic field does no work, $v_\parallel$ must decrease. If the field becomes strong enough, the particle stops and is reflected. This is a *magnetic mirror*. The Earth's dipole field forms a natural magnetic bottle: particles bounce back and forth between the polar regions, forming the *Van Allen radiation belts*. Those whose velocity is aimed too closely along the field escape through the *loss cone* and hit the atmosphere near the poles, producing the aurora.

§17.5 Plasma as a fluid: magnetohydrodynamics

On large scales, a plasma can be treated as a single conducting fluid coupled to the magnetic field. This is *magnetohydrodynamics* (MHD). In a perfectly conducting plasma, the electric field in the fluid's own frame must vanish, $\mathbf{E} + \mathbf{v}\times\mathbf{B} = 0$. Combined with Faraday's law, this gives

Theorem 17.1 (Alfvén's frozen-flux theorem)**.** In a perfectly conducting fluid, magnetic field lines move with the fluid: the magnetic flux through any surface moving with the fluid is constant.

The field is frozen into the plasma. When the Sun's plasma flows out as the solar wind, it drags the Sun's magnetic field into a spiral filling the solar system. When a star collapses into a neutron star, its field is squeezed and amplified to 10^8 T or more.

The magnetic field exerts forces on the plasma that can be split into a *magnetic pressure* $B^2/2\mu_0$, pushing perpendicular to field lines, and a *magnetic tension* along them, as if field lines were stretched elastic bands. Pluck a field line and a wave travels along it, the *Alfvén wave*, at speed

$$v_A = \frac{B}{\sqrt{\mu_0 \rho}},$$

exactly analogous to a wave on a string with tension B^2/μ_0 (Chapter 5). The ratio of plasma pressure to magnetic pressure, $\beta = nk_BT/(B^2/2\mu_0)$, says who is in charge: at low β the field shapes the plasma; at high β the plasma carries the field around.

When the plasma is not perfectly conducting, field lines can break and reconnect into new configurations, releasing stored magnetic energy explosively. *Magnetic reconnection* powers solar flares, drives magnetic storms in the Earth's magnetosphere, and disrupts fusion plasmas.

§17.6 Instabilities

Confining a hot plasma with magnetic fields has been compared to holding jelly with rubber bands. Plasmas are prone to an enormous variety of *instabilities*. Wherever a heavy fluid sits on a light one — or a plasma pushes against a magnetic field curving away from it — the interface ripples and fingers (the *Rayleigh–Taylor* or *interchange* instability). A plasma column carrying current kinks like a twisted garden hose, or pinches off like a string of sausages. Turbulence driven by small-scale instabilities leaks heat out of fusion devices far faster than simple collisions would. Much of the history of fusion research is the history of discovering and taming instabilities.

§17.7 Kinetic theory and Landau damping

The fluid picture misses effects that depend on the detailed distribution of particle velocities. The most famous is **Landau damping**. In 1946 Landau showed that waves in a plasma damp away even with no collisions at all. Particles moving slightly slower than the wave are pushed forward by it and gain energy; particles moving slightly faster are held back and give energy to the wave. In thermal equilibrium there are more slower particles than faster ones at any speed above the peak of the distribution, so the wave loses energy on balance. It is a subtle, collisionless, essentially kinetic effect — like surfers catching a wave — and it was controversial for years until confirmed experimentally in 1965. Run backwards, with more fast particles than slow, the same mechanism makes waves *grow*: an electron beam in a plasma amplifies waves.

§17.8 Fusion

The Sun is powered by fusing hydrogen into helium (Chapter 22), slowly, over billions of years. On Earth we need something faster, and the easiest reaction is deuterium with tritium:

$$\mathrm{D} + \mathrm{T} \to {}^4\mathrm{He}\,(3.5\,\mathrm{MeV}) + n\,(14.1\,\mathrm{MeV}).$$

The ions must collide with enough energy to tunnel through their Coulomb repulsion (Problem 14.13), which requires temperatures around 150 million kelvin, ten times hotter than the Sun's core. A reactor produces net energy only if the plasma is dense enough and holds its heat long enough. The condition is the **Lawson criterion**, usually stated as the *triple product*

$$n\,T\,\tau_E \gtrsim 3\times 10^{21}\ \mathrm{keV\,s/m^3},$$

where τ_E is the energy confinement time. There are two main strategies:

- **Magnetic confinement**: hold a low-density plasma ($n \sim 10^{20}\ \mathrm{m^{-3}}$) for seconds in a magnetic bottle. The leading design is the *tokamak*, a doughnut-shaped chamber with a strong toroidal field and a current flowing in the plasma itself; ITER, under construction in France, is designed to produce ten times more fusion power than the heating power put in. *Stellarators* achieve the necessary twist of the field lines with intricately shaped coils instead.
- **Inertial confinement**: crush a tiny pellet of fuel with lasers so quickly that it fuses before it flies apart, for a fraction of a nanosecond at enormous density. In December 2022 the National Ignition Facility in California produced more fusion energy than the laser energy delivered to the target — the first ignition in the laboratory.

§17.9 Space and astrophysical plasmas

The Sun's corona is a plasma at over a million kelvin, mysteriously hotter than the 5800 K surface beneath it — the *coronal heating problem*, still not fully solved, though Alfvén waves and countless small reconnection events are prime suspects. The corona expands continually as the *solar wind*, streaming past the Earth at about 400 km/s. The Earth's magnetic field deflects it, carving out a cavity called the *magnetosphere*, compressed on the sunward side and stretched into a long tail on the night side. When gusts of solar wind reconnect with the Earth's field, energy is dumped into the magnetosphere, driving geomagnetic storms that can damage satellites and power grids — and lighting the aurora. In accretion disks around black holes (Chapter 27), a magnetic instability drives the turbulence that lets matter lose angular momentum and fall in.

§17.10 Problems

17.1 $O(\epsilon)$ ★ Compute the Debye length and the number of particles in a Debye sphere for (a) the ionosphere ($n = 10^{12}\ \mathrm{m^{-3}}$, $T = 1000\ \mathrm{K}$), (b) a fusion plasma ($n = 10^{20}\ \mathrm{m^{-3}}$, $T = 10^{8}\ \mathrm{K}$), and (c) the Sun's core ($n = 10^{32}\ \mathrm{m^{-3}}$, $T = 1.5\times 10^{7}\ \mathrm{K}$). Which are good plasmas in the sense $n\lambda_D^3 \gg 1$?

17.2 $O(\epsilon)$ ★ The ionosphere's peak electron density is about $10^{12}\ \mathrm{m^{-3}}$. Find its plasma frequency. Explain why AM radio (1 MHz) can be received far beyond the horizon at night, while FM (100 MHz) cannot.

17.3 $O(\epsilon)$ Find the Larmor radius of a solar wind proton moving at 400 km/s perpendicular to the interplanetary magnetic field of 5 nT. Compare with an electron of the same speed, and with the size of the Earth.

17.4 $O(\epsilon^2)$ ★ Derive the $\mathbf{E}\times\mathbf{B}$ drift by transforming to a frame moving with velocity $\mathbf{v}_E$ and showing that the electric field vanishes there (for $v \ll c$). Explain physically why the drift is independent of charge and mass.

17.5 $O(\epsilon^2)$ Copper has 8.5×10^{28} free electrons per cubic metre. Find its plasma frequency and the corresponding photon energy. Explain why copper reflects visible light but is transparent to X-rays, and why gold looks yellow (hint: a bound-electron resonance takes a bite out of the blue).

17.6 $O(\epsilon^2)$ A magnetic mirror has field strength $B_{\min}$ at its centre and $B_{\max}$ at its ends. Show that a particle at the centre is trapped if its pitch angle θ (between velocity and field) satisfies $\sin^2\theta > B_{\min}/B_{\max}$. For a mirror ratio of 4 and an isotropic velocity distribution, what fraction of particles escapes?

17.7 $O(\epsilon^2)$ Estimate the Alfvén speed in the solar corona, with $B = 10^{-3}\,\mathrm{T}$ and a proton density of $10^{15}\,\mathrm{m^{-3}}$. How long does an Alfvén wave take to cross a coronal loop $10^{8}\,\mathrm{m}$ long?

17.8 $O(\epsilon^2)$ ★ ITER is designed to run with $n \approx 10^{20}\,\mathrm{m^{-3}}$ and $T \approx 15\,\mathrm{keV}$. What energy confinement time does the Lawson criterion require? Why does it help to build the reactor bigger?

17.9 $e^{-1/\epsilon}$ ★ (*The photon with mass.*) For waves obeying $\omega^2 = \omega_p^2 + c^2k^2$, compute the phase and group velocities and show that $v_{\mathrm{phase}}v_{\mathrm{group}} = c^2$. Which one exceeds c, and why doesn't that violate relativity? Show that the energy–momentum relation of the corresponding quanta is exactly that of a particle of mass $\hbar\omega_p/c^2$.

17.10 $e^{-1/\epsilon}$ (*Why fusion reactors are doughnuts.*) A purely toroidal magnetic field (field lines running the long way around a doughnut) is weaker on the outer side than the inner side. Show that the gradient drift then pushes ions up and electrons down (or vice versa), creating a vertical electric field, and that the resulting $\mathbf{E}\times\mathbf{B}$ drift carries the whole plasma outward into the wall. Explain how twisting the field lines helically — with a current in the plasma (tokamak) or twisted coils (stellarator) — cancels the effect on average. Why can't a magnetic bottle be a sphere instead?

Hint: You cannot comb the hair on a sphere flat without leaving a cowlick (the hairy ball theorem).

17.11 $e^{-1/\epsilon}$ (*Landau's surfers.*) Consider a wave with phase velocity v_ϕ in a plasma with electron velocity distribution $f(v)$. Argue that electrons with v slightly below v_ϕ are accelerated by the wave and those slightly above are decelerated. Conclude that the wave is damped if $f'(v_\phi) < 0$ and amplified if $f'(v_\phi) > 0$. Sketch a distribution with a "bump on the tail" and explain why such a beam–plasma system is unstable.

Further reading

- **Francis F. Chen,** ***Introduction to Plasma Physics and Controlled Fusion*** — The standard introduction, clear and physical. Start here.
- **Paul M. Bellan,** ***Fundamentals of Plasma Physics*** — A more complete and modern graduate-level treatment.
- **Kivelson and Russell,** ***Introduction to Space Physics*** — The solar wind, magnetospheres, and the aurora.
- **Goldston and Rutherford,** ***Introduction to Plasma Physics*** — Strong on fusion-relevant physics.

CHAPTER 18

Atomic, molecular and optical physics

BUILDS ON optics (Chapter 8), quantum mechanics (Chapter 14)

Why every atom glows in its own private set of colours — and how physicists learned to hold single atoms still, cool them to a billionth of a degree, and build clocks that would not lose a second in the age of the universe.

PROTOTYPE The hydrogen atom: one proton, one electron. Its spectrum of sharp lines was the Rosetta Stone of quantum mechanics, and every refinement of it — fine structure, the Lamb shift, the 21-centimetre line — opened a new chapter of physics.

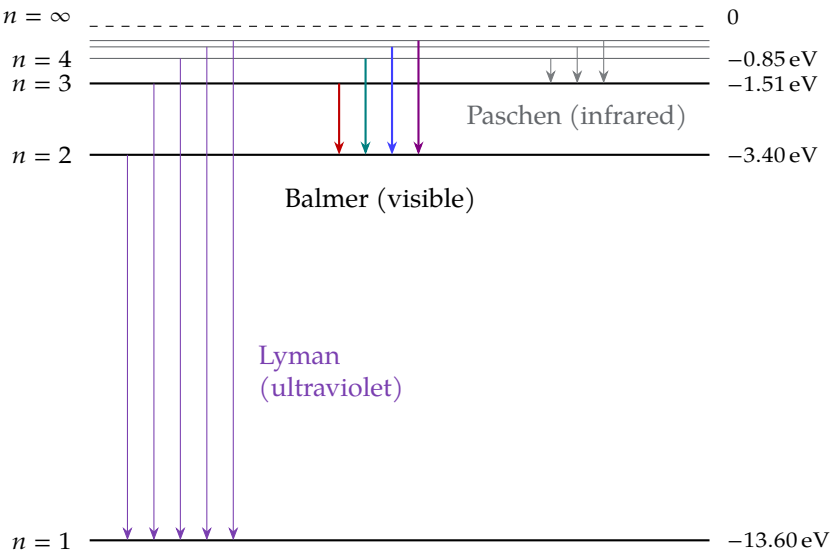
Atomic, molecular, and optical (AMO) physics is quantum mechanics at the scale where it is cleanest. Atoms are simple, identical, and isolated; light can be controlled with exquisite precision. This makes AMO physics the field where quantum mechanics is tested most precisely, where the most accurate measurements in all of science are made, and where many of the strangest quantum phenomena — Bose–Einstein condensates, entangled atoms, squeezed light — are engineered on purpose.

§18.1 The hydrogen spectrum

In 1885 Johann Balmer, a Swiss schoolteacher, found that the four visible lines of hydrogen fit a formula, later generalized by Rydberg:

$$\frac{1}{\lambda} = R_\infty \left(\frac{1}{n'^2} - \frac{1}{n^2} \right), \qquad R_\infty = 1.097 \times 10^7 \,\mathrm{m}^{-1}.$$

With $n' = 2$ it gives the visible *Balmer series*, including the red Hα line at 656 nm that colours glowing nebulae; with $n' = 1$ the ultraviolet *Lyman series*; with $n' = 3$ the infrared *Paschen series*. Quantum mechanics explains it at once: the energies are $E_n = -13.6\,\mathrm{eV}/n^2$ (Chapter 14), and each line is a photon emitted as the electron drops between two levels.



The energy levels of hydrogen, $E_n = -13.6\,\text{eV}/n^2$, drawn to scale. Jumps down to $n = 1$ give the ultraviolet Lyman series, jumps to $n = 2$ the visible Balmer series (in roughly their real colours, from red Hα to violet), and jumps to $n = 3$ the infrared Paschen series.

In 1913, before the Schrödinger equation existed, Niels Bohr got these energies with a spherical cow of genius: a classical electron in a circular orbit, with the single extra rule that its angular momentum must be a whole multiple of $\hbar$. The model is wrong in nearly every detail — electrons do not orbit, and the ground state has zero angular momentum — yet it gets the energies exactly right, and it was the bridge to the real theory.

§18.2 Fine structure and the fine-structure constant

Look closely, with a good spectrometer, and each hydrogen line splits into closely spaced components. This *fine structure* has two sources of the same size. The electron moves at a speed $v \approx c/137$, so relativistic corrections of order $(v/c)^2$ shift the energies. And the electron's spin interacts with the magnetic field it sees from the proton circling around it in its own frame (*spin–orbit coupling*). Both effects are smaller than the main energies by the square of

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137.036},$$

the *fine-structure constant*. It is dimensionless — the same in any system of units — and it measures the strength of electromagnetism. Its value is unexplained, and it is perhaps the most famous number in physics. Dirac's relativistic equation for the electron (Chapter 30) predicts the fine structure exactly.

§18.3 Beyond Dirac: the Lamb shift and the 21-centimetre line

Dirac's theory predicts that the $2s$ and $2p_{1/2}$ states of hydrogen have exactly the same energy. In 1947 Willis Lamb measured them with microwaves and found the $2s$ state higher by about 1 GHz

(a few millionths of an eV). The *Lamb shift* is caused by the vacuum itself: the quantized electromagnetic field is never truly empty, and its zero-point fluctuations jiggle the electron, smearing out its interaction with the nucleus (Problem 18.12). Explaining it quantitatively forced the development of renormalized quantum electrodynamics (Chapter 30), now the most precisely tested theory in science.

The proton has a spin too, and the electron and proton spins interact magnetically. In the ground state, the configuration with spins parallel lies slightly above the antiparallel one. This *hyperfine* splitting is tiny, 5.9×10^{-6} eV, corresponding to radio waves of frequency 1420 MHz and wavelength 21 cm. The transition is so slow that a given atom makes it about once every ten million years. But there is so much hydrogen in space that the 21-cm line is one of the brightest signals in radio astronomy. It passes through dust that blocks visible light, and its Doppler shifts map the motion of gas throughout our galaxy and others — which is how the flat rotation curves pointing to dark matter were measured (Chapter 27).

§18.4 Many electrons and the periodic table

In an atom with many electrons, each electron feels the nucleus partly *screened* by the others. Inner electrons see nearly the full nuclear charge; outer electrons see much less. Screening breaks the hydrogen degeneracy between orbitals with the same n: s orbitals, which penetrate close to the nucleus, lie lower than p, which lie lower than d. Filling orbitals in order of energy, two electrons per orbital by the Pauli principle (Chapter 14), gives the shells of the periodic table, with rows of $2, 8, 8, 18, 18, 32, 32$ elements. Within a shell, electrons spread out with parallel spins where they can (*Hund's rules*), because the antisymmetric spatial wavefunction keeps them apart and reduces their repulsion. Chemistry is mostly the physics of the outermost shell (Chapter 21).

Knock out an inner electron with an electron beam, and an outer electron falls into the hole, emitting an X-ray. In 1913 Henry Moseley found that the X-ray frequencies scale as $(Z-1)^2$, and used them to put the periodic table in order by atomic number rather than mass, predicting missing elements. He was killed at Gallipoli two years later, aged twenty-seven.

§18.5 Atoms in fields

A magnetic field splits each atomic level according to the component of its angular momentum along the field. This is the **Zeeman effect**, with splittings of order $\mu_B B$, where the *Bohr magneton* $\mu_B = e\hbar/2m_e = 5.8 \times 10^{-5}$ eV/T. In 1908 George Ellery Hale saw Zeeman-split lines in the light from sunspots and discovered that sunspots are regions of intense magnetic field, about 0.3 T. Astronomers still measure stellar magnetic fields this way. An electric field also shifts and splits levels (the *Stark effect*), by polarizing the atom.

§18.6 How atoms absorb and emit light

An atom with energy levels $E_1 < E_2$ interacts with light of frequency $\nu = (E_2 - E_1)/h$ in three ways. It can *absorb* a photon and jump up; it can *spontaneously emit* a photon and drop down; and, as Einstein realized in 1917, a passing photon can *stimulate* it to emit a second photon identical to

the first. Einstein described the rates with coefficients A (spontaneous) and B (stimulated and absorbed), and by demanding that atoms in a cavity reach equilibrium with Planck's blackbody radiation (Chapter 9), he showed they must be related:

$$\frac{A}{B} = \frac{8\pi h\nu^3}{c^3}.$$

Spontaneous emission therefore grows steeply with frequency. Visible transitions have lifetimes of nanoseconds; microwave transitions, like the 21-cm line, can take millions of years.

Not every pair of levels is connected. The photon carries one unit of angular momentum, so the strongest (*electric dipole*) transitions obey *selection rules* such as $\Delta\ell = \pm 1$. Transitions that break them are "forbidden" — not impossible, but millions of times slower. In the near-vacuum of nebulae, atoms can sit in such *metastable* states long enough to radiate rather than being knocked out by collisions. The striking green glow of planetary nebulae comes from forbidden lines of doubly ionized oxygen, which baffled astronomers in the nineteenth century so completely that they attributed it to a new element, "nebulium."

§18.7 Lasers

Normally there are more atoms in lower states than upper ones (Boltzmann; Chapter 9), so light passing through matter is absorbed more than it is amplified. A laser needs a *population inversion*: more atoms in the upper level than the lower. This cannot be achieved with two levels (Problem 18.11); practical lasers pump atoms into a high level that decays quickly to a long-lived upper laser level, from which they are stimulated down to a lower level that empties quickly. Put the gain medium between two mirrors, and stimulated emission builds up a coherent beam.

Lasers now come in every colour and power: diode lasers in barcode scanners, helium–neon lasers in classrooms, carbon dioxide lasers that cut steel, and titanium–sapphire lasers that emit pulses a few femtoseconds long — short enough to watch chemical bonds break. An *optical frequency comb*, a laser emitting a train of such pulses, produces a spectrum of millions of perfectly evenly spaced frequencies: a ruler for light, which made optical atomic clocks practical (Nobel Prize, 2005).

§18.8 Molecules

A molecule's nuclei are thousands of times heavier than its electrons, so the electrons rearrange almost instantly as the nuclei move. In the **Born–Oppenheimer approximation**, we first solve for the electrons with the nuclei clamped in place, giving an effective potential in which the nuclei then vibrate and rotate (Chapter 21). This splits molecular energies into a hierarchy:

- **Electronic** transitions: a few eV, in the visible and ultraviolet.
- **Vibrational** levels: harmonic oscillator rungs $(n + \tfrac{1}{2})\hbar\omega$ (Chapter 3), about 0.1 eV apart, in the infrared.
- **Rotational** levels: $E_\ell = \hbar^2\ell(\ell+1)/2I$, about 10^{-3} eV apart, in the microwave.

Each step down is smaller by a factor of about $\sqrt{m_e/M} \sim 1/50$. Molecular spectra are fingerprints: astronomers have identified over two hundred molecules in interstellar clouds by their

rotational lines, and atmospheric scientists measure greenhouse gases by their vibrational ones. Carbon dioxide absorbs strongly at 15 μm, where its bending vibration lies — right near the peak of the Earth's thermal emission (Chapter 25). Nitrogen and oxygen, being symmetric diatomic molecules with no dipole moment to wiggle, are nearly transparent in the infrared.

§18.9 Laser cooling and trapping

Light carries momentum, so an atom that absorbs a photon gets a kick of $\hbar k$. Each kick is tiny, but an atom can scatter millions of photons per second, giving accelerations of 10^5 to $10^6\,g$.

To cool atoms, tune the lasers slightly *below* the atomic resonance. An atom moving toward a laser sees it Doppler-shifted up, closer to resonance, and scatters more photons from it, each pushing it back. An atom moving away sees it shifted further from resonance and scatters fewer. With lasers from all six directions, every atom feels a velocity-dependent friction, an *optical molasses*. This *Doppler cooling* reaches temperatures of about

$$T_D = \frac{\hbar\Gamma}{2k_{\mathrm{B}}},$$

where Γ is the natural linewidth of the transition: a few hundred microkelvin. Cleverer schemes reach microkelvin, and adding a magnetic field gradient turns the molasses into a *magneto-optical trap* that also holds the atoms in place. The final step is *evaporative cooling*: let the hottest atoms escape, as steam carries heat away from a cup of coffee. In 1995 this produced the first Bose–Einstein condensates, at about a hundred nanokelvin (Chapter 9). Laser-cooled atoms can be loaded into *optical lattices*, crystals made of standing waves of light, where they mimic electrons in a solid — a way to simulate quantum materials (Chapter 19). And a tightly focused laser beam can grip a single bead or bacterium as *optical tweezers* (Chapter 23).

§18.10 Clocks and precision

Atoms make perfect clocks: every caesium atom is identical, and its transitions do not wear out. Since 1967 the second has been defined as exactly 9 192 631 770 oscillations of the hyperfine transition of caesium-133. Modern optical clocks, using transitions at visible frequencies a hundred thousand times higher and trapped single ions or lattices of atoms, are accurate to about one part in 10^{18}: they would not have lost a second since the Big Bang. They are so precise that raising one by a centimetre measurably changes its rate, through gravitational time dilation (Chapter 26). Such clocks are being used to search for changes in the fundamental constants, for dark matter, and to measure the shape of the Earth's gravitational field.

The most precise agreement between theory and experiment in all of science is in AMO physics: the magnetic moment of the electron, which quantum electrodynamics predicts and Penning-trap experiments measure, agree to about one part in a trillion.

§18.11 Quantum optics

Ordinary light from a bulb is a jumble of photons arriving at random. Laser light is described by a *coherent state*, the quantum state closest to a classical wave, with photon numbers following a

Poisson distribution. But quantum mechanics allows stranger states of light: single photons on demand, entangled photon pairs (used in tests of Bell's inequality and quantum cryptography; Chapter 20), and *squeezed light*, in which the quantum noise in one property is pushed below the vacuum level at the expense of another. LIGO injects squeezed light into its interferometers to beat the quantum noise limit (Chapter 29). In *cavity quantum electrodynamics*, a single atom and a single photon trapped between mirrors exchange energy back and forth coherently — a clean realization of the light–matter interaction at its most elementary.

§18.12 Problems

18.1 $O(\epsilon)$ ★ Compute the wavelengths of the first three Balmer lines (Hα, Hβ, Hγ) and of the shortest-wavelength line in the series. What colours are they?

18.2 $O(\epsilon)$ Convert the hydrogen hyperfine splitting, 1420.4 MHz, to an energy in electron-volts and to a temperature. Why is the upper level populated in interstellar gas at 100 K, even though the transition is so slow?

18.3 $O(\epsilon)$ Estimate the Zeeman splitting, in eV and in GHz, for an atom in a 1 T field. A sunspot spectral line at 525 nm is split into components 0.01 nm apart. Estimate the sunspot's magnetic field.

18.4 $O(\epsilon^2)$ ★ Derive Einstein's relation $A/B = 8\pi h\nu^3/c^3$. In equilibrium the upward and downward rates must balance, $B\,u(\nu)\,N_1 = A\,N_2 + B\,u(\nu)\,N_2$ (for equal degeneracies), with $N_2/N_1 = e^{-h\nu/k_{\mathrm{B}}T}$. Solve for $u(\nu)$ and demand that it equal Planck's law at all temperatures.

18.5 $O(\epsilon^2)$ The CO molecule has bond length 0.113 nm. Find its moment of inertia and the frequency of its lowest rotational transition, $\ell = 1 \to 0$. (Astronomers use this 115 GHz line to map molecular clouds where stars are born.)

18.6 $O(\epsilon^2)$ The HCl molecule has bond stiffness 516 N/m. Find its vibrational frequency and the wavelength of infrared light it absorbs. By what factor does the frequency change for DCl, with deuterium replacing hydrogen?

18.7 $O(\epsilon^2)$ ★ Estimate the fine structure of hydrogen: in the ground state the electron moves at $v \approx \alpha c$, so relativistic corrections are of relative size $(v/c)^2$. Show that the corrections are of order $\alpha^2 \times 13.6\,\mathrm{eV}$, and evaluate this. What fraction of a 656 nm wavelength does this correspond to?

18.8 $O(\epsilon^2)$ Rubidium's cooling transition has natural linewidth $\Gamma = 2\pi \times 6$ MHz and wavelength 780 nm. Find the Doppler cooling limit, and the *recoil temperature* $T_r = \hbar^2 k^2/mk_{\mathrm{B}}$ set by the kick of a single photon. Which is lower?

18.9 $O(\epsilon^2)$ A sodium atom ($m = 23\,\mathrm{u}$) scatters 589 nm photons at up to $\Gamma/2 = 3 \times 10^7$ photons per second. Find its maximum deceleration. How far and how long does it take to stop an atom from an oven at 600 m/s?

18.10 $O(\epsilon^2)$ ★ Two optical clocks are compared, one 1 cm higher than the other. Show that the higher one runs fast by a fraction $g\,\Delta h/c^2$, and evaluate it. How long must a clock with fractional uncertainty 10^{-18} run to detect this?

18.11 $e^{-1/\epsilon}$ ★ Write rate equations for a two-level system driven by light of intensity I, with absorption and stimulated emission equally likely and spontaneous decay at rate A. Show that as $I \to \infty$ the populations approach equality but never invert. Explain why a laser medium needs at least three levels, and why four-level lasers are easier still.

18.12 $e^{-1/\epsilon}$ (*Welton's estimate of the Lamb shift.*) The zero-point fluctuations of the electromagnetic field make the electron jitter randomly by a small displacement $\delta\mathbf{r}$, with $\langle \delta r^2 \rangle \approx \frac{2\alpha}{\pi}\left(\frac{\hbar}{m_e c}\right)^2 \ln\frac{1}{\alpha}$. Expanding the Coulomb potential to second order, show that the average potential the electron feels shifts by $\frac{1}{6}\langle \delta r^2 \rangle \nabla^2 V$. Since $\nabla^2 V \propto \delta^3(\mathbf{r})$, only s states, which have $\psi(0) \neq 0$, are shifted. Estimate the shift of the $2s$ state and compare with the measured 1.06 GHz.

Further reading

- **Christopher J. Foot, *Atomic Physics*** — A clear modern introduction from the Oxford master's series, strong on laser cooling and precision measurement.
- **Budker, Kimball, and DeMille, *Atomic Physics: An Exploration Through Problems and Solutions*** — Learn the field by working its problems.
- **Gerry and Knight, *Introductory Quantum Optics*** — The standard first book on photons, coherent and squeezed states, and cavity QED.
- **Metcalf and van der Straten, *Laser Cooling and Trapping*** — The field's own textbook.

CHAPTER 19

Condensed matter

BUILDS ON statistical mechanics (Chapter 9), quantum mechanics (Chapter 14)

The emergent physics of matter in bulk — why copper conducts and glass doesn't, why iron is magnetic, how a transistor works, and why a superconductor floats — in the largest field in physics.

PROTOTYPE A one-dimensional chain of identical atoms. Its atoms can vibrate, and the quanta of those vibrations are *phonons*. Its electrons can hop from atom to atom, and their allowed energies spread into *bands* separated by gaps. Almost every idea in this chapter can first be seen on the chain.

There are about 10^{23} atoms in a sugar cube, and condensed matter physics is the study of what they do together. The fundamental laws — the Schrödinger equation and Coulomb's law — are known completely. Yet the collective behaviour they produce is endlessly surprising: metals, insulators, semiconductors, magnets, superconductors, superfluids, liquid crystals, and topological materials whose properties are quantized exactly. Condensed matter is the largest branch of physics, the source of nearly all modern technology, and the home of Anderson's "more is different" (Chapter 16).

§19.1 Crystals

Most solids are *crystals*: the same unit of atoms repeated on a regular lattice. Symmetry restricts the possibilities severely — in three dimensions there are only 14 types of lattice (the *Bravais lattices*) and 230 space groups of full crystal symmetries. A crystal's regularity can be probed by diffraction (Chapter 8). X-rays with wavelength comparable to the atomic spacing reflect strongly from planes of atoms when waves from successive planes add in phase:

$$2d \sin\theta = n\lambda \qquad \text{(Bragg's law)}.$$

In 1912 von Laue first diffracted X-rays from a crystal, proving both that X-rays are waves and that crystals are lattices. Diffraction has since revealed the structures of salt, diamond, DNA, and hundreds of thousands of proteins.

The mathematics of diffraction naturally lives in *reciprocal space*, the space of wavevectors $\mathbf{k}$: the diffraction pattern of a lattice is itself a lattice, the *reciprocal lattice*, the Fourier transform of the real one (Chapter 1). Its unit cell is the *Brillouin zone*, which we met on the beaded string (Chapter 5). Crystals were long believed to need translational symmetry; in 1982 Dan Shechtman found *quasicrystals*, with sharp diffraction but fivefold symmetry no periodic lattice can have, and was ridiculed for a decade before winning the 2011 Nobel Prize.

§19.2 Phonons

Atoms in a crystal vibrate about their equilibrium positions. For small vibrations the crystal is a set of coupled oscillators, and its normal modes (Chapter 4) are waves — sound waves at long wavelengths, with the dispersion relation of the beaded chain, $\omega = 2\sqrt{K/m}\,|\sin(ka/2)|$, as k runs across the Brillouin zone. Crystals with more than one atom per unit cell also have *optical* branches, in which neighbouring atoms move against each other.

Each normal mode is a quantum harmonic oscillator with energies $(n + \frac{1}{2})\hbar\omega$ (Chapter 14). A quantum of vibration is a *phonon*, which behaves in almost every way like a particle: it carries energy and momentum, scatters, and is in thermal equilibrium a boson with a Bose–Einstein distribution. Phonons are the first example of a *quasiparticle*: not a fundamental particle, but an excitation of a material that acts like one.

Phonons explain the thermal properties of solids. At high temperatures every mode is excited and the heat capacity reaches the classical Dulong–Petit value $3Nk_B$ (Chapter 9). At low temperatures, only modes with $\hbar\omega \lesssim k_B T$ can be excited, and since the number of such long-wavelength modes grows as T^3, so does the heat capacity — the **Debye T^3 law** (Problem 19.5). Phonons also carry heat (diamond, whose stiff light lattice has fast phonons, conducts heat five times better than copper), and the anharmonicity of the lattice, which lets phonons scatter, is why solids expand when heated.

§19.3 Electrons in metals

The Drude model. In 1900, just three years after the electron's discovery, Paul Drude modelled a metal as a gas of free electrons bouncing off fixed ions. An electric field accelerates them; collisions every time τ randomize their velocities; the result is a drift velocity proportional to the field, and a conductivity

$$\sigma = \frac{ne^2\tau}{m}.$$

Drude got Ohm's law, and the ratio of thermal to electrical conductivity right. But his model predicted a large electronic heat capacity that metals do not have, and it could not explain why the Hall effect (§19.8) has the wrong sign in some metals.

The Sommerfeld model. The fix was quantum statistics. Electrons are fermions, filling states up to the Fermi energy $E_F \sim 5\,\text{eV}$ (Chapter 9). Only electrons within $k_B T$ of the *Fermi surface* can be excited, which explains the small heat capacity. The electrons that carry current move at the *Fermi velocity*, about $10^6\,\text{m/s}$, not at thermal speeds, so their mean free path between collisions is tens of nanometres. And in a perfect crystal, it turns out, electrons do not scatter off the ions at all; they scatter only off imperfections and phonons. To understand why, we need band theory.

§19.4 Bands and gaps

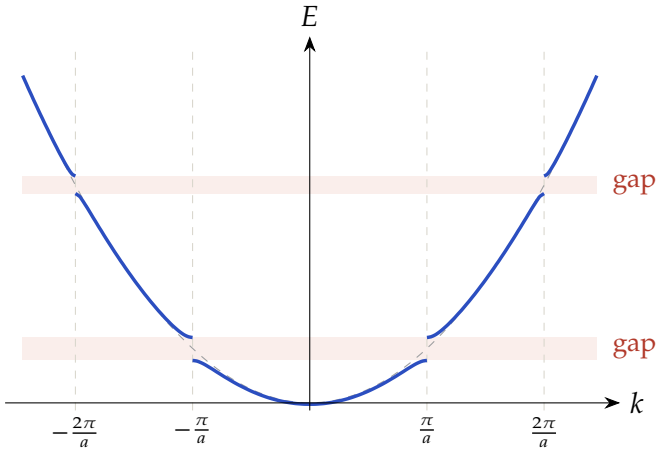
Bloch's theorem. In a periodic potential, the electron eigenstates can be chosen as plane waves modulated by functions with the period of the lattice:

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r}), \qquad u_{\mathbf{k}}(\mathbf{r}+\mathbf{a}) = u_{\mathbf{k}}(\mathbf{r}).$$

These *Bloch waves* extend through the whole crystal. An electron in a perfect crystal moves without resistance; resistance comes only from departures from perfect periodicity.

Two pictures show how *bands* and *gaps* appear.

Nearly free electrons. Start with free electrons, $E = \hbar^2 k^2 / 2m$, and turn on a weak periodic potential. Electrons with wavelengths satisfying the Bragg condition — at the edges of the Brillouin zone — are reflected by the lattice, forming standing waves. One standing wave piles electron density on the ions (lowering its energy), the other between them (raising it). An energy gap opens at the zone boundary.



Nearly free electrons. The dashed parabola is a free electron's energy. A weak periodic potential splits it wherever an electron's wave would be Bragg-reflected, at $k = \pm n\pi/a$, opening forbidden bands of energy (shaded) between the allowed ones.

Tight binding. Start instead from isolated atoms, each with an electron in a level E_0, and let electrons hop between neighbours with amplitude t. On a chain with spacing a, the eigenstates are Bloch waves with energies

$$E(k) = E_0 - 2t\cos(ka).$$

Each sharp atomic level broadens into a *band* of width $4t$. Different atomic levels give different bands, with gaps between them.

KEY IDEA Whether a material conducts depends on how its bands are filled. If the highest occupied band is partly filled, electrons can move into nearby empty states, and the material is a **metal**. If the bands are completely full, with a gap to the next empty band, no electron can change its state without a large energy jump, and the material is an **insulator**. If the gap is small, about 1 eV, a few electrons are thermally excited across it, and the material is a **semiconductor**.

Band theory also explains colour. Photons with energy below the gap pass through: diamond, with a 5.5 eV gap, is transparent; glass too. Photons above the gap are absorbed: silicon, with 1.1 eV, absorbs all visible light and looks grey and opaque.

Near the bottom of a band, $E(k)$ is parabolic, so electrons behave like free particles with an *effective mass* m^* set by the band's curvature — sometimes much lighter than a free electron. Near the top of a nearly full band, it is easier to track the few *missing* electrons. These *holes* act

like particles of positive charge and positive mass, and explain the "wrong-sign" Hall effect that baffled Drude.

§19.5 Semiconductors

Pure silicon at room temperature has only about 10^{16} free electrons per cubic metre — a trillion times fewer than a metal — because the number thermally excited across the gap goes as $e^{-E_g/2k_{\mathrm{B}}T}$. The magic of semiconductors is that this can be controlled. *Doping* with a few phosphorus atoms (five valence electrons) donates extra electrons, making *n*-type silicon; boron (three) creates holes, making *p*-type. One dopant atom per million can increase the conductivity a hundred-thousand-fold.

Put *p*-type and *n*-type silicon in contact and electrons and holes diffuse across and annihilate, leaving a *depletion region* with a built-in electric field. This *p–n junction* conducts easily one way and hardly at all the other: a *diode*. Run current through a junction of the right material and electrons falling across the gap emit light of energy E_g: an LED. Shine light on it and photons create electron–hole pairs that the field separates: a solar cell. And a *transistor*, a junction whose conductivity is controlled by a voltage on a third terminal, is a switch with no moving parts. Several billion of them sit in your phone. The information age is band theory, engineered.

§19.6 Magnetism

All matter responds to magnetic fields. **Diamagnetism** (weak repulsion) is universal: by Lenz's law, electron orbits adjust to oppose an applied field. **Paramagnetism** (weak attraction) occurs when atoms have permanent magnetic moments, which tend to align with a field against thermal agitation; the induced magnetization falls as $1/T$ (*Curie's law*), a direct application of the Boltzmann factor.

Ferromagnetism — the spontaneous alignment of moments in iron, cobalt, and nickel — is quite different, and cannot be explained by magnetic forces between atoms, which are thousands of times too weak. The cause is quantum mechanical: the *exchange interaction*. Two electrons with parallel spins must have an antisymmetric spatial wavefunction (Chapter 14), which keeps them apart and lowers their Coulomb repulsion. So electrostatics plus the Pauli principle makes aligned spins energetically favourable. Above the *Curie temperature* (1043 K for iron) thermal agitation wins and the magnetism vanishes, in a phase transition well described by the Ising model (Chapter 9). A ferromagnet breaks into *domains* of different magnetization to reduce its external field energy; magnetizing a piece of iron means growing the domains aligned with the field. Some materials prefer *antiparallel* neighbours and become antiferromagnets. The low-energy excitations of a magnet are spin waves, quantized as *magnons*.

§19.7 Superconductivity

In 1911 Heike Kamerlingh Onnes, having just liquefied helium, cooled mercury to 4.2 K and watched its resistance vanish. Not decrease: vanish. Currents in superconducting rings have flowed for years without measurable decay.

Zero resistance is only half the story. In 1933 Meissner and Ochsenfeld found that a superconductor *expels* magnetic fields: cool it in a field, and the field is pushed out. A superconductor is a perfect diamagnet, which is why a magnet floats above one. The field penetrates only a thin surface layer of depth $\lambda_L = \sqrt{m/\mu_0 n e^2}$, tens of nanometres. Inside, it is as if the photon has acquired a mass, exactly as in a plasma (Chapter 17) — the condensed matter ancestor of the Higgs mechanism (Chapter 32).

The explanation took until 1957. Bardeen, Cooper, and Schrieffer (BCS) showed that electrons near the Fermi surface feel a weak attraction mediated by phonons: an electron passing through the lattice pulls the positive ions toward it, and a second electron is attracted to the resulting positive wake. Electrons pair up into *Cooper pairs*, which are bosons, and condense into a single coherent quantum state (Chapter 9), separated from excited states by an energy gap. Scattering a pair out of the condensate costs energy, so at low temperature it cannot happen: no resistance.

The evidence for pairs is beautiful: the magnetic flux through a superconducting ring is quantized in units of

$$\Phi_0 = \frac{h}{2e} = 2.07 \times 10^{-15}\,\text{Wb},$$

with $2e$, the charge of a pair. Two superconductors separated by a thin barrier exhibit the *Josephson effect*, a supercurrent that tunnels across; Josephson junctions make SQUIDs, the most sensitive magnetic field detectors in existence, and the superconducting qubits of quantum computers (Chapter 20).

In 1986 Bednorz and Müller discovered superconductivity in copper oxides at 35 K, and within a year the record stood at 93 K, warmer than liquid nitrogen. These *high-T_c* superconductors do not follow BCS theory, and nearly four decades later their mechanism is still not settled. Superconducting magnets run MRI scanners (Chapter 24) and particle accelerators (Chapter 31).

§19.8 The quantum Hall effect and topology

Run a current along a conducting strip in a perpendicular magnetic field, and the Lorentz force pushes carriers to one side, producing a transverse voltage: the classical *Hall effect*. Its sign reveals whether the carriers are electrons or holes, and its size gives their density.

In 1980 Klaus von Klitzing measured the Hall effect in a two-dimensional layer of electrons at low temperature and high field, and found that the Hall resistance does not vary smoothly: it forms flat plateaus at exactly

$$R_H = \frac{h}{\nu e^2}, \qquad \nu = 1, 2, 3, \ldots$$

"Exactly" means to better than one part in a billion, in samples full of impurities and of irregular shape. Such precision demanded a deep explanation, and it was found in *topology*. The Hall conductance turns out to be a topological invariant of the electrons' quantum states — an integer, like the number of holes in a doughnut, which cannot change under small deformations. Dirt cannot change it any more than squeezing a doughnut can remove its hole. The value h/e^2 is now used as a resistance standard.

At even higher fields, plateaus appear at *fractional* ν like $\frac{1}{3}$, where electrons interact so strongly that they collectively form a new state whose quasiparticles carry a fraction of an electron's charge and obey neither Bose nor Fermi statistics (*anyons*). Since 2005, *topological insulators* have been found: materials that are insulating inside but must conduct on their surfaces, protected by topology. The topological classification of phases of matter, recognized by the 2016 Nobel Prize, is one of the great conceptual advances of modern physics.

§19.9 Soft matter

Not all condensed matter is crystalline. *Soft matter* — polymers, liquid crystals, colloids, gels, foams, and granular materials — is held together by interactions comparable to $k_{\mathrm{B}}T$, so it is easily deformed and strongly affected by thermal fluctuations and entropy. Rubber's elasticity is entropic (Chapter 7). Liquid crystals, whose rod-shaped molecules align in direction but not position, are in every LCD screen. Colloidal particles can crystallize purely because of entropy: at high density, ordered packing leaves each particle *more* room to jiggle than disorder does. Sand flows like a liquid but piles like a solid. Living cells are soft matter (Chapter 23).

§19.10 Quasiparticles and emergence

The deepest lesson of condensed matter is that the low-energy excitations of a complex system often behave like simple particles in a new "vacuum." Phonons, holes, magnons, Cooper pairs, and anyons are all quasiparticles. In *graphene*, a single sheet of carbon atoms in a honeycomb isolated in 2004, the electrons near the Fermi level have a linear dispersion $E = \hbar v_F\,|\mathbf{k}|$ — they behave as massless relativistic particles moving at $c/300$, obeying a version of the Dirac equation (Chapter 30). A pencil trace contains a toy universe. The same mathematics, quantum field theory, describes elementary particles and the excitations of solids, and ideas flow constantly in both directions.

§19.11 Problems

19.1 $O(\epsilon)$ ★ X-rays of wavelength 0.154 nm reflect from planes of atoms 0.282 nm apart in a salt crystal. At what angles do the first two Bragg reflections occur? Why can't visible light be used for this?

19.2 $O(\epsilon)$ Copper has conductivity 6×10^{7} S/m and 8.5×10^{28} free electrons per cubic metre. Using the Drude formula, find the collision time τ. With the Fermi velocity 1.6×10^{6} m/s, find the mean free path. How many atomic spacings is it?

19.3 $O(\epsilon)$ ★ Silicon, gallium nitride, cadmium sulfide, and diamond have band gaps of 1.1, 3.4, 2.4, and 5.5 eV. Predict the appearance of each (transparent, coloured, or opaque), and what colour of LED each could make.

19.4 $O(\epsilon^{2})$ ★ For the tight-binding band $E(k) = E_0 - 2t\cos(ka)$, find the bandwidth and the group velocity $v = \hbar^{-1}\,\mathrm{d}E/\,\mathrm{d}k$. Near the band bottom, expand to find the effective mass

$m^* = \hbar^2/(2ta^2)$. What happens to the group velocity at the zone boundary, and to the effective mass near the top of the band?

19.5 $O(\epsilon^2)$ ★ (*The Debye T^3 law.*) At low temperature only phonons with $\hbar\omega \lesssim k_B T$ are excited. With $\omega = vk$, show that the number of such modes in a crystal of volume V scales as $V(k_B T/\hbar v)^3$, each carrying energy $\sim k_B T$. Deduce that the heat capacity scales as T^3. Why does the same argument applied to photons give the Stefan–Boltzmann T^4 law?

19.6 $O(\epsilon^2)$ The intrinsic carrier density of a semiconductor scales roughly as $e^{-E_g/2k_B T}$. For silicon, by what factor does it increase between 300 K and 350 K? Why do electronic devices need cooling, and why is doping essential?

19.7 $O(\epsilon^2)$ An electron spin in a 1 T field at 300 K: compute the ratio $\mu_B B/k_B T$ and hence the fractional net alignment of a paramagnet. At what temperature would the alignment reach about 50%? (This is why magnetic resonance signals are so weak, Chapter 24.)

19.8 $O(\epsilon^2)$ ★ Compute the flux quantum $h/2e$. A SQUID can detect a thousandth of a flux quantum through a loop of area 1 mm^2. What magnetic field does this correspond to? Compare with the Earth's field and with the magnetic fields of the human brain, about 10^{-13} T.

19.9 $O(\epsilon^2)$ Estimate the London penetration depth for a superconductor with a pair density of $10^{28}\ \mathrm{m}^{-3}$ (treat pairs as particles of mass $2m_e$ and charge $2e$).

19.10 $O(\epsilon^2)$ A copper strip 1 cm wide and 0.1 mm thick carries 10 A in a 1 T field. Find the Hall voltage, using $n = 8.5\times10^{28}\ \mathrm{m}^{-3}$. Why is the Hall effect much easier to measure in semiconductors?

19.11 $e^{-1/\epsilon}$ ★ (*A gap from a dimerized chain.*) In a chain of atoms whose hopping amplitudes alternate, $t_1, t_2, t_1, t_2, \ldots$, the unit cell has two atoms. Write Bloch states with amplitudes (a, b) on the two sublattices and show that the energies are

$$E(k) = \pm\left|t_1 + t_2 e^{ik\cdot 2a}\right|,$$

so a gap of size $2\,|t_1 - t_2|$ opens. (Peierls showed a one-dimensional metal will spontaneously distort this way to open a gap and lower its energy. The two dimerized patterns, $t_1 > t_2$ and $t_1 < t_2$, turn out to be topologically distinct, and the boundary between them hosts a zero-energy state.)

19.12 $e^{-1/\epsilon}$ (*Why the Hall conductance is quantized.*) Here is Laughlin's argument. Roll the electron layer into a cylinder, with the magnetic field through its surface, and slowly thread an extra magnetic flux $\Phi_0 = h/e$ through the cylinder's hollow. Argue that this returns the system to an equivalent state while transporting an integer number ν of electrons from one edge to the other. Using Faraday's law for the induced voltage around the cylinder, show that the Hall conductance is $\nu e^2/h$.

Further reading

- **Steven H. Simon,** ***The Oxford Solid State Basics*** — A short, clear, and funny introduction. The best place to start.
- **Charles Kittel,** ***Introduction to Solid State Physics*** — The classic undergraduate text, encyclopaedic in scope.
- **Ashcroft and Mermin,** ***Solid State Physics*** — The graduate standard for half a century, still unmatched for the single-electron theory of solids.
- **Girvin and Yang,** ***Modern Condensed Matter Physics*** — A modern graduate text with topology, the quantum Hall effect, and superconductivity done properly.

CHAPTER 20

Quantum information

BUILDS ON quantum mechanics (Chapter 14)

Turning the weirdest features of quantum mechanics — superposition, entanglement, the impossibility of copying — into resources for computing, communication, and cryptography.

> **PROTOTYPE** The entangled pair: two qubits in the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Measure either one and you get 0 or 1 at random; measure both and they always agree, along any axis, however far apart they are. Neither qubit has a state of its own. All the information is in the correlations.

For most of the twentieth century, the paradoxes of quantum mechanics — superposition, entanglement, measurement — were treated as philosophical puzzles to be explained away. Starting in the 1980s, physicists and computer scientists asked a different question: what can you *do* with them? The answer turned out to be: quite a lot. Quantum information science treats quantum mechanics not as a strange theory of nature but as a new theory of information, with its own units, its own rules, and its own possibilities. It has produced provably secure cryptography, algorithms that would break most of today's encryption, and a global race to build quantum computers.

§20.1 Qubits

A classical bit is 0 or 1. A **qubit** is a two-level quantum system (Chapter 14) in a state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Since an overall phase has no physical meaning, a qubit state can be written as $\cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$, a point on the surface of a sphere, the **Bloch sphere**: $|0\rangle$ at the north pole, $|1\rangle$ at the south, and superpositions everywhere else. The states $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$ sit on the equator. A qubit seems to contain a continuous infinity of information — two real angles — but measuring it yields a single bit, 0 with probability $|\alpha|^2$ or 1 with probability $|\beta|^2$, and destroys the superposition. The richness is in what can be done *before* measurement.

Physically, a qubit can be an electron spin, the polarization of a photon, two energy levels of an atom or ion, or the lowest two states of a superconducting circuit.

§20.2 Many qubits and entanglement

Two qubits have four basis states, $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, and their general state is a superposition of all four. n qubits need 2^n complex amplitudes. Three hundred qubits require more amplitudes than there are atoms in the observable universe. This exponential growth is why quantum systems are so hard to simulate on classical computers — and why Richard Feynman suggested in 1981 that we simulate them with quantum computers instead.

Most of these states are *entangled*: they cannot be written as a product of single-qubit states. The four **Bell states**,

$$|\Phi^{\pm}\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad |\Psi^{\pm}\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}},$$

form a basis of maximally entangled states. Measure the first qubit of $|\Phi^+\rangle$ and you get 0 or 1 at random; the second then always gives the same. This is not like a pair of gloves, one left and one right, separated in boxes: as Bell showed (§20.6), no assignment of pre-existing values reproduces the correlations measured along *different* axes.

§20.3 You cannot copy a qubit

Theorem 20.1 (No-cloning)**.** There is no physical process that takes an unknown quantum state $|\psi\rangle$ and produces two copies, $|\psi\rangle|\psi\rangle$.

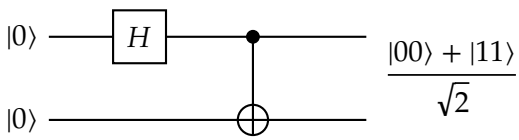
The proof is two lines, and uses only linearity (Problem 20.4). Suppose a machine copies $|0\rangle$ and $|1\rangle$ correctly. Then by linearity it must turn $\alpha|0\rangle + \beta|1\rangle$ into $\alpha|00\rangle + \beta|11\rangle$ — an entangled state, not two copies. No-cloning is why entanglement cannot transmit signals faster than light (Bob cannot make many copies of his qubit to find out what Alice measured), and why quantum cryptography is secure (an eavesdropper cannot copy the key).

§20.4 Gates and circuits

A quantum computation is a sequence of *gates*: unitary operations (Chapter 14) acting on one or two qubits at a time. Because they are unitary, quantum gates are reversible. Important single-qubit gates include the Pauli gates X (bit flip, $|0\rangle \leftrightarrow |1\rangle$) and Z (phase flip, $|1\rangle \to -|1\rangle$), and the **Hadamard** gate

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle,$$

which turns a definite bit into an equal superposition and back again. The essential two-qubit gate is the **controlled-NOT** (CNOT), which flips the second qubit if and only if the first is 1. A Hadamard followed by a CNOT turns two unentangled qubits into a Bell state:



A small set of gates — for example H, CNOT, and a rotation by $\pi/8$ — is *universal*: any unitary operation on any number of qubits can be approximated by a circuit built from them.

§20.5 Protocols

Superdense coding. Alice and Bob share a Bell pair. By applying one of four operations (I, X, Z, or XZ) to her qubit alone, Alice can turn the pair into any of the four Bell states. She sends her qubit to Bob, who measures both in the Bell basis and learns which. One qubit has carried two classical bits.

Teleportation. Alice holds an unknown qubit $|\psi\rangle$ and shares a Bell pair with Bob. She makes a joint Bell measurement on $|\psi\rangle$ and her half of the pair, and phones Bob the result: two classical bits. Bob applies a correction (I, X, Z, or XZ) to his half, which is now exactly $|\psi\rangle$ (Problem 20.6). The original is destroyed in the process, as no-cloning demands; nothing travels faster than light, since Bob must wait for the phone call. Teleportation is routine in laboratories and has been done between the ground and a satellite over 1200 km.

Quantum key distribution. In the BB84 protocol (Bennett and Brassard, 1984), Alice sends single photons, each randomly prepared in one of two bases ($|0\rangle, |1\rangle$ or $|+\rangle, |-\rangle$). Bob measures each in a randomly chosen basis. Afterwards they announce publicly which bases they used — not the results — and keep only the bits where the bases matched, giving a shared secret key. An eavesdropper who measures the photons must guess the basis, and half the time guesses wrong, disturbing the state and introducing errors at a rate of 25% in the kept bits (Problem 20.10). Alice and Bob detect her by comparing a sample. The security rests on the laws of physics, not on the presumed difficulty of a mathematical problem.

§20.6 Bell's inequality, sharpened

Suppose Alice can measure one of two quantities A or A', and Bob one of B or B', each with outcomes ± 1. If the outcomes are determined in advance by some local hidden variables, then for every run, $A(B+B') + A'(B-B')$ equals ± 2 (one bracket is zero and the other is ± 2). Averaging gives the **CHSH inequality**:

$$S = |\langle AB\rangle + \langle AB'\rangle + \langle A'B\rangle - \langle A'B'\rangle| \le 2.$$

For a Bell pair, with spin measurements along suitably chosen axes, quantum mechanics predicts $S = 2\sqrt{2} \approx 2.83$ (Problem 20.8). Experiments find 2.8. No local hidden variable theory can reproduce the predictions of quantum mechanics. In 2015 three groups closed the remaining loopholes, and Aspect, Clauser, and Zeilinger shared the 2022 Nobel Prize for the decades of experiments that led there. Remarkably, quantum mechanics itself cannot exceed $2\sqrt{2}$ (*Tsirelson's bound*, Problem 20.13), though logic alone would allow S up to 4. Why nature stops exactly at $2\sqrt{2}$ is a deep question.

§20.7 Quantum algorithms

A quantum computer is not a faster classical computer. It cannot simply try all answers at once and read out the right one: measurement returns a single random outcome. The trick of every

quantum algorithm is to arrange *interference* so that amplitudes for wrong answers cancel and amplitudes for right answers add up.

Deutsch's problem. Given a black box computing $f : \{0,1\} \to \{0,1\}$, decide whether f is constant or balanced. Classically, you must query it twice. Quantumly, once suffices, by querying a superposition and letting the two branches interfere (Problem 20.12). It is a toy, but the first proof that quantum computers can do something classical ones cannot.

Grover's search. To find a marked item among N unsorted items, a classical computer needs about $N/2$ guesses. Grover's algorithm (1996) needs only about $\frac{\pi}{4}\sqrt{N}$, by repeatedly reflecting the state about the marked item and about the average, rotating it step by step toward the answer. A quadratic speedup, applicable to any search problem.

Shor's algorithm. In 1994 Peter Shor showed that a quantum computer could factor large numbers exponentially faster than any known classical method. Factoring reduces to finding the period of a function, and a quantum computer can find periods efficiently using the *quantum Fourier transform*, which applies the Fourier transform (Chapter 1) to 2^n amplitudes with only about n^2 gates. The security of the RSA encryption that protects most internet traffic rests on factoring being hard. Shor's algorithm launched the field, and governments are already replacing RSA with "post-quantum" cryptography in anticipation.

Quantum simulation. Feynman's original motivation remains perhaps the most important application: simulating molecules (Chapter 21), materials (Chapter 19), and nuclear matter, whose quantum states are exponentially hard to represent classically.

§20.8 Decoherence and mixed states

Real qubits are fragile. Their interactions with the environment — stray fields, vibrations, a single photon — entangle them with the outside world, and the delicate phase relationships that make a superposition quantum leak away. This is *decoherence* (Chapter 14).

To describe a qubit that may be entangled with things we are not tracking, we use the *density matrix* ρ. A pure state $|\psi\rangle$ has $\rho = |\psi\rangle\langle\psi|$. A random mixture of states $|\psi_i\rangle$ with probabilities p_i has $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$. The state $|+\rangle$ and a 50/50 coin-flip mixture of $|0\rangle$ and $|1\rangle$ give identical statistics when measured in the $\{|0\rangle, |1\rangle\}$ basis, but very different ones in the $\{|+\rangle, |-\rangle\}$ basis; their density matrices differ in the off-diagonal elements, the *coherences*. Decoherence is the decay of those off-diagonal elements. The time it takes, T_2, ranges from microseconds for superconducting qubits to many seconds for trapped ions.

§20.9 Quantum error correction

Classical computers fight errors by copying: store each bit three times and take a majority vote. Quantum computers cannot copy (no-cloning), cannot look at their data without destroying it, and suffer continuous rather than discrete errors. For a while it seemed quantum computing was doomed. Then in 1995 Shor showed that quantum error correction is possible after all.

The idea, in the simplest *bit-flip code*, is to encode one logical qubit as $\alpha|000\rangle + \beta|111\rangle$ — entangled, not copied. If one physical qubit flips, measuring the *parities* Z_1Z_2 and Z_2Z_3 reveals which one, without revealing anything about α and β, and the error can be undone (Problem 20.11). More elaborate codes protect against all errors. The **threshold theorem** says that if each phys-

ical gate fails with probability below some threshold (about 1% for the leading *surface code*), then arbitrarily long quantum computations are possible by adding more physical qubits per logical one. In 2023 and 2024, experiments demonstrated logical qubits whose error rates *fall* as the code is made bigger — crossing the threshold for the first time.

§20.10 Building quantum computers

Several physical platforms are in a tight race:

- **Superconducting circuits**: tiny loops of superconductor interrupted by Josephson junctions (Chapter 19), cooled to 10 mK, behaving as artificial atoms. Fast gates, short coherence, fabricated like computer chips.
- **Trapped ions**: individual atoms held in electromagnetic traps and manipulated with lasers (Chapter 18). Superb fidelity and long coherence, but slower gates.
- **Neutral atoms** held in arrays of optical tweezers and entangled via highly excited Rydberg states; hundreds to thousands of qubits.
- **Photons**, natural carriers of quantum information over long distances; and **spins** in silicon, which could ride on existing semiconductor manufacturing.

Today's machines have hundreds to a few thousand physical qubits and the first handfuls of error-corrected logical qubits. Useful fault-tolerant quantum computing will require millions of physical qubits, and its arrival date is an open question. Meanwhile, quantum *sensors* — atomic clocks, magnetometers, interferometers — already outperform classical ones, and quantum networks are being built.

§20.11 Problems

20.1 $O(\epsilon)$ ★ Locate the states $|0\rangle$, $|1\rangle$, $|+\rangle$, $|-\rangle$, and $(|0\rangle + i|1\rangle)/\sqrt{2}$ on the Bloch sphere. Show that $H^2 = I$, and describe H geometrically as a rotation of the Bloch sphere.

20.2 $O(\epsilon)$ Two qubits are in the state $|\Phi^+\rangle$. Show that measuring both in the $\{|0\rangle, |1\rangle\}$ basis always gives equal results. Rewrite $|\Phi^+\rangle$ in the $\{|+\rangle, |-\rangle\}$ basis and show that the results also always agree there.

20.3 $O(\epsilon)$ How many complex amplitudes describe 50 qubits? If each is stored as 16 bytes, how much memory would a classical simulation need? What about 300 qubits?

20.4 $O(\epsilon^2)$ ★ Prove the no-cloning theorem: assume a unitary U satisfies $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for all $|\psi\rangle$, take the inner product of this equation for two states $|\psi\rangle$ and $|\phi\rangle$, and show that $\langle\psi|\phi\rangle$ must be 0 or 1.

20.5 $O(\epsilon^2)$ Verify superdense coding: show that applying I, X, Z, and XZ to the first qubit of $|\Phi^+\rangle$ produces the four distinct Bell states.

20.6 $O(\epsilon^2)$ ★ (*Teleportation.*) Write the three-qubit state $(\alpha|0\rangle + \beta|1\rangle) \otimes |\Phi^+\rangle$, where Alice holds the first two qubits and Bob the third. Rewrite Alice's two qubits in the Bell basis and show that the total state is

$$\tfrac{1}{2}\Big[|\Phi^+\rangle(\alpha|0\rangle + \beta|1\rangle) + |\Phi^-\rangle(\alpha|0\rangle - \beta|1\rangle) + |\Psi^+\rangle(\beta|0\rangle + \alpha|1\rangle) + |\Psi^-\rangle(\alpha|1\rangle - \beta|0\rangle)\Big].$$

For each of Alice's four outcomes, what correction must Bob apply?

20.7 $O(\epsilon^2)$ Run Grover's algorithm by hand for $N = 4$ items: start in the uniform superposition, flip the sign of the marked item's amplitude, then reflect all amplitudes about their average. Show that after one iteration the marked item has probability exactly 1.

20.8 $O(\epsilon^2)$ ★ For the singlet state, the correlation between spin measurements along unit vectors $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ is $\langle AB\rangle = -\hat{\mathbf{a}}\cdot\hat{\mathbf{b}}$. Choose $\hat{\mathbf{a}}$, $\hat{\mathbf{a}}'$, $\hat{\mathbf{b}}$, $\hat{\mathbf{b}}'$ in a plane at angles 0°, 90°, 45°, and -45°, and show that $S = 2\sqrt{2}$. (Try $\hat{\mathbf{b}}'$ at 135° instead: what goes wrong?)

20.9 $O(\epsilon^2)$ Write the density matrices for the pure state $|+\rangle$ and for an equal mixture of $|0\rangle$ and $|1\rangle$. Show that they give the same statistics for Z-measurements but different ones for X-measurements, and that $\operatorname{tr}\rho^2$ is 1 for the first and $\tfrac{1}{2}$ for the second.

20.10 $O(\epsilon^2)$ In BB84, an eavesdropper intercepts every photon, measures it in a randomly chosen basis, and resends what she found. Show that in the bits Alice and Bob keep, the error rate is 25%.

20.11 $e^{-1/\epsilon}$ ★ (*The three-qubit code.*) A logical qubit is encoded as $\alpha|000\rangle + \beta|111\rangle$, and then one of the three qubits may be flipped by X. Show that measuring the observables Z_1Z_2 and Z_2Z_3 (each ±1) identifies which qubit, if any, was flipped, and that the measurement leaves α and β untouched. Why does this code fail if two qubits flip, and why doesn't it protect against phase errors?

20.12 $e^{-1/\epsilon}$ (*Deutsch's algorithm.*) A black box implements $|x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle$. Prepare $|0\rangle|1\rangle$, apply H to both qubits, apply the black box once, then apply H to the first qubit and measure it. Show that the result is 0 if f is constant and 1 if it is balanced.

Hint: Show that the box acting on $|x\rangle|-\rangle$ gives $(-1)^{f(x)}|x\rangle|-\rangle$: the function's value has been "kicked back" into a phase.

20.13 $e^{-1/\epsilon}$ (*Tsirelson's bound.*) Let A, A', B, B' be observables with eigenvalues ±1, so $A^2 = B^2 = 1$, with Alice's operators commuting with Bob's. Define $C = AB + AB' + A'B - A'B'$. Show that $C^2 = 4 - [A, A'][B, B']$, and use $\|[A, A']\| \le 2$ to conclude that $|\langle C\rangle| \le 2\sqrt{2}$.

Further reading

- **Nielsen and Chuang, *Quantum Computation and Quantum Information*** — The field's standard textbook, comprehensive and still the reference.
- **N. David Mermin, *Quantum Computer Science: An Introduction*** — A short, elegant introduction for those who know some linear algebra but no quantum mechanics.

- **John Preskill, lecture notes for Caltech Ph219** — Deep, clear, and free online.
- **Scott Aaronson, *Quantum Computing Since Democritus*** — A witty tour of quantum computing and complexity theory.

CHAPTER 21

Chemical physics

BUILDS ON quantum mechanics (Chapter 14)

Where quantum mechanics turns into chemistry: why atoms stick together, why molecules have the shapes they do, and why reactions happen at the rates they do.

PROTOTYPE The hydrogen molecular ion, H_2^+: two protons and one electron. The electron's wave can be built from an orbital around each proton, added either in phase or out of phase. In phase, the two waves reinforce between the protons, the electron's charge piles up there, and it glues the protons together. That constructive interference *is* the chemical bond.

"The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known," wrote Dirac in 1929, "and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble." Chemistry is quantum mechanics of electrons in the field of nuclei. But the difficulty Dirac mentioned is real: the Schrödinger equation for a molecule with N electrons lives in $3N$ dimensions, and exact solutions are out of reach for anything bigger than a few atoms. Chemical physics is the art of approximations — spherical cows of extraordinary power — that turn the equations into understanding.

§21.1 The chemical bond

Take H_2^+. With the protons far apart, the electron sits in a 1s orbital ψ_A around proton A, or ψ_B around B, with the same energy. Bring the protons together and the electron can hop between them, so — exactly as for the tight-binding chain of Chapter 19 or the coupled pendulums of Chapter 1 — the true states are the symmetric and antisymmetric combinations:

$$\psi_\pm = \frac{\psi_A \pm \psi_B}{\sqrt{2(1 \pm S)}},$$

where $S = \langle \psi_A | \psi_B \rangle$ is their overlap. This is the method of *linear combinations of atomic orbitals*, LCAO.

In the symmetric *bonding orbital* ψ_+, the two waves interfere constructively in the region between the protons. The electron spends more time there, where it is attracted by both nuclei at once, and its energy is lowered below that of a separated atom. In the antisymmetric *antibonding orbital* ψ_-, there is a node between the protons, the electron is pushed to the outside, and the energy rises. Adding the repulsion of the protons gives a potential energy curve for ψ_+ with a

minimum at a definite *bond length* and *bond energy*. A bond is a standing wave that has found a way to lower its energy.

KEY IDEA Combining two orbitals always produces one that is lower in energy and one that is higher, and the antibonding orbital is raised slightly more than the bonding one is lowered. Two electrons (opposite spins) in a bonding orbital make a stable bond. Two more in the antibonding orbital cancel it — which is why H_2 exists and He_2 does not.

The **Born–Oppenheimer approximation** underlies all of this. Nuclei are thousands of times heavier than electrons, so electrons adjust instantly to wherever the nuclei are. Solve for the electrons with the nuclei clamped at positions $\mathbf{R}$; the electronic energy $E(\mathbf{R})$ then serves as the potential energy surface on which the nuclei move. Molecular vibrations are small oscillations in this landscape (Chapters 3 and 18), and a chemical reaction is a journey across it, from one valley to another, usually over a mountain pass called the *transition state*.

§21.2 A zoo of bonds

Chemists distinguish bonds by how the electrons are shared, but they are all electromagnetic in origin, and they span a factor of a thousand in strength.

- **Covalent bonds** (a few eV): electrons shared in a bonding orbital, as in H_2, the C–C bonds of organic chemistry, and diamond.
- **Ionic bonds** (a few eV): one atom gives up an electron entirely to another, and the resulting ions attract electrostatically. Forming Na^+Cl^- costs 5.1 eV to ionize sodium but recovers 3.6 eV when chlorine accepts the electron and about 6 eV from the Coulomb attraction at the bond length: net binding.
- **Metallic bonds**: electrons delocalized over the whole crystal, a sea of shared electrons (Chapter 19).
- **Hydrogen bonds** (about 0.2 eV): a hydrogen atom attached to a strongly electron-hungry atom (oxygen, nitrogen) is left slightly positive and attracts a lone pair of electrons on a neighbouring molecule. Hydrogen bonds hold the two strands of DNA together, fold proteins (Chapter 23), and give water its anomalously high boiling point and the low density of ice.
- **Van der Waals forces** (about 0.01–0.1 eV): even neutral, nonpolar atoms attract. The electron cloud of one atom fluctuates, creating a momentary dipole that polarizes its neighbour, and the two fluctuating dipoles attract with an energy falling as $1/r^6$ (Problem 21.11). These *London dispersion forces* are why noble gases liquefy at all, and why a gecko can walk up a glass wall on millions of microscopic hairs.

At room temperature $k_B T \approx 0.025\,\text{eV}$. Covalent and ionic bonds are far stronger, so molecules persist; hydrogen bonds are only about ten times stronger, so they constantly break and re-form. Life uses both: strong bonds for structure, weak bonds for function.

§21.3 The shapes of molecules

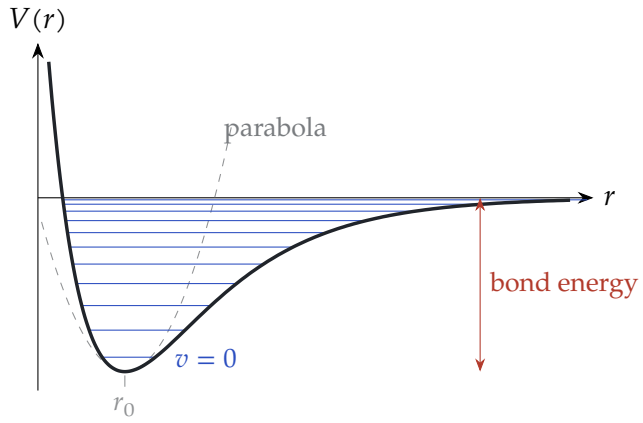
Carbon has four electrons in its outer shell, in one 2*s* and three 2*p* orbitals. In methane, CH_4, all four bonds are identical and point to the corners of a tetrahedron, 109.5° apart. The explanation is *hybridization*: it is energetically favourable to mix the 2*s* and 2*p* orbitals into four equivalent sp^3 hybrids that point away from each other. Mixing *s* with two *p* orbitals gives three sp^2 hybrids in a plane, 120° apart, leaving one *p* orbital perpendicular; this is the geometry of ethylene, benzene, and graphene. Mixing with one gives two linear *sp* hybrids.

The leftover *p* orbitals can overlap sideways to form π bonds. In molecules with alternating single and double bonds, these π electrons are delocalized along the whole chain — they behave like particles in a box (Chapter 14). The longer the box, the closer the energy levels, and the longer the wavelength of light absorbed. Short chains absorb in the ultraviolet and are colourless; β-carotene, with eleven conjugated double bonds, absorbs blue light and is orange, which is why carrots are (Problem 21.6). The retinal molecule in your eye is a close relative.

Atoms differ in how strongly they pull on shared electrons (their *electronegativity*), so bonds can be polar, and the molecule's shape decides whether the polarities cancel. Carbon dioxide is linear and has no dipole moment. Water is bent at 104.5°, with a large dipole moment, which makes it the universal solvent: it surrounds ions with oriented molecules that screen their charges (its dielectric constant is 80; Chapter 6).

§21.4 Spectroscopy

Molecules announce their structure through the light they absorb (Chapter 18). Ultraviolet and visible spectra probe electronic transitions. Infrared spectra probe vibrations, and since each functional group (C–H, O–H, C=O) vibrates at a characteristic frequency, an infrared spectrum is a fingerprint. *Raman* spectroscopy sees vibrations through the slight frequency shifts of scattered laser light. *Nuclear magnetic resonance* (NMR) uses the precession of nuclear spins in a magnetic field (Chapter 14); the precise frequency depends slightly on the electrons around each nucleus, shifting by a few parts per million, so NMR reveals which atoms are bonded to which. The same physics, spatially encoded, is MRI (Chapter 24).



The potential energy of a diatomic molecule against the distance between its nuclei, with its vibrational levels. Near the bottom the well is almost a parabola (dashed), so the levels are evenly spaced, as for a harmonic oscillator. Higher up the well widens, the levels crowd together, and at the top the molecule falls apart.

§21.5 Will it react? Thermodynamics

Whether a reaction can happen is decided by the change in Gibbs free energy (Chapter 7),

$$\Delta G = \Delta H - T\Delta S.$$

A reaction proceeds spontaneously when $\Delta G < 0$. At equilibrium the ratio of products to reactants is fixed by the *equilibrium constant*

$$K = e^{-\Delta G^\circ / RT},$$

where ΔG° is the free energy change under standard conditions. Every 5.7 kJ/mol of ΔG° at room temperature is a factor of ten in K. This is the Boltzmann factor (Chapter 9) applied to whole populations of molecules, and indeed K can be computed from the partition functions of the reactants and products. Raising the temperature favours whichever direction absorbs heat (the *van 't Hoff equation*, Problem 21.10) — one precise form of Le Chatelier's principle.

§21.6 How fast? Kinetics

Thermodynamics says whether; kinetics says how fast. A mixture of hydrogen and oxygen can sit for years, though ΔG is hugely negative, until a spark sets it off. The molecules must first climb over an energy barrier, the *activation energy* E_a, and the fraction with enough energy to do so is a Boltzmann factor. This gives the **Arrhenius law**:

$$k = A\, e^{-E_a/RT}.$$

With a typical activation energy of 50 kJ/mol, reaction rates roughly double for every 10°C rise in temperature — which is why food keeps in the fridge, why fevers matter, and why cold-blooded animals slow down in the cold.

Transition state theory (Eyring, 1935) refines this by treating the molecules at the top of the barrier as a special species in equilibrium with the reactants. *Catalysts* speed reactions by providing

an alternative path with a lower barrier, without changing ΔG or the equilibrium. Enzymes, nature's catalysts, can speed reactions by factors of 10^{10} or more. And because nuclei are quantum particles, some reactions — especially those moving hydrogen — go partly by *tunnelling* through the barrier rather than over it (Chapter 14), which shows up as a large difference in rate between hydrogen and its heavier isotope deuterium (Problem 21.9).

§21.7 Computing chemistry

The many-electron Schrödinger equation cannot be solved exactly, so computational chemistry is a hierarchy of approximations.

- **Hartree–Fock**: each electron moves in the average field of all the others, with antisymmetry built in. A mean-field theory (like Chapter 9's Ising approximation), it gets about 99% of the total energy — but the missing 1%, the *correlation energy*, is the same size as chemical bond energies, so it is not good enough on its own.
- **Correlated methods** add back the correlations systematically, at rapidly growing cost.
- **Density functional theory** (Kohn and Sham, 1965) proves that the ground-state energy is a functional of the electron density alone — a function of three variables instead of $3N$. The exact functional is unknown, but good approximations make DFT accurate enough for most chemistry and materials science at reasonable cost. It is among the most widely used methods in all of science.

Machine learning is now changing the field again: AlphaFold predicts protein structures (Chapter 23), and neural networks trained on quantum calculations can emulate them at a fraction of the cost (Chapter 13). Quantum computers may eventually solve molecular problems that no classical computer can (Chapter 20).

§21.8 Light-driven chemistry

When a molecule absorbs a photon, an electron is promoted to an antibonding orbital, weakening a bond and sometimes changing the molecule's shape. In your eye, absorbing a single photon flips the retinal molecule from a bent to a straight form in about 200 femtoseconds — one of the fastest chemical reactions known — and that change in shape triggers the nerve signal of vision. In photosynthesis, chlorophyll absorbs red and blue light (reflecting the green), and the excitation hops between pigment molecules to a reaction centre with nearly 100% efficiency, where it drives the separation of charge that ultimately powers almost all life on Earth. High in the atmosphere, ozone absorbs ultraviolet light between 200 and 300 nm, shielding the surface (Chapter 25).

§21.9 Problems

21.1 $O(\epsilon)$ The bond energy of H_2 is 4.5 eV. Convert this to kJ/mol. At what temperature would the Boltzmann factor $e^{-E/k_B T}$ for breaking the bond reach 10^{-10}? Why is hydrogen gas stable at room temperature?

21.2 $O(\epsilon)$ ★ A reaction has activation energy 50 kJ/mol. By what factor does its rate increase from 27°C to 37°C? From 0°C to 100°C?

21.3 $O(\epsilon)$ A reaction has $\Delta G^\circ = -20\,\mathrm{kJ/mol}$ at 298 K. Find its equilibrium constant. How negative must ΔG° be for the reaction to go 99.9% to completion (with K defined as a simple product-to-reactant ratio)?

21.4 $O(\epsilon)$ Estimate the binding energy of an NaCl molecule (bond length 0.236 nm) from the Coulomb attraction of the ions, the ionization energy of sodium (5.1 eV), and the electron affinity of chlorine (3.6 eV). Compare with the measured 4.2 eV. What does the difference represent?

21.5 $O(\epsilon^2)$ ★ In the LCAO model of H_2^+, show that the energies of $\psi_\pm$ are $E_\pm = (H_{AA} \pm H_{AB})/(1 \pm S)$, where $H_{AA} = \langle \psi_A | \hat{H} | \psi_A \rangle$ and $H_{AB} = \langle \psi_A | \hat{H} | \psi_B \rangle$. Since $H_{AB} < 0$, show that $E_+ < E_-$ and that the antibonding orbital is raised more than the bonding orbital is lowered.

21.6 $O(\epsilon^2)$ ★ (*The colour of carrots, and a failed cow.*) β-carotene has 22 π electrons delocalized over a conjugated chain about 2.9 nm long. Model them as particles in a one-dimensional box, filling the lowest 11 levels two at a time. Find the wavelength needed to promote an electron from the highest filled level to the lowest empty one. You should find it in the infrared — but carrots are orange, because the true absorption peaks near 450 nm. The box model fails because the chain's bonds alternate between single and double, so the hopping alternates; use Problem 19.11 to explain how this opens an extra gap that pushes the absorption into the visible. (For dyes whose bonds are all equivalent, the box model works well.)

21.7 $O(\epsilon^2)$ Water boils at 100°C, but hydrogen sulfide, a heavier molecule, boils at −60°C. Trouton's rule says the entropy of vaporization is roughly the same, about 88 J/(mol K), for many liquids. Use it to estimate the enthalpies of vaporization of both, and attribute the difference to hydrogen bonding. How many hydrogen bonds per water molecule does this suggest, at about 20 kJ/mol each?

21.8 $O(\epsilon^2)$ Use the Born–Oppenheimer picture to show that the ratio of vibrational to electronic energy spacings in a molecule is about $\sqrt{m_e/M}$, and the ratio of rotational to vibrational spacings is about the same. Evaluate for a typical molecule.

21.9 $O(\epsilon^2)$ (*The kinetic isotope effect.*) A C–H bond vibrates at $3000\,\mathrm{cm^{-1}}$. Replacing hydrogen with deuterium (twice the mass) lowers the frequency by a factor $\sqrt{2}$, and with it the zero-point energy. If the transition state has no zero-point energy along the reaction coordinate, the activation energy is larger for C–D by the difference in zero-point energies. Estimate the ratio of the rates at room temperature. (Observed ratios are typically about 7; much larger ratios signal tunnelling.)

21.10 $O(\epsilon^2)$ From $\ln K = -\Delta G^\circ/RT$ with $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$, derive the van 't Hoff equation $\mathrm{d}\ln K/\mathrm{d}T = \Delta H^\circ/RT^2$. Explain why heating favours endothermic reactions, and why the Haber process for making ammonia (exothermic) faces a dilemma between yield and speed.

21.11 $e^{-1/\epsilon}$ ★ (*Why neutral atoms attract.*) Model two neutral atoms as two three-dimensional harmonic oscillators (an electron of mass m bound to its nucleus with frequency ω_0), separated by r. Their instantaneous dipoles interact with an energy of order $e^2 x_1 x_2/(4\pi\epsilon_0 r^3)$. For simplicity, keep only one dimension: the coupled system has normal modes with $\omega_\pm^2 = \omega_0^2(1 \pm \kappa)$, where $\kappa = 2e^2/(4\pi\epsilon_0 m\omega_0^2 r^3)$. Show that the total zero-point energy $\frac{1}{2}\hbar(\omega_+ + \omega_-)$ is lowered by an amount proportional to $\kappa^2 \propto 1/r^6$. The attraction is a purely quantum effect: classically, at zero temperature, the atoms would not fluctuate at all.

21.12 $e^{-1/\epsilon}$ (*Why helium doesn't bond.*) Use the molecular orbital picture to show that He_2, with four electrons, fills both the bonding and the antibonding orbitals, giving a net *repulsion* because the antibonding orbital is more antibonding than the bonding one is bonding. Then explain why He_2^+, with three electrons, is bound. Generalize: define the *bond order* as half the difference between the numbers of bonding and antibonding electrons, and use it to explain why O_2 has a double bond and N_2 a triple bond.

Further reading

- **Peter Atkins and Julio de Paula, *Physical Chemistry*** — The standard comprehensive text, covering quantum chemistry, spectroscopy, thermodynamics, and kinetics.
- **McQuarrie and Simon, *Physical Chemistry: A Molecular Approach*** — Quantum mechanics first, with a physicist's rigour.
- **Linus Pauling, *The Nature of the Chemical Bond*** — The classic that founded quantum chemistry as chemists practise it.
- **Szabo and Ostlund, *Modern Quantum Chemistry*** — The clearest introduction to Hartree–Fock and beyond, for when you want to compute.

CHAPTER 22

Nuclear physics

BUILDS ON quantum mechanics (Chapter 14)

What binds the nucleus, what happens when it lets go, and how the elements of your body were forged in stars.

PROTOTYPE The liquid-drop nucleus. Treat a nucleus as a droplet of incompressible nuclear fluid: constant density, a volume energy that holds it together, a surface tension that resists deformation, and an electric charge that tries to tear it apart. This almost absurdly simple picture predicts the binding energies of hundreds of nuclei to within a percent, and explains both fission and fusion.

The nucleus is a hundred thousand times smaller than the atom, yet holds more than 99.9% of its mass. Its energies are measured in millions of electron-volts rather than the few eV of chemistry, which is why a kilogram of uranium can power a city and why the Sun can shine for ten billion years. Nuclear physics sits between atomic physics and particle physics, and it needs both: quantum mechanics to describe nucleons in a nucleus, and the strong and weak forces of Chapter 32 to describe what holds them and what transforms them.

§22.1 The discovery of the nucleus

In 1909 Geiger and Marsden, working for Rutherford, fired alpha particles at a thin gold foil. Most passed through with small deflections, as expected if the atom's positive charge were spread out like a pudding. But about one in eight thousand bounced back. "It was quite the most incredible event that has ever happened to me in my life," Rutherford said. "It was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you." The only explanation was that the atom's positive charge and nearly all its mass are concentrated in a tiny *nucleus*.

Nuclei are made of protons and neutrons (*nucleons*; the neutron was found by Chadwick in 1932). A nucleus with Z protons and N neutrons has mass number $A = Z + N$; nuclei with the same Z but different N are *isotopes* of the same element. Scattering experiments show that nuclear radii follow

$$R \approx r_0 A^{1/3}, \qquad r_0 \approx 1.2\,\text{fm}$$

($1\,\text{fm} = 10^{-15}\,\text{m}$). The volume is proportional to A: every nucleus has the same density, about $2 \times 10^{17}\,\text{kg/m}^3$. A teaspoon of nuclear matter would weigh about a billion tonnes. Nucleons pack together like the molecules of a liquid: this is the evidence for the liquid drop.

§22.2 The strong force

What holds a nucleus together against the electric repulsion of its protons? A new force, the *strong nuclear force*, with remarkable properties. It is strong — about a hundred times stronger than electromagnetism at nuclear distances. It is short-ranged, vanishing beyond about 2 fm, which is why nuclei are small and why each nucleon interacts only with its immediate neighbours. It acts almost identically between protons and neutrons. And at very short distances it becomes strongly repulsive, which is why nuclear matter does not collapse.

In 1935 Hideki Yukawa explained the short range. If a force is carried by exchanging a particle of mass m, the uncertainty principle lets the particle exist for a time $\Delta t \sim \hbar/mc^2$, so it can travel at most a distance $\hbar/mc$. Yukawa showed that such a force has potential

$$V(r) \propto -\frac{e^{-r/\lambda}}{r}, \qquad \lambda = \frac{\hbar}{mc},$$

the Coulomb potential cut off exponentially. A range of about 1.4 fm requires $mc^2 \approx 140\,\text{MeV}$, and the *pion*, with exactly this mass, was found in 1947. (The deeper theory of the strong force, quantum chromodynamics, describes it in terms of quarks and gluons; pion exchange is its low-energy spherical cow. See Chapter 32.)

§22.3 Binding energy

A nucleus weighs less than its separated protons and neutrons. The difference, times c^2, is its *binding energy* B, the energy needed to pull it apart (Chapter 10). Plot the binding energy per nucleon, B/A, against A, and you get one of the most important graphs in physics. It rises steeply through the light nuclei (with a sharp spike for helium-4, an exceptionally tightly bound little nucleus), peaks at about 8.8 MeV per nucleon near iron and nickel ($A \approx 56$–62), and falls slowly toward the heaviest nuclei.

KEY IDEA Iron is the most tightly bound nucleus. Fusing light nuclei toward iron releases energy (fusion); splitting heavy nuclei toward iron also releases energy (fission). Every nuclear power source in the universe is a way of rolling toward the peak of this curve.

§22.4 The liquid-drop model

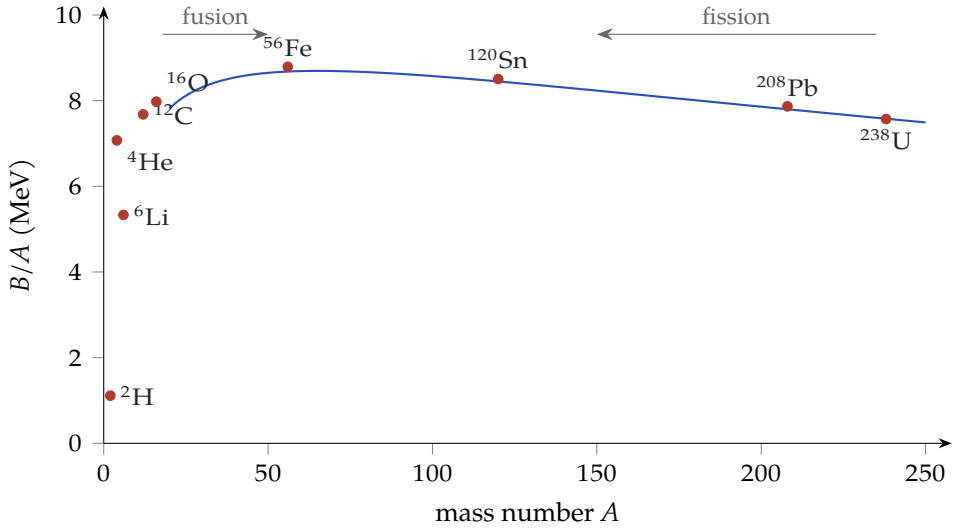
In 1935 von Weizsäcker wrote down a formula for the binding energy based on the liquid drop. Each term is a piece of simple physics:

$$B(A,Z) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} + \delta(A,Z).$$

- **Volume** ($a_V \approx 15.8\,\text{MeV}$): each nucleon is bound to its neighbours, so binding grows in proportion to A, as for a liquid.
- **Surface** ($a_S \approx 18.3\,\text{MeV}$): nucleons at the surface have fewer neighbours, a surface tension proportional to the area $\propto A^{2/3}$. It matters most for small nuclei, which are mostly surface.

- **Coulomb** ($a_C \approx 0.71\,\text{MeV}$): the protons repel each other, an energy $\propto Z^2/R$. It matters most for heavy nuclei.
- **Asymmetry** ($a_A \approx 23.2\,\text{MeV}$): a quantum effect. Protons and neutrons each fill their own ladder of energy levels (they are fermions; Chapter 14). Having more of one than the other means filling one ladder higher, which costs energy.
- **Pairing** (δ, about $\pm 12/\sqrt{A}\,\text{MeV}$): nucleons like to pair up, so even–even nuclei are a little more bound and odd–odd ones a little less.

With five fitted constants, this *semi-empirical mass formula* reproduces the binding energies of hundreds of nuclei to about one percent. It explains why the B/A curve has its shape: the surface term suppresses binding of light nuclei, the Coulomb term that of heavy ones. It predicts the *valley of stability*: for each A, the most stable Z balances the asymmetry term (which wants $Z = A/2$) against the Coulomb term (which wants fewer protons), so heavy nuclei carry extra neutrons — lead-208 has 82 protons and 126 neutrons. And it shows why there is a limit to the periodic table: beyond about $Z^2/A \approx 50$, a nucleus would split spontaneously.



Binding energy per nucleon. The curve is the semi-empirical mass formula with the constants above (drawn from $A = 20$, below which it is unreliable); the dots are measured values. Iron and nickel sit at the top, so fusing light nuclei and splitting heavy ones both release energy. Helium-4 lies well above the curve: it is unusually tightly bound.

§22.5 The shell model

The liquid drop misses something. Nuclei with 2, 8, 20, 28, 50, 82, or 126 protons or neutrons are unusually stable, abundant, and spherical. These *magic numbers* are the nuclear analogue of the noble gases. In 1949 Maria Goeppert Mayer and Hans Jensen explained them with a *shell model*: nucleons move independently in an average potential created by all the others, filling energy levels as electrons do in atoms — but with a strong spin–orbit coupling that reshuffles the levels to produce exactly the observed magic numbers. That nuclei can be described both as a dense liquid and as independent particles in orbits is one of the field's great paradoxes, resolved by the Pauli principle: in a filled Fermi sea, nucleons cannot scatter into occupied states, so they

move almost freely. Doubly magic nuclei like helium-4, oxygen-16, calcium-40, and lead-208 are especially stable, and theorists predict an *island of stability* of superheavy nuclei near $Z = 114$–126, $N = 184$.

§22.6 Radioactivity

Unstable nuclei decay. Each nucleus has a fixed probability per unit time of decaying, independent of its age, so a sample decays exponentially:

$$N(t) = N_0 e^{-t/\tau} = N_0 2^{-t/t_{1/2}}, \qquad t_{1/2} = \tau \ln 2.$$

Half-lives range from less than a microsecond to far longer than the age of the universe. There are three classic kinds of decay.

Alpha decay: heavy nuclei emit a helium-4 nucleus. The alpha particle is trapped behind the Coulomb barrier with less energy than its height, and escapes by quantum tunnelling (Chapter 14). The tunnelling probability depends exponentially on the energy, so tiny differences in energy produce enormous differences in half-life: alpha energies from 4 to 9 MeV correspond to half-lives ranging over more than twenty orders of magnitude (the *Geiger–Nuttall law*, Problem 22.10). Gamow's explanation of this in 1928 was one of the first triumphs of quantum mechanics.

Beta decay: a neutron turns into a proton, emitting an electron, $n \to p + e^- + \bar{\nu}_e$ (or a proton into a neutron, emitting a positron). The electrons come out with a continuous range of energies, which seemed to violate energy conservation. In 1930 Pauli proposed, as a "desperate remedy," that an invisible neutral particle carries off the missing energy. Fermi named it the *neutrino* and built a theory of beta decay around it; it was finally detected in 1956. Beta decay is driven by the *weak force* (Chapter 32). A free neutron decays with a half-life of about ten minutes; inside a stable nucleus, the Pauli principle blocks it.

Gamma decay: an excited nucleus drops to a lower state by emitting a photon of MeV energy, the nuclear analogue of atomic emission (Chapter 18).

Example 22.1 (Radiocarbon dating)**.** Cosmic rays continually make carbon-14 in the upper atmosphere, and living things keep the same ratio of ^{14}C to ^{12}C as the air. When an organism dies, its ^{14}C decays with a half-life of 5730 years. Measuring the remaining fraction dates bones, wood, and cloth up to about fifty thousand years old. Longer-lived isotopes — uranium-238 decaying to lead with a half-life of 4.5 billion years — date rocks, and have established the age of the Earth as 4.54 billion years.

§22.7 Fission

In 1938 Hahn and Strassmann found barium in uranium bombarded with neutrons; Meitner and Frisch, using the liquid-drop model on a walk in the snow, realized the nucleus had split in two. A uranium-235 nucleus absorbing a slow neutron becomes so excited that it oscillates like a wobbling drop, elongates until the Coulomb repulsion of the two ends overcomes the surface tension, and splits, releasing about 200 MeV and two or three new neutrons.

Those neutrons make a *chain reaction* possible. If on average more than one neutron from each fission causes another fission, the reaction grows exponentially. Whether it does depends on how many neutrons escape through the surface before being captured, so there is a *critical mass*: about 50 kg for a bare sphere of pure uranium-235 (Problem 22.11). In a reactor, the chain reaction is held at exactly one new fission per fission. A *moderator* (water or graphite) slows neutrons down so they are more likely to cause fission, and control rods absorb excess neutrons. Control is possible only because a small fraction of neutrons are emitted seconds after the fission rather than instantly; these *delayed neutrons* slow the response enough for mechanical systems to keep up. Remarkably, nature built its own reactors: two billion years ago, when uranium-235 was more abundant, groundwater-moderated uranium deposits at Oklo in Gabon sustained chain reactions for hundreds of thousands of years.

§22.8 Fusion and the stars

Light nuclei release energy by fusing, but they must first overcome their Coulomb repulsion, about 1 MeV for two protons. In the Sun's core, at fifteen million kelvin, typical thermal energies are only about 1 keV. Fusion happens only by tunnelling, and only for the rare protons in the high-energy tail of the Maxwell–Boltzmann distribution (Chapters 9 and 14). The product of the falling Boltzmann factor and the rising tunnelling probability peaks sharply at a few times $k_{\mathrm{B}}T$, the *Gamow peak*.

The Sun fuses hydrogen to helium mainly by the *proton–proton chain*, whose net result is

$$4\,p \to {}^{4}\mathrm{He} + 2e^{+} + 2\nu_e + 26.7\,\mathrm{MeV}.$$

The first step, in which two protons fuse and one turns into a neutron by the weak interaction, is so slow — a given proton waits on average billions of years — that the Sun burns steadily for ten billion years instead of exploding. Heavier stars use carbon, nitrogen, and oxygen as catalysts (the *CNO cycle*). When the hydrogen runs out, helium fuses to carbon via the *triple-alpha* process, which works only because carbon-12 has an excited state at just the right energy — predicted by Fred Hoyle in 1953 from the mere fact that carbon exists. Chapter 27 follows the life of stars; Chapter 17 discusses fusion on Earth.

§22.9 Where the elements came from

Hydrogen, helium, and a little lithium were made in the first three minutes after the Big Bang (Chapter 28). Everything else up to iron was made by fusion in stars. Beyond iron, fusion no longer releases energy, and heavier elements are built by capturing neutrons, which feel no Coulomb barrier. In the *s-process* (slow), neutrons are captured one at a time in giant stars, with time for beta decays in between. In the *r-process* (rapid), a flood of neutrons is captured in seconds, building very neutron-rich nuclei that then decay back to stability. The site of the r-process was long uncertain; in 2017 the collision of two neutron stars, observed in both gravitational waves and light, showed the glow of freshly made heavy elements (Chapter 29). Much of the gold and platinum on Earth was forged in such collisions. As Carl Sagan put it, we are made of star stuff.

§22.10 Nuclear matter and beyond

Protons and neutrons are themselves made of quarks, bound by gluons (Chapter 32). Heat nuclear matter to about two trillion kelvin — as in collisions of gold or lead nuclei at Brookhaven and CERN — and the nucleons melt into a *quark–gluon plasma*, the state of the universe a microsecond after the Big Bang. It turns out to be the most perfect fluid known, with the lowest viscosity relative to its entropy of any substance. At the other extreme, a neutron star is essentially a single nucleus twenty kilometres across, held together by gravity instead of the strong force (Chapter 27).

§22.11 Problems

22.1 $O(\epsilon)$ ★ Use $R = 1.2A^{1/3}\,\text{fm}$ to find the radius of a gold nucleus ($A = 197$) and the density of nuclear matter. What would a teaspoon ($5\,\text{cm}^3$) of it weigh? Compare with the density of a neutron star.

22.2 $O(\epsilon)$ The mass of a helium-4 atom is 4.002602 u; a hydrogen atom is 1.007825 u and a neutron 1.008665 u ($1\,\text{u} = 931.5\,\text{MeV}/c^2$). Find the binding energy of helium-4 in total and per nucleon.

22.3 $O(\epsilon)$ ★ A piece of wood contains 25% of the carbon-14 found in living trees. How old is it? A second sample contains 1%. Why is radiocarbon dating useless for dinosaur bones?

22.4 $O(\epsilon)$ Compare the energy released by fissioning 1 kg of uranium-235 (200 MeV per fission) with burning 1 kg of coal ($3\times10^7\,\text{J}$). How much uranium does a 1 GW power station consume per day?

22.5 $O(\epsilon^2)$ ★ Use Yukawa's argument to predict the mass of the particle that carries the nuclear force, given a range of 1.4 fm. The weak force has a range of about 2×10^{-18} m; what does this predict for the mass of its carrier? (Compare with the W boson, $80\,\text{GeV}/c^2$.)

22.6 $O(\epsilon^2)$ ★ Using the semi-empirical mass formula, find the value of Z that maximizes B for fixed A (neglect pairing and use Z^2 for $Z(Z-1)$). Show that $Z \approx A/(2 + 0.015A^{2/3})$, and check it for $A = 56$ and $A = 238$.

22.7 $O(\epsilon^2)$ ★ (*Solar neutrinos.*) The Sun's luminosity is 3.8×10^{26} W. Each conversion of four protons to helium releases 26.7 MeV and two neutrinos. How many neutrinos does the Sun emit per second, and how many pass through each square centimetre of your body every second? Why don't you notice?

22.8 $O(\epsilon^2)$ Use the surface and Coulomb terms of the mass formula to estimate the energy released when a uranium-238 nucleus splits into two equal halves. (You should get roughly 200 MeV.) Which term makes fission release energy, and which resists it?

22.9 $O(\epsilon^2)$ Marie Curie defined the curie as the activity of one gram of radium-226, whose half-life is 1600 years. Compute the activity in decays per second. Then estimate the

activity of the potassium-40 in your body (0.012% of potassium, with 140 g of potassium and a half-life of 1.25 billion years).

22.10 $e^{-1/\epsilon}$ ★ (*The Geiger–Nuttall law.*) For an alpha particle of energy E tunnelling through the Coulomb barrier of a nucleus of charge Z, the tunnelling probability is approximately $e^{-2\pi\eta}$ with $\eta = 2Ze^2/(4\pi\epsilon_0\hbar v)$, where v is the alpha particle's speed. Show that this gives $\ln t_{1/2} \approx a\,Z/\sqrt{E} + b$. For polonium ($Z \approx 84$), estimate how much the half-life changes between $E = 5$ and $E = 9\,\mathrm{MeV}$. (Real half-lives in this range span from about a microsecond to over a hundred days.)

22.11 $e^{-1/\epsilon}$ (*Critical mass.*) In uranium-235, a fast neutron travels a mean free path of about $\ell \approx 4\,\mathrm{cm}$ between collisions, and each fission makes $\nu \approx 2.5$ neutrons. A neutron diffuses as a random walk. Argue that a sphere is critical when its radius R is large enough that a neutron typically causes another fission before escaping, and that this gives $R_c \sim \ell\,\pi/\sqrt{3(\nu-1)}$. Estimate R_c and the critical mass (density $19\,\mathrm{g/cm^3}$). How does surrounding the sphere with a neutron reflector change things?

22.12 $e^{-1/\epsilon}$ (*Why stars stop at iron.*) Using the binding energy curve, explain why fusion in stellar cores stops at iron. Then explain why the heavier elements must be built by neutron capture rather than by fusing charged nuclei: compare the Coulomb barrier for adding a proton to a nucleus with $Z = 50$ with typical stellar thermal energies. Why can't the s-process build elements beyond bismuth?

Further reading

- **Kenneth S. Krane,** ***Introductory Nuclear Physics*** — The standard undergraduate text, thorough and readable.
- **B. R. Martin,** ***Nuclear and Particle Physics: An Introduction*** — A compact modern introduction to both fields together.
- **Carlos Bertulani,** ***Nuclear Physics in a Nutshell*** — A concise, more advanced survey.
- **Richard Rhodes,** ***The Making of the Atomic Bomb*** — The history of nuclear physics as a gripping human story.

CHAPTER 23

Biophysics

BUILDS ON statistical mechanics (Chapter 9), fluid dynamics (Chapter 11)

Physics turned loose on the machinery of living things: motors that walk, cables that fire, molecules that fold, and why a bacterium cannot coast.

PROTOTYPE The random walk. Inside a cell, every molecule is battered by water molecules trillions of times a second and wanders aimlessly. Life does not fight this chaos; it exploits it. Diffusion delivers its supplies, thermal fluctuations fold its proteins, and its motors ratchet random kicks into directed motion.

Living things obey the laws of physics, and nothing else. But they inhabit a corner of the physical world with unusual properties: everything is wet, warm, small, and crowded; energies are comparable to thermal energy; inertia is irrelevant; and systems are driven far from equilibrium by a constant supply of chemical fuel. Biophysics asks how the physics of this world constrains and enables life. Schrödinger asked "What is life?" in 1944 and helped inspire the discovery of DNA's structure, made by a physicist and a biologist using the physics of X-ray diffraction. The traffic between the fields has never stopped.

§23.1 The world of the cell

Begin with the numbers. A bacterium is about 1 μm long; a human cell about 20 μm; a protein a few nanometres. The forces that matter are piconewtons. And the unit of energy is thermal energy at body temperature:

$$k_{\mathrm{B}}T \approx 4.3 \times 10^{-21}\,\mathrm{J} \approx 4.3\,\mathrm{pN\,nm} \approx \tfrac{1}{37}\,\mathrm{eV}.$$

A force of a few piconewtons acting over a few nanometres does work of order $k_{\mathrm{B}}T$. That is no coincidence: molecular machines must operate just above the thermal noise, strong enough to work reliably but weak enough to be built from floppy molecules and switched by single chemical reactions. The cell's fuel, the hydrolysis of one ATP molecule, releases about $20k_{\mathrm{B}}T$.

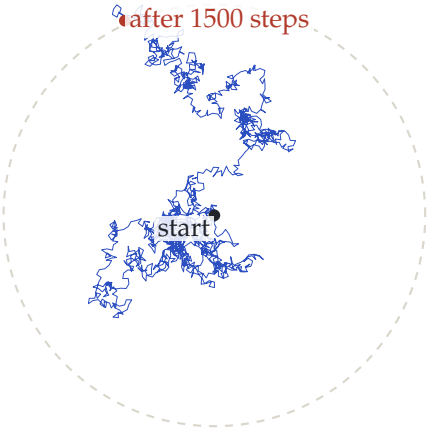
Two more facts shape everything. First, cells live at absurdly low Reynolds number, around 10^{-5} (Chapter 11): water is to a bacterium as honey is to you, and inertia is meaningless. Second, cells are crowded, about 30% protein by volume, more like a gel than a dilute solution.

§23.2 Diffusion

A molecule in water performs a random walk, spreading so that $\langle x^2 \rangle = 2Dt$ in each direction (Chapter 9). For a typical protein, $D \approx 10\,\mu\text{m}^2/\text{s}$. The time to diffuse a distance L is

$$t \sim \frac{L^2}{D}.$$

Across a bacterium ($1\,\mu\text{m}$): a tenth of a second. Across a human cell ($20\,\mu\text{m}$): about a minute. Down a nerve axon a metre long: years. Diffusion is fast over short distances and hopelessly slow over long ones. This single scaling law explains why cells are small, why large cells have internal transport systems, why animals need circulatory systems, and why your lungs branch into 300 million tiny alveoli.



A random walk of 1500 steps of equal length in random directions. The dashed circle has radius $\sqrt{N}$ steps, the typical distance covered: going ten times as far takes a hundred times as long.

§23.3 Molecular machines

Cells are full of machines that turn chemical energy into motion.

- **Kinesin** walks along microtubule tracks, carrying cargo from the cell body down long axons. It takes 8 nm steps, one per ATP, against loads up to about 6 pN, "hand over hand" with two heads. It has been watched taking single steps in optical traps.
- **Myosin** pulls on actin filaments; billions of them acting together contract your muscles.
- **ATP synthase** is a rotary motor. Protons flowing through it across a membrane turn a rotor, and the rotation forces ADP and phosphate together into ATP. Run backwards, it spins by burning ATP. It is nearly 100% efficient.
- **The bacterial flagellar motor**, about 45 nm across, spins a corkscrew flagellum at hundreds of revolutions per second, driven by a flow of ions, and can reverse direction.

How can a motor work when it is constantly kicked about by thermal noise many times larger than its step? One answer is a *Brownian ratchet*: let thermal fluctuations do the moving, and use chemical energy only to rectify them — to prevent backward steps. Feynman famously analysed

a ratchet and pawl and showed that in a single heat bath it cannot extract work, or it would violate the second law (Chapter 7). Molecular motors evade this only because they are driven out of equilibrium by burning ATP (Problem 23.11).

§23.4 Polymers

DNA and proteins are *polymers*: long chains of repeating units. A flexible polymer of N segments of length b adopts a random-walk shape, with end-to-end distance

$$R \approx \sqrt{N}\, b,$$

much smaller than its contour length Nb (Chapter 1). Real polymers resist bending over a *persistence length*; for DNA, about 50 nm, so a short piece is a stiff rod and a long one a random coil.

Pulling the ends of a polymer apart reduces the number of available configurations, i.e. its entropy. The chain therefore resists with a force that is purely entropic, like a gas resisting compression:

$$F = \frac{3k_{\mathrm{B}}T}{Nb^2}\, x.$$

This *entropic spring* is stiffer at higher temperature, which is why a rubber band contracts when heated (Chapter 7). Single DNA molecules have been stretched with optical tweezers and magnetic tweezers, and their force–extension curves match polymer theory beautifully.

Packaging is a feat. The DNA in a single human cell is about 2 m long, yet it fits in a nucleus 6 µm across, wound around protein spools, coiled, and folded — while remaining accessible to the machines that read it.

§23.5 Electricity in cells

A cell membrane is a lipid bilayer 5 nm thick, an excellent insulator. Pumps in the membrane, burning ATP, maintain concentration differences: potassium is concentrated inside, sodium outside. When the membrane is permeable to one ion, that ion diffuses down its concentration gradient until the resulting voltage balances the diffusion. Equilibrium requires the Boltzmann factor $e^{-qV/k_{\mathrm{B}}T}$ to account for the concentration ratio, giving the **Nernst potential**

$$V = \frac{k_{\mathrm{B}}T}{q} \ln \frac{c_{\mathrm{out}}}{c_{\mathrm{in}}}.$$

At body temperature, $k_{\mathrm{B}}T/e \approx 27\,\mathrm{mV}$, and a tenfold concentration ratio gives about 60 mV. A resting neuron sits at about $-70\,\mathrm{mV}$, near the potassium Nernst potential. The field across the thin membrane is about $10^7\,\mathrm{V/m}$, near the breakdown strength of most insulators.

A neuron signals with an **action potential**. A small depolarization opens voltage-gated sodium channels; sodium rushes in, depolarizing the membrane further and opening more channels: positive feedback. Within a millisecond the voltage swings to about $+40\,\mathrm{mV}$; then the sodium channels close and potassium channels open, restoring the resting state. The spike propagates along the axon as a nonlinear travelling wave, regenerated at every point. Hodgkin and Huxley measured the currents in the giant axon of the squid and wrote down, in 1952, a

set of nonlinear differential equations that predicted the shape and speed of the spike — one of the triumphs of quantitative biology. In vertebrates, insulating *myelin* sheaths force the signal to jump between gaps, speeding conduction to about 100 m/s.

§23.6 Protein folding

A protein is a chain of a hundred or more amino acids that folds, in seconds or less, into a precise three-dimensional shape that determines its function. In 1969 Cyrus Levinthal pointed out that if each amino acid could take just three orientations, a chain of 100 would have $3^{100} \approx 10^{48}$ conformations; sampling them at one per picosecond would take far longer than the age of the universe. Yet proteins fold. The *Levinthal paradox* is resolved by the shape of the energy landscape: it is a rugged *funnel*, sloping downhill toward the native state, so the chain is guided rather than searching at random. A major driving force is the *hydrophobic effect*: oily amino acids cluster in the core because water molecules around them are forced into ordered cages, so burying them *increases* the entropy of the water. Misfolded proteins cause diseases from Alzheimer's to prion diseases. Predicting structure from sequence was a fifty-year grand challenge until the machine-learning system AlphaFold largely solved it in 2020.

§23.7 The physics of the senses

Our senses operate at the limits physics allows. A dark-adapted rod cell in your retina responds to a *single photon*: absorbing one photon isomerizes one retinal molecule, triggering a cascade that amplifies it a millionfold. In dim light, what you see is limited by the Poisson statistics of photon arrivals. The ear detects vibrations of the eardrum smaller than the diameter of an atom, near the limit set by thermal noise. The cochlea is a mechanical Fourier analyser (Chapter 1): a tapered membrane resonates at different frequencies at different points along its length, and hair cells actively amplify the vibrations, so actively that healthy ears emit faint sounds of their own. Some birds appear to sense the Earth's magnetic field through a quantum effect in light-activated molecules in their eyes — a leading example of the emerging field of quantum biology.

§23.8 Scaling laws

Across all mammals, from shrews to whales, the metabolic rate scales with body mass as

$$P \propto M^{3/4},$$

Kleiber's law (1932). A pure surface-to-volume argument would give $M^{2/3}$ (heat is lost through the skin); the observed 3/4 is often explained by the geometry of branching supply networks like blood vessels. Many other quantities follow quarter-power laws. Lifespan grows as about $M^{1/4}$ and heart rate falls as $M^{-1/4}$, so the number of heartbeats in a lifetime is roughly the same, about a billion, for a mouse and an elephant. Small animals live fast: a shrew must eat its own weight in food every day.

§23.9 Information

A cell is an information-processing machine. The human genome holds about 3×10^9 base pairs, each worth two bits: about 750 megabytes, less than a film. Copying it requires extraordinary accuracy, about one error per 10^9 bases, yet the free energy differences between correct and incorrect base pairing would by themselves allow only about one error in 10^4. In 1974 John Hopfield showed how cells beat this limit, by *kinetic proofreading*: adding an irreversible, energy-consuming checking step multiplies the selectivity (Problem 23.12). Accuracy costs energy — another case of information being physical (Chapter 7).

§23.10 Tools

Biophysics is also a toolkit. X-ray crystallography revealed the double helix and the structures of hundreds of thousands of proteins (Chapter 19). *Cryo-electron microscopy* images flash-frozen molecules at near-atomic resolution (Nobel Prize, 2017). Fluorescence microscopy lights up specific molecules in living cells, and *super-resolution* methods beat the diffraction limit of Chapter 8 by switching molecules on a few at a time and locating each precisely (Nobel Prize, 2014). *Optical tweezers* grip single molecules and measure their piconewton forces (Nobel Prize, 2018; Chapter 18).

§23.11 Problems

23.1 $O(\epsilon)$ ★ Express k_BT at 37°C in joules, pN nm, and meV. Kinesin takes 8 nm steps against up to 6 pN, using one ATP ($\approx 20k_BT$) per step. What is its maximum efficiency?

23.2 $O(\epsilon)$ ★ A protein has $D = 10\,\mu m^2/s$. How long does it take to diffuse across a bacterium (1 µm), a human cell (20 µm), and a 1 m axon? Kinesin moves at about 1 µm/s. At what distance does active transport beat diffusion?

23.3 $O(\epsilon)$ A human cell holds 6×10^9 base pairs of DNA, each 0.34 nm long. Find the total length and compare it with the diameter of the nucleus, 6 µm. If the DNA were a random coil with persistence length 50 nm (segment length $b = 100$ nm), how big would the coil be?

23.4 $O(\epsilon^2)$ ★ Inside a neuron the potassium concentration is 140 mM; outside it is 5 mM. Find the potassium Nernst potential at 37°C. For sodium, 12 mM inside and 145 mM outside. Explain why opening sodium channels makes the inside positive.

23.5 $O(\epsilon^2)$ ★ (*The entropic spring.*) For a random-walk polymer, the end-to-end vector $\mathbf{R}$ has a Gaussian distribution, $P(\mathbf{R}) \propto e^{-3R^2/2Nb^2}$. Using $S = k_B \ln P + \text{const}$ and $F = -T\,\partial S/\partial R$, derive the entropic spring force. Why does the spring get stiffer as temperature increases?

23.6 $O(\epsilon^2)$ (*A bacterium cannot coast.*) *E. coli* is about 1 µm in size, has mass about 10^{-15} kg, and swims at 30 µm/s in water. If its motor suddenly stops, how far does it coast? Use

Stokes drag, $6\pi\eta r v$, to find the stopping time. Compare the distance with the size of a hydrogen atom.

23.7 $O(\epsilon^2)$ Make Levinthal's estimate for a 100-residue protein with 3 conformations per residue, sampled at 10^{12} per second. How long would a random search take? Real proteins fold in microseconds to seconds. By what factor is the search space effectively reduced?

23.8 $O(\epsilon^2)$ Using Kleiber's law, by what factor is the metabolic rate per kilogram of a 20 g mouse greater than that of a 5000 kg elephant? If both live about a billion heartbeats, and a mouse's heart beats 600 times per minute, estimate the lifespans of both.

23.9 $O(\epsilon^2)$ In an unmyelinated axon, the signal speed scales as the square root of the axon's diameter; in a myelinated one, linearly with diameter. The squid evolved a giant axon 0.5 mm across to escape predators quickly. Why did vertebrates not need giant axons? How much thinner can a myelinated axon be for the same speed?

23.10 $e^{-1/\epsilon}$ ★ (*The physical limit to smelling.*) A bacterium measures the concentration c of an attractant with a receptor of size a, over a time T. Argue that the number of molecules it counts is about $DacT$ (the rate at which molecules diffuse into a sphere of radius a is $\sim Dac$), and that counting statistics give a fractional error $\delta c/c \sim 1/\sqrt{DacT}$. This is the *Berg–Purcell limit*. Estimate it for $c = 1\,\mu\text{M}$, $a = 1\,\mu\text{m}$, $T = 1\,\text{s}$. Real bacteria come within a factor of a few of it.

23.11 $e^{-1/\epsilon}$ (*Feynman's ratchet.*) A ratchet and pawl attached to paddles in a gas would seem to rectify random molecular kicks into one-way rotation, lifting a weight. Explain why it fails if the pawl is at the same temperature as the gas: the pawl itself jiggles and sometimes lets the ratchet slip back, at exactly the rate needed to balance forward steps. Show that the device works as an engine only if the paddles and pawl are at different temperatures, and that its efficiency is then at most Carnot's. How does a kinesin motor get around the problem?

23.12 $e^{-1/\epsilon}$ (*Kinetic proofreading.*) An enzyme must pick the right substrate over a wrong one that binds with free energy weaker by ΔG. In a single equilibrium binding step, the error rate is about $e^{-\Delta G/k_B T}$. Hopfield proposed that after binding, the complex is converted (using energy) into an activated form, from which either the product is made or the substrate falls off. Argue that if falling off happens at the same rate ratio as the original unbinding, the error rate becomes $e^{-2\Delta G/k_B T}$. Why must the activation step consume energy to achieve this?

Further reading

- **Philip Nelson, *Biological Physics: Energy, Information, Life*** — The best introduction for physics students: careful, quantitative, and full of insight.
- **Phillips, Kondev, and Theriot, *Physical Biology of the Cell*** — Order-of-magnitude thinking applied to all of cell biology.

- **Howard C. Berg,** ***Random Walks in Biology*** — A short classic on diffusion and how bacteria swim and sense.
- **William Bialek,** ***Biophysics: Searching for Principles*** — An advanced, inspiring look at life operating at physical limits.
- **Erwin Schrödinger,** ***What Is Life?*** — The 1944 lectures that inspired a generation of physicists to turn to biology.

CHAPTER 24

Medical physics

BUILDS ON quantum mechanics (Chapter 14), nuclear physics (Chapter 22)

Physics pointed at the human body — to see inside it without cutting, and to heal it — built almost entirely from discoveries made for other reasons.

PROTOTYPE The CT scan. Shine a thin fan of X-rays through the body from hundreds of angles, and measure how much gets through along each line. No single picture shows the inside. But from all the shadows together, a computer can reconstruct a three-dimensional map of the body, slice by slice.

Almost every tool in a modern hospital started as a physics experiment. Röntgen's X-rays were radiographing hands within weeks of their discovery in 1895. Nuclear magnetic resonance, developed to measure the magnetic moments of nuclei, became MRI. The positron, predicted by Dirac from pure theory, now finds tumours in PET scanners. Particle accelerators built to probe the nucleus treat cancer in thousands of hospitals. Medical physics is the art of turning fundamental physics into ways of seeing and healing, under the strictest of constraints: the patient must not be harmed.

§24.1 X-rays

X-rays are photons with energies of tens of keV, short enough in wavelength to pass through soft tissue. A beam passing through a thickness x of material is attenuated exponentially,

$$I = I_0\, e^{-\mu x},$$

where the *attenuation coefficient* μ depends on the material and the photon energy. At diagnostic energies two processes dominate: the *photoelectric effect* (Chapter 8), in which the photon is absorbed by ejecting an inner electron, and *Compton scattering* (Chapter 10). The photoelectric effect scales roughly as Z^3/E^3, so it depends steeply on atomic number. Bone, rich in calcium ($Z = 20$), absorbs far more than soft tissue, made of hydrogen, carbon, and oxygen. That contrast is why bones show up white on an X-ray. Iodine and barium ($Z = 53, 56$) are swallowed or injected as *contrast agents* to make blood vessels and the gut visible.

An X-ray picture is a shadow: everything along each line is superimposed. A rib in front of a tumour hides it.

§24.2 Computed tomography

In 1971 Godfrey Hounsfield built a machine that took X-ray projections from many angles and reconstructed a cross-sectional image by computer; Allan Cormack had independently worked out the mathematics. (They shared the 1979 Nobel Prize. Hounsfield worked for the record company EMI, and it is often said, though disputed, that the Beatles' profits paid for the first scanner.) The measurement along each line gives the integral of μ along that line:

$$\ln \frac{I_0}{I} = \int \mu(x,y)\ \mathrm{d}\ell.$$

The set of all such line integrals is the *Radon transform* of the image, and the problem is to invert it. The key fact is the **projection–slice theorem**: the one-dimensional Fourier transform of a projection taken at angle θ equals a slice through the two-dimensional Fourier transform of the image, along the same angle (Chapter 1). So projections from all angles fill in the whole two-dimensional transform, which can then be inverted. In practice, CT scanners use *filtered backprojection*: smear each projection back across the image along the lines it came from, after first sharpening it with a filter that undoes the blur that naive smearing would cause (Problem 24.11). Modern scanners take a full rotation in a third of a second.

§24.3 Magnetic resonance imaging

MRI images the protons in the body's water and fat. A proton has spin $\frac{1}{2}$ and a magnetic moment (Chapter 14). In a strong magnetic field B_0 — typically 1.5 or 3 T, from a superconducting magnet (Chapter 19) — the spins have two energy levels, and they precess at the *Larmor frequency*

$$f = \frac{\gamma}{2\pi} B_0, \qquad \frac{\gamma}{2\pi} = 42.58\,\mathrm{MHz/T}$$

for protons: 128 MHz at 3 T, a radio frequency. At body temperature the energy splitting is a tiny fraction of $k_\mathrm{B}T$, so the excess of spins aligned with the field is only a few parts per million (Problem 24.6). But a cubic millimetre of tissue holds 10^{20} protons, so the net magnetization is measurable.

An MRI scan works like this:

1. A radio-frequency pulse at the Larmor frequency tips the net magnetization away from the field. (This is resonance, Chapter 3, in a quantum system.)
2. The tipped magnetization precesses, inducing a radio signal in a receiver coil (Faraday's law, Chapter 6).
3. The signal decays. The spins relax back toward alignment with the field over a time T_1 (hundreds of milliseconds), by giving energy to their surroundings; and they lose phase coherence with each other over a shorter time T_2, because each spin sees slightly different local fields. Different tissues have different T_1 and T_2, and choosing the timing of pulses and measurements makes different tissues bright or dark. MRI has exquisite soft-tissue contrast.
4. Position is encoded by adding magnetic field *gradients*, so that the Larmor frequency varies across the body. The frequency of a signal then tells you where it came from, and phase

tells you the other coordinate. The raw data are the Fourier transform of the image, and a fast Fourier transform (Chapter 13) recovers it.

MRI uses no ionizing radiation. *Functional* MRI exploits the fact that oxygenated and deoxygenated blood have different magnetic properties, mapping the brain regions that become active during a task.

§24.4 Ultrasound

Ultrasound imaging sends pulses of sound at a few megahertz into the body and listens for echoes, like sonar or a bat. Echoes come from boundaries between tissues of different *acoustic impedance* $Z = \rho v$ (Chapter 5), and the fraction of intensity reflected is

$$R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2 .$$

Soft tissues differ in impedance by a few percent, reflecting weakly and letting most of the sound through to deeper layers — ideal for imaging. Air and tissue differ by a factor of three thousand, so almost everything reflects; this is why a gel is smeared on the skin, and why ultrasound cannot see through the lungs. The time of flight gives depth, since sound travels at about 1540 m/s in tissue. Resolution is limited by the wavelength, a fraction of a millimetre, and higher frequencies give finer detail but are absorbed more strongly, so they cannot see as deep. *Doppler ultrasound* measures the frequency shift of echoes from moving blood cells to map blood flow (Chapter 5). Ultrasound is cheap, portable, real-time, and safe enough to watch a fetus.

§24.5 Nuclear medicine

Instead of shining radiation through the body, nuclear medicine puts a radioactive tracer inside it and watches where it goes. The image shows *function* rather than anatomy.

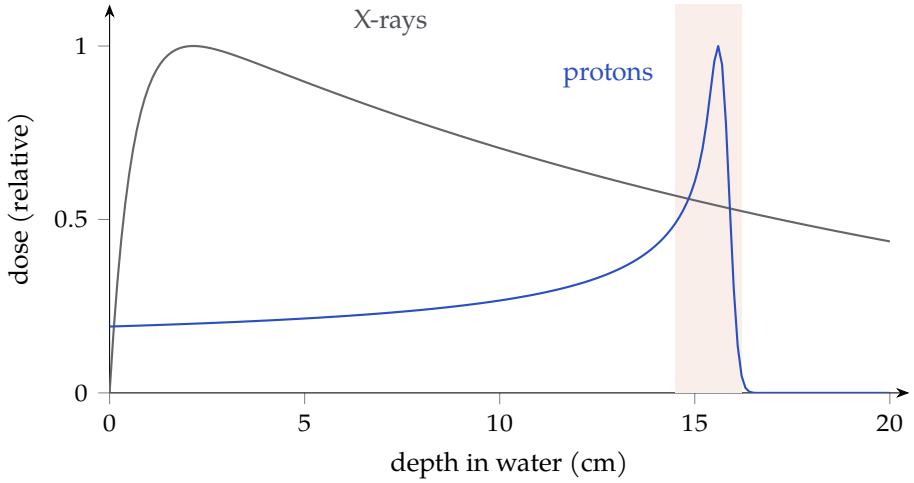
PET (positron emission tomography) uses tracers containing positron-emitting nuclei. The most common is fluorodeoxyglucose (FDG), a sugar molecule tagged with fluorine-18 (half-life 110 minutes); cancer cells, which consume glucose voraciously, light up. Each positron travels a millimetre or so, meets an electron, and annihilates into two gamma-ray photons. Because the positron and electron are nearly at rest, conservation of energy and momentum (Chapter 10) demands that the photons each carry exactly $m_e c^2 = 511$ keV and fly off in opposite directions. A ring of detectors registers pairs arriving in coincidence; the annihilation happened somewhere on the line between them. Many such lines reconstruct the tracer's distribution, as in CT. Antimatter, put to work.

SPECT uses single gamma emitters, most commonly technetium-99m, a metastable nucleus that emits a 140 keV gamma ray with a six-hour half-life — long enough to image, short enough to fade quickly. It is used in tens of millions of procedures a year.

§24.6 Radiation therapy

Ionizing radiation damages DNA. That makes it dangerous, and it also makes it a weapon against cancer. About half of all cancer patients receive radiation therapy. The absorbed dose is measured in *grays* (1 Gy = 1 J/kg); a typical course delivers 60 Gy to a tumour, in daily fractions of about 2 Gy. The energy involved is tiny — 60 Gy would warm tissue by only 0.015°C — but it is delivered as ionizations that break chemical bonds in DNA (Problem 24.8). *Fractionation* exploits the fact that healthy cells repair damage between sessions better than tumour cells do.

Most treatments use X-rays of several MeV from compact electron linear accelerators (Chapter 31), aimed from many angles so the beams overlap on the tumour and spread out elsewhere. **Proton therapy** uses a cleverer piece of physics. A charged particle loses energy in matter at a rate that grows as it slows down (roughly as $1/v^2$; Chapter 31), so it deposits most of its energy in a sharp *Bragg peak* just before it stops. By tuning the proton energy, the peak can be placed in the tumour, with little dose beyond it. For tumours near the spinal cord or in children, sparing the tissue behind the target matters enormously.



How the dose from a beam of X-rays and a beam of 150 MeV protons varies with depth in water. The X-rays deposit the most just under the skin and keep going right through. The protons deposit most of their energy in a sharp Bragg peak at the end of their range, and nothing beyond it.

§24.7 Radiation and safety

Different radiations do different amounts of biological damage per gray, and different organs differ in sensitivity, so radiation protection uses the *effective dose* in *sieverts*. Some benchmarks:

- natural background (radon, cosmic rays, potassium-40 in your body): about 3 mSv per year;
- a chest X-ray: 0.02 mSv; a transatlantic flight: about 0.05 mSv;
- a CT scan of the abdomen: about 10 mSv;
- a dose likely to cause acute radiation sickness: about 1 Sv; lethal to half of those exposed without treatment: about 4–5 Sv.

At low doses, the cancer risk is too small to measure directly. Regulators assume it is proportional to dose (the *linear no-threshold* model), roughly 5% extra lifetime risk per sievert, which is a prudent assumption rather than an established fact.

§24.8 Light and electricity in medicine

Optics is everywhere in a hospital. *Endoscopes* carry images out of the body along bundles of optical fibres that trap light by total internal reflection (Chapter 8). LASIK eye surgery reshapes the cornea with pulses from an ultraviolet excimer laser at 193 nm, which breaks molecular bonds and removes tissue a fraction of a micrometre at a time without heating its surroundings. The *pulse oximeter* clipped to your finger shines red and infrared light through it; oxygenated and deoxygenated haemoglobin absorb these wavelengths differently, so the ratio of transmitted intensities reveals your blood oxygen level (Problem 24.12).

The body is also electrical (Chapter 23). The coordinated firing of heart muscle cells creates a rotating electric dipole whose field reaches the skin as a millivolt signal: the *electrocardiogram*. The brain's activity produces microvolt signals (EEG) and femtotesla magnetic fields, detectable with SQUIDs (Chapter 19) or atomic magnetometers (Chapter 18).

§24.9 Problems

24.1 $O(\epsilon)$ ★ At 60 keV, soft tissue has $\mu \approx 0.2\,\mathrm{cm}^{-1}$ and bone about $0.6\,\mathrm{cm}^{-1}$. What fraction of the beam passes through 20 cm of tissue? Through 18 cm of tissue and 2 cm of bone? Explain why bone appears white on the film.

24.2 $O(\epsilon)$ ★ Find the proton Larmor frequency at 1.5 T and 3 T. What part of the electromagnetic spectrum is this? Why must an MRI room be shielded with a copper cage?

24.3 $O(\epsilon)$ ★ Compute the fraction of ultrasound intensity reflected at a boundary between soft tissue ($Z = 1.6 \times 10^6\,\mathrm{kg/(m^2 s)}$) and bone ($7.8 \times 10^6$), and between air (430) and tissue. Why is a gel used?

24.4 $O(\epsilon)$ What is the wavelength of 5 MHz ultrasound in tissue? Estimate the finest detail it can resolve. Why does imaging the liver use 3 MHz, but imaging the eye uses 20 MHz?

24.5 $O(\epsilon^2)$ Explain, using conservation of energy and momentum, why positron annihilation produces two photons of exactly 511 keV travelling in opposite directions, and why a single photon is impossible. If a PET scanner measures the difference in arrival times of the two photons to within 300 ps, how precisely can it locate the annihilation along the line?

24.6 $O(\epsilon^2)$ At 3 T and body temperature, find the ratio of the proton spin splitting to $k_B T$, and hence the fractional excess of spins aligned with the field. How many excess aligned protons are there in a cubic millimetre of water? Why do MRI manufacturers keep building stronger magnets?

24.7 $O(\epsilon^2)$ In an MRI scanner, a gradient of 10 mT/m is applied. How different are the Larmor frequencies of protons 1 mm apart? If the signal is recorded for 10 ms, can these be distinguished?

Hint: Frequencies can be distinguished if they differ by more than about the inverse of the recording time (Chapter 1).

24.8 $O(\epsilon^2)$ ★ A tumour of mass 1 kg receives a dose of 2 Gy. How much energy is deposited, and what is the temperature rise? A lethal whole-body dose is about 5 Gy, which warms the body by about 0.001°C — less than a sip of hot tea. Explain why such a small energy is so dangerous: estimate how many ionizations 2 Gy produces per cell (a cell has mass about 10^{-12} kg, and each ionization deposits about 30 eV).

24.9 $O(\epsilon^2)$ Technetium-99m has a half-life of 6.0 hours. A patient is injected with 500 MBq. What activity remains after 24 hours? Why is a half-life of a few hours ideal for imaging?

24.10 $O(\epsilon^2)$ Protons of 150 MeV have a range of about 16 cm in water. Since the stopping power grows roughly as $1/v^2$ (and hence as $1/E$ for nonrelativistic protons), sketch the dose deposited as a function of depth, and explain the Bragg peak. Why is it a problem if the patient's anatomy is not exactly as planned?

24.11 $e^{-1/\epsilon}$ ★ (*Why CT needs a filter.*) A single bright point at the origin produces, at every angle, a projection that is a spike at the origin. Show that naively "backprojecting" — smearing each projection back along its lines and adding over all angles — gives an image proportional to $1/r$, a blurry star rather than a point. Using the projection–slice theorem, show that the fix is to multiply each projection's Fourier transform by $|k|$ (the *ramp filter*) before backprojecting.

Hint: In two dimensions, the Fourier transform of $1/r$ is proportional to $1/|k|$.

24.12 $e^{-1/\epsilon}$ (*The pulse oximeter.*) At 660 nm deoxygenated haemoglobin absorbs much more strongly than oxygenated; at 940 nm the reverse is true. Using the Beer–Lambert law with absorption coefficients ϵ_{ox} and ϵ_{de} at each wavelength, show how measuring the attenuation at both wavelengths determines the fraction of haemoglobin that is oxygenated. Then explain why the device uses only the part of the signal that pulses with each heartbeat: what unknown quantities does this remove?

Further reading

- **Hobbie and Roth,** ***Intermediate Physics for Medicine and Biology*** — A physics-first introduction to the body and to medical imaging.
- **Bushberg et al.,** ***The Essential Physics of Medical Imaging*** — The standard reference for X-rays, CT, MRI, ultrasound, and nuclear medicine.
- **Faiz M. Khan,** ***The Physics of Radiation Therapy*** — The standard text for radiation oncology physics.

- **Bettyann Kevles,** ***Naked to the Bone*** — A cultural history of medical imaging in the twentieth century.

CHAPTER 25

Geophysics and climate

BUILDS ON fluid dynamics (Chapter 11), continuum mechanics (Chapter 15)

The physics of the planet we actually live on: its molten heart, its drifting skin, its spinning air and ocean, and the energy balance that sets its temperature.

PROTOTYPE The Earth as a single sphere in balance: sunlight in, infrared out. Equate the two and you get the temperature the Earth would have without an atmosphere — 255 K, a frozen world. The real surface is 33 K warmer. Explaining that difference is the physics of the greenhouse effect, and of climate.

Geophysics applies every branch of classical physics to one particular object: the Earth. It is a spinning, layered ball of rock and iron, wrapped in a thin skin of rock plates floating on a slowly churning mantle, covered by an ocean and an atmosphere that are turbulent rotating fluids, and heated from inside by radioactivity and from outside by the Sun. Almost nothing about it can be put in a laboratory, and almost nothing about it can be seen directly. Yet physicists have mapped its core, measured its age, explained its magnetic field, and built models of its climate that predicted, decades in advance, the warming now being measured.

§25.1 Weighing the Earth

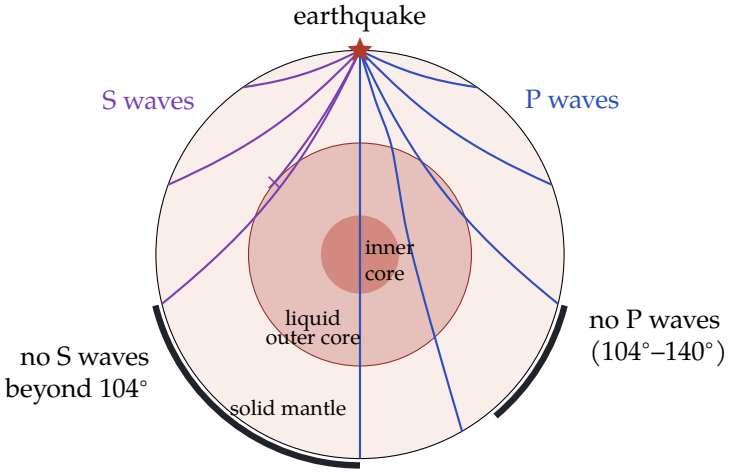
Eratosthenes measured the Earth's radius around 240 BCE from shadow lengths in two Egyptian cities, getting within a few percent of 6371 km. Once Cavendish measured G in 1798 (Chapter 2), $g = GM/R^2$ gave the mass, 6.0×10^{24} kg, and hence the mean density, $5.5\,\mathrm{g/cm^3}$. Surface rocks have only about half that. The interior must be much denser. The Earth's rotation gives a second clue: its moment of inertia is $0.33\,MR^2$, below the $0.4\,MR^2$ of a uniform sphere, so mass is concentrated toward the centre.

§25.2 Seeing inside with earthquakes

Earthquakes are the Earth's X-rays. They send P waves (compressional) and S waves (shear) through the interior (Chapter 15), and because the wave speeds increase with depth, the rays curve back up to the surface, where seismometers around the world record them. Their arrival times map the interior.

In 1906 Richard Oldham noticed that seismic waves arriving on the far side of the Earth were delayed. By 1914 Beno Gutenberg had found a *shadow zone*: between about 104° and 140° from an earthquake, no direct P waves arrive, and beyond 104° no direct S waves arrive at all. P waves

are bent by a central core where their speed suddenly drops; S waves are stopped by it. Since shear waves cannot travel through liquids, the outer core is molten (Problem 25.10). In 1936 Inge Lehmann, studying faint P waves that appeared inside the shadow, deduced a solid *inner core*.



Seismic waves from an earthquake, sketched. They curve as they go, because they travel faster at depth. The liquid outer core stops S waves (left) and bends P waves (right) so sharply that it casts a shadow, a band of the surface that no direct P waves reach.

The Earth's structure, then:

- **Crust**: 5–70 km thick, light silicate rock.
- **Mantle**: to 2900 km depth, hot solid rock that flows slowly over millions of years.
- **Outer core**: to 5150 km, liquid iron and nickel.
- **Inner core**: a solid iron ball 1220 km in radius, solid despite being as hot as the Sun's surface, because the pressure is so high.

Modern *seismic tomography*, using millions of seismograms the way CT uses X-rays (Chapter 24), images hot plumes rising and cold slabs sinking through the mantle.

§25.3 Earthquakes and tsunamis

Earthquakes happen where rock, slowly strained by moving plates, suddenly slips along a fault, releasing stored elastic energy. Their sizes are measured on a logarithmic *magnitude* scale: each step of one magnitude is about 32 times more energy. They obey the Gutenberg–Richter power law (Chapter 16): for every magnitude 6, about ten magnitude 5s. An earthquake under the ocean can lift the seafloor and launch a *tsunami*. With wavelengths of hundreds of kilometres, tsunamis are shallow-water waves even in the deep ocean, travelling at $\sqrt{gh} \approx 200\,\mathrm{m/s}$ — as fast as a jet — with an amplitude of a metre or less. Approaching the coast they slow, and their amplitude grows (Chapter 5).

§25.4 The shape of the Earth and the tides

A rotating fluid ball bulges at its equator. Newton predicted the Earth's oblateness from its rotation; the equatorial radius exceeds the polar radius by 21 km. On smaller scales, the Earth's crust floats on the mantle like ice on water: mountains have deep roots of light crust, so that the weight of every column is balanced (*isostasy*, Archimedes on a planetary scale; Chapter 11). Scandinavia is still rising, rebounding from the weight of ice-age glaciers that melted ten thousand years ago.

Tides arise because gravity weakens with distance. The Moon pulls harder on the near side of the Earth than on its centre, and harder on the centre than on the far side, so the Earth is stretched along the Earth–Moon line, raising two bulges — one facing the Moon and one opposite — and producing two high tides a day. The tidal acceleration scales as M/r^3 rather than M/r^2, so the Moon, despite its smaller mass, beats the Sun by a factor of about two (Problem 25.7). When Sun and Moon align, at full and new moon, tides are larger (*spring tides*); at quarter moons, smaller (*neap tides*).

Tides dissipate energy, about 3.7 TW, mostly in shallow seas. The Earth's rotation drags its tidal bulges slightly ahead of the Moon, and their pull speeds the Moon up in its orbit while braking the Earth's spin. The day lengthens by about 2 milliseconds per century, and the Moon recedes by 3.8 cm per year, as measured by bouncing lasers off reflectors left by Apollo astronauts (Problem 25.11). The same process long ago locked the Moon's rotation to its orbit, so we always see the same face.

§25.5 The Earth's heat and plate tectonics

The Earth is hot inside: the temperature rises about 25 K per kilometre of depth, and 47 TW of heat flows out through the surface, roughly half from the decay of uranium, thorium, and potassium (Chapter 22), the rest primordial heat from the planet's formation. In the 1860s Lord Kelvin used the rate of cooling to estimate the Earth's age at around a hundred million years, too young for the geologists and Darwin. He had made two mistakes, neither of which he could have known: he did not know about radioactive heating, and he assumed heat leaves only by conduction. In fact the mantle convects.

The mantle is solid but, over millions of years, flows like an extremely viscous fluid ($\eta \sim 10^{21}$ Pa s). The relevant dimensionless number is the *Rayleigh number*, the ratio of buoyant driving to viscous and thermal damping, which for the mantle is about 10^7, far above the threshold for convection. Hot rock rises, cold rock sinks, and the cool, rigid outer shell is broken into *plates* that move at a few centimetres per year, about the speed your fingernails grow. Where they pull apart, new ocean floor forms; where they collide, mountains rise or one plate dives beneath the other. Alfred Wegener proposed continental drift in 1912 from the fit of the continents and their fossils; he was dismissed for decades because no one could imagine a mechanism, until the magnetic stripes on the seafloor proved it in the 1960s.

§25.6 The geodynamo

The Earth has a magnetic field, roughly a dipole of about 50 μT at the surface, generated by the convecting, rotating liquid iron of the outer core: a self-sustaining *dynamo*, in which moving conductor and magnetic field continually regenerate each other (Chapter 17). The field is not steady. Every few hundred thousand years, irregularly, it reverses. Lava records the field direction as it cools, so the seafloor spreading away from mid-ocean ridges carries symmetric stripes of alternating magnetization, like a tape recording — the evidence that convinced geologists of plate tectonics. The field also shields the Earth from the solar wind, carving out the magnetosphere (Chapter 17).

§25.7 The atmosphere

The atmosphere is a thin film: half its mass lies below 5.5 km. In an isothermal atmosphere the pressure would fall exponentially with a scale height $k_B T/mg \approx 8\,\mathrm{km}$ (Chapter 9). In reality the lower atmosphere, the *troposphere*, is heated from below by the warm ground and is stirred by convection. Rising air expands and cools adiabatically (Chapter 7) at the *dry adiabatic lapse rate*

$$\Gamma = \frac{g}{c_p} \approx 9.8\,\mathrm{K/km}.$$

Moist air cools more slowly, because condensing water releases latent heat, and the observed average is about 6.5 K/km. Where rising air cools to its dew point, clouds form. Above about 12 km, in the *stratosphere*, temperature rises again, because ozone absorbs ultraviolet sunlight (Chapter 21), and the inverted temperature profile suppresses convection — which is why thunderstorms flatten into anvils at the top of the troposphere.

§25.8 Spinning fluids: weather and ocean currents

On large scales, the atmosphere and ocean are dominated by the Earth's rotation. The importance of rotation is measured by the *Rossby number*, $\mathrm{Ro} = U/(fL)$, where U and L are typical speeds and sizes and $f = 2\Omega\sin\lambda$ is the Coriolis parameter at latitude λ (Chapter 2). For a bathtub, $\mathrm{Ro} \sim 10^3$: the Coriolis force is irrelevant, and the direction of the drain vortex is set by how you filled the tub. For a weather system a thousand kilometres across, $\mathrm{Ro} \sim 0.1$: rotation dominates.

At small Rossby number, a remarkable balance holds: the Coriolis force balances the pressure gradient, and the wind blows not from high pressure to low but *along* the isobars. This *geostrophic balance* is why weather maps show winds circling around lows — counterclockwise in the northern hemisphere, clockwise in the southern — and why hurricanes spin. The global circulation is shaped by it too: warm air rises at the equator and sinks around 30° latitude (the *Hadley cells*), the returning surface flow is deflected into the steady trade winds, and at mid-latitudes the jet streams meander around the globe. Weather is chaotic (Chapter 12), which limits forecasts to about two weeks — but climate, the statistics of weather, is predictable, just as you cannot predict a single roll of a die but can predict the average of a million.

The ocean is a second spinning fluid, a thousand times denser and with a heat capacity so large that its top three metres store as much heat as the entire atmosphere. Winds drive surface

currents like the Gulf Stream; differences in temperature and salinity drive a slow, deep *thermohaline circulation*, in which cold salty water sinks in the North Atlantic and around Antarctica and returns over about a thousand years.

§25.9 Climate

The energy balance. The Sun delivers $S = 1361\,\mathrm{W/m^2}$ at the Earth's distance. The Earth reflects a fraction $A \approx 0.3$ (its *albedo*) and intercepts sunlight over a disc of area πR^2, while radiating infrared from its whole surface, $4\pi R^2$. Balancing gives

$$(1-A)S\,\pi R^2 = 4\pi R^2\,\sigma T_e^4 \quad\Longrightarrow\quad T_e = \left(\frac{(1-A)S}{4\sigma}\right)^{1/4} \approx 255\,\mathrm{K}.$$

That is -18°C. The actual average surface temperature is about 288 K, $+15^\circ$C.

The greenhouse effect. The atmosphere is nearly transparent to visible sunlight but absorbs strongly in parts of the infrared, where water vapour and carbon dioxide molecules vibrate and rotate (Chapter 18). The Earth's surface radiates infrared upward; the atmosphere absorbs much of it and re-radiates, half back down. The planet as seen from space must still radiate at an effective 255 K to balance the sunlight it absorbs, but that radiation now escapes mainly from high, cold layers of the atmosphere. Because temperature falls with height (the lapse rate), the surface beneath must be warmer. Adding greenhouse gases raises the altitude from which radiation escapes, and the surface warms. The basic physics has been known since Fourier (1824), Tyndall (1859), and Arrhenius (1896), who first estimated the warming from doubling CO_2; Syukuro Manabe's 1967 radiative–convective model, which got the numbers right, won a share of the 2021 Nobel Prize.

Forcing, feedbacks, and sensitivity. Because the CO_2 absorption band is already saturated at its centre, adding more CO_2 widens it at the edges, and the extra trapped energy grows only logarithmically:

$$\Delta F \approx 5.35 \ln \frac{C}{C_0}\,\mathrm{W/m^2}.$$

Doubling CO_2 gives a forcing of about $3.7\,\mathrm{W/m^2}$. On its own, this would warm the Earth by a little over 1 K (Problem 25.6). But warming triggers *feedbacks*. Warmer air holds more water vapour, itself a greenhouse gas, roughly doubling the effect. Melting ice exposes darker ground and ocean, lowering the albedo. Clouds respond in complicated ways that remain the largest uncertainty. The net result, the *equilibrium climate sensitivity*, is about 3 K per doubling, likely between 2.5 and 4 K. Since 1850, CO_2 has risen from 280 to over 420 parts per million, and the Earth has warmed by about 1.2 K, consistent with these numbers once the slow warming of the ocean is accounted for.

Climate history. Over the last million years, ice ages have come and gone roughly every 100 000 years, paced by small periodic changes in the Earth's orbit and tilt (*Milankovitch cycles*) and amplified by the feedbacks of ice and CO_2. Further back, the Earth may have frozen over almost completely, at least twice, in "snowball Earth" episodes. Because the ice–albedo feedback can create multiple stable states, the climate can in principle show tipping points and hysteresis — the bifurcations of Chapter 12 on a planetary scale (Problem 25.12).

§25.10 Problems

25.1 $O(\epsilon)$ ★ From $g = 9.8\,\mathrm{m/s^2}$, $R = 6371\,\mathrm{km}$, and G, compute the Earth's mean density. Compare with the density of surface rock, $2.7\,\mathrm{g/cm^3}$, and of iron, $7.9\,\mathrm{g/cm^3}$. What do you conclude?

25.2 $O(\epsilon)$ Using a scale height of 8 km, at what altitude is the pressure half its sea-level value? What fraction of the atmosphere's mass lies above the summit of Everest (8.85 km)?

25.3 $O(\epsilon)$ A tsunami crosses the Pacific Ocean (average depth 4 km) from Chile to Japan, about 17 000 km. How long does it take? Why do tsunami warnings for distant coasts work?

25.4 $O(\epsilon^2)$ ★ Derive the Earth's effective temperature $T_e \approx 255\,\mathrm{K}$ from the energy balance, with $S = 1361\,\mathrm{W/m^2}$ and $A = 0.3$. Repeat for Venus ($S = 2600\,\mathrm{W/m^2}$, $A = 0.77$) and compare with its actual surface temperature of 737 K. What does Venus tell you?

25.5 $O(\epsilon^2)$ (*A one-layer greenhouse.*) Model the atmosphere as a single layer that is transparent to sunlight but absorbs all infrared, radiating equally up and down. Show that the surface temperature is $2^{1/4}T_e \approx 303\,\mathrm{K}$. Why is the real Earth cooler than this model predicts?

25.6 $O(\epsilon^2)$ ★ If the Earth simply radiated more as a blackbody to balance an extra forcing ΔF, show that its effective temperature would rise by $\Delta T = \Delta F/(4\sigma T_e^3)$. Evaluate this for a CO_2 doubling ($\Delta F = 3.7\,\mathrm{W/m^2}$). Then, if water vapour feedback amplifies warming by a factor $1/(1-f)$ with $f \approx 0.5$, what is the warming?

25.7 $O(\epsilon^2)$ ★ Show that the tidal acceleration across the Earth due to a body of mass M at distance r is about $2GMR/r^3$. Compute the ratio of the lunar to the solar tidal effect. Why is the Sun's tide weaker, even though the Sun's *gravitational* pull on the Earth is about 180 times stronger than the Moon's?

25.8 $O(\epsilon^2)$ Derive the dry adiabatic lapse rate $\Gamma = g/c_p$ by combining the hydrostatic equation $\mathrm{d}P/\mathrm{d}z = -\rho g$ with the adiabatic law for an ideal gas. Evaluate it for air, with $c_p \approx 1000\,\mathrm{J/(kg\,K)}$.

25.9 $O(\epsilon^2)$ Compute the Rossby number for draining a bathtub ($L \approx 1\,\mathrm{m}$, $U \approx 0.1\,\mathrm{m/s}$) and for a hurricane ($L \approx 500\,\mathrm{km}$, $U \approx 30\,\mathrm{m/s}$) at latitude 30°. Settle the bathtub question.

25.10 $e^{-1/\epsilon}$ ★ (*The size of the core.*) In a crude model, seismic rays travel in straight lines through the mantle, and the S-wave shadow begins at an angular distance of 104° from the earthquake: this is the ray that just grazes the core. Find the core's radius in this model. (The true value is 3480 km; real rays curve, because seismic speeds increase with depth.)

25.11 $e^{-1/\epsilon}$ (*The lengthening day.*) The Moon recedes from the Earth at 3.8 cm/yr. Using conservation of the total angular momentum of the Earth–Moon system (Earth's spin

plus the Moon's orbit; take the Earth's moment of inertia as $0.33MR^2$), estimate how fast the length of the day is increasing, in milliseconds per century. Ancient eclipse records give about 2 ms per century. How long was the day when the dinosaurs lived, 100 million years ago?

25.12 $e^{-1/\epsilon}$ (*Snowball Earth.*) Let the albedo depend on temperature: $A = 0.65$ for $T < 250\,\mathrm{K}$ (ice-covered), $A = 0.3$ for $T > 280\,\mathrm{K}$, and varying linearly in between. Plot the absorbed sunlight $(1 - A(T))S/4$ and the emitted radiation $\epsilon\sigma T^4$ (with $\epsilon \approx 0.6$ to mimic the greenhouse effect) on the same graph. Show that there can be three equilibria, two of them stable. Explain how the Earth could get stuck in a snowball state, and what it would take to get out (hint: volcanic CO_2 accumulating for millions of years).

Further reading

- **Stacey and Davis, *Physics of the Earth*** — A comprehensive physics-based treatment of the solid Earth.
- **William Lowrie, *Fundamentals of Geophysics*** — A clear introduction to seismology, gravity, magnetism, and heat flow.
- **Raymond T. Pierrehumbert, *Principles of Planetary Climate*** — The physics of climate, on Earth and other planets, from first principles.
- **Marshall and Plumb, *Atmosphere, Ocean, and Climate Dynamics*** — An excellent introduction to geophysical fluid dynamics.
- **David Archer, *Global Warming: Understanding the Forecast*** — A short, quantitative introduction to climate change for non-specialists.

CHAPTER 26

General relativity

BUILDS ON special relativity (Chapter 10)

Gravity reimagined as the bending of spacetime itself — Einstein's masterpiece, which predicted black holes, an expanding universe, and ripples in the fabric of space.

PROTOTYPE A black hole: a region where spacetime is curved so strongly that not even light can escape. It is the simplest macroscopic object in the universe, completely described by its mass and its spin, and it is where general relativity stops being a small correction to Newton and becomes the whole story.

Special relativity left a problem. Newton's gravity acts instantly across any distance, which special relativity forbids; and it was unclear how gravity should treat energy, which special relativity had shown is a form of mass. Einstein spent eight years, from 1907 to 1915, finding a theory of gravity compatible with relativity. What he found was not a new force at all. Gravity, in general relativity, is the curvature of spacetime, and freely falling objects simply follow the straightest possible paths through it. It is widely regarded as the most beautiful theory in physics, and it has passed every test.

§26.1 The equivalence principle

Galileo found that all bodies fall at the same rate, whatever their mass or composition. In Newton's theory this is a coincidence: the mass that appears in $F = ma$ (inertial mass) happens to equal the mass that appears in $F = GMm/r^2$ (gravitational mass). Experiments have confirmed the equality to about one part in 10^{15}. Einstein took it as a clue.

Imagine you are in a closed elevator. If it is at rest on Earth, you feel your weight. If it is in deep space, accelerating upward at $9.8\,\mathrm{m/s^2}$, you feel exactly the same thing, and no experiment inside can tell the difference. If instead the elevator cable snaps and it falls freely, you float: gravity has vanished.

Law 26.1 (The equivalence principle)**.** Locally — in a small enough region over a short enough time — the effects of gravity are indistinguishable from those of acceleration. In particular, a freely falling frame is locally an inertial frame, in which special relativity holds.

Einstein called this "the happiest thought of my life." Two consequences follow immediately. First, *light bends in gravity*: in an accelerating elevator, a light beam crossing the cabin appears to curve downward, so by the equivalence principle it must curve in a gravitational field too. Second, *clocks run slower deeper in a gravitational potential*. Send a light pulse up an accelerating

rocket of height h: by the time it arrives, the receiver is moving faster, so it sees the pulse Doppler-shifted to lower frequency (Problem 26.3). By the equivalence principle, light climbing out of a gravitational field is *redshifted* by

$$\frac{\Delta f}{f} = \frac{gh}{c^2} = \frac{\Delta\Phi}{c^2},$$

where Φ is the Newtonian potential. In 1959 Pound and Rebka measured this shift — two parts in 10^{15} — for gamma rays climbing a 22.5 m tower at Harvard. Today's optical clocks detect it over a height difference of one centimetre (Chapter 18).

§26.2 Gravity is geometry

If gravity can be removed locally by falling freely, then it is not really a force. What remains, and cannot be removed, is *tidal* effects: two balls dropped side by side in a falling elevator drift slowly toward each other, because they fall toward the Earth's centre along converging lines; two balls one above the other drift apart. No choice of frame removes this relative acceleration. Einstein's insight was that tidal effects are *curvature*.

Think of two travellers on the Earth's surface, starting at the equator a few kilometres apart and both heading due north, each walking in the straightest possible line. They gradually approach each other and meet at the pole. No force pulled them together; the surface is curved. In general relativity, freely falling objects follow the straightest possible paths — *geodesics* — through curved spacetime, and they appear to attract because spacetime is curved by mass and energy.

The geometry of spacetime is encoded in the **metric**, which gives the interval between nearby events:

$$\mathrm{d}s^2 = g_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu.$$

For flat spacetime, $g_{\mu\nu}$ is the Minkowski metric of Chapter 10. For weak gravity, the time–time component is $g_{00} \approx -(1 + 2\Phi/c^2)$: the Newtonian potential is just a small warping of time. In fact, almost all of everyday gravity — apples, orbits, tides — comes from the warping of *time*, not space: clocks run at different rates at different heights, and objects drift toward where time runs slowest. A freely falling clock maximizes its proper time between two events (Chapter 10), and that principle alone reproduces Newton's laws in the weak-field limit.

§26.3 Einstein's equations

How does matter curve spacetime? After years of struggle, Einstein found the answer in November 1915:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}\,T_{\mu\nu}.$$

On the right is the *stress–energy tensor* $T_{\mu\nu}$, describing the density and flow of energy and momentum. On the left is the *Einstein tensor* $G_{\mu\nu}$, built from the second derivatives of the metric — a measure of curvature constructed from the *Riemann tensor*, which describes how vectors twist when carried around small loops. The term with Λ, the *cosmological constant*, Einstein added in 1917 and later regretted; it is back, as dark energy (Chapter 28).

John Wheeler summarized: *spacetime tells matter how to move; matter tells spacetime how to curve.* These are ten coupled nonlinear partial differential equations, and exact solutions are rare and precious. For weak fields and slow motions they reduce to Newton's $\nabla^2\Phi = 4\pi G\rho$. The pressure and energy of matter also gravitate, not just its mass, which matters in neutron stars and the early universe.

§26.4 The Schwarzschild solution

Within weeks of Einstein's paper, Karl Schwarzschild, serving at the Russian front, found the exact solution for the spacetime outside a spherical mass:

$$\mathrm{d}s^2 = -\left(1 - \frac{r_s}{r}\right)c^2\,\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{1 - r_s/r} + r^2\left(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2\right), \qquad r_s = \frac{2GM}{c^2}.$$

The *Schwarzschild radius* r_s is about 3 km for the Sun and 9 mm for the Earth. As long as the mass is spread over a radius much larger than r_s, the corrections to Newton are small, of order r_s/r: about 10^{-9} at the Earth's surface and 10^{-6} at the Sun's.

Reading the metric: a clock at rest at radius r ticks off proper time $\mathrm{d}\tau = \sqrt{1 - r_s/r}\;\mathrm{d}t$, slower than a clock far away. Radial distances are stretched by $1/\sqrt{1 - r_s/r}$: the space around a mass is curved.

§26.5 The classic tests

General relativity's corrections are small in the solar system, but measurable.

- **Mercury's perihelion.** Mercury's elliptical orbit slowly rotates, mostly because of tugs from other planets. But after accounting for all of them, 43 arcseconds per century were unexplained. General relativity predicts exactly 43″ (Problem 26.11). Einstein later said the discovery gave him heart palpitations.
- **Light deflection.** Starlight grazing the Sun is bent by 1.75″, twice what a Newtonian argument for light "particles" would give, because space as well as time is curved (Problem 26.5). Eddington's 1919 eclipse expedition confirmed it and made Einstein world-famous overnight.
- **The Shapiro delay.** Radar signals passing near the Sun are delayed, because they travel through curved space. Tracking of the Cassini spacecraft in 2003 confirmed the prediction to about one part in 10^5.
- **Frame dragging.** A spinning mass drags spacetime around with it. Gravity Probe B measured this tiny effect with gyroscopes orbiting the Earth.
- **GPS.** GPS satellite clocks, higher in the Earth's potential, run fast by 45 μs per day relative to the ground; their orbital speed slows them by 7 μs per day (Problem 10.4). Uncorrected, the net 38 μs per day would cause position errors growing by about ten kilometres a day. Your phone is a test of general relativity.

§26.6 Black holes

If a mass is compressed inside its Schwarzschild radius, something extraordinary happens. At $r = r_s$, the time coefficient in the metric vanishes: light emitted from there is infinitely redshifted, and nothing, not even light, can climb out. The sphere $r = r_s$ is the *event horizon*, and the region inside is a *black hole*.

The horizon is not a physical surface. An astronaut falling in notices nothing special as she crosses it (for a large black hole, the tidal forces there are gentle; Problem 26.6). The apparent infinity in the metric at r_s is an artefact of the coordinates, like the apparent singularity of longitude at the North Pole. But once inside, all future-pointing paths lead inward, toward $r = 0$, where the curvature becomes truly infinite — a *singularity*, where general relativity breaks down. A distant observer, meanwhile, never sees her cross: her signals are redshifted ever more and she fades to black, frozen at the horizon.

A few distances characterize the region around a black hole:

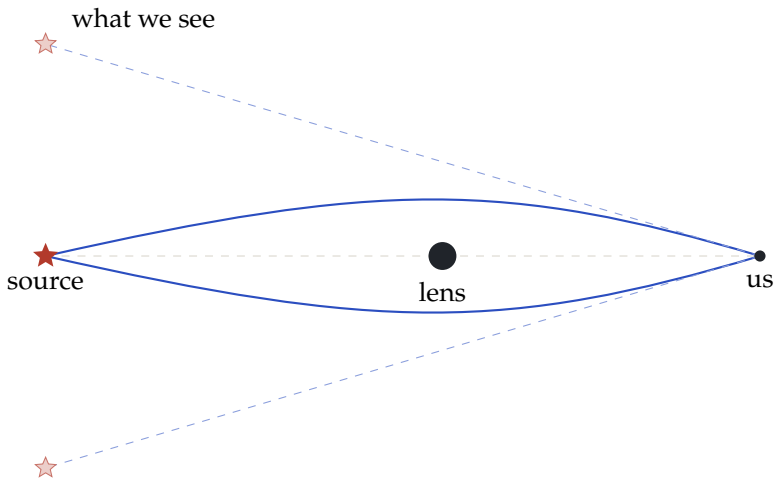
- At $r = \frac{3}{2}r_s$, light can orbit in a circle: the *photon sphere*, which forms the bright ring in images of black holes.
- At $r = 3r_s$ is the *innermost stable circular orbit*. Matter in an accretion disk spirals in to here, then plunges. The energy released on the way down is about 6% of the rest mass energy — far more than the 0.7% from nuclear fusion (Chapter 22) — and up to 42% for a rapidly spinning black hole. Accreting black holes power the brightest objects in the universe, quasars (Chapter 27).

Rotating black holes are described by Roy Kerr's 1963 solution, which includes an *ergosphere* outside the horizon where spacetime is dragged around so fast that nothing can stand still. The *no-hair theorem* says that a black hole in equilibrium is completely described by three numbers: mass, spin, and charge. Everything else about what fell in is hidden.

For decades black holes were regarded as mathematical curiosities. Now they are among the best-observed objects in astronomy. Stars orbiting the centre of our galaxy reveal a black hole of four million solar masses, Sagittarius A* (Nobel Prize, 2020; Problem 26.10). The Event Horizon Telescope, a network of radio dishes the size of the Earth, imaged the shadow of the black hole in the galaxy M87 in 2019, and Sagittarius A* in 2022. And LIGO hears black holes merge (Chapter 29).

§26.7 Gravitational lensing

Because mass bends light, a galaxy or cluster between us and a more distant source acts as a lens, distorting, magnifying, and multiplying its image. A perfectly aligned source appears as an *Einstein ring*. Lensing weighs whatever does the bending, including matter that emits no light, which makes it one of the key tools for mapping dark matter (Chapter 28). Within our galaxy, a star passing in front of another briefly brightens it (*microlensing*), a technique that has found exoplanets.



Gravitational lensing. Light from a distant source bends around a massive galaxy or cluster on its way to us, so we see the source displaced, and here split into two images. If source, lens and observer lined up exactly, the images would spread into an Einstein ring.

§26.8 Hawking radiation

In 1974 Stephen Hawking combined general relativity with quantum field theory (Chapter 30) and found that black holes are not black. Quantum fluctuations near the horizon produce real particles that escape, and a black hole radiates like a blackbody at the *Hawking temperature*

$$T_H = \frac{\hbar c^3}{8\pi G M k_{\mathrm{B}}}.$$

For a solar-mass black hole this is 6×10^{-8} K, far colder than the cosmic microwave background, so real astrophysical black holes absorb far more than they emit. But in principle, a black hole slowly evaporates, faster as it shrinks. Combined with Bekenstein's discovery that black holes have entropy proportional to their horizon area, Hawking's result made black holes thermodynamic objects — and raised the *information paradox*: if a black hole evaporates into featureless thermal radiation, what happens to the information about what fell in? This question is one of the main drivers of research into quantum gravity (Chapter 33).

§26.9 Where general relativity goes

Applied to the universe as a whole, general relativity predicts that space itself must expand or contract (Chapter 28). Its equations have wave solutions, ripples in spacetime that travel at the speed of light (Chapter 29). And its singularities — at the centres of black holes and at the beginning of the universe — are places where the theory predicts its own failure, because it ignores quantum mechanics (Chapter 33).

§26.10 Problems

26.1 $O(\epsilon)$ ★ Compute the Schwarzschild radius of the Sun, the Earth, and a 70 kg person. How dense would each have to be to become a black hole? Why is it easier to make a black hole from a large mass than a small one?

26.2 $O(\epsilon)$ Compute the fractional gravitational redshift for light climbing the 22.5 m tower in the Pound–Rebka experiment, and for light escaping from the surface of the Sun to infinity. How would the Sun's spectral lines be shifted?

26.3 $O(\epsilon^2)$ ★ (*Redshift from the equivalence principle.*) A rocket of height h accelerates upward at g in empty space. A light pulse is emitted from the floor at frequency f and travels to the ceiling in time $\approx h/c$, during which the ceiling's speed increases by gh/c. Use the (non-relativistic) Doppler shift to show that it is received at $f(1-gh/c^2)$. Explain why the equivalence principle then implies that clocks higher in a gravitational field run faster.

26.4 $O(\epsilon^2)$ ★ GPS satellites orbit at an altitude of 20 200 km. Compute the fractional rate difference between their clocks and clocks on the ground due to the difference in gravitational potential, and convert it to microseconds per day. Combine it with the special-relativistic effect from Problem 10.4 to get the net correction.

26.5 $O(\epsilon^2)$ ★ A Newtonian particle moving at speed c past the Sun with impact parameter b is deflected by angle $2GM/(c^2 b)$. Evaluate this for light grazing the Sun's surface. General relativity predicts twice this. Why would Eddington's measurement, if it had found the Newtonian value, have refuted Einstein but not the equivalence principle?

26.6 $O(\epsilon^2)$ ★ (*Spaghettification.*) The tidal acceleration across a body of length L at distance r from mass M is about $2GML/r^3$. Compute it for a 2 m person at the event horizon of a 10-solar-mass black hole and of Sagittarius A* (4×10^6 solar masses). Show that it scales as $1/M^2$ at the horizon. Which black hole could you fall into intact?

26.7 $O(\epsilon^2)$ On a sphere of radius R, two travellers start on the equator a distance d apart and walk due north. Show that their separation is $d\cos(s/R)$ after each has walked a distance s, so their "relative acceleration" is $-d/R^2$ for small s. Compare with the tidal convergence of two balls dropped side by side toward the Earth, and identify the analogue of the sphere's curvature.

26.8 $O(\epsilon^2)$ For light moving in the Schwarzschild geometry, one can show that the radial motion behaves as if in an effective potential proportional to $(1 - r_s/r)/r^2$. Show that this has a maximum at $r = \tfrac{3}{2}r_s$, the photon sphere, and explain why circular light orbits there are unstable.

26.9 $O(\epsilon^2)$ Compute the Hawking temperature of a black hole with the mass of the Sun, and of one with the mass of the Moon (7×10^{22} kg). Which would absorb more from the 2.7 K cosmic background than it emits? The evaporation time scales as M^3; given that a black hole of about 2×10^{11} kg would evaporate in the age of the universe, estimate the evaporation time of a solar-mass black hole.

26.10 $O(\epsilon^2)$ ★ The star S2 orbits Sagittarius A* with a period of 16.0 years and a semi-major axis of about 1000 AU. Use Kepler's third law in solar units ($M = a^3/T^2$, with a in AU

and T in years) to find the mass of the central object. At its closest approach, S2 is about 120 AU from the centre. Why must the central object be a black hole?

26.11 $e^{-1/\epsilon}$ ★ (*Mercury's perihelion.*) In general relativity, the effective potential for a planet has, in addition to the Newtonian terms, a small attractive term $-GML^2/(c^2m^2r^3)$ per unit mass, where L is the angular momentum. Treating it as a perturbation, show that the orbit precesses by

$$\Delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)}$$

per orbit. Evaluate it for Mercury ($a = 5.79\times10^{10}\,\mathrm{m}$, $e = 0.206$, period 88 days) in arcseconds per century.

Hint: Write the orbit equation in terms of $u = 1/r$; the correction adds a term $\propto u^2$. Look for a solution $u \propto 1 + e\cos(\omega\phi)$ with ω slightly less than 1.

26.12 $e^{-1/\epsilon}$ (*The horizon you carry with you.*) An observer accelerates forever at constant proper acceleration a (Problem 10.9 has the formulas). Show that light emitted from more than a distance c^2/a behind her at the moment she starts can never catch up: she has a horizon. Unruh showed in 1976 that such an observer sees the vacuum as a thermal bath at temperature $\hbar a/(2\pi c k_{\mathrm{B}})$. Compare this with the Hawking temperature, using the acceleration needed to hover just outside a black hole's horizon. Why must the equivalence principle make the two effects related?

Further reading

- **James B. Hartle,** ***Gravity: An Introduction to Einstein's General Relativity*** — A physics-first introduction that reaches black holes, cosmology, and gravitational waves before the full tensor machinery.
- **Bernard Schutz,** ***A First Course in General Relativity*** — The standard undergraduate text, developing tensors carefully.
- **Sean Carroll,** ***Spacetime and Geometry*** — The modern graduate standard, clear and comprehensive.
- **Kip Thorne,** ***Black Holes and Time Warps*** — The history and physics of black holes, beautifully told for a general audience.

CHAPTER 27

Astrophysics

BUILDS ON statistical mechanics (Chapter 9), nuclear physics (Chapter 22), general relativity (Chapter 26)

The life of stars and the violent afterlives of what they leave behind — all deduced from light that left its source before we were born.

PROTOTYPE The Sun: a ball of hot gas, held together by its own gravity and held up by its own pressure. Fusion in its core supplies the heat that keeps the pressure up. Nearly everything about it — its temperature, its size, its brightness, its lifetime — follows from that balance, with a few lines of estimation.

Astrophysics is physics without experiments. We cannot touch a star, weigh a galaxy, or rewind a supernova; almost all we have is the light that reaches us, plus a trickle of neutrinos, cosmic rays, and, since 2015, gravitational waves. From these, astrophysicists have worked out how stars are born, live, and die, what powers the brightest objects in the universe, and how much of the universe is made of matter we cannot see. Every branch of physics in this book is needed, and the spherical cow is not a joke here; it is the main tool.

§27.1 What light tells us

Distance. The nearest stars shift slightly against the background as the Earth orbits the Sun. A star with a *parallax* of one arcsecond is one *parsec* away, 3.26 light-years; the nearest star, Proxima Centauri, is 1.3 pc. The Gaia satellite has measured parallaxes for over a billion stars. Farther out, astronomers use *standard candles*, objects of known luminosity L: measuring their flux $F = L/4\pi d^2$ gives their distance. Cepheid variable stars, whose pulsation period reveals their luminosity, reach nearby galaxies; Type Ia supernovae reach across the universe (Chapter 28). Each rung of this *distance ladder* is calibrated by the one below.

Temperature and composition. A star's colour gives its surface temperature through Wien's law (Chapter 9): red stars are about 3000 K, the Sun 5800 K, blue stars 30 000 K. Its spectral lines (Chapter 18) reveal its composition. In 1925 Cecilia Payne-Gaposchkin showed, using the Saha equation (Chapter 17), that stars are made mostly of hydrogen and helium — a conclusion so contrary to expectation that she was pressured to call it "almost certainly not real."

Motion. Doppler shifts of spectral lines give velocities along the line of sight, to better than a metre per second with modern spectrographs.

§27.2 A star is a spherical cow

A star is in *hydrostatic equilibrium*: at every radius, the pressure gradient holds up the weight of the layers above,

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{Gm(r)\rho(r)}{r^2}.$$

The **virial theorem** gives a powerful shortcut: for a gas in equilibrium under its own gravity, twice the thermal kinetic energy equals minus the gravitational energy, $2K = -U$. With $K \sim Nk_{\mathrm{B}}T$ and $U \sim -GM^2/R$, the typical internal temperature is

$$k_{\mathrm{B}}T \sim \frac{GMm_p}{R}.$$

For the Sun this gives about 10^7 K (Problem 27.4) — right at the temperature needed for hydrogen fusion. That is no coincidence: a contracting ball of gas heats up until fusion switches on and halts the contraction. Stars are self-regulating thermostats.

The virial theorem has a strange consequence. A star that loses energy contracts and gets *hotter*: it has a negative heat capacity. This is why stars are stable. If fusion runs too fast, the star expands, cools, and fusion slows down.

In the nineteenth century, before nuclear physics, Kelvin and Helmholtz proposed that the Sun shines by slowly contracting, converting gravitational energy into heat. The available energy is $\sim GM^2/R$, which at the Sun's luminosity lasts only about 30 million years. Geologists and Darwin needed billions. The resolution, found in the 1930s by Bethe and others, was nuclear fusion (Chapter 22): converting 0.7% of the mass of hydrogen into energy, the Sun can shine for about ten billion years.

§27.3 Getting the energy out

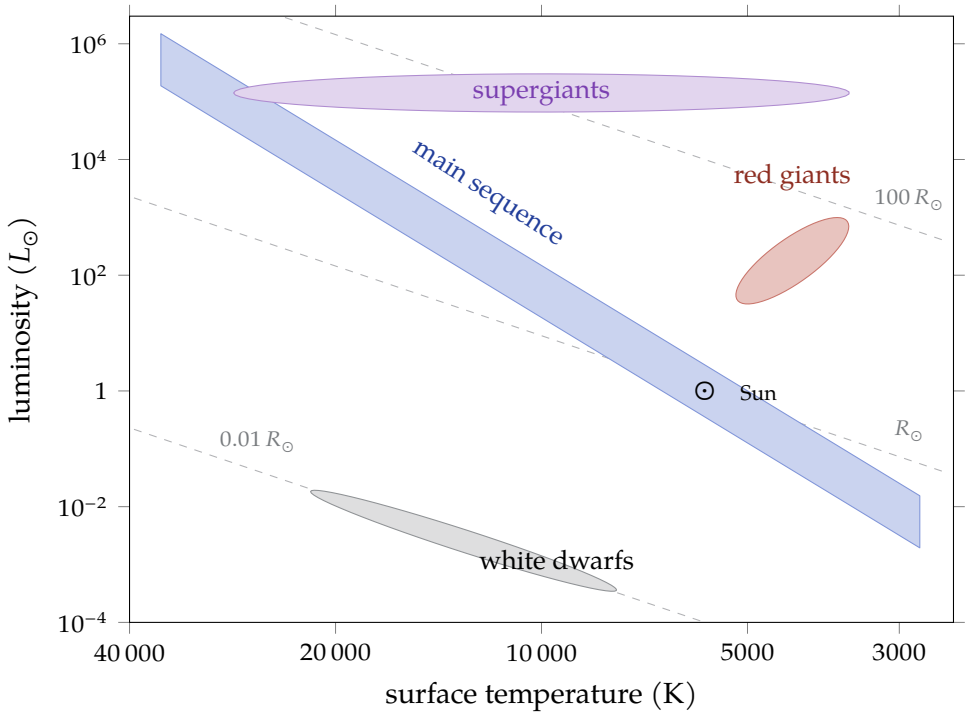
Energy produced in the Sun's core takes a long time to escape. A photon in the dense interior travels only about a millimetre before being absorbed and re-emitted in a random direction. To random-walk 7×10^8 m out takes $(R/\ell)^2 \sim 10^{24}$ steps and tens of thousands of years (Problem 27.5). The energy reaching you as sunlight today left the core before the end of the last ice age. In the outer third of the Sun, the gas becomes opaque enough that convection takes over, carrying heat up in boiling cells visible as the granulation of the surface. Neutrinos, by contrast, escape the core in two seconds and reach us in eight minutes: they show us the Sun's core as it is now.

Combining hydrostatic equilibrium with the physics of radiative diffusion gives a steep *mass–luminosity relation* for main-sequence stars, roughly $L \propto M^{3.5}$. A star's lifetime is its fuel ($\propto M$) divided by its burn rate ($\propto L$), so $t \propto M^{-2.5}$. Massive stars live fast and die young: a star of 10 solar masses lives about 30 million years, while a red dwarf of 0.1 solar masses will shine for trillions.

§27.4 The Hertzsprung–Russell diagram

Plot stars by luminosity against surface temperature and they fall into a few groups, not a random scatter. Most lie along the *main sequence*, the long middle age of hydrogen burning, running

from hot, bright, massive stars to cool, dim, small ones. Above it lie *giants* and *supergiants*, bright but cool, so they must be enormous. Below it lie *white dwarfs*, hot but faint, so they must be tiny. The HR diagram is the map on which stellar evolution is traced, and every star moves across it as it ages.



The Hertzsprung–Russell diagram, sketched. Most stars, the Sun among them, lie on the main sequence, where luminosity rises steeply with temperature. The dashed lines join stars of equal radius, since $L \propto R^2T^4$: giants are big and cool, white dwarfs small and hot. By astronomical tradition, temperature increases to the left.

§27.5 How stars live and die

When a star exhausts the hydrogen in its core, the core contracts and heats, hydrogen burns in a shell around it, and the envelope swells enormously: the star becomes a *red giant*. In about five billion years the Sun will swell past the orbit of Venus. The core eventually becomes hot enough to fuse helium into carbon and oxygen. Then the fates diverge.

Low-mass stars like the Sun never get hot enough to fuse carbon. They puff off their outer layers as a glowing *planetary nebula* and leave behind their exposed core: a *white dwarf*, the mass of the Sun packed into the size of the Earth, which slowly cools forever.

Massive stars, above about eight solar masses, burn through successively heavier fuels — carbon, neon, oxygen, silicon — in shells like an onion, each stage faster than the last: the final silicon burning takes about a day. The core becomes iron, from which no more energy can be extracted by fusion (Chapter 22). With nothing to hold it up, the iron core, more massive than the Sun, collapses in less than a second, from the size of the Earth to a ball of neutrons twenty kilometres across. Infalling matter rebounds off the stiff neutron core, and a flood of neutrinos carrying 99% of the energy helps blast the rest of the star apart: a *core-collapse supernova*, briefly

as bright as a whole galaxy. In 1987, a supernova in the nearby Large Magellanic Cloud was seen first in neutrinos: two dozen were caught by detectors in Japan and the United States, confirming the theory. Supernovae seed the galaxy with the heavy elements made in the star.

§27.6 Compact objects

White dwarfs are held up not by thermal pressure but by *electron degeneracy pressure*, the resistance of electrons to being squeezed into the same states (Chapters 9 and 14). Balancing degeneracy pressure against gravity gives a strange result: more massive white dwarfs are *smaller*, $R \propto M^{-1/3}$ (Problem 27.8). As the mass increases, the electrons are squeezed until they move at nearly the speed of light, and their pressure can no longer keep up with gravity. In 1930, on a ship to England, the nineteen-year-old Subrahmanyan Chandrasekhar found that there is a maximum mass,

$$M_{\text{Ch}} \approx 1.4\, M_\odot,$$

above which no white dwarf can exist (Problem 27.11). It is built from fundamental constants: $M_{\text{Ch}} \sim (\hbar c/G)^{3/2}/m_p^2$. A white dwarf in a binary that accretes matter past this limit is destroyed in a thermonuclear explosion, a *Type Ia supernova*. Because they all explode at nearly the same mass, these are standard candles, and they revealed the accelerating expansion of the universe (Chapter 28).

Neutron stars are what remains after a core collapse, held up by neutron degeneracy and nuclear forces: one and a half solar masses in a sphere about 12 km across, the density of an atomic nucleus. Conservation of angular momentum spins them up (Chapter 2) and conservation of magnetic flux (Chapter 17) amplifies their fields to 10^8 T or more. A rotating neutron star whose magnetic axis sweeps a beam of radio waves past the Earth is seen as a *pulsar*, ticking with a regularity rivalling atomic clocks. The first was found in 1967 by the graduate student Jocelyn Bell Burnell, who noticed a "bit of scruff" on her chart recorder. The fastest spins 716 times a second.

Above about two to three solar masses, nothing can resist gravity, and the core collapses to a **black hole** (Chapter 26).

§27.7 Accretion and the brightest things in the universe

Matter falling onto a compact object of mass M and radius R releases gravitational energy GMm/R per mass m. For a neutron star or black hole this is 10 to 40% of mc^2, dwarfing nuclear fusion. Infalling gas, carrying angular momentum, forms an *accretion disk* that heats up to millions of degrees and glows in X-rays.

There is a limit to how bright an accreting object can be. Radiation pushes outward on the infalling gas, mainly by scattering off electrons, while gravity pulls inward, mainly on protons. The two balance at the **Eddington luminosity**

$$L_{\text{Edd}} = \frac{4\pi GMm_p c}{\sigma_T} \approx 3 \times 10^4\, L_\odot \frac{M}{M_\odot},$$

where σ_T is the Thomson cross-section for scattering light off an electron (Problem 27.7). At the centres of galaxies lurk *supermassive black holes* of millions to billions of solar masses. When they

feed, they become *quasars* and other *active galactic nuclei*, outshining their entire host galaxies from a region the size of the solar system, and often launching jets of plasma at nearly the speed of light across hundreds of thousands of light-years.

§27.8 Galaxies and dark matter

Our galaxy, the Milky Way, is a disk of a few hundred billion stars, 30 kiloparsecs across, with the Sun about 8 kpc from the centre, orbiting at 220 km/s once every 230 million years. If most of the galaxy's mass were in its visible stars, concentrated toward the centre, the orbital speeds of stars and gas far from the centre should fall off like Kepler's planets, $v \propto r^{-1/2}$. In the 1970s Vera Rubin and Kent Ford measured the rotation curves of spiral galaxies, and radio astronomers traced the 21-cm line of hydrogen far beyond the visible disks (Chapter 18). The curves are *flat*: orbital speeds stay constant out to great distances. The enclosed mass must keep growing in proportion to radius (Problem 27.9), far beyond where the stars thin out. Galaxies are embedded in vast halos of invisible *dark matter*, outweighing the visible matter by about five to one. Fritz Zwicky had seen the same thing in clusters of galaxies in 1933. What dark matter is remains one of the great open questions in physics (Chapters 28 and 32).

§27.9 Exoplanets

In 1995 Michel Mayor and Didier Queloz detected the tiny Doppler wobble of the star 51 Pegasi, pulled back and forth by a planet half the mass of Jupiter orbiting in just four days — a "hot Jupiter" that no theory had predicted. Since then over five thousand *exoplanets* have been found. The *radial velocity* method measures a star's wobble; the *transit* method, used by the Kepler and TESS satellites, detects the slight dimming as a planet crosses its star's face, by a fraction $(R_p/R_*)^2$ (Problem 27.10). Combining both gives a planet's mass and radius, hence its density, and transits even allow the light filtering through a planet's atmosphere to be analysed for its molecules. Planets turn out to be common, and many stars have planets in their *habitable zones*, where liquid water could exist.

§27.10 The high-energy universe

The universe also speaks in particles. *Cosmic rays*, mostly protons and nuclei, arrive with energies up to 10^{20} eV — the kinetic energy of a well-hit tennis ball in a single proton — accelerated by shock waves in supernova remnants and other, still uncertain, sources. *Gamma-ray bursts*, flashes of gamma rays that briefly outshine everything else, come from the collapse of massive stars and the mergers of neutron stars. The IceCube detector, a cubic kilometre of Antarctic ice instrumented with light sensors, catches high-energy neutrinos from distant galaxies. And since 2015, gravitational waves (Chapter 29) have added a new sense entirely: *multi-messenger astronomy*.

§27.11 Problems

27.1 $O(\epsilon)$ ★ A star has a parallax of 0.10 arcseconds. How far away is it in parsecs and light-years? Gaia measures parallaxes to about 20 micro-arcseconds. How far can it measure distances to 10% accuracy?

27.2 $O(\epsilon)$ ★ The Sun converts about 10% of its hydrogen to helium during its main-sequence life, releasing 0.7% of the rest mass energy. Estimate its main-sequence lifetime from its mass (2×10^{30} kg, about 70% hydrogen) and luminosity (3.8×10^{26} W).

27.3 $O(\epsilon)$ Estimate the Kelvin–Helmholtz time $GM^2/(RL)$ for the Sun. Why did this estimate cause a crisis in nineteenth-century science?

27.4 $O(\epsilon^2)$ ★ Use the virial theorem to estimate the Sun's central temperature from its mass and radius (7×10^8 m), treating it as a gas of protons and electrons. Compare with the thermal energy needed for protons to fuse classically (~ 1 MeV), and with the quantum-mechanical requirement (Problem 14.13).

27.5 $O(\epsilon^2)$ A photon in the solar interior has a mean free path of about 1 mm. Estimate how many steps it needs to random-walk out of the Sun, and how long that takes. How does this compare with the neutrino travel time?

27.6 $O(\epsilon^2)$ Using $L\propto M^{3.5}$ and the Sun's lifetime of 10^{10} years, estimate the main-sequence lifetimes of stars of 10, 3, and 0.2 solar masses. Which of these could have hosted planets on which life evolved?

27.7 $O(\epsilon^2)$ ★ Derive the Eddington luminosity by balancing the outward radiation force on an electron, $\sigma_T L/(4\pi r^2 c)$, against the inward gravitational force on a proton (the two are bound together electrostatically), using $\sigma_T = 6.65\times10^{-29}$ m^2. Evaluate it for a black hole of 10^9 solar masses, and compare with the luminosity of a large galaxy, about $10^{11}L_\odot$.

27.8 $O(\epsilon^2)$ (*White dwarfs shrink as they grow.*) For non-relativistic degenerate electrons, the pressure scales as $P\sim\hbar^2 n^{5/3}/m_e$ with $n\propto M/R^3$. The pressure needed to support a star scales as $P\sim GM^2/R^4$. Show that $R\propto M^{-1/3}$, and estimate the radius of a one-solar-mass white dwarf (about 2 electrons per proton plus neutron pair, i.e. $n\approx\rho/2m_p$).

27.9 $O(\epsilon^2)$ ★ Stars and gas in a galaxy orbit at a constant $v=220$ km/s from 8 kpc out to 50 kpc. Show that the mass enclosed within radius r is $M(r)=v^2 r/G$, and compute it at both radii. The visible stars and gas total about 6×10^{10} solar masses. What fraction of the mass within 50 kpc is dark?

27.10 $O(\epsilon^2)$ When Jupiter transits the Sun, by what fraction does the Sun dim? When the Earth does? ($R_\odot = 7.0\times10^5$ km, $R_J = 7.0\times10^4$ km, $R_\oplus = 6.4\times10^3$ km.) Why was it so much harder to find Earth-sized planets?

27.11 $e^{-1/\epsilon}$ ★ (*The Chandrasekhar mass.*) For ultra-relativistic degenerate electrons, each electron's energy is pc rather than $p^2/2m$, so the pressure scales as $P\sim\hbar c n^{4/3}$. Show that this pressure and the gravitational requirement GM^2/R^4 now scale the *same* way with R, so they cannot balance at any particular radius. Instead there is a critical mass, $M\sim$

$(\hbar c/G)^{3/2}/m_p^2$. Evaluate this. Explain physically why a white dwarf heavier than this must collapse.

27.12 $e^{-1/\epsilon}$ (*Why stars have the masses they do.*) Below about $0.08\,M_\odot$, a contracting ball of gas becomes electron-degenerate before its core reaches fusion temperatures, so it never ignites: a brown dwarf. Above about 100–$150\,M_\odot$, a star's luminosity would approach the Eddington limit. Using the virial temperature and the mass–luminosity relation, explain both limits qualitatively, and why the range of stellar masses spans only a factor of about 10^3.

27.13 $e^{-1/\epsilon}$ (*Making a pulsar.*) The iron core of a massive star, about the size of the Earth, rotates once per hour and has a magnetic field of about 100 T before collapsing to a neutron star of radius 12 km. Assuming angular momentum and magnetic flux are conserved, estimate the neutron star's spin period and magnetic field. Then estimate the maximum spin rate, at which the equator would move fast enough to fly off, and compare with the fastest known pulsar.

Further reading

- **Carroll and Ostlie,** ***An Introduction to Modern Astrophysics*** — The comprehensive undergraduate text, affectionately known as "the big orange book."
- **Ryden and Peterson,** ***Foundations of Astrophysics*** — A clear, modern, more concise alternative.
- **Dan Maoz,** ***Astrophysics in a Nutshell*** — A compact, physics-driven tour with excellent order-of-magnitude problems.
- **A. C. Phillips,** ***The Physics of Stars*** — A short, careful book on how stars work.

CHAPTER 28

Cosmology

BUILDS ON general relativity (Chapter 26), astrophysics (Chapter 27)

The history of the whole universe — written in the stretching of light, the afterglow of its birth, and the cosmic recipe of how much of everything there is.

PROTOTYPE A loaf of raisin bread rising in the oven. As the dough expands, every raisin moves away from every other, and raisins twice as far apart separate twice as fast. No raisin is at the centre. On the largest scales, that is the universe: not an explosion into space, but space itself expanding, carrying the galaxies apart.

Cosmology is the study of the universe as a whole: where it came from, what it contains, and where it is going. A century ago it was mostly philosophy. Today it is a precision science, with a standard model whose half-dozen parameters are measured to within a percent — and which contains two ingredients, dark matter and dark energy, that make up 95% of the universe and that nobody understands. The universe is the ultimate spherical cow: on large enough scales, it really is the same everywhere.

§28.1 The expanding universe

In 1929 Edwin Hubble, building on Vesto Slipher's measurements of galaxy redshifts and Henrietta Leavitt's Cepheid distance scale (Chapter 27), found that distant galaxies recede from us at speeds proportional to their distance:

$$v = H_0\, d, \qquad H_0 \approx 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}.$$

At first sight this puts us at the centre of an explosion. It does not. If space itself is stretching uniformly, then from *any* galaxy every other galaxy recedes with speed proportional to distance — like the raisins in the bread. There is no centre and no edge.

The light from a receding galaxy is *redshifted*: its wavelength is stretched by the same factor as space has expanded while the light was in transit,

$$1 + z = \frac{\lambda_{\mathrm{obs}}}{\lambda_{\mathrm{emit}}} = \frac{a(t_{\mathrm{now}})}{a(t_{\mathrm{emit}})},$$

where $a(t)$, the *scale factor*, measures the size of the universe relative to today. A galaxy at redshift $z = 1$ emitted its light when the universe was half its present size. Running the expansion backwards, everything was once on top of everything else, a time about $1/H_0 \approx 14$ billion years ago: the *Hubble time*. The universe had a beginning.

§28.2 The cosmological principle

On scales larger than a few hundred million light-years, the universe looks the same in every direction and at every place: it is *homogeneous* and *isotropic*. This **cosmological principle**, once an assumption, is now an observed fact, confirmed by galaxy surveys and the uniformity of the cosmic microwave background. It drastically simplifies general relativity: the only spacetimes with this symmetry are the *Friedmann–Lemaître–Robertson–Walker* metrics,

$$\mathrm{d}s^2 = -c^2\,\mathrm{d}t^2 + a(t)^2\left[\frac{\mathrm{d}r^2}{1-kr^2} + r^2\,\mathrm{d}\Omega^2\right],$$

in which the entire history of the universe is a single function $a(t)$, and the geometry of space is flat ($k=0$), positively curved like a sphere ($k>0$), or negatively curved like a saddle ($k<0$).

§28.3 The Friedmann equation

Einstein's equations for this metric give the **Friedmann equation**. Remarkably, it can be derived from Newtonian physics (Problem 28.5). Consider a sphere of radius $R = ar$ in a uniform universe of density ρ, and a test galaxy on its surface. By the shell theorem (Chapter 2), only the mass inside matters. Its energy per unit mass is $\tfrac{1}{2}\dot{R}^2 - GM/R = \text{constant}$, which gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2}.$$

The left side is H^2, the Hubble parameter squared. The universe is like a ball thrown upward: whether it expands forever or recollapses depends on whether its kinetic energy beats its gravitational energy. The dividing line is the *critical density*

$$\rho_c = \frac{3H^2}{8\pi G} \approx 9\times10^{-27}\,\mathrm{kg/m^3},$$

about five protons per cubic metre (Problem 28.2). If $\rho = \rho_c$, space is flat. It is conventional to measure each component's density as a fraction $\Omega = \rho/\rho_c$ of the critical density. Observations show that the universe is flat to within a fraction of a percent: $\Omega_{\text{total}} = 1$.

§28.4 What is in the universe

Different ingredients dilute differently as the universe expands:

- **Matter** (ordinary and dark): the number of particles in a volume stays fixed while the volume grows as a^3, so $\rho_m \propto a^{-3}$.
- **Radiation**: the number of photons also dilutes as a^{-3}, but each photon's wavelength is stretched by a, losing energy, so $\rho_r \propto a^{-4}$.
- **Dark energy**, if it is a cosmological constant: an energy of empty space itself, which stays the same however much space there is, so ρ_Λ is constant.

So the universe passes through eras. Early on, radiation dominates and $a \propto t^{1/2}$. After about 50 000 years, at a redshift of about 3400, matter takes over and $a \propto t^{2/3}$. About six billion years ago the expansion began to *accelerate*, and about four billion years ago dark energy overtook matter entirely; the universe is now heading toward exponential growth.

KEY IDEA Today's cosmic inventory: about 5% ordinary matter (atoms), 27% dark matter, and 68% dark energy, plus a trace of radiation and neutrinos. Everything we have ever seen or touched — every star, planet, and person — is the 5%.

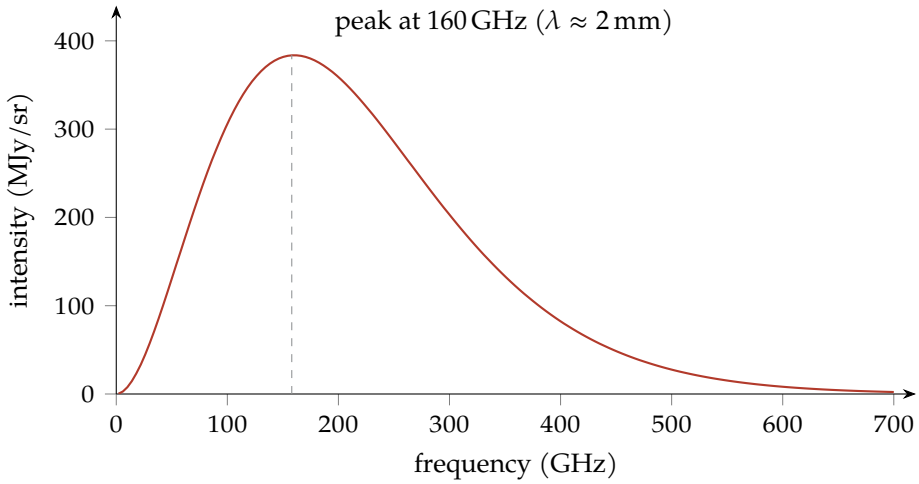
§28.5 The hot Big Bang

As the universe expands, radiation cools as $T \propto 1/a$. Running the film backwards, the early universe was hot and dense, and its history can be read off by asking what was happening at each temperature, using the statistical mechanics of Chapter 9 and the particle physics of Chapter 32.

- **First microsecond**: a quark–gluon plasma (Chapter 22). As it cools below about 10^{12} K, quarks are confined into protons and neutrons.
- **One second** (10^{10} K): neutrinos stop interacting and stream freely; a cosmic neutrino background, at 1.9 K, still fills the universe. The ratio of neutrons to protons freezes out at about 1 : 6, set by a Boltzmann factor and the neutron–proton mass difference.
- **Three minutes** (10^{9} K): **Big Bang nucleosynthesis**. The universe is now cool enough for deuterium to survive, and almost all the remaining neutrons are swept into helium-4. The result: about 25% of the mass in helium, the rest hydrogen, with traces of deuterium, helium-3, and lithium (Problem 28.9). These predictions, which depend on the density of ordinary matter, match the oldest stars and gas clouds — a precise test of physics in the first minutes of time.
- 380 000 **years** (3000 K): **recombination**. Electrons and nuclei combine into neutral atoms (Chapter 17). The universe, until now an opaque plasma, becomes transparent, and the light of that moment travels freely ever after.

§28.6 The cosmic microwave background

In 1965 Arno Penzias and Robert Wilson, testing a horn antenna at Bell Labs, found a faint microwave hiss coming equally from every direction, which no amount of cleaning (including removing pigeon droppings) would remove. It was the light of recombination, redshifted by a factor of 1100 from 3000 K to 2.7 K: the **cosmic microwave background** (CMB). In 1990 the COBE satellite measured its spectrum and found the most perfect blackbody ever observed, at 2.725 K (Chapter 9) — the smoking gun of a hot, dense early universe.



The spectrum of the cosmic microwave background: a blackbody at 2.725 K. COBE's measurements fell on this curve so precisely that their error bars would be invisible at this scale.

The CMB is not quite uniform. Its temperature varies by about one part in 10^5 across the sky, and those ripples are a snapshot of the infant universe. Before recombination, photons and ordinary matter formed a single fluid, in which gravity pulled matter into the slightly denser regions while photon pressure pushed back: sound waves rang through the young universe. At recombination they froze. The pattern of hot and cold spots shows a characteristic angular scale of about one degree — the distance sound could travel before recombination — with harmonics at smaller scales (the *acoustic peaks*). The angular size of that ruler measures the geometry of space (it is flat), and the heights of the peaks measure the amounts of ordinary matter, dark matter, and dark energy. The Planck satellite has pinned down the standard cosmological model to about a percent.

§28.7 Dark matter

The evidence for dark matter is overwhelming and comes from many independent directions: the flat rotation curves of galaxies and the motions of galaxies in clusters (Chapter 27); gravitational lensing by clusters, which weighs them directly (Chapter 26); the *Bullet Cluster*, where two clusters collided and their hot gas, slowed by the collision, separated from their mass, which passed through unimpeded; the acoustic peaks of the CMB; and the formation of galaxies, which could not have grown from the tiny CMB ripples in the available time without the extra gravity of matter that does not feel radiation pressure.

Whatever dark matter is, it is not ordinary matter: Big Bang nucleosynthesis and the CMB both limit ordinary matter to about 5%. It must be cold (moving slowly), interacting with ordinary matter feebly if at all, except through gravity. Leading candidates include weakly interacting massive particles (WIMPs), the extremely light *axion*, and primordial black holes. Enormous detectors deep underground, colliders, and astronomical searches have so far found nothing. This is one of the clearest signs that the Standard Model of particle physics is incomplete (Chapter 32).

§28.8 Dark energy

In 1998, two teams measuring distant Type Ia supernovae (Chapter 27) found them fainter than expected: farther away than they should be if the expansion were slowing down. The expansion of the universe is *accelerating* (Nobel Prize, 2011). Something with negative pressure, filling space uniformly, is pushing it apart. The simplest explanation is Einstein's cosmological constant Λ, a constant energy density of the vacuum.

Here lies the worst prediction in physics. Quantum field theory (Chapter 30) says the vacuum should have an energy density from the zero-point energies of all fields. Estimated naively, it exceeds the observed dark energy by about 120 orders of magnitude. Why the vacuum energy is so small, yet not zero, is the *cosmological constant problem*, and it is completely unsolved. Recent galaxy surveys have hinted that dark energy may change slowly with time, which would rule out a pure cosmological constant; this is being tested intensively.

§28.9 Inflation

The hot Big Bang leaves puzzles. **The horizon problem**: the CMB has the same temperature, to one part in 10^5, in opposite directions of the sky, but in the standard expansion those regions were never close enough to exchange light before recombination (Problem 28.8). How did they reach the same temperature? **The flatness problem**: any departure from flatness grows with time, so for the universe to be flat today it must have been flat to one part in 10^{60} at early times. Who tuned it?

In 1980 Alan Guth proposed **inflation**: a brief period, within the first 10^{-32} seconds, when the universe expanded exponentially, by a factor of at least e^{60}, driven by the vacuum energy of a field called the *inflaton*. Inflation stretches a tiny, causally connected region to a size larger than the observable universe (solving the horizon problem) and flattens any curvature, like inflating a balloon to the size of the Earth (solving the flatness problem). Most strikingly, it explains where the ripples came from: quantum fluctuations of the inflaton (Chapter 14), stretched to astronomical size, became the seeds of all structure. The predicted pattern — nearly, but not exactly, the same amplitude on all scales — matches the CMB beautifully. The galaxies may be quantum fluctuations, blown up. Inflation's most distinctive remaining prediction, a background of primordial gravitational waves imprinted as a twisting "B-mode" polarization of the CMB, is the object of intense searches.

§28.10 Structure and the cosmic web

From the tiny ripples, gravity did the rest. Denser regions pulled in more matter, growing slowly at first, then collapsing into dark matter halos, inside which gas cooled to form the first stars a few hundred million years after the Big Bang, and then galaxies. Computer simulations (Chapter 13) of this process reproduce the observed *cosmic web*: galaxies strung along filaments around vast empty voids, with clusters at the intersections. The James Webb Space Telescope is now seeing galaxies from the first few hundred million years, earlier and brighter than many expected.

§28.11 Open questions

The standard model of cosmology fits an extraordinary range of data with six parameters. But it raises as many questions as it answers. What is dark matter? What is dark energy, and is it constant? Measurements of H_0 from the nearby universe (73) and from the CMB (67) disagree by more than their errors — the *Hubble tension* — which may be an error, or a crack in the model. Why is there more matter than antimatter (Chapter 32)? What drove inflation? And what, if anything, came before?

§28.12 Problems

28.1 $O(\epsilon)$ ★ With $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, find the Hubble time in years. How fast does a galaxy 100 Mpc away recede? Galaxies beyond about 4300 Mpc recede faster than light. Why doesn't this violate relativity?

28.2 $O(\epsilon)$ ★ Compute the critical density for $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, in $\mathrm{kg/m^3}$ and in protons per cubic metre. Compare with the density of air and of interstellar gas in our galaxy (about 10^6 protons per cubic metre).

28.3 $O(\epsilon)$ The CMB was released at $z \approx 1100$. What was its temperature then? At what wavelength did it peak then, and at what wavelength does it peak now? What colour would the sky have been at recombination?

28.4 $O(\epsilon^2)$ ★ In a flat universe containing only matter, $a \propto t^{2/3}$. Show that the age of such a universe is $t_0 = 2/(3H_0)$, and evaluate it. The oldest stars are about 13 billion years old. What does this tell you?

28.5 $O(\epsilon^2)$ ★ Derive the Friedmann equation from Newtonian mechanics: write the energy per unit mass of a galaxy on the surface of a sphere of radius $R(t) = a(t)r$, with mass $M = \frac{4}{3}\pi R^3 \rho$ inside, and identify the constant of integration with the curvature term.

28.6 $O(\epsilon^2)$ Today, radiation has $\Omega_r \approx 9 \times 10^{-5}$ and matter $\Omega_m \approx 0.31$. At what redshift were their densities equal? How old was the universe then, roughly?

28.7 $O(\epsilon^2)$ The CMB contains about 411 photons per cubic centimetre. The average density of ordinary matter is about 0.25 protons per cubic metre. Find the ratio of baryons to photons. Why is this tiny number one of the most important in cosmology?

28.8 $O(\epsilon^2)$ ★ (*The horizon problem.*) At recombination, light could have travelled at most about $3ct$ since the Big Bang (a factor accounts for expansion), with $t = 380\,000$ years. That region has since been stretched by a factor of 1100. Today, the surface of last scattering is about 46 billion light-years away. Estimate the angle on the sky subtended by a causally connected patch at recombination. Why is it a puzzle that the whole sky has the same temperature?

28.9 $e^{-1/\epsilon}$ ★ (*Why a quarter of the universe is helium.*) Neutrons and protons stay in equilibrium, with $n/p = e^{-\Delta mc^2/k_B T}$ ($\Delta mc^2 = 1.29\,\mathrm{MeV}$), until the weak interactions freeze out at $k_B T \approx 0.8\,\mathrm{MeV}$. Find n/p at freeze-out. Free neutrons then decay (lifetime 880 s) for about three minutes until nucleosynthesis begins. Find the final n/p, and assuming all surviving neutrons end up in helium-4, show that the helium mass fraction is $Y = 2(n/p)/(1 + n/p)$, and compare with the observed $Y \approx 0.25$.

28.10 $e^{-1/\epsilon}$ (*How much inflation?*) During the radiation era, $|\Omega - 1|$ grows as a^2, and during the matter era as a. Today $|\Omega - 1| < 0.01$. Estimate how close to 1 the density had to be at the time when the temperature was $10^{15}\,\mathrm{GeV}$ (when a was about 10^{-28}). During inflation, $|\Omega - 1|$ shrinks as e^{-2N}, where N is the number of e-folds. How large must N be to explain the flatness without fine-tuning?

28.11 $e^{-1/\epsilon}$ ★ (*Olbers' paradox.*) Suppose the universe were infinite, eternal, static, and uniformly filled with stars. Show that every line of sight would eventually end on the surface of a star, so the night sky would be as bright as the surface of the Sun. Explain why our night sky is dark. Which matters more, the finite age of the universe or its expansion?

Further reading

- **Barbara Ryden, *Introduction to Cosmology*** — The best undergraduate introduction: clear, concise, and physical.
- **Steven Weinberg, *The First Three Minutes*** — The classic popular account of the early universe, still superb.
- **Daniel Baumann, *Cosmology*** — A modern graduate text covering inflation, the CMB, and structure formation.
- **Dodelson and Schmidt, *Modern Cosmology*** — The standard graduate reference on the CMB and large-scale structure.

CHAPTER 29

Gravitational waves

BUILDS ON general relativity (Chapter 26)

Ripples in spacetime itself, predicted in 1916, thought undetectable for half a century, and finally heard in 2015 as two black holes collided a billion light-years away.

PROTOTYPE An inspiralling binary: two black holes or neutron stars orbiting each other, radiating gravitational waves, losing energy, and spiralling closer and faster until they merge. Its signal is a *chirp*, a wave rising in both frequency and amplitude, ending in a crescendo. Its shape alone encodes the masses, spins, and distance of the pair.

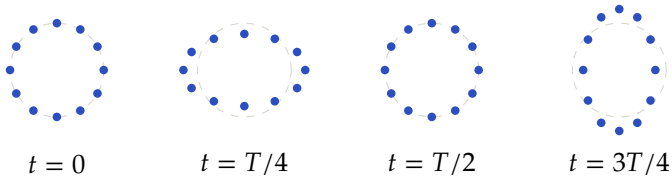
If mass curves spacetime, then moving masses should send ripples through it, spreading outward at the speed of light, the way a moving charge sends out electromagnetic waves (Chapter 6). Einstein predicted these *gravitational waves* in 1916, then doubted they were real, then decided they were too weak ever to detect. For decades physicists argued whether they carried energy at all. On 14 September 2015, the two LIGO detectors in Louisiana and Washington recorded a signal lasting a fifth of a second: the merger of two black holes, 1.3 billion years ago. A new sense had been added to astronomy.

§29.1 Ripples in spacetime

Far from any source, write the spacetime metric as flat spacetime plus a small perturbation, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h| \ll 1$. To first order, Einstein's equations in vacuum become a wave equation,

$$\left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2\right) h_{\mu\nu} = 0,$$

so gravitational waves travel at the speed of light. Like light, they are *transverse*, and they have two polarizations. But their effect is different: a gravitational wave passing through a ring of free particles stretches it in one direction while squeezing it in the perpendicular one, then reverses, twice per cycle. The two polarizations, called "plus" and "cross," do this along axes at 45° to each other.



A ring of free particles as a "plus"-polarized wave passes through the page.

The size of the effect is the dimensionless *strain* h: a length L changes by about $\Delta L = \frac{1}{2}hL$. And h is tiny. Spacetime is extraordinarily stiff, because the coupling between matter and curvature is G/c^4, about 10^{-44} in SI units. The strongest waves reaching the Earth from astronomical sources have $h \sim 10^{-21}$: they change the distance between the Earth and the Sun by about the width of an atom.

§29.2 How gravitational waves are made

In electromagnetism, the dominant radiation comes from an oscillating electric *dipole*. Gravity has no dipole radiation. The "mass dipole" of a system is its total mass times its centre of mass, and its second time derivative is the rate of change of total momentum, which is zero for an isolated system (Problem 29.4). Nor does gravity have monopole radiation, since total mass–energy is conserved. A spherically symmetric collapsing star emits nothing. The leading radiation comes from a changing *quadrupole moment*, a measure of how a mass distribution departs from spherical symmetry. Einstein's **quadrupole formula** gives the radiated power:

$$P = \frac{G}{5c^5}\left\langle \dddot{Q}_{ij}\dddot{Q}_{ij}\right\rangle ,$$

where Q_{ij} is the (traceless) quadrupole moment. The prefactor G/c^5 is $3\times10^{-53}\ \mathrm{W}^{-1}$ in SI units, which is why only enormous, fast-moving masses radiate appreciably. The Earth orbiting the Sun radiates about 200 watts, far too little to matter (Problem 29.9).

For a binary of masses m_1, m_2 in a circular orbit of radius a, the radiated power is

$$P = \frac{32}{5}\frac{G^4}{c^5}\frac{(m_1m_2)^2(m_1+m_2)}{a^5}.$$

The radiation carries away orbital energy, so the orbit shrinks; a smaller orbit means faster motion and stronger radiation, which shrinks it faster still. The gravitational-wave frequency is *twice* the orbital frequency, because the quadrupole pattern of the binary repeats every half orbit. As the stars spiral together, the frequency and amplitude both rise: the *chirp*. The time remaining until merger scales as a^4 (Problem 29.9), and the rate at which the frequency sweeps up depends on the masses almost entirely through a single combination, the *chirp mass*

$$\mathcal{M} = \frac{(m_1m_2)^{3/5}}{(m_1+m_2)^{1/5}},$$

which is therefore the quantity a detector measures best (Problem 29.10).

§29.3 The first evidence: a pulsar clock

In 1974 Russell Hulse and Joseph Taylor discovered a pulsar (Chapter 27) in a tight orbit with another neutron star, with a period of under eight hours. A pulsar is a superb clock, and timing its pulses over years revealed the orbit shrinking — the period decreasing by about 76 microseconds per year — exactly as predicted by the energy lost to gravitational waves. The agreement, now at the level of a fraction of a percent, was the first evidence that gravitational waves exist (Nobel Prize, 1993). In about 300 million years, the two stars will merge.

§29.4 Detecting a change of 10^{-18} metres

LIGO is a Michelson interferometer (Chapter 8) with arms 4 km long. A passing gravitational wave stretches one arm and squeezes the other, changing the interference at the output. For $h = 10^{-21}$, the arm lengths change by about 4×10^{-18} m — a few thousandths of the width of a proton. Measuring that requires heroic engineering:

- The light bounces back and forth about 300 times in each arm (a Fabry–Pérot cavity), making the effective arm length over a thousand kilometres.
- "Power recycling" builds up several hundred kilowatts of laser light circulating in the arms, to beat down *shot noise*: the random arrival of photons, which limits phase measurements to precision $\sim 1/\sqrt{N}$ for N photons (Chapter 1).
- The 40 kg fused-silica mirrors hang from multi-stage pendulum suspensions that filter out ground vibrations by a factor of 10^{10} above 10 Hz (Chapter 3): a driven oscillator far above resonance barely moves.
- The whole beam path is in one of the largest ultra-high vacuums on Earth.
- At its best sensitivity, LIGO is limited by quantum mechanics itself: the photons' shot noise, and the random kicks of radiation pressure on the mirrors. Since 2019 LIGO has injected *squeezed light* (Chapter 18), redistributing quantum uncertainty to beat the shot-noise limit.

Two detectors thousands of kilometres apart are essential: a real signal appears in both within 10 milliseconds (the light travel time between them), while local noise does not. Europe's Virgo and Japan's KAGRA join them, and with three or more detectors, a source's position on the sky can be triangulated.

§29.5 GW150914

The first detection, named for its date, was almost absurdly clean: a chirp sweeping from 35 to 250 Hz in a fifth of a second, visible by eye in the raw data. Matching it against simulated waveforms from numerical relativity (Chapter 13) revealed two black holes of 36 and 29 solar masses, 1.3 billion light-years away, merging into one of 62 solar masses. The missing three solar masses — 5×10^{47} J — were radiated as gravitational waves in a fraction of a second, a peak power greater than that of all the stars in the observable universe combined (Problem 29.3). The final moments, a decaying "ringdown" as the new black hole settled down, matched the predictions of general relativity for a Kerr black hole (Chapter 26). Weiss, Barish, and Thorne received the 2017 Nobel Prize.

§29.6 GW170817: multi-messenger astronomy

On 17 August 2017, LIGO and Virgo detected a longer, slower chirp lasting about 100 seconds: two neutron stars merging 130 million light-years away. Just 1.7 seconds after the merger, the

Fermi satellite recorded a burst of gamma rays from the same direction. Within hours, telescopes found a new point of light in the galaxy NGC 4993, and some seventy observatories followed it across the spectrum. The event delivered a cascade of results:

- **Gravity travels at the speed of light**, to about one part in 10^{15} (Problem 29.5), eliminating many alternative theories of gravity at a stroke.
- **Neutron star mergers make short gamma-ray bursts**, solving a decades-old puzzle (Chapter 27).
- **The glow of a "kilonova"** revealed the radioactive decay of heavy elements freshly forged by the r-process (Chapter 22), some ten Earth masses of gold and platinum among them.
- **A new way to measure the Hubble constant.** The amplitude of a gravitational wave gives the source's distance directly, with no distance ladder, and the host galaxy gives its redshift. Such "standard sirens" (Problem 29.11) may help resolve the Hubble tension (Chapter 28).

§29.7 A new window

LIGO, Virgo, and KAGRA have now detected hundreds of mergers, revealing a population of black holes heavier than anyone expected from stellar evolution, black holes with masses in a supposed "gap" where none should form, and neutron star–black hole pairs. Each event is a test of general relativity in the strong-field regime, where spacetime is wildly curved and moving at nearly the speed of light.

Gravitational waves span a spectrum as wide as electromagnetic ones:

- **Ground-based interferometers** (10 to 1000 Hz): stellar-mass mergers.
- **LISA**, a planned triangle of spacecraft millions of kilometres apart ($\sim$ mHz): mergers of supermassive black holes across the universe, scheduled for the 2030s.
- **Pulsar timing arrays** (nanohertz): dozens of millisecond pulsars, monitored for years, act together as a detector the size of the galaxy. In 2023 several collaborations reported evidence for a background of such waves, probably from the slow inspiral of supermassive black hole pairs throughout the universe.
- **The CMB** (the lowest frequencies of all): primordial gravitational waves from inflation would leave a twisting B-mode pattern in its polarization (Chapter 28). Not yet seen.

§29.8 Problems

29.1 $O(\epsilon)$ ★ A gravitational wave with strain $h = 10^{-21}$ passes through LIGO's 4 km arm. By how much does the arm change length? Express it in proton diameters (1.7×10^{-15} m).

29.2 $O(\epsilon)$ Explain why the gravitational-wave frequency of a binary is twice its orbital frequency. A binary radiates at 100 Hz. What is its orbital period?

29.3 $O(\epsilon)$ ★ GW150914 radiated three solar masses of energy. Compute it in joules and compare with the total energy the Sun will radiate in its 10-billion-year lifetime. If most of it was emitted in about 20 ms, what was the average power? Compare with the total luminosity of the stars in the observable universe, about 10^{49} W.

29.4 $O(\epsilon^2)$ ★ Show that the mass dipole moment $\mathbf{D} = \sum m_i \mathbf{r}_i$ of an isolated system has zero second time derivative. Why does this rule out dipole gravitational radiation, while electromagnetism has dipole radiation? (Hint: what plays the role of charge in gravity, and does every particle have the same charge-to-mass ratio?)

29.5 $O(\epsilon^2)$ ★ The gamma rays from GW170817 arrived 1.7 s after the gravitational-wave merger signal, after travelling about 130 million light-years. Assuming they were emitted within a few seconds of each other, bound the fractional difference between the speed of gravity and the speed of light.

29.6 $O(\epsilon^2)$ An order-of-magnitude estimate of the strain from a binary at distance r is $h \sim \frac{4}{r}\left(\frac{G\mathcal{M}}{c^2}\right)^{5/3}\left(\frac{\pi f}{c}\right)^{2/3}$. Evaluate it for GW150914 ($\mathcal{M} \approx 28 M_\odot, f \approx 150\,\text{Hz}, r = 410\,\text{Mpc}$). Compare with the measured peak strain of 10^{-21}.

29.7 $O(\epsilon^2)$ The Hulse–Taylor pulsar's orbit has a period of 7.75 hours. Using Kepler's third law, estimate the orbital separation for two neutron stars of 1.4 solar masses each. Compare it with the radius of the Sun.

29.8 $O(\epsilon^2)$ The innermost stable circular orbit of a black hole of mass M is at $r = 6GM/c^2$. Find the gravitational-wave frequency of a test particle orbiting there, and evaluate it for a 60-solar-mass black hole and a 10^6-solar-mass one. Which detector would see each?

29.9 $e^{-1/\epsilon}$ ★ (*How long to merge.*) For a circular binary, the orbital energy is $E = -Gm_1m_2/2a$. Setting $\mathrm{d}E/\mathrm{d}t = -P$ with the quadrupole power formula, show that the time to merge from separation a is

$$t = \frac{5}{256}\frac{c^5 a^4}{G^3 m_1 m_2 (m_1 + m_2)}.$$

Compute the power radiated by the Earth–Sun system, and the time for the Earth to spiral into the Sun this way. Compare with the age of the universe.

29.10 $e^{-1/\epsilon}$ (*Weighing with a chirp.*) Using the results of the previous problem and Kepler's law relating a to the gravitational-wave frequency f, show that $\dot{f} \propto \mathcal{M}^{5/3} f^{11/3}$, with no other dependence on the masses. Explain why measuring the frequency and its rate of change determines the chirp mass, and why the individual masses are harder to extract.

29.11 $e^{-1/\epsilon}$ (*Standard sirens.*) The gravitational-wave amplitude of GW170817 implied a distance of about 40 Mpc; its host galaxy NGC 4993 recedes (after correcting for its local motion) at about 3000 km/s. Estimate H_0. Why is a single event's uncertainty large, and how many events would be needed to measure H_0 to 2%?

Further reading

- **Bernard Schutz, *A First Course in General Relativity*** — Chapter 9 is a clear introduction to gravitational waves and their detection.
- **Michele Maggiore, *Gravitational Waves*, Volume 1** — The comprehensive graduate reference on theory and experiment.
- **Peter Saulson, *Fundamentals of Interferometric Gravitational Wave Detectors*** — How LIGO actually works, including all its noise sources.
- **Janna Levin, *Black Hole Blues*** — The human story of LIGO's half-century quest.

CHAPTER 30

Quantum field theory

BUILDS ON analytical mechanics (Chapter 4), special relativity (Chapter 10), quantum mechanics (Chapter 14)

Where quantum mechanics meets relativity, and the particles turn out to be ripples on fields that fill all of space.

PROTOTYPE A field of oscillators. Imagine a mattress of springs filling all of space, one harmonic oscillator at every point, each coupled to its neighbours. Its normal modes are waves; quantize them, and each mode's energy comes in lumps of $\hbar\omega$. Those lumps are particles. A photon is one quantum of one mode of the electromagnetic field. Chapter 3, repeated infinitely many times, is the foundation of all of particle physics.

Quantum mechanics (Chapter 14) describes a fixed number of particles. Special relativity (Chapter 10) says mass is a form of energy, so with enough energy, particles can be created and destroyed. The two are not compatible as they stand. The resolution, worked out between 1927 and the 1970s, is *quantum field theory* (QFT): the fundamental objects are not particles but fields, and particles are their quantized excitations. QFT is the language of the Standard Model of particle physics (Chapter 32), of condensed matter physics (Chapter 19), and of critical phenomena (Chapter 9). It is the most accurately tested theory in the history of science, and also the most demanding to learn. This chapter tries to give the ideas, not the machinery.

§30.1 Why particles are not enough

Try to confine an electron to a region of size Δx. Its momentum becomes uncertain by $\hbar/\Delta x$, and its energy by about $\hbar c/\Delta x$ once that exceeds $m_e c^2$. When Δx is smaller than the *Compton wavelength*

$$\lambda_C = \frac{\hbar}{mc},$$

about 4×10^{-13} m for an electron, the energy uncertainty exceeds the energy needed to create an electron–positron pair. Probe an electron that closely and you will find not one particle but a fluctuating swarm. The number of particles is not a fixed quantity in a relativistic quantum theory, and a theory of a fixed number of particles cannot be right.

There is a second problem: causality. In relativity, events outside each other's light cones have no definite time order (Chapter 10). If a particle could be emitted at one such event and absorbed at another, observers would disagree about which happened first. Consistency requires that what one observer sees as a particle going forward, another sees as an *antiparticle* going the other way. Relativity plus quantum mechanics demands antimatter.

§30.2 Natural units

From here on, we set $\hbar = c = 1$. Then mass, energy, and momentum have the same units, which are also the units of inverse length and inverse time. Everything is measured in powers of energy, usually GeV. The conversion back is via

$$\hbar c \approx 197\,\text{MeV fm},$$

so a mass of 197 MeV corresponds to a length of 1 fm. The proton, at about 1 GeV, has a Compton wavelength of 0.2 fm. In these units the fine-structure constant is simply $\alpha = e^2/4\pi \approx 1/137$.

§30.3 Classical fields

A field $\phi(\mathbf{x}, t)$ assigns a value to every point in spacetime. Its dynamics comes from an action, just as in Chapter 4, but now the Lagrangian is a *density* integrated over space and time, $S = \int \mathcal{L}\, \mathrm{d}^4x$. For the simplest relativistic field, a single real scalar,

$$\mathcal{L} = \tfrac{1}{2}(\partial_t\phi)^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2.$$

The three terms are kinetic energy, the "spring" coupling to neighbouring points, and a restoring force at each point. The Euler–Lagrange equation is the **Klein–Gordon equation**,

$$\left(\partial_t^2 - \nabla^2 + m^2\right)\phi = 0.$$

Plane waves $e^{-i(Et-\mathbf{p}\cdot\mathbf{x})}$ solve it provided $E^2 = \mathbf{p}^2 + m^2$: exactly the relativistic energy–momentum relation (Chapter 10). Each Fourier mode of the field, with wavevector $\mathbf{k}$, oscillates independently as a harmonic oscillator with frequency $\omega_\mathbf{k} = \sqrt{\mathbf{k}^2 + m^2}$. A free field is an infinite collection of uncoupled oscillators.

§30.4 From oscillators to particles

Now quantize. Each mode is a quantum harmonic oscillator (Chapter 14), with ladder operators $a_\mathbf{k}$ and $a_\mathbf{k}^\dagger$ and energy levels $(n_\mathbf{k} + \tfrac{1}{2})\omega_\mathbf{k}$. The interpretation is the heart of QFT:

KEY IDEA The state with $n_\mathbf{k} = 0$ for every mode is the *vacuum*, $|0\rangle$. Acting with $a_\mathbf{k}^\dagger$ adds one quantum of energy $\omega_\mathbf{k} = \sqrt{\mathbf{k}^2 + m^2}$ and momentum $\mathbf{k}$ — that is, it *creates a particle* of mass m and momentum $\mathbf{k}$. The number $n_\mathbf{k}$ is the number of particles with momentum $\mathbf{k}$. Particles are quanta of fields.

Several mysteries dissolve at once. All electrons are identical because they are all excitations of the same field. Bosons come out automatically symmetric, because $a_\mathbf{k}^\dagger a_{\mathbf{k}'}^\dagger = a_{\mathbf{k}'}^\dagger a_\mathbf{k}^\dagger$; for fermions, the creation operators *anticommute*, $b^\dagger b^\dagger = -b^\dagger b^\dagger$, so creating two in the same state gives zero — the Pauli exclusion principle (Chapter 14). Particle creation and destruction are just the raising and lowering of oscillators. The photon, introduced by Einstein as an ad hoc lump of light, is a quantum of the electromagnetic field.

§30.5 The vacuum is not empty

Each mode has zero-point energy $\frac{1}{2}\omega_{\mathbf{k}}$, and there are infinitely many modes, so the vacuum energy is infinite. In most of physics this does not matter, because only energy *differences* are measurable. But changes in the vacuum energy are real. Between two parallel conducting plates, only modes that fit can exist (Chapter 5), so the vacuum energy between them is lower than outside, and the plates are pushed together: the **Casimir effect**, with a pressure

$$P = \frac{\pi^2 \hbar c}{240\, d^4}$$

for plates a distance d apart. It was measured precisely in 1997, and it matters in nanoscale machines. The vacuum also fluctuates: the Lamb shift (Chapter 18) is caused by vacuum fluctuations of the electromagnetic field jiggling the electron. And gravity, unlike everything else, *does* feel the absolute energy of the vacuum; the mismatch between the vast zero-point energy and the tiny observed cosmological constant (Chapter 28) is the worst problem in theoretical physics (Problem 30.10).

§30.6 The Dirac equation and antimatter

In 1928 Paul Dirac sought a relativistic wave equation for the electron that, unlike Klein–Gordon, was first order in time. He found that it required four-component wavefunctions and 4×4 matrices:

$$(i\gamma^\mu \partial_\mu - m)\psi = 0.$$

Out of the mathematics fell the electron's spin $\frac{1}{2}$ and its magnetic moment (with the "g-factor" $g = 2$), neither of which had been put in. So did solutions with negative energy, which Dirac eventually interpreted as a new particle with the electron's mass and the opposite charge. Carl Anderson found the **positron** in cosmic rays in 1932. Every particle has an antiparticle; when they meet they can annihilate into pure energy, as in PET scanners (Chapter 24). Dirac said, "The equation was smarter than I was."

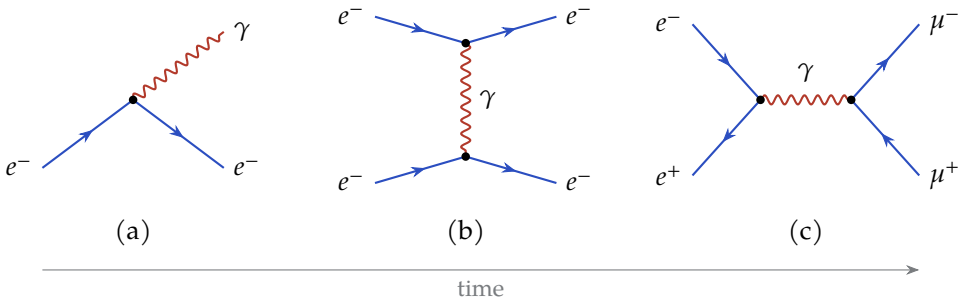
§30.7 Interactions and Feynman diagrams

Free fields are just oscillators. Interactions couple them. In quantum electrodynamics (QED), the Lagrangian contains a term $e\bar{\psi}\gamma^\mu\psi A_\mu$ coupling the electron field ψ to the photon field A_μ. It allows a single basic process: an electron emits or absorbs a photon. Everything in electromagnetism and chemistry is built from this one vertex.

Because e is small ($\alpha \approx 1/137$), we can calculate in perturbation theory, as a power series in α. Richard Feynman invented a way to draw each term as a picture: lines for particles travelling through spacetime, vertices where they interact. Each diagram stands for a precise mathematical expression, and each vertex contributes a factor of $e \propto \sqrt{\alpha}$.

- Two electrons repel by exchanging a photon: one emits it, the other absorbs it. This simplest diagram reproduces Coulomb's law. Exchanging a massive particle instead gives the short-range Yukawa potential e^{-mr}/r (Problem 30.4; Chapter 22).

- The exchanged photon is *virtual*: it need not satisfy $E^2 = p^2 + m^2$, and it exists only as an intermediate step, permitted by the uncertainty principle.
- Diagrams with closed loops — an electron briefly emitting and reabsorbing a photon, or a photon briefly splitting into an electron–positron pair — give corrections suppressed by further powers of α.



Feynman diagrams. (a) The basic vertex of quantum electrodynamics: an electron emits or absorbs a photon. Every other diagram is built from copies of it, and each vertex costs a factor of e. (b) Two electrons repel by exchanging a photon. (c) An electron and a positron annihilate into a photon, which becomes a muon and an antimuon. Antiparticles are drawn with their arrows pointing backwards in time.

The results are staggering. The electron's magnetic moment is predicted to be slightly larger than Dirac's $g = 2$, by an amount $\alpha/2\pi$ plus higher corrections computed to fifth order in α (involving thousands of diagrams). Theory and experiment agree to about one part in a trillion, making QED the most precisely tested theory in all of science.

§30.8 Infinities and renormalization

Loop diagrams have a problem: they involve integrals over the momenta of virtual particles, and those integrals diverge at high momentum. For two decades, QFT seemed to predict infinite answers. In the late 1940s Feynman, Schwinger, Tomonaga, and Dyson found a cure, *renormalization*: the infinities can be absorbed into the definitions of the electron's mass and charge, which are anyway measured rather than predicted. After that, every other prediction is finite and agrees with experiment.

For years renormalization looked like sweeping infinities under the rug. The modern understanding, due largely to Kenneth Wilson in the 1970s, is profound and liberating. Every physical theory is an *effective theory*, valid below some energy scale and ignorant of what happens above it. The divergences are a sign of that ignorance, and renormalization is the statement that physics at low energies depends on physics at high energies only through a few parameters. As the energy scale changes, those parameters *run*. The electromagnetic coupling, screened at long distances by a cloud of virtual electron–positron pairs that polarize the vacuum, grows from $1/137$ at low energy to about $1/128$ at the mass of the Z boson (Problem 30.7). The strong coupling does the opposite (Chapter 32).

KEY IDEA Effective field theory is the spherical cow, made rigorous. You do not need to know what happens at the Planck scale to compute the energy levels of hydrogen, just as you do not need to know about atoms to design a bridge. Each scale of physics has its own effective description, and the renormalization group connects them. The same mathematics explains universality at critical points (Chapter 9).

§30.9 Gauge symmetry

Why does electromagnetism exist at all? Here is the modern answer. The electron field has a phase, and multiplying the field everywhere by the same phase, $\psi \to e^{i\alpha}\psi$, changes nothing observable. By Noether's theorem (Chapter 4), this *global* symmetry gives a conserved quantity: electric charge.

Now demand more: that physics be unchanged even if the phase is changed *differently at every point*, $\psi \to e^{i\alpha(x)}\psi$. Derivatives of ψ now pick up extra terms, and the Lagrangian is no longer invariant. To restore the symmetry, we must introduce a new field A_μ that shifts in just the right way to cancel those terms, $A_\mu \to A_\mu - \frac{1}{e}\partial_\mu\alpha$ — exactly the gauge freedom of the electromagnetic potentials (Chapter 6). The photon is *required* by local phase symmetry. Its coupling to the electron, its masslessness, and Maxwell's equations all follow.

This *gauge principle* is the organizing idea of fundamental physics. In 1954 Yang and Mills generalized it to symmetries in which the "phase" is a matrix, as for the group $SU(2)$ or $SU(3)$. The resulting gauge fields interact with each other as well as with matter. With $SU(3)$, they are the gluons of the strong force; with $SU(2) \times U(1)$, the photon and the W and Z bosons of the electroweak force (Chapter 32). The forces of nature are what you get when you demand that certain symmetries hold locally.

§30.10 Hidden symmetry and the Higgs mechanism

Gauge symmetry seems to forbid the gauge bosons from having mass. Yet the W and Z are heavy. The resolution is *spontaneous symmetry breaking*: the laws can be symmetric while the vacuum is not. Picture a ball in a potential shaped like the bottom of a wine bottle (or a Mexican hat), $V = -\mu^2|\phi|^2 + \lambda|\phi|^4$. The potential is symmetric under rotations, but the lowest energy is on a circle of nonzero $|\phi|$, and the ball must settle somewhere on it, choosing a direction (compare the bead on a hoop, Problem 2.11, and the ferromagnet of Chapter 19). Oscillations around the circle cost no energy and give massless particles (*Goldstone bosons*); oscillations up the side give massive ones.

When the symmetry is a gauge symmetry, something remarkable happens: the would-be Goldstone bosons are absorbed by the gauge fields, which become massive. This is the **Higgs mechanism**, proposed in 1964 by Higgs, Englert and Brout, and others. Its condensed matter ancestors are the massive photon in a plasma (Chapter 17) and in a superconductor (Chapter 19). The leftover oscillation up the side of the hat is the *Higgs boson*, discovered at CERN in 2012 (Chapter 32).

§30.11 Problems

30.1 $O(\epsilon)$ ★ Find the Compton wavelengths $\hbar/mc$ of the electron (0.511 MeV), the proton (938 MeV), and the W boson (80 GeV), using $\hbar c = 197\,\text{MeV fm}$. Interpret each as a length scale in physics.

30.2 $O(\epsilon)$ Show that the plane wave $e^{-i(Et-\mathbf{p}\cdot\mathbf{x})}$ solves the Klein–Gordon equation if and only if $E^2 = \mathbf{p}^2 + m^2$. What happens for $m = 0$?

30.3 $O(\epsilon)$ An electron is confined to a box of size L. Below what L does its kinetic energy exceed the energy needed to create an electron–positron pair? Why does this mean the one-electron picture fails at that scale?

30.4 $O(\epsilon^2)$ ★ The static Klein–Gordon equation with a point source is $(\nabla^2 - m^2)\phi = -g\,\delta^3(\mathbf{r})$. Show that $\phi = \frac{g}{4\pi}\frac{e^{-mr}}{r}$ solves it. With $m = 0$, it reduces to the Coulomb potential. Restore $\hbar$ and c to show the range is $\hbar/mc$, and find the range for a mediator of mass 140 MeV.

30.5 $O(\epsilon^2)$ A process with two QED vertices has an amplitude proportional to $e^2 \propto \alpha$, so a cross-section proportional to α^2. Each extra loop costs roughly another factor of α/π. Estimate the size of one-loop corrections relative to the leading term. Why is perturbation theory so successful for QED?

30.6 $O(\epsilon^2)$ ★ Evaluate the Casimir pressure between two plates 1 μm apart and 10 nm apart. At what separation does it equal atmospheric pressure? Why does the effect matter for microelectromechanical devices?

30.7 $O(\epsilon^2)$ Including only the electron's vacuum polarization, the coupling runs as $\alpha(Q) = \alpha/[1 - (2\alpha/3\pi)\ln(Q/m_e)]$. Evaluate $1/\alpha$ at $Q = 91\,\text{GeV}$. Including all charged fermions lighter than 91 GeV (each weighted by its charge squared and number of colours), the measured value is about 128. Explain physically why the effective charge grows at short distances.

30.8 $O(\epsilon^2)$ ★ (*Gauge invariance in the Schrödinger equation.*) For a particle of charge q, the Schrödinger equation is $i\hbar\partial_t\psi = \frac{1}{2m}(-i\hbar\nabla - q\mathbf{A})^2\psi + q\phi\psi$. Show that it is unchanged in form under the combined transformation $\psi \to e^{iq\chi/\hbar}\psi$, $\mathbf{A} \to \mathbf{A} + \nabla\chi$, $\phi \to \phi - \partial_t\chi$, for any function $\chi(\mathbf{x}, t)$. Why would the equation fail to be invariant without the potentials?

30.9 $O(\epsilon^2)$ For $V = -\mu^2|\phi|^2 + \lambda|\phi|^4$ with complex ϕ, find the radius of the circle of minima. Writing $\phi = (v + h)e^{i\theta}/\sqrt{2}$ near a minimum, show that small oscillations in h have a mass while those in θ do not.

30.10 $e^{-1/\epsilon}$ ★ (*The worst prediction in physics.*) Suppose QFT is valid up to the Planck energy $E_P = \sqrt{\hbar c^5/G} \approx 1.2 \times 10^{19}\,\text{GeV}$, so that zero-point energies are summed up to that cutoff. Estimate the vacuum energy density as $E_P^4/(\hbar c)^3$, in J/m^3. The observed dark energy density is about $6 \times 10^{-10}\,\text{J/m}^3$. By how many orders of magnitude does the estimate miss? What would a cutoff at 1 TeV give?

30.11 $e^{-1/\epsilon}$ ★ (*The weak scale from Fermi's theory.*) Fermi described beta decay with a four-fermion interaction of strength $G_F \approx 1.17 \times 10^{-5}\,\mathrm{GeV}^{-2}$. By dimensional analysis, a cross-section at energy E in this theory must scale as $\sigma \sim G_F^2 E^2$, growing without limit. Quantum mechanics forbids cross-sections from growing faster than $\sim 1/E^2$ (unitarity). Show that the theory must break down below about $E \sim G_F^{-1/2}$. In the Standard Model, $G_F = \sqrt{2}g^2/8m_W^2$ with $g \approx 0.65$; estimate m_W and compare with 80 GeV. The breakdown of an effective theory announced the new particle.

Further reading

- **Lancaster and Blundell, *Quantum Field Theory for the Gifted Amateur*** — The most gentle and physical introduction, in short chapters.
- **A. Zee, *Quantum Field Theory in a Nutshell*** — An inspiring, idea-driven tour that reaches the frontier quickly.
- **David Tong, lecture notes on Quantum Field Theory** — Clear and complete Cambridge lecture notes, free online.
- **Peskin and Schroeder, *An Introduction to Quantum Field Theory*** — The standard graduate text for computing things.
- **Matthew Schwartz, *Quantum Field Theory and the Standard Model*** — A modern graduate text with excellent physical explanations.

CHAPTER 31

Accelerators and detectors

BUILDS ON electromagnetism (Chapter 6), special relativity (Chapter 10)

The machines that make the fundamental visible — by smashing particles together hard enough to see inside them and to make new ones, then catching what flies out.

PROTOTYPE A storage ring: a vacuum pipe bent into a circle, threaded by magnets that steer particles around it and radio-frequency cavities that kick them a little harder each lap. Particles circulate billions of times, gaining energy until, at the collision points, two counter-rotating beams are brought together.

To see small things you need short wavelengths, and by de Broglie's relation $\lambda = h/p$ (Chapter 14), short wavelengths mean high momenta. To resolve the inside of a proton ($\sim 1\,\text{fm}$) you need momenta of order a GeV; the Large Hadron Collider probes distances a thousand times smaller. To create heavy particles you need energy, at least mc^2 of it (Chapter 10): the Higgs boson needed 125 GeV just for its mass. Particle accelerators are microscopes and factories at once. They are also the largest and most precise machines ever built — and, through medicine, materials science, and industry, among the most useful. About thirty thousand accelerators are in use worldwide; only a few hundred are for particle physics.

§31.1 Getting a push

A charge q falling through a potential difference V gains energy qV (Chapter 6). The first accelerators were simply very high voltages. In 1932 Cockcroft and Walton used 700 000 volts to accelerate protons into lithium and split it — the first nuclear reaction made by humans. Van de Graaff generators reach about 20 MV, limited by sparks. To go further, you must use the same voltage many times.

Linear accelerators do this with radio-frequency (RF) electric fields oscillating in a series of metal cavities, timed so that a particle always arrives when the field points forward — like a surfer riding a wave. The Stanford Linear Accelerator, three kilometres long, accelerated electrons to 50 GeV. Modern superconducting RF cavities sustain gradients of 30–40 MV/m. The linear accelerators in hospitals, a metre long, produce the X-rays of radiation therapy (Chapter 24).

Circular accelerators reuse the same cavities on every lap. In 1931 Ernest Lawrence realized that a charged particle in a uniform magnetic field circles at the cyclotron frequency $\omega = qB/m$, independent of its speed (Chapter 6). In his *cyclotron*, particles spiral outward between two D-shaped electrodes, kicked by an alternating voltage at exactly this frequency each half-turn. It fails at high energy because relativity increases the effective mass and the particles fall out of

step. The *synchrotron* solves this by keeping the particles on a fixed circular orbit and ramping up the magnetic field as their momentum grows.

§31.2 Steering and focusing

The field needed to bend a particle of momentum p and charge e into a circle of radius ρ follows from $p = eB\rho$. In practical units,

$$p\,[\mathrm{GeV}/c] = 0.3\,B\,[\mathrm{T}]\,\rho\,[\mathrm{m}],$$

and $B\rho$ is called the *magnetic rigidity*. The LHC, in a 27 km tunnel near Geneva, steers 6.8 TeV protons with 1232 superconducting dipole magnets operating at 8.3 T and 1.9 K, colder than outer space. Higher energy needs either a bigger ring or stronger magnets.

Particles must also be kept in a beam a fraction of a millimetre wide over billions of laps. *Quadrupole* magnets focus the beam, but each one focuses in one direction while defocusing in the perpendicular one. The surprising discovery of 1952 was *strong focusing*: alternating focusing and defocusing quadrupoles give net focusing in both directions (Problem 31.10), the same mathematics as the Kapitza pendulum (Problem 4.10). Longitudinally, particles that arrive early or late in the RF cavity get slightly different kicks that push them back toward the ideal timing, so they oscillate stably about it in bunches.

§31.3 Synchrotron radiation

An accelerating charge radiates (Chapter 6), and a particle on a circle is always accelerating. For relativistic particles, the radiated power grows as γ^4, and the energy lost per turn is

$$\Delta E \propto \frac{\gamma^4}{\rho} \propto \frac{E^4}{m^4 \rho}.$$

For electrons this is brutal. The LEP collider, in the same tunnel that now houses the LHC, accelerated electrons to 104 GeV, at which point each electron lost about 3 GeV per lap to synchrotron radiation, all of which had to be replaced by the RF cavities (Problem 31.5). Doubling the energy would multiply the loss by sixteen. This is why future high-energy electron colliders are either linear or enormous. Protons, 1836 times heavier, lose 10^{13} times less at the same energy, which is why the LHC can reach 6.8 TeV in the same tunnel.

What is a curse for particle physics is a gift for everyone else. *Synchrotron light sources* are storage rings built to produce intense, tightly collimated X-ray beams, used to determine the structures of proteins (Chapter 23) and materials (Chapter 19). *Free-electron lasers* pass electron bunches through rows of alternating magnets to produce X-ray laser pulses a few femtoseconds long, bright enough to image a single molecule before it explodes.

§31.4 Colliders, energy, and luminosity

What matters for making new particles is the energy in the centre-of-mass frame (Chapter 10). For a beam of energy E hitting a stationary target of mass m, $\sqrt{s} \approx \sqrt{2Emc^2}$, growing only as the

square root of the beam energy; for two beams of energy E colliding head-on, $\sqrt{s} = 2E$. Firing the LHC's protons at a fixed target would waste all but 113 GeV of the available energy; colliding them head-on gives 13.6 TeV (Problem 31.4). That is why all energy-frontier machines since the 1970s have been colliders.

The other figure of merit is how often collisions happen. The probability of a given process is described by its *cross-section* σ, an effective target area, measured in *barns* ($1\,\mathrm{b} = 10^{-28}\,\mathrm{m}^2$, roughly the size of a uranium nucleus — "as big as a barn"). The rate of events is

$$\text{rate} = \sigma \mathcal{L},$$

where the *luminosity* $\mathcal{L}$ measures how tightly packed the colliding beams are. The LHC reaches $\mathcal{L} \approx 2{\times}10^{34}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$, with bunches of 10^{11} protons crossing 40 million times a second, squeezed to 10 μm at the collision points. Even so, the Higgs boson, with a cross-section of about 50 picobarns, is produced only about once a second, among a billion other collisions (Problem 31.6).

§31.5 How particles lose energy in matter

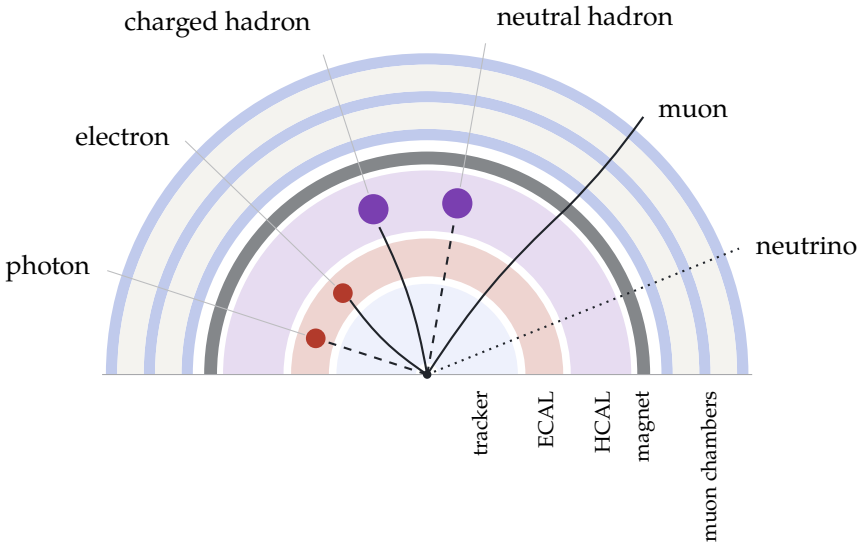
Everything a detector does relies on how particles interact with matter.

- **Charged particles** ionize atoms as they pass, losing energy at a rate described by the *Bethe formula*: roughly proportional to the square of their charge and inversely proportional to the square of their speed. Fast particles lose energy slowly; as a particle slows, its losses rise sharply, producing the Bragg peak at the end of its track (Chapter 24). Near the speed of light, every singly charged particle loses about 2 MeV per gram per square centimetre of material traversed: a *minimum ionizing particle*.
- **Electrons and photons** at high energy produce *showers*. An electron passing near a nucleus radiates a photon, a process called *bremsstrahlung* ("braking radiation"), and a photon with more than 1.022 MeV can convert into an electron–positron pair. Each generation doubles the number of particles and halves their energy, until the energy is absorbed. The characteristic length is the *radiation length*, about 0.56 cm in lead.
- **Hadrons** (protons, pions, neutrons) hit nuclei and produce messier showers over a longer distance, the *nuclear interaction length*.
- **Muons** are heavy electrons that do not shower and do not feel the strong force, so they pass through metres of iron.
- **Neutrinos** interact only by the weak force, and pass through essentially anything: a neutrino from a nuclear reactor would typically cross light-years of lead before interacting (Problem 31.9).

§31.6 The anatomy of a detector

A modern collider detector like ATLAS or CMS is built like an onion around the collision point, each layer identifying different particles:

1. **Silicon pixel and strip detectors** closest to the beam track charged particles to a few micrometres. They can see that some particles were born a few millimetres away from the collision — the decay vertices of particles containing bottom quarks, which live about a picosecond.
2. A large **solenoid magnet** (3.8 T in CMS) bends the tracks. Measuring their curvature gives each particle's momentum, using $p = 0.3BR$, and the direction of bending gives its charge.
3. The **electromagnetic calorimeter** absorbs electrons and photons in showers and measures their energy. (CMS uses 75 000 lead tungstate crystals that scintillate.)
4. The **hadronic calorimeter**, thicker and denser, absorbs and measures hadrons.
5. **Muon chambers** on the outside catch the only charged particles that make it through everything.
6. Particles that escape everything — neutrinos, or perhaps dark matter — reveal themselves by *missing momentum*: in the plane perpendicular to the beam, the momenta of everything seen must add to zero. An imbalance means something invisible carried momentum away.



Half of a slice through a collider detector like CMS, and how each particle shows up. Charged particles leave tracks, bent by the magnet. Electrons and photons stop in the electromagnetic calorimeter (ECAL), hadrons in the hadronic calorimeter (HCAL). Only muons reach the outer chambers, and neutrinos escape unseen, noticed only as missing momentum.

Other tools identify particles by their speed. Particles moving faster than light does in a medium emit *Cherenkov radiation* in a cone of angle $\cos\theta = 1/n\beta$ (Chapter 5), the optical sonic boom that makes nuclear reactor pools glow blue. Combining the measured speed with the measured momentum gives the mass.

§31.7 Too much data

The LHC delivers a billion collisions per second, producing data at a rate far beyond what can be stored. A *trigger*, a cascade of fast electronics and software, decides within microseconds

which collisions look interesting and keeps about one in 40 000. Even so, the experiments record petabytes per year, analysed on a computing grid spanning hundreds of sites around the world. Machine learning (Chapter 13) now pervades every stage, from triggering to reconstructing tracks and classifying events. (The World Wide Web itself was invented at CERN, in 1989, to share physics data.)

§31.8 Listening for whispers

Not all detectors sit at colliders. To catch neutrinos, which almost never interact, you need enormous volumes. Super-Kamiokande in Japan is a tank of 50 000 tonnes of ultrapure water, a kilometre underground, lined with 11 000 photomultiplier tubes that see the Cherenkov rings of particles struck by neutrinos; it discovered that neutrinos have mass (Chapter 32). IceCube instruments a cubic kilometre of Antarctic ice. Dark matter searches look for the tiny recoil of a xenon nucleus struck by a passing dark matter particle, in tonnes of liquid xenon so free of radioactivity that it is among the quietest places in the universe. These detectors live deep underground, where a kilometre of rock filters out the cosmic rays.

§31.9 Problems

31.1 $O(\epsilon)$ ★ To resolve a structure of size 0.1 fm you need a de Broglie wavelength comparable to it. What momentum does that require? What distance scale does the LHC, at a few TeV per parton, resolve?

31.2 $O(\epsilon)$ Find the cyclotron frequency of protons in a 1.5 T field. If the magnet's pole pieces have a radius of 0.5 m, what is the maximum kinetic energy of the protons? At what energy would relativistic effects (a few percent) start to spoil the resonance?

31.3 $O(\epsilon)$ ★ The LHC's dipoles fill about 2/3 of its 26.7 km circumference, giving an effective bending radius of about 2.8 km. Find the magnetic field needed to hold 6.8 TeV protons. What field would be needed in a 100 km ring to reach 50 TeV?

31.4 $O(\epsilon^2)$ ★ Show that a beam of energy $E \gg mc^2$ hitting a stationary particle of mass m has $\sqrt{s} \approx \sqrt{2Emc^2}$. Compare $\sqrt{s}$ for a 6.8 TeV proton hitting a stationary proton, and for two 6.8 TeV protons colliding head-on. How energetic would a fixed-target beam need to be to match the LHC?

31.5 $O(\epsilon^2)$ ★ For electrons, the energy lost per turn to synchrotron radiation is $\Delta E\,[\mathrm{keV}] = 88.5\,E^4[\mathrm{GeV}]/\rho\,[\mathrm{m}]$. Evaluate it for LEP ($E = 104$ GeV, $\rho = 3.1$ km). Scale it to protons in the LHC (multiply by $(m_e/m_p)^4$, at $E = 6.8$ TeV, $\rho = 2.8$ km). Why are LHC protons nonetheless a problem for the magnets, which must absorb this radiation at 1.9 K?

31.6 $O(\epsilon^2)$ The Higgs production cross-section at the LHC is about 55 pb, and the luminosity is $2 \times 10^{34}\,\mathrm{cm^{-2}s^{-1}}$. How many Higgs bosons are made per second, and per year (taking 10^7 seconds of running)? The Higgs decays to two photons only 0.23% of the time. How many such events are produced per year? Why was the discovery still hard?

31.7 $O(\epsilon^2)$ Find the Cherenkov angle for a particle with $\beta \approx 1$ in water ($n = 1.33$). What is the minimum speed for Cherenkov emission? Find the minimum momentum at which a muon ($106\,\mathrm{MeV}/c^2$) and a proton radiate in water.

31.8 $O(\epsilon^2)$ In CMS's 3.8 T field, a muon track has a radius of curvature of 10 m. What is its momentum? A 1 m-long track segment of this muon deviates from a straight line by its *sagitta*, $s \approx L^2/8R$. Evaluate s. If the sagitta is measured to 50 μm, what is the momentum resolution?

31.9 $O(\epsilon^2)$ ★ A reactor antineutrino of a few MeV has an interaction cross-section of about $10^{-47}\,\mathrm{m}^2$ per nucleon. Lead has about 7×10^{30} nucleons per cubic metre. Find the mean free path, in light-years. How did Reines and Cowan detect reactor neutrinos in 1956? (Hint: the flux from a reactor is about $10^{17}\,\mathrm{m}^{-2}\mathrm{s}^{-1}$.)

31.10 $e^{-1/\epsilon}$ ★ (*Strong focusing.*) A thin quadrupole acts as a lens of focal length f in one plane and $-f$ in the other. Two such lenses, one focusing and one defocusing, separated by a distance d, have a combined focal length given by $1/F = 1/f_1 + 1/f_2 - d/(f_1 f_2)$. Show that for $f_1 = f, f_2 = -f$, the combination is focusing ($F = f^2/d > 0$) — in *both* planes. Explain the physical reason: where is the beam wide and where is it narrow as it passes through each lens?

31.11 $e^{-1/\epsilon}$ (*The end of the circular road.*) Suppose you want to collide electrons at $\sqrt{s} = 1\,\mathrm{TeV}$ in a circular ring, and you can afford to replace at most 100 MW of synchrotron power with a beam current of 10 mA. Use the formula in Problem 31.5 to find the minimum ring radius. Compare with a linear collider using cavities at 40 MV/m: how long would it be?

31.12 $e^{-1/\epsilon}$ (*Accelerating on a plasma wave.*) In a plasma wakefield accelerator, a laser pulse or particle bunch pushes the electrons of a plasma aside, setting up a plasma oscillation (Chapter 17) that trails behind it like the wake of a boat. Argue that the maximum electric field is roughly $E \sim m_e c\,\omega_p/e$, where ω_p is the plasma frequency, and show that this is about $100\sqrt{n\,[\mathrm{cm}^{-3}]}\,\mathrm{V/m}$. Evaluate it for $n = 10^{18}\,\mathrm{cm}^{-3}$ and compare with conventional RF cavities. Why might this shrink future colliders from tens of kilometres to tens of metres?

Further reading

- **Edmund Wilson, *An Introduction to Particle Accelerators*** — A short, clear introduction to the physics of accelerators.
- **Klaus Wille, *The Physics of Particle Accelerators*** — A more complete treatment of beam optics, RF, and synchrotron radiation.
- **Kolanoski and Wermes, *Particle Detectors*** — A modern, comprehensive text on how detectors work.
- **Particle Data Group, *Review of Particle Physics*** — Free online. Its review chapters on accelerators and on the passage of particles through matter are the field's handbook.

CHAPTER 32

Particle physics

BUILDS ON quantum field theory (Chapter 30), accelerators and detectors (Chapter 31)

The Standard Model: seventeen particles, three forces, a few symmetries, and the most successful — and most obviously incomplete — theory of nature ever written down.

PROTOTYPE Symmetry becoming force. Demand that the laws of physics be unchanged under a local symmetry, and a force appears, carried by a gauge boson (Chapter 30). The phase symmetry $U(1)$ gives the photon; $SU(2)$ gives the W and Z; $SU(3)$ gives the eight gluons. Hide part of the symmetry with the Higgs field, and some of those carriers become heavy. The whole Standard Model is this idea, applied three times.

By the 1950s, physicists had found far more "elementary" particles than anyone wanted — pions, kaons, lambdas, sigmas, deltas, a whole zoo. "If I could remember the names of all these particles," Fermi supposedly said, "I'd be a botanist." Over the next twenty-five years the zoo was reduced to order, and the result, the *Standard Model of particle physics*, has passed every test thrown at it for half a century, with the discovery of the Higgs boson in 2012 completing its predicted list of particles. And yet we know it cannot be the final theory. This chapter describes what it says, how we know, and where it falls short.

§32.1 The particles

All known matter is built from *fermions* (spin $\frac{1}{2}$) of two kinds, quarks and leptons, which come in three *generations*, each a heavier copy of the one before. Forces are carried by *gauge bosons* (spin 1), and there is one spin-0 particle, the Higgs boson.

Quarks				Leptons			
up	u	$+\frac{2}{3}$	2.2 MeV	electron	e	-1	0.511 MeV
down	d	$-\frac{1}{3}$	4.7 MeV	electron neutrino	ν_e	0	< 1 eV
charm	c	$+\frac{2}{3}$	1.27 GeV	muon	μ	-1	106 MeV
strange	s	$-\frac{1}{3}$	93 MeV	muon neutrino	ν_μ	0	< 1 eV
top	t	$+\frac{2}{3}$	173 GeV	tau	τ	-1	1.78 GeV
bottom	b	$-\frac{1}{3}$	4.18 GeV	tau neutrino	ν_τ	0	< 1 eV

Bosons: photon γ (massless), gluons g (massless), $W^\pm$ (80.4 GeV), Z (91.2 GeV), Higgs H (125 GeV)

Every particle has an antiparticle with opposite charges. Ordinary matter needs only the first generation: up and down quarks (making protons and neutrons), electrons, and electron

neutrinos. Why there are three generations, and why their masses span from below an electron-volt to 173 GeV, a factor of more than 10^{11}, is unknown. When I. I. Rabi heard of the muon, he asked, "Who ordered that?"

§32.2 Forces and conservation laws

The Standard Model has three forces. Gravity is left out; it is far too weak to matter for individual particles (Problem 6.1 compared it with electromagnetism), and nobody knows how to fit it in (Chapter 33).

- **Electromagnetism**, carried by the photon, acts on electric charge, with strength $\alpha \approx 1/137$ and infinite range.
- **The weak force**, carried by the $W^{\pm}$ and Z, acts on all fermions. At low energy it is feeble and short-ranged (10^{-18} m) because its carriers are heavy. It is the only force that changes one kind of particle into another — quarks of one flavour into another, as in beta decay.
- **The strong force**, carried by eight gluons, acts on quarks and gluons via *colour charge*, binding quarks into protons and neutrons and, as a residual effect, nucleons into nuclei (Chapter 22).

Which processes happen is governed by conservation laws (Chapter 4). Energy, momentum, angular momentum, and electric charge are exactly conserved. So, as far as we know, is *baryon number* (quarks count $+\frac{1}{3}$, antiquarks $-\frac{1}{3}$), which is why the proton, the lightest baryon, is stable. In the Standard Model, *lepton number* is also conserved, separately for each generation in practice: a muon can decay to an electron only by also producing a muon neutrino and an electron antineutrino, never as $\mu \to e\gamma$. Other quantities — the numbers of strange or charm quarks — are conserved by the strong and electromagnetic forces but changed by the weak force, which is why strange particles live a billion times longer than particles that decay strongly.

§32.3 Quarks and colour

In 1961 Murray Gell-Mann and Yuval Ne'eman independently noticed that the zoo of hadrons fit into patterns of the symmetry group $SU(3)$ — octets and decuplets — the "eightfold way." One slot in the decuplet was empty, and Gell-Mann predicted a particle to fill it, the Ω^-, with a definite mass and strangeness. It was found in 1964. That year Gell-Mann and George Zweig proposed that the patterns arise because hadrons are made of three types of *quarks* ("three quarks for Muster Mark," from *Finnegans Wake*), with fractional electric charges. *Baryons* are made of three quarks: the proton is uud, the neutron udd. *Mesons* are a quark and an antiquark: the π^+ is $u\bar{d}$.

No one has ever seen a free quark. But in 1968 electrons scattered off protons at SLAC bounced as if from hard, point-like constituents inside — a re-run of Rutherford's experiment (Chapter 22) one level down.

There was a puzzle. The Δ^{++} is uuu, with all three quark spins aligned in the lowest-energy, symmetric spatial state. That is a totally symmetric state of three identical fermions, which the Pauli principle forbids. The resolution is that quarks carry an extra quantum number, *colour*,

with three values ("red," "green," "blue"), and the wavefunction is antisymmetric in colour (Problem 32.10). Colour turned out to be the charge of the strong force.

§32.4 Quantum chromodynamics

The theory of the strong force, *quantum chromodynamics* (QCD), is a gauge theory with symmetry group $SU(3)$ (Chapter 30). Its eight gluons carry colour themselves, so unlike photons they interact with each other. That changes everything.

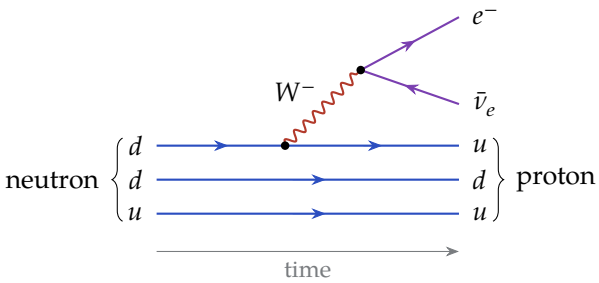
Asymptotic freedom. In QED, the effective charge grows at short distances because virtual pairs screen it (Chapter 30). In QCD, the gluon self-interactions produce the opposite effect: an *anti*-screening, so the strong coupling gets *weaker* at short distances and high energies (Problem 32.8). Inside a proton, quarks probed at high energy behave almost as free particles, which is why the SLAC experiments saw point-like constituents. Gross, Wilczek, and Politzer discovered this in 1973 (Nobel Prize, 2004).

Confinement. At long distances the coupling becomes strong. Pull a quark and antiquark apart and the gluon field between them is squeezed into a narrow tube with constant energy per length, about 1 GeV/fm, like a stretched rubber band that never weakens. Before you could separate them, the tube has so much energy that it snaps, creating a new quark–antiquark pair at the break: two mesons instead of two free quarks. In high-energy collisions, quarks and gluons appear as *jets*, sprays of hadrons collimated along the original direction. Confinement has never been proved mathematically; it is another Millennium Prize Problem.

Where your mass comes from. The up and down quarks in a proton weigh only about 9 MeV together, about 1% of the proton's 938 MeV (Problem 32.7). The rest is the energy of the gluon field and the quarks' kinetic energy, confined to a femtometre, via $E = mc^2$ (Chapter 10). Almost all the mass of everything you can touch is QCD binding energy. Supercomputer calculations of QCD on a spacetime lattice (Chapter 13) reproduce the proton mass to about 1%.

§32.5 The weak force

The weak force is responsible for beta decay, which at the quark level is $d \to u + W^-$, with the virtual W^- turning into an electron and an antineutrino. Its feebleness at low energy is entirely due to the W's large mass: the strength of the effective interaction, Fermi's constant G_F, is proportional to g^2/m_W^2, where the underlying coupling g is actually *larger* than the electromagnetic one (Chapter 30).



Beta decay, quark by quark. One of the neutron's down quarks turns into an up quark by emitting a virtual W^-, which becomes an electron and an electron antineutrino. The other two quarks are spectators. The neutron (udd) has become a proton (uud).

The weak force has three strange properties.

- **It violates parity.** In 1956 Lee and Yang pointed out that no experiment had ever tested whether the weak force respects mirror symmetry. Within months, Chien-Shiung Wu aligned the spins of cobalt-60 nuclei in a magnetic field and found that beta-decay electrons come out preferentially in one direction relative to the spin: the mirror image of the experiment does not happen in nature. The weak force acts only on left-handed particles (and right-handed antiparticles). Pauli had bet that God was not "a weak left-hander"; he lost.

- **It mixes generations.** The quark states that the W couples to are not quite the same as the states of definite mass; they are rotated by a 3×3 unitary matrix, the *CKM matrix*. That is why a strange quark can decay to an up quark, slowly.

- **It violates CP.** The combined symmetry of swapping matter with antimatter and reflecting in a mirror is broken, slightly, as Cronin and Fitch found in neutral kaons in 1964. Kobayashi and Maskawa showed in 1973 that this requires at least three generations of quarks — before the third generation was discovered. CP violation means matter and antimatter behave slightly differently, which is a precondition for the universe ending up with more matter than antimatter (Problem 32.12).

§32.6 Electroweak unification and the Higgs

In the 1960s Glashow, Weinberg, and Salam unified the electromagnetic and weak forces into a single gauge theory with symmetry $SU(2) \times U(1)$. At high energies the theory has four massless gauge bosons. The Higgs mechanism (Chapter 30) breaks the symmetry: a Higgs field fills all of space with a nonzero value, $v = 246\,\text{GeV}$, and three of the four gauge bosons absorb parts of it and become massive — the W^+, W^-, and Z — while one combination stays massless: the photon. The theory predicted the W and Z masses, and CERN found them in 1983, exactly where predicted.

The Higgs field gives mass to the fermions too: each fermion's mass is proportional to the strength of its coupling to the Higgs field, $m_f = y_f v/\sqrt{2}$. Why those couplings range from 10^{-6} for the electron to almost exactly 1 for the top quark is one of the great unexplained patterns. The leftover ripple in the Higgs field is the Higgs boson. It was the last missing piece, sought for decades, and on 4 July 2012 the ATLAS and CMS experiments at the LHC (Chapter 31)

announced its discovery at 125 GeV. Its measured couplings to other particles so far all match the Standard Model's predictions.

§32.7 Neutrinos

Neutrinos are the most elusive particles in the Standard Model, and they have provided its biggest surprise. In the 1960s Ray Davis built a tank of 600 tonnes of cleaning fluid in a South Dakota gold mine to catch neutrinos from the Sun (Chapter 22), counting the argon atoms they made one at a time. He found only a third of the predicted number. For thirty years the *solar neutrino problem* persisted. The answer, confirmed by Super-Kamiokande in 1998 with neutrinos made by cosmic rays and by SNO in 2001 with solar neutrinos, was that neutrinos change flavour as they travel: an electron neutrino can turn into a muon or tau neutrino and back. This *neutrino oscillation* happens only if neutrinos have mass, because it is an interference effect between mass states with slightly different energies (Problem 32.13):

$$P(\nu_e \to \nu_\mu) = \sin^2 2\theta \, \sin^2\!\left(\frac{\Delta m^2 c^4 \, L}{4\hbar c E}\right).$$

The Standard Model had assumed neutrinos massless. Their masses are tiny, below about 1 eV, at least a million times lighter than the electron, and nobody knows why. Nor do we know whether a neutrino is its own antiparticle, a question being probed by searches for an extremely rare nuclear decay.

§32.8 Beyond the Standard Model

The Standard Model is spectacularly successful and certainly incomplete.

- **Gravity** is not included, and combining it with quantum field theory runs into deep trouble (Chapter 33).
- **Dark matter** (Chapter 28) is not any particle in the Standard Model.
- **Neutrino masses** need some extension, perhaps new heavy particles.
- **The matter–antimatter asymmetry**: the CP violation in the Standard Model is far too small to explain why the universe is made of matter (Problem 32.12).
- **The hierarchy problem**: quantum corrections should drag the Higgs mass up toward the highest energy scale in the theory, perhaps the Planck scale. That it sits at 125 GeV seems to require a cancellation to one part in 10^{30}. Supersymmetry and other theories proposed to explain this predicted new particles at the LHC, which have not been found.
- **The strong CP problem**: QCD allows a CP-violating term that experiments limit to below 10^{-10} of its natural size. One proposed explanation predicts a new particle, the *axion*, which is also a dark matter candidate.

Grand unified theories suggest that the three gauge couplings, running with energy (Chapter 30), might meet at about 10^{16} GeV, uniting the forces into one. Most predict that protons decay,

with lifetimes of 10^{34} years or so; enormous detectors have watched for it, and find the proton lives longer than 10^{34} years (Problem 32.11). The muon's magnetic moment was for years a tantalizing hint of new physics; improved theoretical calculations have since brought theory and experiment much closer together. Physics beyond the Standard Model surely exists. Where it will first appear is the field's central question.

§32.9 Problems

32.1 $O(\epsilon)$ ★ Which of these processes are allowed, and which conservation law forbids each of the others? (a) $p \to e^+ + \gamma$; (b) $n \to p + e^- + \bar{\nu}_e$; (c) $\mu^- \to e^- + \gamma$; (d) $\pi^0 \to \gamma + \gamma$; (e) $\Lambda^0 \to p + \pi^-$ (the Λ^0 is uds); (f) $e^+ + e^- \to \gamma$.

32.2 $O(\epsilon)$ ★ Give the quark content and check the electric charge of the proton, neutron, Δ^{++}, Ω^- (sss), π^+, and K^- ($s\bar{u}$).

32.3 $O(\epsilon)$ Estimate the range of the weak force from the W mass. Compare with the size of a proton. Why, then, is the weak force "weak"?

32.4 $O(\epsilon^2)$ ★ (*Predicting the tau's lifetime.*) The muon decays only via $\mu \to e\bar{\nu}_e\nu_\mu$, with lifetime 2.2 μs. Fermi theory predicts a decay rate $\propto G_F^2 m^5$. The tau decays to an electron and two neutrinos in 17.8% of its decays. Predict the tau's lifetime, and compare with the measured 2.9×10^{-13} s. Why the fifth power?

32.5 $O(\epsilon^2)$ ★ (*Counting colours.*) At energies where it can make quark pairs, the ratio $R = \sigma(e^+e^- \to \text{hadrons})/\sigma(e^+e^- \to \mu^+\mu^-)$ is predicted to be $N_c \sum_q Q_q^2$, summed over the quarks light enough to be produced, where N_c is the number of colours. Compute R below the charm threshold, between charm and bottom, and above bottom. The measured values are about 2, 3.3, and 3.7. What is N_c?

32.6 $O(\epsilon^2)$ For atmospheric neutrinos, $\Delta m^2 c^4 \approx 2.5 \times 10^{-3}\,\text{eV}^2$ and the mixing is nearly maximal. At what distance does a 1 GeV muon neutrino have the greatest chance of having changed flavour? Why do neutrinos coming up through the Earth show fewer muon neutrinos than those coming down from the sky?

32.7 $O(\epsilon^2)$ ★ Add up the masses of the quarks in a proton and in a neutron. What fraction of the nucleon's mass is this? Where does the rest come from? If the up and down quarks were massless, roughly what would the proton weigh?

32.8 $O(\epsilon^2)$ The strong coupling runs approximately as $\alpha_s(Q) = \dfrac{12\pi}{(33 - 2n_f)\ln(Q^2/\Lambda^2)}$, with $\Lambda \approx 0.2\,\text{GeV}$ and n_f the number of quark flavours lighter than Q. Evaluate α_s at 91 GeV ($n_f = 5$) and at 2 GeV ($n_f = 3$). What happens as Q approaches Λ? Why does the sign of $33 - 2n_f$ matter, and what would happen with 17 flavours?

32.9 $O(\epsilon^2)$ Fermion masses are $m_f = y_f v/\sqrt{2}$ with $v = 246\,\text{GeV}$. Find the Yukawa couplings y_f of the electron, the bottom quark, and the top quark. Why is the top quark special?

32.10 $e^{-1/\epsilon}$ ★ (*The case for colour.*) The Δ^{++} has spin $\frac{3}{2}$: all three u quarks have spin up, and they are in the spatially symmetric ground state. Show that the combined space–spin wavefunction is symmetric under exchanging any two quarks, violating the Pauli principle. Show that adding a three-valued colour quantum number and requiring the colour part to be totally antisymmetric, $\epsilon_{ijk} q_i q_j q_k$, resolves the problem. Why does this make every baryon colour-neutral?

32.11 $e^{-1/\epsilon}$ Super-Kamiokande contains 50 000 tonnes of water. How many protons does it contain? If the proton lifetime were 10^{34} years, how many decays per year would be expected (ignoring detection efficiency)? Explain how watching for years without seeing any sets a lower limit on the lifetime.

32.12 $e^{-1/\epsilon}$ (*Why is there matter?*) Andrei Sakharov showed in 1967 that a universe starting with equal amounts of matter and antimatter can develop an excess of matter only if three conditions hold: baryon number is violated; C and CP are violated; and there is a departure from thermal equilibrium. Explain why each is necessary. (For the third, use the fact that in equilibrium the populations depend only on energies, and CPT symmetry guarantees that particles and antiparticles have equal masses.) The observed asymmetry is about one extra baryon per billion photons (Chapter 28).

32.13 $e^{-1/\epsilon}$ (*Why oscillation means mass.*) Suppose $\nu_e = \cos\theta\,\nu_1 + \sin\theta\,\nu_2$, where ν_1 and ν_2 have masses m_1 and m_2. Each mass state evolves with phase $e^{-iE_j t/\hbar}$, with $E_j \approx pc + m_j^2 c^4/2pc$ for a relativistic neutrino. Show that the probability of detecting a ν_e after distance L is $1 - \sin^2 2\theta \sin^2(\Delta m^2 c^4 L/4\hbar c E)$. Why does this vanish if $m_1 = m_2$?

Further reading

- **David Griffiths, *Introduction to Elementary Particles*** — The standard undergraduate introduction, especially good for learning to compute.
- **Mark Thomson, *Modern Particle Physics*** — A modern, experimentally grounded undergraduate text.
- **David Tong, lecture notes on Particle Physics** — A qualitative modern overview of the Standard Model, free online.
- **Frank Close, *The Infinity Puzzle*** — The story of how the Standard Model was built, for a general reader.
- **Particle Data Group, *Review of Particle Physics*** — Every measured property of every particle, free online.

CHAPTER 33

String theory and quantum gravity

BUILDS ON general relativity (Chapter 26), quantum field theory (Chapter 30)

The deepest end of the pool: making gravity and the quantum agree — the great unfinished problem of fundamental physics, and where this book ends.

PROTOTYPE A vibrating string. Replace every point-like particle with a tiny loop or strand of string. Like a guitar string, it can vibrate in many modes, and each mode looks, from far away, like a different particle. One of the modes of a closed loop is a massless particle of spin 2 — a graviton. Gravity is not added to string theory by hand; it is unavoidable.

Two theories describe everything we have ever measured. General relativity (Chapter 26) governs gravity, planets, black holes, and the cosmos. Quantum field theory (Chapter 30) governs everything else, from atoms to the Higgs boson. Each is confirmed to extraordinary precision in its own domain. They are mutually incompatible. In nearly all of nature, this does not matter, because gravity is negligible where quantum effects are large and quantum effects are negligible where gravity is strong. But at the centre of a black hole and at the beginning of the universe, both matter, and there we have no theory. This chapter is a tasting of the attempts to build one. It is more speculative than the rest of the book, and it is honest to say so.

§33.1 The Planck scale

Where does quantum gravity become unavoidable? Dimensional analysis (Chapter 1) answers immediately. From $\hbar$ (quantum mechanics), c (relativity), and G (gravity), there is exactly one way to make a length, a mass, and a time:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \approx 1.6 \times 10^{-35}\,\text{m}, \qquad m_P = \sqrt{\frac{\hbar c}{G}} \approx 2.2 \times 10^{-8}\,\text{kg}, \qquad t_P = \frac{\ell_P}{c} \approx 5.4 \times 10^{-44}\,\text{s}.$$

The *Planck energy* $m_P c^2 \approx 1.2 \times 10^{19}\,\text{GeV}$ is a thousand trillion times beyond the LHC. Physically, the Planck mass is the mass whose Compton wavelength (where quantum effects become essential) equals its Schwarzschild radius (where gravity becomes essential; Problem 33.3). To probe a distance ℓ_P you would need to concentrate so much energy into so small a region that it would collapse into a black hole of that size. Below the Planck length, our very notion of distance may cease to make sense.

§33.2 Why gravity resists quantization

At first sight, quantizing gravity should work like quantizing electromagnetism. Write the metric as flat spacetime plus a small fluctuation (Chapter 29), quantize the waves, and you get *gravitons*: massless particles of spin 2. This works at low energies — one can even compute the tiny quantum corrections to Newton's law. But as an effective field theory (Chapter 30) it breaks down at high energies, for the same reason Fermi's theory of the weak force broke down. Newton's constant has dimensions: the strength of gravity between two particles of energy E is effectively $GE^2/\hbar c^5 = (E/E_P)^2$, which becomes larger than 1 near the Planck energy (Problem 33.7). Each order of perturbation theory brings worse divergences, needing infinitely many new parameters to absorb them. The theory is *non-renormalizable*. Something new must take over at the Planck scale, as the W boson took over from Fermi's theory. The question is what.

§33.3 Black holes are thermodynamic

The most important clues so far come from black holes. In 1972 Jacob Bekenstein, a graduate student, asked what happens to the entropy of a hot cup of tea thrown into a black hole. If it simply vanishes, the second law of thermodynamics (Chapter 7) is violated. He proposed that black holes themselves carry entropy, proportional to the area of their horizon. Hawking, initially sceptical, then discovered that black holes radiate at a temperature (Chapter 26), which fixed the constant:

$$S_{\mathrm{BH}} = \frac{k_{\mathrm{B}} A}{4\ell_P^2} = \frac{k_{\mathrm{B}} c^3 A}{4G\hbar}.$$

This formula contains every fundamental constant, and it is enormous. A solar-mass black hole has an entropy of about $10^{77} k_{\mathrm{B}}$, twenty orders of magnitude more than the Sun itself (Problem 33.4). Black holes are the most entropic objects possible.

Two things about this are startling. First, in statistical mechanics, entropy counts microstates (Chapter 9). So a black hole — which, classically, is perfectly described by mass, spin, and charge, with "no hair" — must have $e^{10^{77}}$ hidden microscopic states. What are they? Second, entropy is usually proportional to *volume*; for a black hole it is proportional to the *area* of its boundary. Since a black hole is the most entropic thing that can fit in a region, the maximum information in *any* region of space is bounded by its surface area, measured in Planck units. This **holographic principle** ('t Hooft, Susskind) suggests that a theory of quantum gravity in a region can be described by degrees of freedom living on its boundary, like a hologram, which stores a three-dimensional image on a two-dimensional film.

§33.4 The information paradox

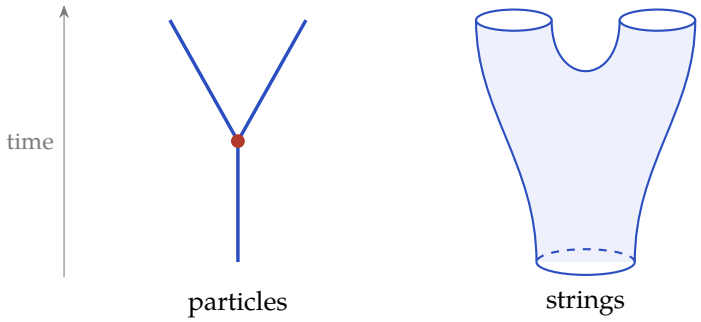
Throw an encyclopaedia into a black hole and wait for the black hole to evaporate. Hawking's calculation says the radiation is exactly thermal, carrying no information about what fell in. Then the information is destroyed, and quantum mechanics, whose evolution is always reversible (unitary), is violated. Either quantum mechanics fails, or Hawking's semiclassical calculation misses something, or the information hides somewhere. This *information paradox* has been argued over for fifty years. Most theorists now believe information is preserved, and

calculations since 2019 using ideas from holography have shown how the entropy of the radiation can follow the *Page curve* expected if information escapes, rising and then falling as the black hole evaporates. Exactly how the information gets out, and what an infalling observer would see, remains under active debate.

§33.5 String theory

String theory began, by accident, as a theory of the strong force in the late 1960s, and was nearly abandoned when QCD (Chapter 32) succeeded. Its fatal flaw for that purpose — it always contained a massless spin-2 particle — turned out, in 1974, to be its greatest virtue: that particle has exactly the properties of the graviton.

The idea is simple to state. The fundamental objects are one-dimensional strings, open or closed, with a tension T. Like a guitar string (Chapter 5), a string vibrates in a set of normal modes, and each mode, seen from a distance much larger than the string, is a particle whose mass is set by its vibrational energy. A single kind of string can thus produce many kinds of particles. The lowest modes are massless and include gauge bosons (from open strings) and the graviton (from closed strings); higher modes have masses of order the string scale, presumably near the Planck scale, far beyond reach.



Left: when point particles interact, their worldlines meet at a point, and the infinities of quantum field theory come from such points. Right: a closed string sweeps out a tube, and one string splitting into two sweeps out a "pair of pants." The surface is smooth everywhere, with no special point where the split happens, which is part of why string theory is free of those infinities.

String theory has remarkable features:

- **It is finite.** Point particles interact at points, and that is the source of the ultraviolet divergences of QFT. Strings interact by joining and splitting smoothly, spread over a region, and the divergences of quantum gravity disappear.
- **It fixes the dimension of spacetime.** Quantum consistency requires 26 dimensions for the simplest (bosonic) strings and 10 for *superstrings*, which include fermions and have a symmetry between bosons and fermions called *supersymmetry*. The extra six dimensions must be curled up too small to see. This idea goes back to Kaluza and Klein in the 1920s: a curled-up dimension of radius R would show up as a tower of heavier copies of every particle, with masses nh/Rc (Problem 33.6). The shape of the hidden dimensions determines the particles and forces we see.

- **It is unique, in a sense.** In the 1980s there seemed to be five different superstring theories. In 1995 Edward Witten and others showed that they are all different limits of a single eleven-dimensional theory, called *M-theory*, linked by surprising equivalences (*dualities*) — for instance, a theory on a small circle of radius R is equivalent to one on a circle of radius ℓ_s^2/R. The theory also contains higher-dimensional objects called *branes*, on which open strings can end.
- **It counts black hole microstates.** In 1996 Strominger and Vafa built certain black holes out of strings and branes, counted their microscopic states, and recovered exactly $S = k_B A/4\ell_P^2$, including the factor of $\frac{1}{4}$. It was the first time anyone had explained the Bekenstein–Hawking entropy microscopically.

§33.6 Holography made precise

In 1997 Juan Maldacena discovered the **AdS/CFT correspondence**: string theory (and hence quantum gravity) in a particular kind of curved spacetime, *anti-de Sitter space*, is exactly equivalent to an ordinary quantum field theory, without gravity, living on that spacetime's boundary, in one fewer dimension. Two theories that look utterly different — one with gravity and black holes in the bulk, one a gauge theory like QCD on the boundary — are the same physics in different languages. A black hole in the bulk corresponds to a hot plasma on the boundary. It is the holographic principle made precise, and it has become one of the most cited ideas in physics.

It has unexpected uses. Hard problems in strongly coupled quantum field theory can become tractable problems in classical gravity. AdS/CFT predicted that strongly coupled fluids have a viscosity-to-entropy ratio near $\hbar/4\pi k_B$ — and the quark–gluon plasma made at heavy-ion colliders (Chapter 22) turns out to be close to this bound, the most perfect fluid known (Problem 33.10). And recent work suggests that the geometry of spacetime in the bulk is built from the quantum *entanglement* of the boundary theory: spacetime itself might emerge from entanglement (Chapter 20).

§33.7 Doubts

String theory has not been tested experimentally, and may not be testable in any foreseeable way. Its natural scale is near the Planck energy. The many ways to curl up the extra dimensions seem to give an enormous *landscape* of possible low-energy worlds — by some estimates 10^{500} or more — with different particles and constants, and no known principle picks out ours. Supersymmetry, which many hoped would appear at the LHC as a sign of string theory and as a solution to the hierarchy problem (Chapter 32), has not appeared. Critics argue that a theory compatible with almost any outcome is not science; defenders reply that string theory has already taught us a great deal about quantum gravity, black holes, and quantum field theory, and that it remains the only framework known to unify all the forces with gravity consistently. Both views are held by serious physicists. You should know that the question is open.

§33.8 Other roads

String theory is the most developed approach, but not the only one.

- **Loop quantum gravity** quantizes the geometry of spacetime directly, without strings or extra dimensions. It predicts that area and volume come in discrete quanta, with space made of a network of "spin networks" at the Planck scale. Recovering smooth spacetime at large scales has proved difficult.
- **Asymptotic safety** proposes that gravity might be a perfectly good quantum field theory after all, if its coupling approaches a fixed value at high energies rather than growing without bound.
- **Causal sets** and **causal dynamical triangulations** model spacetime as a discrete structure, summing over ways of assembling it from building blocks.
- **Emergent gravity** proposals suggest that gravity, like temperature or elasticity, is not fundamental but arises from the collective behaviour of underlying quantum degrees of freedom — perhaps from entanglement itself.

§33.9 Can we ever test it?

Directly probing the Planck scale with today's magnet technology would require a ring hundreds of light-years across — and even that would not work (Problem 33.9). But there are indirect windows. Inflation (Chapter 28) may have stretched quantum fluctuations of the gravitational field itself to cosmic size; detecting them as B-modes in the CMB polarization would be direct evidence that gravity is quantized. Black hole mergers (Chapter 29) test the structure of horizons. The arrival times of gamma rays from distant bursts constrain any Planck-scale graininess of spacetime that would make light's speed depend on its energy; so far, none is seen. And tabletop experiments have been proposed to test whether gravity can entangle two small masses, which would show that gravity is quantum.

§33.10 Epilogue: how to keep going

This is the last plate on the menu, and it is the one with the most left unfinished. That is fitting. Physics is not a completed edifice but a set of very good approximations — spherical cows, all of them, each valid in its domain and each a limit of something deeper. Newton's mechanics is the slow-motion limit of relativity and the large-action limit of quantum mechanics. Thermodynamics is the large-number limit of statistical mechanics. The Standard Model is an effective field theory, valid below some energy we do not yet know. Even general relativity, our deepest theory of space and time, is presumably the low-energy limit of a quantum theory of gravity we do not yet have.

The open questions are enormous, and they are not only at the Planck scale. What are dark matter and dark energy? Why is there more matter than antimatter? What is the mechanism of high-temperature superconductivity? Is turbulence predictable? How do proteins fold so fast, and how did life begin? What does quantum mechanics mean?

If you have tasted a chapter here and want the meal, every chapter has told you where to find it. The standard texts are standard because they work. Read them with a pencil. Do the problems — more than you think you need to. Estimate before you calculate. Learn the prototype, and test everything against it. And when a problem looks hopeless, look for the small parameter, assume a spherical cow, and see how far it gets you. It usually gets you surprisingly far.

§33.11 Problems

33.1 $O(\epsilon)$ ★ Using dimensional analysis, show that the only combinations of $\hbar$, c, and G with dimensions of length, mass, and time are the Planck units (up to numerical factors), and evaluate them. Also find the Planck temperature $m_P c^2/k_B$.

33.2 $O(\epsilon)$ Compare the Planck energy with the collision energy of the LHC (13.6 TeV) and with the most energetic cosmic rays ($\sim 10^{20}$ eV). How many orders of magnitude separate them?

33.3 $O(\epsilon^2)$ ★ Find the mass at which a particle's reduced Compton wavelength $\hbar/mc$ equals its Schwarzschild radius $2Gm/c^2$. Compare it with the Planck mass. Interpret: why can't you localize a particle to better than the Planck length?

33.4 $O(\epsilon^2)$ ★ Compute the Bekenstein–Hawking entropy of a solar-mass black hole, in units of k_B and in bits. The Sun's entropy is roughly $10^{58} k_B$ (about one per particle times a modest factor). Estimate the entropy of the supermassive black hole at the centre of our galaxy, and compare it with the entropy of the cosmic microwave background in the observable universe, about $10^{89} k_B$.

33.5 $O(\epsilon^2)$ A black hole's lifetime scales as M^3; one of about 2×10^{11} kg would be evaporating now if it formed in the Big Bang. Find its Schwarzschild radius and Hawking temperature. What would the final moments of its evaporation look like?

33.6 $O(\epsilon^2)$ ★ (*Kaluza–Klein.*) Suppose space has a fifth dimension curled into a circle of radius R. A particle's momentum around the circle must be quantized, $p_5 = n\hbar/R$ (its wavefunction must fit; Chapter 14). Show that, seen from four dimensions, the particle appears as a tower of particles with masses $m_n = n\hbar/Rc$ (for a massless particle in five dimensions). The LHC has found no such towers below a few TeV. What upper limit does this put on R?

33.7 $O(\epsilon^2)$ Show that the gravitational interaction energy of two particles of energy E at a separation equal to their wavelength $\hbar c/E$, divided by E, is of order $GE^2/\hbar c^5 = (E/E_P)^2$. Evaluate this for two electrons at rest and for two protons at the LHC. Compare with the analogous argument for Fermi's theory (Chapter 30).

33.8 $O(\epsilon^2)$ The holographic bound says the maximum entropy that can fit inside a sphere of radius r is that of a black hole of the same size, $k_B \pi r^2/\ell_P^2$. Evaluate it for $r = 1$ m, and compare with the entropy of a cubic metre of air at room temperature (about 10^3 J/K, i.e. $10^{26} k_B$). Why does ordinary matter come nowhere near the bound?

33.9 $e^{-1/\epsilon}$ ★ (*An accelerator to the Planck scale.*) Using LHC-strength magnets (8 T) and the rigidity formula $p\,[\mathrm{GeV}] = 0.3B\,[\mathrm{T}]\,\rho\,[\mathrm{m}]$, find the radius of a ring that could accelerate protons to the Planck energy. Express it in light-years. Then consider synchrotron radiation (Chapter 31): would even this ring work? Is there any way, even in principle, to probe the Planck length with particle collisions?

Hint: Think about what happens when you concentrate a Planck energy into a Planck-sized region.

33.10 $e^{-1/\epsilon}$ (*The most perfect fluid.*) AdS/CFT predicts that strongly coupled fluids have viscosity-to-entropy-density ratio $\eta/s \approx \hbar/4\pi k_{\mathrm{B}}$. Evaluate this in SI units. For water at room temperature, $\eta \approx 10^{-3}$ Pa s and the entropy density is about $4\times10^{6}\,\mathrm{J/(K\,m^{3})}$. How far above the bound is water? Measurements at heavy-ion colliders place the quark–gluon plasma within a factor of a few of the bound. What does a small η/s mean physically?

33.11 $e^{-1/\epsilon}$ (*The paradox, quantitatively.*) A solar-mass black hole evaporates in about 10^{67} years. Don Page argued that if information escapes, the entanglement entropy of the emitted radiation must rise, peak around the time when the black hole has lost half its entropy (the *Page time*), and then fall to zero. Explain why, using the fact that a pure state that is divided into two parts has equal entanglement entropy in each part (Chapter 20). Why does Hawking's original calculation, in which the entropy only rises, conflict with unitarity?

Further reading

- **Barton Zwiebach, *A First Course in String Theory*** — A remarkably accessible introduction for advanced undergraduates.
- **Carlo Rovelli, *Reality Is Not What It Seems*** — A popular introduction to quantum gravity from the loop quantum gravity perspective.
- **Leonard Susskind, *The Black Hole War*** — The information paradox and holography, told by one of the protagonists.
- **Thomas Hartman, lecture notes on Quantum Gravity and Black Holes** — A modern course on black hole thermodynamics, holography, and the information problem, free online.
- **Joseph Polchinski, *String Theory*, Volumes I and II** — The graduate standard, for when you are ready.

APPENDIX A

Constants and orders of magnitude

A working physicist carries a few dozen numbers in their head. Here are the ones this book uses most. Values are rounded to the precision that is usually useful; look up the exact values when you need them.

Fundamental constants

Quantity	Symbol	Value
Speed of light	c	$2.998 \times 10^{8}\ \mathrm{m/s}$ (exact)
Planck's constant	h	$6.626 \times 10^{-34}\ \mathrm{J\,s}$ (exact)
Reduced Planck constant	$\hbar = h/2\pi$	$1.055 \times 10^{-34}\ \mathrm{J\,s} = 6.58 \times 10^{-16}\ \mathrm{eV\,s}$
Elementary charge	e	$1.602 \times 10^{-19}\ \mathrm{C}$ (exact)
Boltzmann's constant	k_{B}	$1.381 \times 10^{-23}\ \mathrm{J/K} = 8.62 \times 10^{-5}\ \mathrm{eV/K}$
Avogadro's number	N_A	$6.022 \times 10^{23}\ \mathrm{mol^{-1}}$
Gas constant	$R = N_A k_{\mathrm{B}}$	$8.314\ \mathrm{J/(mol\,K)}$
Gravitational constant	G	$6.674 \times 10^{-11}\ \mathrm{N\,m^2/kg^2}$
Vacuum permittivity	ϵ_0	$8.854 \times 10^{-12}\ \mathrm{F/m}$
Coulomb constant	$1/4\pi\epsilon_0$	$8.988 \times 10^{9}\ \mathrm{N\,m^2/C^2}$
Vacuum permeability	μ_0	$1.257 \times 10^{-6}\ \mathrm{N/A^2}$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$1/137.036$
Stefan–Boltzmann constant	σ	$5.670 \times 10^{-8}\ \mathrm{W/(m^2K^4)}$
Electron mass	m_e	$9.109 \times 10^{-31}\ \mathrm{kg} = 0.511\ \mathrm{MeV}/c^2$
Proton mass	m_p	$1.673 \times 10^{-27}\ \mathrm{kg} = 938.3\ \mathrm{MeV}/c^2$
Neutron mass	m_n	$939.6\ \mathrm{MeV}/c^2$
Atomic mass unit	u	$1.661 \times 10^{-27}\ \mathrm{kg} = 931.5\ \mathrm{MeV}/c^2$
Bohr radius	a_0	$0.0529\ \mathrm{nm}$
Hydrogen ground state		$-13.6\ \mathrm{eV}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\ \mathrm{m^{-1}}$
Bohr magneton	μ_B	$9.27 \times 10^{-24}\ \mathrm{J/T} = 5.79 \times 10^{-5}\ \mathrm{eV/T}$
Electron Compton wavelength	$\hbar/m_e c$	$3.86 \times 10^{-13}\ \mathrm{m}$
A handy conversion	$\hbar c$	$197.3\ \mathrm{MeV\,fm} = 197.3\ \mathrm{eV\,nm}$
Another	hc	$1240\ \mathrm{eV\,nm}$
Planck length	ℓ_P	$1.6 \times 10^{-35}\ \mathrm{m}$
Planck mass	m_P	$2.2 \times 10^{-8}\ \mathrm{kg} = 1.2 \times 10^{19}\ \mathrm{GeV}/c^2$
Planck time	t_P	$5.4 \times 10^{-44}\ \mathrm{s}$

The Earth, the Sun, and the universe

Earth's mass	5.97×10^{24} kg	Sun's mass $M_\odot$	1.99×10^{30} kg
Earth's radius	6.37×10^{6} m	Sun's radius $R_\odot$	6.96×10^{8} m
Surface gravity	$9.81\,\mathrm{m/s^2}$	Sun's luminosity $L_\odot$	3.83×10^{26} W
Moon's mass	7.35×10^{22} kg	Sun's surface temperature	5800 K
Earth–Moon distance	3.84×10^{8} m	Solar constant at Earth	$1361\,\mathrm{W/m^2}$
Astronomical unit	1.50×10^{11} m	Hubble constant H_0	$\approx 70\,\mathrm{km\,s^{-1}Mpc^{-1}}$
Light-year	9.46×10^{15} m	Age of the universe	13.8×10^{9} yr
Parsec	3.09×10^{16} m $= 3.26$ ly	CMB temperature	2.725 K
One year	3.16×10^{7} s $\approx \pi \times 10^{7}$ s	Atmospheric pressure	1.01×10^{5} Pa

Numbers worth knowing by heart

- Thermal energy at room temperature: $k_\mathrm{B}T \approx \frac{1}{40}$ eV. At body temperature, $k_\mathrm{B}T \approx 4.3$ pN nm.
- Energy of a visible photon: 2–3 eV ($\lambda = 400$–700 nm; use $E\,[\mathrm{eV}] = 1240/\lambda\,[\mathrm{nm}]$).
- Chemical bond: a few eV. Hydrogen bond: about 0.2 eV. Van der Waals: 0.01–0.1 eV.
- Nuclear binding: about 8 MeV per nucleon. Fission: about 200 MeV per nucleus. Fusion of hydrogen to helium: 0.7% of the rest mass.
- Sizes: atom 10^{-10} m, nucleus 10^{-15} m, proton 0.84×10^{-15} m, cell 10^{-5} m.
- Densities: water $1000\,\mathrm{kg/m^3}$, air $1.2\,\mathrm{kg/m^3}$, rock $3000\,\mathrm{kg/m^3}$, nuclear matter $2{\times}10^{17}\,\mathrm{kg/m^3}$.
- Speeds: sound in air 343 m/s, in water 1500 m/s; molecules in air ~ 500 m/s; escape from Earth 11.2 km/s; Earth's orbit 30 km/s.
- A mole of gas at room temperature and pressure occupies about 24 litres.
- Viscosity of water 10^{-3} Pa s; of air 1.8×10^{-5} Pa s.

A tour in powers of ten

Lengths (m)		Times (s)	
10^{-35}	Planck length	10^{-44}	Planck time
10^{-18}	range of the weak force	10^{-24}	lifetime of a W boson
10^{-15}	proton	10^{-15}	period of visible light
10^{-10}	atom	10^{-13}	molecular vibration
10^{-7}	wavelength of light	10^{-9}	excited atom's lifetime
10^{-5}	living cell	10^{-3}	nerve impulse
10^{0}	you	10^{0}	heartbeat
10^{7}	the Earth	3×10^{7}	one year
10^{11}	Earth–Sun distance	10^{17}	age of the Earth
10^{21}	the Milky Way	4×10^{17}	age of the universe
10^{26}	observable universe	10^{74}	evaporation of a solar-mass black hole

APPENDIX B

Where to go next

Every chapter ended with a short list of books. This appendix steps back and suggests a path. It is built on a simple, widely shared conviction, put especially clearly by Kevin Zhou in his short guide *The Theoretical Minimum*: you can get a deep understanding of physics by working through a small set of canonical textbooks, doing many of their problems yourself. The books below are recommended because they are self-contained, anticipate a learner's misconceptions, and above all contain good problems.

The undergraduate core

If you work through these, doing most of the problems, you will know the core theoretical content of an undergraduate physics degree.

1. **Any calculus-based introductory text** — Halliday, Resnick, and Walker; Young and Freedman; or similar. They are about equally good; what matters is doing the problems.
2. **John R. Taylor, *Classical Mechanics*.** A smooth, thorough introduction to intermediate mechanics, starting from ordinary calculus. (Chapters 2–4, 12.)
3. **Daniel V. Schroeder, *An Introduction to Thermal Physics*.** Clean, focused, and clear. (Chapters 7 and 9.)
4. **David J. Griffiths, *Introduction to Electrodynamics*.** The best undergraduate electromagnetism book by a wide margin, with an excellent crash course on vector calculus. (Chapter 6.)
5. **Griffiths and Schroeter, *Introduction to Quantum Mechanics*.** The most accessible introduction, built on elementary exercises plus a rich variety of harder problems. (Chapter 14.)

Alongside these, **Mary Boas, *Mathematical Methods in the Physical Sciences*** (Chapter 1) supplies the mathematics as you need it.

Specializations at the undergraduate level

Each of these is a standard first serious book in its field.

- **Optics**: Hecht, *Optics*.
- **Special relativity**: Taylor and Wheeler, *Spacetime Physics*; or Morin, *Special Relativity for the Enthusiastic Beginner*.

- **General relativity**: Hartle, *Gravity*; or Schutz, *A First Course in General Relativity*.
- **Fluids and continua**: Lautrup, *Physics of Continuous Matter*.
- **Chaos**: Strogatz, *Nonlinear Dynamics and Chaos*.
- **Computation**: Newman, *Computational Physics*.
- **Condensed matter**: Simon, *The Oxford Solid State Basics*.
- **Atomic physics**: Foot, *Atomic Physics*; quantum optics: Gerry and Knight, *Introductory Quantum Optics*.
- **Quantum information**: Nielsen and Chuang, *Quantum Computation and Quantum Information*.
- **Nuclear and particle physics**: Krane, *Introductory Nuclear Physics*; Griffiths, *Introduction to Elementary Particles*.
- **Plasma physics**: Chen, *Introduction to Plasma Physics and Controlled Fusion*.
- **Astrophysics**: Carroll and Ostlie, *An Introduction to Modern Astrophysics*.
- **Cosmology**: Ryden, *Introduction to Cosmology*.
- **Biophysics**: Nelson, *Biological Physics*.
- **String theory**: Zwiebach, *A First Course in String Theory*.

The graduate core

A physics PhD typically begins by seeing the undergraduate core again at a higher level. Modern texts that improve on the long-standing classics include:

- **Mechanics**: Helliwell and Sahakian, *Modern Classical Mechanics*.
- **Electrodynamics**: Zangwill, *Modern Electrodynamics*.
- **Statistical mechanics**: Kardar, *Statistical Physics of Particles* and *Statistical Physics of Fields*.
- **Quantum mechanics**: Robert Littlejohn's Berkeley lecture notes (free online), or Sakurai.
- **Quantum field theory**: Schwartz, *Quantum Field Theory and the Standard Model*; or Peskin and Schroeder.
- **General relativity**: Carroll, *Spacetime and Geometry*.

Problems, and more problems

Physics is learned by solving problems. Beyond the textbooks above:

- **David Morin**, *Introduction to Classical Mechanics* and *Problems and Solutions in Introductory Mechanics*: hundreds of beautiful problems with full solutions.

- **Kevin Zhou's physics olympiad handouts**: a free, carefully graded collection of challenging problems spanning mechanics, electromagnetism, thermodynamics, optics, and modern physics, with insightful commentary.
- **I. E. Irodov**, *Problems in General Physics*: the classic Soviet problem book, terse and difficult.
- **Mahajan**, *Street-Fighting Mathematics* and *The Art of Insight in Science and Engineering*: problems in estimation and approximation — the spherical cow, as a discipline.

For perspective

- ***The Feynman Lectures on Physics*** (free online): the whole of physics, seen through the eyes of a master.
- **Feynman**, *The Character of Physical Law* and *QED*: short books on how physics works and what it means.
- **Steven Weinberg**, *To Explain the World* and *Dreams of a Final Theory*.
- **Free lecture notes**: David Tong's Cambridge notes on nearly every subject in theoretical physics; MIT OpenCourseWare; and, for what is happening right now, the physics section of the arXiv preprint server.

How to read a physics textbook

Slowly, with a pencil. Before reading a derivation, try to guess the result; afterwards, close the book and reproduce it. Do the examples before reading the solutions. Do more problems than are assigned, and return to the ones you could not do. When you are stuck, estimate, find a limiting case, or draw a picture. And when you finish a book, you will find that the next one is easier than you feared. That is the point of the minimum: the foundation is set.

APPENDIX C

Hints to the nonperturbative problems

The problems marked $e^{-1/\epsilon}$ need an idea that does not come from grinding on the obvious approach. Here is a nudge toward that idea for each of them — a first step, a change of viewpoint, or the quantity worth computing. Try the problem for a while before you look, and look at only one hint at a time. Full solutions are in Appendix D.

1 Mathematical methods

1.10 Write the integrand as $e^{f(x)}$ with $f = N \ln x - x$. Find where f peaks and how sharply: what is f'' there?

1.11 Substitute $u = 1/\epsilon$. What does every derivative of $e^{-1/\epsilon}$ look like in terms of u, and what does e^{-u} do to any polynomial as $u \to \infty$?

1.12 Five quantities and three dimensions leave two dimensionless groups; one is $\rho g r^2/\sigma$. For a small tap this group is tiny. What can the other group then depend on?

2 Newtonian mechanics

2.10 Suppose the top k books balance exactly on the edge of book $k+1$. Where is the centre of mass of the top $k+1$ books, measured from that same edge?

2.11 Go to the rotating frame and add the centrifugal potential $-\frac{1}{2}m\Omega^2(R\sin\theta)^2$ to gravity's. Equilibria and their stability come from the first and second derivatives.

2.12 You will get two coupled linear equations for the small components ω_1 and ω_3. Differentiate one, substitute the other, and look at the sign of the coefficient: it is a product of two differences of moments of inertia.

3 The harmonic oscillator

3.9 Bound $\langle p^2\rangle$ below by $(\Delta p)^2$ and use $\Delta p \geq \hbar/2\Delta x$. Now the energy is a function of Δx alone. Minimize it.

3.10 You need $g\sin\phi = \omega^2 s$, i.e. $s \propto \sin\phi$. Use $\mathrm{d}x = \cos\phi\ \mathrm{d}s$ and $\mathrm{d}y = \sin\phi\ \mathrm{d}s$, and write everything in terms of 2ϕ.

3.11 At the instant of standing up, the force on the child is radial, so angular momentum about the pivot is conserved. Between such instants, energy is conserved. Chain the two to relate one amplitude to the next.

4 Analytical mechanics

4.9 The integrand does not depend on x, which plays the role of time. What is the analogue of the conserved energy for a Lagrangian with no explicit time dependence?

4.10 To leading order the fast jitter is $\xi \approx -(a/\ell)\sin\bar{\theta}\cos\Omega t$. Expand $\sin(\bar{\theta}+\xi)$ to first order in ξ and average over one period; $\langle\cos^2\Omega t\rangle = \frac{1}{2}$.

4.11 Each bounce off a wall approaching at speed u adds $2u$ to the ball's speed. How often does the ball hit that wall? Turn this into a differential equation relating v and ℓ.

5 Waves

5.10 In shallow water the group velocity equals $\sqrt{gh}$, and the wave's period does not change as it travels. Conserve (energy per area)×(group velocity).

5.11 Follow individual points of the wave profile, each moving at its own local speed. When do the paths of a crest and the trough in front of it cross?

5.12 Energy emitted when the boat was a distance D behind its present position, in all directions θ, lies on a circle. Find that circle's centre and radius, then the tangent line from the boat.

6 Electromagnetism

6.11 Laplace's equation implies the mean-value property: V at a point equals its average over any small sphere around it. Can a function with that property have a strict minimum?

6.12 The positive charges move with the test charge, so they are at rest in its frame. Find the negative charges' speed in that frame with the velocity-addition formula, and compare how the two densities change, to order v^2/c^2.

7 Thermodynamics

7.10 How much can the demon lower the gas's entropy for each bit it learns? Compare that with the minimum cost of erasing each bit.

7.11 For radiation $U = u(T)V$, so $(\partial U/\partial V)_T = u$, and $P = u/3$ gives $(\partial P/\partial T)_V = u'(T)/3$. The identity becomes a differential equation for $u(T)$.

8 Optics

8.10 A stationary deviation needs $\mathrm{d}r/\,\mathrm{d}i = \frac{1}{2}$. Differentiate Snell's law to get $\mathrm{d}r/\,\mathrm{d}i$ in terms of i and r, then eliminate r.

8.11 Divide the open wavefront beyond the disc's edge into Fresnel zones, rings whose distances to the screen point increase by half a wavelength each. Their contributions alternate in sign and shrink slowly. What is the sum?

8.12 Optics is reversible: light can retrace any path. If the target were hotter than the Sun, which way would net radiation flow along those paths?

9 Statistical mechanics

9.11 Solve $n\lambda_T^3 = 2.6$ for λ_T, then $\lambda_T = h/\sqrt{2\pi m k_\mathrm{B} T}$ for T. For liquid helium, get n from the mass density.

9.12 Use Stirling's approximation for $\ln \binom{N}{n}$ and differentiate with respect to n. For the heat flow, ask which direction of energy transfer increases the total entropy when one of the two temperatures is negative.

9.13 Compare $S(2N, 2V, T)$ with $2S(N, V, T)$. Then subtract $k_\mathrm{B} \ln N!$ using Stirling, and watch what happens to the $\ln V$.

10 Special relativity

10.9 The turnaround is the event $(5\,\mathrm{yr}, 4\,\mathrm{ly})$ in the Earth's frame. Find where the traveller's lines of simultaneity through that event — one for the outgoing frame, one for the returning frame — cross the Earth's worldline.

10.10 Are the two door-closings simultaneous in the pole's frame? For the shut door: how quickly can the news that the front has stopped reach the back?

10.11 Note that $c^2/g \approx 1\,\mathrm{ly}$ and $c/g \approx 1\,\mathrm{yr}$. Solve for the proper time to reach the midpoint and double it. For the catch, think about the rocket equation with exhaust at the speed of light.

11 Fluid dynamics

11.10 With no inertia, the body's velocity at each instant is a linear function of the rate of change of its shape. What does that imply about the net displacement over a path in shape space that is traced forwards and then backwards?

11.11 By Kutta–Joukowski, lift per length is $\rho v \Gamma$. For a spinning ball, estimate the circulation as (surface speed)×(size). Then compute the acceleration and the sideways drift over about one second of flight.

11.12 Fix the volume first: it forces the mean radius to shrink slightly, by order δ^2. Then expand the area to order δ^2, keeping both the change in mean radius and the slope term $\sqrt{1 + r'^2}$.

12 Nonlinear dynamics and chaos

12.10 Count dimensions of phase space. For the driven pendulum, treat the phase of the drive as a third variable. For maps, ask whether a one-dimensional map must be invertible.

12.11 Subtract the two equations. You get a single equation $\dot{\phi} = \Delta\omega - 2K\sin\phi$ on a circle. When does it have a fixed point?

12.12 Divide successive intervals. For r_∞, sum the geometric series of all remaining intervals after r_4.

13 Computational physics

13.11 Write the step as a 2×2 matrix M. A quadratic form $\mathbf{u}^T S\mathbf{u}$ is conserved exactly when $M^T S M = S$. For the instability, look at the trace of M.

13.12 The difference between runs starts at round-off level, about 10^{-15}, and grows like $e^{0.9t}$. For the second part, ask whether the computed trajectory might be an exact trajectory of the true system, just from a different starting point.

14 Quantum mechanics

14.12 The bound state must decay as $e^{-\kappa|x|}$ on both sides. The delta function produces a kink: integrate the Schrödinger equation from $-\epsilon$ to $+\epsilon$ to find the jump in ψ'.

14.13 Multiply the tunnelling probability by the Boltzmann factor $e^{-E/k_{\mathrm{B}}T}$. One falls with energy and the other rises: find the maximum of their product.

14.14 Follow the amplitudes through each beam splitter. With the bomb in place, one arm acts as a measurement: the photon is either absorbed there or definitely in the other arm, and then there is nothing to interfere with.

15 Continuum mechanics and acoustics

15.10 Write the total energy as a function of crack half-length, $U(a) = 4a\gamma - \pi a^2\sigma^2/E$. Where is its maximum, and what happens to a crack longer than that?

15.11 The modes with frequency below F correspond to lattice points (m, n) inside a circle. For large F, count lattice points by area.

15.12 Divide each measured frequency by that of the $(1, 1)$ mode and compare with $1 : 1.5 : 2 : 2.5$. For the $(0, n)$ modes, think about what they do to the volume of air in the kettle, and where on the head the drum is struck.

16 Complex systems and networks

16.9 Solve the ODE to get $k_i(t)$ in terms of the arrival time t_i. A node has degree greater than k if it arrived early enough; arrival times are uniform.

16.10 Pairwise exchanges conserving a total are like collisions conserving energy. What distribution maximizes the entropy of a fixed total shared among many agents?

16.11 For the code, keep a stack of sites with four or more grains and topple them until none remain. For the second part, ask what happens to the average slope when avalanches are too rare, and when they are too large.

17 Plasma physics

17.9 Compute ω/k and $\mathrm{d}\omega/\,\mathrm{d}k$ directly. Then set $E = \hbar\omega$, $p = \hbar k$ and compare with $E^2 = (mc^2)^2 + (pc)^2$.

17.10 The grad-B drift has opposite signs for opposite charges. What field does the resulting charge separation create, and what drift does that field produce, for every particle?

17.11 Go to the wave's frame. Particles moving slightly slower or faster than the wave are reflected by its potential hills. Which group gains energy, and which loses it?

18 Atomic, molecular and optical physics

18.11 Find the steady state of $\dot{N}_2 = BI(N_1 - N_2) - AN_2$. When $N_1 = N_2$, what do absorption and stimulated emission do to each other?

18.12 Average $V(\mathbf{r} + \delta\mathbf{r})$ over random isotropic jitter. The first-order term averages to zero; the second-order term involves $\nabla^2 V$, which for the Coulomb potential is a delta function.

19 Condensed matter

19.11 Write the two coupled equations for the amplitudes a and b on the two sublattices and multiply them together. Where in k is $\left|t_1 + t_2 e^{ikd}\right|$ smallest?

19.12 Faraday's law turns a change of flux into an EMF around the cylinder; the Hall conductance turns that EMF into a current along it. Integrate over the time it takes to add one flux quantum.

20 Quantum information

20.11 List the four possible states (no flip, or a flip on qubit 1, 2, or 3) and compute Z_1Z_2 and Z_2Z_3 for each. Do $|000\rangle$ and $|111\rangle$ (and their flipped versions) give the same values?

20.12 After the box, the first qubit is $[(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle]/\sqrt{2}$. What is this when $f(0) = f(1)$, and when $f(0) \neq f(1)$?

20.13 Group $C = A(B + B') + A'(B - B')$ before squaring. You will need $(B + B')(B - B') = -[B, B']$.

21 Chemical physics

21.11 Expand $\sqrt{1 \pm \kappa}$ to second order in κ. The first-order terms cancel in the sum.

21.12 Problem 5 told you which orbital moves more. For O_2 and N_2, fill the valence molecular orbitals in the order $\sigma_{2s}, \sigma^*_{2s}, \pi_{2p}, \sigma_{2p}, \pi^*_{2p}, \sigma^*_{2p}$ (for O_2 the σ_{2p} drops below the π_{2p}, but the count is the same).

22 Nuclear physics

22.10 The half-life is the inverse of (attempt rate)×(tunnelling probability). Write v in terms of E; then evaluate $2\pi\eta = 4\pi Z\alpha c/v$ at the two energies.

22.11 Write a diffusion equation with a source: $\partial n/\partial t = D\nabla^2 n + (\nu - 1)n/\tau$, with $D = \ell v/3$ and $\tau = \ell/v$. The slowest-decaying mode in a sphere has $\nabla^2 n = -(\pi/R)^2 n$.

22.12 The Coulomb barrier is $Z_1 Z_2 \times 1.44\,\mathrm{MeV\,fm}/R$, with R the sum of the radii. For the s-process, follow what happens to bismuth-209 after it captures a neutron.

23 Biophysics

23.10 How long does a molecule stay near the receptor before diffusing away? In that time, how many are there, and how many such independent looks fit in T?

23.11 Write the rate of forward steps (the gas, at T_1, supplies ϵ plus the work) and of backward steps (the pawl, at T_2, is lifted by ϵ). Compare. Then find the heat and work per step.

23.12 After activation, the complex either proceeds (rate W) or releases the substrate (rate ℓ'). If $W \ll \ell'$, what fraction proceeds, and how does it differ for right and wrong substrates? For the second part, ask what equilibrium alone can achieve.

24 Medical physics

24.11 For backprojection, write $\delta(x\cos\theta + y\sin\theta)$ in polar coordinates and integrate over θ. For the filter, write the inverse two-dimensional Fourier transform in polar coordinates and look at the Jacobian.

24.12 Write the absorbance at each wavelength as $cL[s\epsilon_{\mathrm{ox}} + (1-s)\epsilon_{\mathrm{de}}]$ and take the ratio: what drops out? Then list everything in the light path that does not change with the heartbeat.

25 Geophysics and climate

25.10 Draw the chord from the earthquake to the edge of the shadow, 104° around the Earth. How close does it come to the centre?

25.11 The Moon's orbital angular momentum is $M_m\sqrt{GM_e a}$. Differentiate with respect to time, set the Earth's loss of spin angular momentum equal to the Moon's gain, and convert $\mathrm{d}\omega/\,\mathrm{d}t$ into a change of the day's length.

25.12 Find the equilibria on each of the three pieces of the absorbed curve. For stability, ask whether a small warming makes the planet emit more than it absorbs.

26 General relativity

26.11 Linearize: $u^2 \approx (GM/h^2)^2(1 + 2e\cos\omega\phi)$. Matching the $\cos\omega\phi$ terms fixes ω; the perihelion recurs when $\omega\phi = 2\pi$.

26.12 Compare the light ray $x = ct - D$ with the asymptote $x = ct - c^2/a$ of the hyperbola. For the second part, the acceleration needed to hover at r is $GM/r^2\sqrt{1 - r_s/r}$; redshift the local Unruh temperature out to infinity.

27 Astrophysics

27.11 Write both pressures in terms of M and R. If they scale the same way with R, their ratio depends only on M. What happens if gravity wins at one radius?

27.12 For the lower limit, add the electron degeneracy energy $\sim \hbar^2 n^{2/3}/m_e$ to the virial balance and find the maximum temperature the ball ever reaches. For the upper limit, compare $L \propto M^{3.5}$ with $L_{\mathrm{Edd}} \propto M$.

27.13 Angular momentum $\propto MR^2/P$ and flux $\propto BR^2$ are conserved. For the maximum spin, set the centripetal acceleration at the equator equal to gravity.

28 Cosmology

28.9 After freeze-out, the neutrons decay as $e^{-t/880\,\mathrm{s}}$. Each helium-4 nucleus holds two neutrons, so all the neutrons, and as many protons, end up in helium.

28.10 Multiply the growth factors of the two eras: $(a_{\mathrm{eq}}/a)^2$ and then $1/a_{\mathrm{eq}}$, with $a_{\mathrm{eq}} \approx 1/3400$. Inflation must shrink $|\Omega - 1|$ from about 1 to this value.

28.11 Find the fraction of the sky covered by the stars in a thin shell at distance r. Does it depend on r? For the second part, compare the distance a line of sight must go to hit a star with the distance light has travelled since the Big Bang.

29 Gravitational waves

29.9 Differentiate $E = -Gm_1m_2/2a$ to get $\dot{a}$, then separate variables and integrate $a^3\ \mathrm{d}a$.

29.10 Write a in terms of f using Kepler's law with $f_{\text{orb}} = f/2$, then $\dot{f}/f = -\frac{3}{2}\dot{a}/a$. Collect the masses into $m_1m_2(m_1+m_2)^{-1/3}$.

29.11 $H_0 = v/d$. For the error, think about what else besides distance changes the amplitude of the wave, and how errors on independent events combine.

30 Quantum field theory

30.10 Convert E_P to joules and use $\hbar c = 3.2\times10^{-26}\,\mathrm{J\,m}$. Count orders of magnitude with $\log_{10}$.

30.11 In natural units, G_F has dimensions of energy^{-2}. Set $G_F^2E^2$ equal to $1/E^2$.

31 Accelerators and detectors

31.10 For the second plane, swap the signs of f_1 and f_2. For the physics, remember that a lens kicks a ray in proportion to its distance from the axis.

31.11 Power is current times energy lost per turn per unit charge, $P = I\,\Delta E/e$. Find the largest allowed ΔE, then solve the synchrotron formula for ρ at 500 GeV per beam.

31.12 If plasma electrons are displaced by x, Gauss's law gives $E = nex/\epsilon_0$. How far can they be displaced before the wave breaks? Try $x \sim c/\omega_p$.

32 Particle physics

32.10 Check the symmetry of each factor — space, spin, flavour — under exchange of two quarks. Then ask what ϵ_{ijk} does when two indices are swapped, and how it transforms under a rotation of the three colours.

32.11 Each water molecule has 10 protons. For the limit: if the expected number of decays were μ, the chance of seeing none would be $e^{-\mu}$.

32.12 For each condition, imagine it fails and show the asymmetry cannot develop. In equilibrium, what determines a population, and how do particle and antiparticle compare?

32.13 Compute the amplitude $\langle\nu_e|\psi(t)\rangle = \cos^2\theta\,e^{-iE_1t/\hbar} + \sin^2\theta\,e^{-iE_2t/\hbar}$ and take its squared magnitude.

33 String theory and quantum gravity

33.9 Rearrange the rigidity formula for ρ. For the synchrotron loss, use the formula of Problem 31.5 scaled by $(m_e/m_p)^4$, and compare with the beam energy.

33.10 In kinetic theory, $\eta \sim np\ell$ and $s \sim nk_B$. What does the uncertainty principle say about $p\ell$?

33.11 If the whole system is pure, the radiation's entanglement entropy equals the black hole's. What is the largest entanglement entropy the black hole can have, and how does it change as the black hole evaporates?

APPENDIX D

Solutions

These are solutions to every problem in the book. They are written to be checked against, not read in place of doing the problem: each gives the key idea, the main steps, and the final numbers, but leaves routine algebra to you. Numerical answers are rounded to the precision the problem's data justify. Where a problem asks you to explain something, the solution gives one good explanation; there are often others.

1 Mathematical methods

1.1 The speed can depend only on λ, g, and ρ. Density is the only quantity containing mass, so it cannot be combined with the others into a speed: it cannot appear. Then $v = C\sqrt{g\lambda}$ for some number C (the full theory gives $C = 1/\sqrt{2\pi}$). Long waves travel faster.

1.2 Heartbeats: 70 per minute for 80 years is $70 \times 60 \times 24 \times 365 \times 80 \approx 3 \times 10^9$, so about 10^9. Seconds in a year: $365.25 \times 86\,400 = 3.16 \times 10^7$; $\pi \times 10^7 = 3.14 \times 10^7$ is within 0.5%. Sand: say 10^6 km of sandy coast, beaches 50 m wide and 10 m deep, gives 5×10^{11} m^3 of sand; a grain of 0.5 mm has volume $\sim 10^{-10}$ m^3, so about 10^{21} grains. (Published estimates range from 10^{19} to 10^{21}; the method matters more than the digit.)

1.3 $(1-x)^{-1/2} \approx 1 + \frac{1}{2}x + \frac{3}{8}x^2$. With $x = 0.02$: $1 + 0.01 + 0.00015 = 1.01015$. The exact value is $1.0101525\ldots$; the error, about 2.5×10^{-6}, matches the next term $\frac{5}{16}x^3$.

1.4 For $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: $\lambda = 3$ with $(1,1)$, and $\lambda = 1$ with $(1,-1)$. For $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$: $\lambda^2 + 1 = 0$, so $\lambda = \pm i$, with eigenvectors $(1, \pm i)$. The matrix sends $(x, y) \mapsto (y, -x)$: a rotation by 90° clockwise. A rotation turns *every* direction, so no real vector is merely stretched.

1.5 $I^2 = \iint e^{-(x^2+y^2)}\,\mathrm{d}x\,\mathrm{d}y = \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,\mathrm{d}r\,\mathrm{d}\theta = 2\pi \cdot \frac{1}{2} = \pi$, so $I = \sqrt{\pi}$. Rescaling, $\int e^{-ax^2}\,\mathrm{d}x = \sqrt{\pi/a}$; differentiating with respect to a gives $\int x^2 e^{-ax^2}\,\mathrm{d}x = \frac{1}{2}\sqrt{\pi}\,a^{-3/2}$.

1.6 The function is odd, so only sines appear: $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f \sin nx\,\mathrm{d}x = \frac{2}{n\pi}(1 - \cos n\pi)$, which is $4/n\pi$ for odd n and 0 for even n:

$$f(x) = \frac{4}{\pi}\left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\right).$$

A smooth function is well described by a few slow sines, and its coefficients fall off fast. A jump changes the function over an arbitrarily short distance, which by $\Delta x\,\Delta k \gtrsim 1$ needs arbitrarily large wavenumbers, so the coefficients fall only as $1/n$. (Truncated sums overshoot near the jump by about 9%: the Gibbs phenomenon.)

1.7 $\nabla \cdot (\nabla \times \mathbf{v}) = \partial_i \varepsilon_{ijk} \partial_j v_k = \varepsilon_{ijk} \partial_i \partial_j v_k = 0$, because ε_{ijk} is antisymmetric in i, j while $\partial_i \partial_j$ is symmetric. For the second,

$$[\nabla \times (\nabla \times \mathbf{v})]_i = \varepsilon_{ijk} \partial_j \varepsilon_{klm} \partial_l v_m = (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) \partial_j \partial_l v_m = \partial_i(\partial_j v_j) - \partial_j \partial_j v_i,$$

using $\varepsilon_{ijk}\varepsilon_{klm} = \varepsilon_{kij}\varepsilon_{klm}$ and the identity in the text.

1.8 The sea surface is a surface of constant gravitational potential. The mountain is an excess mass per area $\Delta\rho H$ (with $\Delta\rho \approx 1700\,\mathrm{kg/m^3}$, rock minus water) spread over a width W, so dimensionally it changes the potential above it by $\Delta\Phi \sim G\Delta\rho HW \approx 6.7 \times 10^{-11} \times 1700 \times 2000 \times 2 \times 10^4 \approx 5\,\mathrm{m^2/s^2}$. The surface rises by $\Delta h = \Delta\Phi/g \approx 0.5\,\mathrm{m}$: closest to a metre. Satellites that measure sea-surface height map the seafloor this way.

1.9 Write $x = \ell \sum_i s_i$ with independent $s_i = \pm 1$. Then $\langle x \rangle = 0$, and $\langle x^2 \rangle = \ell^2 \sum_{i,j} \langle s_i s_j \rangle = N\ell^2$, since the cross terms average to zero. For perfume, $\sqrt{N}\ell = 5\,\mathrm{m}$ needs $N = (5/10^{-7})^2 = 2.5 \times 10^{15}$ steps, or $2.5 \times 10^5\,\mathrm{s}$: about three days. You smell it within a minute because air currents carry the molecules; diffusion is hopeless over metres.

1.10 The integrand is $e^{f(x)}$ with $f = N \ln x - x$. Then $f'(x) = N/x - 1$ vanishes at $x = N$, and $f''(N) = -1/N$. So $f \approx N \ln N - N - (x-N)^2/2N$, and

$$N! \approx N^N e^{-N} \int_{-\infty}^{\infty} e^{-(x-N)^2/2N}\, \mathrm{d}x = \sqrt{2\pi N}\, N^N e^{-N}.$$

Extending the lower limit to $-\infty$ is harmless because the peak has width $\sqrt{N} \ll N$. Even for $N = 10$ the error is under 1%.

1.11 With $u = 1/\epsilon$, $\mathrm{d}/\,\mathrm{d}\epsilon = -u^2\,\mathrm{d}/\,\mathrm{d}u$, so by induction every derivative has the form $P_n(u)e^{-u}$ for a polynomial P_n. As $\epsilon \to 0^+$, $u \to \infty$ and $P_n(u)e^{-u} \to 0$, so every derivative is continuous at 0 with value 0 (and the difference quotient $f^{(n)}(\epsilon)/\epsilon = uP_n(u)e^{-u}$ also tends to 0). The Taylor series is identically zero, yet $f(\epsilon) > 0$ for every $\epsilon > 0$. A perturbative calculation computes exactly these Taylor coefficients, order by order, so it returns zero at every order. Effects like tunnelling need nonperturbative methods: WKB, instantons, or an exact solution.

1.12 The quantities m, σ (units $\mathrm{kg\,s^{-2}}$), ρ, g, r involve three dimensions, giving two groups: $\Pi_1 = mg/\sigma r$ and the *Bond number* $\Pi_2 = \rho g r^2/\sigma$. So $mg/\sigma r = F(\rho g r^2/\sigma)$. For a small tap, $\Pi_2 \ll 1$ and F tends to a constant: $m \propto \sigma r/g$, independent of ρ. Physically, the drop hangs from the rim by surface tension, a force up to $2\pi r\sigma$, and falls when its weight exceeds that: $m \approx 2\pi r\sigma/g$ (Tate's law). For $r \approx 3\,\mathrm{mm}$, $m \approx 2\pi(0.003)(0.072)/9.8 \approx 0.14\,\mathrm{g}$, a drop about 3 mm in radius. (Only part of the hanging drop actually falls, so real drops are a little smaller.)

2 Newtonian mechanics

2.1 $R = v_0^2 \sin 2\theta/g$ is largest when $\sin 2\theta = 1$, i.e. $\theta = 45°$. From a cliff the projectile has extra time in the air while it falls the height h, and that extra time is worth more spent moving horizontally. So the optimum tilts toward flatter throws: less than 45°, approaching 0° for a very high cliff.

2.2 Along the ramp, gravity pulls with $mg \sin\theta$; friction can supply at most $\mu_s N = \mu_s mg \cos\theta$. Sliding starts when $\tan\theta > \mu_s$. The mass cancels. To measure μ_s, tilt the ramp slowly until the block just starts to slide, and take $\mu_s = \tan\theta_c$.

2.3 For each mass along its direction of motion: $m_1 a = m_1 g - T$ and $m_2 a = T - m_2 g$. Adding,

$$a = \frac{m_1 - m_2}{m_1 + m_2}\, g, \qquad T = \frac{2 m_1 m_2}{m_1 + m_2}\, g.$$

As $m_2 \to 0$, $a \to g$ and $T \to 0$: free fall. For $m_1 = m_2$, $a = 0$ and $T = mg$: balance.

2.4 The drop has mass $\frac{4}{3}\pi r^3 \rho = 4.2 \times 10^{-6}\,\mathrm{kg}$ and area $\pi r^2 = 3.1 \times 10^{-6}\,\mathrm{m}^2$. With $C_D \approx 0.5$ for a sphere, $v_t = \sqrt{2mg/C_D \rho_{\mathrm{air}} A} \approx 6.6\,\mathrm{m/s}$ (measured: about 6.5 m/s). Without air it would hit the ground from a kilometre up at 140 m/s. Air drag caps it at a few metres per second, and a 4 mg drop at that speed carries less than 10^{-4} J.

2.5 $GMm/r^2 = m\omega^2 r$ with $\omega = 2\pi/T$ gives $T^2 = (4\pi^2/GM) r^3$. With $GM_\oplus = 3.99 \times 10^{14}\,\mathrm{m}^3/\mathrm{s}^2$ and $T = 86\,164\,\mathrm{s}$, $r = (GMT^2/4\pi^2)^{1/3} = 4.22 \times 10^7\,\mathrm{m} \approx 6.6\,R_\oplus$, an altitude of about 35 800 km.

2.6 Energy conservation gives $Mgh = \frac{1}{2} M v^2 (1 + \beta)$ with $\beta = I/MR^2$, so $v^2 = 2gh/(1 + \beta)$, and the acceleration $g \sin\theta/(1 + \beta)$ is constant. Mass and radius cancel; only the shape factor β matters. Sphere ($\frac{2}{5}$) beats disc ($\frac{1}{2}$) beats hoop (1). The frictionless block has $\beta = 0$, wastes nothing on spinning, and wins.

2.7 Take, for example, payload 1, total structure 10, fuel 89. Single stage: $\Delta v = u \ln(100/11) = 2.21u$. Two stages, each with 44.5 fuel and 5 structure: first burn $u \ln(100/55.5) = 0.59u$, then drop the empty stage and burn $u \ln(50.5/6) = 2.13u$; total $2.72u$. Staging wins because the rocket stops paying to accelerate empty tanks; the logarithm in the rocket equation punishes every kilogram of dead weight.

2.8 In the rotating frame the falling ball (downward speed gt) feels an eastward Coriolis acceleration $2\Omega g t$ at the equator. Integrating twice, $x = \frac{1}{3}\Omega g t^3$. With $t = \sqrt{2h/g} = 4.5\,\mathrm{s}$ and $\Omega = 7.3 \times 10^{-5}\,\mathrm{s}^{-1}$, $x \approx 2.2\,\mathrm{cm}$ east. (The naive inertial estimate, extra speed Ωh times t, gives 3.3 cm; the factor $\frac{2}{3}$ arises because the ball's eastward speed advantage over the ground shrinks as it falls.)

2.9 $v_{\mathrm{esc}} = \sqrt{2GM/R} = \sqrt{2(4.9 \times 10^{12})/1.74 \times 10^6} = 2.4\,\mathrm{km/s}$, so the survival threshold is about 0.4 km/s. Nitrogen at the Moon's ~ 390 K daytime temperature moves at $\sqrt{3k_{\mathrm{B}}T/m} \approx 0.6\,\mathrm{km/s}$: lost. For the Earth, $v_{\mathrm{esc}}/6 \approx 1.9\,\mathrm{km/s}$. In the hot upper atmosphere (~ 1000 K), nitrogen moves at 0.9 km/s and is kept, while hydrogen atoms move at about 5 km/s and escape.

2.10 Build from the top. Suppose the top k books have their combined centre of mass exactly over the edge of book $k+1$. Measure positions inward from that edge: the top k books contribute centre of mass 0, book $k+1$ contributes $L/2$, so the top $k+1$ books have centre of mass $(L/2)/(k+1) = L/2(k+1)$ inside the edge of book $k+1$. That is how far book $k+1$ can overhang book $k+2$. The top book overhangs the second by $L/2$, the second the third by $L/4$, and so on, for a total of $\frac{L}{2} H_n$ with $H_n = \sum_{k \le n} 1/k$. Since $H_n \sim \ln n$ diverges, the overhang is unbounded. The top book clears the table when $H_n > 2$, which first happens at $n = 4$ ($H_4 = 2.08$).

2.11 In the rotating frame, at angle θ from the bottom, the bead has potential energy $U = -mgR\cos\theta - \frac{1}{2} m\Omega^2 R^2 \sin^2\theta$ (gravity plus centrifugal). Then $U' = mR\sin\theta\,(g - \Omega^2 R\cos\theta)$, so equilibria are at $\theta = 0$, $\theta = \pi$, and $\cos\theta = g/\Omega^2 R$ if $\Omega^2 > g/R$. At the bottom, $U''(0) = mR(g - \Omega^2 R)$: stable for $\Omega^2 < g/R$, unstable above. At the new points, $U'' = m\Omega^2 R^2 \sin^2\theta > 0$: stable. The bead must choose one side, though the setup is symmetric — a pitchfork bifurcation.

2.12 Set $\omega = (\epsilon_1, \Omega, \epsilon_3)$ and keep first order: $I_1\dot{\epsilon}_1 = (I_2 - I_3)\Omega\epsilon_3$ and $I_3\dot{\epsilon}_3 = (I_1 - I_2)\Omega\epsilon_1$. Then $\ddot{\epsilon}_1 = \lambda^2\epsilon_1$ with

$$\lambda^2 = \frac{(I_2 - I_3)(I_1 - I_2)}{I_1 I_3}\,\Omega^2 > 0,$$

since both factors are negative. The wobble grows as $e^{\lambda t}$. Repeating the calculation for rotation about axis 3 gives $\lambda^2 \propto (I_2 - I_3)(I_3 - I_1) < 0$, and about axis 1 gives $\lambda^2 \propto (I_1 - I_2)(I_3 - I_1) < 0$: oscillations, so those rotations are stable.

3 The harmonic oscillator

3.1 At rest, $kd = mg$. For a displacement y below equilibrium, $m\ddot{y} = mg - k(d + y) = -ky$, so $\omega = \sqrt{k/m} = \sqrt{g/d}$. Gravity is a constant force: it shifts the equilibrium but does not change how the restoring force varies with displacement, which is all the frequency depends on.

3.2 $\omega = 1/\sqrt{LC}$. Charge Q plays position, inductance L plays mass, $1/C$ plays stiffness, current $I = \dot{Q}$ plays velocity, magnetic energy $\frac{1}{2}LI^2$ plays kinetic energy, and electric energy $Q^2/2C$ plays spring energy. A resistor plays damping.

3.3 $\omega = \sqrt{200/0.5} = 20\,\mathrm{rad/s}$, so $T = 2\pi/\omega = 0.31\,\mathrm{s}$. $E = \frac{1}{2}kA^2 = \frac{1}{2}(200)(0.04)^2 = 0.16\,\mathrm{J}$. $v_{\max} = \omega A = 0.8\,\mathrm{m/s}$.

3.4 Let s be the arc length travelled by the ball's centre, on a circle of radius $R - r \approx R$. Rolling means the ball spins at $\dot{s}/r$, so $T = \frac{1}{2}m\dot{s}^2 + \frac{1}{2}\cdot\frac{2}{5}mr^2(\dot{s}/r)^2 = \frac{7}{10}m\dot{s}^2$. The potential is $V \approx mgs^2/2R$. The equation of motion $\frac{7}{5}m\ddot{s} = -mgs/R$ gives $\omega^2 = 5g/7R$. The $5/7 = 1/(1 + \frac{2}{5})$ is the extra inertia of spinning: the same restoring force must accelerate an effective mass $\frac{7}{5}m$. A frictionless ball does not spin, so its effective mass is m and $\omega = \sqrt{g/R}$.

3.5 Energy conservation gives $\dot{\theta}^2 = (2g/L)(\cos\theta - \cos\theta_0)$, so $T = 4\int_0^{\theta_0} \mathrm{d}\theta/\dot{\theta}$. Substituting $\sin(\theta/2) = k\sin\phi$ with $k = \sin(\theta_0/2)$ turns this into

$$T = 4\sqrt{\frac{L}{g}}\int_0^{\pi/2} \frac{\mathrm{d}\phi}{\sqrt{1 - k^2\sin^2\phi}} \approx 4\sqrt{\frac{L}{g}}\cdot\frac{\pi}{2}\left(1 + \frac{k^2}{4}\right) \approx T_0\left(1 + \frac{\theta_0^2}{16}\right).$$

For $\theta_0 = 5^\circ = 0.087$ rad, the correction is 4.8×10^{-4}, so the clock runs slow by about 41 seconds a day compared with an ideal small-amplitude pendulum. (In practice clocks are regulated at their working amplitude; what hurts accuracy is the *variation* of amplitude.)

3.6 Inside a uniform Earth, the shell theorem gives a gravitational acceleration $-(g/R)\mathbf{r}$, proportional to the distance from the centre. Its component along any straight tunnel is $-(g/R)x$, where x is measured from the tunnel's midpoint. So $\ddot{x} = -(g/R)x$: simple harmonic motion with $\omega = \sqrt{g/R}$, whatever the tunnel. The period is $2\pi\sqrt{R/g} = 84$ minutes, so the one-way trip takes 42 minutes. A satellite skimming the surface has $v^2/R = g$ and the same 84-minute period; the tunnel motion is the projection of that orbit onto a line.

3.7 The amplitude is largest where $D(\omega) = (\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2$ is smallest. Setting $\mathrm{d}D/\,\mathrm{d}(\omega^2) = -2(\omega_0^2 - \omega^2) + 4\gamma^2 = 0$ gives $\omega_{\max}^2 = \omega_0^2 - 2\gamma^2$. There $D = 4\gamma^2\omega_0^2 - 4\gamma^4 \approx 4\gamma^2\omega_0^2$, so $A_{\max} \approx F_0/2m\gamma\omega_0$; dividing by $F_0/k = F_0/m\omega_0^2$ gives $\omega_0/2\gamma = Q$. The average absorbed power is $P = \langle F\dot{x}\rangle = \gamma F_0^2\omega^2/(mD)$, and $D/\omega^2 = (\omega_0^2/\omega - \omega)^2 + 4\gamma^2$ is minimized exactly at $\omega = \omega_0$.

3.8 With fixed spring and damper, $\gamma = b/2m \propto 1/m$ and $\omega_0 = \sqrt{k/m} \propto m^{-1/2}$, so $\gamma/\omega_0 \propto m^{-1/2}$. It starts at 1 (critical) and becomes $1/\sqrt{1.25} = 0.89$: underdamped. The new $Q = \omega_0/2\gamma = \sqrt{1.25}/2 \approx 0.56$ — just barely oscillatory, with one small overshoot.

3.9 $\langle E\rangle = \langle p^2\rangle/2m + \frac{1}{2}m\omega^2\langle x^2\rangle \geq (\Delta p)^2/2m + \frac{1}{2}m\omega^2(\Delta x)^2$. With $\Delta p \geq \hbar/2\Delta x$,

$$\langle E\rangle \geq \frac{\hbar^2}{8m(\Delta x)^2} + \tfrac{1}{2}m\omega^2(\Delta x)^2.$$

Minimizing over Δx gives $(\Delta x)^2 = \hbar/2m\omega$ and $E \geq \frac{1}{4}\hbar\omega + \frac{1}{4}\hbar\omega = \frac{1}{2}\hbar\omega$. The bound is achieved exactly by the true ground state, a minimum-uncertainty Gaussian.

3.10 Isochronism requires the restoring acceleration along the curve to be exactly proportional to the arc length: $g\sin\phi = \omega^2 s$, so $s = (g/\omega^2)\sin\phi$, where ϕ is the slope angle. Then $\mathrm{d}y = \sin\phi\ \mathrm{d}s = (g/2\omega^2)\sin 2\phi\ \mathrm{d}\phi$ and $\mathrm{d}x = \cos\phi\ \mathrm{d}s = (g/2\omega^2)(1+\cos 2\phi)\ \mathrm{d}\phi$. With $\psi = 2\phi$ and $a = g/4\omega^2$,

$$x = a(\psi + \sin\psi), \qquad y = a(1-\cos\psi),$$

a cycloid with its cusps pointing up, traced by a circle of radius a. The period, $2\pi/\omega = 4\pi\sqrt{a/g}$, is exactly independent of amplitude.

3.11 Before standing, at the bottom, $v^2 = 2g(L+\delta)(1-\cos\theta_1)$. The sudden shortening is caused by a radial force, so the angular momentum mrv about the pivot is conserved: $v' = v(L+\delta)/(L-\delta)$. The swing then rises to θ_2 with $v'^2 = 2g(L-\delta)(1-\cos\theta_2)$, so

$$1-\cos\theta_2 = (1-\cos\theta_1)\left(\frac{L+\delta}{L-\delta}\right)^3 \approx (1-\cos\theta_1)\left(1+\frac{6\delta}{L}\right).$$

Squatting at the turning point happens at zero speed, leaving the angle unchanged. So the energy (measured by $1-\cos\theta$ at fixed length) is multiplied by about $1+6\delta/L$ every half period: exponential growth. From rest there is no angular momentum to amplify: multiplying zero gives zero. Parametric pumping needs a seed, which is why children start by pushing off.

4 Analytical mechanics

4.1 Let m_1 descend a distance x, so m_2 rises by x. Then $L = \frac{1}{2}(m_1+m_2)\dot{x}^2 + (m_1-m_2)gx$, and the Euler–Lagrange equation is $(m_1+m_2)\ddot{x} = (m_1-m_2)g$, as Newton's laws gave. The tension never appears because the coordinate already respects the inextensible string.

4.2 In spherical coordinates the velocity has components $\dot{r}$, $r\dot{\theta}$, and $r\sin\theta\,\dot{\phi}$, giving the stated L. It does not depend on ϕ, so $p_\phi = mr^2\sin^2\theta\,\dot{\phi} = L_z$ is conserved. The potential depends only on r, so the Lagrangian is invariant under every rotation, and by Noether's theorem every component of $\mathbf{L}$ is conserved; choosing the polar axis along any direction makes that component the manifest one. (Equivalently, a central force exerts no torque.)

4.3 $H = p^2/2m + \frac{1}{2}m\omega^2x^2$ gives $\dot{x} = p/m$ and $\dot{p} = -m\omega^2x$. Since $H = E$ is conserved, the trajectory is the ellipse $x^2/a^2 + p^2/b^2 = 1$ with $a = \sqrt{2E/m\omega^2}$ and $b = \sqrt{2mE}$. At the top ($x = 0$, $p > 0$) we have $\dot{x} > 0$: moving right, so clockwise. The area is $\pi ab = 2\pi E/\omega$.

4.4 With θ measured from the bottom, the bead's velocity has components $R\dot\theta$ along the hoop and $R\Omega\sin\theta$ around the axis, so $L = \tfrac12 mR^2(\dot\theta^2 + \Omega^2\sin^2\theta) + mgR\cos\theta$. Then

$$H = p_\theta\dot\theta - L = \tfrac12 mR^2\dot\theta^2 - \tfrac12 mR^2\Omega^2\sin^2\theta - mgR\cos\theta,$$

whereas the energy is $E = \tfrac12 mR^2\dot\theta^2 + \tfrac12 mR^2\Omega^2\sin^2\theta - mgR\cos\theta$. They differ in the sign of the rotational term. H is conserved because L has no explicit time dependence. E is not, because the motor keeping the hoop spinning at fixed Ω does work on the bead through the normal force whenever the bead's distance from the axis changes.

4.5 With $x_1 = \ell\sin\theta_1$, $y_1 = -\ell\cos\theta_1$, $x_2 = x_1 + \ell\sin\theta_2$, $y_2 = y_1 - \ell\cos\theta_2$:

$$L = \tfrac12 m\ell^2[2\dot\theta_1^2 + \dot\theta_2^2 + 2\dot\theta_1\dot\theta_2\cos(\theta_1-\theta_2)] + mg\ell(2\cos\theta_1 + \cos\theta_2).$$

For small angles, $M = m\ell^2\begin{pmatrix}2&1\\1&1\end{pmatrix}$ and $K = mg\ell\begin{pmatrix}2&0\\0&1\end{pmatrix}$. With $w = \omega^2\ell/g$, $\det(K-\omega^2 M) = 0$ becomes $(2-2w)(1-w) - w^2 = w^2 - 4w + 2 = 0$, so $w = 2\pm\sqrt2$. The slow mode ($w = 2-\sqrt2$) has $\theta_2 = \sqrt2\,\theta_1$: both rods swing the same way, the lower one further. The fast mode ($w = 2+\sqrt2$) has $\theta_2 = -\sqrt2\,\theta_1$: they swing in opposition.

4.6 With $L_x = yp_z - zp_y$ and $L_y = zp_x - xp_z$, the only nonzero pieces are $\{yp_z, zp_x\} = -yp_x$ and $\{zp_y, xp_z\} = xp_y$, so $\{L_x, L_y\} = xp_y - yp_x = L_z$. By the same token $\{L_z, L_x\} = L_y$ and $\{L_z, L_y\} = -L_x$, so $\{L_z, L_x^2 + L_y^2 + L_z^2\} = 2L_xL_y - 2L_yL_x = 0$. Finally $\{x, L_z\} = \{x, xp_y - yp_x\} = -y$.

4.7 With polar angle θ from the downward vertical and azimuth ϕ, $L = \tfrac12 m\ell^2(\dot\theta^2 + \sin^2\theta\,\dot\phi^2) + mg\ell\cos\theta$. The conserved quantities are the energy and $p_\phi = m\ell^2\sin^2\theta\,\dot\phi$ (the vertical angular momentum). Eliminating $\dot\phi$ leaves $\tfrac12 m\ell^2\dot\theta^2 + U_{\text{eff}}(\theta) = E$ with $U_{\text{eff}} = p_\phi^2/(2m\ell^2\sin^2\theta) - mg\ell\cos\theta$. A steady cone needs $\ddot\theta = 0$: $\sin\theta\cos\theta\,\dot\phi^2 = (g/\ell)\sin\theta$, so $\dot\phi = \sqrt{g/(\ell\cos\theta_0)}$.

4.8 Put the start at height a above the interface and the end at depth b below it, a horizontal distance d apart, and let the ray cross at x. The time is $t = [n_1\sqrt{a^2+x^2} + n_2\sqrt{b^2+(d-x)^2}]/c$. Setting $\mathrm{d}t/\mathrm{d}x = 0$ gives $n_1 x/\sqrt{a^2+x^2} = n_2(d-x)/\sqrt{b^2+(d-x)^2}$, which is $n_1\sin\theta_1 = n_2\sin\theta_2$.

4.9 Starting from rest, $v = \sqrt{2gy}$, and $\mathrm{d}s = \sqrt{1+y'^2}\,\mathrm{d}x$, so $t = \int\sqrt{(1+y'^2)/2gy}\,\mathrm{d}x$. The integrand $F(y,y')$ has no explicit x, so $F - y'\,\partial F/\partial y'$ is constant (the analogue of energy conservation, with x as time). This gives $1/\sqrt{y(1+y'^2)} = \text{const}$, i.e. $y(1+y'^2) = 2a$. Substituting $y = a(1-\cos\psi)$ gives $\mathrm{d}x/\mathrm{d}\psi = a(1-\cos\psi)$, so $x = a(\psi - \sin\psi)$: a cycloid. It is the same curve as the tautochrone of Problem 3.10 — the cycloid is both the fastest path and the isochronous one.

4.10 In the pivot's frame, gravity is effectively $g + \ddot z_p = g - a\Omega^2\cos\Omega t$, so

$$\ell\ddot\theta = -(g - a\Omega^2\cos\Omega t)\sin\theta.$$

Write $\theta = \bar\theta + \xi$. The fast part obeys $\ell\ddot\xi \approx a\Omega^2\cos\Omega t\sin\bar\theta$, so $\xi = -(a/\ell)\sin\bar\theta\cos\Omega t$. Averaging the slow equation over a fast period, the only surviving extra term is $a\Omega^2\langle\cos\Omega t\,\xi\rangle\cos\bar\theta = -(a^2\Omega^2/2\ell)\sin\bar\theta\cos\bar\theta$. So

$$m\ell^2\ddot{\bar\theta} = -mg\ell\sin\bar\theta - \tfrac12 ma^2\Omega^2\sin\bar\theta\cos\bar\theta = -\frac{\mathrm{d}U_{\text{eff}}}{\mathrm{d}\bar\theta},$$

with the stated U_{eff}. At $\theta = \pi$, $U''_{\text{eff}} = -mg\ell + \frac{1}{2}ma^2\Omega^2$, which is positive (a stable minimum) when $a^2\Omega^2 > 2g\ell$.

4.11 A wall approaching at speed u adds $2u$ to the ball's speed at each bounce, and the ball hits it every $2\ell/v$. So $\dot{v} = uv/\ell$, while $\dot{\ell} = -u$. Dividing, $\mathrm{d}v/v = -\,\mathrm{d}\ell/\ell$: $v\ell$ is constant. (It is the action $\oint p\,\mathrm{d}x = 2mv\ell$, an adiabatic invariant.) The energy goes as $v^2 \propto 1/\ell^2$. For N balls the average force on a wall is $Nmv^2/\ell \propto \ell^{-3}$, so $P\ell^3$ is constant: $\gamma = 3$, matching $\gamma = 1 + 2/f$ for $f = 1$ degree of freedom.

5 Waves

5.1 The fundamental has $\lambda = 2L = 1.3\,\mathrm{m}$, so $v = \lambda f = 143\,\mathrm{m/s}$. Then $T = \mu v^2 = 0.006 \times 143^2 \approx 123\,\mathrm{N}$. Since $f \propto \sqrt{T}$, raising the pitch an octave (doubling f) needs four times the tension.

5.2 You hear 3 beats per second at a pitch of 441.5 Hz. Wax adds mass and lowers a fork's frequency. Lowering 443 Hz toward 440 Hz makes the beats slower, as observed, so the wax lowered it.

5.3 The fixed end is a node; the free end (no transverse force, so $\partial y/\partial x = 0$) is an antinode. The string must hold an odd number of quarter wavelengths: $L = (2n-1)\lambda/4$, so $\lambda_n = 4L/(2n-1)$ and $f_n = (2n-1)v/4L$. Only odd multiples of the fundamental appear.

5.4 Approaching: $f = 700 \times 343/(343-30) = 767\,\mathrm{Hz}$. Receding: $700 \times 343/(343+30) = 644\,\mathrm{Hz}$. The ratio 1.19 is $12\log_2(1.19) \approx 3$ semitones — about a minor third.

5.5 Write the incident, reflected, and transmitted waves as $Ae^{i(k_1x-\omega t)}$, $Be^{i(-k_1x-\omega t)}$, $Ce^{i(k_2x-\omega t)}$. Continuity at $x = 0$ gives $A + B = C$. The transverse force $T\,\partial y/\partial x$ must also be continuous; since $Tk = Z\omega$, this gives $Z_1(A - B) = Z_2C$. Solving, $B/A = (Z_1 - Z_2)/(Z_1 + Z_2)$ and $C/A = 2Z_1/(Z_1 + Z_2)$. With power $\frac{1}{2}Z\omega^2\,|\text{amplitude}|^2$,

$$\frac{Z_1B^2 + Z_2C^2}{Z_1A^2} = \frac{(Z_1 - Z_2)^2 + 4Z_1Z_2}{(Z_1 + Z_2)^2} = 1.$$

5.6 From $\omega = \sqrt{gk}$: $v_p = \omega/k = \sqrt{g/k}$ and $v_g = \mathrm{d}\omega/\,\mathrm{d}k = \frac{1}{2}\sqrt{g/k}$. In terms of the period, $v_g = g/2\omega = gT/4\pi$: 15.6 m/s for $T = 20\,\mathrm{s}$ and 6.2 m/s for $T = 8\,\mathrm{s}$. Over 2000 km these take 36 and 89 hours, so the short-period waves arrive about 53 hours, or two days, later.

5.7 $\int e^{-x^2/2\sigma^2}e^{-ikx}\,\mathrm{d}x = \sqrt{2\pi}\,\sigma e^{-\sigma^2k^2/2}$, and the factor $\cos k_0x$ shifts copies to $\pm k_0$: $\tilde{y}(k) = \sqrt{\pi/2}\,\sigma\,[e^{-\sigma^2(k-k_0)^2/2} + e^{-\sigma^2(k+k_0)^2/2}]$, Gaussians of width $1/\sigma$. Measuring widths as standard deviations of $|y|^2$ and $|\tilde{y}|^2$ gives $\Delta x = \sigma/\sqrt{2}$ and $\Delta k = 1/\sqrt{2}\sigma$, so $\Delta x\,\Delta k = \frac{1}{2}$. For any packet, the Cauchy–Schwarz inequality (the same argument that proves Heisenberg's principle) gives $\Delta x\,\Delta k \geq \frac{1}{2}$; the Gaussian is the unique minimizer.

5.8 Substituting $y_j = Ae^{i(kja-\omega t)}$ into $m\ddot{y}_j = K(y_{j+1} - 2y_j + y_{j-1})$ gives $-m\omega^2 = K(e^{ika} - 2 + e^{-ika}) = -4K\sin^2(ka/2)$, so $\omega = 2\sqrt{K/m}\,|\sin(ka/2)|$. Then $v_g = a\sqrt{K/m}\cos(ka/2)$ for $0 < k < \pi/a$, which vanishes at $k = \pi/a$. There, $y_j \propto (-1)^j$: neighbouring beads move exactly opposite, the pattern does not travel, and the chain vibrates at its maximum frequency.

5.9 $\sin\theta = 1/2$, so $\theta = 30°$. You hear the boom when the cone reaches you, when the jet is a horizontal distance $h/\tan 30° = 17\,\text{km}$ past you. At altitude the sound speed is about 300 m/s, so the jet flies at 600 m/s, and the delay is about 29 s.

5.10 The energy per unit area of surface goes as a^2, and in shallow water the energy travels at the group velocity $\sqrt{gh}$ (equal to the phase velocity). The period is unchanged as the wave moves, so conservation of energy flux per unit crest length gives $a^2\sqrt{h} = \text{constant}$, i.e. $a \propto h^{-1/4}$. From 4000 m to 10 m the amplitude grows by $400^{1/4} = 4.5$, from 0.5 m to about 2.2 m (and more as it runs up the shore). In the open ocean, $v = \sqrt{9.8 \times 4000} = 200\,\text{m/s}$, about 700 km/h.

5.11 Each point of the profile moves at its local speed $\sqrt{g(h+\eta)}$, so crests ($\eta > 0$) outrun troughs ($\eta < 0$). The front face, where the surface rises from a trough ahead to a crest behind, is compressed: the crest catches up with the trough in front of it. Following points as they move (along *characteristics*), the paths from crest and trough cross in a finite time, at which point the profile would be multivalued: before that, its front becomes vertical, and it breaks. Approaching a beach, h shrinks and η grows (Problem 5.10), so η/h rises and breaking is inevitable; waves typically break when their height is about 0.8 of the depth.

5.12 Let the boat move at speed u. A wave pattern that is steady relative to the boat, with crest normal at angle θ to the path, must have phase velocity $u\cos\theta$. Consider waves emitted when the boat was at P, a distance $D = ut$ behind its current position. Their energy has moved at the group velocity, half the phase velocity, a distance $\frac{1}{2}D\cos\theta$ in direction θ. With the boat at the origin and $P = (-D, 0)$, these points are $(-D + \frac{1}{4}D(1+\cos 2\theta),\ \frac{1}{4}D\sin 2\theta)$: a circle of radius $D/4$ centred at $(-\frac{3}{4}D, 0)$. All the energy from all earlier times therefore lies inside a cone from the boat tangent to these circles, with half-angle $\arcsin\frac{D/4}{3D/4} = \arcsin\frac{1}{3} = 19.47°$. Both u and D cancel.

6 Electromagnetism

6.1 $F_e/F_g = e^2/(4\pi\epsilon_0 G m_p^2) = (8.99\times10^9)(1.60\times10^{-19})^2/[(6.67\times10^{-11})(1.67\times10^{-27})^2] \approx 1.2\times10^{36}$. Both forces fall as $1/r^2$, so the ratio is independent of distance. Gravity wins in astronomy because large bodies are neutral to extraordinary precision: positive and negative charges cancel, and any imbalance quickly attracts opposite charge, while every mass adds to gravity.

6.2 For $r < R$, the enclosed charge is Qr^3/R^3, so $E = Qr/(4\pi\epsilon_0 R^3)$, rising linearly. For $r > R$, $E = Q/(4\pi\epsilon_0 r^2)$. The field peaks at the surface. Gravity inside a uniform Earth has exactly the same shape, $g(r) = gr/R$, which is what made the tunnel motion simple harmonic.

6.3 $C = \epsilon_0 A/d = 8.85\,\text{nF}$, $Q = CV = 8.85\,\mu\text{C}$, $U = \frac{1}{2}CV^2 = 4.4\,\text{mJ}$. The field is $E = V/d = 10^6\,\text{V/m}$ (a third of air's breakdown field), with energy density $\frac{1}{2}\epsilon_0 E^2 = 4.4\,\text{J/m}^3$; times the gap volume $10^{-3}\,\text{m}^3$ this is again 4.4 mJ.

6.4 $V = \sqrt{2U/C} = \sqrt{400/10^{-4}} = 2000\,\text{V}$. $\tau = RC = 50\times10^{-4} = 5\,\text{ms}$.

6.5 Electron: $\omega = eB/m = 1.76\times10^{11}\,\text{rad/s}$ (28 GHz); $v = \sqrt{2K/m} = 1.9\times10^7\,\text{m/s}$; $r = v/\omega = 0.11\,\text{mm}$. Proton: $\omega = 9.6\times10^7\,\text{rad/s}$ (15 MHz); $v = 4.4\times10^5\,\text{m/s}$; $r = 4.6\,\text{mm}$. Same energy, but the heavier particle circles slower in a bigger orbit.

6.6 Take the curl of the Ampère–Maxwell law in vacuum and use Faraday's law: $\nabla\times(\nabla\times\mathbf{B}) = \mu_0\epsilon_0\,\partial_t(\nabla\times\mathbf{E}) = -\mu_0\epsilon_0\,\partial_t^2\mathbf{B}$. Since $\nabla\cdot\mathbf{B}=0$, the left side is $-\nabla^2\mathbf{B}$, giving the wave equation with speed $1/\sqrt{\mu_0\epsilon_0}$. For plane waves $\propto e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$, the divergence equations give $\mathbf{k}\cdot\mathbf{E}_0 = \mathbf{k}\cdot\mathbf{B}_0 = 0$ (transverse), and Faraday's law gives $\mathbf{k}\times\mathbf{E}_0 = \omega\mathbf{B}_0$, so $\mathbf{B}_0 = \hat{\mathbf{k}}\times\mathbf{E}_0/c$: perpendicular to both, with $B = E/c$.

6.7 As the magnet falls, the flux through rings of pipe below it grows and through rings above it shrinks. The induced EMFs drive circulating eddy currents, which by Lenz's law oppose the change: the ring below repels the magnet and the ring above attracts it — both brake it. The EMF is proportional to the rate of change of flux, hence to v; the current is proportional to the conductivity σ and the wall thickness w; so the braking force is $\propto \sigma w v$. At terminal velocity $mg = k\sigma w v$, so $v_t \propto 1/\sigma w$. The magnet's lost potential energy heats the pipe.

6.8 A perfect reflector feels force $2IA/c = 2(1361)(10^4)/(3\times10^8) = 0.091\,\mathrm{N}$. The Sun pulls with $GM_\odot m/r^2 = 0.59\,\mathrm{N}$. The ratio is 0.15. Both forces fall as $1/r^2$, so the ratio is the same everywhere in the solar system; to beat gravity, the sail would need to weigh less than about $1.5\,\mathrm{g/m^2}$.

6.9 The pair $+q$ at height d and $-q$ at depth d gives zero potential everywhere on the plane, by symmetry, and has the right source (q) in the region above. By the uniqueness theorem, a solution of Poisson's equation with the right sources and boundary values is the only one, so it *is* the field above the plane. The force on q is the attraction to its image: $F = q^2/4\pi\epsilon_0(2d)^2 = q^2/16\pi\epsilon_0 d^2$. The trick works because the fictitious charge lies outside the region where we need the answer.

6.10 For the circular orbit, $a = e^2/(4\pi\epsilon_0 m_e a_0^2) \approx 9\times10^{22}\,\mathrm{m/s^2}$, so $P = e^2a^2/6\pi\epsilon_0c^3 \approx 4.7\times10^{-8}\,\mathrm{W}$. The binding energy is $13.6\,\mathrm{eV} = 2.2\times10^{-18}\,\mathrm{J}$, so a crude estimate is $|E|/P \approx 5\times10^{-11}\,\mathrm{s}$. Doing it properly, $E = -e^2/8\pi\epsilon_0 r$ and $P \propto r^{-4}$, so $\mathrm{d}E/\mathrm{d}t = -P$ integrates to $t = a_0^3/(4r_e^2c) \approx 1.6\times10^{-11}\,\mathrm{s}$, where $r_e = e^2/4\pi\epsilon_0 m_e c^2$ is the classical electron radius. Classical atoms would not last a billionth of a second.

6.11 In empty space V satisfies Laplace's equation, which implies the mean-value property: V at any point equals its average over any small sphere around it. A strict local minimum (or maximum) would be below (or above) all its neighbours, hence below (above) their average — impossible. Equivalently, $\nabla^2V = 0$ forbids all three second derivatives from being positive. So qV has at best saddle points: stable in some directions, unstable in others. Ion traps escape by using oscillating fields (a time-averaged effective potential, as for the Kapitza pendulum) or by adding a magnetic field.

6.12 *Lab frame.* $B = \mu_0 I/2\pi r = \mu_0\lambda v/\pi r$, and $F = qvB = \mu_0 q\lambda v^2/\pi r$, attractive. *Test charge's frame.* The positive charges move with q, so they are at rest: their density is their rest density, λ/γ_v. The negative charges move at $u = 2v/(1+v^2/c^2)$, so their density is $-(\lambda/\gamma_v)\gamma_u$. The net density is $\lambda' = (\lambda/\gamma_v)(1-\gamma_u) \approx -2\lambda v^2/c^2$ to leading order. The wire is negatively charged, and attracts q with

$$F' = q\frac{|\lambda'|}{2\pi\epsilon_0 r} = \frac{q\lambda v^2}{\pi\epsilon_0 c^2 r} = \frac{\mu_0 q\lambda v^2}{\pi r},$$

using $\epsilon_0\mu_0 = 1/c^2$. A transverse force differs between the frames only by a factor $\gamma \approx 1$, so the two agree to leading order. The "magnetic" force is an electric force seen from another frame.

7 Thermodynamics

7.1 $W = nRT\ln 2 = (8.31)(300)(0.693) = 1.73\,\text{kJ}$. Since $\Delta U = 0$, it absorbs $Q = 1.73\,\text{kJ}$ of heat. $\Delta S = Q/T = nR\ln 2 = 5.8\,\text{J/K}$.

7.2 $\eta_{\max} = 1 - 300/800 = 62.5\%$. At 40%, producing 1 GW takes 2.5 GW of heat, of which 1.5 GW is dumped. A river carrying 10^5 kg/s warms by $\Delta T = 1.5\times10^9/(10^5\times4186) \approx 3.6\,\text{K}$.

7.3 $\text{COP} = T_h/(T_h - T_c) = 293/20 \approx 15$. As the outside gets colder the denominator grows, so each joule of work moves less heat; real units also lose efficiency and must defrost their outdoor coils.

7.4 $T_2 = T_1(V_1/V_2)^{\gamma-1} = 300\times20^{0.4} \approx 990\,\text{K}$, and $P_2 = P_1\times20^{1.4} \approx 66\,\text{atm}$. Diesel fuel autoignites at about 500 K; sprayed into air at nearly 1000 K, it ignites on its own.

7.5 By symmetry $T_f = 50^\circ\text{C} = 323.15\,\text{K}$. Then

$$\Delta S = mc\left[\ln\frac{323.15}{353.15} + \ln\frac{323.15}{293.15}\right] = 4186\,(-0.0888 + 0.0974) \approx +36\,\text{J/K}.$$

The cold water gains more entropy than the hot water loses, because the same heat counts for more at lower temperature.

7.6 With $\Delta V \approx V_{\text{gas}} = RT/P$ per mole, $\mathrm{d}P/\,\mathrm{d}T = LP/RT^2$, so $\mathrm{d}\ln P/\,\mathrm{d}T = L/RT^2$ and $P \propto e^{-L/RT}$. Then $\ln(0.33) = -(L/R)(1/T - 1/373)$ gives $1/T = 0.002681 + 0.000227$, so $T \approx 344\,\text{K} \approx 71^\circ\text{C}$.

7.7 A reversible engine creates no entropy, so $C\ln(T_f/T_1) + C\ln(T_f/T_2) = 0$, giving $T_f = \sqrt{T_1T_2}$. The work is the heat lost by the hot body minus the heat gained by the cold one: $C(T_1 - T_f) - C(T_f - T_2) = C(\sqrt{T_1} - \sqrt{T_2})^2$. In direct contact, $T_f = (T_1 + T_2)/2$ (the arithmetic mean, higher than the geometric mean), no work is done, and entropy increases by $C\ln[(T_1+T_2)^2/4T_1T_2] > 0$.

7.8 Connect the same initial and final states by a reversible isothermal expansion, which absorbs heat $Nk_{\text{B}}T\ln 2$: so $\Delta S = Nk_{\text{B}}\ln 2$. The free expansion is irreversible because its reverse — the gas spontaneously gathering into one half — would lower the entropy; it has probability 2^{-N}, never seen for macroscopic N.

7.9 The tension of rubber is mostly entropic: stretching straightens the chains, reducing their number of configurations, so $f \approx -T(\partial S/\partial x)_T$, which grows with T at fixed length. *Quick stretch*: nearly adiabatic, so total entropy is fixed; the configurational entropy drops, so the vibrational (thermal) entropy must rise — the band warms. *Heating under load*: the tension at a given length increases in proportion to T, so the band contracts until it balances the weight again, lifting it. A steel wire, whose elasticity is energetic, does the opposite.

7.10 Each binary decision the demon makes can lower the gas's entropy by at most $k_{\text{B}}\ln 2$ (one bit of information about which side a molecule is on). Its memory is finite, so eventually it must erase each bit, which adds at least $k_{\text{B}}\ln 2$ to the environment. The total entropy change over a full cycle is ≥ 0. (Bennett showed in 1982 that the measurement itself can be made reversible; it is the erasure that cannot.) Erasing a gigabyte, 8×10^9 bits, at 300 K costs at least $8\times10^9\times k_{\text{B}}T\ln 2 \approx 2\times10^{-11}$ J. Real transistors dissipate about 10^{-15} J per bit operation, roughly a million times the Landauer limit.

7.11 From $\mathrm{d}U = T\,\mathrm{d}S - P\,\mathrm{d}V$, $(\partial U/\partial V)_T = T(\partial S/\partial V)_T - P = T(\partial P/\partial T)_V - P$, using the Maxwell relation. For radiation, $U = u(T)V$ gives $(\partial U/\partial V)_T = u$, and $P = u/3$ gives $(\partial P/\partial T)_V = u'/3$. So $u = Tu'/3 - u/3$, i.e. $u'/u = 4/T$, and $u = aT^4$.

8 Optics

8.1 It appears $2.0/1.33 = 1.5$ m deep. Light from the whole sky above the water is squeezed into a cone of half-angle $\theta_c = \arcsin(1/1.33) = 49°$ around the vertical ("Snell's window"). Outside that cone the fish sees the surface as a mirror, totally reflecting the underwater scene.

8.2 Diamond: $\arcsin(1/2.42) = 24°$. Glass: $\arcsin(1/1.5) = 42°$. In the fibre, $\sin\theta_c = 1.46/1.48$ gives $\theta_c = 80.6°$ from the normal to the core wall, so light is trapped if it travels within $9.4°$ of the axis. (Equivalently, the fibre accepts light from air within $\arcsin\sqrt{1.48^2 - 1.46^2} = 14°$ of its axis.)

8.3 $1/s' = 1/10 - 1/30$, so $s' = 15$ cm and $m = -0.5$: a real, inverted image at half size. At 5 cm: $1/s' = 1/10 - 1/5$, so $s' = -10$ cm and $m = +2$: a virtual, upright, magnified image on the object's side — a magnifying glass.

8.4 $\Delta y = \lambda L/d = (633 \times 10^{-9})(2.0)/(10^{-4}) = 1.3$ cm. In water the wavelength shrinks to $\lambda/n = 476$ nm (the frequency is unchanged), so the fringes close up to 0.95 cm.

8.5 With $\theta = 1.22\lambda/D$ at 550 nm: eye (3 mm) 2.2×10^{-4}, resolving 90 km on the Moon; 20 cm telescope 3.4×10^{-6}, about 1.3 km; Hubble 2.8×10^{-7}, about 100 m. A 2.4 m mirror at 250 km resolves 7 cm on the ground: enough to see a car, not to read characters whose strokes are about a centimetre wide. The atmosphere's turbulence blurs things further, though looking *down* through it is less damaging than looking up, because the turbulence lies close to the target rather than close to the telescope.

8.6 Reflection off the top surface (air to film) flips the phase; reflection off the bottom (film to air) does not. Constructive interference needs $2nt = (m + \frac{1}{2})\lambda$, so the thinnest film has $t = \lambda/4n = 550/5.32 \approx 100$ nm. As $t \to 0$, only the half-wavelength flip remains, the two reflections cancel at every wavelength, and the film looks black.

8.7 $\theta_B = \arctan 1.33 = 53°$ from the vertical. Glare is then completely horizontally polarized, and is cut most effectively, when the Sun is $90° - 53° = 37°$ above the horizon.

8.8 Resolving power is $\lambda/\Delta\lambda = mN$. With $\lambda/\Delta\lambda = 589.3/0.6 \approx 980$ and $m = 1$, about 1000 illuminated slits are needed.

8.9 For source A and observer B above a mirror, the path via a point x on the mirror has length $\sqrt{a^2 + x^2} + \sqrt{b^2 + (d - x)^2}$; minimizing gives $\sin\theta_i = \sin\theta_r$. (Equivalently, reflect B in the mirror; the shortest path to the image is a straight line.) Over a hot road the air is hotter, less dense, and has a slightly lower refractive index, so light travels slightly faster near the road. The least-time path from the sky to your eye dips down toward the road and curves back up, like the lifeguard running along the faster beach. Your eye traces the arriving light back along a straight line and sees the sky on the road: a shimmering "puddle."

8.10 Entering, the ray turns by $i - r$; reflecting off the back, by $180° - 2r$; leaving, by $i - r$. Total $D = 180° + 2i - 4r$. Stationary D needs $\mathrm{d}r/\mathrm{d}i = \frac{1}{2}$; differentiating Snell's law, $\mathrm{d}r/\mathrm{d}i =$

$\cos i/(n\cos r)$, so $n\cos r = 2\cos i$. Combined with $n\sin r = \sin i$, this gives $\cos^2 i = (n^2-1)/3$. For $n = 4/3$: $i = 59.4°$, $r = 40.2°$, $D = 138°$, so the bow lies $180° - 138° = 42°$ from the antisolar point. Since $\mathrm{d}D/\mathrm{d}i = 0$ there, a wide band of entry angles all exit at nearly the same angle, so light piles up: a caustic.

8.11 Every point on the disc's circular rim is the same distance from the centre of the shadow, so all the secondary wavelets from the rim arrive there in phase. More carefully, divide the unobstructed wavefront outside the disc into Fresnel zones; their contributions alternate in sign with slowly decreasing size, and the sum is about half the first unobstructed zone's contribution — nearly the same intensity as if no disc were there. Particles would leave a dark shadow. Only waves can make the spot.

8.12 Optics is reversible: any path that light takes from the Sun to the target, light can take from the target back to the Sun. If a passive system could heat the target above the Sun's temperature, the hotter target would then send net heat back to the colder Sun along those same paths — or, run the other way, heat would flow from cold to hot with no work, violating the second law. The optical statement is conservation of *radiance* (power per area per solid angle): a lens can enlarge the solid angle the Sun fills as seen from the target, but cannot make its surface brighter. The best possible system surrounds the target with the Sun's surface on all sides, which delivers exactly the flux the target would get at the solar surface, and equilibrium is reached at $T_\odot$.

9 Statistical mechanics

9.1 Mean 50, standard deviation $\sqrt{Npq} = 5$. $P(50) \approx 1/(\sqrt{2\pi}\cdot 5) = 0.080$ (exact: 0.0796). For 60 or more, use $z = (59.5-50)/5 = 1.9$, giving 2.9% (exact: 2.8%).

9.2 A slab of thickness $\mathrm{d}h$ supports its own weight: $\mathrm{d}P = -nmg\,\mathrm{d}h$. With $P = nk_\mathrm{B}T$ at constant T, $\mathrm{d}n/\mathrm{d}h = -n\,mg/k_\mathrm{B}T$, so $n = n_0 e^{-mgh/k_\mathrm{B}T}$. For N_2 at 250 K, $k_\mathrm{B}T/mg = 7.6$ km. Helium is seven times lighter, so its scale height is seven times larger and it thins out more slowly; above about 100 km, where turbulence no longer mixes the gases, the atmosphere is increasingly helium-rich.

9.3 The ratio is $e^{-\varepsilon/k_\mathrm{B}T}$: $e^{-3.87} = 0.021$ at 300 K, and $e^{-0.39} = 0.68$ at 3000 K. Ten percent occupation means a ratio of 1/9, so $\varepsilon/k_\mathrm{B}T = \ln 9$ and $T = 0.1\,\mathrm{eV}/(2.20\,k_\mathrm{B}) = 530$ K.

9.4 Air is diatomic: three translational and two rotational quadratic terms give $\frac{5}{2}R$. Vibrations need $k_\mathrm{B}T \sim \hbar\omega \approx 0.3$ eV (3000 K) and are frozen. At high temperatures vibrations switch on, raising the heat capacity toward $\frac{7}{2}R$ (until the molecules dissociate). At very low temperatures rotations freeze too, leaving $\frac{3}{2}R$ — seen clearly for hydrogen below about 100 K, whose light molecules have widely spaced rotational levels.

9.5 Maximizing $v^2 e^{-mv^2/2k_\mathrm{B}T}$ gives $v_p = \sqrt{2k_\mathrm{B}T/m}$; Gaussian integrals give $\langle v\rangle = \sqrt{8k_\mathrm{B}T/\pi m}$ and $v_\mathrm{rms} = \sqrt{3k_\mathrm{B}T/m}$. For N_2 at 300 K, $\sqrt{k_\mathrm{B}T/m} = 298$ m/s, so the three speeds are 422, 476, and 517 m/s, in the ratio $\sqrt{2} : \sqrt{8/\pi} : \sqrt{3}$.

9.6 With $\lambda_\mathrm{max} = 2.9\,\mathrm{mm\,K}/T$: Sun 500 nm; body (310 K) 9.4 µm; red-hot element ($\sim$ 1000 K) 2.9 µm, mostly infrared, with only its visible tail seen; CMB 1.06 mm. Hotter stars peak in the blue or ultraviolet and look blue-white; cooler ones peak in the red or infrared and look red. A

blackbody peaking in the green still emits strongly across the whole visible band, and the eye sees that mixture as white.

9.7 $L = 4\pi R^2\sigma T^4 = 4\pi(7.0\times10^8)^2(5.67\times10^{-8})(5800)^4 \approx 3.9\times10^{26}\,\mathrm{W}$. At the Earth, $I = L/4\pi r^2 \approx 1400\,\mathrm{W/m^2}$. A black sphere absorbs $\pi R_E^2 I$ and emits $4\pi R_E^2\sigma T^4$, so $T = (I/4\sigma)^{1/4} \approx 280\,\mathrm{K}$.

9.8 $n = 8.9\times10^3/(63.5\times1.66\times10^{-27}) = 8.4\times10^{28}\,\mathrm{m^{-3}}$, so $E_F = (\hbar^2/2m)(3\pi^2 n)^{2/3} = 7.0\,\mathrm{eV}$ and $T_F = 81\,000\,\mathrm{K}$. Only electrons within about $k_\mathrm{B}T$ of the Fermi surface can be excited — the rest have no empty states nearby — so only a fraction $\sim k_\mathrm{B}T/E_F \approx 0.4\%$ take up heat. (The exact result is $C = \frac{\pi^2}{2}Nk_\mathrm{B}(k_\mathrm{B}T/E_F)$.)

9.9 Plot $y = m$ and $y = \tanh(T_c m/T)$; for $T < T_c$ the tanh starts steeper than 1, so it crosses the line at $m = 0$ and at $\pm m_0$. Expanding $\tanh x \approx x - x^3/3$ with $x = T_c m/T$ gives $m^2 = 3(T_c/T - 1)(T/T_c)^3 \approx 3(T_c - T)/T$. At $T = T_c$ in a small field h, $m = \tanh(m + h/k_\mathrm{B}T_c)$ expands to $h/k_\mathrm{B}T_c \approx m^3/3$, so $m \approx (3h/k_\mathrm{B}T_c)^{1/3}$: the response is nonlinear, $m \propto h^{1/3}$.

9.10 $D = k_\mathrm{B}T/6\pi\eta r = 4.1\times10^{-21}/(6\pi\times10^{-3}\times5\times10^{-7}) = 4.4\times10^{-13}\,\mathrm{m^2/s}$. In one direction, $\sqrt{2Dt}$ is 0.9 μm after a second and 56 μm after an hour. To go 1 mm takes $x^2/2D \approx 1.1\times10^6\,\mathrm{s}$, about two weeks.

9.11 For rubidium, $\lambda_T = (2.6/10^{20})^{1/3} = 3.0\times10^{-7}\,\mathrm{m}$, so $T = h^2/(2\pi m k_\mathrm{B}\lambda_T^2) \approx 0.4\,\mu\mathrm{K}$. For helium, $n = 145/(6.6\times10^{-27}) = 2.2\times10^{28}\,\mathrm{m^{-3}}$, $\lambda_T = 4.9\times10^{-10}\,\mathrm{m}$, and $T \approx 3.1\,\mathrm{K}$, close to the observed 2.17 K (the difference is due to interactions in the liquid). The rubidium gas must stay dilute — a hundred million times less dense than liquid helium — or it would form molecules and solidify. Its atoms are therefore far apart, the thermal wavelength must be correspondingly longer, and since $T \propto n^{2/3}/m$, the gas must be far colder.

9.12 With $n = U/\varepsilon$ excited spins and Stirling, $S = k_\mathrm{B}[N\ln N - n\ln n - (N-n)\ln(N-n)]$, so

$$\frac{1}{T} = \frac{\partial S}{\partial U} = \frac{k_\mathrm{B}}{\varepsilon}\ln\frac{N-n}{n}.$$

For $n < N/2$, $T > 0$; at $n = N/2$, $T = \infty$; for $n > N/2$, $T < 0$. Heat flows in whichever direction increases total entropy. A negative-temperature system *gains* entropy by losing energy, and any positive-temperature system gains entropy by receiving it, so energy always flows from the negative-temperature system. The ordering of hotness is $+0 < \cdots < +\infty = -\infty < \cdots < -0$. This requires an energy spectrum bounded above, which is why it occurs for spins and lasers but not for gases.

9.13 Doubling N and V gives $S' = 2Nk_\mathrm{B}\ln(2V\,T^{3/2}) + \cdots = 2S + 2Nk_\mathrm{B}\ln 2$: not twice S. Likewise, removing a partition between two identical boxes would add $2Nk_\mathrm{B}\ln 2$, though nothing physical has happened. Dividing Ω by $N!$ subtracts $k_\mathrm{B}\ln N! \approx k_\mathrm{B}(N\ln N - N)$, which turns $\ln V$ into $\ln(V/N)$:

$$S = Nk_\mathrm{B}\left[\ln\frac{V}{N} + \tfrac{3}{2}\ln T + \text{const}\right].$$

Now S depends only on the intensive ratio V/N, so it is extensive, and mixing identical gases changes nothing. (This is the Sackur–Tetrode entropy, whose constant contains h — another sign that quantum mechanics is at work.)

10 Special relativity

10.1 The trip takes $15\,\text{km}/(0.998c) = 50\,\mu\text{s}$. Without time dilation the survivors are $e^{-50/2.2} \approx 10^{-10}$. With $\gamma = 15.8$ the lab-frame lifetime is 35 μs and $e^{-50/35} \approx 24\%$ survive. In the muon's frame, the atmosphere is contracted to $15/15.8 = 0.95\,\text{km}$ and rushes past in 3.2 μs — the same 1.4 lifetimes, the same 24%.

10.2 $mc^2 = 10^{-3} \times 9 \times 10^{16} = 9 \times 10^{13}\,\text{J}$, about 21 kilotons of TNT. Fifteen kilotons, $6.3 \times 10^{13}\,\text{J}$, is about 0.7 g of mass.

10.3 $u = (0.8 + 0.8)/(1 + 0.64)\,c = 0.976c$. Galileo would have said $1.6c$.

10.4 $v^2/2c^2 = (3.9 \times 10^3)^2/(1.8 \times 10^{17}) = 8.5 \times 10^{-11}$; times 86 400 s, about 7 μs per day. Light travels 2 km in that time, so uncorrected position errors would grow by about 2 km per day from this effect alone.

10.5 In the pion's rest frame the muon and neutrino have equal and opposite momenta. Energy conservation gives $E_\mu = (m_\pi^2 + m_\mu^2)c^2/2m_\pi = (139.6^2 + 105.7^2)/279.2\,\text{MeV} = 109.8\,\text{MeV}$, so its kinetic energy is 4.1 MeV.

10.6 At threshold the four final particles are at rest in the centre-of-mass frame, so $s = (4m_pc^2)^2$. For a beam of energy E on a proton at rest, $s = 2m_p^2c^4 + 2Em_pc^2$. Setting these equal, $E = 7m_pc^2$, and the kinetic energy is $6m_pc^2 = 5.6\,\text{GeV}$. Most of the energy is not available for making mass: momentum conservation forces the products to move forward, carrying kinetic energy.

10.7 $c^2t'^2 - x'^2 = \gamma^2[(ct - \beta x)^2 - (x - \beta ct)^2] = \gamma^2(1 - \beta^2)(c^2t^2 - x^2) = c^2t^2 - x^2$. Since $\gamma^2 - (\gamma\beta)^2 = 1$, we can write $\gamma = \cosh\phi$ and $\gamma\beta = \sinh\phi$, with $\tanh\phi = \beta$. Two boosts multiply as hyperbolic rotations, and the hyperbolic addition formulas give $\phi = \phi_1 + \phi_2$ — which, written in terms of velocities, is exactly the velocity-addition formula.

10.8 Squaring $p'_e = p_\gamma + p_e - p'_\gamma$ and using $p_\gamma^2 = p'^2_\gamma = 0$, $p_e^2 = p'^2_e = m^2c^2$:

$$0 = p_\gamma \cdot p_e - p_\gamma \cdot p'_\gamma - p_e \cdot p'_\gamma = Em - \frac{EE'}{c^2}(1 - \cos\theta) - E'm.$$

Dividing by mEE': $1/E' - 1/E = (1 - \cos\theta)/mc^2$, which with $E = hc/\lambda$ is $\Delta\lambda = (h/m_ec)(1 - \cos\theta)$.

10.9 The Earth twin waits $8/0.8 = 10$ years. The traveller's clock runs slow by $\gamma = 5/3$, recording 6 years (in its own frame each leg is $4/\gamma = 2.4$ light-years long, crossed in 3 years). The asymmetry: the Earth twin stays in one inertial frame, while the traveller changes frames at the turnaround. The turnaround is the event $(t, x) = (5\,\text{yr}, 4\,\text{ly})$. The outgoing frame's line of simultaneity through it, $t - 0.8x = \text{const}$, meets the Earth's worldline at $t = 1.8$ years; the returning frame's line, $t + 0.8x = \text{const}$, meets it at $t = 8.2$ years. So in the traveller's accounting, 6.4 years of Earth time "happen" during the turnaround, and $1.8 + 6.4 + 1.8 = 10$. On the spacetime diagram, the Earth's straight worldline has the greatest proper time; the bent one has less.

10.10 "Both doors shut at the same time" is a statement about simultaneity, which is frame-dependent. In the pole's frame, the far door shuts and reopens *before* the front of the pole reaches it, and the near door shuts only *after* the back has passed it; at no moment is the pole inside with both doors shut. The two door events are spacelike separated, so their order depends on the frame, and both descriptions are correct. If the far door stays shut, the front of the pole stops; the back cannot learn of this faster than light, so it keeps coming, and the pole is crushed. There are no rigid bodies in relativity. Both frames agree on the wreckage.

10.11 Here $c^2/g = 0.97$ ly and $c/g = 0.97$ yr. After 10 years, $g\tau/c = 10.3$ and $x \approx 0.97\cosh(10.3) \approx 1.4\times10^4$ ly. After 20 years, $x \approx 4\times10^8$ ly — far beyond the Andromeda galaxy. To cover 13 000 ly to the midpoint: $\cosh(g\tau/c) = 1 + 13\,000/0.97$, so $g\tau/c = 10.2$ and $\tau \approx 10$ years; with deceleration, about 20 years of crew time (and 26 000 years on Earth). The catch is fuel: even a perfect photon rocket, with exhaust at c, needs a mass ratio $e^{\Delta\phi}$ for a total rapidity change $\Delta\phi$, here $e^{20} \approx 10^9$. And at $\gamma \approx 10^4$, interstellar hydrogen atoms hit the ship like cosmic rays.

11 Fluid dynamics

11.1 $P = \rho g h = 1025 \times 9.8 \times 11\,000 = 1.1 \times 10^8$ Pa, about 1100 atm. The fractional compression is $P/K = 1.1 \times 10^8/2.2 \times 10^9 = 5\%$, so the water there is 5% denser.

11.2 Floating means weight equals buoyancy: $\rho_{\text{ice}} V = \rho_{\text{sea}} V_{\text{sub}}$, so $V_{\text{sub}}/V = 917/1025 = 89\%$. An ice cube floating in fresh water displaces exactly its own mass of water, and melts into exactly that volume: the level does not change. An iceberg (fresh ice) in seawater displaces seawater, but melts into fresh water, which is 2.5% less dense and so takes up 2.5% more volume than the seawater it displaced: a very slight rise. The serious sea-level rise comes from ice on land.

11.3 With $\nu_{\text{water}} = 10^{-6}$ and $\nu_{\text{air}} = 1.5 \times 10^{-5}$ m^2/s: bacterium $(2\times10^{-6})(3\times10^{-5})/10^{-6} = 6\times10^{-5}$; swimmer (2 m, 1 m/s) 2×10^6; car (4 m, 30 m/s) 8×10^6; aorta $(0.025)(0.3)/(3\times10^{-6}) = 2500$. The swimmer's and car's wakes are fully turbulent. The aorta sits near the transition (about 2000 for pipes) and becomes briefly turbulent at the peak of each heartbeat.

11.4 The level falls as $A\,\mathrm{d}h/\mathrm{d}t = -a\sqrt{2gh}$, so $\mathrm{d}h/\sqrt{h} = -(a/A)\sqrt{2g}\,\mathrm{d}t$. Integrating from h_0 to 0, $2\sqrt{h_0} = (a/A)\sqrt{2g}\,T$, so $T = (A/a)\sqrt{2h_0/g}$. The outflow speed is $\sqrt{2gh}$, which is largest when the water is deepest.

11.5 Balancing weight against Stokes drag, $v = 2r^2\rho g/9\eta = 2(10^{-5})^2(1000)(9.8)/(9\times1.8\times10^{-5}) \approx 1.2$ cm/s. Then $\mathrm{Re} = \rho_{\text{air}} v(2r)/\eta \approx 0.02$: tiny, so Stokes' law applies. Droplets fall about a metre a minute, and the gentlest updraft — a few centimetres per second — holds them up.

11.6 $Q \propto R^4$, so the flow falls to $0.8^4 = 0.41$ of its original value, a 59% drop. Restoring it needs $1/0.41 = 2.4$ times the pressure difference.

11.7 In the rotating frame, the water is at rest under gravity and the centrifugal force: $\partial P/\partial r = \rho\Omega^2 r$ and $\partial P/\partial z = -\rho g$. On the free surface P is constant, so $\rho\Omega^2 r\,\mathrm{d}r = \rho g\,\mathrm{d}z$, giving $z = \Omega^2 r^2/2g$. (Equivalently, the surface is a level set of the effective potential $gz - \frac{1}{2}\Omega^2 r^2$.) Newton argued the water "knows" it rotates relative to absolute space; Mach suggested relative to the distant matter of the universe; general relativity says relative to the local inertial frames, which matter determines.

11.8 Surface tension pulls up around the perimeter with force $2\pi r\gamma$ (for a fully wetting liquid), holding up a column of weight $\rho g\pi r^2 h$, so $h = 2\gamma/\rho g r$. For $r = 0.1$ mm, $h = 2(0.072)/(1000\times9.8\times10^{-4}) = 15$ cm. For xylem, $r = 20\,\mu$m gives only 0.7 m — far short of 100 m.

11.9 For tea, $\varepsilon = 0.01/0.3 = 0.03$ W/kg and $\nu = 10^{-6}$, so $\eta_K = (\nu^3/\varepsilon)^{1/4} \approx 7\times10^{-5}$ m, about 0.1 mm. For the atmosphere, $\nu = 1.5\times10^{-5}$ and $\varepsilon = 10^{-3}$ give $\eta_K \approx 1.4$ mm. Below these sizes, viscosity smooths everything out.

11.10 The Stokes equations are linear and contain no time derivative, so at each instant the flow is fixed uniquely by the velocities of the body's surface at that instant — there is no inertia

and no memory. The body's velocity is therefore a linear function of its rate of change of shape, $\dot{X} = A(s)\,\dot{s}$, where s describes the shape. The net displacement over a stroke, $\Delta X = \int A(s)\,\mathrm{d}s$, depends only on the path through shape space, not on how fast it is traversed. A reciprocal stroke retraces its path, so the forward and backward integrals cancel exactly. A rotating helical flagellum is not reciprocal: it rotates continuously in one direction, a cycle in shape space that does not retrace itself, so its contributions add up stroke after stroke.

11.11 The circulation set up by a spinning ball scales as the surface speed ωr times a length r, and the Kutta–Joukowski force per length is $\rho v \Gamma$; over a length $\sim r$ this gives $F \sim \rho(\omega r) v r^2$, up to geometric factors. With $\omega = 63\,\mathrm{rad/s}$, $\omega r = 6.9\,\mathrm{m/s}$, $\pi r^2 = 0.038\,\mathrm{m}^2$, and a prefactor $\frac{1}{2}$: $F \approx \frac{1}{2}(1.2)(6.9)(25)(0.038) \approx 4\,\mathrm{N}$, an acceleration of $9\,\mathrm{m/s^2}$. Over a flight of about 1 s, the ball bends by $\frac{1}{2}at^2 \approx 4\,\mathrm{m}$ (less in practice, as the ball and its spin slow). That is ample to bend a kick around a wall of defenders.

11.12 With $r = R_0 + \delta\cos kz$, the volume per length is $\pi(R_0^2 + \delta^2/2)$; keeping it equal to πR^2 requires $R_0 \approx R - \delta^2/4R$. The area per length is $\langle 2\pi r\sqrt{1+r'^2}\rangle \approx 2\pi[R_0 + \frac{1}{4}R\delta^2k^2]$. So the change in area per length is

$$2\pi\left(\frac{R\delta^2k^2}{4} - \frac{\delta^2}{4R}\right) = \frac{\pi\delta^2}{2R}\,(k^2R^2 - 1),$$

which is negative when $kR < 1$, i.e. for wavelengths longer than $2\pi R$. Surface tension lowers its energy by amplifying such ripples until the stream pinches into drops. (Which of the unstable wavelengths grows fastest depends on dynamics, and turns out to be about $9R$.)

12 Nonlinear dynamics and chaos

12.1 $x - x^3 = x(1-x^2)$ vanishes at $x = 0, \pm 1$. With $f'(x) = 1 - 3x^2$: $f'(0) = 1$, unstable; $f'(\pm 1) = -2$, stable. Every positive start flows to $+1$, every negative start to -1. A linear equation has a single fixed point, approached or fled exponentially; it cannot produce two stable states (bistability) or saturation.

12.2 For $r < 0$, fixed points at $x = \pm\sqrt{-r}$; since $f'(x) = 2x$, the negative one is stable and the positive one unstable. They merge at $r = 0$ and vanish for $r > 0$, after which x runs off to infinity. For small $r > 0$, $\dot{x}$ is tiny near $x = 0$, the "ghost" of the vanished fixed points, and the time to pass is $\int_{-\infty}^{\infty} \mathrm{d}x/(r + x^2) = \pi/\sqrt{r}$: arbitrarily long as $r \to 0^+$.

12.3 The Jacobian $\begin{pmatrix} 0 & 1 \\ -1 & -0.2 \end{pmatrix}$ has $\lambda^2 + 0.2\lambda + 1 = 0$, so $\lambda = -0.1 \pm 0.995i$: a stable spiral. It is the damped harmonic oscillator $\ddot{x} + 0.2\dot{x} + x = 0$, with $Q = 5$.

12.4 Fixed points: $x = 0$ and $x^* = 1 - 1/r$. Since $F'(x) = r(1-2x)$, $F'(x^*) = 2 - r$, and $|2 - r| < 1$ for $1 < r < 3$. Dividing $F(F(x)) - x$ by $x(x - x^*)$ leaves $r^2x^2 - r(r+1)x + (r+1) = 0$, whose roots $x_\pm = [(r+1) \pm \sqrt{(r+1)(r-3)}]/2r$ are real for $r > 3$: the 2-cycle. Its stability multiplier is $F'(x_+)F'(x_-) = r^2[1 - 2(x_+ + x_-) + 4x_+x_-] = -r^2 + 2r + 4$, which equals 1 at $r = 3$ and reaches -1 when $r^2 - 2r - 5 = 0$, at $r = 1 + \sqrt{6}$.

12.5 Nearby points separate as $\delta_n = 2^n\delta_0$ (while small), so $\lambda = \ln 2$. Writing $x_0 = 0.b_1b_2b_3\ldots$ in binary, the map discards the leading digit and shifts the rest left: after n steps the leading digit

of x_n is b_{n+1}. Knowing x_0 to k bits lets you predict k steps and no more. The future is encoded in ever-deeper digits of the initial condition, which no finite measurement can supply.

12.6 $t \approx (1/\lambda)\ln(10^5) = 1.5{\times}11.5 \approx 17$ days. A thousandfold improvement adds $1.5{\times}\ln 1000 \approx 10$ days. (Real forecasts are good for about ten days, because model errors grow too.)

12.7 Contours of $E = \frac{1}{2}m\ell^2\dot\theta^2 - mg\ell\cos\theta$: closed loops around $\theta = 0$ for $E < mg\ell$ (swinging), wavy open curves for $E > mg\ell$ (rotating), and the separatrix $E = mg\ell$, $\dot\theta = \pm 2\sqrt{g/\ell}\cos(\theta/2)$, through the saddles at $\pm\pi$. Near the saddle, motion is governed by $\ddot\phi = (g/\ell)\phi$, so a trajectory passing within ϵ of it spends a time $\sim \sqrt{\ell/g}\ln(1/\epsilon)$ there. In closed form, as $\theta_0 \to \pi$,

$$T = 4\sqrt{\frac{\ell}{g}}\,K\!\left(\sin\frac{\theta_0}{2}\right) \approx 4\sqrt{\frac{\ell}{g}}\,\ln\frac{4}{\cos(\theta_0/2)},$$

a logarithmic divergence.

12.8 At the origin the Jacobian is block-diagonal: z decays at rate β, and the (x, y) block $\begin{pmatrix} -\sigma & \sigma \\ \rho & -1 \end{pmatrix}$ has negative trace and determinant $\sigma(1-\rho)$, positive for $\rho < 1$, so both eigenvalues have negative real parts: stable. Fixed points require $x = y$, $x^2 = \beta z$, and $x(\rho - 1 - z) = 0$; the nonzero solutions have $z = \rho - 1$ and $x = y = \pm\sqrt{\beta(\rho-1)}$, which exist for $\rho > 1$. At $\rho = 1$ convection begins, in a pitchfork bifurcation.

12.9 The Cantor set is two copies of itself at scale $\frac{1}{3}$: $D = \ln 2/\ln 3 = 0.63$. The Koch curve is four copies at scale $\frac{1}{3}$: $D = \ln 4/\ln 3 = 1.26$. Each step multiplies the perimeter by $\frac{4}{3}$, so it diverges. Starting from a triangle of area A_0, step n adds $3\cdot 4^{n-1}$ triangles each of area $A_0/9^n$, a total of $A_0 \cdot \frac{3}{4}\sum_n (4/9)^n = \frac{3}{5}A_0$, so the final area is $\frac{8}{5}A_0$: finite.

12.10 The damped pendulum's state is $(\theta, \dot\theta)$: a two-dimensional autonomous flow, which by Poincaré–Bendixson can only settle onto fixed points or closed orbits (and friction makes it settle at rest). A periodic drive makes the equations depend explicitly on time; treating the drive's phase Ωt as a third variable gives a three-dimensional autonomous flow, where chaos is allowed — and the driven damped pendulum is indeed chaotic for suitable parameters. Maps are different: invertible maps need at least two dimensions (the Hénon map), but non-invertible maps like the logistic map can be chaotic in one, because they can fold.

12.11 $\dot\phi = (\omega_1 - \omega_2) - 2K\sin\phi$. A fixed point needs $\sin\phi = (\omega_1 - \omega_2)/2K$, possible if and only if $|\omega_1 - \omega_2| \le 2K$. Just outside, there is no fixed point but a bottleneck: ϕ creeps past the ghost of the vanished fixed point, then slips quickly through 2π. These *phase slips* recur with period $2\pi/\sqrt{(\omega_1-\omega_2)^2 - 4K^2}$, which diverges at the locking threshold — a saddle-node bifurcation on a circle.

12.12 The intervals are 0.4495, 0.0946, 0.0203, with ratios 4.75 and 4.66 — already close to 4.669. If the intervals continue to shrink by δ, the remaining sum is a geometric series: $r_\infty \approx r_4 + (r_4 - r_3)/(\delta - 1) = 3.5644 + 0.0055 = 3.5699$. A dripping tap's successive intervals $T_{n+1} = F(T_n)$ obey some one-humped return map. Feigenbaum's renormalization argument shows that the cascade's ratios depend only on the map having a smooth quadratic maximum, not on its details, so every such system shares δ.

13 Computational physics

13.1 `0.1 + 0.2` evaluates to `0.30000000000000004`, so the test fails: none of 0.1, 0.2, 0.3 has a finite binary expansion, and the rounding errors do not cancel. Halving from 1, the last ϵ for which $1+\epsilon > 1$ is $2^{-52} \approx 2.2\times10^{-16}$: double precision stores 52 bits after the leading one.

13.2 Running it:

x	direct	rewritten
10^{-1}	0.499583472	0.499583472
10^{-3}	0.4999999583	0.4999999583
10^{-5}	0.50000004	0.4999999999958
10^{-7}	0.4996	0.5
10^{-8}, 10^{-9}	0	0.5

$1-\cos x \approx x^2/2$ is computed as the difference of two numbers near 1, so its relative error is about $\epsilon/(x^2/2)$, which reaches order one when $x \sim \sqrt{2\epsilon} \approx 2\times10^{-8}$. Below that, $\cos x$ rounds to exactly 1.

13.3 $E' = \frac{1}{2}(v-h\omega^2 x)^2 + \frac{1}{2}\omega^2(x+hv)^2$; the cross terms cancel, leaving $(1+\omega^2h^2)E$. With 100 steps per period, $\omega h = 2\pi/100$ and the energy grows per period by $(1+0.0039)^{100} \approx e^{0.39} = 1.48$. Keeping growth below 1% over 100 periods needs $\exp(100\cdot 4\pi^2/N) < 1.01$, i.e. $N \gtrsim 4\times10^5$ steps per period. Hopeless.

13.4 Taylor expansion gives $[f(x+h)-f(x-h)]/2h = f'(x) + \frac{1}{6}h^2 f'''(x) + \ldots$; each function value carries a rounding error $\sim \epsilon|f|$, contributing $\sim \epsilon|f|/h$. Minimizing $h^2|f'''|/6 + \epsilon|f|/h$ gives $h = (3\epsilon|f|/|f'''|)^{1/3} \sim \epsilon^{1/3} \approx 10^{-5}$, with error $\sim \epsilon^{2/3} \approx 10^{-11}$. On log-log axes the error has slope $+2$ for large h (truncation) and slope -1 for small h (round-off): a V with its point near $h \approx 10^{-5}$.

13.5 $x_{n+1} = x_n - (x_n^2-2)/2x_n = \frac{1}{2}(x_n + 2/x_n)$. From $x_0 = 1$: 1.5, 1.4166667, 1.4142157, 1.414213562375, 1.414213562373095, with errors $9\times10^{-2}, 2\times10^{-3}, 2\times10^{-6}, 2\times10^{-12}$ — roughly 1, 3, 6, 12 correct digits. In general $e_{n+1} = e_n^2/2x_n$: quadratic convergence.

13.6 With step 0.01 yr for 100 years (units of AU and years, $GM = 4\pi^2$), a run of the code gives:

- **Euler**: the energy rises steadily; after 100 years it has risen by 85% of $|E_0|$ and the "Earth" has spiralled out to 6 AU, nearly unbound.
- **RK4**: a tiny but systematic drift, $\Delta E/E \approx -2\times10^{-5}$ after 100 years, growing steadily with time.
- **Velocity Verlet**: the energy error oscillates with the orbital period, staying below 4×10^{-6} throughout, with no long-term drift.

RK4 is more accurate per step, but only the symplectic method stays bounded forever.

13.7 The computed levels, in units of the exact $\pi^2n^2/2$:

N	$n=1$	$n=2$	$n=3$	$n=4$
10	0.9932	3.892	8.463	14.33
100	0.99992	3.9987	8.9935	15.979
1000	0.999999	3.99999	8.99993	15.9998

The discretized $-\frac{1}{2}\,\mathrm{d}^2/\,\mathrm{d}x^2$ has exact eigenvalues

$$\frac{2}{h^2}\sin^2\frac{n\pi h}{2} \approx \frac{(n\pi)^2}{2}\left(1-\frac{(n\pi h)^2}{12}\right),$$

so the relative error of level n is $-(n\pi h)^2/12$, falling a hundredfold for each tenfold increase in N. With $V=\frac{1}{2}x^2$ on $[-10,10]$ and $N=1000$, the levels are 0.49999, 1.49994, 2.49984, 3.49969.

13.8 Substituting the Fourier mode, $g = 1+r(e^{ik\Delta x}-2+e^{-ik\Delta x}) = 1-4r\sin^2(k\Delta x/2)$. The worst case is $\sin^2=1$, giving $g=1-4r$, and $|g|\le 1$ requires $r\le\frac{1}{2}$. The first mode to go unstable has $k\Delta x=\pi$: the sawtooth that alternates sign at every grid point — the checkerboard you see when a simulation explodes.

13.9 The fraction inside is $p=\pi/4$, so the error in π is $4\sqrt{p(1-p)/N}\approx 1.6/\sqrt{N}$. A run gave errors of 0.10, 0.016, and 0.0016 for $N=10^2$, 10^4, 10^6, against predictions of 0.16, 0.016, 0.0016. A grid with 10 points per axis in 10 dimensions needs 10^{10} points; Monte Carlo's $1/\sqrt{N}$ error does not care about dimension, so 10^4 samples give about 1%.

13.10 On a 32×32 lattice (averaging $|m|$ over 2000 sweeps):

T	1.5	2.0	2.2	2.3	2.4	2.6	3.0	3.5
$\langle\lvert m\rvert\rangle$	0.99	0.91	0.75	0.58	0.34	0.15	0.08	0.06

The steepest drop is near $T\approx 2.3$, consistent with Onsager's 2.269 and far below the mean-field 4. A finite lattice cannot have a true singularity: correlations cannot grow larger than the lattice, so the transition is smeared over a range of width $\sim 1/L$, and above T_c, $|m|$ stays nonzero at the level of the fluctuations, $\sim 1/L$.

13.11 With $a=\omega^2h^2/2$, one step is $x'=(1-a)x+hv$, $v'=-h\omega^2(1-a/2)x+(1-a)v$. The determinant is $(1-a)^2+h^2\omega^2(1-a/2)=1$. Writing the matrix as $M=\begin{pmatrix} c & h \\ -q & c\end{pmatrix}$ with $c^2+hq=1$, a direct calculation gives $M^TSM=S$ for $S=\mathrm{diag}(q/h,1)$, i.e. $q/h=\omega^2(1-\omega^2h^2/4)$: so $\tilde H$ is exactly conserved. For $\omega h<2$, $\tilde H$ is a positive quadratic form, the motion stays on a fixed ellipse, and the true energy differs from $\tilde H$ by a bounded oscillating amount of relative size $\sim\omega^2h^2$. For $\omega h>2$, $\tilde H$ is indefinite, $|\mathrm{tr}\,M|=2|1-a|>2$, one eigenvalue exceeds 1 in magnitude, and the numerical solution blows up exponentially.

13.12 A run with $h=10^{-3}$ and $h(1+10^{-12})$ from $(1,1,1)$ gives separations of 2×10^{-11} at $t=1$, 5×10^{-10} at $t=10$, 10^{-5} at $t=30$, 0.2 at $t=40$, and order 10 (the size of the attractor) by $t\approx 45$ — growth at about the maximal Lyapunov exponent, 0.9. So no simulation tracks *the* true trajectory for long. The *shadowing lemma* says that for hyperbolic systems, every slightly erroneous numerical trajectory stays close to some *exact* trajectory from a slightly different initial condition. The simulation is a faithful sample of the attractor, just not of the trajectory you asked for — so averages, Lyapunov exponents, and other statistical properties come out right. (The Lorenz attractor is not uniformly hyperbolic, but shadowing holds in practice for very long times.)

14 Quantum mechanics

14.1 Electron: $\lambda = h/\sqrt{2mE} = 6.63\times10^{-34}/\sqrt{2(9.11\times10^{-31})(1.6\times10^{-17})} = 0.12\,\text{nm}$, comparable to the spacing of atoms in a crystal, so crystals diffract it. Baseball: $\lambda = h/mv = 1.1\times10^{-34}\,\text{m}$; a doorway diffracts it by an angle of 10^{-34} radians.

14.2 $\int A^2e^{-2|x|/a}\,\mathrm{d}x = A^2a = 1$, so $A = 1/\sqrt{a}$. $P(|x|<a) = 1-e^{-2} = 0.86$. $\langle x\rangle = 0$ by symmetry, and $\langle x^2\rangle = (2/a)\int_0^\infty x^2e^{-2x/a}\,\mathrm{d}x = a^2/2$.

14.3 $E_n = n^2h^2/8mL^2$ with $L = 1\,\text{nm}$: 0.38, 1.50, and 3.38 eV. The $2\to1$ photon has 1.13 eV, wavelength $1240/1.13 = 1100\,\text{nm}$: infrared, invisible.

14.4 A 400 nm photon carries 3.10 eV, so $K_{\max} = 0.8\,\text{eV}$. A 600 nm photon carries 2.07 eV $<$ 2.3 eV: no electrons at all, however bright the light.

14.5 $\mathrm{d}E/\mathrm{d}r = -\hbar^2/mr^3 + e^2/4\pi\epsilon_0r^2 = 0$ gives $r = 4\pi\epsilon_0\hbar^2/me^2 = a_0 = 0.053\,\text{nm}$ and $E = -e^2/8\pi\epsilon_0a_0 = -13.6\,\text{eV}$. With a muon, m grows by 207 (more precisely the reduced mass by 186), so muonic hydrogen is about 200 times smaller, $2.8\times10^{-13}\,\text{m}$, and bound by about 2.5 keV. Its size is close enough to the proton's that its energy levels are sensitive to the proton's radius — which is how the proton radius was measured precisely in 2010.

14.6 $\kappa = \sqrt{2m(4\,\text{eV})}/\hbar = 1.0\times10^{10}\,\text{m}^{-1}$, so $T \approx e^{-2\kappa a} = e^{-10.2} \approx 4\times10^{-5}$. Doubling the width squares this, to about 10^{-9}. A proton has κ larger by $\sqrt{1836} = 43$, giving e^{-440}: never.

14.7 $[\hat H,\hat a^\dagger] = \hbar\omega[\hat a^\dagger\hat a,\hat a^\dagger] = \hbar\omega\hat a^\dagger[\hat a,\hat a^\dagger] = \hbar\omega\hat a^\dagger$. So $\hat H\hat a^\dagger|n\rangle = \hat a^\dagger\hat H|n\rangle + \hbar\omega\hat a^\dagger|n\rangle = (E+\hbar\omega)\hat a^\dagger|n\rangle$. In position space, $\hat a|0\rangle = 0$ reads $(x + \frac{\hbar}{m\omega}\frac{\mathrm{d}}{\mathrm{d}x})\psi_0 = 0$, whose solution is the Gaussian $\psi_0 \propto e^{-m\omega x^2/2\hbar}$.

14.8 Each Pauli matrix has eigenvalues ±1, with eigenvectors $|\uparrow\rangle, |\downarrow\rangle$ for σ_z; $(|\uparrow\rangle \pm |\downarrow\rangle)/\sqrt{2}$ for σ_x; and $(|\uparrow\rangle \pm i|\downarrow\rangle)/\sqrt{2}$ for σ_y. A spin up along x gives z-results up or down with probability $\frac{1}{2}$ each. Along an axis at angle θ to x, the probability of "up" is $\cos^2(\theta/2) = (1+\cos\theta)/2$. In the sequence $z\to x\to z$, the x result is random, and the final z result is again random, 50–50, regardless of the first z result: the x measurement erased it.

14.9 $[\hat x,\hat p]f = x(-i\hbar f') + i\hbar(xf)' = i\hbar f$. For angular momentum, the only non-commuting pieces of $[yp_z - zp_y,\ zp_x - xp_z]$ are $[yp_z, zp_x] = yp_x[p_z,z] = -i\hbar yp_x$ and $[zp_y, xp_z] = xp_y[z,p_z] = i\hbar xp_y$. Together, $[\hat L_x,\hat L_y] = i\hbar(xp_y - yp_x) = i\hbar\hat L_z$.

14.10 $\Delta E = \lambda\langle x^4\rangle = 3\lambda\hbar^2/4m^2\omega^2$. First-order perturbation theory is trustworthy when this is much smaller than the level spacing $\hbar\omega$, i.e. when the dimensionless coupling $\lambda\hbar/m^2\omega^3 \ll 1$. (Even then, the full perturbation series in λ diverges — it is only asymptotic — because for $\lambda<0$ the potential has no bottom.)

14.11 In units of Ry = 13.6 eV and $u = a_0/b$, the Gaussian gives $\langle T\rangle = \frac{3}{2}u^2$ and $\langle V\rangle = -(4/\sqrt{\pi})u$. Minimizing $E(u)$ gives $u = 4/3\sqrt{\pi}$ and $E = -\frac{8}{3\pi}\,\text{Ry} = -11.5\,\text{eV}$. The variational principle guarantees any trial energy lies above the true -13.6 eV. The Gaussian is worse because it falls off too fast at large r and is smooth at $r=0$, where the $1/r$ potential demands a cusp; $e^{-r/a}$ has both features right, and is the exact answer.

14.12 Away from the origin, $\psi'' = \kappa^2\psi$ with $E = -\hbar^2\kappa^2/2m$, and normalizability with continuity gives $\psi = Ae^{-\kappa|x|}$. Integrating the Schrödinger equation across $x=0$: $-(\hbar^2/2m)[\psi'(0^+) -$

$\psi'(0^-)] = \alpha\psi(0)$, i.e. $(\hbar^2/2m)(2\kappa A) = \alpha A$, so $\kappa = m\alpha/\hbar^2$ and $E = -m\alpha^2/2\hbar^2$. There is only one such κ, and odd functions vanish at the origin and never feel the delta, so there is exactly one bound state.

14.13 At 1 fm, $e^2/4\pi\epsilon_0 r = 1.44\,\text{MeV fm}/1\,\text{fm} = 1.4\,\text{MeV}$. At 1.5×10^7 K, $k_\text{B}T = 1.3\,\text{keV}$. With $E_G = 2(469\,\text{MeV})(\pi/137)^2 = 490\,\text{keV}$, the tunnelling factor is $e^{-\sqrt{490}} = 2\times10^{-10}$ at 1 keV and $e^{-\sqrt{98}} = 5\times10^{-5}$ at 5 keV. The rate at energy E goes as $e^{-E/k_\text{B}T}e^{-\sqrt{E_G/E}}$: the Boltzmann factor falls with E while tunnelling rises, and their product peaks sharply at the *Gamow peak*, $E_0 = (k_\text{B}T\sqrt{E_G}/2)^{2/3} \approx 6\,\text{keV}$, about $4.5\,k_\text{B}T$. On top of the tiny tunnelling probability, two protons can only stick if one turns into a neutron by the weak interaction during their brief encounter, which almost never happens. A given proton in the Sun's core waits about ten billion years to fuse.

14.14 The first beam splitter puts amplitude $1/\sqrt{2}$ in each arm. With a bomb in arm 2: with probability $\frac{1}{2}$ the photon is found there and the bomb explodes. Otherwise the photon is in arm 1, and since the other path is blocked there is no interference at the second beam splitter, which sends it to A or B with probability $\frac{1}{2}$ each. So 25% of the time B clicks — impossible without a bomb, since interference forbids it — and the bomb is certified live without having absorbed a photon. (25% of the time A clicks and the test is inconclusive; repeat it.) With many weak interrogation steps, the quantum Zeno effect keeps freezing the photon out of the bomb's path, pushing the success rate toward 100%.

15 Continuum mechanics and acoustics

15.1 $\sigma = F/A = 981\,\text{N}/(7.9\times10^{-7}\,\text{m}^2) = 1.25\,\text{GPa}$, five times the yield strength of mild steel: it would yield and fail. Piano wire survives, stretching by $\Delta L = \sigma L/E = (1.25\times10^9)(2)/(2\times10^{11}) = 1.25\,\text{cm}$.

15.2 Isothermal: $\sqrt{P/\rho} = \sqrt{1.013\times10^5/1.20} = 290\,\text{m/s}$. Adiabatic: $\sqrt{\gamma P/\rho} = 343\,\text{m/s}$. Helium ($\gamma = 5/3$, $m = 4\,\text{u}$): $\sqrt{\gamma k_\text{B}T/m} \approx 1000\,\text{m/s}$. The vocal cords set the fundamental, which is unchanged, but the resonances of the vocal tract (the formants, which give a voice its timbre) scale with the sound speed and rise about threefold: the voice sounds higher and thinner.

15.3 One violin is ten times quieter: $70 - 10 = 60\,\text{dB}$. Reaching 90 dB needs 100 times the intensity of 70 dB, so 1000 violins.

15.4 Open at both ends: $f_n = nv/2L = 172, 343, 515\,\text{Hz}$. Closed at one end: $f = (2n-1)v/4L = 86, 257, 429\,\text{Hz}$.

15.5 $f = \frac{v}{2\pi}\sqrt{A/V\ell} = \frac{343}{2\pi}\sqrt{(3\times10^{-4})/(7.5\times10^{-4}\times0.08)} \approx 120\,\text{Hz}$, about the B below the bass clef (end corrections lengthen the effective neck and lower it a little). Drinking increases the air volume V, so the pitch falls as $V^{-1/2}$.

15.6 $E \sim U_0/a^3 = (3\times1.6\times10^{-19}\,\text{J})/(1.5\times10^{-10}\,\text{m})^3 \approx 1.4\times10^{11}\,\text{Pa} = 140\,\text{GPa}$. Diamond is about seven times stiffer — within an order of magnitude, which is all the estimate claims. (Diamond's dense network of short bonds, each atom bonded four ways, adds a geometric factor.)

15.7 $(E/\rho g)^{1/3} = (10^{10}/(700\times9.8))^{1/3} \approx 113\,\text{m}^{1/3}$, so with $D = 1\,\text{m}$, $h_\text{max} \approx 0.8\times113 \approx 90\,\text{m}$. With a safety factor of four, a real tree with a 1 m trunk stands about 25 m tall — about right. The tallest redwoods, about 100 m, have trunks several metres across.

15.8 Weight ×1000; bone cross-section ×100; stress ×10. To keep the stress the same, the cross-section must grow 1000-fold, so the bones must be $\sqrt{10} \approx 3$ times thicker than mere scaling gives. The giant would have legs like an elephant's, mostly bone, and would still struggle to move.

15.9 $\Delta t = d(1/v_S - 1/v_P) = d(0.119\,\mathrm{s/km})$, so $d \approx 67$ km. One station gives only the distance: a circle on the map. Three stations pin down the epicentre; four also fix the depth and the time of the quake.

15.10 The energy per thickness as a function of crack half-length is $U(a) = 4a\gamma - \pi a^2\sigma^2/E$. It increases for small a and decreases for large a, with a maximum at $\mathrm{d}U/\,\mathrm{d}a = 0$: $a_c = 2\gamma E/\pi\sigma^2$. A crack longer than a_c lowers its energy by growing, so it runs. Equivalently, the fracture stress for a crack of half-length a is $\sigma_f = \sqrt{2E\gamma/\pi a}$. For glass with $a = 1\,\mu$m: $\sqrt{2(7\times10^{10})(1)/\pi\times10^{-6}} \approx 2\times10^8$ Pa $= 200$ MPa, against a theoretical strength of $E/10 = 7$ GPa. (Window glass, with larger flaws, breaks at about 50 MPa.)

15.11 A square membrane of side L has modes $f = (c/2L)\sqrt{m^2+n^2}$ with $m, n \geq 1$. Those with $f < F$ are the lattice points with positive coordinates inside a circle of radius $R = 2LF/c$: about a quarter of its area, $\frac{1}{4}\pi R^2 = \pi L^2F^2/c^2 = \pi AF^2/c^2$. Weyl showed the leading term is the same for any shape, with a correction proportional to the perimeter. So counting the frequencies below F for large F determines the area: a perfect ear could hear it (and the perimeter too).

15.12 Measurements on orchestral timpani (Fletcher and Rossing) put the $(1,1)$, $(2,1)$, $(3,1)$, and $(4,1)$ modes at about $1 : 1.50 : 1.98 : 2.44$ times the $(1,1)$ frequency — close to $2 : 3 : 4 : 5$, the second to fifth harmonics of a tone an octave below. Because the strongest partials line up nearly as a harmonic series, the ear fuses them into one definite pitch, heard at the frequency of the principal $(1,1)$ mode. The symmetric $(0,n)$ modes move air in and out of the kettle as a whole, so they radiate efficiently and die within a fraction of a second; and the drum is struck about a quarter of the way in from the rim, where they are excited only weakly. They contribute the initial thud, not the pitch.

16 Complex systems and networks

16.1 $1 - 1/R_0$: 33%, 67%, and 93%. A vaccine that protects 90% of recipients leaves at most 90% of the population immune even if everyone is vaccinated — short of the 93% measles needs. In practice a second dose raises protection to about 97%, which is why the measles vaccine is given twice.

16.2 $H_{100}/H_{50\,000} = (\ln 100 + 0.577)/(\ln 50\,000 + 0.577) = 5.19/11.4 \approx 45\%$. The hundred commonest words make up nearly half of everything you read.

16.3 $\int_1^\infty x \cdot x^{-\alpha}\,\mathrm{d}x$ converges only for $\alpha > 2$, and $\int_1^\infty x^2 \cdot x^{-\alpha}\,\mathrm{d}x$ only for $\alpha > 3$. With infinite variance, the sample variance never settles: it keeps growing as you collect data, dominated by the largest event seen so far. Averages fluctuate wildly, the central limit theorem fails, and events far beyond anything in the historical record are to be expected.

16.4 $\mathrm{d}s/\,\mathrm{d}r = -\beta si/\gamma i = -R_0 s$, so $s = e^{-R_0 r}$ (starting from $s \approx 1$, $r \approx 0$). At the end, $i = 0$ and $s_\infty = 1 - r_\infty$, giving $r_\infty = 1 - e^{-R_0 r_\infty}$. For $R_0 = 2$, iterating gives $r_\infty = 0.80$: 20% are never infected. Note the overshoot: the epidemic stops growing once 50% have been infected, but cases already in progress carry it to 80%.

16.5 A node is outside the giant component if each of its neighbours is, independently, outside it (through the rest of the network). Averaging u^k over a Poisson degree distribution with mean c gives $u = \sum_k e^{-c}c^k u^k/k! = e^{-c(1-u)}$, so $S = 1-u$ obeys $S = 1-e^{-cS}$. The right side has slope c at $S = 0$ and is concave, so it crosses the line $y = S$ at a positive value only if $c > 1$. This is exactly the SIR final-size equation with c in place of R_0: an epidemic is percolation on the network of transmissions.

16.6 Following a random edge reaches a node with probability proportional to its degree, so the expected degree found there is $\sum_k k\cdot kP(k)/\sum_k kP(k) = \langle k^2\rangle/\langle k\rangle$. Since $\langle k^2\rangle - \langle k\rangle^2 = \mathrm{Var}(k) \geq 0$, this is at least $\langle k\rangle$, with equality only if every node has the same degree.

16.7 The energy released by quakes near magnitude M goes as (number)×(energy each) $\propto 10^{-M}\cdot 10^{1.5M} = 10^{0.5M}$, increasing with M. The total is dominated by the largest earthquakes: a single magnitude 9 releases more energy than all the smaller earthquakes of several decades combined.

16.8 Die: $\log_2 6 = 2.58$ bits. Coin: 1 bit. Biased coin: $-0.9\log_2 0.9 - 0.1\log_2 0.1 = 0.47$ bits. Compressing to 25% of 8 bits shows English has at most 2 bits per character; better models, and Shannon's own guessing experiments, suggest about 1.

16.9 $\mathrm{d}k_i/k_i = \mathrm{d}t/2t$ gives $k_i(t) = m(t/t_i)^{1/2}$. A node has degree greater than k if $t_i < m^2t/k^2$; since arrival times are uniform on $[0,t]$, the fraction is m^2/k^2. Differentiating, $P(k) = 2m^2/k^3$. The oldest nodes have had the longest to accumulate links, and since growth is proportional to degree, their head start compounds: rich get richer.

16.10 Random pairwise exchanges that conserve total money are mathematically like elastic collisions that conserve total energy. The equilibrium is the most probable distribution of a fixed total among many agents, which maximizes entropy subject to the constraint and is exponential, $P(m) = e^{-m/T}/T$, with "temperature" equal to the average, $T = \$100$. A simulation with 2000 agents and 400 000 exchanges gives 64% of agents below \$100 (exponential: 63%), 5.1% above \$300 (exponential: 5.0%), and the richest 1% holding 5.5% of the money. In real economies the richest 1% hold a third or more of the wealth: the top follows a Pareto power law, produced by multiplicative effects — returns on capital proportional to wealth — that a pure exchange model lacks.

16.11 A run on a 50×50 grid, discarding the first 15 000 grains and recording 60 000 more, gave a mean height of 2.10 grains per site (the known value for large grids is 2.125); 43% of added grains triggered an avalanche, of sizes up to about 8400 topplings. On log-log axes, the size distribution is a straight line over three decades, with slope about -1.0 to -1.1 (large-system studies give an exponent near 1.1–1.2), cut off at sizes set by the grid. Why critical: if the pile is subcritical (too flat), grains pile up with few avalanches and the slope rises; if it is supercritical (too steep), large avalanches carry grains off the edges and the slope falls. Slow driving and fast relaxation pin the system to the boundary between the two, where avalanches of every size occur.

17 Plasma physics

17.1 With $\lambda_D = \sqrt{\epsilon_0 k_\mathrm{B}T/ne^2}$: (a) ionosphere 2.2 mm, $n\lambda_D^3 \approx 10^4$; (b) fusion plasma 70 µm, $n\lambda_D^3 \approx 3\times 10^7$; (c) solar core 2.7×10^{-11} m, $n\lambda_D^3 \approx 2$. The first two are good plasmas, dominated

by collective effects. The solar core is so dense that each particle has only a couple of neighbours within a Debye length — a mildly "strongly coupled" plasma, where Debye screening is only a rough description.

17.2 $f_p \approx 9\sqrt{10^{12}}\,\text{Hz} = 9\,\text{MHz}$. AM waves at 1 MHz are below the plasma frequency and are reflected, bouncing between ionosphere and ground far beyond the horizon. (By day, a lower layer absorbs them; it disappears at night.) FM at 100 MHz is above f_p and passes into space, so it reaches only as far as line of sight.

17.3 $r_L = mv/eB = (1.67{\times}10^{-27})(4{\times}10^{5})/[(1.6{\times}10^{-19})(5{\times}10^{-9})] \approx 840\,\text{km}$, about an eighth of the Earth's radius. An electron at the same speed has a radius 1836 times smaller, about 0.5 km.

17.4 For $v \ll c$, a frame moving with velocity $\mathbf{u}$ sees the electric field $\mathbf{E}' = \mathbf{E} + \mathbf{u} \times \mathbf{B}$. Choose $\mathbf{u} = \mathbf{E} \times \mathbf{B}/B^2$. For $\mathbf{E} \perp \mathbf{B}$, $(\mathbf{E} \times \mathbf{B}) \times \mathbf{B} = -B^2\mathbf{E}$, so $\mathbf{E}' = 0$. In that frame there is only a magnetic field, and particles simply gyrate about fixed centres; back in the original frame, their centres drift at $\mathbf{u}$. The drift comes from a change of frame, which knows nothing about the particle, so it is the same for every charge and mass. Physically: a positive charge speeds up on the half of its orbit where it moves along $\mathbf{E}$ and so has a larger radius there; a negative charge gyrates the other way but is also pushed the other way, and drifts in the same direction.

17.5 $\omega_p = \sqrt{ne^2/\epsilon_0 m_e} = 1.6 \times 10^{16}\,\text{rad/s}$, so $\hbar\omega_p \approx 11\,\text{eV}$, in the far ultraviolet. Visible photons (1.7–3.1 eV) are below it and reflected; X-rays are far above it, where the electrons cannot follow the field, and pass through. In gold, transitions of bound $5d$ electrons into the conduction band begin at about 2.4 eV, absorbing blue and violet light, so the reflected light is yellow. (Copper's similar threshold, near 2.1 eV, makes it reddish.)

17.6 Energy conservation keeps v fixed, and $\mu = mv_\perp^2/2B$ is conserved, so $v_\perp^2 = v^2 \sin^2\theta\, B/B_{\min}$. The particle reflects where $v_\perp$ reaches v; this happens before $B_{\max}$ if $\sin^2\theta\, B_{\max}/B_{\min} > 1$, i.e. $\sin^2\theta > B_{\min}/B_{\max}$. For a mirror ratio of 4, the loss cone is $\theta < 30^\circ$ (or $> 150^\circ$); for an isotropic distribution the fraction of directions inside it is $1 - \cos 30^\circ = 13\%$.

17.7 $v_A = B/\sqrt{\mu_0\rho} = 10^{-3}/\sqrt{(4\pi \times 10^{-7})(10^{15})(1.67 \times 10^{-27})} \approx 7 \times 10^5\,\text{m/s}$. Crossing 10^8 m takes about 150 s.

17.8 $\tau_E \geq 3 \times 10^{21}/(10^{20} \times 15) = 2\,\text{s}$. Heat leaks out by diffusion across the field, which takes a time $\sim L^2/D$ for a plasma of size L, so confinement improves rapidly with size; also, fusion power grows with volume while losses grow with surface area. This is why ITER is enormous.

17.9 $v_{\text{phase}} = \omega/k = \sqrt{c^2 + \omega_p^2/k^2} > c$, and $v_{\text{group}} = \mathrm{d}\omega/\,\mathrm{d}k = c^2k/\omega < c$, so their product is c^2. The phase velocity exceeds c, but it is the speed of a pattern of crests that carries no energy or information; signals travel at the group velocity, which is below c. With $E = \hbar\omega$ and $p = \hbar k$, the dispersion relation becomes $E^2 = (\hbar\omega_p)^2 + (pc)^2$: a particle of mass $\hbar\omega_p/c^2$.

17.10 By Ampère's law the toroidal field falls as $1/R$, so ∇B points inward. The grad-B drift, $\propto \mathbf{B}{\times}\nabla B/q$, is vertical and has opposite signs for ions and electrons, so charge separates, say ions up and electrons down. The resulting vertical $\mathbf{E}$, crossed with the toroidal $\mathbf{B}$, gives an $\mathbf{E} \times \mathbf{B}$ drift that points outward for every particle, carrying the plasma into the wall within microseconds. Twisting the field lines helically makes each line wind poloidally, visiting the top and bottom and the inside and outside of the torus in turn; electrons flowing along the lines short out the vertical charge separation, and each particle's drifts average away. A sphere won't work because, by the hairy ball theorem, any tangent field on a sphere has a zero somewhere: confining field lines must lie on surfaces with the topology of a torus.

17.11 In the frame moving with the wave, particles moving slightly slower than v_ϕ drift backward slowly and are reflected by the wave's potential hills, gaining speed; those slightly faster are reflected the other way and lose speed. Particles therefore gain energy from the wave in proportion to $f(v_\phi - \delta)$ and give energy to it in proportion to $f(v_\phi + \delta)$, so the net transfer to the wave is $\propto f'(v_\phi)$. For a Maxwellian, $f' < 0$ at any $v_\phi > 0$: the wave damps. A beam adds a bump on the tail of the distribution, creating a range of speeds where $f' > 0$. Waves with phase velocities there grow, drawing energy from the beam until they flatten the bump into a plateau.

18 Atomic, molecular and optical physics

18.1 $1/\lambda = R_\infty(\frac{1}{4} - 1/n^2)$: H$\alpha$ ($n = 3$) 656 nm, red; Hβ ($n = 4$) 486 nm, blue-green; Hγ ($n = 5$) 434 nm, violet-blue. The series limit ($n \to \infty$) is 365 nm, in the ultraviolet.

18.2 $E = hf = 5.9\times10^{-6}$ eV, or $E/k_\mathrm{B} = 0.068$ K. At 100 K, collisions between atoms easily supply this energy and keep the hyperfine levels populated in their statistical ratio of 3 : 1, much faster than the upper level decays radiatively. Each atom emits only once in ten million years, but a galaxy has so many hydrogen atoms that the line is bright.

18.3 $\mu_B B = 5.8 \times 10^{-5}$ eV at 1 T, a frequency of 14 GHz. In the sunspot, $\Delta E = hc\,\Delta\lambda/\lambda^2 = (1240\,\mathrm{eV\,nm})(0.01)/525^2 \approx 4.5 \times 10^{-5}$ eV. If that is the spacing between the outer components of a simple triplet ($2\mu_B B$), $B \approx 0.4$ T; real lines have Landé factors of order one, but the answer is a few tenths of a tesla, as Hale found.

18.4 Solving the balance for u:

$$u(\nu) = \frac{AN_2}{B(N_1 - N_2)} = \frac{A/B}{N_1/N_2 - 1} = \frac{A/B}{e^{h\nu/k_\mathrm{B}T} - 1}.$$

Matching Planck's law, $u = (8\pi h\nu^3/c^3)/(e^{h\nu/k_\mathrm{B}T} - 1)$, at every temperature requires $A/B = 8\pi h\nu^3/c^3$. (Einstein also needed the absorption and stimulated-emission coefficients to be equal, which the high-temperature limit demands.)

18.5 $\mu = \frac{12\times16}{28}$ u $= 1.14 \times 10^{-26}$ kg and $I = \mu r^2 = 1.46 \times 10^{-46}$ kg m^2. The $\ell = 1 \to 0$ transition has energy $\hbar^2/I = 7.6 \times 10^{-23}$ J, a frequency of 115 GHz and wavelength 2.6 mm.

18.6 $\mu = \frac{35}{36}$ u $= 1.61 \times 10^{-27}$ kg, so $\omega = \sqrt{k/\mu} = 5.7 \times 10^{14}$ rad/s, $f = 9.0 \times 10^{13}$ Hz, $\lambda = 3.3\,\mu$m. For DCl, $\mu = \frac{2\times35}{37}$ u, nearly twice as large, so the frequency drops by $\sqrt{0.97/1.89} = 0.72$, close to $1/\sqrt{2}$.

18.7 The kinetic energy is about 13.6 eV, and relativistic corrections are a fraction $(v/c)^2 = \alpha^2$ of it: $\alpha^2 \times 13.6\,\mathrm{eV} = 7 \times 10^{-4}$ eV for the ground state. The splittings that show up in the Balmer lines come from the $n = 2$ and $n = 3$ levels, smaller by about $1/n^3$: the $n = 2$ fine structure is 4.5×10^{-5} eV. Relative to the 1.89 eV Hα photon this is 2×10^{-5}, about 0.015 nm — visible only with a good spectrometer.

18.8 $T_D = \hbar\Gamma/2k_\mathrm{B} = (1.05 \times 10^{-34})(3.8 \times 10^7)/(2 \times 1.38 \times 10^{-23}) = 144\,\mu$K. $T_r = \hbar^2k^2/mk_\mathrm{B} = 0.36\,\mu$K, about 400 times lower. Sub-Doppler cooling methods get down to a few recoil temperatures.

18.9 Each photon carries $\hbar k = h/\lambda = 1.1 \times 10^{-27}$ kg m/s; at 3×10^7 per second the force is 3.4×10^{-20} N, and on a mass of 3.8×10^{-26} kg the deceleration is 9×10^5 m/s^2, about $10^5 g$.

Stopping from 600 m/s takes 0.7 ms over $v^2/2a \approx 0.2$ m. (As the atom slows, its Doppler shift changes, so a "Zeeman slower" uses a tapered magnetic field to keep it in resonance.)

18.10 Light climbing a height Δh is redshifted by $g\Delta h/c^2$ (Chapter 26), so the upper clock runs fast by that fraction: $9.8 \times 0.01/(9 \times 10^{16}) = 1.1 \times 10^{-18}$. The best optical clocks average down to this level in about an hour or two of running; in 2022 the redshift was resolved across a single millimetre-tall cloud of atoms.

18.11 $\dot{N}_2 = BI(N_1 - N_2) - AN_2$. In steady state, $N_2/N_1 = BI/(BI + A)$, which approaches 1 as $I \to \infty$ but never exceeds it: whenever the populations are equal, absorption and stimulated emission cancel and pumping does nothing more. A three-level laser pumps from level 1 to level 3, which decays quickly to a long-lived level 2, and lases from 2 to 1; inversion then needs more than half the atoms moved out of the ground state. A four-level laser lases from level 3 to a level 2 that empties quickly into the ground state, so the lower laser level is almost empty and even a little pumping inverts the population.

18.12 $\langle V(\mathbf{r} + \delta\mathbf{r})\rangle = V + \frac{1}{2}\langle \delta r_i \delta r_j\rangle \partial_i\partial_j V = V + \frac{1}{6}\langle \delta r^2\rangle \nabla^2 V$, since the jitter is isotropic ($\langle \delta r_i \delta r_j\rangle = \delta_{ij}\langle \delta r^2\rangle/3$) and averages to zero to first order. For the Coulomb potential, $\nabla^2 V = (e^2/\epsilon_0)\delta^3(\mathbf{r})$, so the shift is $\Delta E = \frac{1}{6}\langle \delta r^2\rangle (e^2/\epsilon_0)\,|\psi(0)|^2$. With $|\psi_{2s}(0)|^2 = 1/8\pi a_0^3$, $e^2/\epsilon_0 = 4\pi\alpha\hbar c$, and $a_0 = \hbar/(\alpha m_e c)$, this simplifies to

$$\Delta E = \frac{\alpha^5}{6\pi} \ln\frac{1}{\alpha}\, m_e c^2 \approx 3 \times 10^{-6}\,\text{eV},$$

a frequency of about 0.7 GHz — the right order of magnitude next to the measured 1.06 GHz. Bethe's 1947 calculation, with a better estimate of the logarithm, got within a few percent.

19 Condensed matter

19.1 $\sin\theta = n\lambda/2d$: 0.273 for $n = 1$ ($\theta = 15.8°$) and 0.546 for $n = 2$ (33.1°). Visible light has $\lambda \approx 500\,\text{nm} \gg 2d = 0.56\,\text{nm}$, so $\sin\theta > 1$: no Bragg reflection is possible.

19.2 $\tau = m\sigma/ne^2 = (9.11 \times 10^{-31})(6 \times 10^7)/[(8.5 \times 10^{28})(1.6 \times 10^{-19})^2] = 2.5 \times 10^{-14}$ s. The mean free path is $v_F\tau = 40$ nm, about 160 atomic spacings: electrons pass many ions without scattering, as Bloch's theorem says they should.

19.3 Visible photons span 1.7–3.1 eV. *Silicon* (1.1 eV) absorbs all of them and looks grey and opaque; an LED would emit infrared near 1.1 μm (and a poor one, since silicon's gap is indirect). *Gallium nitride* (3.4 eV) is transparent; it emits near-ultraviolet, and alloyed with indium it gives the blue LEDs behind white LED lighting (Nobel Prize, 2014). *Cadmium sulfide* (2.4 eV) absorbs violet and blue and looks yellow-orange; it would emit green. *Diamond* (5.5 eV) is transparent, and would emit deep ultraviolet.

19.4 The band runs from $E_0 - 2t$ to $E_0 + 2t$: width $4t$. $v = \hbar^{-1}\,\mathrm{d}E/\,\mathrm{d}k = (2ta/\hbar)\sin ka$. Near $k = 0$, $E \approx E_0 - 2t + ta^2k^2$, which matches $\hbar^2k^2/2m^*$ for $m^* = \hbar^2/2ta^2$. At the zone boundary $k = \pi/a$, $v = 0$: Bragg reflection turns the electron into a standing wave. Near the top of the band, the curvature is negative, so the effective mass is negative: an electron there accelerates opposite to the applied force. That is why it is simpler to describe the few empty states near the top as positively charged holes with positive mass.

19.5 Modes with $|\mathbf{k}| < k_T = k_\mathrm{B}T/\hbar v$ fill a sphere in k-space containing $V \cdot \frac{4}{3}\pi k_T^3/(2\pi)^3$ wavevectors, times three polarizations: a number $\propto VT^3$. Each holds energy $\sim k_\mathrm{B}T$, so $U \propto VT^4$ and $C = \mathrm{d}U/\mathrm{d}T \propto T^3$. Photons are counted identically (two polarizations, speed c), so their energy density grows as T^4, and so does the flux they radiate: the Stefan–Boltzmann law. A cold crystal is a box of thermal sound.

19.6 $\exp[(E_g/2k_\mathrm{B})(1/300 - 1/350)] = \exp[(6380\,\mathrm{K})(4.8\times10^{-4}\,\mathrm{K}^{-1})] \approx e^{3.0} \approx 20$ (about 26 including the $T^{3/2}$ prefactor). Heat increases the number of intrinsic carriers exponentially, swamping the designed doping and multiplying leakage currents, which heat the device further. Doping fixes the carrier density at a value set by the number of dopant atoms, nearly independent of temperature, which is what makes devices predictable.

19.7 $\mu_B B/k_\mathrm{B}T = 5.8\times10^{-5}/0.0259 = 2.2\times10^{-3}$, and the net alignment is $\tanh(\mu_B B/k_\mathrm{B}T) \approx 0.2\%$. Alignment reaches 50% when $\tanh x = 0.5$, i.e. $x = 0.55$, at $T = 1.2\,\mathrm{K}$. Nuclear moments are about 1000 times smaller, so the nuclear polarization in an MRI scanner is only a few parts per million.

19.8 $h/2e = 2.07\times10^{-15}\,\mathrm{Wb}$. A thousandth of this through $1\,\mathrm{mm}^2$ is $2\times10^{-12}\,\mathrm{T}$: about 10^7 times smaller than the Earth's field, but still twenty times larger than the brain's fields. Magnetoencephalography uses SQUIDs with larger pickup coils, longer averaging, and magnetically shielded rooms to reach the femtotesla level.

19.9 $\lambda_L = \sqrt{m^*/\mu_0 n^* e^{*2}} = \sqrt{2m_e/(\mu_0\cdot10^{28}\cdot4e^2)} \approx 40\,\mathrm{nm}$. A magnetic field dies away within a few tens of nanometres of a superconductor's surface.

19.10 $V_H = IB/(net) = (10)(1)/[(8.5\times10^{28})(1.6\times10^{-19})(10^{-4})] \approx 7\,\mu\mathrm{V}$. In a semiconductor the carrier density is a million to a hundred million times lower, so the Hall voltage is millivolts to volts: easy to measure, and the basis of the Hall sensors in phones and cars.

19.11 Label each cell's two atoms A and B, with hopping t_1 between A and B in the same cell and t_2 between B and the next cell's A. Bloch states $(a, b)e^{ikjd}$, with cell length $d = 2a$, satisfy $Ea = -(t_1 + t_2 e^{-ikd})b$ and $Eb = -(t_1 + t_2 e^{ikd})a$. Multiplying, $E^2 = \left|t_1 + t_2 e^{ikd}\right|^2 = t_1^2 + t_2^2 + 2t_1 t_2 \cos kd$, so $E = \pm\left|t_1 + t_2 e^{ikd}\right|$. The smallest magnitude, $|t_1 - t_2|$, occurs at $kd = \pi$, leaving a gap of $2\,|t_1 - t_2|$ between the bands. With one electron per atom the lower band is exactly full: the chain is an insulator. In one dimension, the electronic energy gained by opening the gap always outweighs the elastic cost of the distortion, so a half-filled chain dimerizes spontaneously.

19.12 Threading flux Φ through the cylinder induces an EMF $\mathcal{E} = -\,\mathrm{d}\Phi/\mathrm{d}t$ around it, which drives a Hall current $I = \sigma_{xy}\mathcal{E}$ along the cylinder, from one edge to the other. Adding exactly $\Phi_0 = h/e$ changes every electron's wavefunction by a phase that can be removed by a gauge transformation, so the Hamiltonian and its spectrum return to their original form. If the bulk has an energy gap and the flux is added slowly, the system returns to its ground state, except that some charge may have been moved from edge to edge — and since electrons are indivisible, it must be an integer number ν of them. So $\nu e = \int I\,\mathrm{d}t = \sigma_{xy}\int \mathcal{E}\,\mathrm{d}t = \sigma_{xy}h/e$, and $\sigma_{xy} = \nu e^2/h$. The quantization comes from the integrality of charge and the gap; disorder and sample shape never enter.

20 Quantum information

20.1 $|0\rangle$ is the north pole and $|1\rangle$ the south pole; $|+\rangle$ and $|-\rangle$ lie on the equator along $+x$ and $-x$, and $(|0\rangle + i|1\rangle)/\sqrt{2}$ along $+y$. Writing $H = (X+Z)/\sqrt{2}$, we get $H^2 = (X^2 + Z^2 + XZ + ZX)/2 = I$, since $X^2 = Z^2 = I$ and $XZ = -ZX$. Geometrically, H is a half-turn about the axis halfway between x and z: it swaps the x and z axes (so $|0\rangle \leftrightarrow |+\rangle$ and $|1\rangle \leftrightarrow |-\rangle$) and reverses y. Doing a half-turn twice gives the identity.

20.2 $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ has no $|01\rangle$ or $|10\rangle$ component, so the results always agree. Substituting $|0\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$ and $|1\rangle = (|+\rangle - |-\rangle)/\sqrt{2}$, the cross terms cancel and $|\Phi^+\rangle = (|++\rangle + |--\rangle)/\sqrt{2}$: the results agree in this basis too. No classical correlation does this in every basis at once.

20.3 $2^{50} \approx 1.1\times10^{15}$ amplitudes, or 18 petabytes — the size of a large supercomputer's memory. For 300 qubits, $2^{300} \approx 2 \times 10^{90}$ amplitudes, more than the number of atoms in the observable universe ($\sim 10^{80}$).

20.4 Take the inner product of $U|\phi\rangle|0\rangle = |\phi\rangle|\phi\rangle$ with $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$. Since U is unitary, the left side gives $\langle\psi|\phi\rangle\langle 0|0\rangle = \langle\psi|\phi\rangle$, and the right side gives $\langle\psi|\phi\rangle^2$. So $x = x^2$ with $x = \langle\psi|\phi\rangle$, which forces $x = 0$ or 1: the machine can clone only states that are identical or orthogonal, i.e. a known set of classical alternatives.

20.5 $I|\Phi^+\rangle = |\Phi^+\rangle$; X_1 gives $(|10\rangle + |01\rangle)/\sqrt{2} = |\Psi^+\rangle$; Z_1 gives $(|00\rangle - |11\rangle)/\sqrt{2} = |\Phi^-\rangle$; and X_1Z_1 gives $(|10\rangle - |01\rangle)/\sqrt{2} = -|\Psi^-\rangle$. The four Bell states are orthogonal, so Bob can distinguish them with certainty: Alice's choice of one of four operations, two bits, arrives on a single qubit. (The other qubit, shared in advance, is the second "bit".)

20.6 Expand: $(\alpha|0\rangle + \beta|1\rangle)(|00\rangle + |11\rangle)/\sqrt{2} = (\alpha|00\rangle|0\rangle + \alpha|01\rangle|1\rangle + \beta|10\rangle|0\rangle + \beta|11\rangle|1\rangle)/\sqrt{2}$, with Alice's two qubits written first. Substitute $|00\rangle = (|\Phi^+\rangle + |\Phi^-\rangle)/\sqrt{2}$, $|11\rangle = (|\Phi^+\rangle - |\Phi^-\rangle)/\sqrt{2}$, $|01\rangle = (|\Psi^+\rangle + |\Psi^-\rangle)/\sqrt{2}$, and $|10\rangle = (|\Psi^+\rangle - |\Psi^-\rangle)/\sqrt{2}$, and collect terms: this gives the stated expression. If Alice finds Φ^+, Bob does nothing; Φ^-, he applies Z; Ψ^+, he applies X; Ψ^-, he applies X and then Z (giving $-(\alpha|0\rangle+\beta|1\rangle)$, an irrelevant overall sign). Until Alice's two classical bits arrive, Bob's qubit is in an equal mixture of the four possibilities, which is maximally mixed and carries no information: nothing travels faster than light.

20.7 All four amplitudes start at $\frac{1}{2}$. Flipping the marked one gives $(-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, with average $\frac{1}{4}$. Reflecting about the average sends $a \mapsto 2\cdot\frac{1}{4} - a$: the marked amplitude becomes 1 and the others 0. One query finds the item with certainty, where a classical search needs 2.25 queries on average.

20.8 The correlation is $-\cos(\theta_a - \theta_b)$. With the given angles: $\langle AB\rangle = -\cos 45°$, $\langle AB'\rangle = -\cos 45°$, $\langle A'B\rangle = -\cos 45°$, and $\langle A'B'\rangle = -\cos 135° = +\cos 45°$. So $S = |-4\cos 45°| = 2\sqrt{2}$. With $\hat{\mathbf{b}}'$ at 135°, $\langle AB'\rangle = +\cos 45°$ and $\langle A'B'\rangle = -\cos 45°$, and the four terms cancel to $S = 0$. The signs in S must line up with the geometry: each pair of axes used with a plus sign must be 45° apart, and the pair with the minus sign 135° apart.

20.9 $\rho_+ = |+\rangle\langle+| = \frac{1}{2}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$ and $\rho_{\text{mix}} = \frac{1}{2}\left(\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right)$. Both have diagonal entries $\frac{1}{2}$, so a Z-measurement gives 0 or 1 with equal probability. For X, $\langle+|\rho_+|+\rangle = 1$ but $\langle+|\rho_{\text{mix}}|+\rangle = \frac{1}{2}$: the pure state always gives +. The off-diagonal entries (the "coherences") are what distinguish a superposition from a mixture. $\rho_+^2 = \rho_+$ has trace 1; $\rho_{\text{mix}}^2 = \frac{1}{4}I$ has trace $\frac{1}{2}$.

20.10 Alice and Bob keep only bits where they used the same basis. Eve guesses that basis right half the time; then she reads the bit correctly and resends it unchanged, causing no error. Half the time she guesses wrong; she then sends a state in the wrong basis, and Bob's measurement in the right one gives a random result, wrong with probability $\frac{1}{2}$. The overall error rate is $\frac{1}{2}\times\frac{1}{2} = 25\%$. Comparing a sample of their bits in public, Alice and Bob would spot this at once.

20.11 Both $|000\rangle$ and $|111\rangle$ have $Z_1Z_2 = Z_2Z_3 = +1$. A flip on qubit 1 makes both $|100\rangle$ and $|011\rangle$, which have $(Z_1Z_2, Z_2Z_3) = (-1,+1)$; similarly qubit 2 gives $(-1,-1)$ and qubit 3 gives $(+1,-1)$. The four syndromes are distinct, so they identify the error, which is then undone by applying the same X. Crucially, both branches of the superposition give the *same* syndrome, so the state is an eigenstate of the measured observables and is not disturbed: the measurement reveals which qubit flipped but nothing about α or β. If two qubits flip, say 1 and 2, the syndrome $(+1,-1)$ blames qubit 3; "correcting" it gives $\alpha|111\rangle + \beta|000\rangle$, a logical error. A phase error Z_1 gives $\alpha|000\rangle - \beta|111\rangle$, which has the same syndrome as no error at all and is itself a logical error. Shor's nine-qubit code protects against both by encoding each of these three qubits again, in the $|\pm\rangle$ basis, where phase flips look like bit flips.

20.12 The box acting on $|x\rangle|-\rangle = |x\rangle(|0\rangle - |1\rangle)/\sqrt{2}$ gives $|x\rangle(|f(x)\rangle - |1\oplus f(x)\rangle)/\sqrt{2}$, which is $|x\rangle|-\rangle$ if $f(x) = 0$ and $-|x\rangle|-\rangle$ if $f(x) = 1$: so it is $(-1)^{f(x)}|x\rangle|-\rangle$. After the two Hadamards, the state is $|+\rangle|-\rangle$; after the box, the first qubit is

$$\tfrac{1}{\sqrt{2}}[(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle],$$

which is $\pm|+\rangle$ if $f(0) = f(1)$ and $\pm|-\rangle$ if not. The final Hadamard turns these into $|0\rangle$ and $|1\rangle$. One query answers a question about a global property of f that classically needs two.

20.13 Write $C = A(B+B') + A'(B-B')$ and square, using $A^2 = A'^2 = B^2 = B'^2 = 1$ and the fact that Alice's operators commute with Bob's:

$$C^2 = (B+B')^2 + (B-B')^2 + AA'(B+B')(B-B') + A'A(B-B')(B+B').$$

The first two terms are $(2 + BB' + B'B) + (2 - BB' - B'B) = 4$. Next, $(B+B')(B-B') = B'B - BB' = -[B,B']$ and $(B-B')(B+B') = [B,B']$, so the last two terms are $-(AA' - A'A)[B,B'] = -[A,A'][B,B']$. Now $\|[A,A']\| \le \|AA'\| + \|A'A\| \le 2$, and likewise for B, so $\|C^2\| \le 4 + 4 = 8$. Hence $\|C\| \le 2\sqrt{2}$ and $|\langle C\rangle| \le 2\sqrt{2}$. In a classical (local hidden variable) theory all the observables commute, the commutators vanish, $C^2 = 4$, and $|\langle C\rangle| \le 2$: Bell's inequality. The violation is made possible by non-commutativity, and limited by it.

21 Chemical physics

21.1 $4.5\,\text{eV}\times 96.5\,(\text{kJ/mol})/\text{eV} = 434\,\text{kJ/mol}$. The Boltzmann factor reaches 10^{-10} when $E/k_\text{B}T = 23$, at $T \approx 2300\,\text{K}$. At room temperature $E/k_\text{B}T = 175$ and $e^{-175} \approx 10^{-76}$: even with 10^{13} attempts per second, a given molecule would wait vastly longer than the age of the universe. (Hydrogen gas does burn readily, but through chain reactions with much lower barriers, not by thermally breaking H_2.)

21.2 $k_2/k_1 = \exp[(E_a/R)(1/T_1 - 1/T_2)]$ with $E_a/R = 6010\,\text{K}$. From 300 to 310 K: a factor of 1.9 — the chemist's rule that rates double every 10°, and the reason a fever speeds up your metabolism. From 273 to 373 K: a factor of about 370, which is why food cooks.

21.3 $K = e^{-\Delta G^\circ/RT} = e^{8.07} \approx 3200$. For 99.9% completion $K = 999$, so $\Delta G^\circ = -RT \ln 999 = -17\,\text{kJ/mol}$. A few tens of kJ/mol, a few times RT in each of several factors, separates "hardly goes" from "goes to completion".

21.4 $e^2/(4\pi\epsilon_0 r) = 1.44\,\text{eV nm}/0.236\,\text{nm} = 6.1\,\text{eV}$. Making the ions costs $5.1 - 3.6 = 1.5\,\text{eV}$, so the binding energy is about 4.6 eV, 10% above the measured 4.2 eV. The difference is the short-range repulsion of the ions' filled electron shells (the Pauli principle), which keeps them apart and is what sets the bond length in the first place.

21.5 Normalize $\psi_\pm = (\psi_A \pm \psi_B)/\sqrt{2(1 \pm S)}$. Then

$$E_\pm = \langle\psi_\pm|\hat{H}|\psi_\pm\rangle = \frac{H_{AA} + H_{BB} \pm 2H_{AB}}{2(1 \pm S)} = \frac{H_{AA} \pm H_{AB}}{1 \pm S},$$

using $H_{BB} = H_{AA}$ by symmetry. Measure energies from H_{AA} (shifting $\hat{H}$ by a constant changes H_{AB} by S times that constant; this choice makes the formulas cleanest). Then $E_\pm = \pm H_{AB}/(1 \pm S)$: with $H_{AB} < 0$, the bonding orbital is lowered by $|H_{AB}|/(1+S)$ and the antibonding orbital raised by $|H_{AB}|/(1-S)$, which is larger since $0 < S < 1$.

21.6 $E_n = n^2h^2/8mL^2$, with $h^2/8mL^2 = 0.045\,\text{eV}$ for $L = 2.9\,\text{nm}$. The highest filled level is $n = 11$ and the lowest empty one $n = 12$, so $\Delta E = 23 \times 0.045 = 1.03\,\text{eV}$, or $\lambda = 1200\,\text{nm}$: infrared, nowhere near the observed 450 nm (2.8 eV). The box model treats the chain as a uniform wire, but its bonds alternate, single and double, so the hopping alternates, t_1 and t_2. With one π electron per carbon, the band is exactly half-filled, and Problem 19.11 showed that alternating hopping opens a gap of $2|t_1 - t_2|$ exactly at half-filling. For a long chain the box-model gap shrinks to zero, but the dimerization gap remains: in polyacetylene, the infinitely long version of the chain, it is about 1.5–2 eV. Added to carotene's finite-length spacing, it pushes the absorption into the blue, and the carrot reflects the orange that is left.

21.7 With $\Delta H = T_b \Delta S$: water $373 \times 88 = 33\,\text{kJ/mol}$; H_2S $213 \times 88 = 19\,\text{kJ/mol}$. (The measured values are 41 and 19 kJ/mol; water breaks Trouton's rule because hydrogen bonds also order the liquid, lowering its entropy.) Without hydrogen bonding, water would be like a lighter H_2S and would boil far below 0°C. The extra 14–22 kJ/mol, at 20 kJ/mol per bond, suggests about one hydrogen bond per molecule — each bond is shared between two molecules, so each molecule takes part in about two. Liquid water actually has nearer 3.5 bonds per molecule, many of them bent and weakened; the estimate catches the order of magnitude.

21.8 Electronic energies are $E_{\text{el}} \sim \hbar^2/m_e a^2$ for a molecule of size a. The bond's spring constant is $k \sim E_{\text{el}}/a^2$, so the vibrational quantum is

$$\hbar\omega = \hbar\sqrt{k/M} \sim \sqrt{\frac{\hbar^2}{Ma^2}E_{\text{el}}} = E_{\text{el}}\sqrt{\frac{m_e}{M}}.$$

The rotational quantum is $\hbar^2/Ma^2 = E_{\text{el}}(m_e/M) = \hbar\omega\sqrt{m_e/M}$. For a reduced mass of 10 u, $\sqrt{m_e/M} \approx 1/135$: electronic $\sim 10\,\text{eV}$ (visible and UV), vibrational $\sim 0.1\,\text{eV}$ (infrared), rotational $\sim 1\,\text{meV}$ (microwave). For CO, the measured values are 8 eV, 0.27 eV, and 0.5 meV. Each level of structure is a small perturbation on the one above, which is why molecular spectra can be organized at all.

21.9 The zero-point energies are $\frac{1}{2}\hbar\omega$: $1500\,\text{cm}^{-1}$ for C–H and $1061\,\text{cm}^{-1}$ for C–D, a difference of $439\,\text{cm}^{-1}$. At 298 K, $k_B T = 207\,\text{cm}^{-1}$, so the ratio of rates is $e^{439/207} = e^{2.1} \approx 8$. The C–H bond starts higher up its well, so it has less far to climb.

21.10 We have $\ln K = -\Delta H^\circ/RT + \Delta S^\circ/R$. Treating ΔH° and ΔS° as constant, differentiate to get $\mathrm{d}\ln K/\mathrm{d}T = \Delta H^\circ/RT^2$. If $\Delta H^\circ > 0$ (endothermic), K grows with T: heat is a reactant, so supplying it pushes the reaction forward (Le Chatelier). The Haber process, $N_2 + 3H_2 \to 2NH_3$, is exothermic, so the yield is best at low temperature, where the reaction is hopelessly slow. The compromise is about 450°C with an iron catalyst, and 200 atmospheres of pressure, which favours the side with fewer molecules (two instead of four); the ammonia is removed and the unreacted gas recycled.

21.11 Expand $\sqrt{1 \pm \kappa} = 1 \pm \frac{1}{2}\kappa - \frac{1}{8}\kappa^2 + \dots$. The first-order terms cancel in the sum, leaving

$$\tfrac{1}{2}\hbar(\omega_+ + \omega_-) = \hbar\omega_0(1 - \tfrac{1}{8}\kappa^2 + \dots), \qquad \Delta E = -\tfrac{1}{8}\hbar\omega_0\kappa^2 = -\frac{\hbar\omega_0}{2}\frac{\alpha^2}{(4\pi\epsilon_0)^2 r^6},$$

where $\alpha = e^2/m\omega_0^2$ is the atom's polarizability. (The three-dimensional calculation replaces $\frac{1}{2}$ by $\frac{3}{4}$; this is London's formula.) The attraction falls off as $1/r^6$. Classically, at zero temperature, both electrons would sit at rest, with no dipoles and no force. Quantum mechanically, their zero-point motions become correlated, so that the instantaneous dipoles tend to line up, and the energy of the correlated motion is slightly lower. This van der Waals attraction holds liquid helium together and helps geckos climb walls.

21.12 He_2: two electrons fill the bonding orbital and two the antibonding orbital. By Problem 5, the antibonding orbital is raised more than the bonding orbital is lowered, so the total energy is *above* that of two separate atoms: net repulsion. (The van der Waals attraction of Problem 11 does bind a helium dimer, but by only about a millikelvin, in a molecule some 5 nm across.) He_2^+ has two bonding electrons and one antibonding one: the net effect is binding, and He_2^+ is bound by about 2.5 eV. Bond orders: He_2 $(2-2)/2 = 0$; He_2^+ $(2-1)/2 = \frac{1}{2}$. For N_2, the ten valence electrons fill $\sigma_{2s}^2\sigma_{2s}^{*2}\pi_{2p}^4\sigma_{2p}^2$: eight bonding and two antibonding, a bond order of 3. O_2 has two more electrons, which go into the antibonding π^* orbitals, giving order $(8-4)/2 = 2$. Because the two π^* orbitals have the same energy, Hund's rule puts these two electrons in different orbitals with parallel spins: O_2 is paramagnetic, and liquid oxygen sticks to a magnet — a striking prediction of the molecular orbital picture that simple pictures of electron-pair bonds miss.

22 Nuclear physics

22.1 $R = 1.2 \times 197^{1/3} = 7.0$ fm. The density is $m_u/(\frac{4}{3}\pi(1.2\,\mathrm{fm})^3) = 2.3 \times 10^{17}\,\mathrm{kg/m^3}$, the same for every nucleus. A teaspoon would weigh 10^{12} kg, a billion tonnes. Neutron stars have similar densities, a few times $10^{17}\,\mathrm{kg/m^3}$: a neutron star is, roughly, one giant nucleus held together by gravity.

22.2 The constituents weigh $2(1.007825) + 2(1.008665) = 4.032980$ u; the deficit is 0.030378 u, or 28.3 MeV (7.07 MeV per nucleon). Using hydrogen atoms instead of bare protons makes the electron masses cancel.

22.3 25% is two half-lives: 11 500 years. 1% is $\log_2 100 = 6.6$ half-lives: 38 000 years. The dinosaurs died 66 million years ago, 11 500 half-lives: the remaining fraction, $2^{-11\,500}$, means not a single atom of the original carbon-14 is left. Dinosaur bones are dated from the uranium and potassium isotopes in the surrounding volcanic rock.

22.4 1 kg of U-235 has 2.6×10^{24} nuclei, which release 8×10^{13} J — nearly three million times as much as 1 kg of coal. A power station producing 1 GW of heat fissions about 1 kg per day; one producing 1 GW of electricity at 33% efficiency, about 3 kg. (A coal plant of the same output burns about 10 000 tonnes a day.)

22.5 The carrier's mass is $mc^2 \approx \hbar c/\text{range}$, which is 197 MeV fm/1.4 fm = 140 MeV: the pion. For the weak force, 197 MeV fm/$(2\times10^{-3}$ fm$) \approx 100$ GeV, in excellent agreement with the W boson.

22.6 $\partial B/\partial Z = -2a_C Z/A^{1/3} + 4a_A(A-2Z)/A = 0$ gives $Z = A/(2 + (a_C/2a_A)A^{2/3})$, with $a_C/2a_A = 0.71/46.4 = 0.015$. For $A = 56$, $Z = 25$ (iron is 26); for $A = 238$, $Z = 92$, exactly uranium. Without the Coulomb term, the asymmetry term would make $Z = A/2$; Coulomb repulsion pushes heavy nuclei toward a neutron excess.

22.7 The Sun completes $L/(26.7\,\text{MeV}) = 8.9\times10^{37}$ reactions per second, emitting 1.8×10^{38} neutrinos per second. Spread over a sphere of radius 1 AU, that is 6×10^{14} per square metre, or 6×10^{10} per square centimetre per second — sixty billion through your fingernail every second, day and night (they pass through the Earth too). Their weak-interaction cross-section is around 10^{-44} cm^2, so the mean free path in lead is of order light-years; the chance that a given neutrino interacts in your body is about 10^{-20}.

22.8 Splitting A into two halves of $A/2$ changes the surface term by $a_S A^{2/3}(2\cdot2^{-2/3}-1) = 0.26\,a_S A^{2/3} = +183$ MeV (a cost: more surface), and the Coulomb term by $a_C Z^2 A^{-1/3}(1-2\cdot2^{-2}\cdot2^{1/3}) = 0.37\,a_C Z^2/A^{1/3} = -359$ MeV (a gain: the halves are farther apart). The energy released is about 180 MeV. Coulomb repulsion drives fission; surface tension resists it, which is what creates the barrier that keeps heavy nuclei from splitting spontaneously all the time.

22.9 One gram of radium is 2.7×10^{21} atoms, and $\lambda = \ln 2/t_{1/2} = 1.37\times10^{-11}$ s^{-1}, so the activity is 3.7×10^{10} decays per second — the curie. Your potassium-40 is 0.017 g, or 2.5×10^{20} atoms, with $\lambda = 1.76\times10^{-17}$ s^{-1}: about 4400 decays per second, all the time, in your own body.

22.10 The half-life is inversely proportional to the tunnelling probability (times an attempt rate that varies little), so $\ln t_{1/2} = 2\pi\eta + \text{const}$. Since $v = \sqrt{2E/m}$, $\eta \propto Z/\sqrt{E}$, which is the Geiger–Nuttall law. Numerically, $2\pi\eta = 4\pi Z\alpha/(v/c)$, with $v/c = \sqrt{2E/m_\alpha c^2}$ and $m_\alpha c^2 = 3727$ MeV. For $Z = 84$: at 5 MeV, $2\pi\eta = 149$; at 9 MeV, 111. The half-lives differ by $e^{38} \approx 10^{16}$. For real polonium isotopes, Po-210 (5.3 MeV, 138 days) and Po-212 (8.95 MeV, 0.3 µs), the formula predicts a ratio of $e^{33} \approx 3\times10^{14}$, against the measured 4×10^{13}: thirteen orders of magnitude captured to within a factor of ten, by a one-line formula. (The daughter nucleus, lead, has $Z = 82$, which makes the agreement slightly better still.)

22.11 Neutrons diffuse with $D = \ell v/3$ and multiply at the rate $(\nu-1)/\tau$, where $\tau = \ell/v$ is the time between collisions (counting each collision as a fission, which is generous). The neutron density obeys

$$\frac{\partial n}{\partial t} = D\nabla^2 n + \frac{\nu-1}{\tau}n.$$

In a sphere with $n = 0$ at the surface, the slowest-decaying mode is $n \propto \sin(\pi r/R)/r$, with $\nabla^2 n = -(\pi/R)^2 n$. It grows if $(\nu-1)/\tau > D\pi^2/R^2$, i.e. for

$$R > R_c = \pi\sqrt{\frac{D\tau}{\nu-1}} = \frac{\pi\ell}{\sqrt{3(\nu-1)}}.$$

For $\ell = 4\,\text{cm}$ and $\nu = 2.5$, $R_c \approx 6\,\text{cm}$ and the mass is $\frac{4}{3}\pi R_c^3 \times 19\,\text{g/cm}^3 \approx 17\,\text{kg}$. (The true bare-sphere value is 52 kg, with $R = 8.5\,\text{cm}$: not every collision causes a fission.) A reflector sends escaping neutrons back in, effectively removing the boundary condition $n = 0$; it can reduce the critical mass by a factor of two or three.

22.12 Binding energy per nucleon peaks near iron and nickel, so fusing iron into anything heavier absorbs energy instead of releasing it. A core of iron can no longer support the star: it collapses, and the result is a supernova. The Coulomb barrier for a proton to enter a $Z = 50$ nucleus ($R \approx 1.2(1 + 120^{1/3})\,\text{fm} = 7\,\text{fm}$) is $50 \times 1.44\,\text{MeV fm}/7\,\text{fm} \approx 10\,\text{MeV}$, while even at $3 \times 10^9\,\text{K}$, $k_\text{B}T \approx 0.26\,\text{MeV}$: tunnelling is hopeless. Neutrons feel no Coulomb barrier, so they are absorbed at any energy. In the slow s-process (in old red giants), each neutron capture is followed by beta decay before the next capture, building up the stable nuclei one step at a time. It stops at bismuth-209: capturing a neutron gives bismuth-210, which decays to polonium-210, which alpha-decays back to lead-206, so the chain cycles. The heaviest elements, thorium and uranium, need the r-process, where neutrons are captured faster than the nuclei can decay — in neutron star mergers (Chapter 29) and perhaps some supernovae.

23 Biophysics

23.1 $k_\text{B}T = 4.3 \times 10^{-21}\,\text{J} = 4.3\,\text{pN nm} = 27\,\text{meV}$. Kinesin does at most $8\,\text{nm} \times 6\,\text{pN} = 48\,\text{pN nm}$ of work per step, from $20k_\text{B}T = 86\,\text{pN nm}$: an efficiency of about 56%, better than a car engine.

23.2 With $t \sim L^2/D$: 0.1 s, 40 s, and 10^{11} s, or 3000 years. Kinesin covers 1 m in 10^6 s, about 12 days (the slowest axonal transport takes months). Active transport wins when $L/v < L^2/D$, for $L > D/v = 10\,\mu\text{m}$: about the size of a cell.

23.3 $6 \times 10^9 \times 0.34\,\text{nm} = 2\,\text{m}$, about 3×10^5 times the nucleus's diameter. As a random coil of $N = 2\,\text{m}/100\,\text{nm} = 2 \times 10^7$ segments, its size would be $\sqrt{N}b \approx 0.45\,\text{mm}$, still 75 times too large. DNA is actively packaged: wound around histone proteins, then coiled and looped into chromatin, in a way that still lets the cell read any gene on demand.

23.4 $V = (k_\text{B}T/e)\ln(c_\text{out}/c_\text{in})$, with $k_\text{B}T/e = 26.7\,\text{mV}$. Potassium: $26.7\ln(5/140) = -89\,\text{mV}$. Sodium: $26.7\ln(145/12) = +67\,\text{mV}$. At rest the membrane is mostly permeable to potassium, so its potential sits near $-70\,\text{mV}$, close to the potassium value. When sodium channels open, sodium becomes the dominant permeability, and it rushes in down both its concentration gradient and the electric field, driving the potential toward +67 mV: the spike.

23.5 $S = -3k_\text{B}R^2/2Nb^2 + \text{const}$, so $F = -T\,\partial S/\partial R = 3k_\text{B}TR/Nb^2$: the force needed to hold the ends a distance R apart. The polymer behaves as a Hookean spring of stiffness $3k_\text{B}T/Nb^2$, which pulls the ends together. The force is entirely entropic: stretching reduces the number of available configurations. The stiffness is proportional to T because the free energy cost of losing entropy is $T\Delta S$. A heated rubber band contracts, unlike almost any other material — try it with a hair dryer and a weight.

23.6 The drag coefficient is $6\pi\eta r = 6\pi(10^{-3})(0.5 \times 10^{-6}) = 9.4 \times 10^{-9}\,\text{kg/s}$, so the stopping time is $m/6\pi\eta r = 10^{-7}\,\text{s}$, and the coasting distance $v\tau = 3 \times 10^{-12}\,\text{m}$: a thirtieth of the diameter of a hydrogen atom. A bacterium lives at a Reynolds number of 10^{-5}, where inertia is irrelevant: it stops dead the instant it stops swimming.

23.7 $3^{100} \approx 5 \times 10^{47}$ conformations, which at 10^{12} per second takes $5 \times 10^{35}\,\text{s}$, or 10^{28} years

— 10^{18} times the age of the universe. A protein that folds in 1 s can sample only about 10^{12} conformations, so the effective search space must be reduced by a factor of about 10^{35} (or 10^{41} for microsecond folders). Folding is not a random search: the energy landscape is a funnel, and the protein slides downhill, forming local structures that guide the rest.

23.8 Metabolic rate per kilogram scales as $M^{-1/4}$, so the mouse's is $(5000/0.02)^{1/4} = 22$ times the elephant's. A mouse at 600 beats per minute uses up 10^9 heartbeats in 3.2 years (real mice live 2–3 years). If heart rate also scales as $M^{-1/4}$, the elephant's is $600/22 = 27$ beats per minute (measured: about 30), and it lives about 70 years (real elephants: 60–70).

23.9 A squid's giant axon conducts at about 25 m/s. Myelinated axons conduct at roughly 6 m/s per micrometre of diameter, so a myelinated axon about 4 µm across matches it: some 100 times thinner, with 10^4 times less cross-section. A vertebrate nerve can pack thousands of fast fibres into the space of one squid axon, which is what makes complex nervous systems — and big brains — affordable.

23.10 A molecule near the receptor wanders away in a time $\sim a^2/D$, and in that time the receptor holds about ca^3 molecules. In a time T it gets TD/a^2 independent looks, so it counts about $ca^3 \cdot TD/a^2 = DacT$ molecules altogether — the same as the diffusive arrival rate $\sim Dac$ times T. Poisson statistics give an error $\sqrt{N}$ in a count N, so $\delta c/c \sim 1/\sqrt{DacT}$. For a small molecule, $D \approx 10^{-9}\,\mathrm{m}^2/\mathrm{s}$, and $c = 1\,\mu\mathrm{M} = 6 \times 10^{20}\,\mathrm{m}^{-3}$. With $a = 1\,\mu\mathrm{m}$ and $T = 1\,\mathrm{s}$, $DacT \approx 6 \times 10^5$ and $\delta c/c \approx 0.1\%$. Measuring for longer helps, but a bacterium swimming in a straight line is knocked off course by rotational diffusion within about 10 s, so it cannot average much longer; and it must swim far enough in that time to outrun diffusion, or it could not tell where the concentration was higher.

23.11 Suppose a forward step requires the gas to supply energy ϵ (to lift the pawl over a tooth) plus $L\theta$ (to lift the weight), with probability $\propto e^{-(\epsilon+L\theta)/k_\mathrm{B}T_1}$ per attempt. A backward step happens when the pawl, at temperature T_2, is thermally lifted by ϵ (probability $\propto e^{-\epsilon/k_\mathrm{B}T_2}$), letting the weight turn the ratchet back. With no load and $T_1 = T_2$, the rates are equal: no net rotation, as the second law requires. In general, the ratchet turns forward when $(\epsilon + L\theta)/T_1 < \epsilon/T_2$. Each forward step takes heat $Q_1 = \epsilon + L\theta$ from the gas, does work $W = L\theta$, and delivers $Q_2 = \epsilon$ to the pawl (when it snaps down). Its efficiency, $W/Q_1 = 1 - \epsilon/(\epsilon + L\theta)$, is less than $1 - T_2/T_1$ exactly when it runs forward: never better than Carnot, and equal to Carnot when the device is balanced and runs infinitely slowly. Kinesin does not run on heat at all. It runs on chemical free energy: the cell keeps ATP far from equilibrium with ADP and phosphate, and each hydrolysis, by changing the motor's shape, switches the potential landscape the motor head sits in. Thermal motion moves the head; the irreversible chemical cycle rectifies it, paying about $20k_\mathrm{B}T$ a step.

23.12 If the right and wrong substrates bind at the same rate but the wrong one falls off faster by $e^{\Delta G/k_\mathrm{B}T}$, equilibrium occupancy discriminates by $e^{-\Delta G/k_\mathrm{B}T}$. After the activation step, the complex either makes product at rate W or releases the substrate, at a rate ℓ' that again is $e^{\Delta G/k_\mathrm{B}T}$ times larger for the wrong one. If $W \ll \ell'$, the fraction making product is W/ℓ', a second factor of $e^{-\Delta G/k_\mathrm{B}T}$: the error rate becomes $e^{-2\Delta G/k_\mathrm{B}T}$. A substrate that falls off the activated complex cannot climb directly back into it: it must start over with binding. This one-way flow is the point. If the activation step were reversible and at equilibrium, the activated complex would be just one more bound state, whose population is fixed by its free energy alone — and equilibrium can never discriminate by more than $e^{-\Delta G/k_\mathrm{B}T}$, however many states there are. Driving the cycle by GTP or ATP hydrolysis breaks detailed balance. The price is energy, and some correct substrates

discarded: ribosomes and DNA polymerases trade both for accuracy.

24 Medical physics

24.1 $e^{-0.2\times20} = e^{-4} = 1.8\%$; $e^{-0.2\times18-0.6\times2} = e^{-4.8} = 0.8\%$. Behind bone the film receives less than half the exposure. Film darkens where X-rays hit it, so a radiograph is a negative: bone, which casts an X-ray shadow, shows up white.

24.2 $f = 42.58\,\text{MHz/T} \times B$: 64 MHz at 1.5 T and 128 MHz at 3 T, in the radio band, right beside FM stations (88–108 MHz). The signal from the body is minute, and any stray radio transmission at these frequencies would swamp it, so the room is a Faraday cage. (It also keeps the scanner's powerful radio pulses inside.)

24.3 $R = ((Z_2 - Z_1)/(Z_2 + Z_1))^2$. Tissue–bone: $(6.2/9.4)^2 = 43\%$ reflected. Air–tissue: 99.9% reflected. Without gel, a thin layer of air between the probe and the skin would reflect almost all the sound before it entered the body; the gel matches impedances and squeezes the air out.

24.4 $\lambda = v/f = 1540/(5 \times 10^6) = 0.3\,\text{mm}$, which is about the finest detail resolvable. Attenuation in tissue grows roughly in proportion to frequency (about 0.5 dB/cm per MHz), so higher frequencies give finer detail but penetrate less. The liver lies 10–20 cm deep and needs 3 MHz (0.5 mm resolution); the eye is only a few centimetres deep, so 20 MHz (0.08 mm) can be used.

24.5 For a positron and electron annihilating at rest, the total momentum is zero and the total energy $2m_ec^2 = 1022\,\text{keV}$. A single photon has momentum $E/c \neq 0$, so it cannot carry away zero momentum. Two photons can if they travel in opposite directions with equal momenta, so equal energies, 511 keV each. (The pair's small residual motion tilts them by about a quarter of a degree.) A timing difference Δt locates the annihilation to $\Delta x = c\Delta t/2 = 4.5\,\text{cm}$ along the line. This "time of flight" information doesn't replace the reconstruction, but it reduces the noise in it.

24.6 $\Delta E/k_\text{B}T = h(128\,\text{MHz})/k_\text{B}(310\,\text{K}) = 2.0 \times 10^{-5}$. The fractional excess is half of this, 1.0×10^{-5}: ten parts per million. A cubic millimetre of water has 6.7×10^{19} protons, so the excess is 7×10^{14}. That's still a large number, which is why MRI works at all. Doubling B doubles the polarization and also doubles the frequency, and so the voltage induced in the coils: the signal grows roughly as B^2, and the signal-to-noise ratio about linearly with B, buying finer resolution or faster scans.

24.7 $\Delta B = (10\,\text{mT/m})(1\,\text{mm}) = 10^{-5}\,\text{T}$, so $\Delta f = 426\,\text{Hz}$. A 10 ms recording resolves frequency differences of about $1/T = 100\,\text{Hz}$, so yes, comfortably — down to about 0.25 mm.

24.8 $2\,\text{Gy} \times 1\,\text{kg} = 2\,\text{J}$, warming the tumour by $2/4186 \approx 5 \times 10^{-4}\,\text{K}$. Per cell, $2 \times 10^{-12}\,\text{J}/30\,\text{eV} \approx 4 \times 10^5$ ionizations. The energy is delivered not as heat, a few hundredths of an electron volt per molecule, but in packets of tens of electron volts, each enough to break several chemical bonds. A few of these hit DNA: 2 Gy causes about 80 double-strand breaks per cell, and a cell that cannot repair them dies when it next tries to divide — which is why fast-dividing tumours are vulnerable, and why the therapy is split into many sessions to let healthy tissue repair.

24.9 24 hours is 4 half-lives: $500/16 = 31\,\text{MBq}$. A few hours is long enough to inject the patient and image them, but short enough that the radioactivity (and the dose) is soon gone. Hospitals make Tc-99m on site from the decay of molybdenum-99 (half-life 66 hours), whose "generators" are shipped weekly.

24.10 With $\mathrm{d}E/\,\mathrm{d}x \propto -1/E$, $E\,\mathrm{d}E \propto -\mathrm{d}x$, so $E^2 = E_0^2(1 - x/R)$. The dose is proportional to the stopping power,

$$\frac{\mathrm{d}E}{\mathrm{d}x} \propto \frac{1}{E} \propto \frac{1}{\sqrt{1 - x/R}},$$

nearly flat for most of the path, rising steeply in the last centimetre, and zero beyond the range R. This is the Bragg peak (in reality, random fluctuations in energy loss smear it over a few millimetres). Tumours are irradiated by stacking peaks of different energies, while sparing tissue behind. But if the anatomy differs from the plan by a centimetre (a filled bowel, weight loss, a shrunken tumour), the peak lands in the wrong place, possibly in a critical organ right behind the tumour. X-rays, whose dose falls off gradually, are much more forgiving.

24.11 Write $s = x\cos\theta + y\sin\theta$, and at each angle the projection is $\delta(s)$. Backprojecting gives

$$b(\mathbf{r}) = \int_0^\pi \delta(x\cos\theta + y\sin\theta)\;\mathrm{d}\theta = \int_0^\pi \delta(r\cos(\theta - \phi))\;\mathrm{d}\theta = \frac{1}{r},$$

since the argument vanishes once in the range, where its derivative has magnitude r. Now the projection–slice theorem says the one-dimensional Fourier transform of the projection at angle θ is the two-dimensional transform $F(\mathbf{k})$ of the image along the line at angle θ. Inverting in polar coordinates,

$$f(\mathbf{r}) = \int \frac{\mathrm{d}^2k}{(2\pi)^2} F(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}} = \frac{1}{(2\pi)^2}\int_0^\pi \mathrm{d}\theta \int_{-\infty}^{\infty} |k|\;\mathrm{d}k\; F(k,\theta) e^{iks}.$$

The inner integral is the projection with its Fourier transform weighted by $|k|$, and the outer integral is a backprojection. Without the factor $|k|$, the Jacobian of polar coordinates, low spatial frequencies are overcounted (the slices all cross at $k = 0$), which is the $1/r$ blur: its Fourier transform is $\propto 1/|k|$. The ramp filter boosts high frequencies and with them noise, so real scanners roll it off at the highest frequencies.

24.12 With saturation s and total haemoglobin concentration c along a path L, the absorbance at wavelength λ is $A_\lambda = cL[s\epsilon_{\mathrm{ox}}(\lambda) + (1-s)\epsilon_{\mathrm{de}}(\lambda)]$. The ratio $R = A_{660}/A_{940}$ is independent of c and L, and solving for s,

$$s = \frac{\epsilon_{\mathrm{de}}(660) - R\,\epsilon_{\mathrm{de}}(940)}{R[\epsilon_{\mathrm{ox}}(940) - \epsilon_{\mathrm{de}}(940)] - [\epsilon_{\mathrm{ox}}(660) - \epsilon_{\mathrm{de}}(660)]}.$$

But the finger also contains skin, bone, tissue, and venous blood, all absorbing; and the LED brightness and detector sensitivity are unknown. None of these change with the heartbeat. Only the arterial blood volume pulses, so the pulsing part of the signal (divided by the steady part, at each wavelength) isolates the arterial blood. Taking the ratio between wavelengths of these normalized pulses removes everything else. Real devices are calibrated empirically against blood samples, since light scattering in tissue complicates the simple Beer–Lambert picture — and skin pigmentation can bias the reading, a problem that came to wide attention during the COVID pandemic.

25 Geophysics and climate

25.1 $g = GM/R^2$ with $M = \frac{4}{3}\pi R^3\rho$ gives $\rho = 3g/4\pi GR = 5.5\,\mathrm{g/cm^3}$, twice the density of surface rock. The interior must be much denser than the crust: a core rich in iron, which seismology later confirmed (Problem 25.10).

25.2 $P = P_0 e^{-z/H}$ is halved at $z = H\ln 2 = 5.5\,\text{km}$. Since the pressure at any height is the weight of the air above it, the fraction of the atmosphere above Everest is $e^{-8.85/8} = 33\%$. Climbers on the summit are above two-thirds of the air.

25.3 $v = \sqrt{gh} = \sqrt{9.8 \times 4000} = 200\,\text{m/s}$, the speed of a jet airliner. Crossing 17 000 km takes about 24 hours. (The 1960 Chilean tsunami reached Japan in 22 hours and killed 140 people there.) Seismic waves travel at kilometres per second and warnings at the speed of light, so distant coasts can be warned hours in advance.

25.4 $\sigma T_e^4 = S(1-A)/4$. Earth: $T_e = (1361 \times 0.7/4\sigma)^{1/4} = 255\,\text{K}$. Venus: $T_e = (2600 \times 0.23/4\sigma)^{1/4} = 227\,\text{K}$ — *colder* than Earth, because its clouds reflect most of the sunlight. Yet its surface is at 737 K. The difference of 500 K is all greenhouse effect, from an atmosphere of 90 bar of CO_2.

25.5 At the top of the atmosphere, only the layer radiates to space, so $\sigma T_a^4 = S(1-A)/4 = \sigma T_e^4$. The surface receives sunlight plus the layer's downward radiation and emits σT_s^4, so $\sigma T_s^4 = \sigma T_e^4 + \sigma T_a^4 = 2\sigma T_e^4$, and $T_s = 2^{1/4} T_e = 303\,\text{K}$. The real Earth is cooler, at 288 K, because the atmosphere is transparent in an infrared "window" near 10 μm through which the surface radiates directly to space, and because convection and evaporation carry heat from the surface up to where it is radiated more easily.

25.6 $F = \sigma T^4$ gives $\Delta F = 4\sigma T^3 \Delta T$, so $\Delta T = 3.7/(4\sigma(255)^3) = 1.0\,\text{K}$. With $f = 0.5$, $\Delta T = 2\,\text{K}$. Including ice–albedo and cloud feedbacks, the best current estimate of the warming from doubling CO_2 is about 3 K, likely between 2.5 and 4 K.

25.7 The difference between the pulls at the near side and the centre is $GM/(r-R)^2 - GM/r^2 \approx 2GMR/r^3$. The ratio of lunar to solar tides is $(M_{\text{Moon}}/M_{\text{Sun}})(r_{\text{Sun}}/r_{\text{Moon}})^3 = (3.7\times10^{-8})(389)^3 = 2.2$. The pull itself falls as $1/r^2$, and the Sun's is 180 times stronger. But the tide falls as $1/r^3$, and the extra factor of r favours the much closer Moon by another factor of 389.

25.8 For a parcel rising adiabatically, the first law per unit mass gives $c_p\,\mathrm{d}T = \mathrm{d}P/\rho$. With $\mathrm{d}P = -\rho g\,\mathrm{d}z$, this is $c_p\,\mathrm{d}T = -g\,\mathrm{d}z$, so $\Gamma = g/c_p = 9.8\,\text{K/km}$. The observed average, about 6.5 K/km, is lower because rising air condenses water vapour, releasing latent heat.

25.9 At 30°, $f = 2\Omega\sin 30° = 7.3\times10^{-5}\,\text{s}^{-1}$. Bathtub: $\text{Ro} = U/fL = 0.1/(7.3\times10^{-5}\times1) \approx 1400$. Hurricane: $30/(7.3\times10^{-5}\times5\times10^5) \approx 0.8$. Coriolis forces dominate the hurricane and are negligible in the bathtub, where the direction of swirl is set by leftover motion and the tub's shape. (In careful experiments, with the water left to settle for a day, the Coriolis effect can just be seen.)

25.10 A straight chord between two surface points separated by angle Δ passes the centre at a distance $R\cos(\Delta/2)$. Setting this equal to the core radius at the shadow edge, $R_c = 6371\cos 52° = 3920\,\text{km}$, within 13% of the true value.

25.11 The Moon's orbital angular momentum is $L_{\text{orb}} = M_m\sqrt{GM_e a} = 2.9\times10^{34}\,\text{kg m}^2/\text{s}$, and it grows as $\mathrm{d}L_{\text{orb}}/\mathrm{d}t = (L_{\text{orb}}/2a)\,\mathrm{d}a/\mathrm{d}t = 4.5\times10^{16}\,\text{N m}$. The Earth loses spin at that rate: $I\,\mathrm{d}\omega/\mathrm{d}t = -\mathrm{d}L_{\text{orb}}/\mathrm{d}t$, with $I = 0.33M_eR^2 = 8.0\times10^{37}\,\text{kg m}^2$. Since the day is $T = 2\pi/\omega$, $\mathrm{d}T/\mathrm{d}t = (T^2/2\pi I)\,\mathrm{d}L_{\text{orb}}/\mathrm{d}t = 6.7\times10^{-13}$, or 2.1 ms per century — in excellent agreement with the eclipse records. Over 100 million years, at the same rate, the day would have lengthened by 35 minutes: dinosaurs had days of about 23.4 hours, as growth bands in fossil corals and shells confirm.

25.12 The absorbed sunlight is $(1-A(T))S/4$: $119\,\text{W/m}^2$ below 250 K, $238\,\text{W/m}^2$ above 280 K,

and rising linearly in between. The emitted curve, $0.6\sigma T^4$, is a smooth rising curve. They cross three times: at 243 K on the icy branch, at 258 K on the slope, and at 289 K on the warm branch. At the outer two, a small warming makes emission exceed absorption, and the planet cools back: stable. At the middle one, a small warming melts ice, lowers the albedo, and runs away to the warm state; a small cooling runs away to the snowball. A dip in sunlight or greenhouse gas can push the climate past the middle point, into the snowball. To escape, the emission curve must drop until the snowball equilibrium disappears: absorption must exceed emission even at 250 K, which needs $\epsilon < 119/(\sigma \cdot 250^4) = 0.54$. Volcanoes keep emitting CO_2, and on a frozen planet there is no rain to remove it by weathering rocks, so over millions of years it builds up. When the threshold is passed, the Earth jumps abruptly to the warm branch — now with a strong greenhouse effect, at around 300 K — and the ice melts in a geological instant. Geologists find the evidence: glacial deposits in the tropics, capped by thick layers of carbonate rock laid down in the hot aftermath. The path out is not the path in: hysteresis.

26 General relativity

26.1 $r_s = 2GM/c^2$: Sun 3.0 km, Earth 9 mm, person 10^{-25} m (a ten-billionth of a proton). Squeezed inside r_s, their densities would be 2×10^{19}, 2×10^{30}, and 10^{76} kg/m^3. Since $r_s \propto M$, the required density M/r_s^3 falls as $1/M^2$: a black hole of 10^8 solar masses has the average density of water. Small black holes need impossible compression; big ones can form from ordinary matter that just keeps piling up.

26.2 Pound–Rebka: $gh/c^2 = 2.5\times10^{-15}$, which they measured using the Mössbauer effect. Sun: $GM/Rc^2 = 2.1\times10^{-6}$. The Sun's lines are redshifted by two parts per million, equivalent to a Doppler shift of 0.64 km/s — hard to disentangle from the motions of convecting gas, but measured.

26.3 In the time h/c that the pulse takes to reach the ceiling, the ceiling has gained speed gh/c relative to the floor's speed at emission, moving away from the source. The Doppler shift is $f(1-v/c) = f(1-gh/c^2)$. By the equivalence principle, the same holds in a gravitational field g: light climbing a height h arrives with lower frequency. The field is static, so crests cannot pile up between floor and ceiling: as many arrive per second (by the upper clock) as leave per second (by the lower clock), yet the upper observer counts fewer. Each second of the lower clock must last longer than a second of the upper one: higher clocks run faster.

26.4 The fractional rate difference is $\Delta\Phi/c^2 = (GM/c^2)(1/R_\oplus - 1/r)$, with $r = 26\,600$ km: 5.3×10^{-10}, or +46 µs per day (satellite clocks fast). Special relativity gives −7 µs per day. The net is +38 µs per day; uncorrected, positions would drift by about 10 km a day. The clocks are deliberately tuned slow before launch.

26.5 $2GM/c^2R_\odot = 4.2\times10^{-6}$ rad $= 0.87''$; general relativity predicts $1.75''$. In a freely falling frame light travels straight, so by the equivalence principle it must fall in a gravitational field like anything else: that gives the Newtonian half, and Einstein himself predicted $0.87''$ in 1911. The other half comes from the curvature of space, which depends on the field equations of general relativity, not on the equivalence principle alone. Eddington's $1.75''$ (within large errors) confirmed the full theory.

26.6 At the horizon $r_s = 2GM/c^2$, the tidal acceleration is $2GML/r_s^3 = Lc^6/(4G^2M^2) \propto 1/M^2$. For $10M_\odot$ ($r_s = 30$ km): 2×10^8 m/s^2, twenty million g — you would be torn apart well outside

the horizon. For Sagittarius A* ($r_s = 1.2\times10^{10}$ m): 1.3×10^{-3} m/s^2, a ten-thousandth of g. You could cross the horizon of the giant black hole without noticing anything — and then, inevitably, reach the singularity.

26.7 Two meridians a distance d apart at the equator are an angle d/R apart. At latitude s/R, they are a distance $d\cos(s/R)$ apart, with second derivative $-(d/R^2)\cos(s/R) \to -d/R^2$ at $s = 0$. Two balls dropped a horizontal distance d apart at distance r from the Earth's centre both fall toward the centre, so they converge with relative acceleration $-(GM/r^3)d$. The analogue of $1/R^2$ is GM/r^3 — with time converted into distance by c, a curvature of GM/c^2r^3. At the Earth's surface this is $(2.4\times10^{11}\text{ m})^{-2}$: spacetime here is curved on the scale of an astronomical unit.

26.8 With $V \propto 1/r^2 - r_s/r^3$, $V' \propto -2/r^3 + 3r_s/r^4$, which vanishes at $r = \frac{3}{2}r_s$. Since $V = 0$ at $r = r_s$ and $V \to 0$ as $r \to \infty$, while it is positive in between, this point is a maximum. A circular orbit on top of a potential hill is unstable: nudge the light slightly inward and it spirals into the black hole; outward, and it escapes. The photon sphere marks the edge of the black hole's "shadow" imaged by the Event Horizon Telescope.

26.9 $T = \hbar c^3/(8\pi GMk_B)$: 6×10^{-8} K for the Sun's mass and 1.8 K for the Moon's. Both are colder than the 2.7 K background, so both absorb more than they emit and grow; only black holes lighter than 4.5×10^{22} kg (about 60% of the Moon's mass) are currently shrinking. Evaporation time: $1.4\times10^{10}\text{ yr}\times(2\times10^{30}/2\times10^{11})^3 \approx 10^{67}$ years.

26.10 $M = 1000^3/16^2 = 3.9\times10^6 M_\odot$. S2's closest approach, 120 AU, is only four times the radius of Neptune's orbit. Four million solar masses within this region, yet emitting almost no light, cannot be a cluster of stars (it would shine brightly and, at that density, collide and merge within a short time), nor of dark remnants such as neutron stars (it would evaporate or collapse). The only long-lived possibility is a single black hole — the conclusion of the 2020 Nobel Prize. (Its Schwarzschild radius is 0.08 AU, and the Event Horizon Telescope has since imaged its shadow.)

26.11 With $h = L/m$, the radial equation is $\ddot r - h^2/r^3 = -GM/r^2 - 3GMh^2/c^2r^4$. Changing variables to $u = 1/r$ and ϕ ($\ddot r = -h^2u^2u''$) gives

$$u'' + u = \frac{GM}{h^2} + \frac{3GM}{c^2}u^2.$$

Try $u = (GM/h^2)(1 + e\cos\omega\phi)$, keeping only terms first order in e and in the small correction: $u^2 \approx (GM/h^2)^2(1 + 2e\cos\omega\phi)$. Matching the coefficients of $\cos\omega\phi$ gives

$$1 - \omega^2 = \frac{6G^2M^2}{c^2h^2}, \qquad \omega \approx 1 - \frac{3G^2M^2}{c^2h^2}.$$

The perihelion recurs when $\omega\phi = 2\pi$, so each orbit takes it a little further, $\Delta\phi = 2\pi/\omega - 2\pi \approx 6\pi G^2M^2/c^2h^2$. With $h^2 = GMa(1 - e^2)$, $\Delta\phi = 6\pi GM/c^2a(1 - e^2)$. For Mercury, $\Delta\phi = 5.0\times10^{-7}$ rad per orbit; over 415 orbits in a century, that is 43″. This was the anomaly Le Verrier had found in 1859, unexplained for half a century; Einstein said that getting it right gave him heart palpitations.

26.12 Starting from rest at $x = 0$, the observer's worldline is $x = (c^2/a)(\sqrt{1 + (at/c)^2} - 1)$, which approaches the line $x = ct - c^2/a$ from above but never reaches it. A light ray $x = ct - D$ emitted at $t = 0$ from a distance D behind her catches up only if it eventually lies ahead of her worldline. Her lead over the ray is $(c^2/a)\sqrt{1 + (at/c)^2} - ct + D - c^2/a$; the first two terms together decrease

steadily toward zero, so the ray catches up if and only if $D < c^2/a$. Events farther behind are forever hidden: a horizon. (For $a = g$, $c^2/g \approx 1$ light-year.) To hover at radius r outside a black hole requires proper acceleration $a = GM/r^2\sqrt{1 - r_s/r}$, which diverges at the horizon. The local Unruh temperature $\hbar a/2\pi c k_B$, redshifted to infinity by the factor $\sqrt{1 - r_s/r}$, becomes

$$T_\infty = \frac{\hbar GM}{2\pi c k_B r^2} \xrightarrow{r \to r_s} \frac{\hbar c^3}{8\pi GM k_B},$$

exactly Hawking's temperature. By the equivalence principle, a small region just outside the horizon looks like flat spacetime; the freely falling observer sees empty vacuum there, and the hovering observer is accelerating relative to it. So the hovering observer must see Unruh's thermal bath — and what climbs out to infinity is Hawking radiation. (The Unruh temperature for $a = g$ is 4×10^{-20} K.)

27 Astrophysics

27.1 $d = 1/p = 10\,\text{pc} = 33$ light-years. For 10% accuracy, the parallax must be ten times the error, 200 μas, giving $d = 5$ kpc: a large part of the Milky Way. (That accuracy is reached only for bright stars; for faint ones the errors are several times larger.)

27.2 $0.1 \times 0.7 \times (2 \times 10^{30}) \times 0.007 \times c^2 = 9 \times 10^{43}$ J, which at 3.8×10^{26} W lasts 2.3×10^{17} s, or 7 billion years. Detailed models give about 10 billion.

27.3 $GM^2/RL \approx 10^{15}$ s ≈ 30 million years. Geologists and Darwin needed hundreds of millions of years for erosion and evolution; Kelvin insisted, on the authority of physics, that the Sun could not be that old. Physics was missing a source of energy: the nucleus, discovered decades later.

27.4 For a gas of $N = 2M/m_p$ particles, the virial theorem gives $3Nk_BT = |U| \approx \frac{3}{5}GM^2/R$ (uniform sphere), so $k_BT \approx 0.1\,GMm_p/R$: $T \approx 2 \times 10^6$ K. This is a mean; the centre is hotter, and detailed models give 1.5×10^7 K. Even so, $k_BT \approx 1$ keV, a thousand times short of the 1 MeV Coulomb barrier. Classically the Sun could not shine. Quantum tunnelling makes fusion possible for the fastest protons at a few times k_BT (Problem 14.13).

27.5 $N = (R/\ell)^2 = (7 \times 10^8/10^{-3})^2 \approx 5 \times 10^{23}$ steps, a total path of $N\ell = 5 \times 10^{20}$ m, which takes about 50 000 years. (The opacity actually varies hugely with depth, and detailed estimates range from about 10^4 to 10^5 years.) Neutrinos fly straight out in 2.3 s: the light we see today left the core in the Stone Age, but the neutrinos are fresh.

27.6 $t \propto M/L \propto M^{-2.5}$: $10M_\odot$, 3×10^7 years; $3M_\odot$, 6×10^8 years; $0.2M_\odot$, 6×10^{11} years, much longer than the age of the universe (every red dwarf ever formed is still shining). On Earth, single-celled life appeared within a billion years, but complex life took four. The $10M_\odot$ star is hopeless; the $3M_\odot$ star might at best host microbes. The red dwarf lives long enough, but its habitable zone is so close that planets there are tidally locked, and exposed to violent flares.

27.7 Setting $\sigma_T L/4\pi r^2 c = GMm_p/r^2$, the r cancels:

$$L_{\text{Edd}} = \frac{4\pi GMm_p c}{\sigma_T} = 3.3 \times 10^4 L_\odot \frac{M}{M_\odot}.$$

For $10^9 M_\odot$, $L_{\text{Edd}} = 1.3 \times 10^{40}\,\text{W} = 3 \times 10^{13} L_\odot$, three hundred times a large galaxy. This is how a quasar outshines its entire host galaxy from a region the size of the solar system.

27.8 Equating $\hbar^2 n^{5/3}/m_e$ with GM^2/R^4 and using $n \sim M/2m_pR^3$:

$$\frac{\hbar^2}{m_e}\left(\frac{M}{2m_p}\right)^{5/3}\frac{1}{R^5} \sim \frac{GM^2}{R^4}, \qquad R \sim \frac{\hbar^2}{Gm_e(2m_p)^{5/3}}M^{-1/3}.$$

For $M = M_\odot$, $R \sim 2000$ km; a careful calculation gives about 6000 km, the size of the Earth. More mass compresses the star to a smaller radius, and at some point the electrons become relativistic (Problem 11).

27.9 For a circular orbit, $v^2/r = GM(r)/r^2$, so $M(r) = v^2 r/G$: $9 \times 10^{10} M_\odot$ at 8 kpc and $5.6 \times 10^{11} M_\odot$ at 50 kpc. The visible matter is only about 11% of the mass inside 50 kpc: nearly 90% is dark. Since $M \propto r$ and no edge is in sight, the dark halo extends farther still.

27.10 Jupiter: $(R_J/R_\odot)^2 = 1\%$. Earth: $(R_\oplus/R_\odot)^2 = 8.4 \times 10^{-5}$, about 84 parts per million. The Earth's atmosphere makes stars twinkle at the 0.1% level, so Earth-sized transits are hidden from the ground; they need space telescopes such as Kepler, which measured brightness to about 20 parts per million. And an Earth-like orbit transits only once a year, so several years are needed to confirm one.

27.11 With $n \sim M/m_pR^3$, $P \sim \hbar c(M/m_p)^{4/3}R^{-4}$, while gravity requires GM^2/R^4. Both scale as R^{-4}, so the ratio of the two is independent of R: if gravity wins at one radius, it wins at every smaller radius too. They balance only for one particular mass, $\hbar c(M/m_p)^{4/3} = GM^2$, i.e.

$$M \sim \left(\frac{\hbar c}{G}\right)^{3/2}\frac{1}{m_p^2} = 3.7 \times 10^{30}\,\text{kg} \approx 1.9 M_\odot.$$

The careful calculation gives $1.4M_\odot$. As a white dwarf gains mass, it shrinks, and its electrons are squeezed into higher momenta until they are relativistic. Then an electron's energy grows only as p, not as p^2, so squeezing raises the pressure only as fast as gravity's demand grows. Above the Chandrasekhar mass, gravity wins at every radius: the star collapses until electrons are forced into protons and it becomes a neutron star — or, for an accreting white dwarf, carbon ignites and the star explodes as a type Ia supernova (Chapter 28).

27.12 A contracting ball of gas heats up as the virial theorem requires, $k_BT \sim GMm_p/R$. But as R shrinks, the electron degeneracy energy $\sim (\hbar^2/m_e)n^{2/3} \propto 1/R^2$ grows faster, and eventually supports the star, which then cools. Balancing the two, the highest temperature reached is $k_BT_{\max} \sim G^2 m_e m_p^{8/3} M^{4/3}/\hbar^2$: it grows with mass. Setting it equal to the few million kelvin needed for hydrogen fusion gives a minimum mass of a few hundredths of a solar mass, within a factor of a few of the true $0.08M_\odot$. At the top, the mass–luminosity relation $L \propto M^{3.5}$ grows faster than the Eddington luminosity $\propto M$; the Sun radiates at 1/33 000 of its Eddington limit, and the two would meet at $33\,000^{1/2.5} \approx 60M_\odot$. Near the limit, radiation pressure drives the outer layers off in winds, and the relation flattens, so the real cutoff is 100–$150M_\odot$. Between these two limits lies a factor of about 10^3, set by fundamental constants: both can be traced to the Chandrasekhar combination $(\hbar c/G)^{3/2}/m_p^2 \approx 2M_\odot$ of Problem 11, multiplied by modest powers of dimensionless numbers such as α and m_e/m_p.

27.13 Conservation of angular momentum, $I\omega \propto MR^2/P$, gives $P_{\text{ns}} = 3600\,\text{s} \times (12/6400)^2 = 13\,\text{ms}$. Conservation of flux, BR^2, gives $B = 100\,\text{T} \times (6400/12)^2 = 3 \times 10^7\,\text{T}$, close to typical pulsar fields. The equator flies off when $\omega^2 R = GM/R^2$, so the shortest period is $2\pi\sqrt{R^3/GM} = 0.6\,\text{ms}$ for $1.4M_\odot$, a spin of 1650 revolutions per second. The fastest known pulsar spins at

716 Hz, less than half the break-up rate. (Real neutron stars bulge as they spin, lowering the limit somewhat.)

28 Cosmology

28.1 $1/H_0 = 4.4 \times 10^{17}\,\mathrm{s} = 14$ billion years. A galaxy at 100 Mpc recedes at 7000 km/s. The limit $c/H_0 = 4300$ Mpc is the Hubble radius. Relativity forbids anything from overtaking a light ray *locally*; it says nothing about how fast the distance between distant objects can grow when space itself expands. Nothing ever moves past light in its own neighbourhood, so no signal ever outruns light.

28.2 $\rho_c = 3H_0^2/8\pi G = 9.2 \times 10^{-27}\,\mathrm{kg/m^3}$, or 5.5 protons per cubic metre. Air ($1.2\,\mathrm{kg/m^3}$) is 10^{26} times denser, and even the interstellar gas of our galaxy is 2×10^5 times denser. The universe is mostly emptier than the best laboratory vacuum.

28.3 $T = 2.725\,\mathrm{K} \times 1101 = 3000\,\mathrm{K}$. By Wien's law it peaked at $\lambda = (2.9\,\mathrm{mm\,K})/T = 970\,\mathrm{nm}$, in the near-infrared; today the peak is at 1.06 mm, in the microwave. A 3000 K blackbody glows a dull orange, like a candle flame or an old incandescent bulb, and the whole sky would have been that colour.

28.4 $a \propto t^{2/3}$ gives $H = \dot{a}/a = 2/3t$, so $t_0 = 2/3H_0 = 9.3$ billion years. That is younger than the oldest stars: a matter-only universe is ruled out. With dark energy, the expansion was slower in the past than now, so for the same H_0 the universe is older, 13.8 billion years.

28.5 By Newton's shell theorem, only the mass M inside affects a galaxy on the sphere, so its energy per unit mass is constant:

$$\tfrac{1}{2}\dot{R}^2 - \frac{GM}{R} = E.$$

With $R = ar$ and $M = \frac{4}{3}\pi a^3 r^3 \rho$, this is $\frac{1}{2}\dot{a}^2 r^2 - \frac{4}{3}\pi G\rho a^2 r^2 = E$. Divide by $\frac{1}{2}a^2 r^2$:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho + \frac{2E}{r^2 a^2}.$$

Since the expansion is uniform, every shell must give the same equation, so $2E/r^2$ is the same constant for all of them; call it $-kc^2$. This is the Friedmann equation. Positive energy ($k < 0$) means expansion forever; negative energy ($k > 0$) means the universe is bound, and eventually recollapses (without dark energy). General relativity reinterprets k as the curvature of space.

28.6 $\rho_r \propto a^{-4}$ and $\rho_m \propto a^{-3}$, so the ratio ρ_r/ρ_m grows as $1/a = 1 + z$ going back in time. They were equal at $1 + z = 0.31/(9 \times 10^{-5}) = 3400$. Integrating the Friedmann equation gives an age of about 50 000 years.

28.7 $n_b/n_\gamma = 0.25\,\mathrm{m^{-3}}/(4.11 \times 10^8\,\mathrm{m^{-3}}) = 6 \times 10^{-10}$. This tiny number is what remains of the slight excess of matter over antimatter in the early universe: for every billion antiprotons there were a billion and one protons, and after annihilation the one survived. It also controls much of later history. It sets the abundances of deuterium and helium made in the first minutes, and the enormous number of photons per baryon delays recombination until 3000 K, far below the 1.6×10^5 K that corresponds to hydrogen's 13.6 eV.

28.8 $3ct = 1.1 \times 10^6$ light-years at recombination, stretched to 1.3×10^9 light-years today. At 46 billion light-years, it subtends 0.027 rad $\approx 1.6°$, about three times the width of the full Moon. In a standard Big Bang, patches of sky more than a couple of degrees apart were never in causal contact, yet they have the same temperature to one part in 10^5. Nothing could have evened them out. Inflation solves the problem by having the whole observable universe emerge from a single tiny patch that was in contact before inflation stretched it.

28.9 At freeze-out, $n/p = e^{-1.29/0.8} = 0.20$, about 1/5. After 180 s of decay, it is $0.20\, e^{-180/880} = 0.16$. Every helium-4 nucleus contains 2 neutrons and 2 protons, so if all the neutrons end up in helium, the mass in helium is $2n$ out of $n+p$ nucleons: $Y = 2n/(n+p) = 2(n/p)/(1+n/p) = 0.28$. This is close to the observed 0.245; a careful treatment, with gradual freeze-out, gives $n/p \approx 1/7$ and $Y \approx 0.25$. A two-line calculation from the weak interaction, the neutron lifetime, and the expansion rate predicts the composition of the universe.

28.10 From $a = 10^{-28}$ to matter–radiation equality at $a \approx 3{\times}10^{-4}$, $|\Omega - 1|$ grows by $(3{\times}10^{24})^2 \approx 10^{49}$; from then to now, by another 3000. So $|\Omega - 1|$ must have been less than $0.01/(3 \times 10^{52}) \approx 3 \times 10^{-55}$ at the time: fine-tuning to fifty-five decimal places. (Dark energy has recently begun to reduce $|\Omega - 1|$, but only slightly.) If inflation starts from $|\Omega - 1| \sim 1$, it must shrink it by $e^{-2N} \lesssim 3 \times 10^{-55}$, so $N \gtrsim \frac{1}{2}\ln(3 \times 10^{54}) \approx 63$. Inflation models are designed to give at least 60 e-folds, and it costs them little to give many more.

28.11 Stars of radius R and number density n in a shell of thickness dr at distance r cover a fraction $n\,4\pi r^2\,\mathrm{d}r \cdot \pi R^2/4\pi r^2 = n\pi R^2\,\mathrm{d}r$ of the sky, independent of r: farther shells contain more stars, each looking smaller. Adding infinitely many shells covers the sky, and a typical line of sight ends on a star after a distance $\ell = 1/n\pi R^2$. A star's surface brightness does not depend on its distance, so the sky would be uniformly as bright as the Sun's surface. With about 10^9 stars per cubic megaparsec on average, $\ell \sim 10^{24}$ light-years. The universe is only 14 billion years old, so we see stars only out to about 10^{10} light-years, and they cover only about 10^{-14} of the sky. (And stars live only billions of years, so they could never fill the sky with light even given time.) The finite age matters far more; expansion dims distant starlight by redshift, but only by a modest factor. The dark night sky tells you the universe had a beginning.

29 Gravitational waves

29.1 $\Delta L = \frac{1}{2}hL = 2 \times 10^{-18}$ m, about a thousandth of a proton's diameter. LIGO can detect this because it averages over the trillions of photons bouncing in its arms, each measurement of position limited by the light's wavelength divided by the square root of the number of photons.

29.2 The radiation comes from the mass quadrupole moment, which depends on the positions squared: rotating a binary by half a turn leaves its mass distribution unchanged (exactly, for equal masses), so the pattern repeats twice per orbit. At 100 Hz, the orbital frequency is 50 Hz and the period 20 ms.

29.3 $3M_\odot c^2 = 5.4 \times 10^{47}$ J. The Sun will emit 3.8×10^{26} W $\times\, 10^{10}$ yr $= 1.2 \times 10^{44}$ J, so this is 4500 Suns' lifetimes of output. Over 20 ms, the power was 3×10^{49} W — briefly more than all the stars in the observable universe combined.

29.4 $\ddot{\mathbf{D}} = \sum m_i \ddot{\mathbf{r}}_i = \sum \mathbf{F}_i = 0$, because internal forces cancel in pairs (Newton's third law, or conservation of momentum). In electromagnetism, the dipole is $\mathbf{p} = \sum q_i \mathbf{r}_i$, and $\ddot{\mathbf{p}} = \sum (q_i/m_i)\mathbf{F}_i$,

which is nonzero unless all particles have the same charge-to-mass ratio: an electron and proton orbiting each other radiate. In gravity, the "charge" is the mass itself, so every particle has the same ratio — this is the equivalence principle. The dipole is just the total mass times the centre of mass, which moves uniformly. (Similarly, the magnetic-dipole analogue is the total angular momentum, also conserved.) Gravitational radiation starts at the quadrupole, which is why it is so weak.

29.5 The light travel time was $130 \times 10^6\,\mathrm{yr} \times 3.16 \times 10^7\,\mathrm{s/yr} = 4 \times 10^{15}\,\mathrm{s}$. A delay of a few seconds over that time implies $\left|v_{\mathrm{gw}} - c\right|/c \lesssim 10^{-15}$. In a single night, this ruled out a large class of alternative theories of gravity.

29.6 With $GM/c^2 = 41\,\mathrm{km}$, $\pi f/c = 1.6 \times 10^{-6}\,\mathrm{m^{-1}}$, and $r = 1.3 \times 10^{25}\,\mathrm{m}$, $h \approx 2 \times 10^{-21}$. That's within a factor of two of the measured value, from a formula derived by dimensional analysis and the quadrupole formula.

29.7 $a = (GMP^2/4\pi^2)^{1/3}$ with $M = 2.8M_\odot$ and $P = 2.8 \times 10^4\,\mathrm{s}$: $a = 1.9 \times 10^9\,\mathrm{m}$, about three solar radii. Two neutron stars orbit in a space that would fit inside a large star, in less than eight hours, with speeds of hundreds of kilometres per second.

29.8 The orbital frequency is $\sqrt{GM/r^3}/2\pi$, so with $r = 6GM/c^2$ the gravitational-wave frequency (twice the orbital) is $f = c^3/(6^{3/2}\pi GM)$, about $4400\,\mathrm{Hz} \times M_\odot/M$. For $60M_\odot$, $f = 73\,\mathrm{Hz}$, in LIGO's band. For $10^6 M_\odot$, $f = 4.4\,\mathrm{mHz}$, far below what ground-based detectors can reach (seismic noise swamps them below about 10 Hz), and squarely in the band of the planned space detector LISA.

29.9 The quadrupole formula gives the power

$$P = \frac{32}{5}\frac{G^4}{c^5}\frac{(m_1 m_2)^2(m_1 + m_2)}{a^5},$$

and setting $\mathrm{d}E/\,\mathrm{d}t = (Gm_1m_2/2a^2)\,\dot{a} = -P$,

$$\dot{a} = -\frac{64}{5}\frac{G^3 m_1 m_2 (m_1 + m_2)}{c^5 a^3}.$$

Separating variables, $a^3\,\mathrm{d}a = -K\,\mathrm{d}t$, where K collects the constants; integrating from a down to 0 gives $t = a^4/4K$, which is the stated result. For the Earth and Sun, $P \approx 200\,\mathrm{W}$ — a couple of light bulbs, radiated by a planet. The inspiral time is 10^{23} years, ten trillion times the age of the universe. The Sun will die long before.

29.10 The gravitational-wave frequency is twice the orbital frequency, so Kepler's law gives $(\pi f)^2 = GM/a^3$ with $M = m_1 + m_2$, i.e. $a \propto (GM)^{1/3} f^{-2/3}$, and $\dot{f}/f = -\frac{3}{2}\dot{a}/a$. Substituting $\dot{a}$ from the previous problem,

$$\dot{f} = \frac{96}{5}\frac{f}{c^5}\frac{G^3 m_1 m_2 M}{a^4} = \frac{96}{5}\pi^{8/3}\left(\frac{GM}{c^3}\right)^{5/3} f^{11/3},$$

since $a^4 = (GM)^{4/3}(\pi f)^{-8/3}$ and $m_1 m_2 M^{-1/3} = \mathcal{M}^{5/3}$. So measuring f and $\dot{f}$ at any moment gives the chirp mass directly. The individual masses affect the signal only through relativistic corrections that become important near the merger, where the signal is briefest, and they are partly degenerate with the black holes' spins: the chirp mass of GW150914 was measured to about 5%, the individual masses only to about 15%.

29.11 $H_0 = v/d = 3000/40 = 75\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. A single event's distance is uncertain by about 15%, mostly because the amplitude depends on how the orbit is tilted to our line of sight, which is hard to separate from distance. The galaxy's local velocity, a few hundred km/s, adds another 10%. Since errors on independent measurements fall as $1/\sqrt{N}$, reaching 2% needs about $(15/2)^2 \approx 50$ events with identified host galaxies. The method needs no cosmic distance ladder, so it can independently test the current tension between measurements of H_0.

30 Quantum field theory

30.1 $\hbar c/mc^2$: electron 386 fm (3.9×10^{-13} m), the scale below which QED's pair creation matters and the electron cannot be localized as a single particle; proton 0.21 fm, a few times smaller than the proton's actual size (0.84 fm, set by the strong force instead); W boson 2.5×10^{-3} fm, the range of the weak force.

30.2 In natural units the Klein–Gordon equation is $(\partial_t^2 - \nabla^2 + m^2)\phi = 0$. Acting on the plane wave, $\partial_t \to -iE$ and $\nabla \to i\mathbf{p}$, so the equation becomes $(-E^2 + \mathbf{p}^2 + m^2)\phi = 0$, which holds if and only if $E^2 = \mathbf{p}^2 + m^2$. For $m = 0$ it is the ordinary wave equation, with $E = |\mathbf{p}|$: waves that travel at the speed of light.

30.3 Confinement to L gives momentum $p \sim \hbar/L$ and, once relativistic, energy $\sim \hbar c/L$. This exceeds $2m_ec^2$ for $L \lesssim \hbar/2m_ec \approx 200$ fm, about half the Compton wavelength. Squeezing the electron harder creates electron–positron pairs, so the number of particles in the box is no longer fixed: the one-particle wavefunction stops making sense, and you need fields.

30.4 For $r > 0$, $\nabla^2 f = r^{-1}(rf)''$ for a spherically symmetric f; with $f = e^{-mr}/r$, $(rf)'' = m^2 e^{-mr}$, so $\nabla^2 f = m^2 f$ and the equation holds. Near $r = 0$, $f \approx 1/r$, and $\nabla^2(1/r) = -4\pi\delta^3(\mathbf{r})$, which supplies the source: $(g/4\pi)(-4\pi\delta^3) = -g\delta^3$. Restoring units turns m in the exponent into $mc/\hbar$, so the range is $\hbar/mc$; for 140 MeV, $197/140 = 1.4$ fm.

30.5 $\alpha/\pi \approx 0.0023$: one-loop corrections are a quarter of a percent, two-loop corrections a few parts per million. Each order is about 400 times smaller than the last, so a few orders give extraordinary precision. The electron's magnetic moment has been computed through five loops, a correction of order $(\alpha/\pi)^5 \sim 10^{-13}$, and agrees with experiment to about one part in 10^{12}.

30.6 $P = \pi^2\hbar c/240d^4$: 1.3×10^{-3} Pa at 1 μm, and 1.3×10^5 Pa — more than an atmosphere — at 10 nm. It equals one atmosphere at $d \approx 11$ nm. Micromachines have surfaces separated by such gaps, and the Casimir and van der Waals forces can make moving parts stick together permanently (*stiction*).

30.7 $1/\alpha(91\,\mathrm{GeV}) = 137.04 - (2/3\pi)\ln(91\,000/0.511) = 137.04 - 2.57 = 134.5$. Heavier charged particles screen too, and bring it to 128. A charge polarizes the vacuum around it: virtual electron–positron pairs orient themselves, positrons pushed away and electrons pulled toward it (for a negative charge, the reverse), partly screening it, as a dielectric screens a charge. From far away, you see the screened charge, $\alpha \approx 1/137$. Probing at shorter distances, inside the cloud, you see more of the bare charge.

30.8 With $\psi' = e^{iq\chi/\hbar}\psi$ and $\mathbf{A}' = \mathbf{A} + \nabla\chi$,

$$(-i\hbar\nabla - q\mathbf{A}')\psi' = e^{iq\chi/\hbar}(-i\hbar\nabla\psi + q\nabla\chi\,\psi - q\mathbf{A}\psi - q\nabla\chi\,\psi) = e^{iq\chi/\hbar}(-i\hbar\nabla - q\mathbf{A})\psi.$$

Applying this twice, the kinetic term transforms into $e^{iq\chi/\hbar}$ times its old value. On the left, $i\hbar\partial_t\psi' = e^{iq\chi/\hbar}(i\hbar\partial_t - q\,\partial_t\chi)\psi$, and the extra term is cancelled by the change in $q\phi\psi$. Every term picks up the same phase, so the equation is unchanged. Without the potentials, the derivatives of χ would be left over: a phase that varies from place to place would change the physics. Demanding that it not — *local* phase invariance — forces you to introduce $\mathbf{A}$ and ϕ, coupled in exactly this way. The electromagnetic field is the price of a local symmetry.

30.9 V depends only on $x = |\phi|^2$: $\mathrm{d}V/\mathrm{d}x = -\mu^2 + 2\lambda x = 0$ at $|\phi| = \mu/\sqrt{2\lambda}$, i.e. $v = \mu/\sqrt{\lambda}$. Substituting, $V = -\frac{1}{2}\mu^2(v+h)^2 + \frac{1}{4}\lambda(v+h)^4$, independent of θ. Then $\partial^2 V/\partial h^2 = -\mu^2 + 3\lambda v^2 = 2\mu^2$ at $h = 0$: a particle of mass $\sqrt{2}\mu$ (the Higgs). θ does not appear in V at all, so moving along the circle of minima costs no energy: a massless Goldstone boson. In a gauge theory, as in the Standard Model, the Goldstone boson becomes the extra polarization of a massive W or Z.

30.10 $E_P = 1.2 \times 10^{19}\,\mathrm{GeV} = 2.0 \times 10^{9}\,\mathrm{J}$ and $\hbar c = 3.2 \times 10^{-26}\,\mathrm{J\,m}$, so $E_P^4/(\hbar c)^3 \approx 5 \times 10^{113}\,\mathrm{J/m^3}$. That is 10^{123} times the observed value. A 1 TeV cutoff gives $2 \times 10^{49}\,\mathrm{J/m^3}$, still 10^{59} times too big. Either some unknown principle cancels these zero-point energies to 123 (or at least 59) decimal places, leaving a tiny remainder, or we do not understand how the vacuum gravitates. This is the cosmological constant problem.

30.11 In natural units G_F has dimensions of energy^{-2} and a cross-section has dimensions of energy^{-2}, so $\sigma \sim G_F^2 E^2$. This exceeds the unitarity limit $\sim 1/E^2$ when $G_F^2 E^4 \gtrsim 1$, at $E \sim G_F^{-1/2} \approx 300\,\mathrm{GeV}$. Something new must appear by then. Solving $G_F = \sqrt{2}g^2/8m_W^2$ gives $m_W = \sqrt{\sqrt{2}g^2/8G_F} = 80\,\mathrm{GeV}$, matching the measured 80.4 GeV. Fermi's contact interaction is the low-energy limit of W exchange, just as a pointlike Coulomb interaction would be the low-energy limit of a massless photon's.

31 Accelerators and detectors

31.1 $p = h/\lambda$; with $\lambda = 0.1\,\mathrm{fm}$, $pc = 2\pi\hbar c/\lambda = 12\,\mathrm{GeV}$ (or 2 GeV using $\hbar/\lambda$; the factor of 2π depends on what "comparable" means). At a few TeV per parton, $\hbar c/E \approx 10^{-4}\,\mathrm{fm} = 10^{-19}\,\mathrm{m}$: ten thousand times smaller than a proton.

31.2 $f = eB/2\pi m_p = 23\,\mathrm{MHz}$. At the edge, $p = eBR$, so $K = (eBR)^2/2m_p = 27\,\mathrm{MeV}$. A few percent of m_pc^2 is 20–30 MeV, so this cyclotron is right at the limit. The proton's frequency drops as $1/\gamma$ and the particles fall out of step; synchrocyclotrons fix this by lowering the frequency as the protons speed up.

31.3 $B = p/0.3\rho = 6800/(0.3 \times 2800) = 8.1\,\mathrm{T}$. A 100 km ring with the same fill factor has $\rho = \frac{2}{3}(100\,\mathrm{km})/2\pi = 10.6\,\mathrm{km}$, and 50 TeV needs $B = 50\,000/(0.3 \times 10\,600) = 16\,\mathrm{T}$ — the design goal of the proposed Future Circular Collider.

31.4 The invariant is $s = (E + mc^2)^2 - p^2c^2$, which equals $2Emc^2 + 2m^2c^4 \approx 2Emc^2$. Fixed target: $\sqrt{2(6800)(0.938)} = 113\,\mathrm{GeV}$. Head-on: 13.6 TeV, 120 times more. To match the LHC, a fixed-target beam would need $E = s/2mc^2 = (13\,600)^2/(2 \times 0.938)\,\mathrm{GeV} \approx 10^{17}\,\mathrm{eV}$: such energies are found only in cosmic rays.

31.5 LEP: $88.5 \times 104^4/3100\,\mathrm{keV} = 3.3\,\mathrm{GeV}$ per turn, 3% of the beam energy, replaced every turn by the RF cavities. LHC: $88.5 \times 6800^4/2800 \times (1/1836)^4\,\mathrm{keV} = 6\,\mathrm{keV}$ per turn, utterly negligible as a fraction of the energy. But the LHC has 3×10^{14} protons per beam, each circulating 11 000

times a second: the radiated power is about 3.5 kW per beam. Removing one watt at 1.9 K takes about a kilowatt of refrigeration power at room temperature, so the radiation is intercepted by a "beam screen" at 5–20 K lining the beam pipe.

31.6 $\sigma L = 55 \times 10^{-36}\,\mathrm{cm}^2 \times 2 \times 10^{34}\,\mathrm{cm}^{-2}\mathrm{s}^{-1} = 1.1$ per second, or 10^7 per year; 0.23% of these is 25 000 diphoton events per year. But there are about a billion proton collisions per second, and ordinary processes make pairs of photons (and jets that mimic photons) at a rate thousands of times higher. The Higgs appeared as a small bump on a large, smooth background in the two-photon mass spectrum, which needed years of data, excellent energy resolution, and careful rejection of fakes. (A billion collisions a second also have to be filtered down to about a thousand recorded per second, in real time.)

31.7 $\cos\theta = 1/n\beta$ gives $\theta = 41^\circ$ for $\beta \approx 1$. Emission requires $\beta > 1/n = 0.75$, i.e. $\gamma > 1.52$ and $\beta\gamma > 1.14$: $p = \beta\gamma mc$ is 121 MeV/c for a muon and 1.07 GeV/c for a proton.

31.8 $p = 0.3 \times 3.8 \times 10 = 11.4\,\mathrm{GeV}/c$. $s = L^2/8R = 1/80\,\mathrm{m} = 12.5\,\mathrm{mm}$. Since $s \propto 1/R \propto 1/p$, $\delta p/p = \delta s/s = 50\,\mu\mathrm{m}/12.5\,\mathrm{mm} = 0.4\%$. At higher momenta the sagitta shrinks and the resolution worsens in proportion to p, which is why muon spectrometers are huge and their magnets strong.

31.9 $\ell = 1/(n\sigma) = 1/(7 \times 10^{30} \times 10^{-47}) = 1.4 \times 10^{16}\,\mathrm{m}$, one and a half light-years of lead. Reines and Cowan placed a few hundred litres of water, with cadmium dissolved in it, next to the Savannah River reactor. With about 3×10^{28} protons, the interaction rate $\Phi\sigma N \approx 10^{17} \times 10^{-47} \times 3 \times 10^{28}$ is a few per hundred seconds. The reaction $\bar{\nu}_e + p \to n + e^+$ produces a distinctive pair of signals: a prompt flash from the positron annihilating into two 511 keV photons, then a second flash a few microseconds later as the neutron is captured by cadmium. Requiring this delayed coincidence picked a few events per hour out of the background.

31.10 With $f_1 = f, f_2 = -f$: $1/F = 1/f - 1/f + d/f^2 = d/f^2$. In the other plane, $f_1 = -f, f_2 = f$, and again $1/F = d/f^2$. Either way, $F = f^2/d > 0$. A thin lens deflects a ray in proportion to its distance from the axis. In the plane where the first lens focuses, the beam arrives at the second (defocusing) lens narrower, so it gets a weaker defocusing kick. In the other plane, the first lens defocuses, so the beam arrives wider at the focusing lens and gets a stronger focusing kick. In both planes the focusing lens acts where the beam is wide and the defocusing lens where it is narrow, so focusing wins. Alternating gradients made accelerators like the LHC possible: the beam pipe can be centimetres wide instead of metres.

31.11 Each beam has $E = 500\,\mathrm{GeV}$. The power lost by a beam of current I is $I \times \Delta E/e$, so 100 MW at 10 mA allows $\Delta E = 10\,\mathrm{GeV}$ per turn. Then $88.5 \times 500^4/\rho\,\mathrm{keV} = 10^7\,\mathrm{keV}$ gives $\rho = 550\,\mathrm{km}$: a ring over 3500 km around, for one beam. A linear collider at 40 MV/m needs $500\,\mathrm{GeV}/(40\,\mathrm{MV/m}) = 12.5\,\mathrm{km}$ per beam, 25 km in all. Because synchrotron losses grow as E^4/ρ, the cost of a circular electron machine grows as E^2 (balancing ring size against power), while a linear collider's grows only as E. Above a few hundred GeV, electrons must go straight.

31.12 If the plasma electrons, density n, are displaced by a distance x, Gauss's law gives a field $E = nex/\epsilon_0$. The wave cannot displace them by much more than its own wavelength over 2π, about c/ω_p, before it breaks. So $E_{\max} \sim nec/\epsilon_0\omega_p = m_e c\,\omega_p/e$, using $\omega_p^2 = ne^2/\epsilon_0 m_e$. Numerically, $m_e c\,\omega_p/e = 96\sqrt{n[\mathrm{cm}^{-3}]}\,\mathrm{V/m}$. For $n = 10^{18}\,\mathrm{cm}^{-3}$: $10^{11}\,\mathrm{V/m} = 100\,\mathrm{GV/m}$, a thousand times more than RF cavities (30–100 MV/m, limited by sparking at the metal walls; a plasma is already ionized and cannot break down). A TeV would take ten metres instead of ten kilometres. The hard parts are keeping the accelerated bunch small and monoenergetic,

chaining stages together, and doing it efficiently, thousands of times a second.

32 Particle physics

32.1 (a) Forbidden: baryon number ($1 \to 0$) and lepton number ($0 \to -1$) change. Proton decay has never been seen (Problem 32.11). (b) Allowed: beta decay. (c) Forbidden: muon number ($1 \to 0$) and electron number ($0 \to 1$) change separately. (It is allowed in principle by neutrino mixing, but at a rate far too small ever to observe; seeing it would mean new physics.) (d) Allowed: this is how the π^0 decays, in 10^{-16} s. (e) Allowed by the weak interaction, which can change strangeness; it is the Λ's main decay, with a lifetime of 2.6×10^{-10} s, long because it is weak. (f) Forbidden by energy and momentum conservation: in the centre-of-mass frame the total momentum is zero, which a single photon cannot have.

32.2 $p = uud$: $\frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +1$. $n = udd$: $\frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0$. $\Delta^{++} = uuu$: $+2$. $\Omega^- = sss$: -1. $\pi^+ = u\bar{d}$: $\frac{2}{3} + \frac{1}{3} = +1$. $K^- = s\bar{u}$: $-\frac{1}{3} - \frac{2}{3} = -1$.

32.3 $\hbar c/m_W c^2 = 197\,\text{MeV fm}/80\,400\,\text{MeV} = 2.5{\times}10^{-3}$ fm, about 1/300 of a proton's radius. The weak coupling $g \approx 0.65$ is actually twice the electromagnetic $e \approx 0.3$. But at energies $E \ll m_W c^2$, the exchanged W is far off its mass shell, and the amplitude is suppressed by about $(E/m_W c^2)^2$. At LHC energies, the weak force is as strong as electromagnetism, as electroweak unification says it should be.

32.4 $\Gamma(\tau \to e\nu\bar{\nu}) = \Gamma_\mu (m_\tau/m_\mu)^5$, and this is 17.8% of the total, so

$$\tau_\tau = 0.178\,\tau_\mu \left(\frac{m_\mu}{m_\tau}\right)^5 = 0.178 \times (2.20 \times 10^{-6}\,\text{s}) \times \left(\frac{105.7}{1777}\right)^5 = 2.9 \times 10^{-13}\,\text{s},$$

in excellent agreement. In natural units a rate is an energy; G_F^2 has dimensions of energy^{-4}, so the rate must be $G_F^2 m^5$ times a number. (Physically: the amplitude is $\propto G_F$ and the phase space for three light particles sharing energy m grows as m^5.) The agreement checks that the tau couples to the W with the same strength as the muon and electron: *lepton universality*.

32.5 Below charm (u, d, s): $N_c(\frac{4}{9} + \frac{1}{9} + \frac{1}{9}) = \frac{2}{3}N_c$. With charm: $\frac{10}{9}N_c$. With bottom: $\frac{11}{9}N_c$. For $N_c = 3$ these are 2, 3.33, and 3.67, matching the data. (With $N_c = 1$ the predictions would be three times too small.) The measured values are a few percent higher still, because of gluon emission, a correction of α_s/π.

32.6 The flavour-change probability is maximal when $\Delta m^2 c^4 L/4\hbar c E = \pi/2$; in practical units $1.27\,\Delta m^2[\text{eV}^2]\,L[\text{km}]/E[\text{GeV}] = \pi/2$, giving $L \approx 500$ km. Neutrinos from the sky above have travelled only 10–20 km from where cosmic rays made them, too short to oscillate. Those from below have crossed the Earth, up to 13 000 km, many oscillation lengths; averaged over energies, half of their muon neutrinos have turned into tau neutrinos, which Super-Kamiokande could not see. This up–down asymmetry, measured in 1998, was the discovery of neutrino mass.

32.7 With $m_u \approx 2.2$ MeV and $m_d \approx 4.7$ MeV: proton 9 MeV, neutron 12 MeV, about 1% of their masses. The other 99% is the energy of the quarks' motion and of the gluon field confining them, as $E = mc^2$ requires. With massless quarks, the proton would still weigh about 900 MeV. Nearly all the mass of ordinary matter, including you, is QCD binding energy, not Higgs-given mass. (The neutron outweighs the proton partly because $m_d > m_u$, which is why hydrogen is stable.)

32.8 At 91 GeV: $12\pi/(23\ln(91^2/0.2^2)) = 0.13$ (measured 0.118; the one-loop formula and the value of Λ are approximate). At 2 GeV: $12\pi/(27\ln 100) = 0.30$. As $Q \to \Lambda$, the logarithm goes to zero and α_s blows up: perturbation theory fails and quarks are confined. The positive sign of $33 - 2n_f$ means the coupling *decreases* at high energy, so quarks behave almost as free particles inside a proton probed at high energy: asymptotic freedom. The 33 comes from gluons, which carry colour charge themselves and anti-screen; the $-2n_f$ from quark loops, which screen as electrons do in QED. With 17 flavours, $33 - 34 < 0$: screening wins, and QCD would behave like QED, weak at low energies and strong at high.

32.9 $y_f = \sqrt{2}m_f/v$: electron 2.9×10^{-6}, bottom 0.024, top 0.99. The top quark's coupling is almost exactly 1, the only fermion mass of the natural size; the others are puzzlingly small. The top is so heavy that it decays (in 5×10^{-25} s) before it can form hadrons, so it is the only quark studied bare, and its loops dominate the quantum corrections to the Higgs mass and the stability of the vacuum.

32.10 The spatial wavefunction is symmetric (all in the ground state), the spin wavefunction ↑↑↑ is symmetric, and so is the flavour *uuu*: exchanging any two quarks leaves the state unchanged, but identical fermions require it to change sign. Multiplying by the colour wavefunction $\epsilon_{ijk}q_iq_jq_k$, which changes sign when any two indices are swapped, makes the total antisymmetric, as Pauli demands. The combination $\epsilon_{ijk}q_iq_jq_k$ is unchanged by any $SU(3)$ rotation of the colours (it is a determinant, and these rotations have determinant 1), so it is colour-neutral, "white". The only other simple neutral combination is $\bar{q}^iq_i$, a meson. If only colour-neutral states can exist freely, this explains why hadrons come as qqq and $q\bar{q}$, and never as free quarks or qq pairs.

32.11 5×10^7 kg of water is 2.8×10^9 moles, or 1.7×10^{33} molecules, each with 10 protons: 1.7×10^{34} protons. With $\tau = 10^{34}$ years, about 1.7 decay per year. If no decays are seen in T years, Poisson statistics say the expected number NT/τ is probably less than about 2.3 (since $e^{-2.3} = 0.1$, a larger expected number would make seeing none less than 10% likely). So $\tau \gtrsim NT\epsilon/2.3$, with ϵ the detection efficiency. Super-Kamiokande's limit for $p \to e^+\pi^0$ is over 10^{34} years, which has ruled out the simplest grand unified theories.

32.12 *Baryon number violation*: starting from $B = 0$, you cannot reach $B \neq 0$ if every process conserves B. *C and CP violation*: if C were exact, every process making baryons would be matched by its mirror process making antibaryons at the same rate, and nothing would be gained. The same is true with CP, which also flips the handedness of particles: both must fail, or for every process producing an excess, its conjugate produces an equal deficit. *Departure from equilibrium*: in thermal equilibrium, the population of each state depends only on its energy, and by CPT symmetry a particle and its antiparticle have exactly the same mass. If baryon number is not conserved, there is no chemical potential to favour it either, so equal numbers of baryons and antibaryons result, whatever processes are at work. The expanding universe provides departures from equilibrium (heavy particles decaying out of equilibrium, or phase transitions). The CP violation measured in the Standard Model is about ten orders of magnitude too weak to produce one baryon per billion photons: our existence is evidence of physics we do not yet know.

32.13 At time t, the state is $\cos\theta\, e^{-iE_1t/\hbar}\nu_1 + \sin\theta\, e^{-iE_2t/\hbar}\nu_2$, and the amplitude to find ν_e is $\cos^2\theta\, e^{-iE_1t/\hbar} + \sin^2\theta\, e^{-iE_2t/\hbar}$. Its squared magnitude is

$$\cos^4\theta + \sin^4\theta + 2\sin^2\theta\cos^2\theta\cos(\Delta Et/\hbar) = 1 - \sin^2 2\theta\sin^2(\Delta Et/2\hbar),$$

using $\cos^4\theta + \sin^4\theta = 1 - 2\sin^2\theta\cos^2\theta$ and $1 - \cos x = 2\sin^2(x/2)$. With $\Delta E = \Delta m^2c^4/2pc \approx \Delta m^2c^4/2E$ and $t = L/c$, the phase is $\Delta m^2c^4L/4\hbar cE$. If $m_1 = m_2$, the two components keep the same relative phase forever and the neutrino stays a ν_e. Oscillation measures differences of squared masses, so seeing it proves that at least one neutrino has mass — but not how much.

33 String theory and quantum gravity

33.1 Write $\hbar^x c^y G^z$, with $[\hbar] = \mathrm{ML^2T^{-1}}$, $[c] = \mathrm{LT^{-1}}$, $[G] = \mathrm{M^{-1}L^3T^{-2}}$. For a length, the powers of M, L, and T must be 0, 1, and 0: $x - z = 0$, $2x + y + 3z = 1$, $-x - y - 2z = 0$, with the unique solution $x = z = \frac{1}{2}$, $y = -\frac{3}{2}$. Similarly for mass and time. Hence $\ell_P = \sqrt{\hbar G/c^3} = 1.6 \times 10^{-35}\,\mathrm{m}$, $m_P = \sqrt{\hbar c/G} = 2.2 \times 10^{-8}\,\mathrm{kg}$ (about a flea's egg), $t_P = \sqrt{\hbar G/c^5} = 5.4 \times 10^{-44}\,\mathrm{s}$, and $T_P = m_Pc^2/k_\mathrm{B} = 1.4 \times 10^{32}\,\mathrm{K}$.

33.2 $E_P = 1.2 \times 10^{19}\,\mathrm{GeV} = 1.2 \times 10^{28}\,\mathrm{eV}$: 15 orders of magnitude above the LHC, and 8 above the most energetic cosmic rays.

33.3 $\hbar/mc = 2Gm/c^2$ gives $m = \sqrt{\hbar c/2G} = m_P/\sqrt{2} = 1.5 \times 10^{-8}\,\mathrm{kg}$. To localize a particle within Δx, you need a probe of wavelength Δx, carrying energy $\sim \hbar c/\Delta x$. Below the Planck length, that energy in that region forms a black hole larger than Δx: the harder you try to look, the bigger the black hole that hides what you are looking at. Distances below ℓ_P may not mean anything.

33.4 $S = k_\mathrm{B}A/4\ell_P^2 = \pi k_\mathrm{B}r_s^2/\ell_P^2$, with $r_s = 3.0\,\mathrm{km}$: $S = 1.0 \times 10^{77}k_\mathrm{B}$, or 1.5×10^{77} bits — 10^{19} times the entropy of the Sun. Since $S \propto M^2$, Sagittarius A* ($4 \times 10^6 M_\odot$) has $S \approx 2 \times 10^{90}k_\mathrm{B}$, more than the entire cosmic microwave background. The entropy of the universe is dominated by its black holes.

33.5 $r_s = 2GM/c^2 = 3 \times 10^{-16}\,\mathrm{m}$, about a third of a proton's radius. $T = \hbar c^3/8\pi GMk_\mathrm{B} = 6 \times 10^{11}\,\mathrm{K}$, or $k_\mathrm{B}T \approx 50\,\mathrm{MeV}$: it emits not only photons but electrons, muons, and pions. As it shrinks it heats up and radiates faster, and the end is an explosion. Since lifetime scales as M^3, the black hole in its last second has $2.6 \times 10^5\,\mathrm{kg}$, releasing $2 \times 10^{22}\,\mathrm{J}$ — about five million megatons — in a final burst of gamma rays. Telescopes have looked for such bursts and seen none, which limits how many primordial black holes there could be.

33.6 A massless particle in five dimensions has $E^2 = (p^2 + p_5^2)c^2$. Seen in four dimensions, p_5c plays the role of a rest energy mc^2, so $m_n = |p_5|\,/c = n\hbar/Rc$: a tower of equally spaced masses. No tower below a few TeV means $\hbar c/R \gtrsim 3\,\mathrm{TeV}$, so $R \lesssim \hbar c/(3\,\mathrm{TeV}) = 197\,\mathrm{MeV\,fm}/(3 \times 10^6\,\mathrm{MeV}) \approx 7 \times 10^{-20}\,\mathrm{m}$: a ten-thousandth of a proton.

33.7 The gravitational energy is $G(E/c^2)^2/r$ with $r = \hbar c/E$, i.e. $GE^3/\hbar c^5$; dividing by E gives $GE^2/\hbar c^5 = (E/E_P)^2$, using $E_P^2 = \hbar c^5/G$. Two electrons at rest: $(0.511\,\mathrm{MeV}/E_P)^2 = 2 \times 10^{-45}$. Two LHC protons: $(6.8\,\mathrm{TeV}/E_P)^2 = 3 \times 10^{-31}$. Gravity is weak, but like Fermi's theory, its effective strength grows as E^2 and reaches 1 at E_P, where perturbation theory breaks down. Fermi's theory broke down at 300 GeV and the W appeared; something new must appear by the Planck energy — perhaps strings.

33.8 $\pi r^2/\ell_P^2 = 1.2 \times 10^{70}$ for $r = 1\,\mathrm{m}$. A sphere of air of that radius ($4\,\mathrm{m}^3$) has about $4 \times 10^{26}k_\mathrm{B}$: 10^{43} below the bound. Ordinary matter has entropy proportional to its volume, roughly one k_B per particle. To approach the bound you would have to pack in so much matter that it would collapse into a black hole first. Only black holes saturate the bound.

33.9 $\rho = p/0.3B = 1.2 \times 10^{19}/(0.3 \times 8)\,\mathrm{m} = 5 \times 10^{18}\,\mathrm{m}$, about 540 light-years, comparable to the distances of many of the stars you can see with the naked eye. The synchrotron loss per turn, $88.5\,E^4/\rho \times (m_e/m_p)^4$ in keV, comes to $3 \times 10^{40}\,\mathrm{GeV}$ — 10^{21} times the beam's energy. The protons would radiate their energy away almost at once. Even in principle, there is no escape: concentrating the Planck energy in a Planck-sized region makes a Planck-mass black hole. Colliding particles at still higher energies makes bigger black holes, which probe *larger* distances, not smaller. Gravity itself seems to forbid seeing below the Planck length.

33.10 $\hbar/4\pi k_\mathrm{B} = 6.1 \times 10^{-13}\,\mathrm{K\,s}$. Water: $10^{-3}/(4 \times 10^6) = 2.5 \times 10^{-10}\,\mathrm{K\,s}$, about 400 times the bound. In kinetic theory, $\eta \sim np\ell$ (momentum carried across a distance ℓ, the mean free path) and $s \sim nk_\mathrm{B}$, so $\eta/s \sim p\ell/k_\mathrm{B}$; the uncertainty principle requires $p\ell \gtrsim \hbar$. A fluid near the bound is one whose particles scatter so strongly that their mean free path is about their own quantum wavelength: there are no well-defined particles at all, and disturbances are damped as fast as quantum mechanics allows. The quark–gluon plasma flows with almost no friction: the most perfect fluid known.

33.11 If the black hole forms from matter in a pure state, then black hole plus radiation is always pure (if evolution is unitary), and the entanglement entropy of the radiation equals that of the black hole. Early on, each Hawking quantum is entangled with a partner behind the horizon, and the radiation's entropy rises. But the black hole's entanglement entropy cannot exceed its Bekenstein–Hawking entropy, the logarithm of the number of its states, and this shrinks as it evaporates. So the radiation's entropy must turn over when the two curves cross, when the black hole has lost about half its entropy (since $S \propto M^2$, at $M = M_0/\sqrt{2}$, about 65% of the way through its life). When the black hole is gone the radiation, now everything, is in a pure state, with entropy zero. In Hawking's calculation the radiation is exactly thermal and its entropy rises monotonically to the end, leaving a mixed state from a pure one: a violation of unitarity. In 2019, "island" calculations in semiclassical gravity reproduced Page's curve, though how the information gets out remains debated.